IV

VOLUME FOUR OF FOUR

The Question Bank

Second series — whether it has stuck

1,023 examination questions in thirty blocks — 2,043 marks
768 single best answers and 255 multiple true/false
Every answer explained, every question referenced
247 memory devices, on the answers that carry a list
A complete answer-key grid for rapid marking
A mark sheet: one row per block, three attempts
207 viva questions with model answers, and a viva tracker
An index by question number

A separate volume, not a replacement: Volume III stands unchanged. This one is pitched at examination standard, and every question carries a reference to the unit of Volume I and the section of Volume II that cover it, so a question you get wrong names its own remedy.

For candidates sitting the examination of 5 November 2026

Dr. Kam O Man, Owen

BDS (HK) · MFDS RCSEd · MFDS RCPS (Glasg)
MClinDent (Fixed and Removable Prosthodontics) (KCL)
MImpDent (RCSEd) · MGDS RCSI

An independent revision aid. Not an official publication of, nor endorsed by, the Hong Kong Institute of Orthodontics, Shanghai Jiao Tong University, or the Royal College of Surgeons of Edinburgh.

The set

Four volumes, in the order you should use them — **III** and **IV** being two question banks over the same ground, not a sequence. Each is cross-referenced to the others, so you can enter from whichever you happen to have open.

I	**Notes In Orthodontics** Revised edition · 277 pp	Teaches the material. A crash notebook of forty-two schematics and four number cards; the twenty essay topics; the sixteen chapters; ethics, radiation, infection control and governance.
II	**A Candidate's Revision Handbook** 164 pp	Maps the material onto the examination. Thirty sections against the sixty-five curriculum items, six number cards, the diagnostic cases, the case presentation, the oral, and the dated countdown.
III	**The Question Bank** 302 pp	The first question bank. 1,016 single best answers in thirty blocks, a mark sheet, 207 vivas. **Unchanged and still usable** — but its stems are short and it largely tests recall.
IV	**The Question Bank, Second Series** **This volume** · 460 pp	A separate, harder bank on the same thirty-block spine. 1,023 questions worth 2,043 marks: 768 vignette single best answers and 255 multiple true/false, every one explained, referenced to Volumes I and II and often given a memory device. Index by question number.

Which way to move

Forwards to learn, backwards to repair. Read the **Volume I** unit, follow its teal strap to the **Volume II** section, then sit the matching block *here*. Under 70 per cent and the flow reverses: the mark sheet names the section, and its rust strap names the unit.

Volumes III and IV are alternatives, not a sequence. Both run one-to-one against Volume II's thirty sections — Block A is §1, Block AD is §30 — so a week's reading feeds either. **Work this one**; keep Volume III for a fast second pass. And because the weeks follow **II** and **IV**, *Volume I is read out of its own order* — never front to back.

When	Which book
Learning something cold	**I**
Checking you are covered against the syllabus	**II**
The end of every session, without exception	**IV**
Repairing a block scored under 70 per cent	**IV** → **II** → **I**
A fast second pass over a block already repaired	**III**
The last three days	**I**'s crash notebook and **II**'s number cards
The case presentations, the diagnostic exercises, the oral	**II** only

Volume I **Part I**, despite its number — **Last, not first** — you cannot compress what you have not yet learned

The plan

Seven working weeks and a weekend, ending **Thursday 5 November 2026**. The written paper is the engine — revising for it is simultaneously revising for the oral and the diagnostic cases, because all three test the same knowledge. The case submissions depend on other people, so their *requests* go out first.

Before Monday — one session, about three hours

1. Pick your three write-up cases and your two presentation cases. For each, list what is missing: pre-treatment records, radiographs, photographs, models or scans, progress notes, debond records, PAR scoring. **2.** Send every request the same afternoon — patient recalls, radiograph retrieval, trainer sign-off. **3.** Ask the Programme Management Office when the three written-up cases are due. If that date falls before 5 November, it outranks everything below. **4.** Sit Blocks **H** and **AD** of this volume cold. 55 questions, ethics and governance — the likeliest thin spot and the cheapest marks in the paper.

The spine is the curriculum, not the book. Volume II's thirty sections and this volume's thirty blocks run one-to-one — §1 is Block A, §30 is Block AD — so the weeks below advance through them in order. The Volume I column then names the units whose *main* treatment sits in that week's sections. **Volume I has two separate series and they are not the same thing**: the twenty essay **topics** of Part II and the sixteen **chapters** of Part III. Chapter 1 is Evidence-Based Orthodontics; Topic 1 is the biological basis of tooth movement. Every chapter and every topic appears exactly once below.

Week	Dates	Curriculum	Read in Volume I	Alongside
1	14–20 Sep	§1–§5 · Blocks A–E	Chapters 2–6 · Topic 1	Viva V1–V20 aloud
2	21–27 Sep	§6–§10 · Blocks F–J	Chapters 1, 7 · Topic 2 · Part V.1	V21–V48. First diagnostic drill
3	28 Sep–4 Oct	§11–§15 · Blocks K–O	Chapters 8, 11 · Topics 3–5, 7	V49–V80. Write up case 1
4	5–11 Oct	§16–§20 · Blocks P–T	Chapters 9, 10, 12–14 · Topics 6, 8, 13–15, 20	V81–V111. Write up case 2
5	12–18 Oct	§21–§25 · Blocks U–Y	Topics 11, 12, 16, 17, 19	V112–V151. Write up case 3
6	19–25 Oct	§26–§30 · Blocks Z–AD	Chapters 15, 16 · Topics 9, 10, 18 · Part V.2–V.4	V152–V207
7	26 Oct–1 Nov	Re-sit every block scored under 70%	Only what the mark sheet sends you back to	Mock oral. Present both cases aloud
Final	2–4 Nov	Two or three blocks a session at examination pace	Crash notebook · number cards	Nothing new

Week 5 lists no chapters because §21–§25 are Chapter 13 territory, read in week 4; that week is topic-led. Part V has no chapter or topic of its own — it is new in this edition.

The daily loop — about two and a half hours

One **Volume I** unit (45 min) → the **Volume II** section its teal strap names (20 min) → that **Volume IV** block, timed at 75 seconds a question (40 min) → read the explanation for *every* question, including the ones you got right (30 min).

Log the score on the mark sheet. Anything under 70 per cent goes on the week 7 list. Close each session with one viva section, aloud, to the end, before reading the model.

THREE RULES THAT DECIDE THIS

1. **Speak from week one, not week seven.** The oral tests whether you can produce a structured spoken answer under pressure — a separate skill from knowing the material. Question blocks are measurable and satisfying, which is exactly why they expand to fill the time and crowd out the one thing you cannot cram.
2. **Protect the case deadline.** You cannot revise your way out of a missing pre-treatment photograph. A patient recall is a three-to-four week loop you do not control.
3. **Do not re-read Volume I cover to cover.** Use it as a textbook — when an explanation here does not land, follow the strap back to that one unit.

Contents

How to use this volume vii
Finding your way around x
Concordance with *Notes In Orthodontics* xii

Part 1 — 1023 examination questions **1**
Block A — Concepts of growth and development 5
Block B — Craniofacial growth, prenatal and postnatal 11
Block C — Development of the dentition 17
Block D — The biology of tooth movement, eruption and root resorption 26
Block E — Occlusion, malocclusion and aetiology 35
Block F — Treatment need, indices and epidemiology 41
Block G — Evidence-based orthodontics 47
Block H — Ethics, consent, communication and the psychosocial patient 54
Block I — Clinical examination and records 61
Block J — Imaging and cephalometrics 67
Block K — Study models, space analysis and the digital set-up 76
Block L — Diagnosis, the problem list and treatment planning 82
Block M — Extraction decisions and space management 92
Block N — Biomechanics, force control and anchorage 102
Block O — Orthodontic materials, wires, bonding and instruments 114
Block P — Removable and functional appliances 124
Block Q — Fixed appliances — labial, lingual and aligners 133
Block R — Treatment mechanics: Stage I, II and III 142
Block S — Retention, relapse and stability 150
Block T — Interceptive orthodontics and the developing dentition 158
Block U — Class I malocclusion and bimaxillary protrusion 168
Block V — Class II malocclusion 174
Block W — Class III malocclusion 183
Block X — Transverse problems and facial asymmetry 189
Block Y — Vertical problems: deep bite and open bite 194
Block Z — TMD, condylar trauma, the airway and myofunctional therapy 200
Block AA — Adult orthodontics and orthognathic surgery 206

Block AB — Cleft lip and palate and craniofacial syndromes 215
Block AC — Risks of orthodontic treatment and the medical history 224
Block AD — Orthodontic practice, infection control and governance 232

Part 2 — Answers, explanations and references **239**
Answer key grid 378
Mark sheet 381

Part 3 — 207 viva questions with model answers **383**
Growth, development and the craniofacial complex 386
Development of the dentition 389
The biology of tooth movement and root resorption 392
Occlusion, aetiology and epidemiology 395
Treatment need, indices and the evidence base 397
Ethics, consent and communication 400
Clinical examination and records 403
Space analysis, diagnosis and treatment planning 406
Extraction decisions and space management 409
Biomechanics and anchorage 412
Materials, wires and bonding 415
Removable, functional and fixed appliances 418
Treatment mechanics and finishing 421
Retention and stability 423
Interceptive orthodontics and impacted teeth 426
Class II and Class III management 429
Transverse and vertical problems 432
TMD, the airway and the medically complex patient 435
Adults, orthognathic surgery, clefts and governance 438
The questions that test judgement 441
Viva tracker 444
Index 445

How to use this volume

This is the volume that tells you whether the material has stuck. It contains **1,023 questions worth 2,043 marks**, arranged in thirty blocks that run one-to-one against the thirty sections of Volume II; answers with explanations, references and memory devices; 207 viva questions with model answers; and an index by question number.

Why there are two question banks

Volume III is not superseded and has not been altered. It remains exactly as it was: 1,016 single best answers, shorter stems, one mark each. What it tests, though, is largely *recall* — its mean stem ran to 82 characters and only one question in ten carried a clinical scenario. That is useful early, when you are checking whether a fact has gone in, and it is much faster to work.

This volume was written to a different standard. The mean stem is 199 characters, two questions in three are set in a clinical situation, and the paper is mixed: some questions ask for the single best answer, others give five statements to judge independently. **Nothing here can be answered from a glossary.** Every question requires a decision, an inference, or a number applied to a situation.

	Volume III	Volume IV — this one
Questions	1,016, all single best answer	1,023, of two types
Marks	1,016	2,043
Mean stem	82 characters	199 characters
Clinical scenario	1 question in 10	2 questions in 3
Memory devices	none	247
Back matter	mark sheet, viva tracker	mark sheet, viva tracker, index
Use it	for a fast pass over a block you have already repaired	**at the end of every study session**

Both banks use the same thirty blocks against the same thirty Volume II sections, so a week's reading feeds either. Work this one; keep Volume III for speed.

The two question types

Type	Marks	How it is answered
Single best answer	1	Five options, one of which is the *best* — not the only defensible one. 768 in this bank. Most are set in a clinical situation: the sifting of relevant findings from irrelevant ones is part of what is being tested.
Multiple true/false	5	Five statements judged *independently*, one mark each. Any number may be true, **including all five and none of them**. 255 in this bank. Do not assume a comfortable two or three.

Because a true/false question carries five marks, the mark totals on the mark sheet exceed the number of questions. Score every block out of its own mark total, not out of its question count.

The three-pass method

Pass	What to do
First	Work one block at a time, under timed conditions, at 75 seconds per question — the vignettes are longer than a recall question and reading them properly is part of the skill. Mark the whole block before reading any answer. Read the explanation for *every* question in the block, including the ones you got right.
Second	Two weeks later, repeat any block you scored under 70% on. If you score under 70% again, stop and re-read the corresponding Handbook section before trying a third time.
Third	In the final fortnight, work the answer-key grid backwards: cover the key, sit two or three blocks in one session at examination pace, and mark them together. This trains stamina, which is the part of a two-hour paper nobody practises.

Reading the answers

Each answer gives the key, the reasoning, and a reference. **The reasoning is more examinable than the fact**: a question you answered correctly for the wrong reason will be answered wrongly when the stem is rephrased. For a single best answer the explanation also names the most tempting wrong option and says why it fails — by its *content*, never by its letter, so it still reads correctly whichever slot the option ended up in. For a true/false question there is one note per statement, in order.

The memory devices

Where an answer turns on a *list*, an ordered sequence, a classification, or a set of numbers that go together, a one-line device follows it, flagged MNEMONIC. There are **201 in the answers and 46 in the viva bank**. They exist for the oral rather than the paper: a spoken answer with no structure to hang on falls apart under pressure, and the commonest way to lose a mark is to produce four items of a five-item list.

Where a device is already in circulation — **MOCDO** for ranking the IOTN dental health component, **MJTHREADS** for the medical history, **PARQ** for the consent discussion, **SLOB** for parallax, the **rule of tens**, **ALARP** — it is named, so you recognise it if an examiner uses it. The rest were written for this volume. **Roughly one answer in five carries one**; the others do not, because a device attached to a single fact or to a judgement is padding, and padding is what makes a candidate distrust the whole book.

One warning about these

A mnemonic is a *handle*, not an answer. It will not get you a mark on its own, and an examiner who hears the acronym without the reasoning will ask for the reasoning next. Learn the list first; use the device to check you have not dropped an item.

References, and how to use them

Every question ends with a reference in square brackets to **your own earlier volumes** — the unit of *Notes In Orthodontics* whose main treatment covers it, and the Handbook section, to the subsection where one can be justified:

`[Vol I Topic 12 · Vol II §22.3]`

Nothing points outside the set, so a question you got wrong resolves to a page you already own. The index at the back runs the other way: look a term up and it names the block to sit and the individual questions elsewhere that turn on it.

The viva bank

Read the question, close the book, and answer it **out loud** to the end before reading the model. Answering in your head is not practice. The model answers show the shape of a good spoken answer — a direct opening sentence, the reasoning, the number where a number exists, and a close that says what you would actually do — and the italic note beneath each says what the question is really testing. These are the same 207 questions as in Volume III, with the memory devices added.

TWO NOTES ON GUESSING

The key has been distributed as evenly across A to E as 768 questions divide — 153 or 154 on each letter — so a run of the same letter means nothing, and letter frequency will not help you. Nor will option length: the options within each question have been written to comparable lengths, so the longest is not the answer. In the true/false questions the number of true statements ranges across all six possibilities, from none to all five, roughly one in eight being wholly true or wholly false. There is no pattern in this bank to find.

Finding your way around

This is **Volume IV** of four, and it is built to be used *with* Volumes I and II on your desk, not instead of them. It is a second, harder question bank on the same thirty-block spine as **Volume III**, which stands unchanged — the two are alternatives, not a sequence. Every block of questions names the **Volume II** section it tests and the part of **Volume I**, *Notes In Orthodontics*, that the material came from, so a block you score badly on tells you exactly what to re-read — and in which book.

If you know what you want, start here

What you want to do	Where to go
Test one curriculum topic	Part 1, the block named in the index overleaf
Test one *Notes In Orthodontics* essay topic or chapter	The concordance overleaf
Mark a block quickly	Part 2, the answer-key grid at the end
Understand why you got one wrong	Part 2, the explanation for that question
Find which Handbook section to re-read	The last column of the mark sheet, or the strap at the head of the block
Look a term up rather than a block	The index at the very back, by question number
Practise speaking	Part 3, the viva bank
Rehearse the set-piece answers	Revision Handbook Part 5 first, then Part 3 here
Track what you have done	The mark sheet at the end of Part 2, and the viva tracker at the end of Part 3

How each block is laid out

Element	What it is for
Block head	Each block starts on a fresh page and its name runs along the top of every page, so you can flip to it
Rust strap	Where this material sits in *Notes In Orthodontics*; which Revision Handbook section to re-read; how many questions and how long to allow
Questions	Numbered continuously 1–1023 across the whole bank, so a question number is unambiguous. A question marked **TRUE OR FALSE** carries five statements and five marks; the rest carry one
Answers, Part 2	In the same block order. **The answer number is a link back to the question** in the electronic version. Each answer gives the key, the reasoning, and a reference to the Volume I unit and Volume II section that cover it
Answer-key grid	Eight answers to a row, for marking a block in under a minute. A true/false answer appears as its five-letter string
MNEMONIC	A one-line memory device, on the answers whose content is a list, a sequence or a set of numbers — about one in five. It is a handle for the oral, not a substitute for the reasoning
Mark sheet	One row per block, three attempts, with the question count, the mark total and the Handbook section
Index	At the very back, by *question number*: look a term up and it names the block to sit

THE ONE HABIT THAT MATTERS

Mark a whole block before reading any answer, then read the explanation for *every* question in it — including the ones you got right. A question answered correctly for the wrong reason will be answered wrongly when the stem is rephrased, and rephrasing is what examiners do.

Concordance with *Notes In Orthodontics*

If you have been revising topic by topic from the Companion, use these two tables to find the questions that test what you have just read. Question ranges are generated, so they stay correct.

The twenty essay topics

	Essay topic	Questions	Viva
1	The biological basis of tooth movement	Block D, Q97–141	V21–V30
2	Cephalometry: uses and limitations	Block J, Q288–327	V59–V68
3	Factors in formulating a treatment plan	Blocks K and L, Q328–352, Q353–392	V69–V80
4	Factors affecting the choice of extraction	Block M, Q393–432	V81–V90
5	Anchorage and its clinical significance	Block N, Q433–485	V91–V102
6	The three orders and the evolution of fixed appliances	Block Q, Q572–612	V112–V121
7	The ideal archwire; steel versus nickel titanium	Block O, Q486–530	V103–V111
8	Functional appliances	Block P, Q531–571	V112–V121
9	Benefits and complications of treatment	Block AC, Q969–998	V49–V58, V174–V183
10	Medical conditions relevant to orthodontics	Block AC, Q969–998	V174–V183
11	Overbite and methods of overbite reduction	Block Y, Q848–877	V162–V173
12	Reducing overjet in a growing child	Block V, Q753–792	V152–V161
13	Management of a missing maxillary central incisor	Block T, Q683–727	V140–V151
14	Management of an impacted maxillary canine	Block T, Q683–727	V140–V151
15	Management of an impacted lower second molar	Block T, Q683–727	V140–V151
16	Methods of crossbite correction	Block X, Q823–847	V162–V173
17	Correcting a Class III incisor relationship	Block W, Q793–822	V152–V161
18	Malocclusion and temporomandibular disorder	Block Z, Q878–907	V174–V183
19	Mild facial asymmetry in a growing child	Block X, Q823–847	V162–V173
20	Late lower incisor crowding	Block S, Q648–682	V130–V139

The sixteen chapters

	Chapter	Questions	Viva
1	Evidence-Based Orthodontics	Block G, Q197–226	V39–V48
2	Occlusion and Malocclusion	Blocks E and F, Q142–171, Q172–196	V31–V48
3	Prenatal Development of the Craniofacial Region	Block B, Q27–56	V1–V10
4	Postnatal Growth of the Craniofacial Region	Blocks A and B, Q1–26, Q27–56	V1–V10
5	Development of the Dentition	Block C, Q57–96	V11–V20

6	Orthodontic Tooth Movement	Blocks D and N, Q97–141, Q433–485	V21–V30, V91–V102
7	The Orthodontic Patient: Examination and Diagnosis	Blocks I and J, Q258–287, Q288–327	V59–V68
8	The Orthodontic Patient: Treatment Planning	Blocks K–M, Q328–352–Q393–432	V69–V90
9	Functional Jaw Orthopaedics	Blocks P and X, Q531–571, Q823–847	V112–V121, V162–V173
10	Contemporary Fixed Appliances	Blocks O, Q, R, Q486–530, Q572–612–Q613–647	V103–V129
11	Skeletal Anchorage	Block N, Q433–485	V91–V102
12	Management of the Developing Dentition	Block T, Q683–727	V140–V151
13	Management of the Permanent Dentition	Blocks R–Y, Q613–647–Q848–877	V122–V173
14	Clear Aligners	Block Q, Q572–612	V112–V121
15	Orthodontics and Orthognathic Surgery	Block AA, Q908–937	V184–V193
16	Cleft Lip and Palate, and Syndromes	Block AB, Q938–968	V184–V193

THE TWO BLOCKS THE FIRST EDITION LEFT UNCOVERED

Block **AD** (Q999–1023, orthodontic practice, infection control and governance) had no equivalent unit in the first edition of *Notes In Orthodontics*, and much of Block **H** (Q227–257, ethics, consent and communication) was only touched on there. Both are explicit curriculum items — items 63, and 9 and 10 — and both are cheap marks in the written paper. The revised Companion now covers them in **Part V**: V.1 ethics and consent, V.2 radiation protection, V.3 infection control, V.4 governance and audit. Sit these two blocks early rather than last, and read Part V alongside them.

Part 1 — 1023 examination questions

How to use this bank. Work one block at a time under timed conditions. Allow **75 seconds per question** — the vignettes are longer than a recall question and reading them properly is part of the skill. Mark a whole block before reading any answer, then read the explanation for *every* question in it, including the ones you got right: the reasoning is more examinable than the fact.

Each block maps to one section of Volume II, so a block you score poorly on names its own remedy.

TWO QUESTION TYPES, AND HOW THEY ARE MARKED

Single best answer. A clinical vignette or a judgement question, then five options. Exactly one is the best answer — others may be partly defensible, which is the point. One mark.
True or false. A scenario or proposition, then five statements judged *independently*. Any number may be true, **including all five and none of them**. Do not assume a comfortable mixture. One mark per statement, so each of these questions is worth five — which is why the totals on the mark sheet exceed the number of questions.

The blocks

Block	Questions	Marks	Vol II	Subject, and where it sits in *Notes In Orthodontics*
A	1–26	50	§1	Concepts of growth and development Chapter 4 (Postnatal Growth)
B	27–56	58	§2	Craniofacial growth, prenatal and postnatal Chapters 3 and 4 (Prenatal Development; Postnatal Growth)
C	57–96	80	§3	Development of the dentition Chapter 5 (Development of the Dentition)
D	97–141	89	§4	The biology of tooth movement, eruption and root resorption Topic 1 (The biological basis of tooth movement); Chapter 6
E	142–171	58	§5	Occlusion, malocclusion and aetiology Chapter 2 (Occlusion and Malocclusion)
F	172–196	49	§6	Treatment need, indices and epidemiology Chapter 2, pp. 106–112 (IOTN and PAR)
G	197–226	62	§7	Evidence-based orthodontics Chapter 1 (Evidence-Based Orthodontics); Appendix D
H	227–257	63	§8	Ethics, consent, communication and the psychosocial patient Part V, unit V.1 (Ethics, consent and capacity); Topic 9 in part; Chapter 7
I	258–287	58	§9	Clinical examination and records Chapter 7 (Examination and Diagnosis)
J	288–327	80	§10	Imaging and cephalometrics Topic 2 (Cephalometry); Chapter 7, pp. 411–482

K	328–352	49	§11	Study models, space analysis and the digital set-up Chapter 7 (study models); Chapter 8, Box 8.6
L	353–392	80	§12	Diagnosis, the problem list and treatment planning Topic 3 (Formulating a treatment plan); Chapter 8
M	393–432	80	§13	Extraction decisions and space management Topic 4 (Choice of extraction); Chapter 8, pp. 505–530
N	433–485	109	§14	Biomechanics, force control and anchorage Topics 5 and 6 (Anchorage; the three orders); Chapters 6 and 11
O	486–530	89	§15	Orthodontic materials, wires, bonding and instruments Topic 7 (The ideal archwire); Chapter 10, pp. 645–665
P	531–571	81	§16	Removable and functional appliances Topic 8 (Functional appliances); Chapter 9
Q	572–612	81	§17	Fixed appliances — labial, lingual and aligners Topic 6 (Evolution of fixed appliances); Chapters 10 and 14
R	613–647	71	§18	Treatment mechanics: Stage I, II and III Chapters 10 and 13 (mechanics; definitive treatment)
S	648–682	71	§19	Retention, relapse and stability Topic 20 (Late lower incisor crowding); Chapter 13, pp. 975–992
T	683–727	89	§20	Interceptive orthodontics and the developing dentition Topics 13, 14 and 15 (missing central; impacted canine; impacted 7); Chapter 12
U	728–752	49	§21	Class I malocclusion and bimaxillary protrusion Chapter 13 (Class I and the permanent dentition)
V	753–792	80	§22	Class II malocclusion Topic 12 (Reducing overjet in a growing child); Chapters 9, 12 and 13
W	793–822	62	§23	Class III malocclusion Topic 17 (Correcting a Class III incisor relationship); Chapters 8, 9 and 13
X	823–847	49	§24	Transverse problems and facial asymmetry Topics 16 and 19 (Crossbite correction; facial asymmetry); Chapters 9, 12, 13 and 15
Y	848–877	58	§25	Vertical problems: deep bite and open bite Topic 11 (Overbite and overbite reduction); Chapter 13, pp. 928–945
Z	878–907	62	§26	TMD, condylar trauma, the airway and myofunctional therapy Topic 18 (Malocclusion and TMD); Chapter 2, pp. 102–105
AA	908–937	62	§27	Adult orthodontics and orthognathic surgery Chapters 13 and 15 (adult treatment; orthognathic surgery)
AB	938–968	63	§28	Cleft lip and palate and craniofacial syndromes

				Chapter 16 (Cleft Lip and Palate, and Syndromes)
AC	969–998	62	§29	Risks of orthodontic treatment and the medical history
				Topics 9 and 10 (Benefits and complications; Medical conditions); Chapters 2 and 7
AD	999–1023	49	§30	Orthodontic practice, infection control and governance
				Part V, units V.2–V.4 (radiation protection; infection control; governance and audit)

Block A — Concepts of growth and development (Q1–26)

Volume I: Chapter 4 (Postnatal Growth)
Volume II: §1 — read it again if you score under 70%
This block: 26 questions (6 of them true/false) · 50 marks · allow 32 minutes

1 A 13-year-old girl attends with a 7 mm overjet on a mild Class II skeletal base. A lateral cephalogram taken for planning shows concavities at the lower borders of C2, C3 and C4, with C3 and C4 both horizontally rectangular. Her mother asks whether a functional appliance would work. Which single statement best guides the decision?

A Peak mandibular growth velocity passed one to two years ago, so little useful skeletal change remains
B Peak mandibular growth velocity is expected within the coming year, so a functional appliance should begin now
C Peak mandibular growth velocity is about two years away, so treatment should be deferred
D The cervical vertebrae are not a valid maturity indicator in girls, in whom the peak is earlier
E No conclusion about timing is possible without a hand-wrist radiograph for comparison

2 An 11-year-old boy attends with mouth breathing and grossly enlarged tonsils. His parents have been told the tonsils will simply grow with him and they want them removed before orthodontic treatment begins. Which single statement about the growth of lymphoid tissue best informs the advice given?

A Lymphoid tissue follows the general body curve and will enlarge again at puberty
B Lymphoid tissue is already at adult size and will not change further
C Lymphoid tissue peaks at about twice adult size around this age and then involutes
D Lymphoid tissue completes 90 per cent of its growth by six years of age
E Lymphoid tissue enlarges steadily until the end of the second decade

3 Scammon's curves are used to explain why different tissues are treated at different ages. Regarding their clinical application:

TRUE OR FALSE — judge each statement independently; any number may be true

A The neural curve explains why the cranial vault is near adult size long before the facial skeleton is
B The genital curve rises steeply in the first two years of life and then plateaus
C Facial growth follows a curve intermediate between the neural and the general curves
D The general curve reaches its adult value earlier in boys than in girls
E The overshoot and involution of the lymphoid curve is a reason to observe mild adenoidal symptoms rather than operate

4 A department compares two ways of describing the adolescent growth spurt. Study 1 measures the height of 800 children on one occasion and groups them by age. Study 2 measures 60 children every six months for ten years. Which single conclusion is best supported?

A Study 1 gives the better estimate of the timing of the individual growth spurt
B Study 2 gives the better estimate of mean height at each age in the population
C The two designs give equivalent information about growth velocity and timing
D Study 2 is invalid because 60 children cannot represent the population
E Study 1 will flatten the spurt because individual peaks occur at different ages

5 A 7-year-old boy sustained bilateral condylar fractures at four years of age. He now has a retrusive chin, marked antegonial notching and limited mouth opening. Which single explanation for his deficient mandibular growth is best supported by current growth theory?

A The condylar cartilage was the primary growth centre and its loss removed the mandible's growth potential
B Reduced function and scarring diminished the growth of the soft-tissue matrix that the mandible follows
C The mandibular symphysis fused prematurely following the injury to the joints
D The nasal septum failed to displace the mandible forwards during growth
E Sutural growth of the mandible was arrested by the scarring after injury

6 Regarding the classical theories of craniofacial growth control:

TRUE OR FALSE — judge each statement independently; any number may be true

A Sicher's sutural theory is supported by the observation that transplanted suture tissue continues to grow autonomously
B Scott held that the sutures were the primary determinant of maxillary displacement
C Midfacial deficiency after early resection of the nasal septum is cited in support of Scott's theory
D Moss regarded the cranial base synchondroses as growth centres independent of their surrounding tissues
E Petrovic regarded condylar growth as wholly independent of the occlusal relationship

7 A 12-year-old girl is to have a functional appliance and the clinician wishes to confirm she is close to her pubertal peak. A lateral cephalogram is already justified for treatment planning. Which single course is most appropriate?

A Take a hand-wrist radiograph because it is the reference standard for skeletal maturity
B Rely on chronological age because maturity indicators add nothing useful
C Take both a hand-wrist radiograph and a cephalogram to improve accuracy
D Stage the cervical vertebrae on the cephalogram that is already being taken
E Take serial hand-wrist radiographs at six-month intervals

8 An 8-year-old boy has retained all four primary incisors, no permanent incisors visible, and a panoramic radiograph showing permanent incisor crowns with roots only one-third formed. His height and weight are both on the 50th centile. Which single conclusion is best supported?

A He has generalised developmental delay and orthodontic assessment should be deferred
B He has hypodontia of the permanent incisors, which explains the retention
C His dental development is delayed but his somatic development is not
D He requires endocrine investigation for generalised growth failure
E His skeletal maturity can be assumed to be delayed by the same amount

9 Serial standing heights recorded for a boy are 142 cm at 12 years, 149 cm at 13 years, 158 cm at 14 years and 164 cm at 15 years. Which single statement about his growth is best supported?

A Peak height velocity occurred between 13 and 14 years
B He has not yet entered the pubertal growth spurt
C Peak height velocity occurred between 14 and 15 years
D His growth has been linear and no peak can be identified
E Velocity cannot be derived from distance measurements

10 Growth and development are distinct concepts that are often used interchangeably in clinical notes. Regarding their application to treatment timing:

TRUE OR FALSE — judge each statement independently; any number may be true

A Development includes the acquisition of function and so continues after increase in size has ceased
B A distance curve can continue to rise while the corresponding velocity curve is falling
C Two children of identical chronological age may differ by more than two years in skeletal maturity
D Slow mandibular growth continues into the third decade in some individuals
E Timing decisions in orthodontics depend on maturity rather than on chronological age

11 A trial of a functional appliance reports a mean increase in mandibular length of 2.1 mm over 12 months compared with untreated controls, with a 95 per cent confidence interval of 0.4 to 3.8 mm. Which single conclusion is best supported?

A The appliance produces a clinically important skeletal change in most patients
B Mandibular length is not a valid outcome measure for this comparison
C The result is not statistically significant at the 5 per cent level
D The result confirms that the condyle is a primary growth centre
E The appliance produces a small mean change whose lower bound may be negligible

12 A brother and sister attend together. Both are 11 years 6 months old, both have a mild Class II skeletal pattern, she has a 6 mm overjet and he has a 7 mm overjet, and both are keen to start. Which single statement best guides the timing of functional appliance treatment?

A Both should start now because they are the same age
B Neither should start before 14 years of age
C He is likely to be nearer his pubertal peak and should start first
D She is likely to be nearer her pubertal peak and should start first
E Timing is unimportant because functional appliances work at any age

13 Regarding the methods by which craniofacial growth has been studied:

TRUE OR FALSE — judge each statement independently; any number may be true

A Bjork's metallic implants permit superimposition on structures that do not remodel, so true rotation can be measured
B Vital staining with madder or alizarin marks the bone that was being laid down when the dye was given
C Superimposition on the mandibular outline measures true rotation as accurately as implants do
D Natural reference structures can be used for superimposition where implants are unavailable, at some cost in accuracy
E Cross-sectional data can identify the age at which an individual child's growth peaked

14 A 12-year-old boy wearing a twin block postures his mandible forwards and, after nine months, has a Class I molar relationship. A colleague states that the appliance has grown his mandible. Which single explanation is most consistent with Petrovic's model of condylar growth?

A The appliance stimulated an autonomous growth centre within the condyle
B Sustained anterior positioning altered the feedback between occlusion and condylar growth
C The appliance acted by stretching the maxillary circumzygomatic sutures
D The change is entirely dental tipping with no skeletal component whatever
E The nasal septum was displaced forwards by the appliance, carrying the maxilla and the

attached mandible forwards

15 A clinician proposes to plan a Class III case definitively at nine years of age using a computerised growth prediction. Which single statement about growth prediction is best supported by the evidence?

A Prediction from a single cephalogram is accurate enough to plan surgery
B Serial records add nothing useful to a single computerised prediction
C Prediction is reliable in Class III cases because the pattern is established early and does not change
D Prediction is unnecessary because facial growth ceases by 16 years in both sexes
E Prediction of the group mean is reasonable but prediction of the individual is unreliable

16 A 17-year-old man with a Class III skeletal pattern is being assessed for orthognathic surgery. Which single approach best establishes that his facial growth has ceased?

A A hand-wrist radiograph showing fusion of the distal radial epiphysis, which is often taken as a general marker of skeletal maturity
B Two lateral cephalograms 12 months apart showing no change on superimposition
C Attainment of 18 years of age, by which time facial growth has ceased
D A cervical stage of CS4 on the planning cephalogram
E No increase in standing height over the preceding three months

17 A 12-year-old girl is being staged to decide when to begin growth modification. Regarding maturity indicators:

TRUE OR FALSE — judge each statement independently; any number may be true

A Appearance of the adductor sesamoid of the thumb indicates that the pubertal peak has already passed
B Capping of the epiphysis over the diaphysis of the middle phalanx of the third finger occurs close to peak height velocity
C Menarche characteristically follows peak height velocity
D Dental age derived from stages of root formation is a reliable guide to skeletal maturity
E Standing height velocity from serial records is the reference against which other indicators are validated

18 A mother reports that her 8-year-old daughter grew unusually quickly over the past year and asks whether treatment should begin to take advantage of it. Which single response is most appropriate?

A This is the pubertal growth spurt and treatment should begin without delay
B Velocity is constant before puberty, so the observation must be a measurement error
C Acceleration at this age indicates precocious puberty and paediatric referral is required
D A modest acceleration at seven to nine years is a recognised juvenile spurt
E The pubertal peak will therefore be early and treatment should be planned for next year

19 At birth the cranium is large relative to the face, and by adult life the proportions have altered markedly. Which single statement best explains the change and its clinical consequence?

A The face grows for longer than the vault, so facial discrepancies declare themselves later
B The cranial vault reduces in size during childhood while the face enlarges
C Face and vault grow at the same rate and the face is simply displaced downwards
D The cranial base is the only part of the skull that grows after six years of age, so facial growth must also have ceased by then
E Facial proportions at birth predict adult facial proportions accurately

20 Regarding cervical vertebral maturation staging as used for treatment timing:

TRUE OR FALSE — judge each statement independently; any number may be true

A At CS1 the lower borders of C2, C3 and C4 are already concave
B CS2 indicates that the pubertal peak occurred approximately one year earlier
C CS3 is the stage at which a functional appliance has its best prospect of a skeletal effect
D The method requires a radiograph taken specifically for the purpose
E Staging depends on the shape of the vertebral bodies rather than on their size

21 A 9-year-old girl has a bilateral posterior crossbite with a 3 mm mandibular displacement on closing and a narrow maxilla. A colleague argues that expansion should be deferred to 14 years so the result will be more stable. Which single statement best guides the decision?

A Expansion should be deferred because the midpalatal suture responds better after puberty
B Expansion at this age produces a dental change with no skeletal component
C Expansion should be carried out now, because the suture becomes progressively interdigitated
D The displacement will resolve spontaneously as the mandible grows
E Expansion is contraindicated until the permanent dentition is complete

22 Four indicators are proposed for deciding when to begin a functional appliance: chronological age, dental age, cervical stage, and standing height velocity from serial records. Which single appraisal is best supported?

A Chronological age is the most useful because it is the most reproducible measurement
B Dental age is the most useful because it is read from a radiograph that is taken anyway
C Cervical staging is invalid because the shape assessment is subjective
D Height velocity is most valid but least practical, and cervical staging is the compromise
E All four indicators are equivalent in validity for this decision

23 A 15-year-old boy presents with an 8 mm overjet on a moderate Class II skeletal base. His cervical stage is CS5 and his standing height has not changed over 12 months. He is anxious to avoid surgery. Which single management is most appropriate?

A Twin block therapy followed by fixed appliances to detail the occlusion
B Fixed appliances with upper first premolar extractions and planned anchorage support
C Headgear to restrain forward maxillary growth and reduce the overjet
D Defer all treatment until growth is complete and then reassess for surgery, avoiding any interim compromise
E Rapid maxillary expansion followed by a functional appliance

24 A 10-year-old girl has a standing height on the 90th centile, a dental age of 8 years and a cervical stage of CS1. Which single interpretation is best supported?

A The records are internally contradictory and must contain a measurement error
B She has a growth disorder and requires paediatric referral
C She must be close to her pubertal peak because she is tall for her age
D Dental age should be used to set her treatment timing instead
E Her size is advanced but her maturity is not, so the peak is about two years away

25 A girl attends at 14 years of age, two years after menarche, with a Class II division 1 malocclusion. Which single statement about her remaining facial growth is best supported?

A She has a small amount of remaining growth, insufficient to plan a skeletal correction around
B She has no remaining facial growth and may be treated exactly as an adult
C She has substantial remaining mandibular growth and a functional appliance is indicated

D Her remaining growth cannot be estimated without a hand-wrist radiograph
E Facial growth in girls continues at the pubertal rate until 18 years

26 A parent asks why her son's face will go on changing after his hands have finished growing. Which statement best represents the modern understanding of what controls craniofacial growth?

A The nasal septum is the primary determinant, and the midface is carried forward passively behind it
B The sutures are genetically programmed growth centres whose output is independent of the tissues around them
C The skeletal units enlarge in response to the demands of the soft tissue matrices they serve
D The condylar cartilage is a primary growth centre that displaces the mandible down and forwards
E The cranial base synchondroses pace facial growth and set the final position of both jaws

Block B — Craniofacial growth, prenatal and postnatal (Q27–56)

Volume I: Chapters 3 and 4 (Prenatal Development; Postnatal Growth)
Volume II: §2 — read it again if you score under 70%
This block: 30 questions (7 of them true/false) · 58 marks · allow 38 minutes

27 A newborn has bilateral malar hypoplasia, downward-sloping palpebral fissures, malformed external ears, conductive deafness and a small mandible. Which single developmental event best accounts for this combination?

A Failure of fusion of the medial nasal and maxillary processes
B Deficient neural crest population of the first and second arches
C Failure of elevation of the palatal shelves at eight weeks
D Premature fusion of the coronal suture in early infancy
E Arrested growth of the spheno-occipital synchondrosis

28 Regarding the pharyngeal arches and their derivatives:

TRUE OR FALSE — judge each statement independently; any number may be true

A The maxillary and mandibular processes are both derivatives of the first pharyngeal arch
B The muscles of mastication are innervated by the nerve of the first arch
C The muscles of facial expression are first arch derivatives
D Meckel's cartilage ossifies directly to form the body of the mandible
E The stapes is derived in part from the second arch

29 A baby is born with a complete cleft of the lip immediately to the left of the midline, extending into the nostril floor and through the alveolus as far back as the incisive foramen. Which single failure best explains this defect?

A Failure of fusion of the two medial nasal processes with each other
B Failure of fusion of the left lateral nasal process with the left maxillary process
C Failure of fusion of the left and right secondary palatal shelves
D Failure of fusion of the left maxillary process with the merged medial nasal process
E Failure of the frontonasal process to develop on the left

30 A baby is born with an intact lip and alveolus but a cleft of the soft palate and the posterior third of the hard palate. Which single statement about the underlying embryology is best supported?

A Failure of fusion of the secondary palatal shelves, which proceeds from front to back
B The defect arose before the sixth week when the primary palate forms, well before the palatal shelves undergo fusion
C The defect represents failure of the medial nasal processes to merge
D The defect indicates that the tongue descended too early in gestation
E The lip must also be affected at a submucous level in every case

31 A couple attend with two affected children: a son with a complete unilateral cleft lip and palate and a daughter with an isolated cleft of the palate. They ask why the same condition has appeared differently. Which single explanation is best supported?

A The two are the same condition and the difference is one of severity only
B The daughter's cleft is necessarily syndromic because isolated cleft palate is always part of a syndrome

C They are embryologically distinct malformations with different timing and frequencies
D The son's cleft arose later in gestation than his sister's did
E Both arose from failure of fusion of the secondary palatal shelves

32 Regarding the timing of prenatal craniofacial development:

TRUE OR FALSE — judge each statement independently; any number may be true

A The human face is essentially formed by the end of the fourth week of intrauterine life
B The nasal placodes appear during the fifth week
C Fusion of the secondary palatal shelves proceeds from anterior to posterior
D Elevation of the palatal shelves to the horizontal requires the tongue to drop away from between them
E The nasal septum fuses with the palatal shelves as they close

33 A nine-month-old infant has a long narrow head, a palpable ridge along the sagittal suture and a normal head circumference for age. Which single statement best explains the head shape?

A Growth has ceased in all directions because a suture has fused
B The defect lies in the cranial base synchondroses rather than the vault
C The cranial vault sutures are primary growth centres and their loss stops vault growth
D The brain has stopped growing and the vault has simply followed it
E Growth is restricted perpendicular to the fused suture and compensates parallel to it

34 A 14-year-old boy with a Class III skeletal pattern is told that his cranial base contributes to the discrepancy. Which single statement about postnatal cranial base growth is best supported?

A The spheno-occipital synchondrosis remains active into the mid-teens and moves the fossa
B The anterior cranial base continues to lengthen significantly until 18 years, matching mandibular growth
C The cranial base grows entirely by apposition at its sutures
D A reduced cranial base angle tends to produce a Class II skeletal pattern
E The cranial base has no bearing on the jaw relationship at all

35 Implant studies show that during growth the anterior surface of the maxilla becomes resorptive while the whole bone moves downwards and forwards. Which single statement best reconciles these two observations?

A The maxilla moves backwards and the apparent forward movement is an artefact of superimposition
B Surface resorption anteriorly means the maxilla cannot increase in length
C Displacement carries the whole bone forwards while its anterior surface resorbs
D The forward movement is produced entirely by apposition at the anterior surface
E Remodelling and displacement are alternative descriptions of the same process

36 Which single account of postnatal mandibular growth best explains how the ramus keeps pace with the lengthening of the corpus?

A Apposition at the anterior border of the ramus with resorption at the posterior border
B Sutural growth between the ramus and the corpus of the mandible
C Interstitial growth within the body of the mandible lengthens it
D Apposition at the symphysis with resorption at the condylar head
E Resorption at the anterior border of the ramus with apposition at the posterior border

37 Regarding the mechanisms available for postnatal craniofacial growth:

TRUE OR FALSE — judge each statement independently; any number may be true

A The midpalatal suture is a synchondrosis
B The mandibular condyle contains a growth plate histologically identical to that of a long bone
C Cartilage in the cranial base grows by appositional growth only
D The cranial vault grows principally by endochondral ossification
E Growth at the maxillary sutures is the autonomous primary determinant of midfacial position

38 Metallic implants placed in the mandible of a growing child show that the implant line rotates forwards by 15 degrees over six years, while the mandibular plane traced from the lower border rotates forwards by only 3 degrees. Which single conclusion is best supported?

A The implants must have migrated within the growing bone
B Remodelling of the lower border has masked most of the true rotation
C The mandibular plane is the more accurate record of the rotation
D The mandible has rotated backwards over the six years
E True rotation cannot be measured in a growing patient

39 A patient shows marked forward rotation of the mandibular corpus measured on implants, yet the mandibular plane angle on serial cephalograms is almost unchanged. Which single term best describes the difference between the two measurements?

A Total rotation
B Matrix rotation
C Remodelling rotation of the condyle
D Intramatrix rotation
E Displacement

40 A 10-year-old girl has a reduced lower anterior face height, a low mandibular plane angle, a deep overbite and a prominent chin point. Which single growth pattern best accounts for these findings and its main treatment implication?

A Backward mandibular rotation, so overbite will reduce spontaneously
B Backward mandibular rotation, so an anterior open bite is likely to develop
C Forward mandibular rotation, so the deep overbite will tend to relapse
D Pure vertical maxillary excess, so surgery will be unavoidable
E A neutral growth pattern with no implication for stability

41 Regarding rotation of the mandible during growth:

TRUE OR FALSE — judge each statement independently; any number may be true

A Forward rotation of the mandible tends to deepen the overbite
B Marked antegonial notching is one of the structural signs associated with backward rotation
C Total rotation is the rotation of the mandibular corpus relative to the anterior cranial base
D Matrix rotation is measured between metallic implants placed within the mandibular body
E Remodelling of the lower border conceals a large part of the true rotation on a conventional cephalogram

42 A 12-year-old boy has a Class III incisor relationship, a mandible of normal length and a maxilla of normal length, but a reduced saddle angle on the cephalogram. Which single explanation for his jaw relationship is best supported?

A A more acute cranial base angle placing the fossa forwards
B Excessive growth of the mandible at the condylar cartilage
C Deficient sutural growth of the nasomaxillary complex
D Backward rotation of the mandible during growth
E Premature fusion of the midpalatal suture in childhood

43 An implant-retained prosthesis is proposed for a missing upper central incisor in a 15-year-old girl. Which single statement about residual facial growth best guides the timing?

A Maxillary growth is complete at 15 years in girls, so placement may proceed
B Only implants placed in the mandible are affected by residual growth
C Implants continue to erupt with the adjacent teeth, so timing is unimportant for this treatment decision
D The implant will be displaced palatally by continuing sutural growth
E Continuing vertical alveolar growth will leave the implant progressively infraoccluded

44 Enlow's V principle is used to explain how the ramus and the maxillary tuberosity enlarge. Which single statement best captures the principle as it applies to the ramus?

A Bone is deposited on the outer surface and resorbed from the inner, so the V moves towards its open end
B The V moves towards its apex as it enlarges, shortening the corpus
C The V enlarges by interstitial growth of bone within its walls
D Bone is deposited on the inner surface and resorbed from the outer, so the V moves towards its wide end
E Deposition and resorption occur on the same surface simultaneously

45 Regarding postnatal growth of the cranial vault and cranial base:

TRUE OR FALSE — judge each statement independently; any number may be true

A The anterior cranial base is essentially complete by about seven years of age, which is why it is used as a superimposition structure
B The spheno-occipital synchondrosis fuses at about eight years of age
C The cranial vault sutures generate a primary growth impulse that separates the vault bones
D Growth of the cranial vault is complete at birth
E Enlargement of the vault proceeds independently of the growth of the brain

46 A 9-year-old girl has an increased lower anterior face height, a steep mandibular plane, an anterior open bite of 2 mm and lip incompetence. Her mother had similar features. Which single statement about her likely growth best guides treatment planning?

A Her open bite will close spontaneously as the mandible rotates forwards with growth
B Continued backward rotation will maintain the open bite, so vertical control is needed
C Her growth pattern makes overbite reduction the main difficulty
D Her pattern predicts a developing Class III incisor relationship
E Growth direction cannot be inferred from the findings described

47 On cephalometric superimposition of a growing child on the anterior cranial base, point A moves 4 mm downwards and 2 mm forwards over three years, while the upper incisor tip moves 4 mm downwards and 2 mm forwards relative to the maxilla. Which single conclusion is best supported?

A The maxilla has not grown because point A and the incisor moved identically

B Superimposition on the cranial base is invalid for assessing maxillary change
C The incisor has not erupted at all during the three years
D The measurements are contradictory and cannot both be correct
E The maxilla has been displaced and the incisor has additionally erupted within it

48 A 13-year-old boy has completed his overjet reduction with a twin block. His parents ask whether his profile will change further. Which single statement about the remaining soft and hard tissue change is best supported?

A The facial profile is now fixed and will not change further
B The nose and chin grow more than the lips, so the profile straightens
C The lips grow more than the nose and chin, so the profile becomes fuller
D Soft tissue change ceases when the dentition is complete
E Profile change after 13 years of age occurs only in girls

49 Regarding facial growth patterns and the structural signs used to recognise them:

TRUE OR FALSE — judge each statement independently; any number may be true

A A low mandibular plane angle with a deep overbite suggests a forward rotating pattern
B Backward rotation tends to increase the lower anterior face height
C Forward rotation in some patients continues to reduce the lower face height into early adult life
D Overbite reduction is more stable in a forward rotator than in a backward rotator
E The structural signs allow the direction of rotation to be predicted reliably in the individual patient

50 During adolescence the maxilla and the mandible are separated by 12 mm of vertical growth, yet the overbite in an untreated Class I child remains 3 mm throughout. Which single mechanism best accounts for this?

A The teeth do not move at all during this period of growth
B The vertical growth measurement must therefore be erroneous, since the overbite has stayed unchanged
C The mandible rotates backwards by exactly the same amount
D Compensatory eruption with alveolar apposition matches the separation
E The condyle grows only in the anteroposterior direction

51 Which single statement about the tissue of the mandibular condyle best explains why it can adapt to a functional appliance?

A It is secondary cartilage with a proliferative layer responsive to local demand
B It is primary hyaline cartilage with an organised growth plate that responds to loading
C It is intramembranous bone with no cartilage present at any stage
D It is fibrous tissue only and cannot contribute to growth in length
E It behaves exactly as the spheno-occipital synchondrosis does

52 A woman of 26 attends four weeks after a positive pregnancy test and asks whether a cleft could still occur. Which single statement is most accurate?

A The lip and palate are both formed after 16 weeks so the risk period is still ahead
B Craniofacial development is complete by the end of the second week
C The lip forms at about six weeks and the palate closes by ten to twelve weeks
D Only the palate remains at risk after the end of the first trimester
E Teratogen exposure has no effect on facial development at any stage

53 Regarding neural crest and early craniofacial development:

TRUE OR FALSE — judge each statement independently; any number may be true

A Neural crest cells give rise to the enamel organ
B Neural crest cells contribute the dental papilla and the dental follicle
C The pharyngeal arches are composed of endoderm alone
D Excess retinoid exposure in the first trimester protects against craniofacial malformation
E Failure of neural crest migration underlies the craniofacial features of several first arch syndromes

54 A 10-year-old boy has a Class III incisor relationship that his parents say has become steadily worse since he was six. Which single statement about the differential growth of the jaws best explains the deterioration?

A The maxilla grows for longer and faster than the mandible, so Class III improves with age
B Class III deterioration is entirely dental and not skeletal in origin
C Both jaws stop growing at the same time, so the relationship is stable
D The mandible grows later and for longer, so a Class III pattern tends to worsen
E Growth in Class III patients ceases earlier than in Class I patients

55 An infant with untreated hydrocephalus has a cranial circumference far above the 99.6th centile with widely separated vault sutures and a tense fontanelle, while the facial skeleton is of normal size. Which single concept best explains the finding?

A The vault bones possess an intrinsic growth potential that has been unmasked
B The vault is a skeletal unit responding to enlargement of its functional matrix
C The sutures have become primary growth centres in the presence of raised pressure
D The facial skeleton is protected from growth by the cranial base
E Vault size is determined by the cranial base synchondroses alone

56 A 9-year-old boy needs space for his second and third permanent molars. Which single site contributes most to the increase in maxillary arch length that accommodates them?

A Apposition at the alveolar crest
B Apposition at the anterior surface of the maxilla
C Growth at the midpalatal suture
D Growth at the incisive suture
E Apposition at the maxillary tuberosity

Block C — Development of the dentition (Q57–96)

Volume I: Chapter 5 (Development of the Dentition)
Volume II: §3 — read it again if you score under 70%
This block: 40 questions (10 of them true/false) · 80 marks · allow 50 minutes

57 A 4-year-old girl attends for a routine examination. The primary dentition is complete, the contact points are all tight and there is no spacing anywhere in either arch. Her mother has had four premolars extracted for crowding. Which single statement is most appropriate?

A Crowding of the permanent incisors is very likely because they are much wider
B The absence of spacing is normal at this age and carries no implication for the permanent dentition
C Crowding is unlikely because the arches will widen enough to accommodate the permanent teeth
D Extraction of the primary canines should be arranged now
E The finding indicates a tooth size discrepancy requiring Bolton analysis

58 A 5-year-old boy has generalised spacing in both arches and, in addition, a conspicuous 2 mm space in a constant position in each quadrant. Which single description of these additional spaces is correct?

A Mesial to the canine in both the upper and lower arches
B Distal to the canine in the maxilla and mesial to the canine in the mandible
C Mesial to the canine in the maxilla and distal to the canine in the mandible
D Distal to the second primary molar in both arches
E Between the central incisors in the upper and lower arches

59 A 5-year-old girl has a complete primary dentition with generalised spacing, an overbite of 1 mm and edge-to-edge incisors. Regarding the primary dentition:

TRUE OR FALSE — judge each statement independently; any number may be true

A Generalised spacing at this age is a favourable finding for the permanent incisors
B The primary dentition is normally complete by about 30 months of age
C An edge-to-edge incisor relationship in the primary dentition reliably predicts a Class III permanent incisor relationship
D Attrition of the primary incisors producing an edge-to-edge relationship is a normal finding
E The primary molar relationship is described in terms of the Angle classification

60 A 6-year-old boy has just erupted all four first permanent molars. The distal surfaces of the upper and lower second primary molars are in the same vertical plane and there is a 2 mm space distal to each lower primary canine. Which single outcome is most likely if no treatment is given?

A The molars will remain end-to-end permanently and a Class II molar relationship will result
B The lower molars will drift forwards into the primate space, producing a Class I molar relationship
C The upper molars will drift forwards more than the lower, producing a Class II molar relationship
D A Class III molar relationship will develop as the molars drift

E The relationship cannot change without extractions being carried out

61 A 9-year-old boy has a lower labial segment crowded by 4 mm. His lower second primary molars are sound and firm, and his lower canines and premolars are all present and normally positioned on the radiograph. Which single statement about the space available is most accurate?

A Exfoliation of the primary canines and molars will release about 5 mm per quadrant
B The leeway space in the lower arch is about 1.5 mm per quadrant
C No further space will become available and extractions are inevitable
D Exfoliation of the lower canines and molars releases about 2.5 mm of leeway per quadrant
E The leeway space becomes available only after the second permanent molars erupt

62 A 7-year-old girl has erupted upper central incisors that are rotated and appear too large for the arch, and her father asks whether they should be reduced in width. Which single explanation is most appropriate?

A The upper permanent incisors need about 7 mm more room than their predecessors
B The incisors are macrodontic and interproximal reduction is indicated
C The appearance indicates a tooth size discrepancy that will need premolar extractions
D The incisors will look smaller in proportion as the face grows
E The finding is pathological and warrants referral for a radiograph of the roots

63 A mixed dentition space analysis is being carried out for a 9-year-old girl. Regarding the method and its interpretation:

TRUE OR FALSE — judge each statement independently; any number may be true

A Space analysis in the mixed dentition is more reliable than in the permanent dentition
B Prediction tables and radiographic methods give identical estimates of unerupted tooth width
C Space required is derived from the sum of the mesiodistal widths of the teeth in the segment
D Space available is measured along the line of the arch that the treatment plan intends to produce
E A positive space analysis means that extractions are required

64 An 8-year-old boy has erupted upper central and lateral incisors that are distally tipped with a 3 mm midline diastema. There is no supernumerary tooth on the radiograph, the labial frenum is of normal form and the canine crowns lie high and distal to the lateral incisor roots. Which single management is most appropriate?

A Close the diastema with a fixed appliance now and retain it
B Refer for a frenectomy before the canines erupt into the arch
C Reassure and review, as the appearance resolves as the canines erupt
D Extract the primary canines to allow spontaneous space closure
E Take a cone beam scan to exclude an unerupted supernumerary

65 A 10-year-old girl has a lower arch with 1 mm of crowding. The radiograph shows the lower second permanent molars at an advanced stage of development, with crowns fully formed, while the lower second premolars have roots only one-third formed. Which single consequence is most likely?

A The second premolars will erupt normally before the second molars and no space will be lost
B The eruption sequence has no bearing on space in the lower arch
C The second premolars will be impacted against the first molars

D The crowding will resolve as the second molars erupt into place
E The second molars will erupt first and drive the first molars forwards

66 A 6-year-old boy has both lower permanent central incisors erupting lingual to the retained lower primary central incisors, which are firm. Which single management is most appropriate?

A Extract both lower primary central incisors and review the position
B Fit a lower removable appliance to move the permanent incisors labially
C Leave both primary and permanent incisors and review in two years
D Refer for a lower fixed appliance to align the incisors
E Take a radiograph to exclude supernumerary teeth before any action

67 A 7-year-old girl attends and her mother asks whether her teeth are erupting on time. Regarding the eruption of the permanent dentition:

TRUE OR FALSE — judge each statement independently; any number may be true

A The lower first permanent molar and lower central incisor are usually the first permanent teeth to erupt
B Girls tend to erupt their permanent teeth slightly earlier than boys
C The upper permanent canine usually erupts after the upper first premolar
D Eruption of a permanent tooth normally occurs when about two-thirds to three-quarters of its root is formed
E An asymmetry of six months between contralateral teeth warrants immediate radiographic investigation

68 A 12-year-old girl has retained lower second primary molars, both firm and in good condition, and a panoramic radiograph confirms that both lower second premolars are absent. There is 3 mm of crowding in the lower labial segment. Which single further finding would most strongly favour planning their extraction rather than retention?

A The primary molars are caries free and well restored
B The patient is 12 years old and near the end of growth
C The occlusal surfaces are 2 mm below the plane and still descending
D The third molars are both present on the radiograph
E The upper arch is well aligned and requires no treatment, so the lower arch can be managed independently

69 A 9-year-old boy has an unerupted upper left permanent central incisor. The contralateral tooth erupted 18 months ago. A periapical radiograph shows a conical supernumerary tooth lying between the roots of the upper central incisors, with the unerupted incisor displaced upwards and its root two-thirds formed. Which single management is most appropriate?

A Review for a further 12 months, as most such incisors erupt spontaneously
B Extract the contralateral incisor to maintain arch symmetry
C Remove the supernumerary and bond a gold chain to the incisor at the same visit
D Extract the unerupted incisor and plan orthodontic space closure
E Remove the supernumerary, maintain space, and review for eruption

70 An 8-year-old boy has all four first permanent molars affected by molar incisor hypomineralisation, with post-eruptive breakdown and sensitivity in three of them. His incisors show only small creamy opacities. He has mild lower arch crowding and all premolars and third molars are present radiographically. Which single management is most appropriate?

A Restore all four molars and plan orthodontic treatment conventionally, deferring any extraction decision
B Extract the four first permanent molars when the second molar bifurcations calcify

C Extract the four first permanent molars without further delay
D Extract only the worst affected molar and restore the others
E Defer any decision until the second permanent molars have erupted

71 Regarding supernumerary teeth:

TRUE OR FALSE — judge each statement independently; any number may be true

A Supernumerary teeth occur more commonly in the mandible than in the maxilla
B A conical supernumerary is the type most likely to obstruct the eruption of a permanent incisor
C Supplemental teeth are most commonly found in the lower premolar region
D A supernumerary tooth in the premaxilla may be associated with cleidocranial dysplasia
E Supernumerary teeth are equally common in males and females

72 A 9-year-old girl has a lower right second primary molar 1 mm below the occlusal plane. The successor is present and normally positioned on radiograph, the adjacent teeth are upright and the contact points are intact. Which single management is most appropriate?

A Extract the primary molar now and fit a space maintainer
B Restore the primary molar to the occlusal plane with a stainless steel crown
C Refer for surgical luxation of the ankylosed tooth
D Review at six-monthly intervals, as most such teeth exfoliate normally
E Begin orthodontic extrusion of the primary molar

73 A 7-year-old boy has an upper right first permanent molar that has erupted with its mesial marginal ridge locked beneath the distal bulbosity of the upper second primary molar. There is 2 mm of resorption of the distal root of the primary molar. Which single management is most appropriate?

A Extract the second primary molar and fit a space maintainer
B Attempt disimpaction with a separating elastomeric module or brass wire and review
C Leave and review, as two-thirds of such molars disimpact by eight years
D Refer for surgical exposure of the ectopic first permanent molar
E Begin upper fixed appliance therapy to distalise the molar

74 A 12-year-old girl has an upper left canine crown lying between the roots of the first and second premolars, with the canine root apex in its normal position. The canine is unerupted and the arch is otherwise well aligned. Which single term and implication is correct?

A Transposition, and the usual site in the maxilla is between the canine and the first premolar
B Ectopic eruption, and interceptive extraction of the primary canine is indicated
C Transposition, and the usual site in the maxilla is between the canine and the lateral incisor
D Ankylosis, and surgical removal of the canine is indicated
E Primary failure of eruption, and orthodontic traction will not work

75 Regarding anomalies of tooth number in the permanent dentition:

TRUE OR FALSE — judge each statement independently; any number may be true

A Excluding third molars, hypodontia affects roughly 5 per cent of the population
B Hypodontia is more common in the primary than in the permanent dentition
C Excluding third molars, the lower second premolar and the upper lateral incisor are the teeth most commonly absent
D Absence of a permanent tooth is often accompanied by reduced size of the teeth that are present
E Hypodontia shows a marked male preponderance

76 A 7-year-old boy has to have a lower right second primary molar extracted because of an abscess. The first permanent molar has erupted, the second premolar is present with one-third of its root formed, and the arch is just crowded by 2 mm. Which single management is most appropriate?

A Extract and review, because space loss after loss of a lower second primary molar is minimal
B Retain the root remnants in situ to preserve the space
C Extract and balance with the contralateral second primary molar
D Extract and compensate by removing the upper second primary molar
E Extract and fit a band and loop space maintainer to hold the space

77 An 8-year-old girl loses the lower left primary canine to resorption by the erupting lateral incisor. The lower labial segment is crowded by 5 mm. Which single consequence is most likely if nothing further is done?

A Space will be lost as the lower first permanent molar drifts mesially
B The crowding will resolve spontaneously
C The lower right primary canine will exfoliate early to compensate
D The lower centreline will shift towards the side of the loss
E The lower incisors will procline

78 Regarding the use of space maintainers in the mixed dentition:

TRUE OR FALSE — judge each statement independently; any number may be true

A A space maintainer is indicated after every primary tooth extraction
B A lower lingual arch placed before exfoliation of the primary molars preserves the leeway space and can relieve mild crowding
C A band and loop appliance is suitable for maintaining space in a bilateral posterior loss
D Space maintainers reduce the need for oral hygiene instruction because they are fixed
E Space maintenance is unnecessary if the successor is absent

79 An 8-year-old boy has a single wide anterior tooth in the position of the upper right central and lateral incisors, with a groove on its labial surface and a notched incisal edge. Radiographically there are two pulp chambers and two root canals within one root outline, and a normal complement of teeth is present elsewhere. Which single conclusion about the tooth count is best supported?

A The arch contains a normal number of tooth germs and the anomaly is gemination
B The quadrant contains one fewer tooth germ than normal, so this is fusion
C The quadrant contains one more tooth germ than normal
D The anomaly is a supernumerary tooth lying in the midline
E The anomaly cannot be classified without extracting the tooth

80 A 13-year-old girl has a peg-shaped upper left lateral incisor and an absent upper right lateral incisor. The canines are erupted and the arch is spaced by 4 mm. Which single statement about this presentation is best supported?

A The two findings are unrelated and have occurred by coincidence
B The peg lateral indicates that the canine is likely to be ectopic
C The peg lateral should be extracted so that both spaces can be closed symmetrically
D The absent lateral incisor indicates a syndromic diagnosis
E Peg and absent laterals are expressions of the same variable genetic trait

81 A 9-year-old girl is being assessed in the mixed dentition. Regarding the figures used in planning:

TRUE OR FALSE — judge each statement independently; any number may be true

A The incisor liability is about 7 mm in the maxilla and about 5 mm in the mandible
B The leeway space is about 1.5 mm per quadrant in the maxilla
C Intercanine width increases by about 10 mm between the primary and permanent dentitions
D The upper permanent canine is palpable buccally in most children by 10 to 11 years of age
E A permanent canine that is not palpable by 11 years warrants radiographic localisation

82 A 9-year-old boy has 8 mm of crowding in each arch, a Class I skeletal pattern, an average overbite, and all permanent teeth present radiographically including third molars. His parents want the crowding managed with the fewest appliances. Which single approach is most appropriate to discuss?

A Immediate extraction of all four first premolars under general anaesthesia, relying on spontaneous alignment
B Serial extraction of the primary canines, first primary molars, then first premolars
C A lower lingual arch to preserve the leeway space in both quadrants
D Expansion of both arches with removable appliances and retention
E Review until the permanent dentition is complete and then plan

83 An 8-year-old girl has 2 mm of crowding of the lower permanent incisors, which her mother says looks worse than it did a year ago. The primary canines are still present and the arch is otherwise normal. Which single management is most appropriate?

A Extract the lower primary canines to align the incisors now
B Refer for lower fixed appliances to align the labial segment, accepting the additional appliance burden
C Fit a lower removable appliance to align the lower incisors
D Reassure and review, as mild crowding improves with intercanine growth
E Plan extraction of the lower first premolars at this stage

84 A screening programme reports that 23 per cent of 11-year-olds have at least one absent permanent tooth. Which single explanation for this unexpectedly high figure is most likely?

A Third molars have been included in the count
B Hypodontia is genuinely this common in the population
C The children were selected from an orthodontic clinic population only
D Panoramic radiography over-diagnoses absence
E Hypodontia has increased in prevalence in recent decades

85 Regarding disturbances of eruption:

TRUE OR FALSE — judge each statement independently; any number may be true

A Primary failure of eruption affects the anterior teeth more often than the posterior teeth
B An affected tooth in primary failure of eruption becomes ankylosed when orthodontic traction is applied
C Primary failure of eruption has been associated with mutations in the parathyroid hormone receptor 1 gene
D An ankylosed permanent tooth in a growing child will move occlusally with the alveolus as growth continues
E A tooth with a dilacerated root usually erupts normally once the obstruction is removed

86 An 11-year-old girl has upper permanent canines that are not palpable buccally, with the upper lateral incisors slightly mobile. A panoramic radiograph shows both canine crowns overlapping the distal half of the lateral incisor roots. The arch is uncrowded. Which single management is most appropriate?

A Extract both upper first premolars to create space
B Expose and bond gold chains to both canines now
C Extract both upper primary canines and review in 12 months
D Review clinically in 12 months without intervening
E Refer for surgical removal of both permanent canines

87 Two periapical radiographs of an unerupted upper canine are taken with the tube shifted mesially for the second film. The canine appears to move mesially relative to the incisor roots between the two films. Which single conclusion is correct?

A The canine is buccally positioned
B Parallax cannot be applied to canines
C The canine is in the line of the arch
D The canine is palatally positioned
E A second film in the vertical plane is required before any conclusion

88 A 14-year-old boy has a retained lower right second primary molar with no successor. The tooth is 1 mm below the occlusal plane, has a sound crown, and the lower arch is uncrowded with a well aligned labial segment. Which single management is most appropriate?

A Extract it now and close the space orthodontically
B Retain, monitor the infraocclusion, and review the plan later
C Extract now and plan an implant at 18 years of age
D Extract and allow spontaneous drift of the first permanent molar
E Build the tooth up to the occlusal plane and plan no further review

89 Regarding changes in arch dimension through the mixed and early permanent dentitions:

TRUE OR FALSE — judge each statement independently; any number may be true

A Intercanine width increases during eruption of the permanent incisors and canines
B Arch length typically increases from the mixed to the permanent dentition
C Maxillary intercanine width shows no change after the permanent canines have erupted
D Late lower incisor crowding occurs in treated and untreated subjects alike
E A reduction in intercanine width after the mid-teens contributes to late lower incisor crowding

90 A 16-year-old girl who had no orthodontic treatment returns with 3 mm of lower incisor crowding that was not present at 12 years. Her third molars are unerupted and mesioangular. Which single explanation is best supported?

A The third molars have pushed the arch forwards and caused the crowding
B The crowding will resolve once the third molars erupt
C The crowding indicates an undiagnosed skeletal discrepancy
D The crowding proves that her arch was expanded in childhood
E Late lower arch crowding is a maturational change of uncertain cause

91 A clinician proposes fitting lower lingual arches routinely to all crowded 9-year-olds to preserve leeway space. Which single appraisal of this proposal is best supported?

A It relieves mild to moderate crowding of up to about 5 mm across the arch
B It will resolve crowding of any severity because the leeway space is large
C It has no effect on space because the leeway space is not usable

D It should be used only after the second permanent molars have erupted
E It is equivalent in effect to extracting four first premolars

92 A 6-year-old boy's mother reports that a new large back tooth has appeared and asks whether it will be replaced. Regarding the first permanent molar:

TRUE OR FALSE — judge each statement independently; any number may be true

A It erupts without a primary predecessor
B It is the tooth most often mistaken by parents for a primary molar
C Its position determines the molar relationship that is recorded in the mixed dentition
D Its early loss in a crowded arch may be used to relieve crowding if the timing is right
E It is more susceptible to caries and to hypomineralisation defects than the second permanent molar

93 A 10-year-old boy has a heavily restored upper right first permanent molar with a poor prognosis. His malocclusion is Class I with 3 mm of upper arch crowding, and all other teeth including third molars are present. Which single statement about extracting the molar is most accurate?

A Compensating extraction of the lower first molar is mandatory in every case
B Extraction at this age will cause the upper canine to become impacted
C An upper first molar gives better spontaneous space closure than a lower
D Balancing extraction of the contralateral upper first molar is mandatory
E The extraction should be deferred until the third molars erupt

94 A mother is concerned that her 8-year-old daughter still has no upper lateral incisors while her friends of the same age have them. The upper central incisors erupted at 7 years, the laterals are visible radiographically with half their roots formed, and there is no supernumerary. Which single response is most appropriate?

A Arrange surgical exposure of both upper lateral incisors now
B Reassure, as the laterals erupt at about eight years and root formation fits
C Extract both upper primary lateral incisors without delay
D Refer urgently, because a delay of more than a year is pathological and warrants specialist assessment
E Take a cone beam scan to assess the position of both incisors

95 A 3-year-old boy fell and intruded his upper right primary central incisor 18 months ago. The tooth was left in situ and re-erupted. Which single sequel to the permanent successor is most likely to be seen, and what is its implication?

A Enamel hypoplasia of the permanent incisor, which will erupt normally but need restoration
B Ankylosis of the permanent incisor at the time of eruption
C Hypodontia of the permanent successor incisor
D Dilaceration of the permanent crown, which may prevent eruption
E Accelerated eruption of the permanent successor incisor

96 A 10-year-old girl has an upper arch with 6 mm of crowding, an upper left permanent canine palpable buccally and in the line of the arch, and both upper first premolars erupting. Her lower arch is well aligned. A colleague suggests extracting the upper primary canines to relieve the crowding. Which single appraisal is best supported?

A The incisors will align but the space will be lost unless it is held
B This is appropriate because it will allow the permanent canines to erupt into alignment
C This is contraindicated because the permanent canines are already palpable

D This will produce a lasting increase in the arch length
E This will have no effect on the position of the incisors

Block D — The biology of tooth movement, eruption and root resorption (Q97–141)

Volume I: Topic 1 (The biological basis of tooth movement); Chapter 6
Volume II: §4 — read it again if you score under 70%
This block: 45 questions (11 of them true/false) · 89 marks · allow 56 minutes

97 A 14-year-old girl returns four weeks after her first upper archwire. She reports that her teeth felt loose and sore for the first few days, that nothing appeared to change for the next three weeks, and she is disappointed. Which single explanation is most accurate?

A The archwire was too light and a heavier one should be fitted
B Tooth movement is linear from the day of activation and the impression of no change is subjective
C An initial displacement over two days is followed by a lag of 20 to 30 days
D The lag represents a failure of the bracket bonds on those teeth
E Movement will not begin at all until the archwire is changed

98 Regarding the phases of orthodontic tooth movement:

TRUE OR FALSE — judge each statement independently; any number may be true

A The initial phase largely reflects displacement of the tooth within the periodontal ligament space
B The lag phase typically lasts about 20 to 30 days
C Increasing the applied force shortens the lag phase
D Tooth movement during the post-lag phase is slower than during the initial phase
E Hyalinisation is a pathological finding that indicates excessive force in every case

99 Histology of a tooth three weeks after the application of a heavy force shows an acellular zone in the compressed periodontal ligament with resorption lacunae on the marrow side of the adjacent alveolar bone. Which single statement best explains this appearance?

A Direct frontal resorption is occurring because the force was appropriate
B The appearance indicates that movement has ceased permanently
C The tooth has become ankylosed to the adjacent socket wall
D The lacunae represent root resorption rather than bone resorption
E The compressed ligament is avascular, so resorption comes from the marrow

100 An upper canine is to be retracted bodily along a rectangular archwire in an adult. Which single force is most appropriate for this movement?

A About 50 to 100 cN
B About 20 to 30 cN
C About 150 to 200 cN
D About 250 to 300 cN
E About 450 to 500 cN

101 A classical study of canine retraction found that the rate of movement rose with force up to about 272 cN and then fell away at higher forces. Which single interpretation of this finding is best supported?

A Forces above 272 cN should be used to retract canines because more force gives more movement

B The finding shows that force magnitude is irrelevant to the rate of movement
C Movement slows at higher forces because widespread hyalinisation delays cellular recruitment
D The optimal clinical force for canine retraction is therefore 272 cN
E Rate of movement is proportional to force at all levels

102 Regarding the RANKL, RANK and osteoprotegerin system in orthodontic tooth movement:

TRUE OR FALSE — judge each statement independently; any number may be true

A Osteoprotegerin binds RANK on the osteoclast precursor and activates it
B RANKL expression in the compressed periodontal ligament peaks at about three days
C Osteoprotegerin acts as a decoy receptor that reduces osteoclast formation
D Osteoblasts and periodontal ligament cells are important sources of RANKL
E The balance between RANKL and osteoprotegerin determines the amount of bone resorption on the pressure side

103 A patient with 8 mm of upper arch space to close asks how long the space closure will take. Assuming uncomplicated bodily space closure with good anchorage control, which single estimate is most realistic?

A About 2 months
B About 4 months
C About 24 months
D About 15 months
E About 8 months

104 Which single statement best explains why the width of the periodontal ligament is relevant to the choice of orthodontic force?

A The ligament is about 2 mm wide, so large initial displacements are possible before bone remodelling is needed
B The ligament is about 0.2 mm wide, so only a small displacement is possible
C The ligament width increases with age, so adults require heavier forces
D The ligament transmits force only in tension and not in compression
E Ligament width determines the centre of resistance of the tooth

105 A colleague argues that tooth movement is explained by piezoelectric signals generated when alveolar bone bends, and therefore that intermittent forces should be more effective than continuous ones. Which single appraisal is best supported?

A Bone bending and piezoelectricity fully explain orthodontic tooth movement
B Intermittent forces produce faster tooth movement than continuous forces in clinical practice
C Piezoelectric signals persist for the duration of a continuous force
D Bone bending may give an early signal, but the sustained response is cellular
E The pressure-tension theory has been disproved by this finding

106 Regarding mechanotransduction in the periodontal ligament and alveolar bone:

TRUE OR FALSE — judge each statement independently; any number may be true

A The osteocyte network plays no part in signalling mechanical strain
B Prostaglandin E2 concentrations fall in the compressed periodontal ligament after force application
C Interstitial fluid flow within the ligament and bone is one proposed mechanism for converting strain into a cellular signal

D Cytokine release is confined to the tension side of the ligament
E The tension side shows osteoblast recruitment and bone deposition alongside ligament fibre remodelling

107 A single upper canine is to be tipped distally with a light force from a removable appliance. Which single moment to force ratio applied at the bracket best describes the requirement for this movement?

A A ratio of about 10 to 1 at the bracket
B A ratio of about 7 to 1 at the bracket
C Zero, with a single force applied at the bracket
D A ratio of about 12 to 1 at the bracket
E A ratio greater than 12 to 1 at the bracket

108 For a single-rooted tooth of normal periodontal support, where does the centre of resistance lie?

A Within the root, about one-third of its length from the crest
B At the apex of the root, whatever the bone level, since force is thought to act around the root tip
C At the incisal edge, where the force is applied
D At the bracket slot, where the archwire engages
E Midway between the bracket and the root apex

109 A patient with moderate horizontal bone loss is to have an incisor moved bodily. Which single consequence of the bone loss is most important for the mechanics?

A The centre of resistance moves coronally, so a smaller moment is required
B Bodily movement cannot be achieved in a reduced periodontium
C The centre of resistance is unchanged because it depends on root length alone
D The centre of rotation coincides with the centre of resistance during bodily movement
E The centre of resistance moves apically, so a greater moment is required

110 Regarding types of orthodontic tooth movement and the forces that produce them:

TRUE OR FALSE — judge each statement independently; any number may be true

A In bodily movement the centre of rotation lies at infinity
B Root movement requires a higher moment to force ratio than bodily movement
C Rotation of a tooth about its long axis is produced by a couple
D Intrusion requires heavier forces than bodily movement because the force is concentrated at the apex
E Extrusion produces areas of compression in the periodontal ligament comparable to those in intrusion

111 Four upper incisors are to be intruded in an adult with a deep overbite. Which single combination is most appropriate?

A About 100 cN in total, accepting some risk of root resorption
B About 10 cN in total, which is below the threshold for movement
C About 200 cN in total, to achieve a faster intrusion
D About 40 to 80 cN in total, directed through the centre of resistance
E A force applied labial to the centre of resistance to produce simultaneous proclination

112 A patient asks about the risk of root shortening before starting fixed appliances. Which single set of figures is most accurate for orthodontically induced inflammatory root resorption?

A Detectable radiographically in about 5 per cent, severe in under 1 per cent
B Radiographic in up to 73 per cent, severe in about 1 to 5 per cent
C Detectable radiographically in about 20 per cent, severe in about 20 per cent
D Detectable histologically in about 10 per cent of patients
E Severe resorption in about 30 per cent of patients treated

113 Which single patient factor is most strongly associated with severe orthodontically induced inflammatory root resorption?

A Age over 30 years at the start of active treatment
B A Class II division 2 malocclusion at presentation
C Female sex, particularly in the second decade
D Self-ligating rather than conventional brackets
E Blunt or shortened root form with previous resorption

114 A 15-year-old boy is 12 months into fixed appliance treatment and a review radiograph shows 2 mm of apical shortening of both upper central incisors. Regarding this finding:

TRUE OR FALSE — judge each statement independently; any number may be true

A Root resorption of this degree will regenerate completely once the appliance is removed
B Continuing treatment unchanged is appropriate because resorption ceases spontaneously
C Prolonged treatment duration is unrelated to the amount of resorption
D A pause of 3 to 6 months allows the resorption lacunae to repair with cementum
E The patient and parent should be informed and the finding recorded

115 The same patient's resorption is discussed with the parents. Which single management is most appropriate?

A Remove the appliances immediately and accept the current occlusion
B Pause for 3 to 6 months with passive archwires, then reassess
C Continue treatment unchanged and repeat the radiograph at debond
D Continue treatment but increase the forces to finish sooner
E Refer for endodontic treatment of both central incisors

116 At what point during fixed appliance treatment is a screening radiograph of the upper incisors most informative for detecting root resorption?

A Before bonding, as a baseline only
B Only at the debond appointment
C At 24 months, when most movement is done
D At 6 to 9 months after active movement begins
E Only if the patient reports symptoms

117 An upper first premolar has failed to respond to eight weeks of extrusive force. It gives a high-pitched note on percussion, has no mobility, and a periapical radiograph shows loss of the lamina dura over part of the root with the adjacent teeth erupting normally. Which single diagnosis is most likely?

A Ankylosis
B Primary failure of eruption
C Hypercementosis
D External cervical resorption
E Periapical periodontitis

118 Regarding ankylosis and primary failure of eruption:

TRUE OR FALSE — judge each statement independently; any number may be true

A A tooth affected by primary failure of eruption typically becomes ankylosed when orthodontic traction is applied
B Ankylosis in a growing child causes the tooth to move occlusally with the alveolus
C Primary failure of eruption characteristically affects a tooth and all the teeth distal to it in the quadrant
D An ankylosed permanent tooth reliably responds to intrusive forces on the adjacent teeth
E Percussion is of no value in distinguishing an ankylosed tooth from a normal one

119 A 13-year-old girl has a lower right first permanent molar 4 mm below the occlusal plane. The second and third molars on that side are also infraoccluded, while the left side is normal. Her father had teeth removed for the same problem. Which single conclusion has the greatest bearing on treatment planning?

A Orthodontic traction will bring the molars into occlusion if sufficient force is used
B Surgical luxation will allow the teeth to erupt normally
C This is likely to be primary failure of eruption, and traction will fail
D The infraocclusion is caused by the opposing teeth overerupting
E Extraction of the third molar alone will resolve the problem

120 In an experimental preparation the dental follicle around an unerupted premolar is removed while the tooth germ is left in place. Which single outcome is expected?

A The tooth erupts normally because eruption is driven by root formation
B The tooth erupts but its root subsequently fails to form
C The tooth erupts more rapidly because the follicle restrains it
D The tooth fails to erupt because the follicle clears the overlying bone
E The tooth becomes ankylosed to the overlying bone and then erupts

121 Which single observation argues most strongly against root elongation being the mechanism that drives tooth eruption?

A Teeth are observed to erupt faster at night than by day
B Eruption continues after the root is complete, and rootless teeth erupt
C Teeth with incomplete root formation can be seen erupting, so root growth cannot be excluded as a cause
D Eruption rate varies considerably between individuals
E Eruption is accompanied by growth of the alveolar bone

122 Regarding the mechanism of tooth eruption:

TRUE OR FALSE — judge each statement independently; any number may be true

A A tooth will not erupt if its dental follicle is removed
B Osteoclastic resorption creates the eruption pathway through the overlying bone
C Eruption continues after root formation is complete
D Compensatory eruption maintains occlusal contact when an opposing tooth is lost
E An impacted tooth without a follicular pathway may remain unerupted indefinitely

123 A 58-year-old woman requesting alignment of her lower incisors has taken an oral bisphosphonate for six years for osteoporosis. Which single statement about her orthodontic treatment is most accurate?

A Tooth movement will be faster because bone turnover is increased
B Bisphosphonate therapy prevents root resorption and is therefore advantageous

C Bisphosphonates affect the maxilla but not the mandible
D Treatment may proceed exactly as for an untreated patient of the same age
E Movement may be slowed or arrested because osteoclasts are suppressed

124 A 35-year-old man is compared with a 14-year-old boy, both having identical upper premolar extraction spaces closed with identical mechanics. Which single difference is most likely?

A The adult shows a longer lag phase but a similar eventual rate
B The adult's teeth will not move at all under these forces
C The adult will move twice as fast because the bone is denser
D The adult will require forces about three times as heavy
E Root resorption will be absent in the adult patient

125 Regarding methods proposed to accelerate orthodontic tooth movement:

TRUE OR FALSE — judge each statement independently; any number may be true

A Low-level vibration devices have been shown in randomised trials to shorten treatment substantially
B Corticotomy produces a permanent increase in the rate of tooth movement for the duration of treatment
C Piezocision carries no surgical morbidity
D Self-ligating brackets shorten overall treatment time compared with conventional brackets
E Systemic administration of prostaglandin is an accepted clinical method of acceleration

126 A patient taking ibuprofen regularly for a sports injury is about to begin fixed appliance treatment and asks what to take for orthodontic pain. Which single piece of advice is best supported?

A Continue ibuprofen, as non-steroidal anti-inflammatory drugs have no effect on tooth movement
B Avoid all analgesics, as pain relief prevents tooth movement
C Use paracetamol instead, as anti-inflammatories may slow movement
D Use aspirin instead, which has no effect on prostaglandins
E Analgesia is unnecessary because orthodontic pain is negligible

127 A 13-year-old girl telephones two days after her first archwire, in considerable discomfort, wanting the appliance removed. Which single response is most appropriate?

A Remove the appliance, as this degree of pain is abnormal for a newly placed archwire
B Explain that the pain peaks at 48 hours and settles within a week
C Reduce the archwire size at an emergency appointment
D Prescribe an antibiotic and a chlorhexidine rinse
E Advise that the pain will persist throughout treatment

128 A 14-year-old boy is having upper first premolar extraction spaces closed to reduce a 7 mm overjet, and maximum anchorage is required. Regarding force levels and anchorage:

TRUE OR FALSE — judge each statement independently; any number may be true

A If the reactive unit is not reinforced, roughly two-thirds of the extraction space is lost to forward movement of the posterior teeth
B Headgear used for anchorage reinforcement is conventionally worn 10 to 12 hours per day at 250 to 350 g per side
C Anchorage value is increased by allowing the molars to tip mesially
D Heavier retraction forces increase the rate of anchorage loss relative to canine movement
E Two independent safety mechanisms are mandatory on any headgear

129 In an extraction case where no anchorage reinforcement is used, approximately what proportion of the extraction space is typically consumed by mesial movement of the posterior teeth?

A About one-tenth
B About one-quarter
C About one-third
D About two-thirds
E The whole space

130 Headgear is to be used to distalise upper molars rather than merely to reinforce anchorage. Which single regimen is most appropriate?

A 450 to 500 g per side for 14 hours per day
B 250 to 350 g per side for 10 to 12 hours per day
C 150 g per side worn at night only
D 700 g per side for 8 hours per day
E 450 to 500 g per side for 6 hours per day

131 Regarding the safety and biomechanics of extraoral traction:

TRUE OR FALSE — judge each statement independently; any number may be true

A A single safety mechanism is sufficient provided the patient is warned about the risks
B The direction of extraoral pull relative to the centre of resistance of the upper first molar determines whether the molar tips or moves bodily
C Headgear may be worn during contact sport provided a mouthguard is used
D A high pull direction tends to intrude as well as distalise the upper molars
E Intraoral injury is the principal reported serious hazard of headgear

132 An orthodontic miniscrew is being placed in the buccal attached mucosa between the upper second premolar and first molar. Which single insertion torque range is most appropriate?

A 1 to 3 Ncm
B 15 to 20 Ncm
C 5 to 10 Ncm
D 25 to 30 Ncm
E Torque is irrelevant provided the screw is stable

133 A clinician reviewing miniscrew failure rates wishes to select the most reliable site for a screw providing absolute anchorage for upper molar distalisation. Which single site is best supported by the reported failure rates?

A Mandibular buccal shelf
B Maxillary buccal interradicular bone
C The mandibular symphysis
D The retromolar area
E The median palate

134 Which single statement best describes the biological advantage of a continuous light force over an intermittent heavy force for orthodontic tooth movement?

A A continuous light force maintains a stress that permits frontal resorption
B A continuous light force produces more hyalinisation and therefore faster undermining resorption
C An intermittent heavy force avoids root resorption altogether
D A continuous light force is more comfortable but biologically identical
E An intermittent heavy force maintains the ligament blood supply throughout

135 Regarding the periodontal ligament and alveolar bone during orthodontic movement:

TRUE OR FALSE — judge each statement independently; any number may be true

A Alveolar bone is deposited on the tension side while the ligament fibres are remodelled to restore their original width
B The periodontal ligament fibres remodel more slowly than the supracrestal gingival fibres
C The lamina dura is remodelled during tooth movement
D Root cementum is more resistant to resorption than alveolar bone
E Attachment level is normally lost during uncomplicated orthodontic tooth movement

136 A severely rotated upper lateral incisor has been derotated and the appliance is due for removal. Which single measure most directly addresses the tissue responsible for rotational relapse?

A A bonded lingual retainer alone
B Interproximal enamel reduction
C Circumferential supracrestal fibrotomy
D A period of overcorrection of two weeks
E A vacuum-formed retainer worn full time for six months

137 A patient asks whether clear aligners will avoid the root shortening she has read about with fixed appliances. Which single response is best supported by the evidence?

A Aligners do not move roots and therefore cannot cause resorption
B The question cannot be answered because no comparative data exist
C Aligners cause more severe resorption than fixed appliances
D Resorption with aligners occurs only in extraction cases
E Resorption occurs with aligners too, and the difference is small

138 A manufacturer claims its bracket system delivers the optimal force for every tooth. Which single appraisal of the concept of an optimal force is best supported?

A Optimal force is a single precisely known value for each tooth
B Optimal force is a range that differs with the movement and root area
C The concept of optimal force has no real biological basis
D Optimal force is the force that produces the most rapid movement regardless of other effects
E Optimal force is determined solely by the archwire dimension

139 A 45-year-old man with treated chronic periodontitis has 40 per cent horizontal bone loss, no residual pocketing over 4 mm and good plaque control. He wishes to have his proclined upper incisors retracted. Which single statement is most accurate?

A Orthodontic movement is contraindicated in any reduced periodontium
B Orthodontic movement will regenerate some of the lost attachment
C Heavier forces are required because the remaining bone is denser
D Movement may proceed with light forces provided inflammation is controlled
E Movement will be safe even if bleeding on probing persists

140 A 16-year-old girl is 20 months into fixed appliance treatment for a 9 mm overjet with upper first premolar extractions. Which single tooth is most likely to show the greatest apical shortening on her debond radiograph?

A Upper first molar
B Upper canine
C Upper central incisor
D Lower first molar

E Lower second premolar

141 A lower first molar is to be moved mesially into an edentulous space from which the second premolar was removed 15 years ago. The ridge is narrow buccolingually and the crest is markedly resorbed. Which single statement about the movement is most accurate?

A Movement is possible but slow, and the root may dehisce in a narrow ridge
B Movement will be faster because the ridge contains no tooth
C Movement into an atrophic ridge is biologically impossible
D The ridge will widen ahead of the tooth without any additional measure
E Heavy forces will overcome the cortical bone of the atrophic ridge

Block E — Occlusion, malocclusion and aetiology (Q142–171)

Volume I: Chapter 2 (Occlusion and Malocclusion)
Volume II: §5 — read it again if you score under 70%
This block: 30 questions (7 of them true/false) · 58 marks · allow 38 minutes

142 A patient at debond has Class I molar and canine relationships, no rotations, no spaces, correct crown angulations and inclinations, but a residual curve of Spee of 3 mm in the lower arch. Which single statement is most accurate?

A All six of Andrews' keys are satisfied
B The finding is of no consequence because the curve of Spee is self-correcting
C The molar key is not satisfied because the curve of Spee is part of it
D Andrews' keys do not include the occlusal plane
E Five keys are satisfied and the sixth, a flat occlusal plane, is not

143 A 13-year-old girl has a Class I molar relationship bilaterally. Her upper central incisors are retroclined, the lower incisor edges lie posterior to the cingulum plateau of the upper central incisors, and the overjet is 2 mm with an overbite of 7 mm that is complete to the palatal mucosa. Which single classification is correct?

A Angle Class I, incisors Class I
B Angle Class I, incisors Class II division 1
C Angle Class II division 2, incisors Class II division 2
D Angle Class I, incisors Class II division 2
E Angle Class III, incisors Class II division 2

144 An examiner is presented with a case in which the lower incisor edges lie immediately below the cingulum plateau of the upper central incisors on the right, and 2 mm posterior to it on the left, with an overjet of 3 mm on the right and 5 mm on the left. Which single approach to classification is most appropriate?

A Record the average of the two sides and classify as Class I
B Classify by the more severe side and record the asymmetry separately
C Classification is impossible where the two sides differ
D Record as Class II division 2 because the overjet differs between the sides
E Classify by the molar relationship on each side instead

145 Regarding the classification of occlusion and malocclusion:

TRUE OR FALSE — judge each statement independently; any number may be true

A Angle's classification assumes that the upper first permanent molar is in a constant position relative to the craniofacial skeleton
B The British Standards incisor classification refers the lower incisor edge to the cingulum plateau of the upper central incisors
C In a Class II division 2 incisor relationship the overjet may occasionally be increased
D Angle's classification can be applied where the first permanent molars have been extracted
E A Class III incisor relationship is defined by the lower incisor edges lying anterior to the cingulum plateau of the upper central incisors

146 A 14-year-old boy has a Class I molar relationship on the right and a half-unit Class II molar relationship on the left. The canine relationship is Class I on both sides, and the lower centreline is displaced 3 mm to the left. Which single explanation is most likely?

A A unilateral skeletal asymmetry of the mandibular body
B Congenital absence of the lower left second premolar with space closure
C Overdevelopment of the upper left buccal segment
D A functional displacement of the mandible to the left
E Early loss of a lower left primary molar with mesial drift of the molar

147 A patient has an overbite in which the lower incisor edges contact the palatal mucosa but there is no contact with the upper incisors. Which single description is correct?

A Increased and complete to tooth
B Increased and complete to soft tissue
C Increased and incomplete
D Reduced and complete
E Traumatic and therefore by definition complete to tooth

148 A 12-year-old girl has an upper left second premolar whose palatal cusp occludes buccal to the buccal cusp of the lower left second premolar. Which single description is correct?

A Buccal crossbite of the upper left second premolar
B Unilateral posterior crossbite with a mandibular displacement
C An anterior crossbite extending into the buccal segment
D Lingual crossbite, or scissors bite, of the upper second premolar
E Bilateral posterior crossbite without displacement

149 An 11-year-old boy has a unilateral posterior crossbite on the right. On closing from the retruded contact position the mandible slides 2 mm to the right into the intercuspal position. Regarding this finding:

TRUE OR FALSE — judge each statement independently; any number may be true

A The crossbite is a true unilateral crossbite of skeletal origin
B The path of closure is of no consequence provided the teeth intercuspate
C The underlying transverse discrepancy is likely to be bilateral, with the mandible displacing to one side to achieve intercuspation
D A unilateral posterior crossbite with displacement is less likely to warrant treatment than one without displacement
E Displacement on closing has no association with temporomandibular signs

150 A 12-year-old boy has an 8 mm overjet, an incompetent lip seal, and on smiling the lower lip rests behind the upper incisors at rest. Which single statement about the soft tissue contribution is best supported?

A The trapped lower lip proclines the uppers and retroclines the lowers
B The lower lip position is a consequence of the overjet alone and has no effect on it
C Lip competence is determined only by the skeletal pattern
D The soft tissues will correct the overjet spontaneously at puberty
E The lower lip exerts no measurable force on the incisors

151 A 9-year-old girl sucks her right thumb for several hours each evening and overnight. Which single combination of findings is most characteristic?

A A symmetrical anterior open bite with a Class III incisor relationship
B A deep overbite with retroclined upper incisors and crowding
C An asymmetrical open bite, an increased overjet and a posterior crossbite

D A reduced overjet with a bilateral buccal segment crossbite
E An anterior open bite with no transverse effect at all

152 A 15-year-old boy has an anterior open bite and the tongue is seen to come forward between the incisors on swallowing. He has no digit habit, an increased lower face height and a steep mandibular plane. Which single interpretation is best supported?

A The tongue thrust is the primary cause and myofunctional therapy is indicated
B The finding is normal and has no bearing on the treatment plan
C The tongue thrust proves an endogenous tongue thrust and predicts relapse of any correction
D The tongue thrust is most likely adaptive to the open bite, which is of skeletal origin
E The open bite is caused by a genuinely enlarged tongue

153 Regarding the equilibrium theory of tooth position:

TRUE OR FALSE — judge each statement independently; any number may be true

A The resting pressures of the lips, cheeks and tongue are more important in determining tooth position than the brief forces of swallowing
B Swallowing forces are of long enough duration to move teeth because they are of high magnitude
C Forces from an appliance need to act for only a few hours a day to produce tooth movement
D The forces of eruption and of the periodontal and gingival tissues form part of the equilibrium
E Expansion beyond the soft tissue equilibrium tends to relapse

154 A 15-year-old girl has an ANB angle of 6 degrees. Her SNA is 88 degrees and her SNB is 82 degrees, and the maxillary and mandibular lengths are both within normal limits. Which single interpretation is best supported?

A The mandible is retrognathic relative to the cranial base
B The maxilla is prognathic and the mandible is within normal limits
C The discrepancy is entirely dentoalveolar in origin
D An ANB of 6 degrees lies within normal limits for this age, given the normal skeletal proportions described
E The cranial base must be abnormal in this patient

155 A 10-year-old boy has a single upper central incisor unerupted 20 months after its partner, with a retained primary predecessor and a tuberculate supernumerary on the radiograph. Which single category of aetiology does this best illustrate?

A Skeletal
B Soft tissue
C Generalised dental
D A habit
E A local cause

156 A public health team plans orthodontic services for a British population of 12-year-olds. Which single set of prevalence estimates is most realistic?

A Class II division 1 about 15 to 20 per cent and Class III about 3 per cent
B Class II division 1 about 3 per cent and Class III about 20 per cent
C Class II division 1 about 50 per cent and Class III about 15 per cent
D Class II division 1 about 5 per cent and Class III about 5 per cent
E Class II division 1 about 30 per cent and Class III about 10 per cent

157 Regarding the prevalence of malocclusion traits:

TRUE OR FALSE — judge each statement independently; any number may be true

A Class III malocclusion is more common in white European populations than in Chinese populations
B Crowding is the commonest single reason for orthodontic referral in the United Kingdom
C Prevalence figures for malocclusion vary substantially with the population studied and the index used
D Anterior open bite in the permanent dentition affects about 25 per cent of British adolescents
E Posterior crossbite affects under 1 per cent of British children

158 A father with a marked Class III skeletal pattern asks how likely it is that his 6-year-old son will need jaw surgery. Which single response is best supported?

A Malocclusion is entirely inherited, so surgery is inevitable
B Malocclusion is entirely environmental, so the family history is irrelevant
C Skeletal pattern is substantially inherited, so the risk is raised but not certain
D The risk can be quantified precisely from the father's cephalogram
E Only the mother's skeletal pattern is relevant to the child's

159 A candidate states that the aim of orthodontic treatment is an ideal occlusion. Which single appraisal is best supported?

A Ideal occlusion is the realistic goal of every orthodontic treatment
B Ideal occlusion is a theoretical standard and normal occlusion the practical goal
C Ideal occlusion and normal occlusion mean the same thing
D Normal occlusion means the commonest occlusion in the population, which is Class I
E Ideal occlusion is defined by Angle's classification alone

160 A 14-year-old girl has retroclined upper central incisors, proclined upper lateral incisors, a deep traumatic overbite and a high lower lip line covering more than one-third of the upper central incisor crowns. Which single explanation of the incisor position is best supported?

A The lower lip position is a consequence of the incisor inclination, having no independent effect on tooth position
B The finding is entirely skeletal and unrelated to the soft tissues
C The tongue retroclines the upper central incisors at rest
D A high resting lower lip line retroclines the centrals but not the laterals
E The deep overbite has retroclined the upper incisors

161 Regarding the Class II division 2 incisor relationship:

TRUE OR FALSE — judge each statement independently; any number may be true

A A deep overbite is characteristic and is often complete to soft tissue
B The skeletal pattern is commonly Class I or mild Class II with a reduced lower face height
C An increased overjet excludes the diagnosis
D Forward mandibular rotation is commonly associated with the pattern
E Correction of the overbite in these patients is prone to relapse

162 A 7-year-old boy has an anterior open bite of 4 mm, an asymmetrical appearance to the open bite, proclined upper incisors and a normal lower face height. He stopped sucking his thumb three months ago. Which single management is most appropriate?

A Reassure and review, as a habit open bite resolves once the habit stops
B Begin fixed appliance therapy to close the open bite now

C Refer for a tongue thrust assessment and myofunctional therapy, since the pattern suggests an ongoing habit
D Fit a posterior bite plane to allow the incisors to erupt
E Plan for surgical correction of the open bite in adolescence

163 A 10-year-old girl closes into a bilateral edge-to-edge incisor relationship in the retruded contact position and then slides forward 3 mm into a reverse overjet of 2 mm. Which single conclusion is best supported?

A She has a true skeletal Class III relationship requiring surgical planning
B The displacement is of no clinical significance here
C She has a forward displacement, and the true relationship is edge to edge
D The finding indicates a temporomandibular disorder
E The retruded contact position cannot be recorded reliably in a child

164 A clinician argues that the sixth of Andrews' keys is unimportant because a curve of Spee has no functional consequence. Which single response is best supported?

A The key is unimportant and may safely be disregarded
B A deeper curve of Spee improves posterior occlusal contacts
C The curve of Spee bears no relation to the overbite
D Andrews' keys were derived from treated cases and therefore do not apply to untreated occlusions
E A flat occlusal plane is required because a residual curve deepens the overbite

165 Regarding the aetiology of malocclusion:

TRUE OR FALSE — judge each statement independently; any number may be true

A A Class II skeletal pattern is always the result of a retrognathic mandible
B Early loss of a primary incisor characteristically causes significant space loss in the labial segment
C Mouth breathing has been shown to be a proven cause of increased lower face height
D A single aetiological factor can usually be identified for a given malocclusion
E Crowding in the permanent dentition is principally determined by the environment rather than by inheritance

166 A patient closes on a premature contact between the upper and lower second molars and then slides anteriorly and laterally into the intercuspal position. Which single statement about recording the malocclusion is most accurate?

A Both positions should be assessed, since a displacement conceals the truth
B The malocclusion should be recorded in the intercuspal position only
C The retruded contact position is a theoretical construct of no clinical use
D A displacement of less than 4 mm is not worth recording
E The premature contact should be adjusted before any assessment is made

167 A 13-year-old boy has 9 mm of crowding in the lower arch. His parents both had orthodontic treatment with extractions. Which single account of the aetiology of his crowding is best supported?

A Crowding is caused by a soft diet in childhood, which is why it has increased in modern populations
B Crowding is caused by the forward pressure of the third molars
C Crowding is a mismatch of tooth size and arch size, both largely inherited
D Crowding results from premature loss of primary teeth in every case
E Crowding is caused by an abnormal tongue posture at rest

168 On completing space closure in a Class I extraction case, a Class I canine relationship is achieved but a 2 mm overjet persists with the lower incisors well aligned and the upper incisors in good torque. Measurement shows the summed width of the lower anterior teeth is a smaller proportion of the upper than expected. Which single conclusion is best supported?

A The upper incisors require further palatal retraction
B The finding indicates an underlying skeletal discrepancy
C The canine relationship must have been recorded incorrectly
D The lower incisors should be proclined to close the overjet, accepting the resulting change in inclination
E There is an anterior tooth size discrepancy, with excess upper material

169 Regarding the description of an occlusion in clinical records:

TRUE OR FALSE — judge each statement independently; any number may be true

A Overjet is measured from the labial surface of the lower incisor to the incisal edge of the most prominent upper incisor, parallel to the occlusal plane
B An overbite is described as reduced when the lower incisors contact the palatal mucosa
C Canine relationship is classified by the position of the lower canine cusp tip relative to the embrasure between the upper canine and lateral incisor
D A crossbite is defined by the relationship of the upper and lower incisors only
E The molar relationship and the canine relationship may differ within the same patient

170 A 9-year-old boy with a Class III incisor relationship and a mild Class III skeletal pattern has treatment planned. His parents ask whether the problem will improve as he grows. Which single response is best supported?

A Class III skeletal patterns typically improve with growth as the maxilla catches up
B Class III patterns tend to worsen because mandibular growth continues later
C Growth has no influence on a Class III skeletal pattern
D Improvement is likely if the patient turns out to be female
E The pattern will improve once the third molars are removed

171 An examiner asks why the British Standards incisor classification is preferred to Angle's classification in United Kingdom practice. Which single justification is strongest?

A It is simpler to apply in routine clinical practice
B It was published more recently than Angle's system
C It correlates better with the underlying skeletal pattern
D It classifies the presenting problem and does not rely on molars
E It can be applied without recourse to study models

Block F — Treatment need, indices and epidemiology (Q172–196)

Volume I: Chapter 2, pp. 106–112 (IOTN and PAR)
Volume II: **§6** — read it again if you score under 70%
This block: 25 questions (6 of them true/false) · 49 marks · allow 31 minutes

172 A 12-year-old boy is referred for an opinion. The overjet measures 9 mm to the most prominent upper central incisor, the lips are competent, there is no crossbite and the largest contact point displacement is 1 mm. Which single dental health component grade should be recorded?

A Grade 2a
B Grade 3a
C Grade 4a
D Grade 5a
E Grade 5i

173 A 13-year-old girl has an unerupted upper left permanent canine lying palatally, with a supernumerary tooth interposed between it and the alveolar crest. Her overjet is 3 mm, her overbite is average and the largest contact point displacement is 2 mm. Which single dental health component grade applies?

A Grade 5i
B Grade 4t
C Grade 4x
D Grade 3d
E Grade 5s

174 An examiner is calibrating a team in the use of the aesthetic component of IOTN before a school screening survey. Which of the following are true?

TRUE OR FALSE — judge each statement independently; any number may be true

A The ten photographs are ranked for attractiveness, so the patient is matched to the photograph of equivalent aesthetic impairment rather than of equivalent malocclusion
B A score of 8 indicates a definite need for treatment on aesthetic grounds
C A dental health component of grade 4 qualifies for treatment whatever the aesthetic score
D Ratings made by patients themselves agree closely with those made by the clinician
E The scale was constructed from photographs of the buccal segments in occlusion

175 A unit audits its outcomes. One case scores 24 on the weighted PAR index before treatment and 18 afterwards. Which single classification of the outcome is correct?

A Not classifiable because the case was treated without extractions
B Improved
C Greatly improved
D Improved only if the starting score exceeded 30
E Worse or no different

176 An 11-year-old girl has a unilateral posterior crossbite with a 3 mm discrepancy between the retruded contact position and the intercuspal position, an overjet of 4 mm with competent lips and contact point displacements of 2 mm. Which single dental health component grade should be recorded?

A Grade 2c
B Grade 3c
C Grade 4l
D Grade 4c
E Grade 5m

177 A 15-year-old girl has a dental health component of grade 1 and an aesthetic component of 2. She is distressed by a 1 mm rotation of an upper lateral incisor, has consulted three practitioners in a year and says she covers her mouth when speaking. Which single course is most appropriate?

A Decline treatment and discharge her, since the index records no objective need for an appliance
B Explore the concern and screen for features of body dysmorphic disorder before committing to any appliance
C Fit aligners as requested, since perceived need alone is what determines whether any treatment is provided
D Take full records so that the minor nature of the discrepancy can be demonstrated objectively
E Refer directly for orthognathic assessment, since her distress implies a facial disproportion

178 A trainee is recording IOTN for the first time. Regarding the index, which of the following are true?

TRUE OR FALSE — judge each statement independently; any number may be true

A The dental health component is scored by summing the scores of every occlusal trait present
B A reverse overjet of 2 mm with no reported functional difficulty is a grade 4 trait
C The aesthetic component is recorded from the alignment of the lower labial segment
D The index incorporates an assessment of caries risk and oral hygiene
E A contact point displacement of 3 mm is recorded as grade 4d

179 A 14-year-old boy has an increased and complete overbite with the lower incisors contacting the palatal mucosa, producing visible indentation and erythema. His overjet is 5 mm with competent lips. Which single dental health component grade applies?

A Grade 2f
B Grade 3f
C Grade 4e
D Grade 5a
E Grade 4f

180 A 14-year-old has 9 mm of crowding confined to the upper buccal segments, an aesthetic component of 3, a Class I buccal relationship and no crossbite. The ICON score is 38, below the cut-off of 43. Which single explanation best accounts for the low score?

A Crowding is not among the five components that contribute to the ICON score
B The aesthetic component carries the heaviest weighting in ICON and is low in this case
C The cut-off of 43 applies only to adults and is set lower for adolescents
D The buccal segment relationship carries the heaviest weighting and is normal here
E ICON measures the complexity of treatment and records nothing about need

181 A commissioner asks a hospital department how its waiting list for a limited number of funded treatment places should be ordered. Which single basis is most defensible?

A Order of referral, because a first come first served rule treats every applicant identically
B Strength of the patient's expressed wish, because demand should determine provision
C Predicted compliance, because places should go to those who will benefit most
D Recorded severity of occlusal need, because the same standard is applied to every applicant
E Distance travelled to the unit, because access is itself a component of equity

182 A 13-year-old girl has a 5 mm contact point displacement between the upper left lateral incisor and canine, an overjet of 4 mm with competent lips and a 3 mm anterior open bite. Which single dental health component grade should be recorded?

A Grade 4d
B Grade 3d
C Grade 3e
D Grade 2a
E Grade 4e

183 A department is planning a prevalence survey and reviews what is known about the distribution of malocclusion traits. Which of the following are true?

TRUE OR FALSE — judge each statement independently; any number may be true

A Class III incisor relationships are considerably more common in Chinese and Japanese populations than in white European populations
B Hypodontia excluding third molars affects about 4 to 6 per cent of the population
C Anterior open bite is less common in people of African ancestry than in white Europeans
D The majority of impacted maxillary canines lie palatal to the arch
E Maxillary canine impaction affects roughly 2 to 3 per cent of the population

184 A 14-year-old boy has developmental absence of the upper left lateral incisor and the upper left second premolar, both confirmed radiographically. A restorative opinion confirms that space will need to be redistributed before prosthetic replacement. All other traits are grade 2 or less. Which single dental health component grade applies?

A Grade 3d
B Grade 4d
C Grade 5h
D Grade 4h
E Grade 5i

185 Two 13-year-olds are both recorded as dental health component grade 4d. One has a single 5 mm displacement between an upper lateral incisor and canine. The other has severe crowding in all four quadrants with an ectopic canine and an 8 mm displacement. Which single conclusion is best supported?

A The index has correctly identified them as equal in need, since the worst trait is the same
B The index is not valid where crowding is generalised and should not be applied
C The second case should be rescored by adding the grades for each quadrant
D The index records only the worst trait, so it cannot express severity beyond that threshold
E The difference between them is captured by the aesthetic component instead

186 A dental public health team compares the malocclusion traits expected in a screening programme in Hong Kong with published United Kingdom figures. Which single trait would be expected to be substantially more prevalent in the Hong Kong cohort?

A Class II division 1 incisor relationship
B Class III incisor relationship
C Hypodontia of the second premolars
D Palatal displacement of the maxillary canine
E Increased and complete overbite

187 A postgraduate is asked to distinguish need, demand and provision when planning a service. Which of the following are true?

TRUE OR FALSE — judge each statement independently; any number may be true

A Normative need is need as judged by a professional against a defined standard
B Perceived need and normative need agree closely in adolescent populations
C A patient may have a high normative need and no perceived need at all
D Demand for treatment is influenced by socioeconomic factors independently of severity
E Allocating limited public funding by recorded grade of need is an application of distributive justice

188 A 9-year-old boy is screened at school and recorded as dental health component grade 3d on the basis of a 3 mm displacement between the erupting upper central incisors. Which single statement should most influence how this grade is used?

A The grade should be acted on now, since the displacement is already established and measurable
B A grade of 3 always requires the aesthetic component to be recorded from a photograph
C The grade should be doubled, since further crowding will develop as the canines arrive
D Screening at this age is invalid and the record should be discarded without further action
E The index is calibrated for the established permanent dentition, so a grade recorded now may change

189 A 12-year-old girl has an overjet of 5 mm. Her lips meet at rest without circumoral muscular activity. There is no crossbite, the overbite is average and the largest displacement is 1 mm. Which single dental health component grade applies?

A Grade 2a
B Grade 1
C Grade 3a
D Grade 4a
E Grade 4b

190 A case is scored on the weighted PAR index before treatment. Which single change during treatment would produce the largest fall in the weighted score?

A Alignment of 4 mm of cumulative upper anterior irregularity
B Correction of a centreline discrepancy of more than half a lower incisor width
C Reduction of an 8 mm overjet to 3 mm
D Reduction of a complete overbite to a 2 mm overbite
E Correction of a half unit Class II buccal segment relationship bilaterally

191 A case scores 42 on the weighted PAR index before treatment and 8 at debond. Which single classification of the outcome is correct?

A Worse or no different
B Greatly improved
C Improved
D Improved, because the percentage change alone determines the category
E Not classifiable without the aesthetic component score

192 A unit is introducing routine PAR scoring for audit. Regarding the index, which of the following are true?

TRUE OR FALSE — judge each statement independently; any number may be true

A The score is recorded on the patient at the chairside during the clinical examination
B The overjet component carries the heaviest weighting in the United Kingdom weightings
C A score of zero confirms that root angulation and torque are also ideal
D The percentage reduction is a valid measure of the standard of care delivered to an individual patient
E A reduction of less than 30 per cent places the case in the worse or no different category

193 A 45-year-old man has drifted and spaced upper incisors following generalised periodontitis, now stable after treatment. The dental health component records grade 4d on the basis of a 6 mm displacement. Which single statement should most influence the use of this grade?

A The grade confirms that treatment is indicated and arrangements should be made to provide it
B The aesthetic component should be used instead, as it is the valid measure in adults
C The grade should be reduced by one because index thresholds were set for children
D The index takes no account of the periodontal cause, so the grade should not drive the decision
E Periodontal disease automatically raises the recorded grade to 5 in an adult patient

194 A survey of 12-year-olds finds that 35 per cent record a dental health component of grade 4 or 5, but only 20 per cent of the cohort has sought an orthodontic opinion. Which single conclusion is best supported?

A Demand is lower than normative need, as awareness, access and parental attitude all intervene
B The survey examiners have over-recorded the grades and should now be recalibrated against a reference standard
C The index threshold for grade 4 has been set too low for this particular population
D The 15 percentage point difference represents cases that will resolve spontaneously
E Normative need and demand are equivalent, so one of the two figures must be erroneous

195 A committee is reviewing whether an occlusal index is fit for commissioning treatment. Which of the following are true?

TRUE OR FALSE — judge each statement independently; any number may be true

A Occlusal indices measure the impact of the malocclusion on quality of life
B Reproducibility between examiners is essential if an index is to be used for commissioning
C Indices record the malocclusion and not the difficulty of correcting it
D The dental health component was derived from direct evidence that each trait causes dental disease
E A case with a dental health component of grade 3 and an aesthetic component of 8 qualifies where the aesthetic threshold is 6

196 A team wishes to estimate the prevalence of definite treatment need among 12-year-olds in a defined city. Which single study design is most appropriate?

A A case-control study comparing treated and untreated children
B A cohort study following children from 8 to 16 years
C A cross-sectional survey of a random sample of the age group
D A randomised trial of screening against no screening
E A review of consecutive new patients attending the orthodontic department

Block G — Evidence-based orthodontics (Q197–226)

Volume I: Chapter 1 (Evidence-Based Orthodontics); Appendix D
Volume II: §7 — read it again if you score under 70%
This block: 30 questions (8 of them true/false) · 62 marks · allow 38 minutes

197 A clinician wishes to know whether treating a 9 mm overjet at 9 years of age reduces incisor trauma compared with waiting until 12. Which single design would give the least biased estimate of the effect?

A A retrospective review of trauma in children treated early in the unit
B A prospective cohort comparing children whose parents chose early treatment with those who waited
C A case-control study comparing children with and without incisor trauma
D A consensus statement from an expert panel of specialists
E A randomised controlled trial allocating children to early or delayed treatment

198 A trial protocol describes computer generation of the allocation sequence, central telephone allocation and stratification by centre. Which of the following are true?

TRUE OR FALSE — judge each statement independently; any number may be true

A Randomisation distributes unknown as well as known prognostic factors between the groups
B Allocation concealment prevents the recruiting clinician from knowing which group the next patient will enter
C Block randomisation with a small fixed block size can allow the final allocation in each block to be predicted
D Stratification is used to balance a prognostic variable such as centre across the groups
E Allocation concealment remains possible in a trial in which the operator cannot be blinded

199 In a trial of two functional appliances the allocation sequence was generated by computer, but the assignments were placed in sealed white envelopes which a recruiter was observed holding up to the window light. Which single criticism is most important?

A Allocation concealment was broken, permitting selection bias at entry
B The randomisation sequence was not truly random
C The trial was not blinded, so performance bias is inevitable
D The sample size calculation must have been performed after recruitment
E Detection bias has been introduced at the outcome assessment

200 A trial compares a twin block with an untreated control group, with mandibular length and self-esteem as outcomes. Which single approach to blinding is most realistic?

A Blind the patients by providing a placebo appliance to the control group
B Blind the operators by concealing the appliance design until fitting
C Blind the assessors who measure the outcome on anonymised records
D Accept that blinding is impossible and abandon the attempt entirely
E Blind the statistician only, since all other blinding is cosmetic

201 In a trial of early Class II treatment, 18 of the 90 children randomised to the early arm never wore the appliance. The investigators propose to analyse only those who complied. Which single approach is most appropriate?

A Analyse only the compliant children, since only they received the intervention
B Report both analyses and treat the compliant analysis as the primary result
C Exclude the 18 children and reduce the control group by the same number
D Re-randomise the 18 children into the control group
E Analyse all children in the groups to which they were randomised

202 A reviewer is appraising how a trial handled participants who deviated from protocol. Which of the following are true?

TRUE OR FALSE — judge each statement independently; any number may be true

A An intention to treat analysis excludes participants who did not receive the allocated intervention
B Intention to treat generally gives a more conservative estimate of effect than a per protocol analysis
C A per protocol analysis preserves the prognostic balance achieved by randomisation
D Loss to follow-up of 30 per cent threatens the validity of an intention to treat analysis
E Intention to treat is applicable only to placebo-controlled drug trials

203 A trial of 24 patients comparing two bracket systems reports a mean difference in alignment time of 1.2 months in favour of one system, with a 95 per cent confidence interval of minus 0.8 to 3.2 months and p equal to 0.21. Which single conclusion is best supported?

A The two systems are equivalent in alignment time
B The trial cannot exclude a difference that would matter clinically
C The result proves that the difference found is due to chance alone
D A larger p value would have made the finding more convincing
E The confidence interval should be recalculated at 99 per cent to reach significance

204 A trial of 400 patients finds that one adhesive gives a mean bond failure rate 0.4 percentage points lower than another, with p equal to 0.002. Which single interpretation is best supported?

A The difference is large enough to justify changing the material used
B The trial must have been analysed incorrectly to produce so small a p value
C The p value shows the difference is clinically as well as statistically important
D The difference is real but too small to influence clinical practice
E The result cannot be interpreted without knowing the confidence interval width

205 A cohort study reports that patients treated with a particular retainer have a risk ratio for clinically significant relapse of 0.92, with a 95 per cent confidence interval of 0.78 to 1.09. Which single conclusion is best supported?

A The retainer reduces the risk of relapse by 8 per cent in these patients
B The retainer has been shown to have no effect whatever on relapse
C The data are compatible with a modest benefit and with a modest harm
D The confidence interval is too narrow for the result to be credible
E The result is significant because the point estimate lies below 1

206 A journal club is discussing how results should be reported. Which of the following are true?

TRUE OR FALSE — judge each statement independently; any number may be true

A A 95 per cent confidence interval that excludes zero corresponds to a p value below 0.05 for a difference between two means
B A p value of 0.04 means there is a 4 per cent probability that the null hypothesis is true
C A confidence interval expresses the precision of the estimate in the units of measurement
D Narrowing a confidence interval requires a larger effect rather than a larger sample
E A result may be statistically significant and clinically unimportant at the same time

207 A meta-analysis of six trials of an appliance reports a pooled mean difference of 1.8 mm with an I squared value of 82 per cent. Which single response is most appropriate?

A Investigate the sources of the inconsistency and pool only if it can be explained
B Refit the data with a fixed effect model to obtain a tighter interval
C Report the pooled estimate as the best available summary without qualification
D Discard the three smallest trials so the remainder become homogeneous
E Conclude that the appliance is ineffective because the trials disagree

208 A funnel plot in a systematic review of an adjunct to accelerate tooth movement shows a gap in the lower left of the plot, with small studies present only where the effect is large and positive. Which single interpretation is best supported?

A Small trials are inherently more accurate and the pooled estimate is conservative
B The plot shows that the intervention works better in the smaller centres
C The asymmetry is expected in any funnel plot and carries no implication here
D Heterogeneity has been correctly handled by the use of a random effects model
E Small trials with unfavourable results are missing, so the pooled effect may be inflated

209 In a cohort study of white spot lesions, 40 per cent of the comparison group and 62 per cent of the exposed group develop a lesion. The authors report an odds ratio of 2.45. Which single statement is most accurate?

A The odds ratio of 2.45 can be read directly as the risk ratio
B Odds ratios and risk ratios always differ by a factor of two
C The odds ratio understates the risk ratio because the outcome is common
D The odds ratio overstates the risk ratio because the outcome is common
E A risk ratio cannot be calculated from a cohort study

210 A trainee is asked how a systematic review differs from a traditional review article. Which of the following are true?

TRUE OR FALSE — judge each statement independently; any number may be true

A A systematic review has a protocol and an explicit, reproducible search strategy specified in advance
B A meta-analysis may be inappropriate even where several trials address the same question
C A narrative review written by an authority ranks above a systematic review in the hierarchy of evidence
D Grey literature and non-English publications are sought in order to reduce publication and language bias
E Pooling trials of low methodological quality increases the reliability of the summary estimate

211 In a trial of early overjet reduction, incisor trauma occurred in 30 per cent of the untreated group and 20 per cent of the treated group. How many children must be treated early to prevent one episode of trauma?

A 3
B 10
C 5
D 20
E 50

212 A parent asks for her 9-year-old son's 8 mm overjet to be treated now, citing a newspaper report. The trials by O'Brien and colleagues found no difference in final occlusal outcome between one-phase and two-phase treatment, an early improvement in self-esteem that was not maintained, and a reduction in incisor trauma with a risk ratio of 0.58. Which single response is best supported?

A Early treatment should be started now because it improves the final occlusal result achieved
B Early treatment is indicated because it shortens the total time in appliances
C Early treatment should be avoided because the trials showed no benefit of any kind
D Early treatment should be started because the gain in self-esteem is durable
E Early treatment is justified here on the grounds of trauma risk rather than occlusal outcome

213 In the multicentre trial of protraction facemask reported by Mandall and colleagues, 60 per cent of the untreated group and 36 per cent of the treated group were judged to need orthognathic surgery at follow-up. How many children must be treated with a facemask to avoid one surgical case?

A 2
B 4
C 6
D 12
E 24

214 A practice principal is considering investing in self-ligating brackets and a rapid maxillary expansion programme on the strength of manufacturers' claims. Regarding the systematic review evidence, which of the following are true?

TRUE OR FALSE — judge each statement independently; any number may be true

A Self-ligating brackets have been shown to shorten total treatment duration by about four months
B Self-ligating brackets align the lower labial segment measurably faster than conventional brackets in randomised trials
C Randomised evidence shows that rapid maxillary expansion resolves obstructive sleep apnoea in adults
D Cochrane reviews of orthodontic interventions typically rest on high quality evidence
E A review that concludes the evidence is insufficient has thereby shown the intervention to be ineffective

215 A five-year follow-up of a randomised trial retains 65 per cent of participants, with 22 per cent lost from one arm and 48 per cent from the other. Which single response is most appropriate?

A Analyse the available cases and report the result without further qualification
B Exclude the arm with greater loss and report the remaining arm descriptively
C Replace the missing participants with new patients recruited at five years
D Treat the differential attrition as a threat to validity and present a sensitivity analysis

E Conclude that the longer follow-up makes the result more reliable than the original report

216 A study asks 200 adults who completed fixed appliance treatment between five and fifteen years ago to rate the pain they experienced in the first week. Which single source of error most threatens the findings?

A Recall bias, as remembered pain decays and is coloured by the outcome
B Selection bias arising from who was invited to take part in the study
C Detection bias in the measurement of the final occlusal result
D Confounding by the operator who provided the treatment concerned
E Regression to the mean across repeated pain scores in the week

217 A retrospective study reports that patients treated with aligners finished with lower PAR scores than those treated with fixed appliances in the same practice. Which single explanation must be excluded before the finding can be accepted?

A Random error arising from the modest size of the two samples
B Recall bias in the retrospective collection of the scores
C Confounding by indication, as milder cases are selected for aligners
D Detection bias in the blinded scoring of the study models, which would inflate the gap
E Regression to the mean between the two measurement points

218 An examiner asks a candidate to classify the threats to validity in a trial. Which of the following are true?

TRUE OR FALSE — judge each statement independently; any number may be true

A Selection bias arises when the groups differ systematically at the point of entry
B Performance bias arises when the care given differs between the groups other than in the intervention itself
C Detection bias is reduced by blinding the person who measures the outcome
D Selective reporting of the more favourable of two pre-specified outcomes is a form of reporting bias
E Bias and random error are both reduced by increasing the sample size

219 A clinician wanting to know whether bonded or vacuum-formed retainers give less relapse searches a database for the term orthodontics and retrieves 94000 records. Which single next step is most useful?

A Read the first fifty abstracts in the order in which they happen to have been returned
B Ask a colleague which retainer they prefer and search for supporting papers
C Restrict the search to the five journals the clinician already reads
D Restructure the question to specify population, intervention, comparator and outcome
E Limit the results to papers published within the last twelve months

220 A trial proposes to compare two bonding adhesives using a split-mouth design, with one adhesive on the right and the other on the left in every patient. Which single circumstance would make this design invalid?

A Bond failure rates are expected to be low in both of the groups being compared
B The outcome on one side can be influenced by the material used on the other
C Patients cannot be blinded to which side received which adhesive
D The two adhesives have different handling characteristics
E The trial is being conducted in more than one clinical centre

221 A trial reports that an appliance produces a 12 point greater reduction in PAR score than its comparator, but no difference in patient-reported satisfaction or oral health related quality of life. Which single interpretation is best supported?

A The PAR difference establishes that the appliance is clinically superior
B The trial has been underpowered for the PAR outcome and should be repeated
C The quality of life instrument must lack sensitivity and should be disregarded
D Patient-reported outcomes are irrelevant when comparing two appliances
E An occlusal index is a surrogate endpoint and correlates weakly with patient reports

222 A research committee is reviewing four draft trial protocols. Which of the following are true?

TRUE OR FALSE — judge each statement independently; any number may be true

A A pilot study is designed to provide a precise estimate of the treatment effect
B A crossover design is suitable for comparing two fixed appliance systems for alignment
C A superiority trial that fails to show a difference has not thereby established equivalence
D Registering a trial before recruitment begins reduces selective outcome reporting
E The primary outcome may be chosen after inspection of the data provided the choice is declared

223 A manufacturer supports a claim that a new bracket shortens treatment by citing a series of 20 consecutive cases treated in one practice, all finishing in under 18 months. Which single criticism is most fundamental?

A There is no comparison group, so the claim of shortening rests on nothing
B The sample of 20 is too small for statistical significance to be judged
C The practice may be unrepresentative of general practice
D The cases were not consecutive throughout the whole period
E Treatment duration is a poor measure of treatment quality

224 A trial of a functional appliance recruited from three specialist units, excluded children with poor attendance records and achieved 88 per cent compliance. A general practitioner wishes to apply the result. Which single consideration is most important?

A The internal validity of the trial is poor because of the exclusions
B The effect size should be doubled to allow for the general practice setting
C The result may not generalise, as the trial population was selected for compliance
D The trial cannot be applied outside the units in which it was conducted
E Randomisation guarantees that the result applies to every population

225 A candidate is asked to rank the evidence available on a clinical question. Which of the following are true?

TRUE OR FALSE — judge each statement independently; any number may be true

A A well conducted systematic review of randomised trials ranks above a single randomised trial
B A large cohort study provides stronger evidence of causation than a small randomised trial of the same question
C Expert opinion is excluded altogether from the hierarchy of evidence
D The hierarchy applies with equal force to questions of therapy, harm, diagnosis and prognosis
E Evidence quality may be downgraded for imprecision, inconsistency or risk of bias

226 A team designing a trial of overbite reduction must decide what difference the study should be able to detect. Which single approach is most appropriate?

A Choose the difference reported in the largest of the previously published trials
B Set the smallest difference that would be regarded as worthwhile and size the trial to it
C Recruit as many patients as the units can supply and analyse whatever difference emerges
D Set the difference after the data have been collected, to avoid an arbitrary choice
E Use a difference of one standard deviation, as this is the usual convention

Block H — Ethics, consent, communication and the psychosocial patient (Q227–257)

Volume I: Part V, unit V.1 (Ethics, consent and capacity); Topic 9 in part; Chapter 7
Volume II: §8 — read it again if you score under 70%
This block: 31 questions (8 of them true/false) · 63 marks · allow 39 minutes

227 A 28-year-old professional flautist is offered lingual fixed appliances. The clinician's routine consent discussion covers decalcification, root resorption, relapse and treatment duration. She mentions in passing that her embouchure is her livelihood. Which single addition to the discussion is most clearly required?

A A statement that all appliances carry some risk of discomfort during treatment
B A recommendation that she obtain a second opinion before consenting to treatment
C A written warning about the general risk of temporomandibular symptoms arising
D Discussion of the effect of lingual attachments on her playing and the alternatives
E A reminder that she may withdraw from treatment at any stage she wishes

228 A 22-year-old man with a Class III skeletal pattern is listed for bimaxillary surgery. He asks what could go wrong. Which single set of matters must the consent discussion cover?

A The material risks including altered lower lip sensation, and the reasonable alternatives
B The surgical technique in anatomical detail, so that he fully understands the operation
C The published complication rates of the unit compared with the national figures
D The risks that the surgeon considers most likely to concern a typical patient
E The financial consequences of the procedure for the commissioning body involved

229 A trainee is asked how the decision in Montgomery in 2015 changed consent in orthodontic practice. Which of the following are true?

TRUE OR FALSE — judge each statement independently; any number may be true

A The test of what must be disclosed is what a reasonable person in the patient's position would attach significance to
B A risk that is statistically rare may nonetheless be material to a particular patient
C Reasonable alternative treatments, including no treatment at all, must be discussed
D The judgment displaced the professional standard previously applied to the disclosure of risk
E A signed consent form is evidence that consent was sought but is not itself consent

230 A 15-year-old boy attends alone and asks for fixed appliances with extraction of two premolars. He explains the plan, the alternatives and the risks accurately in his own words. His mother telephones to say she does not want extractions. Which single course is most appropriate?

A Proceed on the boy's consent alone without further contact with his mother
B Proceed only once the mother has signed the consent form herself
C Assess and record his competence, proceed on his consent and involve his mother
D Defer all treatment until he reaches 16 years of age and can consent
E Refer the disagreement for a legal opinion before any treatment begins

231 A unit sees new patients, discusses the plan and fits appliances at the same appointment, asking the patient to sign the consent form immediately before bonding. Which single criticism of this practice is most important?

A The form should be countersigned by a second clinician who was present
B Written information alone would satisfy the requirement without discussion
C A verbal record would be preferable to a written form in these circumstances
D The form should be signed after bonding rather than immediately before
E Consent is a process, and information given minutes beforehand allows no reflection

232 A 13-year-old girl is brought by her father, who has consented to a functional appliance. She is quiet throughout and finally says she does not want anything in her mouth. Which single course is most appropriate?

A Explore her concerns, and do not begin while she is unwilling
B Fit the appliance and rely on her adapting once she has worn it
C Fit the appliance, since the father's consent is legally sufficient
D Discharge her from the clinic and advise the father to seek another opinion
E Ask the father to sign an additional form accepting responsibility for her cooperation

233 An adult patient with a mild learning disability attends for assessment and the team discusses her capacity to consent. Which of the following are true?

TRUE OR FALSE — judge each statement independently; any number may be true

A Capacity is decision specific and may fluctuate over time
B The patient must be able to understand, retain, weigh and communicate the decision
C A patient who makes a decision the clinician considers unwise thereby lacks capacity
D Capacity is presumed in an adult unless the contrary is demonstrated
E A diagnosis of learning disability establishes that a patient cannot consent to orthodontic treatment

234 A 19-year-old woman attends requesting treatment. Her occlusion is well aligned with a 0.5 mm centreline discrepancy. She has seen four clinicians in 18 months, describes checking her teeth in a mirror for long periods, has stopped attending her university tutorials and says her life will be unbearable until the discrepancy is corrected. Which single course is most appropriate?

A Begin a short course of aligners to address the discrepancy that she describes
B Decline treatment and discharge her without any further discussion
C Explain the findings, decline appliance treatment and offer psychological referral
D Take a full set of records and review her in six months to see whether she changes her mind
E Refer her for orthognathic assessment since her distress implies an underlying skeletal problem

235 Body dysmorphic disorder is reported in about 2 per cent of the general population but in roughly 5 to 13 per cent of those seeking orthodontic or cosmetic treatment. Which single conclusion is best supported?

A The condition is rare enough in practice that routine consideration is unnecessary
B Patients with the condition are usually satisfied once the perceived defect is corrected
C The figures show that orthodontic treatment causes the condition in susceptible patients
D The condition should be diagnosed and treated by the orthodontist who identifies it
E The elevated prevalence makes active consideration of the diagnosis part of assessment

236 A 14-year-old boy has worn fixed appliances for six months. Plaque covers the gingival third of most brackets, the gingivae bleed on probing and early white lesions are visible around four upper brackets. He says he cannot be bothered to brush. Which single course is most appropriate?

A Continue treatment unchanged and review the hygiene at the next routine visit
B Institute an intensive preventive regime, set an explicit standard and a review date
C Debond the appliances immediately and without any further discussion
D Apply fluoride varnish and prescribe a high fluoride toothpaste, continuing treatment to completion regardless
E Refer him to the hygienist and leave the decision about continuing with her

237 A patient is being consented for upper and lower fixed appliances. Regarding the risks that should be discussed, which of the following are true?

TRUE OR FALSE — judge each statement independently; any number may be true

A White spot lesions may appear within six months of bonding and affect up to half of patients
B Severe root resorption amounting to a third of root length or more affects about 1 to 5 per cent of patients
C Some relapse is expected and retention may need to be lifelong
D Pain typically peaks at about 48 hours and has largely resolved by seven days
E There is good evidence that fixed appliance treatment causes temporomandibular disorder

238 A 34-year-old woman asks for aligners to correct a 9 mm overjet in six months without extractions, having seen an advertisement. Her skeletal pattern is moderately Class II with a reduced lower face height. Which single course is most appropriate?

A Provide the aligners as requested, since she is an adult entitled to choose
B Begin aligner treatment and convert to fixed appliances when it fails
C Decline to see her further, since her expectations cannot be met
D Explain what is achievable, set out the realistic options and their compromises
E Refer her directly for orthognathic surgery as the only means of meeting her request

239 A 16-year-old girl attends alone and asks that her mother, who pays for the treatment, should not be told that she has stopped wearing her retainers. Which single course is most appropriate?

A Telephone the mother, since she is the person paying for the treatment
B Record the request and inform the mother at the next appointment regardless
C Respect the request, explore why she has stopped and encourage her to tell her mother
D Discharge the patient, as treatment cannot continue without parental involvement
E Disclose to the mother the fact of non-wear but not the reason for it

240 An adult with generalised bone loss and 6 mm pockets insists on fixed appliances to align his upper incisors, having been told the periodontal risk. He says the decision is his to make. Which single statement best resolves the conflict?

A Autonomy is a right to refuse rather than to demand, and non-maleficence limits offers
B Respect for autonomy requires the clinician to provide the treatment requested
C Justice requires that he be treated in the same way as any other patient asking for alignment
D Beneficence requires the clinician to provide whatever the patient believes will benefit him
E A signed disclaimer transfers responsibility for the outcome to the patient

241 A clinical governance session is examining how the four principles apply in orthodontic decisions. Which of the following are true?

TRUE OR FALSE — judge each statement independently; any number may be true

A Autonomy entitles a patient to demand any treatment they request
B Beneficence and non-maleficence are weighed against each other in every treatment decision
C Justice in this context refers principally to the avoidance of litigation
D The four principles are ranked, so that autonomy overrides the others where they conflict
E Respect for autonomy removes the clinician's responsibility for the outcome

242 A bonded lower retainer fractured at some point after debond and was not detected at review. The lower incisors have relapsed by 3 mm. Which single course is most appropriate?

A Repair the retainer and make no comment, since relapse is a recognised outcome
B Refer the patient to a colleague so that the discussion can be held independently
C Record the finding in the notes and await any complaint from the patient herself
D Attribute the change to natural late lower incisor crowding when explaining it
E Tell the patient what has happened and why, apologise and discuss the options

243 An 11-year-old boy is brought by his mother, who does all the talking. He looks at the floor and gives one-word answers. Which single approach to the consultation is most appropriate?

A Take the history from the mother, since she is the more reliable historian
B Postpone the consultation until the boy is old enough to participate
C Ask the mother to leave the room so that the boy can speak more freely
D Address the boy directly with age-appropriate explanation, involving his mother
E Provide written information for the mother to explain to him at home

244 A 17-year-old girl has agreed to fixed appliances but refuses the two premolar extractions the plan requires. Which single course is most appropriate?

A Proceed with the planned extractions, since she has consented to the treatment as a whole
B Explore her reasons, explain what would be compromised and offer the alternative plan
C Discharge her, since a treatment plan cannot be modified once it is agreed
D Extract one premolar as a compromise and reassess the space required
E Refer her to a colleague who does not use extractions in such cases

245 A patient has requested copies of her orthodontic records and a solicitor has written separately about a different matter. Which of the following are true?

TRUE OR FALSE — judge each statement independently; any number may be true

A A patient is entitled to access her own orthodontic records including photographs
B Records may be released to a solicitor acting for the patient without the patient's consent
C Information may be disclosed without consent where there is a risk of serious harm to others
D Images used for teaching require the patient's consent even when anonymised
E Orthodontic records should be retained for a defined minimum period after treatment ends

246 A clinician wishes to use a patient's pre-treatment and post-treatment photographs in a lecture and in a departmental teaching file. Which single course is most appropriate?

A Rely on the consent given for the clinical photographs at the start of treatment
B Obtain consent only if the patient can be recognised by name in the images
C Crop the eyes from the images, after which consent is no longer required

D Use the images without consent, since teaching is a benefit to the public
E Obtain separate recorded consent for use beyond direct care, naming the audiences

247 A patient alleges after debond that she was never told relapse could occur. The notes record the appliances fitted and the adjustments made but nothing of the discussions. Which single omission is most significant?

A The absence of a signed consent form anywhere in the clinical notes
B The absence of any record of the risks discussed and the alternatives offered
C The absence of study models taken at the start of the treatment
D The absence of a written treatment plan copied to the referring practitioner
E The absence of photographs taken at the debond appointment itself

248 A candidate is asked about the recognition and management of body dysmorphic disorder in orthodontic practice. Which of the following are true?

TRUE OR FALSE — judge each statement independently; any number may be true

A Prevalence among patients seeking orthodontic or cosmetic treatment is reported at about 5 to 13 per cent
B Patients characteristically focus on a defect that is slight or not observable to others
C Repeated consultations with different clinicians is a recognised feature
D Correcting the perceived defect usually resolves the preoccupation
E Referral for psychological or psychiatric assessment is the appropriate management

249 An adult patient six months into fixed appliance treatment develops joint clicking and intermittent masticatory muscle pain, and states that the appliances have caused it and that she intends to complain. Which single course is most appropriate?

A Debond immediately, since the appliances are the most likely cause
B Continue treatment unchanged and record that she has been told the symptoms are unrelated
C Refer her for temporomandibular joint arthroscopy before continuing treatment
D Examine and record the joint findings, explain the evidence and manage conservatively
E Advise her to seek an independent opinion before any further treatment is given

250 A patient shows a clinician a bonded retainer placed elsewhere which has an unpolished composite ledge and marked plaque retention. He asks whether the previous clinician was negligent. Which single response is most appropriate?

A Describe the clinical findings factually and explain what should now be done
B Tell him the work is negligent and that he should make a complaint
C Decline to discuss the retainer at all and refer him back to the original clinician
D Remove the retainer without comment and make no record of the finding
E Contact the previous clinician before saying anything to the patient

251 A team is revising its consent policy for child patients. Which of the following are true?

TRUE OR FALSE — judge each statement independently; any number may be true

A A child under 16 can never give valid consent to orthodontic treatment
B A person with parental responsibility must always countersign for a competent 15-year-old
C Proceeding on parental consent alone where a competent young person refuses is ethically problematic
D Assent given by a young child has the same legal standing as consent
E A step-parent acquires parental responsibility automatically on marrying the child's parent

252 A 19-year-old man listed for a mandibular advancement brings a photograph of an actor and says this is the profile he expects. His skeletal discrepancy would allow an 8 mm advancement. Which single course is most appropriate?

A Proceed, since the planned advancement is close to what the photograph shows
B Plan a larger advancement in order to come closer to the photograph
C Defer surgery, explore his expectations and arrange psychological assessment
D Proceed with orthodontic decompensation and revisit the discussion at the surgical planning stage
E Decline all further involvement and discharge him from the clinic

253 A clinician tells a patient that root resorption is rare, that decalcification sometimes happens and that relapse is possible. Which single improvement to this discussion is most important?

A Provide the same information in a printed leaflet as well as saying it verbally
B Record in the notes that the risks were discussed with her in full
C Ask the patient to repeat the information back before she signs
D Replace the descriptive words with numerical frequencies, which are read alike
E Restrict the discussion to the risks that the patient asks about

254 A 12-year-old girl attends with her mother, who speaks little English. The mother's 15-year-old nephew has come to interpret. Which single course is most appropriate?

A Proceed with the nephew interpreting, as he is a family member the mother trusts
B Arrange a professional interpreter and rearrange the consent discussion
C Give the mother written information in her own language and obtain the signature today
D Proceed using simplified English and short sentences
E Ask the 12-year-old patient to interpret for her mother

255 A practice is reviewing how it handles complaints. Which of the following are true?

TRUE OR FALSE — judge each statement independently; any number may be true

A An early, personal and unhurried response resolves most complaints without escalation
B An apology amounts to an admission of legal liability and should therefore be avoided
C A complaint should be answered only after a defence organisation has drafted the reply
D A patient who complains should be removed from the list to avoid further difficulty
E A complaint about the outcome is best answered by citing the index score recorded at assessment

256 A 16-year-old boy requires upper and lower fixed appliances for 20 months. His family will emigrate in four months and no arrangements have been made for care abroad. His mother asks that treatment be started now so that he is not further delayed. Which single course is most appropriate?

A Bond the appliances now and provide a letter describing the plan for a future clinician
B Refuse any further involvement and discharge him from the clinic list
C Bond the upper arch only, as this can be managed more easily by another clinician
D Begin treatment and remove the appliances immediately before their departure
E Do not begin, but arrange records and a written plan for whoever takes over

257 A 14-year-old girl attends alone and asks to start fixed appliance treatment for crowding. She can explain what bonding involves, the risk of decalcification, the likely duration and the retention that will follow, and why she wants treatment. Her mother has previously said she is against it. Which is the single most appropriate course?

A Proceed on the girl's consent alone, having judged and recorded that she is Gillick competent
B Decline to treat until the mother's written consent has been obtained, since the girl is under 16
C Treat only once a second clinician has independently confirmed that the girl is competent to decide
D Record the mother's objection in the notes and place the girl on review until she reaches 16
E Apply to the court for authority to treat, since parent and child are in open disagreement

Block I — Clinical examination and records (Q258–287)

Volume I: Chapter 7 (Examination and Diagnosis)
Volume II: §9 — read it again if you score under 70%
This block: 30 questions (7 of them true/false) · 58 marks · allow 38 minutes

258 A 10-year-old girl is seen for the first time. On casual closure she has a left buccal segment crossbite and her lower centreline is 2 mm to the left. Which single step should be taken before the occlusion is recorded?

A Take an alginate impression so that the occlusion can be studied on models
B Record the buccal segment relationship in maximum intercuspation and note the centreline shift
C Guide the mandible into retruded contact and observe the path of closure into intercuspation
D Measure the intercanine widths in both arches with a pair of dividers and record them
E Ask the patient to swallow so that the tongue position at rest can be assessed

259 A candidate is asked to describe extra-oral assessment. Which of the following are true?

TRUE OR FALSE — judge each statement independently; any number may be true

A Skeletal pattern should be assessed with the patient seated upright and the head in natural head position or with the Frankfort plane horizontal
B Lower anterior face height is normally about 55 per cent of total anterior face height
C Transverse asymmetry is best judged by sitting directly in front of the patient at eye level
D A mandibular displacement can make a face appear asymmetric when the skeletal bases are symmetric
E The antero-posterior, vertical and transverse planes should each be assessed in turn

260 On clinical assessment of a 13-year-old boy, lines drawn along the Frankfort and mandibular planes are estimated to intersect well behind the occiput. The overbite is 7 mm and complete. Which single interpretation is best supported?

A The maxillary and mandibular planes diverge markedly, giving an increased vertical proportion
B The planes are close to parallel, giving a reduced vertical proportion
C The assessment is unreliable and only a cephalogram can indicate the vertical proportion
D The finding indicates an anterior open bite tendency with a high angle face
E The finding indicates a Class III skeletal pattern with mandibular excess

261 A 12-year-old boy has a 10 mm overjet with the lips apart at rest and the lower lip resting behind the upper incisors. Which single soft tissue finding at the end of overjet reduction would most favour stability?

A The lips remaining apart at rest by about 3 mm after reduction
B The upper lip remaining short relative to the incisor crowns
C The mentalis remaining active on closure once the overjet is reduced
D The lower lip coming to cover the incisal third of the upper incisor crowns
E The tongue coming to rest between the incisors on swallowing

262 A 14-year-old girl has an instanding upper left lateral incisor in crossbite. The upper right central incisor is the most prominent upper tooth. Which single method of recording the overjet is correct?

A Measure to the most prominent upper incisor parallel to the occlusal plane
B Measure to the instanding lateral incisor, as this is the tooth in crossbite
C Measure the mean of the four upper incisors to the lower incisal edges
D Measure perpendicular to the long axis of the upper central incisor
E Record the overjet as negative because one incisor is in crossbite

263 A 15-year-old boy has lower incisors that are not visible when the teeth are in maximum intercuspation and whose incisal edges contact the palatal mucosa behind the upper incisors. Which single description of the overbite is correct?

A Increased and incomplete
B Increased and complete to tooth
C Increased and complete to soft tissue
D Reduced and complete to tooth
E Average and complete

264 A 19-year-old woman is dissatisfied with her smile although her occlusion is well aligned. Regarding smile analysis, which of the following are true?

TRUE OR FALSE — judge each statement independently; any number may be true

A A consonant smile arc is one in which the upper incisal edges are flatter than the curvature of the lower lip
B Gingival display of more than about 2 to 3 mm on smiling is generally regarded as excessive
C Buccal corridor width is influenced by the transverse dimension of the upper arch
D Upper incisor show at rest increases with advancing age
E Upper incisor show at rest of about 3 to 4 mm is typical in a young adult

265 A 14-year-old girl has a 7 mm complete overbite and shows 2 mm of upper incisor at rest. Which single approach to overbite reduction is most appropriate for her?

A Intrude the upper incisors, since this is the most direct means of reducing the overbite
B Use anterior bite planes to intrude both upper and lower incisors equally
C Extrude the upper incisors to increase the show and accept the deeper overbite
D Accept the overbite, as it cannot be reduced without compromising the smile
E Reduce the overbite by intruding the lower incisors and allowing posterior eruption

266 An 11-year-old boy has a right buccal segment crossbite. In maximum intercuspation the lower centreline is 3 mm to the right. Guided into retruded contact, the first contact is on the upper and lower right canines and the centrelines are coincident. Which single conclusion is best supported?

A He has a bilateral transverse discrepancy with a displacement into a unilateral crossbite
B He has a true unilateral transverse discrepancy requiring unilateral expansion
C He has a skeletal mandibular asymmetry requiring a posteroanterior cephalogram
D The centreline discrepancy is dental and should be corrected by asymmetric elastics
E The crossbite is of no significance because the retruded position is normal

267 A clinician wishes to establish whether a patient has a discrepancy between the retruded contact position and the intercuspal position. Which single technique is most appropriate?

A Ask the patient to bite together hard and observe the final position
B Take a wax registration in the intercuspal position and articulate the models
C Guide the mandible into retruded contact and watch the first contact and the slide

D Ask the patient to protrude and then retrude the mandible rapidly several times
E Palpate the condyles while the patient opens and closes the mouth widely

268 A 15-year-old girl has the upper centreline 2 mm to the right of the facial midline and the lower centreline coincident with the upper. There is no displacement and the buccal segments are Class I on both sides. Which single treatment aim is most appropriate?

A Move the lower centreline to the right so that it matches the face
B Correct the discrepancy surgically, as a 2 mm shift indicates skeletal asymmetry
C Accept both centrelines as they are, since they coincide with each other
D Correct the upper centreline using asymmetric extraction of an upper premolar on the left
E Move the upper centreline to the left to the facial midline, taking the lower with it

269 A trainee is recording the occlusion of a new patient. Which of the following are true?

TRUE OR FALSE — judge each statement independently; any number may be true

A Incisor classification depends on the relationship of the lower incisor edges to the cingulum plateau of the upper incisors
B Molar relationship is classified from the relationship of the upper and lower permanent canines
C A posterior crossbite is described by the position of the lower buccal cusps relative to the upper
D Overbite is recorded as the vertical overlap of the lower incisors by the uppers, in millimetres or as a proportion
E Angle's classification describes the antero-posterior relationship of the skeletal bases

270 A 16-year-old boy has a chin point 5 mm to the left of the facial midline. The centrelines are coincident in both the intercuspal and the retruded contact positions, there is no crossbite and his mother says his face has looked this way since early childhood without progression. Which single conclusion is best supported?

A The asymmetry is produced by a mandibular displacement and will resolve on correcting the occlusion
B The asymmetry is skeletal and non-progressive, and a posteroanterior film is indicated
C The asymmetry indicates active condylar hyperplasia requiring isotope bone scanning
D The finding is within normal variation and requires no record or action
E The asymmetry is dental and can be corrected with asymmetric elastics alone

271 A department is deciding whether to replace plaster study models with intra-oral scans for all new patients. Which single consideration is most relevant?

A Plaster models must be retained because digital records have no medicolegal standing
B Plaster models are more accurate for overjet and overbite measurement in every case
C Digital models cannot be used for PAR scoring and would end the audit programme
D Digital models are accurate enough for routine measurement and easier to store
E Scans require more clinical time than impressions in all circumstances

272 A 13-year-old girl declines to have clinical photographs taken, saying she dislikes being photographed. She is otherwise willing to proceed with fixed appliances. Which single course is most appropriate?

A Take the photographs anyway, since they form part of the standard clinical record
B Decline to treat her until she agrees to have photographs taken
C Record the refusal, explain their purpose, take the remaining records and proceed
D Take extra-oral photographs only, as these are the less intrusive
E Ask her mother to consent to the photographs on her behalf instead

273 A practice is reviewing its clinical photography protocol. Which of the following are true?

TRUE OR FALSE — judge each statement independently; any number may be true

A Photographs form part of the clinical record and are subject to the same duties of confidentiality
B A standardised head position and constant magnification allow before and after comparison
C Photographs taken at debond are valuable if relapse is later disputed
D Consent for clinical photography should be recorded
E Retracted buccal views should be taken with the teeth in the intercuspal position

274 A 25-year-old woman has bimaxillary proclination, a nasolabial angle of about 115 degrees and an upper lip that appears thin and unsupported. She asks for her teeth to be made straighter. Which single consideration should most influence the plan?

A Retraction of the upper incisors would open the angle further and flatten the lip
B Premolar extraction with retraction will improve the profile by reducing the incisor prominence
C The nasolabial angle is unaffected by the position of the upper incisors
D An obtuse nasolabial angle indicates that more retraction is required
E The soft tissue profile cannot be assessed until after alignment is complete

275 A 9-year-old girl has a symmetrical anterior open bite of 3 mm, proclined upper incisors and the tongue visibly placed between the incisors on swallowing. She stopped sucking her thumb six months ago. Which single interpretation is best supported?

A The tongue thrust is the primary cause and will maintain the open bite indefinitely
B A tongue crib should be fitted before any other treatment is considered
C Speech therapy directed at the swallowing pattern is the treatment of choice
D The open bite indicates a skeletal vertical excess requiring surgical planning
E The tongue placement is adaptive and is likely to resolve as the bite closes

276 An 8-year-old boy has an asymmetric anterior open bite, proclined upper incisors, retroclined lower incisors and a unilateral posterior crossbite. Which single conclusion is best supported by this combination?

A The findings indicate a skeletal Class II pattern with vertical excess
B The combination indicates early mandibular condylar pathology
C The findings indicate a primary tongue thrust of endogenous type
D The combination is characteristic of a persisting digit sucking habit
E The asymmetry indicates that the crossbite is a true unilateral discrepancy

277 A 15-year-old boy has 2 mm of gingival recession on the labial aspect of the lower left central incisor, with a thin gingival biotype. The plan under consideration involves 3 mm of lower incisor proclination to relieve crowding. Which single course is most appropriate?

A Proceed as planned, since proclination does not affect the periodontal tissues
B Discuss the increased risk of recession, obtain a periodontal opinion and reconsider
C Proclination should be increased to move the tooth away from the recession
D Defer all treatment until the recession has resolved spontaneously
E Photograph the recession and proceed without any further discussion

278 A candidate is asked about findings on intra-oral examination. Which of the following are true?

TRUE OR FALSE — judge each statement independently; any number may be true

A Mobility of a permanent central incisor in a 13-year-old is a normal finding
B Blanching of the tissue on traction of the labial frenum confirms that the frenum is the cause of a midline diastema
C Absence of a palpable canine bulge at nine years of age indicates an ectopic canine
D Oral hygiene need not be recorded before a removable appliance because plaque does not accumulate around them
E An ankylosed tooth gives a duller percussion note than its neighbours

279 An 11-year-old girl has no palpable canine bulge in either buccal sulcus, and both upper lateral incisors are tipped distally. The primary canines are firm. Which single course is most appropriate?

A Reassure and review in two years, as canines erupt late
B Palpate again in six months before taking any radiograph
C Refer for surgical exposure of both canines
D Extract both primary canines immediately without radiographic assessment
E Take radiographs now to establish the position of the permanent canines

280 A 9-year-old boy has a lower right second primary molar whose occlusal surface is 3 mm below the adjacent teeth. Which single combination of findings would most strongly support ankylosis?

A A duller percussion note with a widened periodontal ligament space radiographically
B A sharper percussion note with loss of the periodontal ligament space radiographically
C Increased mobility with a widened periodontal ligament space radiographically
D A normal percussion note with an intact periodontal ligament space radiographically
E Tenderness to percussion with a periapical radiolucency present on the film

281 An 18-year-old woman has a painless click from the right temporomandibular joint on opening, an interincisal opening of 45 mm, no deviation and no muscle tenderness. She is about to start fixed appliances. Which single course is most appropriate?

A Defer treatment until the click has been investigated by magnetic resonance imaging
B Abandon fixed appliance treatment because of the risk of worsening the joint
C Provide a stabilisation splint for three months before any bonding
D Record the findings, explain that the click is common and not a contraindication
E Refer for arthroscopy before any orthodontic treatment is started

282 A candidate is asked about mandibular displacements. Which of the following are true?

TRUE OR FALSE — judge each statement independently; any number may be true

A A displacement is detected by guiding the mandible into retruded contact and observing the path of closure into maximum intercuspation
B A displacement of 2 mm or more is recorded within the aesthetic component of IOTN
C An anterior displacement into an anterior crossbite is a normal finding requiring no record
D A lateral displacement may create an apparent skeletal asymmetry that disappears in the retruded position
E Displacements are confined to the mixed dentition and resolve when the permanent dentition is complete

283 A clinician is assessing a patient's profile for orthognathic planning and must choose a head position for the clinical assessment and the photographs. Which single choice is most appropriate?

A Natural head position, because it is the orientation in which the patient is seen
B Frankfort plane horizontal, because it is the anatomical reference used in cephalometrics
C Sella nasion plane horizontal, because it is reproducible between films
D Occlusal plane horizontal, because it relates the face to the dentition
E Any position at all, provided that it is the same on every occasion

284 A 14-year-old girl has a Class II division 2 incisor relationship with a deep and complete overbite. The lower lip line covers the incisal half of the upper central incisor crowns. Which single implication for treatment is most important?

A The lip position is irrelevant once the incisors have been proclined
B The lip position indicates that surgery will be needed to correct the overbite
C The high lower lip line is likely to retrocline the upper incisors again
D The high lower lip line will help to retain the corrected incisor position
E The finding indicates that the lower incisors should be proclined instead

285 A clinician is deciding which records to take for a 12-year-old with mild lower incisor crowding, all teeth present and erupted and no other clinical concern. Which single principle should govern the decision?

A A complete set of records including a lateral cephalogram should be taken for every new patient
B Radiographs are always required before any appliance is fitted at all
C Records should be taken only if the patient asks for them to be taken
D Records should be taken where they will alter the diagnosis or the plan
E The same record set should be used for every patient to allow audit comparisons

286 A unit is auditing its record keeping. Which of the following are true?

TRUE OR FALSE — judge each statement independently; any number may be true

A A full set of records including radiographs should be taken at the first visit for every new patient
B Records should be taken when they will change the diagnosis or the treatment plan
C Study models must be retained in plaster form for the statutory retention period
D A lateral cephalogram is required before every course of fixed appliance treatment
E Photographs carry no medicolegal value and are therefore optional in every case

287 Two years after debond a patient alleges that gingival recession on her lower incisors was caused by her treatment. Which single record would be most useful in resolving the allegation?

A The consent form signed by her at the start of the treatment
B Dated pre-treatment photographs and models of the lower labial segment
C The appointment history showing her attendance record throughout
D The panoramic radiograph taken immediately before bonding
E The written retainer instructions issued to her at debond

Block J — Imaging and cephalometrics (Q288–327)

> **Volume I:** Topic 2 (Cephalometry); Chapter 7, pp. 411–482
> **Volume II:** **§10** — read it again if you score under 70%
> **This block:** 40 questions (10 of them true/false) · 80 marks · allow 50 minutes

288 A 9-year-old boy attends for a routine orthodontic screening. All teeth expected at his age are present and erupting symmetrically, there is no caries, no displacement and no clinical concern. The practice protocol is to take a panoramic radiograph on every new orthodontic patient. Which single course is most appropriate?

A Take the panoramic radiograph in accordance with the practice protocol
B Defer the panoramic radiograph until the patient is 12 years old and then take it routinely
C Take a panoramic radiograph but with a reduced exposure setting
D Take two bitewings instead, as these deliver a lower dose
E Take no radiograph, as no clinical question has been raised that a film would answer

289 A 12-year-old girl has no palpable upper canines and distally tipped upper lateral incisors. The unerupted canines must be localised buccopalatally. Which single combination of films is most appropriate?

A A panoramic radiograph with an upper anterior occlusal view
B A single panoramic radiograph of the whole dentition
C A lateral cephalogram with a panoramic radiograph
D A single upper anterior occlusal view of the maxilla
E A posteroanterior cephalogram with a panoramic radiograph

290 A practice is reviewing its radiographic protocols against the regulations. Which of the following are true?

TRUE OR FALSE — judge each statement independently; any number may be true

A Every exposure must be justified by a net benefit to the individual patient
B The referrer must provide sufficient clinical information for the exposure to be justified
C Radiographs taken at fixed intervals during treatment are justified by the value of a complete record
D Optimisation requires the dose to be as low as reasonably practicable consistent with the diagnostic purpose
E The whole of every radiograph must be examined and reported, not only the region of interest

291 Two periapical radiographs are taken of an unerupted upper left canine. Between the two films the tube head is moved 20 mm mesially with the patient and film unchanged. On the second film the canine crown appears to have moved mesially relative to the root of the lateral incisor. Which single conclusion is correct?

A The canine lies buccal to the arch, since it has moved with the tube
B The canine lies buccal to the arch, since it has moved against the tube
C The canine lies palatal to the arch, since it has moved with the tube
D The canine lies in the line of the arch, as movement in the same direction indicates no displacement
E No conclusion is possible without a third film at a different vertical angle

292 A panoramic radiograph and an upper anterior occlusal view are compared to localise an unerupted upper right canine. The tube position is lower for the panoramic and higher for the occlusal view. Relative to the roots of the incisors, the canine crown appears more superior on the occlusal view than on the panoramic. Which single conclusion is correct?

A The canine lies buccally, since it moved in the same direction as the tube
B The canine lies palatally, since it moved in the same direction as the tube
C The canine lies buccally, since it moved in the opposite direction to the tube
D The films cannot be compared because their magnifications differ
E Vertical parallax is unreliable and a horizontal shift must be used instead

293 A 13-year-old has a palatally displaced upper left canine localised on two conventional films. The upper left lateral incisor is now mobile and tender, and resorption of its root is suspected. Which single investigation is most appropriate?

A Repeat the panoramic radiograph in six months to compare
B Take a further periapical film at a different horizontal angle
C Take a lateral cephalogram to assess the canine position in the sagittal plane
D Take a small field of view cone beam computed tomographic scan of the region
E Take a posteroanterior cephalogram to assess the transverse position

294 A trainee is asked which radiograph answers which clinical question. Which of the following are true?

TRUE OR FALSE — judge each statement independently; any number may be true

A A panoramic radiograph is the best film for assessing the periodontal bone level around the lower incisors
B An upper anterior occlusal view gives better detail of the upper incisor region than a panoramic radiograph
C A lateral cephalogram is used to establish the buccopalatal position of an unerupted maxillary canine
D A posteroanterior cephalogram is indicated where there is marked transverse facial asymmetry
E A panoramic radiograph reliably demonstrates early proximal caries

295 Eight months into fixed appliance treatment an upper central incisor becomes mobile and the clinician suspects root resorption. Which single radiographic approach is most appropriate?

A Take a periapical radiograph of the incisors using a paralleling technique
B Repeat the panoramic radiograph and compare with the pre-treatment film
C Take an upper anterior occlusal view of the incisor region
D Take a cone beam computed tomographic scan of the anterior maxilla
E Take a lateral cephalogram and measure the incisor root length

296 A clinician is comparing the dose implications of imaging options for localising an ectopic canine. Which single statement is most accurate?

A A cone beam scan delivers a dose comparable with that of a single periapical film
B Cone beam scans deliver no ionising radiation and need not be justified
C A large field cone beam scan may deliver many times the dose of a panoramic
D The dose from a panoramic radiograph exceeds that from a full mouth series of periapicals
E A lateral cephalogram delivers a greater dose than a panoramic radiograph

297 An 8-year-old boy has an erupted upper left permanent central incisor but the upper right central has not appeared 14 months later. The upper right primary central incisor was exfoliated a year ago. Which single radiographic approach is most appropriate?

A No radiograph, as eruption of the contralateral tooth may be delayed by up to two years
B A cone beam scan as the first line investigation
C A lateral cephalogram to assess the incisor angulation
D Bitewing radiographs to exclude a carious cause
E A panoramic radiograph together with an upper anterior occlusal view

298 A department is drafting guidance on the use of cone beam computed tomography. Which of the following are true?

TRUE OR FALSE — judge each statement independently; any number may be true

A A cone beam image is geometrically accurate and free of the magnification of a conventional radiograph
B A scan is justified where an impacted canine is suspected of resorbing the root of an adjacent incisor
C A small field of view examination delivers a higher dose than a large field of view examination
D The whole of the acquired volume must be examined and reported by a suitably trained person
E The resolution is insufficient for cone beam to be a first line investigation for proximal caries

299 A lateral cephalogram of a 14-year-old boy gives SNA 86 degrees, SNB 79 degrees and a sella nasion to maxillary plane angle of 8 degrees. What is the Eastman corrected ANB value?

A 4.5 degrees
B 2.0 degrees
C 7.0 degrees
D 9.5 degrees
E 11.5 degrees

300 A lateral cephalogram gives SNA 87 degrees, SNB 82 degrees, ANB 5 degrees and a sella nasion to maxillary plane angle of 15 degrees. Which single course is most appropriate for assessing the antero-posterior skeletal relationship?

A Apply the Eastman correction and report a corrected ANB of 2 degrees
B Report the raw ANB of 5 degrees and disregard the maxillary plane angle
C Use the Wits appraisal, since the Eastman correction assumes an angle of 8 degrees
D Repeat the cephalogram, as the maxillary plane angle indicates a positioning error
E Apply the Eastman correction in reverse because the maxillary plane angle is increased

301 A Wits appraisal is performed on a 17-year-old male patient. The perpendicular from point A meets the functional occlusal plane 4 mm ahead of the perpendicular from point B. Which single interpretation is best supported?

A The reading is within the normal range for a male patient
B The reading indicates a vertical rather than an antero-posterior discrepancy
C The reading indicates a skeletal Class III relationship
D The reading cannot be interpreted without the ANB value
E The reading indicates a skeletal Class II relationship

302 A candidate is asked to identify cephalometric landmarks on a tracing. Which of the following are true?

TRUE OR FALSE — judge each statement independently; any number may be true

A Sella is the most anterior point on the outline of the sella turcica
B Point A is the deepest point on the anterior outline of the mandibular symphysis
C Gonion is constructed by bisecting the angle formed by tangents to the posterior border of the ramus and the lower border of the mandible
D Nasion is the most inferior point of the frontonasal suture
E Porion is among the most reliably identified landmarks on a lateral cephalogram

303 A 13-year-old girl has a maxillary mandibular planes angle of 34 degrees, an overbite of 1 mm and an increased lower anterior face height. Which single implication for her treatment is most important?

A Overbite reduction will be simple and stable because the bite is already shallow
B Extrusive mechanics will rotate the mandible backwards and open the bite further
C The high angle indicates that the overbite will deepen with growth
D Anterior bite planes are indicated to reduce the overbite quickly
E The angle has no bearing on the choice of mechanics in this case

304 A cephalometric analysis of a 14-year-old boy gives an upper incisor to maxillary plane angle of 95 degrees, a lower incisor to mandibular plane angle of 92 degrees, an interincisal angle of 145 degrees and an overbite that is increased and complete. Which single conclusion is best supported?

A The upper incisors are proclined and should therefore be retracted
B The interincisal angle is within the normal range and requires no change
C The lower incisors are retroclined and are the cause of the deep overbite
D The upper incisors are retroclined, and proclining them will create space
E The values indicate a Class III incisor relationship overall

305 A 16-year-old girl has 5 mm of lower labial segment crowding and a lower incisor to mandibular plane angle of 103 degrees. The plan under consideration is non-extraction with further proclination. Which single consideration is most important?

A Proclination is safe because the lower incisors are already in a stable position
B The angle of 103 degrees is within the normal range for a girl of this age
C The lower incisors are already beyond the mean, so further proclination is unstable
D Proclination will reduce the risk of relapse by increasing arch length
E The lower incisor angle cannot be used to guide this decision at all

306 A candidate is asked about the values used in a standard cephalometric analysis. Which of the following are true?

TRUE OR FALSE — judge each statement independently; any number may be true

A ANB is influenced by the antero-posterior position of nasion as well as by the jaw relationship
B The Wits appraisal is independent of the occlusal plane
C The maxillary mandibular planes angle and the lower anterior face height proportion usually but not invariably agree
D An interincisal angle of 135 degrees lies within the normal range
E The Holdaway ratio compares the position of the lower incisor with the maxillary plane

307 On a lateral cephalogram the lower incisor tip lies 6 mm anterior to the nasion to point B line and pogonion lies 1 mm anterior to the same line. Which single interpretation is best supported?

A The lower incisors are excessively prominent relative to the chin
B The relationship is ideal, since both measurements are positive
C The chin is excessively prominent relative to the lower incisors
D The measurement indicates a Class III skeletal pattern
E The Holdaway ratio cannot be calculated without the ANB value

308 A patient has an unusually low nasion and a cranial base that on tracing gives an SNA of 74 degrees despite a clinically normal maxillary prominence. Which single approach to describing the antero-posterior skeletal relationship is most appropriate?

A Report SNA and SNB as measured and accept the Class III impression they give
B Report the raw ANB, which is unaffected by the position of nasion
C Use the Frankfort plane in place of sella nasion, since porion is easily identified
D Repeat the film, as the values indicate an error in the tracing
E Use a measurement independent of the cranial base, such as the Wits appraisal

309 A lateral cephalogram shows two clearly separated outlines of the lower border of the mandible and two outlines of the posterior ramus, with the orbital rims also duplicated. Which single explanation is most likely?

A A true facial asymmetry that the film has correctly demonstrated
B An error in the processing of the digital receptor
C Movement of the patient during the exposure producing a blurred image
D Rotation of the head within the cephalostat during the exposure
E Superimposition of the cervical spine on the mandible

310 Regarding the sources of error in cephalometric measurement described by Houston in 1989, which of the following are true?

TRUE OR FALSE — judge each statement independently; any number may be true

A Landmark identification error is purely random and therefore cancels out when a group mean is calculated
B Projection error arises because the beam diverges and the structures lie at differing distances from the receptor
C Landmark identification is less reproducible for porion and gonion than for sella
D Remeasuring a completed tracing detects errors of landmark identification
E Random error shifts the mean of a group in one direction

311 A unit wishes to quantify the reliability of the cephalometric measurements used in a research project. Which single approach is most appropriate?

A Have a second observer measure the same tracings and report the level of agreement
B Retrace and remeasure a sample of films after an interval to find the method error
C Repeat the radiographs on a sample of patients and compare the values
D Compare the measured values with published population mean values
E Measure each film three times in the same sitting and take the mean

312 Serial cephalograms of a 12-year-old taken 18 months apart are superimposed on the anterior cranial base. The upper incisor tip has moved 5 mm downwards and 3 mm forwards. The clinician wishes to know how much of this is tooth movement. Which single further step is required?

A Repeat the superimposition using the mandibular plane as the reference

B Conclude that all the change represents tooth movement, since the cranial base is stable
C Measure the change in SNA between the two films
D Superimpose on the occlusal plane and remeasure
E Superimpose regionally on stable maxillary structures such as the anterior palate

313 A 12-year-old boy with a Class III skeletal pattern is to have a computerised growth prediction performed so that a definitive surgical plan can be made now. Which single statement is best supported by the evidence?

A Prediction from a single cephalogram is sufficiently accurate to plan surgery in an individual
B Prediction performs acceptably for group means but unreliably for the individual
C Prediction is more reliable in Class III cases because the pattern is established early
D Prediction becomes reliable once two films a year apart are available
E Prediction is unnecessary because facial growth ceases at 14 years in boys

314 A researcher is planning to measure treatment change on serial cephalograms. Which of the following are true?

TRUE OR FALSE — judge each statement independently; any number may be true

A The mandibular plane is a stable structure on which to superimpose serial mandibular tracings
B Bjork's structural method uses the inner cortical outline of the symphysis and the outline of the mandibular canal
C Superimposition on the anterior cranial base demonstrates tooth movement relative to the maxilla
D Metallic implants are required before any clinically useful superimposition can be performed
E Growth change and treatment change cannot be separated without an untreated comparison group

315 A panoramic radiograph taken to assess the developing dentition of a 12-year-old shows a well defined 15 mm radiolucency at the left mandibular angle, remote from the teeth being assessed. Which single course is most appropriate?

A Confine the report to the dentition, as this was the reason the film was taken
B Disregard the finding, as lesions at the angle are almost always artefacts
C Repeat the film in two years to see whether the lesion has changed
D Record the finding, arrange appropriate further assessment and inform the patient
E Refer the patient back to the general dental practitioner without comment

316 A new patient brings a panoramic radiograph of diagnostic quality taken by another practice seven months ago. The clinical examination has raised no new question since. Which single course is most appropriate?

A Obtain and use the existing radiograph, copying it into the patient's record
B Take a new panoramic radiograph, since the previous film belongs to another practice
C Take a new panoramic radiograph because films older than six months are invalid
D Take periapical films of the whole dentition instead
E Take a cone beam scan to obtain more information than the existing film provides

317 A 29-year-old woman who is 10 weeks pregnant attends for an orthodontic assessment. A panoramic radiograph would be taken as part of the routine assessment but no urgent clinical question depends on it. Which single course is most appropriate?

A Take the film with a lead apron over the abdomen, which makes the exposure safe
B Take the film only after obtaining written consent from her obstetrician
C Defer the non-urgent radiograph and proceed with the clinical assessment
D Take a cone beam scan instead, as it involves no risk to the fetus
E Decline any orthodontic assessment until after she has delivered

318 A candidate is asked to place dental radiographic doses in context. Which of the following are true?

TRUE OR FALSE — judge each statement independently; any number may be true

A A panoramic radiograph delivers an effective dose of the order of 0.01 mSv
B A lateral cephalogram delivers a lower effective dose than a panoramic radiograph
C Rectangular collimation substantially reduces the dose from an intra-oral radiograph
D For the same exposure the lifetime risk is greater in a child than in an adult
E Digital receptors generally require less exposure than conventional film

319 A cephalometric analysis gives SNA 78 degrees, SNB 82 degrees, ANB minus 4 degrees, maxillary mandibular planes angle 30 degrees and a sella nasion to maxillary plane angle of 8 degrees. Which single description of the skeletal pattern is most accurate?

A A Class III pattern caused solely by mandibular prognathism, the maxilla being normal
B A Class I skeletal pattern with a vertical discrepancy alone
C A Class III pattern caused solely by retrusion of the maxilla
D A Class III pattern with both a retrusive maxilla and a prognathic mandible
E A Class II pattern misreported because of the position of nasion

320 A lateral cephalogram gives SNA 90 degrees, SNB 87 degrees, ANB 3 degrees and a sella nasion to maxillary plane angle of 8 degrees. The patient has a concave profile. Which single interpretation is best supported?

A The raw ANB of 3 degrees confirms a Class I skeletal pattern consistent with the profile
B The Eastman corrected ANB is minus 1.5 degrees, a Class III tendency
C The Eastman correction cannot be applied because both values are raised
D The profile must have been assessed in an incorrect head position
E The raw ANB should be doubled because both angles are raised

321 A lateral cephalogram taken for orthodontic planning shows a prominent soft tissue shadow in the nasopharynx. The parents ask whether this proves their son has an obstructed airway. Which single response is most appropriate?

A The film confirms obstruction and adenoidectomy should be arranged
B A cone beam scan should be taken to measure the airway volume and settle the question
C The film excludes obstruction because an airway column is visible
D The film should be repeated with the mouth closed to confirm the finding
E The film is a single static image and cannot establish airway function

322 A trainee is asked about the limitations of the panoramic radiograph. Which of the following are true?

TRUE OR FALSE — judge each statement independently; any number may be true

A Mesiodistal angulation of teeth in the anterior region is accurately represented
B Magnification is uniform in the horizontal and vertical directions

C A ghost shadow appears on the opposite side of the image and higher than the real structure
D Structures in the midline are unaffected by superimposition of the cervical spine
E The buccopalatal position of an unerupted canine can be determined from a single panoramic view

323 A 13-year-old girl requires upper and lower fixed appliances for crowding. All permanent teeth to the second molars are present and erupted in good position, the canines were palpable buccally, there is no caries and no clinical concern. Which single radiographic approach is most appropriate?

A No radiograph, unless a question arises that the examination has not answered
B A panoramic radiograph and a lateral cephalogram as a standard pre-treatment set
C A panoramic radiograph alone, to document the roots before bonding
D A lateral cephalogram alone, since the incisor position will change
E A cone beam scan in order to plan the positions of the brackets

324 A lateral cephalogram is taken at a magnification of about 8 per cent. A researcher wishes to compare linear and angular measurements from this film with those from a film taken on a different machine. Which single statement is most accurate?

A Both linear and angular measurements are unaffected by the difference in magnification
B Linear measurements are unaffected, whereas angular measurements must be corrected
C Angular measurements are unaffected, whereas linear measurements must be corrected
D Neither type of measurement can be compared between machines under any circumstances
E Magnification affects only measurements made in the vertical dimension

325 A 15-year-old Chinese girl has an upper incisor to maxillary plane angle of 118 degrees and a lower incisor to mandibular plane angle of 99 degrees, with a pleasing profile and competent lips. Which single interpretation is most appropriate?

A Both labial segments are proclined and should be retracted to the Eastman values
B Cephalometric norms are population specific and bimaxillary proclination may be normal
C The values indicate a Class II division 1 skeletal pattern requiring functional appliance therapy
D The tracing must contain an error, as both values cannot exceed the mean simultaneously
E The values are irrelevant here, since the profile is already acceptable

326 A candidate is asked how a cephalometric analysis should be used. Which of the following are true?

TRUE OR FALSE — judge each statement independently; any number may be true

A A cephalometric analysis establishes the diagnosis independently of the clinical examination
B An ANB of 5 degrees establishes that the mandible is retrognathic
C A film taken with the teeth apart alters the recorded mandibular plane angle
D Cephalometric norms derived from white European populations apply to all ethnic groups
E Vertical facial proportions can be assessed from the film as well as angular relationships

327 A clinician is deciding whether a lateral cephalogram is justified for a 14-year-old boy with an 8 mm overjet on a moderate Class II skeletal base who is to have a functional appliance. Which single justification is most appropriate?

A It provides a record against which any later complaint can be answered
B It will allow the growth pattern of the individual to be predicted accurately
C It is required by convention before any functional appliance is fitted

D It will quantify the skeletal and incisor relationships that govern the decision
E It will demonstrate the state of the airway before and after treatment

Block K — Study models, space analysis and the digital set-up (Q328–352)

Volume I: Chapter 7 (study models); Chapter 8, Box 8.6
Volume II: §11 — read it again if you score under 70%
This block: 25 questions (6 of them true/false) · 49 marks · allow 31 minutes

328 A 14-year-old girl is to have her lower arch aligned without any change in the position of her lower incisors. Measured on the study models there is 5 mm of crowding in the lower labial segment and the curve of Spee is 2 mm deep, and one millimetre of arch circumference is allowed for each millimetre of depth to be levelled. Which single figure best represents the total space requirement in the lower arch?

A 7 mm
B 5 mm
C 3 mm
D 9 mm
E 11 mm

329 A 15-year-old boy has completed alignment. The summed mesiodistal widths of his six upper anterior teeth measure 45.0 mm and of his six lower anterior teeth 37.5 mm, and the upper incisors are of normal width for the population. Which single conclusion is best supported?

A The anterior ratio is within normal limits and no tooth size discrepancy exists
B There is a maxillary anterior excess of about 2.8 mm, which will appear as upper spacing, since the ratio compares upper tooth mass against the lower arch
C There is a mandibular anterior excess of about 2.8 mm, which will appear as residual overjet or lower crowding
D The discrepancy cannot be identified until the upper arch has been fully retracted
E A ratio above 77.2 per cent indicates that upper incisor reduction is required to restore it

330 Regarding the Bolton analysis and its use in planning:

TRUE OR FALSE — judge each statement independently; any number may be true

A The overall ratio is calculated from the twelve teeth in each arch from first molar to first molar
B The mean anterior ratio is 77.2 per cent
C A discrepancy beyond two standard deviations from the mean is conventionally regarded as significant
D The analysis gives the size of the discrepancy but not which teeth are responsible for it
E A mandibular excess of tooth material appears as residual overjet, lower crowding or both once the arches are aligned

331 A 9-year-old boy has his four lower permanent incisors erupted with a summed mesiodistal width of 22.0 mm. Using the Tanaka and Johnston prediction, which single figure best represents the predicted combined width of the canine and two premolars in one lower quadrant?

A 19.5 mm
B 20.5 mm
C 24.0 mm

D 22.5 mm
E 21.5 mm

332 A clinician reads the Moyers probability tables at the 75th percentile rather than the 50th when predicting unerupted canine and premolar widths. Which single consequence is best supported?

A The prediction tends to overestimate tooth size, so that crowding is not understated
B The prediction becomes more accurate for the individual patient because the wider range is used
C The prediction tends to underestimate tooth size, so that fewer extractions are planned
D The percentile chosen has no systematic effect, since the tables are symmetrical about the mean
E The tables become transferable to populations other than the one from which they were derived

333 A 16-year-old boy has a lower curve of Spee 5 mm deep, lower incisors already proclined to 100 degrees to the mandibular plane and 1 mm of lower labial crowding. A colleague proposes to level the arch with a continuous archwire alone. Which single consequence should be anticipated?

A Levelling will occur by intrusion of the lower incisors, with no change in their inclination, which is the mechanism a continuous archwire is designed to produce
B The curve will level spontaneously once the lower second molars have erupted into occlusion
C Levelling with a continuous archwire will procline the lower incisors further and increase the overjet
D Levelling will release about 5 mm of space in the lower arch, relieving the crowding
E The depth of the curve has no bearing on the position of the lower labial segment

334 Regarding mixed dentition space analysis:

TRUE OR FALSE — judge each statement independently; any number may be true

A Prediction equations transfer between populations without systematic error
B The radiographic method of predicting unerupted tooth width needs no correction for magnification
C Space available is measured along the existing arch form whatever arch form the plan intends to produce
D Leeway space counts towards the space available whether or not the first permanent molars are prevented from migrating mesially
E The analysis is valid only once all four permanent canines have erupted

335 A 13-year-old girl has 4 mm of lower labial crowding and a colleague proposes to relieve all of it by increasing the lower intercanine width. Which single statement best guides the decision?

A Lower intercanine width can be increased by 4 mm with a fixed appliance and the result is stable
B Lower arch expansion is stable provided that the case is retained for a full 12 months
C Each millimetre of intercanine expansion gains about 2 mm of arch circumference, so 2 mm of expansion suffices
D Lower intercanine width increases naturally until 16 years, so the crowding will resolve without treatment
E Lower intercanine width is among the least stable dimensions to alter and tends to return towards its original value

336 A 15-year-old boy has upper central incisors retroclined to 95 degrees to the maxillary plane, and the plan is to torque their roots palatally so that the crowns come forwards to a normal inclination. Which single effect on the upper arch space analysis best follows?

A The manoeuvre is space neutral because no tooth is moved bodily along the arch
B Advancing the incisal edges increases arch circumference and so releases space
C Advancing the incisal edges reduces arch circumference and so consumes space
D Torque alters root position only and therefore cannot affect arch circumference
E The effect depends entirely on how much crowding is present in the labial segment

337 A digital set-up produced from an intra-oral scan shows a Class I finish with all spaces closed for a 14-year-old girl with 7 mm of lower crowding treated without extractions. Which single limitation most affects its value as a plan?

A The set-up cannot display the final occlusal contacts between the arches
B Digital set-ups systematically overestimate the amount of crowding that is present, because the scan software rounds contact points outward
C The set-up cannot be produced at all without a cone beam scan of both arches
D The set-up rearranges crowns without regard to root position, alveolar boundaries or the soft tissue limit
E A set-up is not valid unless the second permanent molars are included in it

338 Regarding intra-oral scanning compared with alginate impressions for study models:

TRUE OR FALSE — judge each statement independently; any number may be true

A A full-arch scan accumulates error along its length, so the largest deviation from a plaster model is between the most widely separated teeth
B Digital models cannot be used for a Bolton analysis because the interproximal contacts are inaccessible
C Scanning removes the airway and gagging risks associated with impression materials
D A single scan can be re-used for appliance manufacture without recalling the patient
E Because it records the soft tissue drape, a scan removes the need to assess lip competence clinically

339 A 12-year-old girl has a bilateral buccal crossbite. On the study models her upper intermolar width measured between the central fossae of the first permanent molars is 34 mm, and her lower intermolar width measured between the buccal cusp tips is 36 mm. Which single conclusion is best supported?

A The mandible is transversely excessive and surgical narrowing will be required
B The two figures are not comparable because different landmarks were used for each arch
C The maxilla is narrow relative to the mandible by about 2 mm at first molar level
D The crossbite must be dental in origin because a difference of only 2 mm is small
E No transverse conclusion can be drawn without a posteroanterior cephalogram

340 A 13-year-old girl has 6 mm of crowding in the lower labial segment. The plan is to procline the lower incisors by 2 mm, and each millimetre of labial movement is taken to yield 2 mm of arch circumference. Which single statement about the residual crowding is correct?

A About 2 mm of crowding will remain and must be found elsewhere
B No crowding will remain, because the proclination fully compensates for it
C About 4 mm of crowding will remain and must be found elsewhere
D The full 6 mm of crowding will remain, since proclination does not create space
E Crowding will increase by 2 mm as a result of the proclination

341 A finished case has a mandibular anterior tooth material excess of 3 mm on the Bolton analysis. The upper arch is well aligned, the upper incisors are of normal width for the population and their inclination is ideal. Which single approach is most appropriate?

A Procline the lower incisors to accommodate the excess and accept the resulting inclination
B Further retraction of the upper labial segment until the overjet closes onto the lower incisors
C Composite build-up of the upper lateral incisors until the tooth material ratio is restored
D Extraction of a lower incisor to remove the surplus tooth material in one step
E Interproximal reduction of the lower anterior teeth distributed across several contact points

342 A space analysis of the lower arch is being carried out under the conventional headings of crowding, curve of Spee, arch width, incisor anteroposterior position, tip and torque. Regarding the arithmetic:

TRUE OR FALSE — judge each statement independently; any number may be true

A Levelling a curve of Spee 3 mm deep adds about 3 mm to the space requirement
B Retracting the lower incisors by 1 mm releases about 2 mm of space
C Expanding the arch by 1 mm across the premolars gains less than 1 mm of arch circumference
D The requirements of the separate headings are summed to give a single figure for the arch
E The analysis is performed on the mandibular arch first because it is the template for the plan

343 Two clinicians measure the crowding in the same irregular lower arch. One lays a length of soft brass wire along the contact points and reads its length; the other measures each segment between contact points with dividers as a series of straight lines and sums them. Which single explanation best accounts for the difference between their figures?

A The two methods measure the same quantity, so any difference is observer error
B The segmental method cannot be applied where the arch is irregular in form
C The brass wire is distorted in handling and tends to over-read the perimeter by about 3 mm, which is why it consistently shows less crowding
D The brass wire follows the curve of the arch and records a longer perimeter, showing less crowding
E Neither method is valid unless the second permanent molars are included in the measurement

344 A finished case is scored with the American Board of Orthodontics objective grading system. The largest deductions are 6 points for marginal ridges and 5 points for buccolingual inclination, with 2 points for alignment and 1 point for overjet. Which single change in finishing would most reduce the score?

A Further detailing of the labial segments to refine the alignment of the incisors
B Correction of the vertical and transverse position of the posterior teeth
C Reduction of the overjet by a further millimetre with anterior torque
D Extension of retention to allow the anterior occlusal contacts to settle
E Retaking the records with the models trimmed to a common base

345 The displacements between the anatomical contact points of the five adjacent pairs of teeth in a lower labial segment measure 1.5, 2.0, 0.5, 2.5 and 1.5 mm. Which single figure is the Little irregularity index for this segment?

A 1.6 mm
B 2.5 mm
C 4.0 mm

D 16.0 mm
E 8.0 mm

346 Regarding the digital set-up as used in aligner and appliance planning:

TRUE OR FALSE — judge each statement independently; any number may be true

A The set-up shows the planned position of the crowns, and any roots displayed are extrapolated rather than measured
B Interproximal reduction planned in a set-up is routinely delivered clinically to within 0.1 mm
C Expansion shown in a set-up is achieved principally by buccal crown tipping when it is delivered with aligners
D The set-up can demonstrate whether the planned tooth positions lie within the alveolar housing
E About 6 per cent of aligner cases are completed without any refinement stage

347 A 9-year-old boy has 3 mm of crowding in the lower labial segment measured on the models. In each buccal segment the space from the distal of the lateral incisor to the mesial of the first permanent molar measures 22.0 mm, and the predicted width of the canine and two premolars is 21.5 mm. Which single conclusion about the lower arch is best supported?

A There is a net surplus of about 1 mm across the arch
B There is a net deficiency of about 2 mm across the arch
C There is a net deficiency of about 3 mm across the arch
D There is a net deficiency of about 4 mm across the arch
E No conclusion is possible without a lateral cephalogram

348 A study compares tooth width measurements taken from digital models and from plaster models in 60 patients and reports a mean difference of 0.08 mm with a 95 per cent confidence interval of 0.02 to 0.14 mm, which is statistically significant. Which single conclusion is best supported?

A The difference is statistically significant and is therefore clinically important
B No conclusion can be drawn because a sample of 60 patients is too small
C The study shows that digital models are unsuitable for space analysis
D The difference is real but lies far below the threshold at which a decision would change
E The difference would disappear if a larger sample were studied

349 A 13-year-old girl has an upper centre line 2 mm to the right of the facial midline. On the models, measured against a grid referenced to the midpalatal raphe, the upper right first permanent molar lies 3 mm mesial to the upper left. The lower arch is symmetrical and well aligned. Which single interpretation is best supported?

A The upper right segment has drifted mesially, so space is to be regained on the right alone
B The lower centre line is at fault and the lower arch should be treated to correct it
C The maxilla is rotated within the face and surgical correction will be required
D The centre line discrepancy is dental in the lower arch and will resolve with alignment
E No conclusion is possible without a posteroanterior cephalogram to assess symmetry

350 Regarding the American Board of Orthodontics objective grading system:

TRUE OR FALSE — judge each statement independently; any number may be true

A The system is applied to the pre-treatment records to establish the difficulty of the case
B Marginal ridge height and buccolingual inclination are both among the criteria scored
C The patient's own rating of the result contributes to the total score

D A case is scored more favourably when its models are trimmed to a common base
E Root angulation is scored from the lateral cephalogram

351 A 16-year-old girl has a peg-shaped upper right lateral incisor 4 mm wide, an upper left lateral incisor 7 mm wide and residual spacing around the peg tooth. The remaining teeth are well aligned and the lower arch is sound. Which single statement best guides the use of the Bolton analysis here?

A The analysis will show a mandibular excess, so lower interproximal reduction is indicated
B The analysis is invalid wherever any tooth is of abnormal form or size
C The analysis quantifies a maxillary deficiency, and space is opened to match the contralateral tooth before restoration
D The peg tooth should be extracted and the canine substituted so that the ratio is restored
E The residual space should simply be closed, as a discrepancy of this size is not clinically significant, since a few millimetres of tooth size mismatch rarely affects the final occlusion

352 A 15-year-old boy has 4 mm of lower labial crowding and a curve of Spee 3 mm deep, allowed at one millimetre of circumference per millimetre of depth. The plan is to retract the lower incisors by 1 mm, each millimetre of retraction costing 2 mm of arch circumference, with no change in arch width or torque. Which single figure best represents the total space requirement?

A 5 mm
B 7 mm
C 11 mm
D 9 mm
E 13 mm

Block L — Diagnosis, the problem list and treatment planning (Q353–392)

Volume I: Topic 3 (Formulating a treatment plan); Chapter 8
Volume II: §12 — read it again if you score under 70%
This block: 40 questions (10 of them true/false) · 80 marks · allow 50 minutes

353 A 14-year-old girl has a Class II division 1 incisor relationship, 4 mm of upper crowding, 5 mm of lower crowding and a reduced overbite on a mild Class II skeletal base with average vertical proportions. Her lower incisors are upright at 90 degrees to the mandibular plane. Which single step should come first in formulating her plan?

A Decide the extraction pattern in the upper arch, since the overjet is the presenting complaint
B Decide the final anteroposterior position of the lower labial segment
C Decide whether the upper molars will finish Class I or Class II
D Decide the appliance system, since it determines what movements are available
E Decide the retention regime, since stability governs everything that precedes it

354 A 13-year-old boy has a Class I skeletal base, 8 mm of lower crowding, lower incisors at 92 degrees to the mandibular plane, competent lips and a thin labial biotype with 1 mm of keratinised tissue labial to the lower central incisors. A colleague proposes to relieve the crowding by proclining the lower labial segment. Which single consideration most strongly argues against this?

A Proclination cannot relieve crowding, since arch circumference is fixed by the skeletal base
B Proclination is not technically achievable with a fixed appliance in a lower arch this crowded
C Proclination would shorten the total treatment time and so compromise the quality of the finish, because faster tooth movement is associated with poorer root paralleling
D Proclination always increases the overbite and would create a traumatic occlusion
E Proclination would leave the lower incisors outside the zone of soft tissue balance and over a thin labial plate

355 A problem list is being constructed for a 15-year-old girl before her plan is written. Regarding the construction and use of the problem list:

TRUE OR FALSE — judge each statement independently; any number may be true

A Problems are recorded in the anteroposterior, vertical and transverse planes and at both skeletal and dental levels
B A problem the patient has not complained of is still recorded if it is present
C The list is separated from the plan so that the problems are defined before any solution is proposed
D Problems are prioritised, because a plan that addresses every problem equally addresses none of them well
E The patient's own concern is recorded on the list even when it is not the largest measured discrepancy

356 A 16-year-old boy has a 10 mm overjet, a Class II skeletal base with ANB 8 degrees, lips that are incompetent at rest with a lip gap of 6 mm, and a lower lip that rests behind the upper incisors. He wants his teeth straightened but declines surgery. Which single statement best describes the limit on what orthodontic camouflage can achieve for him?

A Camouflage is limited by the retraction the alveolus and the soft tissues will tolerate before the profile flattens
B Camouflage is limited only by the anchorage available, which miniscrews now make effectively unlimited
C Camouflage is limited by the time the patient will accept in fixed appliances, which caps the movement achievable
D Camouflage is limited by the depth of the overbite, which determines the achievable overjet
E Camouflage is limited by the width of the upper arch, which determines incisor position

357 The envelope of discrepancy is used to decide between orthodontics alone, orthodontics with growth modification, and orthodontics with surgery. Which single statement about the envelope is best supported?

A It is drawn for the adult dentition only and has no application to a growing patient
B Its zones are symmetrical, so equal movement is obtainable in every direction
C Its three zones are concentric, and the range obtainable by orthodontics alone is the smallest of the three
D It defines limits of movement measured in degrees of tooth inclination rather than in millimetres, since the envelope was constructed from cephalometric angular data
E It applies to the maxillary arch only, the mandibular limits being set by the symphysis

358 Regarding the thresholds between growth modification, camouflage and orthognathic surgery:

TRUE OR FALSE — judge each statement independently; any number may be true

A Growth modification remains a realistic option for a patient at cervical stage CS4 or later
B A patient's refusal of surgery converts an otherwise surgical discrepancy into one that camouflage can correct
C The decision rests on the size of the overjet alone, the skeletal pattern being a secondary consideration
D Camouflage improves the skeletal relationship as well as the occlusion
E Where a discrepancy lies within the surgical zone of the envelope, attempting camouflage first carries no penalty

359 An 11-year-old girl has a 9 mm overjet, a Class II skeletal base, average vertical proportions, well-aligned arches and is at cervical stage CS2. She is teased at school and her mother asks for the quickest route to a normal overjet. Which single plan is most appropriate?

A Begin fixed appliances with Class II elastics now and accept the dental compensation
B Begin a functional appliance now, since the pubertal peak is expected within about a year
C Extract upper first premolars now and retract the upper labial segment on headgear anchorage
D Defer all treatment until the permanent dentition is complete and plan camouflage then
E Refer now for an orthognathic opinion, since a 9 mm overjet exceeds the camouflage range

360 A 34-year-old woman requests alignment of a crowded lower arch. She has generalised horizontal bone loss of about 40 per cent, probing depths of 4 to 5 mm with bleeding on probing at several sites, and plaque scores of 60 per cent. Which single course is most appropriate?

A Begin fixed appliances now and manage the periodontal condition alongside the tooth movement

B Proceed with aligners rather than fixed appliances, since they carry no periodontal risk
C Decline treatment, since orthodontics is contraindicated where bone loss exceeds a third of root length
D Achieve periodontal stability with no bleeding and low plaque scores, then reassess for treatment
E Extract the most mobile lower incisor and align the remainder without periodontal therapy

361 A 15-year-old boy with well-controlled epilepsy and a history of two generalised seizures in the past year attends for assessment of a Class II division 1 malocclusion. He takes sodium valproate. Which single modification to the plan is most appropriate?

A Prefer a fixed appliance to a removable one and ensure all components are securely bonded
B Avoid fixed appliances altogether because of the risk of injury during a seizure
C Defer all orthodontic treatment until he has been seizure-free for five years
D Treat with a removable appliance only, which can be removed at the onset of a seizure
E Avoid extractions because of the bleeding risk associated with his medication

362 A 12-year-old boy has a plaque score of 70 per cent, two untreated carious lesions in first permanent molars and an unrestored fracture of an upper central incisor. He is keen to start fixed appliances. Regarding the patient factors bearing on his plan:

TRUE OR FALSE — judge each statement independently; any number may be true

A Establishing oral hygiene and completing restorative care before bonding is a condition of treatment rather than a preference
B White spot lesions appear in up to half of fixed appliance patients and can be seen within six months of bonding
C His keenness to start is sufficient evidence of the compliance the treatment will require
D A traumatised incisor should be assessed for pulpal and root status before any force is applied to it
E Motivation in an adolescent is best judged by the parent's account of how much the child wants treatment

363 A 14-year-old girl has attended four of her last six appointments, has a plaque score of 55 per cent after two sessions of oral hygiene instruction, and her headgear wear diary is blank. Her malocclusion requires maximum anchorage in the upper arch. Which single modification to the plan is most appropriate?

A Continue as planned and reinforce the headgear instruction at each visit
B Extract upper second premolars instead of first premolars to reduce the anchorage demand
C Change to an anchorage strategy that does not depend on her cooperation
D Discontinue treatment and discharge her to her general dental practitioner
E Accept a Class II molar finish and close the residual space in the lower arch

364 A 27-year-old man asks for treatment to give him a straight profile. He has a Class I incisor relationship, mild lower crowding, a slightly retrusive chin and a nose that he describes as prominent. Which single response is most appropriate?

A Agree to the profile change, which alignment of the lower arch will deliver
B Decline to treat, since a patient who mentions his appearance has unrealistic expectations, and such concerns rarely respond to any orthodontic explanation
C Plan extraction of four premolars with maximum retraction to flatten the profile
D Refer him directly for a genioplasty, since the chin is the source of his concern
E Explain that his occlusal problem is minor and that his profile concern arises from the chin and the nose

365 A 15-year-old girl has a Class II division 2 incisor relationship, a deep and complete overbite, retroclined upper central incisors, and a high smile line showing 4 mm of gingiva. Her chief complaint is that her gums show when she smiles. Which single statement best reconciles the occlusal and facial aims?

A Levelling by incisor intrusion addresses both the overbite and the gingival display, and is therefore the aim to state first
B Levelling by molar extrusion addresses the overbite and is preferable because it is mechanically simpler, and any increase in facial height is negligible over a normal course of treatment
C The gingival display cannot be improved orthodontically and only the overbite should be addressed
D The overbite should be accepted and the gingival display treated by crown lengthening alone
E Proclining the upper incisors will reduce both the overbite and the gingival display

366 Treatment objectives are being written for a Class II division 1 patient. Regarding the statement of aims:

TRUE OR FALSE — judge each statement independently; any number may be true

A Aims are written as the intended end position of the teeth and jaws rather than as a list of appliances
B An aim to accept an existing Class II molar relationship is legitimate provided the interdigitation and the incisor relationship are correct
C Facial aims and occlusal aims may conflict, and the plan records which has been given priority
D Where growth is to be used, the aim states the growth assumed and what happens to the plan if it does not occur
E A well-written set of aims allows a different clinician to judge whether the finished case met them

367 A 9-year-old boy has an anterior crossbite of two upper incisors with a 3 mm mandibular displacement on closing, a mild Class III skeletal base, and 6 mm of crowding predicted in the lower arch. Which single approach to phasing is most appropriate?

A Defer all treatment to the permanent dentition and accept the displacement in the interim, since displacements resolve spontaneously with growth
B Treat everything in one phase of fixed appliances beginning now
C Treat the displacement now in a first phase and defer the crowding to a second phase in the permanent dentition
D Treat the crowding now by serial extraction and defer the crossbite
E Treat in one phase but begin only when the second permanent molars have erupted

368 A 16-year-old girl is being consented for extraction of four first premolars and fixed appliances for a 6 mm overjet. Which single element is most essential to the consent discussion for this specific plan?

A The general risk of decalcification, root resorption and relapse common to all fixed appliance treatment, which applies regardless of the extraction pattern chosen
B The type of bracket to be used and the sequence of archwires planned
C The number of appointments and the likely duration of treatment in months
D The cost of the treatment and the arrangements for payment
E The effect of the premolar extractions and retraction on her lip support and profile, and the alternatives including no treatment

369 Regarding consent as it bears on treatment planning:

TRUE OR FALSE — judge each statement independently; any number may be true

A The alternatives that must be discussed include accepting the malocclusion untreated
B A plan involving extractions requires the extraction pattern and its facial consequence to be discussed before the teeth are removed
C Consent obtained at the first consultation covers a change of plan made 18 months later
D Where two plans are reasonable, the clinician records which the patient chose and why
E Consent is a single event completed by a signature rather than a process continuing through treatment

370 A 13-year-old boy has a Class I incisor relationship, an anterior open bite of 4 mm, a lower facial height 4 mm greater than the upper, a steep mandibular plane and an anterior tongue position at rest. Which single feature of his problem list most constrains the plan?

A The tongue position, since myofunctional therapy will correct the open bite
B The vertical skeletal pattern, since it limits what tooth movement can achieve and threatens stability
C The Class I incisor relationship, since it removes the need for anteroposterior correction
D The anterior open bite of 4 mm, since that figure alone determines whether surgery is required, regardless of the skeletal pattern producing it
E The lower facial height, which can be reduced by extraction of four premolars

371 A 12-year-old girl has a unilateral buccal crossbite of the upper left first permanent molar and premolars, with a 3 mm shift of the mandible to the left on closing into occlusion. In the retruded contact position the arches are symmetrical and the buccal segments meet edge to edge on both sides. Which single conclusion is best supported?

A She has a true unilateral transverse discrepancy requiring asymmetric expansion of the left side alone
B No transverse problem exists, since the arches meet symmetrically in the retruded position
C The crossbite is dental and confined to the left buccal segment
D She has a bilateral maxillary narrowing presenting as a unilateral crossbite because of the displacement
E The mandible is asymmetric and surgical correction of the asymmetry is required

372 A plan lists nine problems for a single patient and proposes to correct all of them. Which single principle should guide the clinician?

A Every problem on the list must be corrected or the treatment has failed
B The list should be shortened by removing the problems that cannot be corrected
C The problems are ranked, and those that will not be treated are recorded as accepted compromises
D The patient's chief complaint is addressed last, once the measurable problems have been dealt with
E A list of nine problems indicates that the case is beyond orthodontic management

373 A 10-year-old girl has a Class III incisor relationship with a reverse overjet of 2 mm, a mild Class III skeletal base, a maxilla that is retrusive rather than a mandible that is prognathic, and she can achieve an edge-to-edge incisor position. Which single feature of her presentation most favours a good outcome from protraction headgear?

A Her age, since maxillary protraction is most effective before about ten years
B The reverse overjet of only 2 mm, which is within the range of dental camouflage
C Her ability to achieve an edge-to-edge position, which indicates a displacement
D The Class III skeletal base, which indicates that surgery will ultimately be needed

E Her sex, since girls respond better to protraction than boys

374 Regarding the principle that the mandibular arch is the template for the treatment plan:

TRUE OR FALSE — judge each statement independently; any number may be true

A The final position of the lower labial segment is fixed first and the upper arch is then built to match it
B The template principle means that the lower arch is always treated before the upper arch is bonded
C Where the lower incisors are to be left where they are found, the upper incisor position is thereby determined for a Class I result
D The principle applies to Class I cases but is abandoned in Class II division 1, where the overjet dictates the plan
E Lower arch form is preserved because the existing form represents the soft tissue balance the patient has achieved

375 A 17-year-old girl has a Class II division 1 malocclusion with a 7 mm overjet, ANB of 6 degrees, a maxillary-mandibular planes angle of 23 degrees, lower incisors at 96 degrees to the mandibular plane, and a well-supported upper lip. Growth is complete. Which single plan is most appropriate?

A Orthognathic mandibular advancement with presurgical decompensation
B Camouflage with lower first premolar extractions and lower incisor retraction, which closes the overjet from the lower arch instead
C A functional appliance to attempt residual mandibular growth
D Acceptance of the overjet with alignment only
E Camouflage with upper first premolar extractions and retraction on reinforced anchorage

376 A 19-year-old man has a Class III incisor relationship with a reverse overjet of 5 mm, ANB of minus 5 degrees, a maxillary-mandibular planes angle of 36 degrees, lower incisors already retroclined to 80 degrees to the mandibular plane, and an anterior open bite tendency. Which single plan is most appropriate?

A Camouflage by proclining the upper incisors and retroclining the lower incisors further
B Extraction of lower first premolars with retraction of the lower labial segment
C Protraction headgear to advance the maxilla
D Orthognathic surgery with presurgical decompensation of both arches
E Acceptance of the reverse overjet with alignment of both arches only

377 A retrospective study compares Class II camouflage with premolar extractions against orthognathic advancement in patients matched for ANB, and reports equivalent occlusal outcomes by PAR but significantly greater improvement in soft tissue profile in the surgical group. Which single conclusion is best supported?

A Camouflage and surgery are interchangeable, since the occlusal results are equivalent and profile is not a criterion for choosing between them
B The occlusal result does not discriminate between the two, and the choice turns on the facial change sought
C Surgery should be offered to all patients with a Class II skeletal discrepancy
D PAR is an invalid measure of outcome in skeletal Class II patients
E Camouflage is inferior and should be abandoned in favour of surgery

378 A trainee has been taught a fixed sequence for formulating a plan. Regarding that sequence:

TRUE OR FALSE — judge each statement independently; any number may be true

A The problem list is completed before any treatment option is considered
B The final lower incisor position is decided before the space analysis is performed
C The anchorage requirement is determined before the extraction pattern is chosen
D The retention plan is decided before treatment begins rather than at debond
E The appliance system is chosen after the required tooth movements are known

379 A 45-year-old woman with a heavily restored dentition, three crowned teeth, an implant replacing an upper first molar and generalised recession requests correction of a 5 mm overjet and upper labial segment spacing. Which single feature most alters the plan compared with the same malocclusion in an adolescent?

A The implant, which can be moved orthodontically but only slowly
B Her age, which makes tooth movement impossible without surgical assistance, since adult periodontal ligament no longer remodels
C The recession, which makes any orthodontic tooth movement contraindicated
D The crowned teeth, which cannot be bonded and so cannot be moved
E The absence of growth, which removes growth modification as an option and fixes the skeletal pattern

380 A 13-year-old boy has a Class I malocclusion with 7 mm of upper crowding, an upper right first permanent molar with a large occlusal restoration and a furcation-involved distal root, and sound premolars throughout. Which single extraction decision best reflects the planning principle of therapeutic extraction?

A Extract four first premolars, since this is the standard pattern for crowding of this severity
B Extract the compromised first permanent molar on the right and plan the mechanics around the resulting asymmetry
C Extract the compromised first permanent molar on the right and balance it with the sound first molar on the left
D Extract no teeth and relieve the crowding by expansion and proclination
E Extract the upper second molars to allow distal movement of the upper arch

381 Regarding the soft tissues as the limit of orthodontic treatment:

TRUE OR FALSE — judge each statement independently; any number may be true

A Lip competence achieved by muscular effort at rest is as sound a basis for planning as true lip competence
B The resting pressures of the lips and tongue determine tooth position more than the brief forces of swallowing
C Upper incisor retraction of 3 mm produces a predictable and proportionate 3 mm of upper lip retraction
D The thickness of the gingival biotype is irrelevant to how far a lower incisor may safely be proclined
E Teeth carried outside the zone of soft tissue balance can be held there indefinitely by a bonded retainer with no tendency to relapse

382 A 12-year-old boy has an 8 mm overjet, incompetent lips at rest, and the lower lip rests behind the upper central incisors. His mother asks whether the lip will correct itself once the overjet is reduced. Which single response is best supported?

A The lip trap is the cause of the overjet and will persist unchanged after treatment

B Lip competence can be predicted from the lip gap at rest alone, without reference to overjet, since the resting gap alone determines whether the lips will meet
C The lip trap has no bearing on the stability of the corrected overjet
D Reducing the overjet allows the lower lip to come to lie in front of the upper incisors, which is stabilising
E The lip will only become competent if lip exercises are performed during treatment

383 A 14-year-old girl has a Class II division 1 incisor relationship with a 6 mm overjet, bimaxillary proclination, full lips and a convex profile. Her chief complaint is the prominence of her upper front teeth. The plan under consideration is extraction of four first premolars with maximum anterior retraction. Which single consequence should most influence the discussion with her?

A Retraction of both labial segments will flatten the lip support and may not suit a face of this form
B Retraction of both labial segments will increase the overjet unless anchorage is reinforced, which is the main risk to weigh against her request
C Four premolar extractions will lengthen the treatment by about six months
D Bimaxillary proclination is a contraindication to any extraction
E The overjet of 6 mm cannot be reduced without a functional appliance

384 A treatment objective is written as follows: reduce the overjet to 3 mm by retraction of the upper labial segment, maintain the lower incisors in their present position, accept a full unit Class II molar relationship and maintain the existing lower facial height. Which single criticism of this objective is best founded?

A It fails to state the vertical objective, which any plan must include
B It is written in terms of appliances rather than tooth positions
C It states no transverse objective, so one plane of space is unaddressed
D It is unauditable because none of its elements can be measured at debond
E It is internally inconsistent, since a Class II molar relationship cannot coexist with a 3 mm overjet

385 An 11-year-old boy attends with his father, who wants treatment started immediately for a 5 mm overjet and mild crowding. The boy is at cervical stage CS1, has excellent oral hygiene and says he has not been troubled by his teeth. Which single course is most appropriate?

A Begin a functional appliance now to take advantage of the whole of the growth period
B Discharge him, since a boy who is untroubled by his teeth has no indication for treatment
C Begin fixed appliances now, since the oral hygiene is good
D Review in 12 months, by which time maturity and the boy's own wishes can be reassessed
E Begin serial extraction now to address the crowding before the canines erupt

386 Regarding the phasing of orthodontic treatment:

TRUE OR FALSE — judge each statement independently; any number may be true

A An interceptive first phase is justified where the problem will worsen or become harder to treat if left
B Two-phase treatment for Class II division 1 produces a better final occlusal result than a single phase started later
C Early treatment for a large overjet has been shown to make no difference to the incidence of incisor trauma
D Two-phase treatment lengthens rather than shortens the total time a child spends in appliances
E A first phase that achieves its stated aim spares most patients a second phase

387 A 15-year-old boy is six months into fixed appliance treatment. His plaque score has risen to 65 per cent, there are early white spot lesions on four upper labial surfaces, and he has missed two appointments. Which single course is most appropriate?

A Continue treatment and add a daily fluoride mouthrinse to the regime
B Suspend active treatment, remove or passivate the appliance and re-establish oral health before continuing
C Accelerate the mechanics so that the appliance can be removed sooner
D Accept the lesions as an inevitable consequence of fixed appliances and continue unchanged, since white spot lesions typically remineralise once the appliance is removed
E Debond permanently and discharge him without further review

388 A 13-year-old girl has the following problems recorded: a Class II skeletal base with ANB 5 degrees, a 7 mm overjet, an increased and complete overbite, 3 mm of upper crowding, 1 mm of lower crowding, a 2 mm centre line discrepancy in the upper arch, and a rotated upper right second premolar. She is at cervical stage CS3. Which single problem should be placed highest in the treatment plan?

A The rotated upper right second premolar, since rotations are the most relapse-prone correction
B The 2 mm upper centre line discrepancy, since centre lines are corrected before space closure begins
C The increased overbite, since it must be reduced before the overjet can be addressed
D The 3 mm of upper crowding, since space must be created before anything else can be moved
E The anteroposterior skeletal and incisor relationship, since growth is available now and will not be later

389 A clinician states that a 4 mm reverse overjet in a 9-year-old with maxillary retrusion should be planned definitively now as a surgical case, on the basis of a computerised growth prediction. Which single objection is best founded?

A Individual growth prediction is unreliable, and Class III mandibular growth is the least predictable of all
B Computerised prediction systematically underestimates maxillary growth and so overcalls surgery, making early surgical planning the safer default in Class III cases
C Surgery can never be planned before the permanent dentition is complete
D A 4 mm reverse overjet is always correctable by camouflage, so the prediction is irrelevant
E Growth prediction is valid only in Class II cases and cannot be applied to Class III

390 Regarding the assessment of the three planes of space in diagnosis:

TRUE OR FALSE — judge each statement independently; any number may be true

A A centre line discrepancy is recorded as a transverse problem and localised to the arch in which it arises
B The overbite is recorded as a vertical problem at the dental level and the facial proportions as a vertical problem at the skeletal level
C An increased overjet is by definition evidence of a Class II skeletal base
D A buccal crossbite is recorded as transverse and requires no distinction between a true and a displaced presentation
E The anteroposterior relationship is assessed at the skeletal level by the incisor relationship alone

391 A 12-year-old girl has a 10 mm overjet on a moderate Class II skeletal base, competent lips, average vertical proportions, and is at cervical stage CS3. She is enthusiastic and has excellent oral hygiene. Her mother asks what proportion of the correction will come from her jaws rather than her teeth. Which single response is best supported by the evidence?

A Almost all of the correction will be skeletal, since the appliance advances the mandible
B The correction is entirely dentoalveolar, since functional appliances produce no skeletal change
C The larger part of the correction will be dentoalveolar, with a modest skeletal contribution
D About three-quarters of the correction will be skeletal at this stage of maturity
E No estimate is possible, since the skeletal and dental components cannot be separated

392 A 16-year-old boy has 9 mm of upper crowding with both upper canines palatally displaced and unerupted, a Class I skeletal base and a well-aligned lower arch. The plan proposes extraction of four first premolars, exposure and alignment of both canines, and a 30-month treatment time. He is reluctant and says he does not mind how his teeth look. Which single consideration should most influence whether treatment proceeds?

A The severity of the crowding, which places him in the high need category of the dental health component
B The absence of any wish for treatment on his part, weighed against the risks of a long and demanding course
C The technical difficulty of aligning two palatally displaced canines
D The cost to the service of a 30-month course of treatment
E The risk that the canines will resorb the incisor roots if left unaligned

Block M — Extraction decisions and space management (Q393–432)

> **Volume I:** Topic 4 (Choice of extraction); Chapter 8, pp. 505–530
> **Volume II:** §13 — read it again if you score under 70%
> **This block:** 40 questions (10 of them true/false) · 80 marks · allow 50 minutes

393 A 14-year-old girl has 3 mm of crowding in each arch, a Class I incisor relationship, lower incisors at 94 degrees to the mandibular plane, competent lips and an average vertical pattern. The arches are of normal width and there is no crossbite. Which single method of making space is most appropriate?

A Extraction of four first premolars, which supplies space of the right order for crowding of this severity
B Extraction of four second premolars, which places the space further back
C Rapid maxillary expansion in the upper arch and slow expansion in the lower
D Interproximal reduction and a small amount of proclination, which supplies the space needed without extraction
E Distalisation of both upper buccal segments with headgear

394 A 13-year-old boy has 9 mm of crowding in the upper arch, a Class II division 1 incisor relationship with a 7 mm overjet and a moderate Class II skeletal base. The lower arch is well aligned and the lower incisors are correctly positioned. Which single extraction pattern is most appropriate?

A Upper first premolars only
B Upper and lower first premolars
C Upper second premolars only
D Upper first premolars and lower second premolars
E Upper and lower second premolars

395 Regarding the four ways in which space may be created in an arch:

TRUE OR FALSE — judge each statement independently; any number may be true

A Extraction, expansion, distalisation and a combination of interproximal reduction with proclination are the recognised means
B Expansion of the upper arch across the premolars gains less arch circumference than the transverse increase achieved
C Distalisation of an upper buccal segment converts an anchorage problem in the anterior segment into one in the posterior segment
D Proclination of the lower labial segment by 1 mm releases about 2 mm of arch circumference
E Interproximal reduction is irreversible and its extent is limited by the thickness of the interproximal enamel

396 A 12-year-old girl has 8 mm of crowding in the lower arch with both lower canines excluded buccally, and 6 mm of crowding in the upper arch. The incisor relationship is Class I on a Class I skeletal base with average vertical proportions. Which single extraction unit is most appropriate in each quadrant?

A The first permanent molars, because their long-term prognosis is the poorest
B The second premolars, because this preserves the contact between the first premolar and the canine

C The first premolars, because the space is delivered adjacent to the crowding the canines must enter
D The second permanent molars, because their extraction relieves crowding from behind
E The lower incisors, because the crowding is concentrated in the labial segment

397 A 15-year-old boy requires 4 mm of space in each upper quadrant for overjet reduction and has minimal crowding. Anchorage demand is maximum. A colleague proposes upper second premolar rather than first premolar extraction to make the case easier. Which single statement about this proposal is correct?

A Second premolar extraction reduces the anchorage demand because the space is closer to the molars
B Second premolar extraction is contraindicated wherever the overjet exceeds 4 mm
C Second premolar extraction has no effect on anchorage, since the same amount of space is created
D Second premolar extraction reduces the anchorage demand only when combined with headgear
E Second premolar extraction increases the anchorage demand, since more space must be closed from behind

398 A 9-year-old boy has an upper right primary canine extracted because the permanent lateral incisor is erupting into its place. Regarding balancing and compensating extractions:

TRUE OR FALSE — judge each statement independently; any number may be true

A A balancing extraction is the removal of the equivalent tooth from the opposing arch on the same side
B A compensating extraction is the removal of a tooth from the opposite side of the same arch
C A balancing extraction of a primary canine is undertaken to create space for the lateral incisor of the opposite side
D Loss of a primary incisor routinely requires a balancing extraction
E Balancing and compensating extractions are performed as a matter of routine whenever a primary tooth is lost early

399 An 8-year-old girl has four first permanent molars of poor prognosis with gross caries and pulpal involvement in two of them. A panoramic radiograph shows the crowns of all four second permanent molars formed and calcification just beginning at the bifurcation of their roots. There is mild crowding. Which single course is most appropriate?

A Extract all four first permanent molars now, since this is the optimum time for spontaneous space closure
B Restore all four and defer extraction until the second molars have erupted
C Extract all four first permanent molars at 12 years of age when the second molars erupt
D Extract only the two symptomatic molars and review the others
E Extract all four and plan fixed appliance space closure in the permanent dentition, since fixed appliances give more reliable control than spontaneous drift

400 A 22-year-old woman has a Class I incisor relationship, 5 mm of lower labial crowding, a heavily restored lower right first molar with a cracked cusp and a root-filled lower left first molar with periapical radiolucency. All premolars are sound. Which single approach best reflects the principle of therapeutic extraction?

A Extract four first premolars, the standard pattern for crowding of this degree
B Extract the two compromised first molars and balance each with the sound contralateral premolar, since balancing extractions keeps the arches symmetrical

C Extract the two compromised first molars and use their space, accepting a more demanding mechanical plan
D Restore both first molars and relieve the crowding by interproximal reduction alone
E Extract the two compromised first molars and leave the crowding untreated

401 A 16-year-old girl has bilateral absence of the upper lateral incisors, a Class I skeletal base, mild lower crowding, canines of light colour and a low smile line. There is 6 mm of space on each side. Which single consideration most favours space closure with canine substitution over space opening for implants?

A Her age, since implants cannot be placed until growth has ceased, which would require prolonged retention of the spaces
B The colour of her canines, which makes them easy to reshape to resemble lateral incisors
C The width of 6 mm on each side, which exactly matches a lateral incisor
D The mild lower crowding, which will resolve as a consequence of upper space closure
E The low smile line, which conceals the gingival margins of the substituted canines

402 A Class I case with 10 mm of crowding in each arch is planned with four first premolar extractions and no anchorage reinforcement. Regarding the space that results:

TRUE OR FALSE — judge each statement independently; any number may be true

A Roughly two-thirds of an extraction space is lost to mesial movement of the posterior segments if the anchorage is unsupported
B The figure of two-thirds comes from a systematic review of anchorage loss in fixed appliance treatment
C Anchorage loss arises only in the upper arch, the lower molars being effectively immovable under space closure forces
D Anchorage loss can be reduced by dividing the space closure into canine retraction followed by incisor retraction
E Where the crowding is 10 mm per arch, most of the extraction space is legitimately consumed by relief of crowding rather than lost to anchorage

403 An 8-year-old girl has a Class I skeletal base, well-aligned buccal segments and 9 mm of crowding in each arch with both upper lateral incisors palatally displaced. A radiograph confirms a full complement of permanent teeth in good position. Her parents ask about serial extraction. Which single sequence is correct?

A Extract the first premolars, then the primary canines, then the primary first molars
B Extract the primary canines, then the primary first molars, then the first premolars
C Extract the primary first molars, then the first premolars, then the primary canines
D Extract the first premolars and primary canines together, then the primary first molars
E Extract the primary canines and primary first molars together, then review without further extraction

404 A 25-year-old man has 2.5 mm of residual lower labial crowding after alignment, a thin labial biotype and a Bolton mandibular anterior excess of 2 mm. The plan is interproximal reduction of the lower labial segment. Which single statement best guides how much enamel may be removed?

A Up to half the thickness of the enamel at each contact, irrespective of how many contacts are used
B About 1.5 mm at each of the five anterior contacts, giving ample space in a single session
C Reduction of the lower central incisors only, removing 1.25 mm from each of the two mesial surfaces
D About 0.5 mm at each of the five anterior contacts, which is within the interproximal

enamel available

E Reduction is contraindicated in the lower labial segment because the enamel there is too thin

405 A Class I case with 4 mm of crowding per arch is to be treated with four first premolar extractions. Regarding the choice between sliding mechanics and loop mechanics for closing the residual spaces:

TRUE OR FALSE — judge each statement independently; any number may be true

A Sliding mechanics moves teeth along a continuous archwire, and the resistance to sliding is negligible in a well-finished stainless steel slot

B Loop mechanics delivers a defined force system at the loop and is independent of friction at the bracket slot

C A closing loop can be designed to deliver a moment-to-force ratio high enough for bodily movement

D Sliding mechanics is contraindicated wherever anchorage is critical, since friction always favours the anterior segment

E Loop mechanics is quicker chairside than sliding mechanics because no archwire adjustment is required

406 A 14-year-old boy has absent upper second premolars bilaterally with retained primary second molars showing no infraocclusion, mild ankylosis on neither side, and no crowding in either arch. The incisor relationship is Class I and the buccal segments interdigitate well. Which single approach is most appropriate?

A Extract both retained primary molars and close the spaces orthodontically

B Extract both retained primary molars and open the spaces for implants at maturity

C Retain and monitor the primary molars, which may survive for many years

D Extract both retained primary molars and allow the first permanent molars to drift forward unaided

E Extract both retained primary molars and plan autotransplantation of the lower second premolars

407 A 15-year-old girl has a Class III incisor relationship with a reverse overjet of 1 mm, a mild Class III skeletal base, 5 mm of crowding in the upper arch and 5 mm of crowding in the lower arch. The plan is dental camouflage with fixed appliances. Which single extraction pattern is most appropriate?

A Upper second premolars and lower first premolars

B Upper first premolars and lower first premolars

C Upper first premolars and lower second premolars

D Upper second premolars and lower second premolars

E Lower first premolars only

408 A 16-year-old boy has a Class II division 2 incisor relationship, a deep overbite, retroclined upper central incisors, 2 mm of upper crowding and 4 mm of lower crowding on a mild Class II skeletal base with reduced vertical proportions. Which single statement about extractions in this case is best founded?

A Extractions are mandatory because the overbite cannot be reduced in a non-extraction case

B No extractions are needed, since a deep overbite of itself indicates a non-extraction plan in a short face

C Four first premolars should be extracted to allow retraction of both labial segments

D Upper second molars should be extracted to allow the upper arch to move distally

E Extraction should be approached cautiously, since proclining the upper central incisors

itself consumes space

409 A 13-year-old girl requires 4 mm of space in each upper quadrant, and distalisation of the upper buccal segments is being considered as an alternative to extraction. Regarding this choice:

TRUE OR FALSE — judge each statement independently; any number may be true

A Headgear used for distalisation is worn for about 14 hours a day at 450 to 500 g per side
B Distalisation is equally effective once the upper second permanent molars are fully erupted
C Intra-oral distalising appliances tend to procline the upper incisors unless the reactive unit is supported
D Distalisation makes greater demands on cooperation than an extraction plan does
E Miniscrew-supported distalisation removes the reciprocal forward force on the anterior teeth

410 A 12-year-old boy has a bilateral buccal crossbite with a narrow maxilla, 6 mm of upper crowding and 2 mm of lower crowding. The midpalatal suture is patent and the lower arch is of normal width. Which single plan best uses expansion as a source of space?

A Slow expansion at 0.25 mm twice weekly, which is the only method that produces skeletal change
B Extraction of four first premolars first, followed by expansion to correct the crossbite
C Rapid maxillary expansion to correct the crossbite, with the understanding that it contributes no arch space, since only transverse tooth movement occurs
D Rapid maxillary expansion first, which will correct the crossbite and release upper arch space that reduces the extraction requirement
E Expansion of the lower arch to match the upper, followed by reassessment of the crowding

411 A 34-year-old woman has 5 mm of lower labial crowding, a Class I incisor relationship, a normal overbite of 3 mm and sound well-aligned buccal segments with good interdigitation. Extraction of a single lower incisor is proposed. Which single consequence should be anticipated and discussed?

A A centre line shift of 5 mm towards the extraction site
B A residual overjet or a deepening of the overbite, because the upper and lower tooth material will no longer match
C Loss of the buccal interdigitation, which will require realignment of both buccal segments even though the extraction was confined to the anterior segment
D An increase in the lower intercanine width as the remaining incisors realign
E Spontaneous space closure without appliances in an adult of this age

412 A 14-year-old girl has a Class I incisor relationship, well-aligned labial segments and bilaterally impacted lower second permanent molars lying at 40 degrees to the occlusal plane against the distal of the first molars. The lower third molars are present and developing in good position. Which single approach is most appropriate?

A Extraction of the lower third molars and surgical uprighting of the second molars
B No treatment, since impacted second molars invariably upright spontaneously
C Extraction of the lower first molars to relieve the impaction from in front
D Extraction of four first premolars to provide space for the second molars to upright
E Extraction of the lower second molars, allowing the third molars to erupt into their place

413 A 9-year-old boy is being assessed for extraction of first permanent molars of poor prognosis. Regarding the decision:

TRUE OR FALSE — judge each statement independently; any number may be true

A Spontaneous space closure after lower first molar extraction is as reliable as after upper first molar extraction
B The presence or absence of the third permanent molars is irrelevant to the decision
C A lower first molar of poor prognosis is routinely balanced by extraction of the sound contralateral first molar
D Extraction timed to the beginning of calcification at the bifurcation of the second molar roots gives the best spontaneous result
E A first permanent molar of poor prognosis should be retained until the permanent dentition is complete so that the space can be closed with appliances

414 A 15-year-old boy has a single absent upper right lateral incisor, a retained primary lateral incisor in good condition, a Class I skeletal base, well-aligned arches with no crowding and a high smile line showing the full gingival margins. Which single approach is most appropriate?

A Close the space by mesial movement of the upper right canine and reshape it as a lateral incisor, which is a well established option regardless of the smile line
B Open the space to the width of the contralateral lateral incisor and plan a resin-bonded bridge or implant
C Extract the contralateral lateral incisor to restore symmetry and close both spaces
D Leave the primary lateral incisor in place indefinitely without further planning
E Extract the upper right canine and bring the first premolar forward

415 A 14-year-old girl requires 7 mm of space in each upper quadrant, of which 6 mm is needed to reduce a 6 mm overjet and 1 mm to relieve crowding. Upper first premolars are to be extracted. Which single statement about the anchorage requirement is correct?

A The anchorage demand is minimum, since the extraction space is available at the front of the arch
B The anchorage demand cannot be classified until the appliance system has been chosen
C The anchorage demand is moderate, since the space is divided between crowding and overjet
D The anchorage demand is maximum, since almost the whole space must be used for anterior retraction
E The anchorage demand is minimum, since first premolar extraction always reduces it

416 A clinician plans first premolar extractions in a case requiring 5 mm of anterior retraction per quadrant and intends to use no anchorage reinforcement, on the grounds that the extraction space is 7 mm per quadrant. Which single criticism is best founded?

A Unsupported anchorage typically loses about two-thirds of the space, leaving far less than the 5 mm required
B Two millimetres of surplus is ample, since anchorage loss rarely exceeds 1 mm per quadrant
C Anchorage loss can be corrected at the finishing stage by Class II elastics without further cost, since elastics reliably regain space lost earlier in treatment
D The calculation is sound, since the molars will not move if a rigid rectangular archwire is used
E Anchorage loss is unpredictable, so no figure can be applied at the planning stage

417 A 13-year-old girl has 6 mm of crowding in the upper left quadrant with the canine excluded buccally, 1 mm of crowding in the upper right quadrant, a Class I incisor relationship and a coincident upper centre line. Which single extraction plan is most appropriate?

A Extract the upper left first premolar and the upper right second premolar
B Extract both upper first premolars, since symmetrical extraction protects the centre line, which is the safer default whenever a centre line must be maintained
C Extract the upper left first premolar alone and control the centre line with asymmetric mechanics
D Extract the upper left and right second premolars
E Extract no teeth and relieve the left-sided crowding by expansion

418 Interproximal reduction is being planned in the lower labial segment of an adult patient. Regarding the technique and its limits:

TRUE OR FALSE — judge each statement independently; any number may be true

A The enamel available for removal in the lower labial segment is of the order of 0.25 mm per surface
B Reduction performed without restoring a smooth contour leaves a surface more retentive of plaque than the original
C Reduction changes the shape of the contact from a point to a broader area, which is often cited as a stability advantage
D The reduction delivered in aligner treatment commonly differs from that planned in the digital set-up
E Enamel removed by interproximal reduction regenerates over about 12 months provided fluoride is used

419 A 14-year-old boy has a Class II division 1 incisor relationship with a 5 mm overjet, 2 mm of upper crowding and 7 mm of lower crowding on a mild Class II skeletal base. The lower incisors are at 90 degrees to the mandibular plane. Which single extraction pattern is most appropriate?

A Lower first premolars only
B Upper second premolars and lower first premolars
C Upper first premolars and lower second premolars
D Upper first premolars and lower first premolars
E Upper first premolars only

420 A 13-year-old girl needs 3 mm of space in each upper quadrant to relieve crowding, has a Class I molar relationship that the plan intends to preserve, and has upper second molars that are erupted and upper third molars developing. Extraction of upper second molars is proposed to allow distal movement of the upper arch. Which single objection is best founded?

A Extraction of second molars is contraindicated wherever the third molars are already present and developing
B Distal movement behind erupted second molars is slow and depends on third molars that may not erupt well
C Second molar extraction will inevitably open the bite and lengthen the lower face
D Second molar extraction provides no space at all in the upper arch
E Second molar extraction will cause the first molars to tip distally beyond correction

421 A systematic review comparing extraction with non-extraction treatment in comparable Class I crowded cases reports a mean difference in upper lip position of 0.9 mm, favouring less prominence in the extraction group, with wide confidence intervals and high heterogeneity. Which single conclusion is best supported?

A Extraction reliably flattens the profile and should be avoided where lip support matters, since the pooled mean shows a consistent effect
B Non-extraction treatment should be preferred in all patients concerned about their profile
C Extraction has no effect on the soft tissue profile in any patient
D The review is invalid because profile is a subjective outcome
E The mean effect on the lip is small and variable, so the profile cannot be predicted from the extraction decision alone

422 A 16-year-old girl has four absent second premolars, retained primary second molars with 3 mm of infraocclusion, and 4 mm of crowding in each arch. Regarding the space decisions:

TRUE OR FALSE — judge each statement independently; any number may be true

A Infraocclusion of this degree indicates that the primary molars are sound and can be retained indefinitely
B The presence of crowding in every quadrant favours closing the spaces rather than opening them
C Closing a second premolar space requires the first permanent molar to move forward, so anchorage is planned to permit rather than to resist that movement
D An infraoccluded primary molar will re-erupt into occlusion once the adjacent teeth are aligned
E Autotransplantation is an alternative where a suitable donor tooth of appropriate root development is available

423 A 15-year-old boy has 8 mm of crowding in the lower arch concentrated around both canines, a Class I incisor relationship and a Class I molar relationship that the plan intends to preserve. Lower first premolars are to be extracted. Which single anchorage strategy is most appropriate?

A Moderate anchorage, since the crowding will consume most of the space and only a little anterior retraction is planned
B Maximum anchorage with a lingual arch and miniscrews, since 8 mm is a large space to control and any anchorage loss would be unacceptable
C Minimum anchorage, allowing the molars to come forward freely into the extraction sites
D No anchorage planning is required, since relief of crowding does not involve anchorage
E Maximum anchorage with headgear worn 14 hours a day at 450 g per side

424 A 13-year-old girl has 4 mm of lower labial crowding, lower incisors at 88 degrees to the mandibular plane, a Class I skeletal base, competent lips and a broad band of attached gingiva labial to the lower incisors. A colleague proposes lower first premolar extraction. Which single alternative is best supported?

A Extraction of a lower incisor, which removes the crowded tooth directly
B Rapid maxillary expansion followed by reassessment of the lower arch
C Proclination of the lower labial segment by about 2 mm, which is well within the soft tissue limit here
D Interproximal reduction of the lower buccal segments to bring space forward, since this avoids any change in incisor inclination
E Distalisation of the lower buccal segments with a lip bumper worn full time

425 A 16-year-old boy has an upper left central incisor with a horizontal root fracture in the middle third sustained four years ago, now with a negative response to pulp testing, apical radiolucency and grade two mobility. He also has 6 mm of upper crowding requiring extraction. Which single plan best reflects the principle of extracting the tooth of poorest prognosis?

A Extract four first premolars and retain the central incisor, moving it with light forces
B Extract the upper left central incisor and manage the space and symmetry within the orthodontic plan
C Extract the upper left central incisor and its contralateral to preserve symmetry
D Root treat the central incisor and proceed with premolar extractions as planned
E Defer all orthodontic treatment until the central incisor has been replaced by an implant, since implant placement should precede any orthodontic space management

426 Regarding the choice of which tooth to extract when space is required:

TRUE OR FALSE — judge each statement independently; any number may be true

A The tooth of poorest long-term prognosis is preferred, even where its position makes the mechanics more demanding
B The extraction site should always be as close as possible to the site of the crowding, irrespective of the anchorage plan
C Second premolar extraction is preferred where the plan requires maximum retraction of the anterior segment
D Extraction of a first permanent molar in a 15-year-old is followed by reliable spontaneous space closure
E A tooth that is severely displaced out of the arch may be the unit of choice even where it is otherwise sound

427 A 15-year-old girl has bilateral absence of the upper lateral incisors, a half unit Class II buccal segment relationship, no lower crowding and a lower arch that the plan intends to leave unchanged. Which single argument most favours closing the upper spaces with canine substitution?

A The bilateral nature of the absence, which makes either approach symmetrical in its result
B The absence of lower crowding, which means no lower extractions are needed
C Her age, since substituted canines are best brought forward before facial growth has ceased, making early treatment the deciding factor here
D The Class II buccal relationship, since closing the spaces brings the buccal segments towards Class I
E The half unit discrepancy, which is too small to require any correction

428 A 14-year-old boy is to have four first premolars extracted for a maximum anchorage Class II case with a transpalatal arch and palatal miniscrews planned in the upper arch. Which single sequence is most appropriate?

A Place the anchorage units, bond the appliance and align before the premolars are extracted
B Extract all four premolars at once and bond the appliance six months later
C Extract the premolars first and place the anchorage units only when anchorage loss becomes apparent
D Extract the upper premolars first and the lower premolars at the end of alignment
E Extract the premolars and place the anchorage units at the same visit, bonding afterwards

429 Anchorage demand in an extraction case is conventionally classified as minimum, moderate or maximum. Which single criterion best defines the classification?

A The number of teeth extracted in the arch
B The absolute amount of extraction space created, measured in millimetres per quadrant, since a larger space in millimetres directly sets the anchorage category
C The proportion of the extraction space that is to be closed by movement of the anterior segment
D The type of appliance selected to close the space
E The combined root surface area of the teeth in the posterior anchorage unit

430 An 8-year-old girl with severe crowding in both arches is being considered for serial extraction. Regarding the technique:

TRUE OR FALSE — judge each statement independently; any number may be true

A Serial extraction is appropriate where the crowding is mild and a borderline extraction decision is anticipated
B A Class I skeletal base with an average overbite and no anteroposterior discrepancy is the classical indication
C A full complement of permanent teeth in favourable position must be confirmed radiographically before starting
D Fixed appliance treatment is usually still required at the end, to detail the result and close residual space
E Serial extraction reduces the total period during which the child is under orthodontic supervision

431 A 15-year-old girl has an upper right canine lying horizontally across the roots of the upper right central and lateral incisors, with resorption of the lateral incisor root to the middle third, and a well-aligned arch with the primary canine in place and of good form. Which single approach is most appropriate?

A Surgical exposure and orthodontic alignment of the canine over a period of about two years, which preserves the tooth and is the standard first choice
B Extraction of the upper right first premolar to create space for the canine to align
C Extraction of the resorbing lateral incisor and alignment of the canine into its place
D Observation with annual radiographs, since resorption of this degree is self-limiting
E Extraction of the ectopic canine to arrest the resorption, with the space then managed on its merits

432 A 16-year-old boy has an anterior open bite of 3 mm, increased vertical proportions with a steep mandibular plane, a Class I incisor relationship and 7 mm of crowding in each arch. Which single extraction pattern is most appropriate?

A Upper and lower second premolars, whose more posterior spaces favour a degree of molar mesial movement and bite closure
B Upper and lower first premolars, accepting that space closure will tend to extrude the posterior teeth, which is an acceptable trade-off for the extra space it provides
C Upper and lower second molars, to reduce the vertical dimension from the back of the arch
D No extractions, relieving the crowding by expansion in both arches
E Upper first premolars only, with the lower crowding accepted

Block N — Biomechanics, force control and anchorage (Q433–485)

Volume I: Topics 5 and 6 (Anchorage; the three orders); Chapters 6 and 11
Volume II: §14 — read it again if you score under 70%
This block: 53 questions (14 of them true/false) · 109 marks · allow 66 minutes

433 A 14-year-old girl is having her upper canines retracted into first premolar extraction spaces with nickel-titanium coil springs delivering 150 cN on each side, tied back to the first molars. After four months the canines have moved 3 mm distally and the molars 2 mm mesially. Which single principle best explains the molar movement?

A The coil spring has exceeded the optimum force level and produced hyalinisation in the molar periodontal ligament, which then moved under the sustained load
B The molars have drifted mesially because of the natural tendency to mesial migration, independently of the spring
C Every force applied to the canine is matched by an equal and opposite force on the molar, and the molar moves because nothing prevents it
D The molar roots are shorter than the canine roots, which makes them move faster under the same force
E The spring has delivered a couple to the molars in addition to a force

434 A 15-year-old boy requires 6 mm of upper incisor retraction out of a 7 mm first premolar extraction space on each side. The upper molars must not move forward. Which single description of his anchorage requirement is correct?

A Minimum anchorage, since the extraction space exceeds the retraction required
B Stationary anchorage, since the molars are to be moved bodily rather than tipped
C Moderate anchorage, since 1 mm of the space on each side is available for molar movement, which is enough margin to call the anchorage requirement only moderate
D Reciprocal anchorage, since the anterior and posterior units are of similar size
E Maximum anchorage, since more than three-quarters of the space is to be used by the anterior segment

435 A single 100 cN force is applied at the bracket of an upper central incisor in a distal direction, with no couple. Regarding the resulting force system:

TRUE OR FALSE — judge each statement independently; any number may be true

A The crown moves distally and the root apex moves mesially, which is uncontrolled tipping
B A moment is generated about the centre of resistance equal to the force multiplied by the perpendicular distance from the line of action to that centre
C The centre of rotation for pure uncontrolled tipping lies apical to the centre of resistance
D Adding a counter-moment at the bracket can move the centre of rotation towards the apex and produce controlled tipping
E A moment-to-force ratio of about 10 to 1 is required at the bracket to produce bodily movement of an incisor

436 A 13-year-old boy is prescribed headgear to reinforce anchorage during upper canine retraction. Which single prescription is most appropriate?

A 450 to 500 g per side, worn 14 hours a day
B 250 to 350 g per side, worn 10 to 12 hours a day
C 100 to 150 g per side, worn 8 hours a day

D 250 to 350 g per side, worn 24 hours a day including school
E 450 to 500 g per side, worn 10 to 12 hours a day

437 A 12-year-old girl is to be fitted with cervical pull headgear. Which single combination of safety features is mandatory?

A A single snap-away facebow release mechanism
B Written instructions to the parent and a signed consent form, without a mechanical device
C A rigid non-releasing facebow with a padded neckstrap
D Two independent mechanisms, such as a locking facebow with a snap-release neckstrap
E A facebow with recurved inner arms and no additional device

438 A 14-year-old boy wearing high pull headgear to the upper first molars develops a progressive reduction in his lower facial height over nine months of treatment, which was not intended. Which single explanation is best supported?

A Cervical pull was inadvertently used, and its line of action passes below the centre of resistance, which would also explain the reduction in lower facial height
B The force level is too low to produce distal movement and has instead produced extrusion
C The direction of pull passes above the centre of resistance of the molars, producing intrusion as well as distal force
D High pull headgear has no vertical effect and the change must be growth alone
E The facebow inner arms were too long, which reverses the vertical component of the force

439 A transpalatal arch is fitted between the upper first permanent molars of a 14-year-old to help resist mesial movement during space closure. Regarding its action:

TRUE OR FALSE — judge each statement independently; any number may be true

A A transpalatal arch alone provides absolute anchorage against mesial molar movement
B A transpalatal arch prevents the molars from rotating mesiopalatally and from moving buccolingually
C A Nance appliance adds anchorage value over a plain transpalatal arch by bearing against the anterior palate
D A transpalatal arch can be activated to produce differential expansion or rotation of one molar relative to the other
E A lingual arch performs the corresponding function in the lower arch and can preserve leeway space

440 An upper central incisor is to be moved bodily 3 mm distally. The bracket is 9 mm from the centre of resistance along the long axis. A distal force of 80 cN is applied at the bracket. Which single counter-moment is required at the bracket for bodily movement?

A About 720 newton-millimetres
B About 240 newton-millimetres
C About 8 newton-millimetres
D About 80 newton-millimetres
E About 0.8 newton-millimetres

441 A clinician wishes to produce pure root movement of an upper central incisor, holding the incisal edge still while the apex moves palatally. Which single moment-to-force ratio at the bracket is required?

A About 2 to 1
B About 5 to 1
C About 7 to 1
D About 10 to 1

E Greater than 12 to 1

442 Regarding the centre of resistance of a tooth and the factors that alter it:

TRUE OR FALSE — judge each statement independently; any number may be true

A The centre of resistance of a single-rooted tooth lies at about one-third to two-fifths of the root length from the alveolar crest
B The centre of resistance moves coronally as alveolar bone is lost
C A force whose line of action passes through the centre of resistance produces pure translation
D Two teeth splinted together share a single centre of resistance lying between them
E The centre of resistance of a tooth is the same point as its centre of rotation

443 A 16-year-old girl requires absolute anchorage in the upper arch for en masse retraction of the labial segment. Her plan calls for miniscrews, and the clinician is choosing between the mid-palate, the maxillary buccal interradicular area and the mandibular buccal shelf. Which single site offers the lowest reported failure rate?

A The mandibular buccal shelf
B The retromolar area
C The maxillary buccal interradicular area between the second premolar and first molar
D The mid-palate
E The infrazygomatic crest

444 A miniscrew is being placed in the maxillary buccal interradicular space between the second premolar and first molar. Which single insertion torque range is most appropriate?

A 1 to 3 Ncm
B 5 to 10 Ncm
C 15 to 20 Ncm
D 25 to 30 Ncm
E Torque should be as high as the driver permits, to maximise primary stability

445 A miniscrew placed between the roots of the upper second premolar and first molar becomes mobile at six weeks and a radiograph shows its tip in contact with the mesial root of the first molar. Which single statement best describes the situation?

A Root contact causes permanent damage to the molar, which will require endodontic treatment
B Root contact is harmless and the screw should be left and loaded
C The mobility is due to insufficient insertion torque and the screw should be tightened further
D The screw has osseointegrated and the mobility must be an artefact of the examination
E Root contact is the single strongest predictor of miniscrew failure and the screw should be removed

446 Regarding the loading of orthodontic miniscrews:

TRUE OR FALSE — judge each statement independently; any number may be true

A Miniscrews depend on primary mechanical retention rather than on osseointegration, so they may be loaded immediately or after a short delay
B A screw must be left unloaded for three months before any force is applied, as with a dental implant
C Overall failure rates for orthodontic miniscrews are reported at about 12 per cent
D Failure rates are higher in the mandible than in the maxilla

E Force levels of about 50 to 200 cN are appropriate for a loaded miniscrew

447 A 15-year-old boy has a Class II division 1 malocclusion. A miniscrew placed in the palate supports retraction of the upper labial segment, and the clinician states that no reactive force acts on the posterior teeth. Which single statement is correct?

A No reactive force exists, since the screw is not part of the dentition
B The reactive force exists but is transferred to the bone rather than to the posterior teeth
C The reactive force is halved because the screw is smaller than a molar
D The reactive force acts on the screw and causes it to migrate mesially by about 2 mm
E The reactive force is absorbed by the archwire and does not reach any anchorage unit

448 A 14-year-old girl has an upper left first molar that has tipped mesially into the space of an early-lost second premolar. The plan is to upright it distally. A distal force alone is applied at the molar tube. Which single result should be expected?

A The molar will translate distally into a corrected position
B The molar will intrude without any distal movement
C The crown and root will move distally at equal rates because the molar has two roots, since two roots resist tipping more than a single-rooted tooth
D The crown will move distally and the roots will move mesially, worsening the root angulation
E The molar will rotate mesiobuccally without any change in angulation

449 Two teeth of unequal size are connected by a closing loop that delivers equal and opposite forces and equal and opposite moments. Regarding the resulting movements:

TRUE OR FALSE — judge each statement independently; any number may be true

A The two teeth move by equal distances in opposite directions, whatever their root surface areas
B The tooth with the smaller root surface area moves further, because the stress in its periodontal ligament is greater
C Differential moments can be used to make one segment tip while the other remains upright
D An anchorage unit of several teeth splinted together resists movement better than a single tooth of the same total root area
E Increasing the moment on the posterior segment relative to the anterior segment protects posterior anchorage

450 A canine is being retracted along an archwire. Which single force best reflects the level reported to produce the most rapid canine retraction?

A 272 cN
B 100 cN
C 150 cN
D 50 cN
E 400 cN

451 A 45-year-old man has generalised horizontal bone loss of 50 per cent and requires retraction of his upper labial segment. Which single modification to the mechanics is most important?

A Increase the force, because reduced bone offers less resistance to movement
B Use the same force and the same ratio, since the centre of resistance is unaffected by bone level, as its position is fixed by root length rather than by the bone that surrounds it
C Reduce the force and recalculate the moment-to-force ratio, because the centre of resistance has moved apically

D Avoid all anteroposterior movement, since retraction is contraindicated after bone loss
E Use heavier forces with a shorter duration to limit the total load on the periodontium

452 A 13-year-old girl has an upper first molar rotated mesiopalatally by 20 degrees. The clinician plans to correct the rotation using a transpalatal arch. Which single force system is required?

A A single force applied at the palatal aspect of the molar
B An intrusive force applied through the centre of resistance
C A moment applied at the buccal tube alone, with no palatal component, since the buccal tube alone can deliver a full couple
D A couple, consisting of two equal and opposite non-collinear forces, delivered through the arch
E A distal force applied at the buccal tube of the same magnitude on both sides

453 Regarding the three orders of tooth position and the way a fixed appliance delivers them:

TRUE OR FALSE — judge each statement independently; any number may be true

A First order describes labiopalatal crown inclination and is delivered by torque placed in the archwire
B Second order describes rotation of a tooth about its own long axis
C Third order bends are known as tip bends and control mesiodistal angulation
D In a preadjusted appliance the three orders are expressed only by bends placed in the archwire by the operator
E Full third order expression is obtained as soon as a rectangular archwire is engaged in the slot, whatever the play between them

454 A 0.019 by 0.025 inch stainless steel archwire is placed in a 0.022 inch slot to express a prescribed 12 degrees of palatal root torque on an upper central incisor. Which single consequence should be anticipated?

A The full 12 degrees will be expressed, since the wire is rectangular
B Several degrees of the prescription will be lost to play between the wire and the slot
C The wire will express more torque than prescribed because stainless steel is stiff
D Torque expression is independent of wire dimension and depends only on the bracket
E The wire will bind and no torque will be expressed at all

455 A 15-year-old boy has both upper central incisors requiring 3 mm of true intrusion. Which single force per tooth is most appropriate?

A Above 300 cN, so that the movement is completed quickly
B 50 to 60 cN
C 100 to 150 cN
D 200 to 250 cN
E 10 to 20 cN

456 An elastomeric chain is used to retract a canine and is reported by the manufacturer to deliver 200 cN on placement. Which single statement about its force delivery is best supported?

A The force decays substantially within the first day and continues to fall over the following weeks
B The force remains within 10 per cent of its initial value for eight weeks, consistent with the manufacturer's rated force at placement
C The force increases over the first week as the elastomer absorbs water
D The force is unaffected by the oral environment, since the material is inert
E The force is delivered as a couple rather than as a single force

457 A 13-year-old girl is prescribed headgear. Regarding the direction of pull and its effects:

TRUE OR FALSE — judge each statement independently; any number may be true

A The line of action of cervical pull headgear passes below the centre of resistance of the upper first molar
B High pull headgear extrudes the upper first molars and increases lower facial height
C Straight pull headgear is indicated where marked intrusion of the upper molars is required
D Headgear used purely for anchorage requires 14 hours a day at 450 to 500 g per side
E A single snap-release mechanism is sufficient provided that written instructions are given to the parent

458 A clinician wishes to retract a canine bodily along a 0.019 by 0.025 inch archwire using a force of 100 cN at the bracket. The canine bracket lies 10 mm from the centre of resistance. Which single feature of the mechanics supplies the necessary counter-moment?

A The elastomeric module holding the wire in the bracket
B The friction between the archwire and the bracket slot
C The couple generated between the archwire and the walls of the bracket slot
D The ligature wire tied from the canine to the molar
E The intermaxillary elastic worn from the canine to the opposing arch

459 A 14-year-old boy has a Class II division 1 malocclusion treated with upper first premolar extractions and Class II intermaxillary elastics worn from the upper canines to the lower molars at 150 cN per side. After six months the overjet has reduced but the lower incisors have proclined by 4 degrees and the occlusal plane has steepened. Which single explanation is best supported?

A The elastics were too light and have produced only dental tipping rather than skeletal change, since heavier elastics would have prevented the lower incisor proclination
B The elastic force acts on both arches, proclining the lower labial segment and extruding the lower molars
C The lower incisors proclined because of mandibular growth rather than the elastics
D Class II elastics have no effect on the lower arch, so the change must be growth
E The steepening of the occlusal plane is an artefact of cephalometric superimposition

460 A 15-year-old girl needs 3 mm of space in each upper quadrant and has excellent cooperation. The plan is to distalise the upper buccal segments with headgear. Which single regime is most appropriate?

A 250 to 350 g per side for 10 to 12 hours a day
B 150 g per side worn continuously
C 450 to 500 g per side for 8 hours a day
D 450 to 500 g per side for 14 hours a day
E 700 g per side for 6 hours a day

461 Regarding intra-oral methods of reinforcing anchorage:

TRUE OR FALSE — judge each statement independently; any number may be true

A A Nance appliance is fitted in the lower arch and bears against the lingual mucosa
B A lower lingual arch raises the anchorage value of the lower molars to the point of absolute anchorage
C Preserving the leeway space with a lower lingual arch gains about 2.5 mm per quadrant
D Class II intermaxillary elastics reinforce upper anterior anchorage without any consequence for the lower arch
E Increasing the number of teeth incorporated into the anchorage unit makes no difference

to the resistance it offers

462 A 16-year-old boy has an upper canine that has been retracted 4 mm and now shows a distally tipped crown with the root apex still mesial. The clinician plans to upright the root. Which single statement about the anchorage consequence is correct?

A Uprighting the root generates a reciprocal moment that tends to tip the anchor unit and must be resisted
B Root uprighting requires no anchorage consideration, since the crown does not move
C Root uprighting reduces the anchorage demand because the crown is already in position, since only the root remains to be moved
D The anchorage demand is the same as for the original retraction, no more and no less
E Root uprighting can only be performed after the appliance has been removed

463 A 15-year-old girl is to have en masse retraction of the upper labial segment against palatal miniscrews, with the retraction force applied from the screw head through a transpalatal wire to a hook on the archwire between the lateral incisor and canine. The hook is 4 mm long and the force is 200 cN. Which single effect of lengthening the hook to 8 mm should be anticipated?

A The force magnitude is halved, so the rate of movement falls
B The line of action moves further from the centre of resistance, increasing lingual crown tipping, because a longer hook increases the moment arm without changing the force
C The line of action moves closer to the centre of resistance of the anterior segment, reducing lingual crown tipping
D The hook length has no bearing on the force system, only on patient comfort
E The anterior segment will intrude without any anteroposterior movement

464 A miniscrew is planned for the infrazygomatic crest of a 16-year-old boy. Which single anatomical consideration most governs the site chosen?

A The thickness of the attached mucosa over the crest, which should be minimised
B The thickness of the midpalatal suture
C The proximity of the greater palatine foramen
D The course of the inferior alveolar nerve
E The position of the maxillary sinus and the buccal roots of the first molar

465 Regarding complications associated with orthodontic miniscrews:

TRUE OR FALSE — judge each statement independently; any number may be true

A Perforation of the maxillary sinus by a short screw is usually of no clinical consequence and heals without intervention
B Soft tissue overgrowth around the head of a screw placed in non-keratinised mucosa is a recognised cause of inflammation and failure
C Fracture of a screw on insertion or removal is more likely where the diameter is below 1.5 mm
D Palatal screws are placed medial to the greater palatine neurovascular bundle to avoid damaging it
E A screw that loosens within the first few weeks can often be replaced at an adjacent site and loaded again

466 A 14-year-old boy has an upper arch aligned on a 0.019 by 0.025 inch stainless steel archwire, and en masse retraction is begun with 150 cN per side. Over the following months the upper incisors retrocline by 8 degrees while retracting by 3 mm. Which single mechanical change would best preserve incisor inclination?

A Add palatal root torque to the archwire in the anterior segment

B Increase the retraction force to 300 cN per side
C Change to a round 0.018 inch archwire to reduce friction
D Apply the retraction force from a shorter hook on the archwire
E Change from sliding mechanics to an elastomeric chain of higher initial force

467 A segmented arch technique is used to intrude the upper labial segment of a 17-year-old girl. An intrusion arch is inserted into the molar auxiliary tubes and tied as a single point contact to the anterior segment. Which single reactive effect on the posterior segment should be anticipated?

A A purely distal force with no vertical component
B An intrusive force and a mesial crown tipping moment on the molars
C An extrusive force and a distal crown tipping moment on the molars
D No reactive effect, since the anterior segment is the only unit engaged
E A rotational couple in the transverse plane only

468 A 13-year-old girl has an upper arch in which the right first molar must be held absolutely while the left first molar is brought forward 4 mm into a second premolar extraction space. Which single approach best delivers this asymmetry?

A A transpalatal arch alone, activated to the left
B Extraction of the right second premolar as well, to restore symmetry, since symmetrical spaces simplify anchorage control
C Class II elastics on the right side only
D Headgear worn to the right side only at 250 g
E A palatal miniscrew supporting the right molar, with the left molar protracted against it

469 Regarding force magnitude and the tissue response to it:

TRUE OR FALSE — judge each statement independently; any number may be true

A Optimal force for bodily tooth movement is in the range 300 to 400 cN
B Heavier forces produce proportionally faster tooth movement throughout the range tested
C The lag phase of tooth movement lasts about three days
D The periodontal ligament space is about 2 mm wide
E Expression of RANKL peaks about 14 days after the application of force

470 A 17-year-old boy has an anterior open bite of 3 mm with increased lower facial height. Miniscrews are placed bilaterally in the infrazygomatic crests and used to intrude the upper posterior teeth by 2 mm. Which single consequence should be anticipated at the chin?

A The mandible rotates backwards and the chin moves downwards and backwards
B The mandible rotates forwards and upwards, closing the open bite and improving chin projection
C No change at the chin, since only the maxillary teeth have been moved
D The chin moves forwards but lower facial height increases
E The open bite widens because the posterior teeth no longer support the occlusion, since posterior support is what keeps the bite closed anteriorly

471 A clinician states that a moment-to-force ratio of about 7 to 1 has been established at the bracket of an upper central incisor during retraction. Which single movement should be expected?

A Uncontrolled tipping, with the centre of rotation near the centre of resistance
B Pure root movement about the incisal edge
C Pure translation of the whole tooth
D Controlled tipping, with the centre of rotation at or near the root apex

E Pure rotation about the long axis of the tooth

472 A 14-year-old girl requires 5 mm of upper incisor retraction. A Nance appliance is fitted and the acrylic button is noted after four months to have embedded itself into the palatal mucosa, which is inflamed and bleeding. Which single course is most appropriate?

A Continue, since the embedding indicates that the appliance is bearing the reactive load effectively, and inflamed tissue typically settles once it adapts to the appliance
B Reduce the retraction force by half and leave the appliance undisturbed
C Remove or remake the appliance and manage the soft tissue, since continued embedding risks ulceration and necrosis
D Replace the acrylic button with a larger one without removing the appliance
E Accept the inflammation as an unavoidable consequence of palatal anchorage

473 A 15-year-old boy has a Class II division 1 malocclusion with a deep overbite, 6 mm of upper crowding and a narrow upper arch in crossbite on the left. The plan requires anchorage to be considered in all three planes. Which single statement best describes the vertical anchorage requirement here?

A Molar extrusion during levelling must be resisted, because it would rotate the mandible backwards and worsen the facial pattern
B The vertical plane requires no anchorage planning, since overbite reduction is achieved by levelling alone, and posterior extrusion during levelling has no effect on mandibular rotation
C Molar extrusion should be encouraged, since it is the most efficient way to reduce a deep overbite
D Vertical anchorage is relevant only in open bite cases
E The vertical plane is controlled entirely by the archwire sequence and requires no separate decision

474 Regarding Newton's third law as it applies to orthodontic anchorage:

TRUE OR FALSE — judge each statement independently; any number may be true

A An anchorage unit can be made absolute by splinting a sufficient number of teeth together
B Reciprocal space closure moves the two units by equal distances whatever their root surface areas
C The reactive force ceases once the active unit has reached its intended position
D Anchorage needs to be considered only in the anteroposterior plane
E Where a large anchorage unit is opposed by a small one, the small unit moves the further under the same force

475 A 16-year-old girl requires 4 mm of retraction of the lower labial segment against a reduced periodontium with 40 per cent horizontal bone loss. Which single combination of force and moment-to-force ratio is most appropriate?

A A force of 100 cN with a ratio of 10 to 1 measured to the original centre of resistance
B A force of 250 cN applied intermittently to allow periodontal recovery between activations
C A force of 150 cN with a ratio of 7 to 1, accepting some controlled tipping
D The same force and ratio as for a patient with normal support, since the bracket position is unchanged
E A reduced force of about 40 cN with a ratio recalculated to the new, more apical centre of resistance

476 A 12-year-old boy attends with a painful, red and swollen left eye two days after his facebow came out of the molar tubes during sleep. Which single course is most appropriate?

A Reassure the parents and review at the next routine appointment
B Reduce the force to 150 g per side and continue as before
C Refit the facebow with a shorter outer bow and continue wear at night only
D Refer urgently for ophthalmological assessment and discontinue the headgear
E Advise a course of topical antibiotic drops and continue headgear wear

477 A 16-year-old girl has an upper lateral incisor rotated 30 degrees mesiolabially. Which single statement about the correction and its retention is best supported?

A Rotational correction requires a single force applied at the mesiolabial line angle and relapses because of bone remodelling
B Rotational correction is achieved by a couple, and the supracrestal fibres are the principal cause of its relapse
C Rotational correction is the most stable of all tooth movements and requires no specific retention
D Rotational relapse is prevented by over-rotation alone, so retention is unnecessary
E Rotational correction is achieved by torque delivered in a rectangular archwire

478 Regarding reported failure rates of orthodontic miniscrews by site:

TRUE OR FALSE — judge each statement independently; any number may be true

A The median failure rate reported for palatal miniscrews is about 1.3 per cent
B The mandibular buccal shelf has a failure rate comparable with that of the palate
C Failure rates are independent of the experience of the operator placing the screw
D The overall failure rate for orthodontic miniscrews pooled across sites is about 12 per cent
E Screws that fail do so almost exclusively after twelve months of loading

479 A distal force of 50 cN is applied to an upper canine at a point 12 mm from its centre of resistance, together with a counter-moment of 300 newton-millimetres delivered by the bracket. Which single movement should be expected?

A Pure translation of the canine
B Intrusion with no anteroposterior component
C Uncontrolled tipping with the centre of rotation near the centre of resistance
D Pure root movement about the incisal tip
E Controlled tipping with the centre of rotation near the apex

480 A 14-year-old boy has an upper first molar that requires 3 mm of distal movement while the upper incisors must not procline. A pendulum appliance with a Nance acrylic button is proposed. Which single limitation of this design is most important?

A The appliance cannot deliver sufficient force to move a molar distally
B The reactive force is delivered to the palate and the premolars, proclining the incisors
C The appliance requires 14 hours of daily patient cooperation to be effective, making compliance the main limitation to its success
D The appliance produces distal bodily movement of the molar with no tipping
E The appliance is contraindicated where the second molars are unerupted

481 A clinician wishes to protract a lower left first molar 5 mm forwards into an extraction space while keeping the lower incisors exactly where they are. Which single approach is most appropriate?

A Sliding mechanics from the lower incisors, using an elastomeric chain
B A lower lingual arch activated to push the molar forwards from the incisors, using the incisors as the sole source of reactive anchorage
C Class III intermaxillary elastics from the upper arch to the lower molar
D A miniscrew placed in the retromolar or buccal shelf region, pulling the molar forwards towards it
E A push coil spring between the lower canine and the molar

482 Regarding the delivery of the three orders by a preadjusted edgewise appliance:

TRUE OR FALSE — judge each statement independently; any number may be true

A Every preadjusted prescription specifies the same tip and torque values for a given tooth
B The vertical and mesiodistal position of the bracket on the crown alters the tip and torque actually delivered
C Torque expressed on an upper central incisor is unaffected by the labiopalatal thickness of the bracket base
D Rotation of a tooth about its long axis is a third order correction
E A canine bracket bonded 1 mm too far gingivally will leave the tooth in an incorrect vertical position at the end of treatment

483 A 17-year-old girl finishing fixed appliance treatment has upper lateral incisors that appear too upright and too short, with the incisal edges 1 mm above the level of the centrals. Review of the bond-up photographs shows the lateral brackets placed slightly gingival and rotated. Which single explanation best accounts for the finish?

A Bracket placement error has altered the delivery of all three orders for those teeth
B The archwire sequence was too stiff, preventing full expression of the prescription
C The bracket prescription for lateral incisors is inherently unreliable
D The lateral incisors are intrinsically resistant to torque because their roots are short
E Insufficient treatment time was allowed for the prescription to express itself

484 An upper arch requires maximum anchorage for the retraction of a 7 mm overjet after the loss of upper first premolars. Working up Proffit's ladder of increasing anchorage reinforcement, which represents the step immediately below skeletal anchorage?

A Adding the second molars to the appliance to increase the root surface area resisting movement
B Reciprocal anchorage between the two buccal segments carried across a transpalatal arch
C Extraoral traction from headgear worn 10 to 12 hours a day at 250 to 350 g a side
D Differential force, a lighter retraction force being applied against a heavier posterior unit
E Tip back bends placed in the archwire to resist mesial movement of the posterior teeth

485 A 13-year-old Class III patient is to have bone anchored maxillary protraction using infrazygomatic and mandibular symphyseal miniplates. Regarding miniplates as skeletal anchorage:

TRUE OR FALSE — judge each statement independently; any number may be true

A The plate body rests on basal bone away from the roots, so root proximity does not constrain siting as it does for miniscrews
B They require a mucoperiosteal flap at placement and a second procedure to remove them
C Reported failure rates are higher than those for palatal miniscrews

D They allow protraction force without the dental side effects of a tooth borne facemask
E They can be loaded immediately because they osseointegrate within 48 hours

Block O — Orthodontic materials, wires, bonding and instruments

(Q486–530)

Volume I: Topic 7 (The ideal archwire); Chapter 10, pp. 645–665
Volume II: **§15** — read it again if you score under 70%
This block: 45 questions (11 of them true/false) · 89 marks · allow 56 minutes

486 A colleague setting up a new practice asks which single archwire he should stock if he could buy only one, since every property in the manufacturer's list sounds desirable. Which single statement best explains why no such wire exists?

A Patent restrictions held by different manufacturers prevent the useful properties from being combined in any one product
B Manufacturing tolerances are too wide for any alloy to be produced to a reliable specification
C Corrosion in the oral environment limits the working life of every alloy to a few weeks
D The demands of initial alignment and of finishing mechanics are in direct opposition, so no one alloy meets both
E Only nickel titanium has been tested adequately and the other alloys remain unproven

487 A 14-year-old girl has upper labial segment crowding of 6 mm with the upper left lateral incisor rotated and displaced palatally by about 3 mm. Brackets have just been bonded in a 0.022 inch slot. Which single archwire is most appropriate as the first wire?

A 0.019 by 0.025 inch stainless steel because it establishes arch form immediately
B 0.014 inch superelastic nickel titanium because it delivers a light force over a large deflection
C 0.018 inch stainless steel because it resists permanent deformation when tied into a displaced tooth
D 0.021 by 0.025 inch beta titanium because its formability allows bends around the displaced tooth
E 0.016 inch cobalt chromium because it can be heat treated after placement

488 Regarding the stress-strain curve of an orthodontic wire and the terms taken from it:

TRUE OR FALSE — judge each statement independently; any number may be true

A The gradient of the initial straight portion of the curve is the modulus of elasticity and determines the stiffness of the wire
B The elastic limit is the greatest stress the wire can sustain without permanent deformation
C Resilience corresponds to the area under the curve up to the elastic limit and represents the energy stored
D Formability is the amount of permanent deformation the wire will accept before it fractures
E Range is the distance the wire can be deflected without exceeding its elastic limit

489 A registrar compares a 0.016 inch round stainless steel wire with a 0.020 inch round stainless steel wire of the same alloy, spanning the same interbracket distance. Which single statement about the larger wire is correct?

A It is about one and a half times as stiff and has a correspondingly shorter range
B It is stiffer only in torsion, the bending behaviour being governed by length alone
C It is about two and a half times as stiff in bending and its range is unchanged
D It is about four times as stiff in bending and its range is increased in proportion

E It is about two and a half times as stiff in bending and its range is reduced

490 An examiner asks how the same archwire can be made to deliver a lighter force without changing its alloy or its cross-section. Which single manipulation is most effective?

A Increase the length of wire between the points of attachment, for example by incorporating a loop
B Increase the interbracket span by bonding the brackets closer to the incisal edges
C Reduce the length of wire between the points of attachment to concentrate the deflection, believing a shorter span softens the force
D Anneal the wire at the chairside immediately before placing it
E Tie the wire with stainless steel ligatures rather than elastomeric modules

491 A 22-year-old man is in the finishing stage. Individual root torque adjustments and a small step to level two marginal ridges are required, and the wire must hold its shape but not deliver steel-like forces. Which single wire is most appropriate?

A 0.019 by 0.025 inch superelastic nickel titanium
B 0.018 inch multistranded stainless steel
C 0.017 by 0.025 inch beta titanium
D 0.019 by 0.025 inch heat-activated nickel titanium
E 0.016 inch round stainless steel

492 Regarding austenitic stainless steel archwires used in orthodontics:

TRUE OR FALSE — judge each statement independently; any number may be true

A The alloy is typically 18 per cent chromium and 8 per cent nickel, the chromium forming the passivating oxide layer
B Stainless steel has a lower modulus of elasticity than beta titanium
C Cold working during drawing increases the yield strength of the wire and reduces its formability
D Stainless steel archwires cannot be soldered and must be joined by welding alone
E A multistranded stainless steel wire is less stiff than a solid wire of the same overall diameter

493 A 13-year-old boy has mild lower labial segment crowding of 2 mm and two lower incisors rotated by about 20 degrees. The practice does not stock nickel titanium. Which single alternative first wire is most appropriate?

A 0.016 inch solid stainless steel, which is the standard first steel wire
B 0.0175 inch multistranded stainless steel, which is flexible yet has a high yield strength
C 0.020 inch solid stainless steel, tied with elastomeric modules to soften the force
D 0.016 inch cobalt chromium in its hardest temper
E 0.021 by 0.025 inch beta titanium, which is formable enough to accommodate the rotations

494 A heat-activated nickel titanium archwire has been placed in a patient who reports that the teeth feel tight after a hot drink and loose again later. Which single explanation is correct?

A The wire has undergone permanent plastic deformation on warming
B Heating has annealed the wire and reduced its modulus of elasticity
C The wire has corroded and released nickel ions that irritate the periodontium
D Warming takes the wire above its transition temperature, so more of it is austenitic and it works harder
E The sensation is unrelated to the wire and reflects the normal pain peak at 48 hours after an adjustment

495 A clinician wants a wire that can be bent easily into complex loops at the chairside and then made stiffer without altering those bends. Which single alloy allows this?

A Cobalt chromium in its softest temper, heat treated after forming
B Superelastic nickel titanium
C Austenitic stainless steel in its hardest temper
D Beta titanium, which stiffens on exposure to saliva
E Multistranded stainless steel, which stiffens as the strands bed down

496 Regarding superelasticity in nickel titanium archwires:

TRUE OR FALSE — judge each statement independently; any number may be true

A Superelastic behaviour arises because the wire work hardens progressively as it is deflected
B The unloading plateau lies at a higher force than the loading plateau
C Superelastic nickel titanium delivers a force that increases in proportion to the deflection across its working range
D Superelastic nickel titanium can be reliably soldered to auxiliaries at the chairside
E The clinical advantage of the plateau is that a shorter wire can be used to reach a displaced tooth

497 A 15-year-old girl is having brackets bonded. The enamel of the upper right central incisor has been etched with 37 per cent phosphoric acid for 30 seconds, washed and dried, and appears glossy rather than frosty. Which single action is most appropriate?

A Proceed with primer and adhesive, since the appearance of etched enamel is unreliable
B Increase the etch time to 120 seconds at the next attempt on all teeth
C Re-isolate, re-etch the surface and re-dry before proceeding
D Abandon bonding on this tooth and place a band instead
E Apply a self-etching primer over the glossy surface without further etching

498 A 12-year-old boy has had four bracket failures on lower premolars in eight weeks, while his upper arch has been stable. He has a deep overbite of 6 mm and eats an unrestricted diet. Which single explanation is most likely?

A Inadequate etching of premolar enamel, which is more highly mineralised than incisor enamel
B Nickel sensitivity causing localised failure of the adhesive interface
C Salivary contamination, which is unavoidable in the lower arch and affects all lower teeth equally
D A faulty batch of adhesive, since the failures are clustered in time
E Occlusal trauma from the opposing upper teeth striking the lower premolar brackets

499 A 10-year-old girl with poor oral hygiene and moderate salivary flow needs a band cementing on a partially erupted lower first molar where absolute moisture control is not achievable. Which single cement is most appropriate?

A Conventional glass ionomer cement, for its fluoride release and tolerance of a moist field
B Chemically cured composite resin, for its high bond strength
C Zinc phosphate cement, for its long clinical record
D Resin cement with a self-etching primer, for the speed with which it can be placed on a partly erupted tooth
E Calcium hydroxide lining cement, for its pulpal protection

500 Regarding the bonding of orthodontic brackets to enamel:

TRUE OR FALSE — judge each statement independently; any number may be true

A Etching with 37 per cent phosphoric acid for 15 to 30 seconds removes of the order of 10 micrometres of enamel
B Retention of the adhesive depends principally on resin tags formed in the etched surface
C A chemically cured adhesive gives the operator more working time than a light-cured adhesive
D Moisture-active primers allow bonding to proceed on a surface that is not perfectly dry
E Bond strength continues to rise for about a month after placement in a light-cured adhesive

501 A 17-year-old woman has just completed fixed appliance treatment. At debond the operator uses a tungsten carbide bur at low speed to remove residual adhesive, and the patient asks whether her enamel has been harmed. Which single reply is most accurate?

A No enamel is lost, because the adhesive separates cleanly at the resin-enamel interface
B Around 1 mm of enamel is removed across the treatment, which is why the teeth appear smaller after debond
C A layer of enamel of the order of tens of micrometres is lost across etching, debonding and clean-up
D Enamel loss is confined to patients in whom a ceramic bracket has been used
E Enamel is lost only if a high-speed handpiece is used for clean-up

502 A 14-year-old boy attends four months after bonding with chalky white opacities on the cervical third of both upper lateral incisors that do not disappear on wetting. His plaque control is poor and he drinks a sports drink daily. Which single course of action is most appropriate?

A Continue treatment unchanged and review the lesions at debond, when remineralisation may have occurred
B Bond glass ionomer over the lesions to seal them
C Microabrade the lesions immediately so that they do not deepen while the appliance is in place
D Debond all appliances at once and restart in adulthood
E Reinforce oral hygiene, prescribe a high-fluoride toothpaste and review the appliance load

503 A 13-year-old girl needs an attachment on a lower first permanent molar that will carry a transpalatal-type auxiliary, resist heavy occlusal load and survive two years of treatment. The tooth has a large amalgam restoration on its occlusal and buccal surfaces. Which single statement best supports banding this tooth rather than bonding it?

A A band is quicker to place and requires no moisture control
B A band does not depend on an enamel bond and so is unaffected by the restored buccal surface
C A band produces no space between adjacent teeth and so does not need separators
D Banding avoids any risk of demineralisation of the tooth
E Bands give better torque control than bonded tubes because the slot lies further from the occlusal surface

504 A 16-year-old boy is having a canine retracted along a 0.019 by 0.025 inch stainless steel archwire in a 0.022 inch slot. Regarding the resistance to sliding that is being generated:

TRUE OR FALSE — judge each statement independently; any number may be true

A Resistance to sliding in the clinical situation is dominated by classical friction rather than by binding

B Binding begins once the angle between the wire and the bracket slot exceeds a critical contact angle
C Notching describes an elastic deformation of the wire that resolves once the ligature is removed
D Classical friction rises in proportion to the apparent area of contact between the wire and the bracket
E An elastomeric module produces a lower normal force against the archwire than a loosely tied stainless steel ligature

505 A practice changes from replacing elastomeric modules at every visit to replacing them at alternate visits, extending the interval from six to twelve weeks. Which single consequence is most likely?

A The modules will deliver a more constant force over the longer interval because they have settled into the slot
B There will be no change, as elastomerics are dimensionally stable in the mouth
C The modules will stiffen with age and increase the force applied to the teeth
D Ligation force will have fallen substantially, with loss of rotational control and slower sliding
E Bond failure rates will rise because the modules hold the brackets in place

506 A 15-year-old girl has an upper lateral incisor rotated by 30 degrees that is failing to derotate on a 0.018 inch nickel titanium wire tied with elastomeric modules. The bracket is correctly positioned and the wire is fully engaged. Which single change is most appropriate?

A Change to a stiffer round stainless steel wire of the same size, which will seat more firmly in the slot
B Remove the wire for four weeks to allow the tissues to adapt
C Tie the tooth with a figure-of-eight steel ligature to seat the wire fully and add a rotation wedge
D Rebond the bracket 1 mm further gingivally
E Place a band on the tooth in place of the bracket

507 A 15-year-old girl reports itching and erythema of the perioral skin and lips two weeks after starting Class II elastics. Regarding allergy in orthodontic practice:

TRUE OR FALSE — judge each statement independently; any number may be true

A Latex allergy affects about 4.3 per cent of the general population
B Nickel allergy is commoner in females, with reported prevalences of 10 to 30 per cent in women and 1 to 3 per cent in men
C Intraoral nickel sensitisation is common and usually presents as widespread oral ulceration
D A type I latex reaction can present as urticaria, rhinitis or in severe cases anaphylaxis
E Non-latex elastics are available and are the appropriate substitution once latex allergy is suspected

508 A nurse is asked to lay out an instrument that will hold the end of an archwire while it is cut distal to the last molar tube, so that the cut segment is not lost in the mouth. Which single instrument is required?

A Distal end cutter
B Weingart pliers
C Mitchell's trimmer
D Adams universal pliers
E Band seater

509 A 14-year-old boy is four months into fixed appliance treatment in a 0.022 inch slot. The arch is fully aligned on a 0.018 inch nickel titanium wire and space closure is planned next. Which single wire should follow?

A 0.020 inch round stainless steel, to keep the force light while the arch form settles
B 0.017 by 0.025 inch beta titanium, to allow torque bends during retraction
C 0.019 by 0.025 inch superelastic nickel titanium, to continue levelling while space closes
D 0.014 inch multistranded stainless steel, to maintain alignment with minimal force
E 0.019 by 0.025 inch stainless steel, to fill the slot and provide a rigid track for sliding

510 An examiner asks what is lost by aligning and finishing an entire case on round wires alone. Which single answer is best?

A Nothing, provided the round wire is of adequate diameter
B Control of arch width, because round wires cannot be expanded
C Control of mesiodistal angulation, because tip is a function of the wire cross-section
D Control of labiolingual root inclination, because a round wire expresses no third order
E The ability to close space, because round wires bind in the slot and cannot slide

511 Regarding the effect of wire dimension and cross-section on clinical behaviour:

TRUE OR FALSE — judge each statement independently; any number may be true

A Range varies directly with the diameter of a round wire
B Halving the interbracket span makes a wire about twice as stiff
C Doubling the diameter of a round wire increases its bending stiffness roughly sixteenfold
D For a rectangular wire in bending, the dimension lying in the plane of bending has the greater influence on stiffness
E A 0.019 by 0.025 inch wire in a 0.022 inch slot is a full-size wire with negligible torsional play

512 Two wires of identical dimension are compared. Wire A has a high yield strength and a high modulus. Wire B has a moderate yield strength and a low modulus. Which single statement about springback is correct?

A Wire A has the greater springback because its yield strength is higher
B Wire B has the greater springback because springback depends on the ratio of yield strength to modulus
C Springback is independent of modulus and is determined by yield strength alone
D Springback is determined by formability alone and cannot be predicted from either of the values given
E Both wires have the same springback because they have the same dimension

513 A 16-year-old boy has an upper archwire that has been in place for ten weeks and shows a permanent bend where it crosses a previously displaced canine bracket. Which single property of the wire has been exceeded?

A Its resilience
B Its corrosion resistance
C Its formability
D Its range of activation in torsion only
E Its elastic limit

514 A clinician wishes to attach a hook to a nickel titanium archwire by soldering it at the chairside. Which single statement is most accurate?

A Silver soldering is satisfactory provided the wire is first pickled
B Nickel titanium cannot be reliably soldered or welded, so a crimpable hook should be used instead
C Laser welding is contraindicated on all orthodontic alloys
D The wire should be heat treated to 480 degrees Celsius before soldering
E Soldering is acceptable because the wire loses its superelastic behaviour only in the heated zone

515 Regarding auxiliaries used with fixed appliances:

TRUE OR FALSE — judge each statement independently; any number may be true

A Nickel titanium closed coil springs should be activated to at least twice their resting length to deliver a useful force
B An open coil spring placed between two brackets and compressed will push the teeth apart to open space
C A closed coil spring in nickel titanium delivers a more constant force over its working range than an elastomeric chain
D Elastomeric chain loses a substantial proportion of its force within the first 24 hours
E A stainless steel ligature tied conventionally exerts a higher and more sustained seating force than an elastomeric module

516 A 12-year-old girl with a high caries rate and several white spot lesions is to have brackets bonded. Which single bonding material offers the best compromise between retention and caries protection?

A Conventional glass ionomer cement, for the high fluoride release its setting reaction sustains, making it preferable despite its lower bond strength
B Zinc polycarboxylate cement, for its chemical adhesion
C Chemically cured composite resin with no fluoride
D Resin-modified glass ionomer cement, which releases fluoride while approaching composite bond strengths
E Light-cured composite resin without a primer

517 A colleague has moved to a self-etching primer to save chair time and asks what he has given up. Which single statement is most accurate?

A The etch pattern is shallower, so bond strength is lower, but failure rates stay acceptable
B Nothing, since bond strengths and failure rates are identical to those of the conventional technique
C Self-etching primers do not etch the enamel at all and rely on chemical adhesion to hydroxyapatite
D Self-etching primers require a longer rinse and so save no time at all
E Self-etching primers can only be used with chemically cured adhesives

518 A randomised trial reports a first-time bracket failure rate of 6 per cent with a conventional two-step technique and 8 per cent with a self-etching primer over 12 months, a difference that does not reach statistical significance in a trial of 60 patients. Which single conclusion is best supported?

A Self-etching primers should be abandoned as inferior
B The trial shows the two techniques are equivalent
C The trial is compatible with a real difference of meaningful size and is too small to show equivalence

D The failure rates quoted are far above the range normally accepted in practice and the trial must be flawed
E Failure rate is not a valid outcome measure for bonding studies

519 Regarding debonding at the end of fixed appliance treatment:

TRUE OR FALSE — judge each statement independently; any number may be true

A Debonding pliers are designed to peel the bracket base rather than to crush the bracket
B Polishing after clean-up restores the enamel surface exactly to its pre-treatment state
C The bond should ideally fail at the bracket-adhesive interface so that the enamel is protected
D Adhesive remnant scores are used to record where the failure occurred
E A ceramic bracket carries a greater risk of enamel fracture at debond than a stainless steel bracket

520 A 16-year-old girl has been wearing 3 ounce Class II elastics changed once a week as instructed by a previous clinician. Which single consequence of this instruction is most important?

A The elastics will have stiffened and delivered increasing force through the week
B The elastics will have caused root resorption because the force applied was continuous and unrelieved
C Weekly changes are correct and the problem must lie elsewhere
D Force decay means most of the useful force is gone within a day, so the mechanics have been ineffective
E Weekly change reduces compliance problems and is the preferred regimen

521 A 13-year-old boy is to have bands fitted on all four first permanent molars in a week. The contacts are tight and no space is available. Which single preparatory measure is most appropriate?

A Interproximal reduction of the mesial surfaces of the molars
B Placement of elastomeric separators mesially and distally to each molar
C Placement of a passive transpalatal arch to allow the molars to drift
D Extraction of the second primary molars to release space
E No preparation, since bands are seated with a band pusher regardless of contact tightness

522 A 24-year-old woman is in the last months of treatment. The upper right lateral incisor requires 5 degrees of additional palatal root torque and a small first order in-and-out correction, and the operator wishes to place both bends in one wire that will hold them. Which single wire is most suitable?

A 0.019 by 0.025 inch superelastic nickel titanium
B 0.016 inch heat-activated nickel titanium
C 0.020 inch round stainless steel
D 0.0175 inch multistranded stainless steel
E 0.017 by 0.025 inch beta titanium

523 Regarding instruments in routine fixed appliance practice:

TRUE OR FALSE — judge each statement independently; any number may be true

A A Mitchell's trimmer is used to place archwires into molar tubes
B Adams universal pliers have one square and one round beak and are used for forming arrowheads and adjusting clasps
C Mauns heavy wire cutters should be used to cut hard stainless steel of 0.7 mm and above

D Light wire cutters are appropriate for cutting a 0.019 by 0.025 inch stainless steel archwire
E Bracket removing pliers apply a peeling force to the bracket base

524 A 15-year-old girl in a 0.018 inch slot appliance has completed alignment on a 0.016 inch nickel titanium wire. She has 4 mm of residual space to close by sliding and requires good torque control of the upper incisors. Which single wire is most appropriate?

A 0.016 by 0.022 inch stainless steel
B 0.016 inch round stainless steel
C 0.018 by 0.025 inch beta titanium
D 0.016 by 0.022 inch superelastic nickel titanium
E 0.018 inch multistranded stainless steel

525 A patient with a documented type I latex allergy requires intermaxillary elastics. Which single management is most appropriate?

A Use latex elastics but change them at each meal to reduce exposure
B Abandon elastics and complete the case with headgear instead
C Use synthetic non-latex elastics and ensure the whole surgery is latex free during her appointments
D Premedicate with an antihistamine before each appointment and continue with latex elastics, since prophylaxis is assumed to blunt the reaction
E Use latex elastics coated with a barrier layer

526 An examiner asks why a 0.019 by 0.025 inch stainless steel wire in a 0.022 by 0.028 inch slot does not deliver the full torque prescribed in the bracket. Which single explanation is best?

A The prescription is expressed for a round wire and must be converted before a rectangular wire is used
B Torsional play between the wire and the slot must be taken up before any torque is expressed
C Stainless steel has too low a modulus to transmit a torquing couple
D The bracket base flexes and absorbs the torque
E The ligation method determines torque expression entirely

527 Regarding the sequence of archwires through a course of fixed appliance treatment:

TRUE OR FALSE — judge each statement independently; any number may be true

A The first archwire is selected principally for its ability to establish and hold the planned arch form
B A wire of low stiffness and long range will also resist the tipping moments generated during retraction
C A rectangular wire is required before torque can be expressed
D A working wire for sliding mechanics should resist deformation and fill a substantial part of the slot
E Finishing bends are most easily placed in a wire of high stiffness and low formability

528 A colleague argues that because nickel titanium delivers a light constant force it should be used for every stage of treatment. Which single argument against this is strongest?

A Nickel titanium releases nickel ions in quantities large enough to cause systemic harm over a long course
B Nickel titanium loses its superelasticity within a fortnight in the mouth
C Nickel titanium is too expensive for routine use in a health service practice
D Nickel titanium cannot be bent, cannot hold arch form against heavy mechanics and has a high surface friction

E Nickel titanium can only be used in a 0.018 inch slot appliance

529 A department is asked to justify a change from stainless steel ligatures to elastomeric modules for routine ligation on grounds of chair time. Which single trade-off should be set against the time saved?

A Elastomeric modules decay, so seating force and therefore rotational and torque control fall between visits
B Elastomeric modules cannot be used with rectangular wires
C Elastomeric modules retain more plaque and cause more enamel demineralisation than steel ligatures do
D Elastomeric modules increase the risk of bracket failure
E Elastomeric modules cannot be used in the lower arch

530 Alignment is complete but a triangular cuff of hyperplastic gingiva covers 2 mm of the upper left central incisor crown and prevents an accurate bracket position for torque finishing. The tissue is firm, oral hygiene is now good, and there is neither pocketing nor bone loss. Which is the single most appropriate step?

A Continue with the existing bracket position and accept the torque discrepancy at debond
B Refer for periodontal surgery with a flap and osseous recontouring before finishing continues
C Excise the excess tissue with a diode laser at the chairside and rebond once haemostasis is achieved
D Prescribe a course of chlorhexidine and review in three months before making any decision
E Debond the tooth and let the tissue resolve before rebonding at the correct height

Block P — Removable and functional appliances (Q531–571)

Volume I: Topic 8 (Functional appliances); Chapter 9
Volume II: **§16** — read it again if you score under 70%
This block: 41 questions (10 of them true/false) · 81 marks · allow 51 minutes

531 A trainee submits a prescription for an upper removable appliance that reads simply "appliance to retract the canines". The laboratory returns it with a query. Which single omission is the most serious?

A The shade of the acrylic has not been stated
B The impression material has not been named
C The patient's age has not been recorded on the prescription
D No instruction has been given about the thickness of the baseplate or the extent of its palatal coverage
E No retention has been specified, so the appliance cannot resist the reaction to the active components

532 An upper removable appliance with two palatal finger springs is repeatedly displaced during eating and the patient reports that the clasps on the first permanent molars feel loose after a fortnight. The clasps are Adams clasps in 0.6 mm hard stainless steel wire. Which single change is most appropriate?

A Remake the molar clasps in 0.7 mm hard stainless steel wire
B Add a labial bow in 0.5 mm wire to improve anterior retention
C Reduce the activation of the finger springs to lower the reaction force
D Replace the Adams clasps with ball-ended clasps in the same gauge
E Add a second baseplate layer to increase bulk and retention

533 Regarding the components of an upper removable appliance:

TRUE OR FALSE — judge each statement independently; any number may be true

A The Adams clasp obtains its retention from the mesiolingual and distolingual undercuts of the abutment tooth
B A labial bow in 0.7 mm wire is an efficient means of retracting incisors through more than 3 mm
C The baseplate contributes nothing to anchorage and serves only to connect the components
D A Southend clasp is constructed in 0.5 mm wire and engages the buccal surfaces of the first premolars
E Upper removable appliances are capable of bodily movement of a tooth when two points of force application are used

534 A 10-year-old girl requires distal movement of the upper right canine by 3 mm. A palatal finger spring in 0.5 mm wire with a guard is planned. Which single activation is most appropriate?

A Activate by 0.25 mm, since any greater deflection will deform the wire
B Activate by 4 mm, which shortens the treatment time proportionately
C Activate by 1 to 2 mm, which delivers of the order of 25 to 40 cN
D Activate by 3 mm so that the full movement is achieved from one adjustment
E Activation is not required because the spring works by the patient's occlusal forces

535 A 9-year-old boy has the upper left central incisor in crossbite with a displacement on closing. The tooth is retroclined, there is no crowding, and the overbite after correction would be 3 mm. Which single active component is most appropriate?

A A T-spring in 0.5 mm wire placed palatal to the upper left canine
B A labial bow in 0.7 mm wire
C A palatal finger spring in 0.5 mm wire with a guard
D A midline expansion screw turned twice weekly
E A Z-spring in 0.5 mm wire behind the incisor, with posterior bite planes

536 A 12-year-old girl has the upper right second premolar in buccal crossbite and requires 2 mm of palatal movement of the lower tooth and buccal movement of the upper. The upper premolar alone is to be moved buccally with a removable appliance. Which single component is most appropriate?

A A buccal canine retractor in 0.7 mm wire
B A T-spring in 0.5 mm wire placed palatal to the tooth
C A Z-spring placed behind the upper incisors
D A midline screw with the baseplate divided sagittally
E A Roberts retractor in 0.5 mm wire with tubing

537 Regarding springs used in removable appliances:

TRUE OR FALSE — judge each statement independently; any number may be true

A The force delivered by a cantilever spring varies inversely with the cube of its length
B Increasing the diameter of a spring from 0.5 mm to 0.7 mm roughly doubles the force it delivers at a given deflection
C Coiling a palatal finger spring increases its effective length and so reduces the force at a given activation
D A guard wire is added to a palatal finger spring to protect it from distortion and to guide its path
E Springs in removable appliances are designed to deliver of the order of 150 cN for tipping a single tooth

538 An upper removable appliance incorporates a midline expansion screw. The patient has been instructed to turn it and attends at four weeks with a 3 mm diastema and a sore palate. Which single instruction was most probably given in error?

A One quarter turn once a week
B One quarter turn twice a week
C One complete turn once a month
D One quarter turn twice a day
E The screw should be turned only by the clinician

539 An 11-year-old boy has a Class II division 1 incisor relationship with an overjet of 6 mm and a deep traumatic overbite. He is still growing. An upper removable appliance with a flat anterior bite plane is planned. Which single mechanism reduces his overbite?

A Intrusion of the lower incisors by the bite plane
B Proclination of the lower incisors against the acrylic
C Eruption of the posterior teeth while the incisors are held apart
D Intrusion of the upper incisors under the force of the labial bow
E Distal movement of the upper molars by the reaction force

540 A 9-year-old girl has a bilateral posterior crossbite with a lateral mandibular displacement of 2 mm and a deep overbite. An upper removable appliance with a midline screw is planned. Which single additional design feature is essential?

A Posterior bite planes
B A flat anterior bite plane
C A Southend clasp on the upper central incisors
D A labial bow in 0.7 mm wire
E An acrylic extension over the palatal gingivae

541 Regarding the indications for and limitations of an upper removable appliance:

TRUE OR FALSE — judge each statement independently; any number may be true

A A removable appliance can tip a tooth about a fulcrum near the middle of its root but cannot move it bodily
B An overbite that is to be reduced by a bite plane requires continuing vertical growth of the alveolus
C A removable appliance can act as a space maintainer and as a passive retainer as well as an active appliance
D Anchorage in a removable appliance is largely provided by the baseplate and the clasped teeth acting together
E Poor compliance limits removable appliance treatment in a way that it does not limit fixed appliance treatment

542 A 13-year-old girl needs additional anterior retention on an upper removable appliance because the appliance drops when she speaks. Her upper central incisors are upright with good cervical undercuts. Which single component is most appropriate?

A A buccal capping of acrylic over the incisal edges
B A labial bow in 0.5 mm wire activated by 1 mm
C A Z-spring in 0.5 mm wire extended over the palatal surfaces of both central incisors
D An Adams clasp in 0.7 mm wire on each central incisor
E A Southend clasp in 0.7 mm wire engaging the cervical undercuts of both central incisors

543 A patient's upper removable appliance fractures across the midline twice within two months. The baseplate is of conventional thickness and the fractures run through the palatal vault. Which single modification is most appropriate?

A Reduce the thickness of the baseplate so that it flexes under load rather than fracturing
B Replace the acrylic with a soft lining material
C Advise the patient to wear the appliance at night only
D Incorporate a heavy stainless steel strengthening wire across the palate within the acrylic
E Remove the anterior bite plane to reduce the load

544 A 14-year-old boy has an upper arch crowded by 7 mm with the upper canines buccally excluded and a Class I molar relationship. He has been offered an upper removable appliance with finger springs. Which single objection is strongest?

A Removable appliances cannot be used where the molar relationship is already Class I and is to be kept
B The crowding requires bodily movement and space closure, which a tipping appliance cannot achieve
C Finger springs cannot be used on canines
D Removable appliances are contraindicated above the age of 12 years
E Crowding of this severity requires rapid maxillary expansion first

545 Regarding anchorage during removable appliance treatment:

TRUE OR FALSE — judge each statement independently; any number may be true

A Anchorage demand falls as the number of teeth being moved at one time increases
B Extending the baseplate to cover the palatal vault increases the anchorage available
C Anchorage loss in a removable appliance presents as forward movement of the anchor teeth, with the appliance no longer seating correctly
D Increasing the activation of the active components is an effective way of overcoming anchorage loss
E Class II elastics can be added to a removable appliance to reinforce anchorage without any effect on the lower arch

546 A patient returns two weeks after fitting an upper removable appliance. The appliance has clearly not been worn: there is no speech adaptation, no clasp wear facets and the acrylic is unmarked. She says she has worn it every day. Which single response is most appropriate?

A Accept the account and continue with the same design
B Convert immediately to a fixed appliance without further discussion
C Add a second appliance to be worn at night
D Increase the activation of the springs so that progress is faster during the hours it is worn
E Discuss the evidence of non-wear openly, review motivation and reconsider whether to continue

547 A 12-year-old boy is to start twin block treatment for a Class II division 1 malocclusion with an 8 mm overjet. His mother asks how many hours a day the appliance should be worn. Which single instruction is correct?

A Fourteen hours a day, concentrated at night
B Full time, including during meals, removing it only for contact sports and swimming
C Ten to twelve hours a day, as for headgear
D Night-time only until the bite is comfortable, and then full time apart from meals
E Full time but removed for all meals to avoid breakage

548 A 13-year-old girl has a Class II division 1 malocclusion with a 9 mm overjet. Her lateral cephalogram shows concavities at the lower borders of C2 and C3 with C3 and C4 still trapezoid in shape. Which single statement best guides the timing of functional appliance treatment?

A Cervical staging is not applicable to girls and a hand-wrist film is required
B She is at cervical stage CS4 and the peak has passed, so a functional appliance will achieve little
C She is at cervical stage CS1 and treatment should be deferred for at least two years
D She is at cervical stage CS2, so the peak is about a year away and treatment should begin now or shortly
E The stage is irrelevant because functional appliances act dentoalveolarly at any age

549 A 12-year-old boy is six months into twin block treatment. Regarding the changes that are occurring:

TRUE OR FALSE — judge each statement independently; any number may be true

A The correction of the overjet is achieved mainly by dentoalveolar change rather than by skeletal change
B Proclination of the lower incisors is an expected and usually unwanted effect
C The upper incisors are typically proclined during treatment
D A posterior open bite commonly appears when the blocks are removed

E Trimming the upper block distally allows the lower molars to erupt and helps to close that open bite

550 An examiner asks how a twin block actually produces its effect. Which single account is most accurate?

A It postures the mandible forwards, unlocking the occlusion and allowing dentoalveolar and modest skeletal change
B It stimulates a true increase in the growth of the mandibular condyle that is fully retained into adulthood
C It applies a distal force to the maxilla equivalent to that of headgear worn for 14 hours a day
D It works by restraining the eruption of the lower incisors
E It expands the upper arch, allowing the mandible to come forward spontaneously

551 A 10-year-old boy with a Class II division 1 malocclusion, a low tongue posture and habitual lip trapping is to have a functional appliance. The clinician wishes to use an appliance that works primarily by holding the cheeks and lips away from the dentition. Which single appliance is intended?

A Bionator
B Herbst appliance
C Frankel appliance
D Twin block
E Medium opening activator

552 A 14-year-old boy has a Class II division 1 malocclusion with a 10 mm overjet. Two previous attempts at twin block treatment failed because he would not wear the appliance, and he remains keen to avoid surgery. Which single appliance is most appropriate?

A A second twin block with a wear diary
B A medium opening activator
C A Frankel appliance
D A Herbst appliance
E A removable bionator worn at night

553 Regarding the evidence from randomised trials of functional appliance treatment for Class II division 1 malocclusion:

TRUE OR FALSE — judge each statement independently; any number may be true

A Early treatment at 8 to 10 years produces a better final occlusal result than treatment started in early adolescence
B Early treatment reduces the incidence of incisor trauma, with a reported risk ratio of about 0.58
C A course of early treatment followed by a second course of adolescent treatment is more cost-effective than a single adolescent course
D Skeletal change accounts for the majority of the overjet reduction achieved
E Self-esteem gains demonstrated at the end of early treatment are maintained through to the end of the second phase

554 A 13-year-old girl has completed nine months of twin block treatment. The overjet is 2 mm, the incisors are Class I and there is a 3 mm posterior open bite bilaterally with only the second molars in contact. Which single next step is most appropriate?

A Remove the appliance and review in six months to allow the occlusion to settle spontaneously

B Trim the upper block progressively so the lower posterior teeth erupt, then use fixed appliances
C Debond and place a retainer immediately to hold the corrected overjet
D Place posterior composite build-ups to restore the occlusion
E Restart the twin block with the blocks made thicker

555 A 12-year-old boy finishing functional appliance treatment has lower incisors that were upright at the start and are now proclined by about 8 degrees, with 2 mm of lower labial segment spacing. Which single statement about this finding is most accurate?

A It is an unexpected complication indicating that the appliance was poorly constructed
B It indicates that the patient has been posturing forward voluntarily rather than wearing the appliance
C It is a desirable effect that should be maximised to reduce the overjet further
D It has no bearing on retention planning
E It is an expected side effect that contributes to overjet reduction but reduces long-term stability

556 A 13-year-old girl has finished twin block treatment with the overjet reduced from 9 mm to 2 mm. Fixed appliances are not to be started for six months. Which single retention arrangement is most appropriate?

A Part-time wear of the functional appliance, or an appliance with an anterior inclined plane
B No retention, since an overjet corrected by growth and dentoalveolar change is inherently stable
C A bonded lower canine-to-canine retainer alone
D A functional appliance worn one night a week
E Headgear worn for ten hours a day

557 Regarding the design and mode of action of functional appliances other than the twin block:

TRUE OR FALSE — judge each statement independently; any number may be true

A The Frankel appliance is largely tissue borne and its buccal shields relieve cheek pressure on the buccal segments
B The bionator is a fixed appliance cemented to the first permanent molars
C The Herbst appliance uses a telescopic arm between an upper and a lower attachment to hold the mandible forward
D The Frankel FR-3 is designed for the correction of Class III malocclusion
E The medium opening activator is constructed at a bite opening of about 15 mm to reduce bulk and improve comfort

558 A 12-year-old boy has a Class II division 1 malocclusion with a 7 mm overjet, an increased lower face height and a reduced overbite. Which single functional appliance design feature is most important in his case?

A Blocks trimmed progressively to encourage eruption of the lower posterior teeth, assumed to reduce the increased lower face height
B A flat anterior bite plane to encourage posterior eruption
C Acrylic capping of the lower posterior teeth to control eruption, with high-pull headgear
D A midline expansion screw turned twice daily
E Lip pads placed in the lower labial sulcus

559 A clinician proposes to combine a functional appliance with headgear in a growing Class II patient with a maxillary component to the discrepancy. Which single statement about the safety requirements is correct?

A Headgear safety measures are unnecessary when the headgear is attached to a functional appliance
B Two independent safety mechanisms are required whatever the appliance to which the headgear is attached
C A single safety release is sufficient provided the appliance is removable
D Safety measures are required only for cervical pull headgear
E Safety requirements apply only where the force exceeds 500 g per side

560 A systematic review reports that functional appliance treatment for Class II division 1 produces a mean increase in mandibular length of about 2 mm compared with untreated controls, with wide confidence intervals and no difference in final occlusal outcome between early and adolescent treatment. Which single conclusion is best supported?

A Functional appliances produce clinically important skeletal change and should be started as early as possible in the mixed dentition
B Mandibular length is an invalid outcome measure
C The review shows functional appliances to be ineffective and they should be abandoned
D The skeletal effect is small and the main benefits of the appliance are dentoalveolar and, in high-risk cases, reduced trauma
E The difference of 2 mm justifies treatment in every Class II child regardless of overjet

561 Regarding unwanted effects of functional appliance treatment:

TRUE OR FALSE — judge each statement independently; any number may be true

A A posterior open bite at the end of the active phase usually requires surgical correction
B Proclination of the lower incisors has no bearing on the long-term stability of the correction
C Temporomandibular joint symptoms are a recognised early complaint as the patient adapts to posturing
D Proclination of the upper incisors accompanies and contributes to the reduction in overjet
E The appliance reliably restrains anteroposterior maxillary growth to the same degree as headgear

562 A 12-year-old girl about to start twin block treatment asks what she can do to make the first fortnight easier. Which single piece of advice is most useful?

A Persist with full-time wear, expecting excess salivation and altered speech that settle within a week or two
B Wear the appliance at night only for the first month, then increase
C Remove the appliance whenever speech is difficult and replace it afterwards
D Take a non-steroidal anti-inflammatory drug continuously for the first fortnight of appliance wear, since analgesia is expected to settle the salivation and speech changes
E Wear the appliance only when at home, to avoid embarrassment at school

563 A 9-year-old boy has a Class III incisor relationship with a mild Class III skeletal pattern, an average lower face height and an ability to achieve an edge-to-edge incisor relationship. A removable functional approach is preferred over a facemask. Which single appliance is most appropriate?

A A Herbst appliance
B A twin block constructed with the mandible postured forwards
C A Frankel FR-3 appliance
D An upper removable appliance with a Z-spring

E A bionator constructed to the habitual bite

564 A department must choose between starting functional appliance treatment at 8 years and waiting until 11 years for children with Class II division 1 malocclusions and overjets of 7 to 9 mm. Which single policy is best supported by the trial evidence?

A Treat all at 8 years, because the skeletal response to the appliance is greater in younger children
B Randomise every patient, since the question remains entirely unresolved
C Treat all at 8 years, because two phases produce a better final occlusion
D Defer all treatment until growth is complete and plan surgically
E Treat at 11 years as a rule, treating earlier only where the overjet puts the incisors at risk of trauma

565 Regarding headgear used with removable and functional appliances:

TRUE OR FALSE — judge each statement independently; any number may be true

A For anchorage reinforcement, headgear is worn 10 to 12 hours a day at 250 to 350 g per side
B For distal movement of the upper buccal segments, 14 hours a day at 450 to 500 g per side is required
C A high-pull direction is preferred where the vertical dimension is already increased
D Headgear attached to an upper removable appliance can move the molars bodily as well as tipping them
E Two independent safety mechanisms are required on every headgear

566 A 27-year-old woman has a deep traumatic overbite and asks for an upper removable appliance with a flat anterior bite plane, which worked for her younger brother. Which single explanation of why this is inappropriate is best?

A Overbite reduction by a bite plane depends on posterior eruption, which needs alveolar growth
B Removable appliances cannot be constructed for adult patients
C Adult enamel is too hard for the acrylic to contact safely
D The bite plane would cause temporomandibular joint damage in an adult with an established occlusion
E Adults cannot tolerate the bulk of a bite plane

567 A patient attends with an upper removable appliance in which a palatal finger spring has been distorted and now lies against the palatal mucosa, which is inflamed. Which single action is most appropriate?

A Reactivate the spring in situ and instruct the patient to continue wearing the appliance
B Advise the patient to stop wearing the appliance permanently and refer her for fixed appliance treatment, since one distorted spring is taken to indicate the appliance has failed
C Remove the spring from service, adjust or replace it so that it lies clear of the mucosa, and reassess the guard
D Prescribe an antifungal mouthrinse and review in a month
E Thin the baseplate to relieve the pressure on the mucosa

568 A 13-year-old boy wearing an upper removable appliance full time develops a red, slightly painful palate that corresponds exactly to the outline of the baseplate. There is no ulceration. Which single management is most appropriate?

A Immediate remake of the appliance in a different acrylic
B Advise night-time wear only for the remainder of treatment
C Stop treatment altogether and refer for investigation of an allergy to residual methyl methacrylate monomer
D Relieve the baseplate over the whole palate
E Reinforce appliance and oral hygiene, advise cleaning after meals, and treat the candidal infection if it persists

569 Regarding screws and expansion in removable appliances:

TRUE OR FALSE — judge each statement independently; any number may be true

A The screw should be turned by the clinician at each visit rather than by the patient at home
B Screw-driven expansion in a removable appliance is predominantly skeletal
C A screw cannot be used to move a single tooth, because the baseplate always moves as one unit
D Screws provide better control of the direction of movement than a comparable spring
E Slow dental expansion is conventionally carried out at about 0.25 mm twice a week

570 A clinician lists cases for which he intends to use an upper removable appliance. Which single case is genuinely appropriate?

A A 15-year-old with 6 mm of upper arch crowding requiring bodily retraction of both canines
B A 9-year-old with one upper central incisor in crossbite, a favourable overbite and no crowding
C A 12-year-old requiring correction of upper incisor rotations of 40 degrees
D An adult with a 4 mm anterior open bite and a tongue thrust
E A 13-year-old requiring 4 mm of space closure after loss of an upper first molar

571 A 12-year-old girl in twin block treatment has an overjet reduced from 8 mm to 4 mm after five months, but the buccal segments remain half a unit Class II and she reports that the blocks no longer meet. Which single action is most appropriate?

A Abandon the appliance and proceed directly to fixed appliances
B Add Class II elastics between the upper and lower parts of the appliance to increase the force
C Trim the lower block to allow the upper molars to erupt
D Reactivate the appliance by advancing the lower block to restore the postured position
E Reduce wear to night time so that the blocks are stressed less

Block Q — Fixed appliances — labial, lingual and aligners (Q572–612)

Volume I: Topic 6 (Evolution of fixed appliances); Chapters 10 and 14
Volume II: **§17** — read it again if you score under 70%
This block: 41 questions (10 of them true/false) · 81 marks · allow 51 minutes

572 A 15-year-old boy has an upper right central incisor whose crown is correctly angulated and correctly positioned labiolingually, but whose root apex lies too far palatally so that the crown appears too upright. Which single order of tooth movement must be addressed?

A First order, because the labiolingual thickness of the bracket base must be altered
B Second order, because the mesiodistal angulation is incorrect
C Third order, because the labiolingual inclination of the long axis is incorrect
D First order, because the tooth is rotated about its long axis
E Second order, because the incisal edge lies in the wrong vertical position

573 Regarding the three orders of tooth movement and the way they are delivered:

TRUE OR FALSE — judge each statement independently; any number may be true

A First order describes the labiolingual position of the tooth and is expressed in the standard edgewise appliance by in-and-out bends in the archwire
B Second order describes mesiodistal angulation and is expressed in a pre-adjusted appliance by the angulation built into the bracket slot
C Third order cannot be expressed by a round archwire whatever its diameter
D In the standard edgewise appliance all three orders are built into the wire rather than into the bracket
E The pre-adjusted appliance moves the information from the wire into the bracket but does not remove the need for individual adjustment

574 An examiner asks what fundamentally distinguishes the pre-adjusted edgewise appliance from the standard edgewise appliance. Which single statement is best?

A The information needed for the finished position is built into the bracket rather than bent into the wire
B The pre-adjusted appliance uses a larger slot
C The pre-adjusted appliance uses nickel titanium archwires whereas the standard edgewise appliance does not
D The pre-adjusted appliance is bonded whereas standard edgewise is banded
E The pre-adjusted appliance requires no archwire sequence

575 A candidate is asked to state the basis on which Andrews derived his prescription. Which single account is correct?

A He recorded the positions preferred by 120 experienced orthodontists
B He surveyed 120 orthodontically treated cases and averaged their final positions
C He derived the values mathematically from the average crown dimensions of a European population
D He measured the inclinations of teeth on 120 lateral cephalograms
E He measured 120 untreated naturally ideal occlusions and described six features common to them

576 A practice is changing from a Roth prescription to an MBT prescription. Which single difference should the clinicians expect to matter most in a case requiring maximum anchorage with four premolar extractions?

A MBT prescribes greater crown angulation on the canines, which increases the anchorage demand during retraction
B MBT prescribes greater palatal root torque on the lower incisors than Roth
C MBT prescribes the same angulation as Roth but a larger slot
D MBT prescribes reduced crown angulation on the canines, which lowers the anchorage demand during retraction
E MBT abolishes third order information from the bracket altogether

577 Regarding pre-adjusted edgewise prescriptions:

TRUE OR FALSE — judge each statement independently; any number may be true

A Roth built over-correction into the prescription in anticipation of settling after appliance removal
B MBT prescribes more palatal root torque on the upper central incisors than Roth
C A prescription is a set of average values and cannot suit every patient
D The prescription values are expressed relative to the long axis of the root rather than to the facial axis of the clinical crown
E A single prescription removes the need to consider bracket position on the tooth

578 A department must choose between a 0.018 inch and a 0.022 inch slot appliance for a practice whose work is mostly extraction cases closed by sliding mechanics. Which single consideration most favours the 0.022 inch slot?

A It expresses the prescribed torque more completely at any given wire dimension, making it the deciding factor for slot selection here
B It permits a rectangular working wire rigid enough to resist deformation during sliding
C It allows smaller initial aligning wires to be used
D It reduces the number of archwire changes needed
E It eliminates the need for anchorage reinforcement

579 A 16-year-old girl has 0.022 inch slot brackets with a prescription of 17 degrees of palatal root torque on the upper central incisors. A 0.019 by 0.025 inch stainless steel wire is in place and the incisors remain proclined. Which single explanation is most likely?

A The prescription has been manufactured incorrectly
B Torque is expressed only by nickel titanium wires
C Stainless steel cannot transmit a torquing couple
D The brackets have been bonded too far gingivally to express the prescription
E Torsional play of about 10 degrees loses much of the prescribed torque

580 A manufacturer's representative states that self-ligating brackets will reduce treatment time by four months, require fewer appointments, cause less pain and expand the arches so that extractions can be avoided. Which single claim is best supported by systematic review evidence?

A Treatment time is reduced by about four months
B Appointment intervals may be lengthened and chairside time per visit is shorter
C Pain in the first week is significantly less
D Arch expansion is greater and extraction rates are lower
E Final occlusal outcomes measured by objective grading are consistently better

581 Regarding self-ligating brackets:

TRUE OR FALSE — judge each statement independently; any number may be true

A An active self-ligating bracket has a clip that intrudes into the slot and presses on the archwire
B Self-ligating brackets abolish resistance to sliding because no ligature presses on the wire
C Randomised evidence shows self-ligating appliances complete treatment significantly faster than conventional appliances
D Objective grading scores at debond are similar for self-ligating and conventionally ligated appliances
E Reduced ligation time is a real and measurable advantage of the design

582 A 25-year-old woman with ceramic brackets on the upper labial segment attends for debond. The brackets are monocrystalline alumina. Which single precaution is most important?

A Use a high-speed handpiece to section each bracket before lifting it from the tooth
B Remove the brackets with band removing pliers
C Soak the brackets in acid to weaken the bond before removal
D Use the manufacturer's debonding instrument, warning of the risk of enamel fracture
E Leave the brackets in place and polish them flat

583 A 30-year-old woman requests ceramic brackets. She has a deep overbite of 6 mm with the lower incisors contacting the palatal surfaces of the upper incisors. Which single objection is most important?

A The lower incisors will occlude on the ceramic brackets and may be abraded or the brackets fractured
B Ceramic brackets cannot be bonded to enamel
C Ceramic brackets have a lower coefficient of friction than steel and will close space too quickly
D Ceramic brackets cannot accept a rectangular archwire
E Ceramic brackets discolour within weeks and will look worse than steel

584 A trial randomises 60 patients to self-ligating or conventional brackets and reports mean treatment duration of 19.2 months against 20.1 months, a difference of 0.9 months with a confidence interval from minus 1.8 to plus 3.6 months. Which single interpretation is best?

A The self-ligating appliance is faster and the finding should change practice
B The trial proves the two systems are identical
C The difference is small and imprecise and of no clinical importance
D The confidence interval shows the result is statistically significant
E Treatment duration is not a legitimate outcome in appliance trials

585 Regarding lingual fixed appliances:

TRUE OR FALSE — judge each statement independently; any number may be true

A Lingual brackets are bonded to a flat, regular lingual enamel surface that makes direct bonding straightforward
B Speech is unaffected by lingual appliances after the first few days
C Lingual mechanics behave identically to labial mechanics because the same prescriptions are used
D The reduced interbracket span on the lingual surface makes the archwire less stiff between brackets
E Lingual appliances are indicated principally where the patient has a deep overbite and heavy occlusal contacts

586 A 34-year-old professional violinist requests treatment for mild upper labial segment crowding but will not accept visible attachments and is unwilling to wear removable appliances for 22 hours a day. Which single appliance is most appropriate?

A Labial ceramic brackets with tooth-coloured ligatures
B An upper removable appliance with finger springs
C Clear aligners
D A lingual fixed appliance
E A bonded upper retainer only

587 A patient two weeks into lingual appliance treatment reports a persistent lisp and tongue soreness. Which single response is most appropriate?

A Remove the appliance and convert to labial brackets at once
B Explain that these are recognised effects, provide wax and review, since most patients adapt
C Prescribe a topical steroid for the tongue
D Reduce the archwire size so that the appliance feels less bulky against the tongue, since a thinner wire is assumed to resolve the lisp
E Refer for speech therapy before continuing

588 A candidate is asked how a clear aligner moves a tooth. Which single account is correct?

A The aligner is manufactured passively and the tooth follows a programmed path by the shape memory of the thermoplastic, requiring no elastic deformation to move the tooth
B The aligner delivers force through the attachments alone, the shell being passive
C The aligner applies a continuous force through a nickel titanium insert
D The aligner works by occlusal loading during function
E The aligner is made to a position slightly ahead of the present one, and its elastic deformation on seating generates the force

589 A 28-year-old woman is considering clear aligner treatment for mild crowding. Regarding what aligners do well and badly:

TRUE OR FALSE — judge each statement independently; any number may be true

A Aligners are effective for mild to moderate crowding relieved by tipping and interproximal reduction
B Aligners predictably extrude an upper lateral incisor without an attachment
C Rotation of a conical premolar or canine is among the least predictable movements for aligners
D Root torque of an upper incisor is delivered less predictably by aligners than by a full-dimension rectangular wire in a fixed appliance
E Posterior intrusion caused by the thickness of the shell can produce an unwanted anterior open bite

590 A treatment plan for clear aligners includes derotation of an upper canine by 25 degrees and extrusion of an upper lateral incisor by 1.5 mm. Which single measure is most likely to improve the chance of achieving these movements?

A Place optimised composite attachments on both teeth and stage the movements more slowly
B Reduce the wear time to 14 hours a day to lengthen the force decay period
C Increase the thickness of the aligner material
D Change the aligner every three days so that the number of activations is increased
E Ask the patient to bite on chewies only at night

591 A patient wearing aligners for eight months attends with the anterior teeth well aligned but the aligner no longer seating fully over the upper first molars, with a visible gap at the distal. Which single term describes this and what should follow?

A Attachment failure, which requires the whole aligner series to be remade from new records
B Relapse, which requires immediate retention
C Loss of tracking, which requires assessment of fit and new records for refinement
D Anchorage loss, which requires a transpalatal arch
E Overexpansion, which requires interproximal reduction

592 A service compares outcomes for clear aligners and fixed appliances in comparable cases. Which single pattern of results is most consistent with the published evidence?

A Aligners and fixed appliances score equally on both peer assessment rating and objective grading
B Aligners achieve comparable peer assessment rating improvement but score worse on objective grading
C Aligners score better on objective grading because the finish is digitally planned
D Aligners score worse on peer assessment rating but better on objective grading
E Neither index can be applied to aligner cases

593 Regarding the reported performance of clear aligner treatment:

TRUE OR FALSE — judge each statement independently; any number may be true

A About 6 per cent of aligner cases are completed without any refinement series
B More than 17 per cent of cases started with aligners are converted to fixed appliances before completion
C Aligners deliver bodily movement of a canine into an extraction space as predictably as a fixed appliance
D Interproximal reduction is a routine part of aligner planning in crowded non-extraction cases
E A refinement series indicates a failure of the laboratory rather than of the biological prediction

594 A 22-year-old man asks how likely it is that his aligner treatment will be finished with the series he is given at the start, without additional aligners. Which single figure should be quoted?

A Refinement is never required if the patient wears the aligners as instructed
B About 60 per cent of patients need no refinement
C About 90 per cent of patients need no refinement
D About 6 per cent of patients need no refinement
E Refinement is required in every case by design

595 A practice audits 200 consecutive aligner cases and finds that 36 were transferred to fixed appliances before completion. Which single interpretation is most appropriate?

A The conversion rate of about 18 per cent is consistent with published figures of more than 17 per cent
B The conversion rate is unusually high and indicates a case selection problem in the practice, nearly double the published average for aligner treatment
C The conversion rate indicates that aligners should not be offered at all
D The audit is invalid because conversion is not a recognised outcome
E The conversion rate should be compared with the bond failure rate for fixed appliances

596 A 14-year-old girl finishes fixed appliance treatment with the upper right first premolar visibly higher than its neighbours and a marginal ridge discrepancy of 1 mm with the second premolar. The archwire is a full-dimension steel wire and is passive. Which single explanation is most likely?

A The bracket was bonded too far gingivally on the first premolar
B The prescription torque on the premolar was incorrect
C The bracket was bonded too far occlusally on the first premolar
D The wire was insufficiently rigid to level the marginal ridges
E The patient has a tooth size discrepancy in the buccal segment

597 Regarding bracket positioning and its consequences:

TRUE OR FALSE — judge each statement independently; any number may be true

A A bracket placed too far mesially or distally on the crown will introduce a rotation as the wire is engaged
B Indirect bonding removes the possibility of bracket positioning error altogether
C Bracket height determines the vertical position at which the tooth finishes when the archwire is flat
D An error in the angulation of the bracket on the crown alters the second order information delivered
E Errors in bracket position are corrected by changing to a heavier archwire

598 A 16-year-old boy is in the finishing stage. The upper right canine is rotated mesiolabially by about 10 degrees despite a full-dimension wire being fully engaged and passive for three months. The bracket lies slightly distal to the centre of the clinical crown. Which single action is most appropriate?

A Add a rotation wedge and continue with the same bracket
B Extract and replace with a restoration
C Change to a beta titanium wire and place a rotation bend at the canine
D Accept the position as within normal variation
E Reposition the bracket at the centre of the clinical crown and re-level

599 An examiner asks why some manufacturers place third order information in the bracket base rather than in the slot. Which single consequence of torque in the base is most important clinically?

A It alters the labiolingual position of the crown as well as its inclination
B It increases the friction during sliding mechanics
C It prevents the use of rectangular archwires
D It removes the need for first order compensation
E It makes the bracket incompatible with self-ligating clip designs, since base torque is thought to alter the clip's seating geometry

600 Regarding digital planning and indirect bonding:

TRUE OR FALSE — judge each statement independently; any number may be true

A A digital set-up allows bracket positions to be determined on the planned final tooth positions rather than on the present ones
B Transfer trays carry the brackets to the teeth in positions determined before the patient is in the chair
C Indirect bonding eliminates the need for moisture control
D A digitally planned set-up guarantees that the predicted final position will be achieved
E Indirect bonding reduces chairside bonding time compared with direct bonding of a full

arch

601 A clinician wishes to bond an entire upper arch indirectly. Which single step is most critical to the accuracy of the result?

A The choice of adhesive colour
B The use of a self-ligating bracket system
C Accurate seating of the transfer tray without rocking, with full control of moisture
D The temperature of the laboratory at the time the transfer tray is fabricated
E The use of a heavy initial archwire

602 A candidate is asked what Andrews contributed in 1972 as distinct from what he contributed later in the decade. Which single statement is correct?

A In 1972 he introduced the 0.022 inch slot
B In 1972 he introduced the pre-adjusted appliance, and later described the six keys in order to justify it
C In 1972 he described the edgewise bracket, which Angle later modified
D In 1972 he published the first randomised trial of fixed appliances
E In 1972 he described the six keys to normal occlusion, later applying them in the pre-adjusted appliance

603 A clinician moving from labial to lingual mechanics finds that retraction of the upper labial segment produces more lingual crown tipping than expected. Which single explanation is best?

A The lingual bracket lies closer to the centre of resistance, so the moment is smaller and torque control is harder
B The lingual bracket lies further from the centre of resistance, so the retraction force generates a larger tipping moment
C Lingual appliances cannot accept rectangular archwires
D The lingual archwire is longer and therefore more flexible between brackets
E Lingual brackets carry no third order prescription

604 Regarding the practical conduct of lingual appliance treatment:

TRUE OR FALSE — judge each statement independently; any number may be true

A Indirect bonding with a laboratory or digital set-up is the normal method of placing lingual brackets
B Interbracket distance is shorter in the lingual arch than in the labial arch
C Additional palatal root torque is commonly prescribed in the upper labial segment to offset the tipping tendency during retraction
D Oral hygiene is easier with a lingual appliance because the attachments are away from the lips
E Emergency management of a displaced lingual archwire is more difficult than for a labial appliance

605 A 29-year-old woman with a Class I malocclusion, 5 mm of upper crowding and 3 mm of lower crowding wants the least visible appliance and is highly motivated and reliable. Bodily movement is not required and interproximal reduction is acceptable to her. Which single appliance is most appropriate?

A An upper removable appliance with a screw
B A lingual fixed appliance
C Labial ceramic brackets
D Clear aligners with planned interproximal reduction

E Fixed appliances with extraction of four first premolars

606 A patient reports wearing her aligners 22 hours a day but the aligners appear clear and unmarked, seat loosely and the planned movements have not occurred. Which single interpretation is most appropriate?

A The aligners were manufactured incorrectly and should be remade
B The patient has an unusually rapid biological response
C The objective signs are inconsistent with the reported wear, which should be discussed first
D The aligners should be changed every five days instead of seven
E A refinement scan should be taken immediately and a new series ordered without discussion

607 A 15-year-old boy has completed alignment with fixed appliances. On the panoramic radiograph the roots of the upper right lateral incisor and canine converge markedly, though the crowns appear well aligned. Which single explanation is most likely?

A The brackets were bonded at incorrect angulations relative to the long axes of the crowns
B The archwire was too flexible to express the prescription
C The patient has dilaceration of both roots
D The prescription contains insufficient first order compensation for these teeth, which manifests as roots angled toward each other on the radiograph
E The radiograph is distorted and no conclusion can be drawn

608 Regarding the components of a labial fixed appliance:

TRUE OR FALSE — judge each statement independently; any number may be true

A A convertible molar tube allows a headgear tube to be converted into a bracket when the second molar erupts
B A single molar tube provides less rotational control of the molar than a bracket engaged by a full-dimension wire
C Auxiliary tubes on molar bands are used for headgear facebows and for utility or transpalatal auxiliaries
D Bracket width has no influence on the interbracket span or on the stiffness of the wire between attachments
E A bonded molar tube gives better resistance to occlusal load than a cemented band on a heavily restored molar

609 A colleague argues that because self-ligating brackets have lower classical friction, extraction cases will close space more quickly with them. Which single counter-argument is strongest?

A Classical friction is irrelevant because space closure does not involve sliding
B Self-ligating brackets have higher classical friction than conventionally ligated brackets
C Self-ligating brackets cannot be used with rectangular wires
D The rate of space closure is determined entirely by the force applied
E Once a tooth tips, binding and notching dominate, and reducing ligation friction changes little

610 An examiner asks what is most often lost when a case is finished with aligners rather than with fixed appliances. Which single answer is best?

A Control of the anteroposterior incisor relationship and of the overjet
B The ability to maintain arch form
C The ability to align crowded incisors

D Detailed control of marginal ridges, buccolingual inclination and occlusal contacts
E The ability to retain the finished result after the appliance is removed, since no retainer is issued once the last aligner is worn

611 A clinician working in a 0.018 inch slot wishes to explain to a trainee why he can achieve good incisor torque with a 0.017 by 0.025 inch steel wire while his colleague in a 0.022 inch slot struggles with a 0.019 by 0.025 inch wire. Which single explanation is correct?

A Steel wires of smaller dimension have a higher modulus of elasticity
B The smaller slot leaves a smaller angle of torsional play before the wire corners engage
C The smaller slot generates less friction and so transmits more of the torquing couple
D The larger slot cannot accept a rectangular wire of adequate dimension
E Torque is expressed by the bracket base and is independent of the wire

612 A candidate is asked to explain how the Begg appliance achieved rapid alignment and space closure on light forces. Which statement best describes the principle on which it worked?

A Heavy rectangular archwires were used from the start to move the teeth bodily at a controlled rate
B A preadjusted bracket slot delivered tip and torque directly from the outset, so neither auxiliary uprighting springs nor a separate third stage were needed
C A vertical slot took uprighting springs while the main archwire delivered full torque throughout treatment
D A wide twin bracket controlled rotations from the outset and light round wires did the aligning
E A ribbon arch bracket with a single point of contact let the crowns tip freely, root position being corrected later by auxiliary springs

Block R — Treatment mechanics: Stage I, II and III (Q613–647)

> **Volume I:** Chapters 10 and 13 (mechanics; definitive treatment)
> **Volume II: §18** — read it again if you score under 70%
> **This block:** 35 questions (9 of them true/false) · 71 marks · allow 44 minutes

613 A trainee asks why alignment and levelling are always completed before overjet reduction and space closure are begun. Which single justification is strongest?

A Patients tolerate retraction better once they have adapted to the appliance
B Retraction forces are transmitted predictably only along a levelled, rigid archwire
C The extraction spaces close spontaneously during alignment and need no further mechanics
D Anchorage is not required during the alignment stage
E Alignment takes longer than retraction and should therefore be done first

614 Regarding the aims of the first stage of fixed appliance treatment:

TRUE OR FALSE — judge each statement independently; any number may be true

A Levelling the curve of Spee requires no arch circumference and is therefore free of space cost
B A round aligning wire is capable of correcting the labiolingual inclination of a rotated incisor
C Anchorage is not at risk during alignment because the forces used are light
D Arch form should be individualised to the maxillary arch, which is the more stable of the two
E A heavy rectangular steel wire is the appropriate first wire where the teeth are severely displaced

615 A 14-year-old girl with upper first premolars extracted has just had brackets bonded. The canines are mesially angulated and the labial segment is crowded by 5 mm. Which single first archwire and auxiliary combination is most appropriate?

A 0.018 inch stainless steel with open coil between the canine and the premolar
B 0.019 by 0.025 inch stainless steel with elastomeric chain from molar to canine
C 0.014 inch superelastic nickel titanium with Class II elastics
D 0.014 inch superelastic nickel titanium with lacebacks from the molars to the canines
E 0.016 by 0.022 inch beta titanium with closing loops

616 An examiner asks what a laceback is intended to do. Which single account is correct?

A It ties the canine back to the molar to prevent mesial crown movement as the wire levels
B It ties the archwire back against the molar tube to prevent the wire sliding forward
C It applies an intrusive force to the labial segment
D It expands the arch across the canines
E It provides direct anchorage from the molar to the incisors for overjet reduction

617 A 15-year-old boy in the first weeks of treatment shows 2 mm of increase in overjet and visible proclination of the upper labial segment on a 0.014 inch nickel titanium wire. The wire ends protrude 3 mm distal to the molar tubes. Which single omission best explains the finding?

A The wire was too flexible for the degree of crowding
B No lacebacks were placed from molar to canine

C No bendback was placed distal to the molar tubes
D The brackets were bonded too far gingivally
E Class II elastics were not prescribed

618 Regarding lacebacks and bendbacks:

TRUE OR FALSE — judge each statement independently; any number may be true

A A bendback is placed distal to the molar tube and prevents the archwire from sliding mesially
B A laceback is conventionally made in soft stainless steel ligature wire from the molar to the canine
C Randomised evidence shows that lacebacks produce a large reduction in incisor proclination during alignment
D Lacebacks are of most value where the canine is mesially angulated and the arch is being aligned after extractions
E Bendbacks have to be checked and re-formed as the arch unravels and the wire length changes

619 A 16-year-old girl has 3 mm of lower labial segment crowding and a curve of Spee 2 mm deep, and her lower incisors must not be proclined. One millimetre of arch circumference is required for each millimetre of curve depth to be levelled. Which single figure best represents her total lower arch space requirement?

A 2 mm
B 3 mm
C 7 mm
D 5 mm
E 8 mm

620 A clinician plans the arch form for a 15-year-old boy with a Class I malocclusion and mild crowding in both arches. Which single principle should guide the choice?

A The upper arch form should be selected first and the lower built to match it
B The lower arch form should follow the patient's existing mandibular arch, which acts as the template
C A preformed ovoid arch should be used in every case for consistency
D Arch form should be expanded transversely by 3 mm in both arches in order to relieve crowding
E Arch form is irrelevant provided the teeth are aligned

621 A 17-year-old girl has a deep overbite of 6 mm with the lower incisors 3 mm above the occlusal plane. Her lower face height is average and she is no longer growing. Which single approach to levelling is most appropriate?

A Relative intrusion by extrusion of the posterior teeth using a flat anterior bite plane
B Acceptance of the overbite, since it cannot be changed after growth
C Levelling with a continuous reverse curve nickel titanium wire alone, accepting the incisor extrusion
D Proclination of the lower labial segment to reduce the overbite
E True intrusion of the lower labial segment using a segmental base arch or intrusion auxiliary

622 Regarding levelling of the arches:

TRUE OR FALSE — judge each statement independently; any number may be true

A A continuous reverse curve archwire in the lower arch tends to procline the lower incisors as well as levelling them
B Levelling by extrusion of the posterior teeth reduces the lower anterior face height
C Levelling by extrusion of the posterior teeth is a reliable method in the adult patient
D A segmental intrusion arch applies its force anterior to the centre of resistance of the labial segment and therefore tends to procline it
E Levelling the curve of Spee releases arch circumference and therefore creates space

623 A 15-year-old boy has completed alignment on a 0.018 inch nickel titanium wire in a 0.022 inch slot. Four first premolars have been extracted and maximum anchorage is required. Which single arrangement is most appropriate for the next stage?

A 0.019 by 0.025 inch stainless steel with closed coil springs and a transpalatal arch
B 0.018 inch nickel titanium with elastomeric chain from the molars to the canines
C 0.019 by 0.025 inch nickel titanium with Class II elastics
D 0.016 inch stainless steel with closing loops
E 0.021 by 0.025 inch beta titanium with elastomeric chain

624 A clinician must choose between sliding mechanics and loop mechanics for closing 7 mm of extraction space in each upper quadrant. Which single statement best distinguishes them?

A Loop mechanics are faster because the force delivered is higher
B Sliding mechanics require wire bending at every visit whereas loop mechanics require none
C Loop mechanics remove resistance to sliding but require wire bending and control of the moments
D Sliding mechanics cannot be used where anchorage is critical
E Loop mechanics deliver a constant force irrespective of activation

625 A 14-year-old girl is having 5 mm of extraction space closed by sliding mechanics with nickel titanium closed coil springs. Which single force per side is most appropriate for bodily movement?

A About 25 cN
B About 150 cN
C About 400 cN
D About 600 cN
E About 1000 cN

626 Regarding anchorage during the space closure stage:

TRUE OR FALSE — judge each statement independently; any number may be true

A If the posterior anchorage is not reinforced, roughly two-thirds of an extraction space is lost to mesial movement of the buccal segments
B Increasing the number of teeth in the anchor unit increases the resistance to unwanted movement
C Retracting the canines separately before the incisors reduces the anchorage demand at any one moment
D A transpalatal arch, a Nance button and a palatal miniscrew all reinforce anchorage but by different mechanisms
E The anchorage requirement should be determined at the planning stage, before the extraction pattern is settled

627 A 15-year-old boy has been wearing Class II elastics full time for four months. His overjet is now 2 mm but his lower incisors have proclined by 6 degrees, the occlusal plane has steepened and he has developed a 1 mm open bite between the second premolars and first molars. Which single explanation accounts for all three findings?

A The elastics have restrained maxillary growth excessively
B The brackets were bonded too far incisally in the lower labial segment at the outset
C The lower archwire was too rigid to allow the teeth to settle
D The elastics have extruded the incisors and molars and proclined the lower labial segment
E The patient has an undiagnosed skeletal Class III tendency

628 A 16-year-old girl is wearing Class III elastics from the upper molars to the lower canines to correct a residual reverse overjet. Which single side effect should be anticipated?

A Proclination of the upper labial segment, retroclination of the lower and extrusion of the upper molars
B Retroclination of the upper labial segment with proclination of the lower labial segment and no vertical change
C Intrusion of the upper molars and lower incisors
D Expansion of the lower intercanine width
E Distal movement of the lower molars with no vertical effect

629 A 14-year-old boy has four first premolars extracted for 6 mm of crowding in each arch, with no anchorage reinforcement planned and a Class I molar relationship to be maintained. Which single outcome should be anticipated?

A The molars will remain in position because the crowding will absorb the space, provided the anchorage value of the crowding equals that of the extraction space
B The space will close entirely by retraction of the labial segments
C Roughly two-thirds of each space will be lost to mesial movement of the buccal segments
D The molars will move distally as the labial segments retract
E The space will remain open at the end of treatment

630 Regarding intermaxillary elastics:

TRUE OR FALSE — judge each statement independently; any number may be true

A Class II elastics extrude the lower molars and the upper incisors
B Class II elastics are well suited to a patient with an increased lower face height and a reduced overbite
C Anterior box or vertical elastics are used at the finishing stage to settle the occlusion
D Latex elastics should be changed at least once a day because of force decay
E Unilateral Class II elastics can be used deliberately to correct a centreline discrepancy

631 A 17-year-old boy requires retraction of the upper labial segment with maximum anchorage and the clinician elects to use a closing loop in 0.017 by 0.025 inch beta titanium rather than sliding mechanics. Which single design feature is most important to preserve anchorage?

A Placing the loop closer to the incisors than to the molars
B Making the loop as small as possible to reduce its range
C Activating the loop by 3 mm at each visit
D Using an elastomeric chain across the loop to supplement the closing force, to speed up space closure without altering the moments
E Creating a greater moment at the posterior end of the loop than at the anterior end

632 A 15-year-old girl is midway through space closure after four first premolar extractions. The teeth adjacent to each extraction site have tipped into the space, the lower curve of Spee has deepened and there are 1 mm open bites in both premolar regions. Which single term describes this and what is the appropriate response?

A The rollercoaster effect, requiring a return to a rigid levelling archwire
B Anchorage loss, requiring extraoral traction
C Relapse, requiring immediate retention
D Failure of the elastomeric chain, requiring stronger elastics and shorter intervals
E Occlusal interference, requiring selective grinding

633 A 16-year-old girl wearing Class II elastics for five months has developed a 2 mm anterior open bite. Her lower face height was already increased at the start. Which single course of action is best supported?

A Continue the elastics and accept the open bite as a transient finding
B Increase the elastic force to close the bite more quickly
C Stop the elastics and change to a mechanic that does not extrude the posterior teeth
D Add anterior vertical elastics while continuing with the Class II elastics as before
E Debond and retain, accepting the result

634 A space closure has stalled: 3 mm of extraction space remains unchanged over three months on a 0.019 by 0.025 inch steel wire with elastomeric chain. Regarding the assessment:

TRUE OR FALSE — judge each statement independently; any number may be true

A The wire should be checked for notching or permanent deformation at the bracket corners
B An elastomeric chain replaced every eight weeks will have delivered a near-constant force throughout
C Gingival tissue in the extraction site cannot impede space closure and need not be examined
D A previously extracted site more than a year old is no more likely to be sclerosed than a recent one
E Failure of a canine to move should prompt no radiographic assessment

635 A 15-year-old boy has an upper centreline 2 mm to the right of the facial midline at the end of space closure. The lower centreline is coincident with the face, the buccal segments are Class I on the left and half a unit Class II on the right, and 2 mm of space remains in the upper right quadrant. Which single management is most appropriate?

A Expand the upper arch to move the centreline
B Close the remaining space by protracting the upper right buccal segment forwards, which is assumed to correct the centreline as a side effect
C Accept the centreline, since 2 mm is within normal variation
D Extract a lower incisor to bring the lower centreline across
E Retract the upper right labial segment into the residual space with asymmetric mechanics

636 A 17-year-old girl in the finishing stage has upper incisors that appear well aligned but insufficiently torqued, with the crowns upright and the roots palatal. She is in 0.022 inch slot brackets with a 0.019 by 0.025 inch steel wire. Which single action is most appropriate?

A Rebond the brackets more incisally
B Place additional palatal root torque in the archwire or change to a wire of larger dimension
C Change to a 0.016 inch round steel wire and use elastics
D Accept the position, since the prescription torque in the bracket cannot be supplemented
E Extract and restore the incisors prosthetically

637 A 16-year-old boy is ready for debond. The panoramic radiograph shows the roots of the upper right canine and second premolar converging over the healed first premolar extraction site. Which single action is most appropriate?

A Debond and retain, since crown alignment is satisfactory
B Extract the second premolar
C Accept the finding as an artefact of panoramic projection and proceed to debond
D Place uprighting bends or reposition the brackets until the roots are parallel
E Place a bonded retainer across the site and debond

638 Regarding the detailing and finishing stage:

TRUE OR FALSE — judge each statement independently; any number may be true

A Settling elastics are worn on flexible or sectional archwires so that the teeth can move vertically into intercuspation
B Root parallelism is assessed on a panoramic radiograph taken towards the end of treatment
C Torque expression is complete as soon as a rectangular wire is placed, so no further adjustment is possible
D Finishing bends are most easily placed in a wire of high formability such as beta titanium
E Interproximal contacts should be checked and reopened where they have been closed by rotation correction

639 An examiner asks which criteria the American Board objective grading system applies to a finished case. Which single list is correct?

A Peer assessment rating weights applied to the final study models with a subjective assessment of the smile
B Alignment, marginal ridges, arch form, overjet, overbite, centrelines, interproximal contacts and the gingival contour of the finished case, the list often confused with the true criteria
C Alignment, marginal ridges, buccolingual inclination, occlusal relationship, occlusal contacts, overjet, interproximal contacts and root angulation
D Overjet, overbite, crowding, centreline and the aesthetic and dental health components combined
E Incisor irregularity, curve of Spee, overjet, overbite and buccal segment relationship on models

640 A department audits finished cases using the peer assessment rating and finds a mean reduction of 26 points and a mean percentage reduction of 78 per cent. Which single conclusion is correct?

A The cases fall into the greatly improved category on both the absolute and the percentage criterion
B The cases fall below the accepted standard on the percentage criterion but not the absolute one
C The peer assessment rating cannot be expressed as a percentage
D A reduction of 26 points indicates no improvement
E The audit shows the detailing of the finish to be of a high standard

641 Regarding the peer assessment rating and the objective grading system as measures of outcome:

TRUE OR FALSE — judge each statement independently; any number may be true

A A peer assessment rating reduction of at least 22 points or at least 30 per cent is classified as greatly improved

B The objective grading system examines features of the finish that the peer assessment rating does not weight heavily
C A case can achieve a large peer assessment rating reduction and still score poorly on objective grading
D The peer assessment rating includes an assessment of root angulation
E Both indices are applied before treatment to determine need

642 A 24-year-old woman is approaching debond. Her occlusion is Class I with good intercuspation, but on smiling the upper incisal edges run flat against a curved lower lip and there is a wide dark space between the corners of her mouth and the upper premolars. Which single finding should be recorded?

A A reverse smile arc requiring surgical correction
B An excessive incisor display at rest
C A cant of the maxillary occlusal plane
D A gingival smile requiring crown lengthening
E A flat smile arc with excessive buccal corridors

643 A clinician wishes to place a small step to correct a residual 1 mm discrepancy between two marginal ridges at the finishing stage without disturbing the rest of the arch. Which single approach is most appropriate?

A A first order offset bend in the archwire
B A change to a round wire with an elastomeric chain across the segment
C A third order torquing bend placed across the whole of the upper arch
D A second order step bend in a formable wire, or repositioning of the bracket
E A change to a heavier nickel titanium wire

644 A 15-year-old boy has been in fixed appliances for 28 months. The occlusion is Class I with a 2 mm overjet and 2 mm overbite, but two premolar marginal ridges differ by 1 mm, a lower incisor remains rotated by about 10 degrees and his oral hygiene has deteriorated with early demineralisation on four upper teeth. Which single decision is best?

A Continue for a further six months until the detailing is perfect
B Debond now and retain, since the risk of further enamel damage outweighs the remaining detail
C Debond the upper arch only and continue in the lower
D Continue but reduce appointments to three-monthly
E Refer for restorative management of the demineralisation before the appliances are debonded

645 Regarding the rationale for the conventional sequence of treatment stages:

TRUE OR FALSE — judge each statement independently; any number may be true

A Overjet reduction is attempted before the arches are levelled so that the anchorage demand is spread more evenly
B Alignment is completed first because a flexible wire can deliver the light forces that displaced teeth require
C Space closure precedes levelling because closed spaces make levelling easier
D Detailing is left until last because it requires a formable wire and a settled occlusion
E Attempting all three stages simultaneously reduces total treatment time without penalty

646 A colleague proposes to begin Class II elastics at the first appointment, on a 0.014 inch nickel titanium wire, in a patient with 8 mm of upper crowding and an 8 mm overjet. Which single objection is strongest?

A Class II elastics are contraindicated in crowded arches
B Elastics at this stage would cause root resorption
C Class II elastics cannot be worn at the same time as nickel titanium aligning wires, since the elastic force is thought to interfere with wire memory
D The overjet is too large for elastics to have any effect
E The flexible wire cannot resist the moments the elastics generate and the segments will tip

647 A 16-year-old girl has completed space closure and the occlusion is Class I, but the buccal segments intercuspate poorly, with visible spaces between opposing cusps on both sides. She is in full-dimension rectangular wires in both arches. Which single next step is most appropriate?

A Debond and allow the occlusion to settle naturally under a removable retainer
B Change to light flexible finishing wires and prescribe vertical settling elastics
C Increase the archwire dimension to seat the teeth more firmly
D Reduce the opposing cusps selectively to remove the interferences
E Place Class II elastics on both sides for a further two months

Block S — Retention, relapse and stability (Q648–682)

Volume I: Topic 20 (Late lower incisor crowding); Chapter 13, pp. 975–992
Volume II: §19 — read it again if you score under 70%
This block: 35 questions (9 of them true/false) · 71 marks · allow 44 minutes

648 A 14-year-old girl has finished fixed appliance treatment. The upper right lateral incisor was rotated by about 70 degrees at the outset and is now aligned, the periodontium is healthy and the overbite is 3 mm. Which single measure is most likely to reduce relapse of that rotation?

A Extend the fixed appliance by six further months so that the fibres reorganise on the wire
B Fit a Hawley retainer worn at night only for six months to hold the labial segment
C Overcorrect the rotation by 20 degrees and debond without fitting any retainer at all
D Divide the supracrestal fibres of that tooth by circumferential fiberotomy before debond
E Reduce the tooth interproximally on both sides so that the contact points are broadened

649 A patient who has just been debonded asks why teeth tend to move back. Regarding the periodontal and gingival tissues after active tooth movement:

TRUE OR FALSE — judge each statement independently; any number may be true

A The principal periodontal ligament fibres reorganise within three to four months of debond
B The supracrestal and transseptal fibre networks may take more than a year to remodel
C Rotational relapse tends to be greatest in the first few months after the appliance is removed
D Dividing the supracrestal fibres removes the need for any subsequent retention
E Alveolar bone around a moved tooth remodels more quickly than the gingival fibres do

650 A 15-year-old boy had a 7 mm overbite reduced to 3 mm with fixed appliances. At debond the lower incisor edges contact the cingulum plateau of the upper incisors in centric occlusion. Which single feature best predicts that the overbite reduction will hold?

A The lower incisor edges meet the upper cingulum plateau, giving a positive occlusal stop
B The upper incisors were intruded rather than the lower incisors being proclined forwards
C The interincisal angle was opened beyond 140 degrees during the finishing stage
D A vacuum-formed retainer covering every tooth will be worn full time for the first year
E The lower intercanine width was expanded to accommodate the aligned labial segment

651 A 16-year-old has had four first premolars extracted and the spaces fully closed with fixed appliances. The clinician wants the retainer that most reliably holds the closed extraction sites in the first six months. Which single choice is most appropriate?

A A Hawley retainer with an adjustable labial bow, which allows the buccal segments to settle
B A bonded canine-to-canine retainer, which controls the labial segment but not the sites
C A vacuum-formed retainer covering all teeth, which resists reopening of the closed sites
D A night-only removable retainer from the outset, which reduces the burden of wear
E No retainer at all, since closed extraction sites are stable once the contacts are tight

652 A 24-year-old woman has worn a bonded canine-to-canine lower retainer for five years. She reports that one canine has twisted. The retainer is intact at every pad, the canine crown sits lingually with the root displaced labially, and the remaining teeth are unchanged. Which single explanation is most likely?

A Partial debond of one composite pad has left the tooth free to drift while the wire looks sound

B Late mandibular growth has rotated the whole labial segment including that canine
C Occlusal interference from the opposing canine has tipped the tooth out of the arch
D An unintended twist in the wire is actively torquing that tooth, the so-called wire syndrome
E Localised attachment loss has allowed the tooth to drift under resting lip pressure

653 A general practitioner reviews a patient with a bonded lower retainer placed three years earlier. Regarding bonded retainers:

TRUE OR FALSE — judge each statement independently; any number may be true

A Bonding removes any need to discuss a long-term commitment with the patient
B A wire bonded only to the canines holds each of the four incisors individually
C Failure is often unnoticed by the patient and is detected at a review appointment
D Calculus accumulates on the lingual wire and is associated with localised gingival inflammation
E A bonded retainer guarantees stability of the lower labial segment for life

654 A follow-up study finds that five years after debond fewer than one in five patients are still wearing their removable retainers as instructed, although most still possess them. Which single conclusion should most influence the way retention is planned?

A Bonded retainers should replace removable ones in every case, removing any reliance on wear
B Patients should be told before treatment that alignment lasts only as long as retainers are worn
C Retention should be stopped at two years, since sustained wear beyond that is unrealistic
D Retainers are ineffective in the long term, so the emphasis should be on stable finishing
E Retainers should be issued only to patients judged at the outset to be likely to comply

655 A 34-year-old man who had fixed appliances at 14 attends with 4 mm of lower labial segment crowding. He wore his retainers for two years only. His third molars are unerupted and he has been told they are pushing his teeth forward. Which single explanation is best supported?

A The third molars are generating a mesial force that has crowded the labial segment
B The lower intercanine width has increased with age and rotated the incisors lingually
C The crowding is direct relapse of the original malocclusion and would not have occurred untreated
D Loss of the retainers allowed the supracrestal fibres to recoil after twenty years
E This is a physiological age change seen in untreated dentitions as well as treated ones

656 A 30-year-old asks whether her lower front teeth will crowd as she ages. Regarding late lower incisor crowding:

TRUE OR FALSE — judge each statement independently; any number may be true

A It occurs in dentitions that have never received orthodontic treatment
B Removal of the third molars does not reliably prevent it
C The change continues into the third and fourth decades of life
D Indefinite bonded or removable retention is the only reliable way of preventing it
E It is better described as a feature of dental ageing than as relapse of treatment

657 An 11-year-old girl has had a bilateral posterior crossbite corrected by rapid maxillary expansion, with the screw turned once daily for three weeks. The appliance is now passive and the buccal segments interdigitate. Which single retention plan is most appropriate?

A Leave the appliance in place passively for three to six months before removal
B Remove the appliance now and fit a vacuum-formed retainer for night-time wear only
C Remove the appliance now, as a corrected buccal interdigitation is self-retaining
D Leave the appliance for two further weeks and then fit a removable expansion plate
E Remove the appliance and place a transpalatal arch adjusted to maintain the width

658 A 19-year-old man has had a 4 mm midline diastema closed with fixed appliances. The upper labial frenum has a low fibrous attachment that blanches the incisive papilla on traction. Which single retention plan is most appropriate?

A A vacuum-formed retainer worn at night for twelve months and then discontinued
B A Hawley retainer with a labial bow tightened at each review for two years
C A bonded retainer across the upper labial segment maintained indefinitely
D No retainer, since the contact point between the central incisors is now tight
E A removable retainer with an anterior bite plane to control the overbite

659 Two clinicians disagree about how to record post-retention change in the lower labial segment. One proposes Little's Irregularity Index, the other a simple statement of whether crowding is present. Which single appraisal of the index is best supported?

A It measures rotation and labiolingual inclination as well as displacement, so it is comprehensive
B It converts contact point displacement into a single reproducible millimetre score for comparison
C It requires a standardised radiograph and so cannot be applied to study models or scans
D It is scored from zero to ten and expresses crowding as a proportion of arch length
E It cannot distinguish an aligned segment from a crowded one without a baseline record

660 Regarding the measurement of alignment and relapse in the lower labial segment:

TRUE OR FALSE — judge each statement independently; any number may be true

A The Irregularity Index is the sum of the five anterior contact point displacements measured perpendicular to the occlusal plane
B A score of zero denotes an ideally aligned mandibular labial segment
C The index records rotations and labiolingual inclinations in addition to displacement
D It is measured with calipers between adjacent anatomical contact points on models or scans
E It is widely used to express post-retention change in long-term follow-up studies

661 A 13-year-old boy with a Class III incisor relationship and a mild Class III skeletal pattern has had his reverse overjet corrected to a 2 mm positive overjet with an upper fixed appliance. He is at cervical stage CS3. Which single statement best guides retention?

A Retention can be discontinued after twelve months because the incisors are now in a positive overjet
B Class III corrections are inherently stable once a positive overbite and overjet are achieved
C A bonded upper retainer will prevent any further mandibular growth from affecting the result
D Continued mandibular growth remains the principal threat and retention must extend beyond the growth period
E Relapse after Class III correction is driven by the supracrestal fibres rather than by growth

662 A clinician is planning treatment for a 15-year-old who will need a bonded upper retainer indefinitely after diastema closure. Which single approach to consent is most appropriate?

A Explain the need for indefinite retention and its costs and risks before treatment begins
B Explain the need for indefinite retention at the debond appointment when the retainer is fitted
C Record the plan in the notes and raise retention only if the patient asks about it
D Obtain consent for the appliance now and for the retainer at the first review after debond
E Defer the discussion until relapse occurs, when the need will be evident to the patient

663 A 17-year-old has finished fixed appliance treatment. The buccal segments show slightly open contacts where the archwire has held the teeth, and the clinician wants them to settle while the labial segments are held. Which single retainer is most appropriate?

A A vacuum-formed retainer covering all teeth, which holds every crown in its finished position
B A bonded canine-to-canine retainer in both arches, with no removable component at all
C A Hawley retainer with a labial bow, which leaves the buccal occlusal surfaces free to settle
D A vacuum-formed retainer trimmed clear of the incisal edges to allow anterior settling
E A soft thermoplastic positioner worn for two hours a day and overnight for three months

664 A patient is offered a choice between a Hawley retainer and a vacuum-formed retainer after debond. Regarding the comparison:

TRUE OR FALSE — judge each statement independently; any number may be true

A A vacuum-formed retainer allows the buccal segments to settle more readily than a Hawley
B A Hawley holds closed extraction spaces more reliably than a vacuum-formed retainer
C Vacuum-formed retainers have been shown to survive longer in service than Hawley retainers
D A Hawley is contraindicated where any tooth has been rotated by more than 30 degrees
E Two years of full-time vacuum-formed wear is supported by trial evidence as better than night-only wear

665 A 21-year-old woman attends with generalised bleeding on probing of the lower incisors. A bonded lower retainer placed two years ago is intact, with visible calculus along its length and no pocketing beyond 3 mm. Which single management is most appropriate?

A Remove the retainer permanently and accept the risk of lower labial segment relapse
B Prescribe a chlorhexidine mouthrinse for three months and review the gingival condition
C Replace the retainer with a night-only vacuum-formed retainer worn indefinitely
D Section the retainer between the canines to allow the incisors to be cleaned separately
E Debride the retainer and teeth and reinforce interdental cleaning, keeping the retainer

666 A 16-year-old girl had 3 mm of lower crowding relieved by proclining the lower incisors from 90 to 99 degrees to the mandibular plane. Her lower lip rests over the incisal third of the upper incisors at rest. Which single statement best predicts the stability of the result?

A The lower incisors have been moved out of the soft tissue balance and will tend to relapse
B The result is stable because the crowding was relieved without extractions being needed
C Stability depends on the intercanine width, which has not been changed by proclination
D Lip pressure acts only on the upper incisors, so lower proclination is unaffected by it
E Proclination is stable provided the interincisal angle remains greater than 130 degrees

667 A 13-year-old boy has completed twin block therapy that reduced a 9 mm overjet to 2 mm, and is about to begin fixed appliances. His cervical stage is CS3. Which single measure best protects the corrected sagittal relationship during the fixed phase?

A Use Class II elastics continuously throughout the fixed phase to hold the correction
B Fit a bonded upper canine-to-canine retainer before the fixed appliances are placed
C Continue night-time wear of the functional appliance or an equivalent during fixed treatment
D Reduce the lower arch by extracting two lower premolars to prevent forward relapse
E Accept some relapse of overjet and correct it at the finishing stage of fixed treatment

668 A department is revising its retention protocol. Regarding retention regimens and their duration:

TRUE OR FALSE — judge each statement independently; any number may be true

A Most contemporary protocols move to night-only removable wear within the first year
B Trial evidence shows six months of full-time vacuum-formed wear gives a better result at two years than night-only wear from the start
C Patients should be told at the outset that retention is indefinite if alignment is to be kept
D Reported retainer wear falls substantially by four to five years after debond
E A review appointment is required to detect silent failure of a bonded retainer

669 A 28-year-old woman with a unilateral posterior crossbite and no mandibular displacement has had the crossbite corrected by dental expansion with fixed appliances. The corrected buccal segments meet cusp to cusp. Which single statement best predicts stability?

A The correction is stable because the midpalatal suture has been opened at this age
B Relapse after expansion occurs only where a mandibular displacement was present
C Dental expansion in adults is more stable than skeletal expansion in children
D Stability depends on the intercanine width rather than on the buccal interdigitation
E The cusp-to-cusp relationship provides no interdigitation and relapse is likely

670 A post-retention study reports that ten years after debond the mean Irregularity Index of the lower labial segment has risen from 0.5 mm to 3.1 mm, and that only a third of cases remain clinically well aligned. Which single conclusion is best supported?

A Fixed appliance treatment of lower crowding should generally be avoided in adolescents
B Long-term alignment of the lower labial segment requires retention that does not stop
C The study shows that the original treatment was technically inadequate in most cases
D An Irregularity Index of 3.1 mm represents an acceptable and clinically stable outcome
E Relapse of this magnitude is confined to cases treated without premolar extractions

671 A 14-year-old girl with an anterior open bite and a tongue-to-lip resting posture has had the open bite closed with fixed appliances and a posterior bite block. The lips are now competent at rest but the tongue still contacts the lower lip on swallowing. Which single factor most threatens the result?

A The supracrestal fibres, which recoil vertically and reopen an anterior open bite
B The absence of a bonded retainer in the upper labial segment after debond
C The residual vertical growth of the maxilla, which is complete by this age in girls
D The persisting soft tissue resting behaviour, which will tend to reopen the anterior bite
E The intercanine width, which was increased during alignment of the arches

672 Regarding the mechanisms that cause orthodontic relapse:

TRUE OR FALSE — judge each statement independently; any number may be true

A Continued mandibular growth in a late-growing male can crowd a corrected lower labial segment
B Teeth placed outside the soft tissue envelope of equilibrium are as stable as those within it
C Relapse after rotation correction is driven mainly by bone that has failed to remodel
D Proclining the lower incisors beyond their pre-treatment position predisposes to relapse
E An unfavourable interincisal angle after overbite reduction predisposes the bite to deepen

673 A 17-year-old boy debonded six months ago has been wearing vacuum-formed retainers at night. He attends with a retainer that no longer seats over the upper left canine and reports two weeks of not wearing it. Which single management is most appropriate?

A Reassure him and ask him to force the retainer into place each night until it seats
B Rebond fixed appliances to the upper arch and realign before making a new retainer
C Take a new impression or scan and make a fresh retainer to the current tooth position
D Advise that the tooth has settled into its natural position and discontinue retention
E Trim the retainer clear of the canine and continue wearing it over the other teeth

674 A trial randomises 400 patients at debond to a bonded lower retainer or a night-only vacuum-formed retainer and reports failure requiring intervention in 30 per cent of the bonded group and 25 per cent of the removable group at 18 months. Which single conclusion is best supported?

A Both approaches need scheduled review, since roughly a quarter to a third need attention
B Bonded retainers should not be used, as they fail more often than removable retainers
C Removable retainers are significantly superior and should be the default after debond
D Failure rates of this magnitude show that retention is not worthwhile after 18 months
E The trial is invalid because failure means different things in the two groups

675 Regarding the stability of particular tooth movements after treatment:

TRUE OR FALSE — judge each statement independently; any number may be true

A Expansion of a posterior crossbite is more stable when a positive buccal interdigitation results
B Closure of a median diastema is prone to relapse and commonly requires bonded retention
C Corrected rotations relapse less readily than intruded incisors do
D Expanding the lower intercanine width beyond its original value is unstable
E Class II correction in a patient with a persistent lip trap is stable once the overjet is reduced

676 A 15-year-old girl has had the upper central incisors derotated by 30 degrees each. Their mesiodistal crowns are broad and the contact points are small and rounded. Which single adjunct is most likely to improve the stability of the derotation?

A Reduction of the incisal edges to equalise the crown heights of the two central incisors
B Overcorrection of each rotation by a further 30 degrees before the appliance is removed
C Bonding attachments to the incisors for a further three months on a rectangular wire
D Proclination of both incisors by 5 degrees to increase the interincisal contact area
E Interproximal reduction to flatten and broaden the contact points between the incisors

677 A 52-year-old man with treated chronic periodontitis has had pathologically migrated upper incisors realigned. Bone levels are reduced to the middle third of the roots but are stable, and the teeth have grade one mobility. Which single retention plan is most appropriate?

A A night-only vacuum-formed retainer for two years followed by a gradual reduction in wear
B No retention, since the periodontal treatment has removed the cause of the migration
C A Hawley retainer worn full time for six months and then discontinued altogether
D A bonded retainer splinting the upper labial segment indefinitely, with maintenance review
E Extraction of the mobile incisors and replacement with an immediate bonded bridge

678 A clinician proposes reducing all patients to night-only retainer wear from the day of debond, citing patient preference. Which single appraisal of this policy is best supported by current evidence?

A It is unsafe, since full-time wear for at least twelve months is required in every case
B It is reasonable, since trials have not shown full-time wear to give better alignment
C It is unsafe, since night-only wear does not hold closed extraction spaces at any stage
D It is reasonable only where a bonded retainer has also been placed in both arches
E It cannot be judged, since no randomised comparison of wear regimens has been performed

679 Regarding patient factors and the success of retention:

TRUE OR FALSE — judge each statement independently; any number may be true

A Non-compliance with removable retainers becomes uncommon after the first year
B A patient with a bonded retainer needs no further review once the wire has been placed
C Consent for retention is best taken at debond rather than before treatment begins
D A fee charged for replacement retainers has no bearing on continued wear
E A patient who understands that alignment depends on wear is more likely to keep wearing the retainer

680 A 19-year-old man asks whether removing his unerupted lower third molars will protect the alignment of his lower incisors after debond. Which single reply is best supported by the evidence?

A Removal is justified, since third molars generate the mesial force that crowds the incisors
B Removal is justified because it allows the lower arch to be distalised during retention
C Removal is justified only where the third molars are mesioangularly impacted against the sevens
D Removal should be deferred until crowding appears and then performed without delay
E Removal is not justified for this purpose, since it does not reliably prevent late crowding

681 A 20-year-old woman attends two years after debond. Her bonded lower retainer has detached from one canine pad without her noticing, and that canine has rotated by about 15 degrees. Which single management is most appropriate?

A Rebond the existing wire to the canine in its present rotated position and review, expecting no further relapse
B Remove the wire, realign the canine with a short course of appliance therapy, then re-retain
C Rebond the wire and accept the rotation, warning the patient that it may worsen
D Fit a vacuum-formed retainer over the rotated canine to hold it where it now sits
E Refer for extraction of the canine and prosthetic replacement given the rotation

682 A 22-year-old woman debonded four years ago has worn her night-time retainers throughout and now asks when she may stop. Her alignment is unchanged and the periodontium is healthy. Which single reply is most appropriate?

A She may stop now, since four years of retention has allowed the tissues to reorganise fully
B She should switch to a bonded retainer, which removes the need for any further review
C She should stop and return only if she notices the teeth beginning to crowd again
D She should continue indefinitely, reducing to one or two nights a week only if alignment holds
E She should continue full-time wear for two more years and then stop retention altogether

Block T — Interceptive orthodontics and the developing dentition (Q683–727)

Volume I: Topics 13, 14 and 15 (missing central; impacted canine; impacted 7); Chapter 12
Volume II: **§20** — read it again if you score under 70%
This block: 45 questions (11 of them true/false) · 89 marks · allow 56 minutes

683 A 10-year-old boy attends for a routine examination. The upper permanent canines cannot be palpated in the buccal sulci, the deciduous canines are firm and the lateral incisors are slightly proclined and mobile. Which single next step is most appropriate?

A Take radiographs to localise the canines, since absent palpation at this age is abnormal
B Reassure the parents and review in two years when the canines would normally erupt
C Extract both deciduous canines immediately without any radiographic assessment first
D Refer directly for surgical exposure of both canines on the clinical findings alone
E Fit an upper removable appliance to create space in the canine region and then review

684 Two periapical radiographs of an unerupted upper canine are taken, the second with the tube shifted distally. Between the two films the crown of the canine appears to move distally relative to the lateral incisor root. Which single interpretation is correct?

A The canine lies in the line of the arch, since movement indicates no displacement at all
B The canine lies buccally, because an object moving with the tube shift lies lingual to the film
C The canine lies palatally, because an object that moves with the tube lies further from the film
D The canine lies palatally, because objects closer to the film move against the tube shift
E No conclusion is possible from two periapicals and a cone beam scan is required

685 A 10-year-old girl is found to have an unerupted palatally displaced upper canine. Regarding maxillary canine impaction:

TRUE OR FALSE — judge each statement independently; any number may be true

A Impaction affects about 10 per cent of the population
B Up to 85 per cent of impacted canines lie palatally
C Palpation of the buccal sulcus should begin at about nine to ten years of age
D Impaction is more common in boys than in girls
E Absence of the upper lateral incisor is associated with palatal canine displacement

686 An 11-year-old girl has a palatally displaced upper right canine. It lies with its crown medial to the long axis of the lateral incisor, at the level of the middle third of the incisor root, and angled 25 degrees to the midsagittal plane. There is no crowding. Which single management is most appropriate?

A Leave both canines and review in three years when growth has finished
B Expose the permanent canine surgically and bond a gold chain for traction now
C Extract the permanent canine and close the space with fixed appliances
D Extract the deciduous canine and review radiographically in twelve months
E Fit a removable appliance with a palatal finger spring to push the canine buccally

687 A 12-year-old boy has a palatally displaced upper left canine. Which single radiographic feature is most strongly associated with a poor prospect of spontaneous improvement after extraction of the deciduous canine?

A A crown lying at the level of the cervical third of the lateral incisor root
B An angulation of the canine of more than 30 degrees to the midsagittal plane
C A crown overlapping less than half the width of the lateral incisor root
D An open apex with root formation approximately two thirds complete
E A follicular space of normal width around the crown of the canine

688 A clinician is advised that extraction of the deciduous canine in a ten-year-old improves the position of a palatally displaced permanent canine in a majority of cases. Which single qualification is most important when applying this to practice?

A The benefit has not been demonstrated in any controlled study of interceptive extraction
B The benefit applies to buccally displaced canines rather than to palatal displacements
C The benefit is confined to children under eight years, in whom the follicle is still active
D The benefit disappears if the contralateral deciduous canine is retained in the arch
E The benefit applies only where the arch is uncrowded and the canine is not severely angulated

689 A 13-year-old has a palatally impacted canine that has not improved after removal of the deciduous canine. Regarding the options now:

TRUE OR FALSE — judge each statement independently; any number may be true

A Surgical exposure with bonded traction is the usual treatment where the canine is to be aligned
B Adequate space must be created in the arch before traction is applied
C Extraction of the canine may be appropriate where it is grossly displaced and the arch is otherwise acceptable
D Autotransplantation is an option where there is insufficient space or time for orthodontic alignment
E Resorption of the adjacent incisor root may influence which tooth is removed

690 A cone beam scan of a 12-year-old with a palatally displaced canine shows resorption of the distopalatal surface of the upper left lateral incisor root amounting to about one third of its length. The canine crown lies over the middle of the incisor root. Which single management is most appropriate?

A Expose the canine and apply traction away from the lateral incisor without delay
B Monitor with annual radiographs until the canine erupts spontaneously into the arch
C Extract the lateral incisor and bring the canine into its place with fixed appliances
D Extract the deciduous canine and review the position in a further twelve months
E Root treat the lateral incisor and leave the canine unerupted under observation

691 An 11-year-old girl requires surgical management of a palatally displaced canine that lies high and close to the midline, with adequate space available in the arch. Which single surgical approach is most appropriate?

A Open exposure with a palatal window left to granulate and spontaneous eruption awaited
B Removal of the overlying bone with no attachment placed, followed by review in six months
C Closed exposure with a gold chain bonded at operation and traction applied orthodontically
D Apically repositioned flap to bring attached gingiva with the tooth as it erupts
E Transalveolar transplantation of the canine into a surgically prepared socket

692 A 15-year-old boy has a grossly displaced upper canine lying horizontally above the apices of the incisors. The arch is well aligned with adequate space and he is unwilling to accept two years of fixed appliances. Which single option is most appropriate to discuss first?

A Surgical exposure with bonded traction over an estimated two to three years
B Autotransplantation of the canine into a surgically prepared socket in the arch
C Leaving the canine in situ with long-term radiographic surveillance for cyst formation
D Extraction of the canine with restorative replacement of the resulting space
E Extraction of the first premolar to create a path for spontaneous canine eruption

693 A 7-year-old boy has not erupted his upper left central incisor although the right erupted a year ago. A supernumerary is suspected. Regarding supernumerary teeth:

TRUE OR FALSE — judge each statement independently; any number may be true

A A mesiodens is the least common supernumerary in the permanent dentition
B Supernumeraries are more common in the mandible than in the maxilla
C A conical supernumerary may erupt, whereas a tuberculate one usually does not
D Supernumerary teeth are more common in girls than in boys
E Removal of the supernumerary always requires simultaneous exposure of the incisor

694 An 8-year-old girl has an unerupted upper right central incisor. A tuberculate supernumerary lies palatal to it, the incisor is vertical with two thirds of its root formed, and there is 8 mm of space in the arch. Which single management is most appropriate?

A Remove the supernumerary and the incisor, planning prosthetic replacement in adult life
B Remove the supernumerary and expose the incisor with bonded traction at the same visit
C Leave the supernumerary and open space with a removable appliance before reviewing
D Remove the supernumerary and maintain the space, reviewing eruption over twelve months
E Review for a further two years before intervening, as many such incisors erupt eventually

695 A 7-year-old boy has an upper first permanent molar that has failed to erupt fully, with its mesial marginal ridge caught beneath the distal surface of the second deciduous molar. Radiographs show minor resorption of the distal root of the deciduous tooth. Which single management should be attempted first?

A Place a separating elastomeric module or brass wire between the two teeth and review
B Extract the second deciduous molar and allow the first molar to erupt into the space
C Fit a removable appliance with a palatal finger spring to push the molar distally
D Fit a fixed distalising appliance anchored on the palate to disimpact the molar
E Review in twelve months, as most such impactions resolve without any intervention

696 A 9-year-old girl has an upper first permanent molar locked beneath the second deciduous molar, whose distal root is extensively resorbed and which is now mobile and tender. Separation has been attempted for six months without success. Which single management is most appropriate?

A Continue with a larger separator for a further six months before reassessing the position
B Extract the first permanent molar and allow the second molar to erupt into its place
C Extract the deciduous molar and plan space management as the first molar drifts mesially
D Leave both teeth and review in two years when the second premolar approaches eruption
E Fit a quadhelix to expand the arch and so release the impaction of the first molar

697 Regarding ectopic eruption of the first permanent molar:

TRUE OR FALSE — judge each statement independently; any number may be true

A It affects a small percentage of children and is more common in the maxilla
B About two thirds of cases disimpact spontaneously
C Resorption of the deciduous second molar distal root is a characteristic finding
D A separating module or brass wire is a reasonable first measure in a reversible case
E Loss of the deciduous second molar in such a case requires no space management

698 A 7-year-old boy has had the lower right second deciduous molar extracted because of caries. The first permanent molars are erupted and in a Class I relationship, and there is no crowding in the lower arch. Which single management is most appropriate?

A Fit a lower lingual arch and extract the remaining deciduous molars at nine years
B Extract the contralateral second deciduous molar to keep the centreline correct
C Extract the upper right second deciduous molar to compensate for the loss
D Leave the space, as the second premolar will erupt into it without hindrance
E Fit a band and loop space maintainer on the first permanent molar and review

699 An 8-year-old girl has lost the lower left first deciduous molar to caries. The lower arch is crowded by about 4 mm and the centreline is already 1 mm to the left. Which single measure best addresses the centreline?

A Extract the lower right first deciduous molar as a balancing extraction to preserve symmetry
B Fit a band and loop space maintainer on the left side and accept the current centreline
C Extract the upper left first deciduous molar as a compensating extraction to match the arches
D Fit a lower lingual arch to hold the arch length until the permanent teeth erupt
E Begin fixed appliance treatment now to correct the centreline before it deteriorates

700 A clinician plans space management in the mixed dentition and needs to estimate the space available as the deciduous molars are replaced. Which single set of figures is correct?

A Leeway space is about 5 mm per quadrant in both arches at the time of the second transition
B Leeway space is about 2.5 mm per quadrant in the maxilla and about 1.5 mm in the mandible
C Leeway space is about 1.5 mm per quadrant in the maxilla and about 2.5 mm in the mandible
D Incisor liability is about 5 mm in the maxilla and about 7 mm in the mandible
E Leeway space is about 1.5 mm per arch in the maxilla and about 2.5 mm per arch in the mandible

701 Regarding space maintenance in the mixed dentition:

TRUE OR FALSE — judge each statement independently; any number may be true

A A band and loop is suitable where a single deciduous molar has been lost in one quadrant
B A lower lingual arch placed before the deciduous molars exfoliate preserves the leeway space
C A space maintainer is unnecessary after early loss of a deciduous incisor
D Space maintainers require no review once fitted, as they are passive appliances
E Early loss of a deciduous canine from a crowded arch tends to shift the centreline

702 A 9-year-old boy has a Class I skeletal and incisor relationship, an average lower face height and 8 mm of crowding in each arch, with all four deciduous canines lost spontaneously to the erupting lateral incisors. The overbite is average. Which single management is most appropriate to discuss?

A Extraction of all four first permanent molars to create space in each quadrant
B Immediate fixed appliance treatment with expansion of both arches to relieve crowding
C Lower lingual arch and upper transpalatal arch to preserve all remaining leeway space
D Observation until the permanent dentition is complete, then fixed appliances with extractions
E A serial extraction programme beginning with the remaining deciduous canines

703 A serial extraction programme has been agreed for a 9-year-old with severe Class I crowding. Which single sequence is correct?

A First deciduous molars first, then deciduous canines, then second premolars
B Deciduous canines first, then first deciduous molars, then first premolars
C First premolars first, then deciduous canines, then first deciduous molars
D Deciduous canines and first premolars together, then the first deciduous molars
E Second deciduous molars first, then deciduous canines, then first premolars

704 A colleague proposes serial extraction for a 9-year-old with 3 mm of lower crowding, a Class II division 2 incisor relationship and a deep overbite. Which single objection is most important?

A The crowding is too severe for the space that four premolar extractions would provide
B The total treatment time will be shorter than with a single course of fixed appliances
C Serial extraction cannot be undertaken before the permanent canines are radiographically visible
D The overbite will deepen further as the buccal segments drift mesially and the incisors retrocline
E Serial extraction is contraindicated in any patient under the age of eleven years

705 Regarding serial extraction as an interceptive measure:

TRUE OR FALSE — judge each statement independently; any number may be true

A It is indicated in Class II division 2 cases with a deep overbite
B Mild crowding of 2 to 3 mm is the ideal indication
C It shortens the total period of orthodontic supervision
D It avoids the need for any subsequent fixed appliance treatment in most cases
E It is appropriate where the lower face height is markedly reduced

706 A 13-year-old boy has a lower left second permanent molar that is mesioangularly impacted against the distal of the first molar at about 25 degrees. The third molar is present with an open apex and the arch is otherwise well aligned. Which single management should be attempted first?

A Extract the third molar and review the second molar radiographically in twelve months
B Bond a tube to the second molar and upright it with a nickel titanium uprighting spring
C Place a separating brass wire or elastomeric module between the first and second molars
D Extract the second molar and allow the third molar to erupt into the space
E Surgically expose the second molar and apply traction from a miniscrew in the retromolar area

707 A 15-year-old girl has a lower right second molar impacted at 45 degrees beneath the distal bulbosity of the first molar. Separation and an uprighting spring have both failed, the first molar has a large restoration and the third molar is well formed and vertical. Which single management is most appropriate?

A Extract the second molar and allow the third molar to erupt into its position
B Extract the third molar to provide room and repeat the uprighting attempt

C Surgically reposition the second molar and splint it for six weeks
D Accept the impaction and monitor the distal of the first molar for caries
E Extract the first molar and upright the second molar into the space

708 A trainee states that impaction of the lower second permanent molar is a rare anomaly encountered perhaps once in a career. Which single statement is the best correction?

A It is confined to the maxilla, where the tuberosity obstructs eruption
B It affects about 10 per cent of individuals and is one of the commonest eruption failures
C It affects fewer than one in a thousand individuals and is genuinely rare
D It occurs only in association with an absent third molar in the same quadrant
E It affects just over 2 per cent of individuals and is encountered regularly in practice

709 A 14-year-old has an impacted lower second molar. Regarding this condition:

TRUE OR FALSE — judge each statement independently; any number may be true

A The prevalence is a little over 2 per cent
B The mandible is affected far more often than the maxilla
C The first line of management is always surgical exposure
D Caries of the distal surface of the first molar is a recognised consequence
E Extraction of the second molar to allow the third molar to erupt is a legitimate endpoint

710 A 9-year-old boy avulsed his upper right central incisor, which was re-implanted within 30 minutes and root treated. Two years later the tooth is 2 mm below the occlusal level of its neighbour, has a high metallic percussion note and does not respond to orthodontic force. Which single explanation is correct?

A The tooth has over-erupted and the discrepancy is in the adjacent incisor
B The periodontal ligament is intact and the tooth requires a heavier continuous force
C The tooth has undergone inflammatory resorption and will exfoliate within months
D The tooth is ankylosed and its position will worsen as the alveolus continues to grow
E The root treatment has stiffened the tooth and orthodontic movement will resume in time

711 A 12-year-old girl lost her upper left central incisor after trauma at the age of nine. The space has closed by 3 mm, she has a Class I incisor relationship, mild upper crowding and an unremarkable smile line. Which single plan is most appropriate?

A Close the space fully and reshape the lateral incisor and canine to substitute for the missing tooth
B Open the space to its full width, maintain it with a bonded pontic and plan an implant at maturity
C Place an immediate implant now to preserve the alveolar bone before further resorption occurs
D Accept the reduced space and restore the gap with a resin retained bridge next year
E Extract the contralateral central incisor to allow symmetrical space closure in the labial segment

712 A 13-year-old boy with a missing upper central incisor asks why he cannot have an implant now rather than wearing a retainer with a pontic for several more years. Which single explanation is best?

A Implants cannot be placed until the third molars have erupted into the arch
B Implant surfaces are damaged by orthodontic forces applied during the finishing stage
C The bone at his age is too dense for successful osseointegration to occur reliably
D An implant placed now would drift mesially with the rest of the dental arch and need frequent adjustment

E An implant behaves like an ankylosed tooth and will become infra-occluded as the alveolus grows

713 A 10-year-old has lost an upper central incisor through trauma. Regarding management of the resulting space:

TRUE OR FALSE — judge each statement independently; any number may be true

A The decision between opening and closing the space can be deferred until the permanent dentition is complete without any consequence
B Space closure with canine and lateral incisor substitution is technically harder for a central incisor than for a lateral
C An implant may be placed as soon as the root of the adjacent incisor is fully formed
D A removable appliance carrying a pontic is an acceptable interim measure during growth
E Ankylosis of a re-implanted incisor is an indication to plan early decoronation rather than orthodontic extrusion

714 A 12-year-old girl suffered a lateral luxation of an upper central incisor six weeks ago. The tooth has been repositioned and splinted, responds to sensibility testing and shows no periapical change. She is due to start fixed appliances. Which single plan is most appropriate?

A Begin appliance therapy now, since the tooth responds normally to sensibility testing
B Defer force on that tooth for at least three months and then reassess it radiographically
C Defer all orthodontic treatment until the tooth has been root treated as a precaution
D Begin treatment but bypass the traumatised tooth in the archwire indefinitely
E Extract the traumatised incisor before treatment to avoid the risk of resorption

715 An 11-year-old boy has hypodontia with absent lower second premolars. The lower second deciduous molars are firm, caries free, in good occlusion and not infra-occluded. There is mild lower crowding of 2 mm. Which single plan is most appropriate?

A Extract both deciduous molars and use the space to relieve the lower crowding by distal movement
B Extract both deciduous molars now and close the spaces with fixed appliances
C Extract both deciduous molars now and plan implants once growth is complete
D Retain the deciduous molars and monitor them, reducing them mesiodistally if space is later needed
E Retain the deciduous molars and plan onlay build-ups to match the occlusal level immediately

716 A 13-year-old girl has bilaterally absent upper lateral incisors. She has a Class I skeletal pattern, 4 mm of upper crowding, canines of good colour that are slightly narrow, and a low smile line. Which single plan is most appropriate?

A Close the spaces with canine substitution and reshape the canines to resemble lateral incisors
B Open the spaces bilaterally and plan resin retained bridges once growth has finished
C Open the spaces bilaterally and plan implants in the lateral incisor positions at maturity
D Accept the spaces and restore the canines with composite to widen them in situ
E Extract the upper first premolars and retract the labial segment to consolidate the space

717 Regarding hypodontia and its orthodontic management:

TRUE OR FALSE — judge each statement independently; any number may be true

A The second premolars and upper lateral incisors are among the most commonly absent teeth

B Retained deciduous molars in hypodontia are commonly infra-occluded by adolescence
C Canine substitution requires reshaping of the canine and often of the first premolar
D Space closure is generally favoured where the arch is crowded and the smile line is low
E Prosthetic replacement of a missing lateral incisor is best completed before the pubertal growth spurt

718 An 8-year-old boy has a lower second deciduous molar that lies about 1 mm below the occlusal plane. The successor is present radiographically, the contact points with the neighbouring teeth are intact and there is no tipping of adjacent teeth. Which single management is most appropriate?

A Build the tooth up with a composite onlay to restore it to the occlusal plane
B Extract the deciduous molar now and fit a band and loop space maintainer
C Monitor at each recall, as most such teeth exfoliate normally when the successor erupts
D Extract the deciduous molar and allow the adjacent teeth to drift into the space
E Refer for surgical luxation of the tooth to restore its eruptive potential

719 A 12-year-old girl has a lower second deciduous molar submerged to the level of the gingival margin. There is no permanent successor, the adjacent teeth have tipped over the space and the opposing upper molar has over-erupted. Which single management is most appropriate?

A Extract the tooth and place an immediate implant in the resulting socket
B Leave the tooth in place and monitor it radiographically at yearly intervals
C Build the tooth up to the occlusal plane with a stainless steel crown and review
D Extract the submerged tooth and plan orthodontic management of the space that results
E Decoronate the tooth and leave the root to resorb without raising a flap

720 A clinician asks at what point an infra-occluded deciduous molar should be removed rather than monitored. Which single criterion is most useful?

A When the tooth is more than 1 mm below the occlusal plane at any age
B When the tooth has submerged to the level of the contact points of the adjacent teeth
C When the patient reaches the age of ten years, regardless of the degree of submergence
D When the permanent successor is present, since exfoliation will then be delayed
E When the tooth shows any radiographic evidence of root resorption at its apex

721 Regarding infra-occlusion of deciduous molars:

TRUE OR FALSE — judge each statement independently; any number may be true

A Infra-occlusion is caused by continued eruption of the adjacent teeth rather than by ankylosis of the affected tooth
B Infra-occlusion is more common in the maxilla than in the mandible
C A deciduous molar with a successor present usually exfoliates despite mild infra-occlusion
D Tipping of the adjacent teeth over the submerged tooth is an indication for its removal
E Infra-occlusion of a deciduous molar excludes the possibility of hypodontia elsewhere in the arch

722 A 9-year-old girl has an anterior crossbite of the upper left central incisor with a 2 mm mandibular displacement on closing. The overbite after correction would be 3 mm, there is 2 mm of space in the upper labial segment and the skeletal pattern is Class I. Which single management is most appropriate?

A Extraction of the opposing lower incisor to relieve the interference on closing
B Fixed appliances in both arches to correct the crossbite, since a removable appliance cannot tip a single incisor
C Rapid maxillary expansion to correct the transverse deficiency underlying the crossbite

D Review until the permanent dentition is complete and then plan comprehensive treatment
E An upper removable appliance with a Z-spring and posterior bite planes to tip the incisor labially

723 A 10-year-old boy has a unilateral posterior crossbite involving the deciduous and permanent molars on the right, with a 3 mm mandibular displacement to that side on closing. The upper arch is narrow and the skeletal pattern is Class I. Which single management is most appropriate?

A Expand the upper arch symmetrically with a quadhelix and eliminate the displacement
B Expand only the crossbite side with a removable appliance carrying a midline screw
C Equilibrate the deciduous canines to remove the interference and review the displacement
D Defer expansion until the permanent dentition, when a bonded expander can be used
E Accept the crossbite, as unilateral posterior crossbites in the mixed dentition self-correct

724 A 9-year-old girl sucks her right thumb for several hours each evening. She has an asymmetrical anterior open bite, proclined upper incisors, a 6 mm overjet and a unilateral posterior crossbite with displacement. She says she wants to stop. Which single management is most appropriate?

A Do nothing, since all the features described resolve spontaneously once the habit ceases
B Fit a fixed appliance to close the open bite and correct the overjet immediately
C Begin with a habit deterrent appliance combined with expansion, having secured her co-operation
D Apply a bitter tasting preparation to the thumb as the sole measure and review in a year
E Refer for psychological assessment before any orthodontic intervention is considered

725 An 8-year-old boy has a persistent digit sucking habit. Regarding digit habits and their effects:

TRUE OR FALSE — judge each statement independently; any number may be true

A The typical features are an asymmetrical anterior open bite with proclined upper incisors
B A posterior crossbite associated with a digit habit reliably self-corrects when the habit ceases
C Habit deterrent appliances should be fitted without discussion where the child is reluctant
D The duration of sucking each day matters less than the force applied during each episode
E Digit habits ceasing before the permanent incisors erupt leave no lasting occlusal effect

726 A clinician reviewing a 7-year-old is deciding which findings justify intervention now rather than at the permanent dentition. Which single finding most strongly justifies acting now?

A A 7 mm overjet with competent lips in a Class II skeletal pattern
B An anterior crossbite with a mandibular displacement on closing
C Spacing between the upper central incisors of 2 mm with divergent crowns
D A midline diastema of 2 mm in the absence of an abnormal frenal attachment
E Mild rotation of an upper central incisor that has recently erupted

727 The parents of an 8-year-old boy are worried that his upper permanent central incisors are spaced and their crowns are tilted distally, with the lateral incisors just erupting. The occlusion is otherwise Class I and there are no other abnormalities. Which single management is most appropriate?

A Refer for frenectomy before the permanent canines erupt into the arch
B Fit an upper removable appliance with a labial bow to close the spaces now
C Take an upper standard occlusal radiograph to exclude a midline supernumerary tooth

D Reassure and review, as the appearance is a normal transient stage of development
E Begin fixed appliances on the upper labial segment to align the incisors early

Block U — Class I malocclusion and bimaxillary protrusion (Q728–752)

Volume I: Chapter 13 (Class I and the permanent dentition)
Volume II: **§21** — read it again if you score under 70%
This block: 25 questions (6 of them true/false) · 49 marks · allow 31 minutes

728 A 14-year-old girl has upper incisors inclined at 118 degrees to the maxillary plane and lower incisors at 98 degrees to the mandibular plane. The lower incisor edges occlude with the cingulum plateau of the upper incisors, and both arches are crowded by 6 mm. Which single classification of her incisor relationship is correct?

A Class I, because the lower incisor edges occlude with the cingulum plateau of the uppers
B Class II division 1, because both labial segments are proclined beyond the normal range
C Class III, because the lower labial segment is proclined relative to the mandibular plane
D Class II division 2, because the incisors are crowded and the overbite is complete
E Unclassifiable, because incisor classification cannot be applied to a crowded dentition

729 A 22-year-old woman has a Class I incisor relationship with 3 mm of crowding confined to the lower labial segment. Her incisors are upright, the periodontium is healthy, the overbite is normal and she has broad flat contact points. Which single management is most appropriate?

A Extraction of two lower first premolars with fixed appliance space closure
B Extraction of one lower incisor followed by alignment and retention of the segment
C Interproximal reduction of the lower labial segment with alignment and bonded retention
D Proclination of the lower incisors by 4 mm to create the space required for alignment
E Expansion of the lower intercanine width to accommodate the crowded incisors

730 Regarding Class I malocclusion and its assessment:

TRUE OR FALSE — judge each statement independently; any number may be true

A A Class I incisor relationship may coexist with a Class II molar relationship
B Crowding in Class I is commonly described as mild below about 4 mm and severe above about 8 mm
C A Class I incisor relationship excludes a significant skeletal discrepancy
D Crowding may be present in one arch and spacing in the other in the same patient
E Class I malocclusions may require extractions as often as Class II division 1 malocclusions

731 A 15-year-old boy has a Class I incisor relationship, an average lower face height and 10 mm of crowding in each arch, with both canines displaced buccally and high. The molar relationship is Class I and the profile is straight. Which single extraction pattern is most appropriate?

A Two lower incisors, with the upper arch aligned without any extraction at all
B Four second premolars, with anchorage reinforced to preserve the buccal segments
C Two upper first premolars only, accepting a Class II buccal segment relationship
D Four first permanent molars, with the second molars brought forward into their place
E Four first premolars, with space closure planned to maintain the incisor position

732 A 9-year-old girl has a 3 mm midline diastema between the upper central incisors. The lateral incisors have just erupted, the canines are palpable buccally and unerupted, and the frenal attachment does not blanch the incisive papilla. Which single management is most appropriate?

A Reassure and review, since the space commonly closes as the canines erupt
B Close the space now with an upper removable appliance and retain with a bonded wire
C Refer for frenectomy before the permanent canines erupt into the arch
D Bond fixed appliances to the four incisors and close the space immediately
E Take a cone beam scan to exclude a midline supernumerary before any treatment

733 A 27-year-old man has a 3 mm midline diastema. The upper labial frenum inserts into the incisive papilla and blanches it on traction, and the canines have been fully erupted for many years. He wants the space closed. Which single plan is most appropriate?

A Close the space orthodontically and rely on a night-time removable retainer alone
B Perform frenectomy first, then close the space and retain with a removable retainer
C Close the space orthodontically, then perform frenectomy, then retain with a bonded wire
D Perform frenectomy alone and allow the space to close spontaneously afterwards
E Restore the space with composite additions to the mesial surfaces of both incisors

734 Regarding the upper labial frenum and the median diastema:

TRUE OR FALSE — judge each statement independently; any number may be true

A A median diastema in a 9-year-old with unerupted canines is an indication for frenectomy
B Frenectomy performed alone reliably closes an established median diastema
C A closed median diastema can be retained with a removable retainer worn for six months
D A blanching test of the incisive papilla is unhelpful in assessing the frenal attachment
E Diastema closure in adults is among the most stable of orthodontic movements

735 A 19-year-old woman of Chinese descent has upper incisors at 122 degrees to the maxillary plane and lower incisors at 101 degrees to the mandibular plane, with a Class I incisor relationship, competent lips at rest and a convex profile with prominent lips. Which single interpretation is best?

A Her incisor inclinations are pathological and indicate an underlying skeletal discrepancy needing surgery
B The finding is an artefact of head positioning and the radiograph should be repeated
C Her cephalometric values are within the normal range for all populations and need no comment
D The proclination is a consequence of incompetent lips allowing the incisors to drift forward
E This is bimaxillary proclination, and the cephalometric norms used were derived from other populations

736 A 24-year-old woman with bimaxillary proclination is to have four first premolars extracted and her upper incisors retracted by 5 mm. She asks how much her lip prominence will change. Which single estimate is best supported?

A Upper lip retraction of about 5 mm, since the lip follows the incisor movement exactly
B Upper lip retraction of about 3.5 mm, on a soft tissue response of roughly 0.7 mm per mm
C Upper lip retraction of about 1 mm, since the lip is largely independent of incisor position
D Upper lip retraction of about 8 mm, as the lip is unsupported once the incisors move back
E No change in lip position, since the change occurs only in the lower lip after retraction

737 A clinician plans incisor retraction in a patient with prominent lips and asks how reliable the predicted soft tissue change is. Which single appraisal is best supported?

A Thick lips follow incisor retraction more closely than thin lips do
B The ratio is fixed and predicts the individual soft tissue outcome to within half a millimetre
C The soft tissue profile cannot be predicted at all from planned incisor movement
D The ratio is a population average and individual lip thickness and tone alter the result appreciably
E The ratio applies only to the lower lip, since the upper lip is supported by the nose

738 A 30-year-old with bimaxillary protrusion is considering premolar extractions and incisor retraction. Regarding this treatment:

TRUE OR FALSE — judge each statement independently; any number may be true

A Retraction of proclined incisors in a thin periodontal phenotype carries a risk of gingival recession
B Bimaxillary proclination is always pathological and should be corrected wherever it is diagnosed
C The incisor inclinations should be compared against norms derived from white European samples
D Relapse is more likely if the incisors are retracted beyond the position their soft tissues will accept
E Lip competence at rest is irrelevant when planning the amount of incisor retraction

739 A 32-year-old man has 2 mm of gingival recession on a lower central incisor with a thin phenotype and minimal attached gingiva. His lower incisors are proclined at 102 degrees to the mandibular plane and he has 4 mm of lower crowding. Which single plan is most appropriate?

A Align by proclining the lower incisors further, since the recession is unrelated to inclination
B Extract the affected incisor and close the space to eliminate the recession altogether
C Consider a mucogingival graft before treatment, and relieve the crowding without further proclination
D Expand the lower intercanine width to relieve the crowding without moving the incisors labially
E Defer all treatment indefinitely, since orthodontics is contraindicated where recession exists

740 A 13-year-old girl has an upper canine and first premolar that have exchanged positions, confirmed radiographically as a complete transposition with both roots in the transposed sites. The arch is well aligned and the occlusion is Class I. Which single management is most appropriate?

A Accept the transposition and reshape the crowns to disguise the altered positions
B Correct the transposition by moving both teeth through each other's positions
C Extract the transposed canine and close the space with fixed appliances
D Extract the transposed premolar and substitute the canine into its correct position
E Refer for surgical repositioning of both teeth into their correct sockets

741 A 16-year-old has a severely rotated upper second premolar in an otherwise well aligned Class I arch with no crowding. Which single consideration is most important in planning its correction?

A Rotational correction is best achieved by proclining the tooth before derotating it
B Rotations correct most readily on a rigid stainless steel archwire engaged at the outset
C Rotational correction of a premolar carries a higher risk of root resorption than intrusion
D A rotated premolar should be extracted, as rotational correction is inherently unstable

E Rotational correction requires a couple, and space must be available on both sides of the tooth

742 Regarding rotations and transpositions in Class I malocclusion:

TRUE OR FALSE — judge each statement independently; any number may be true

A A rotated tooth occupies more arch space than the same tooth once it has been aligned
B Complete transposition is more safely accepted than corrected in most cases
C Transposition of the upper canine and central incisor is the commonest form reported
D Rotational relapse is unaffected by the shape of the interproximal contact points
E Overcorrection of rotations has no place in contemporary practice

743 A 48-year-old woman with treated chronic periodontitis has spaced and labially migrated upper incisors, 5 mm of attachment loss around them and grade one mobility. Disease control has been achieved and the tissues are stable. Which single statement best guides orthodontic planning?

A Treatment may proceed only after regenerative surgery has restored the lost attachment
B Orthodontic treatment is contraindicated wherever attachment loss exceeds 4 mm
C Forces should be increased to compensate for the reduced root surface in the bone
D Light forces may be used once inflammation is controlled, with permanent retention planned
E Fixed retention is unnecessary because the periodontal fibres are no longer intact

744 A 25-year-old woman is dissatisfied with her smile despite a well aligned Class I occlusion. On smiling she shows wide dark spaces between the buccal teeth and the corners of the mouth. Which single feature best describes this and its usual management?

A A reverse smile arc, addressed by differential intrusion of the upper labial segment
B Wide buccal corridors, addressed by transverse development of the upper arch
C Excessive gingival display, addressed by intrusion of the upper labial segment
D An occlusal cant, addressed by asymmetrical vertical correction of the upper arch
E Inadequate incisor display, addressed by extrusion of the upper labial segment

745 A 20-year-old woman shows 5 mm of gingiva above her upper central incisors on full smile, with a normal crown height of 10 mm, a Class I occlusion and no vertical maxillary excess on cephalometric assessment. Which single management is most appropriate?

A Acceptance and reassurance, since 5 mm of gingival display is within the normal range
B Crown lengthening surgery of the upper labial segment to increase the clinical crown height
C Le Fort I impaction of the maxilla to reduce the vertical dimension of the midface
D Extraction of the upper first premolars and retraction to reduce the gingival display
E Orthodontic intrusion of the upper labial segment, with retention of the intruded position

746 Regarding smile analysis in the Class I patient:

TRUE OR FALSE — judge each statement independently; any number may be true

A The smile arc ideally follows the curvature of the lower lip on smiling
B Buccal corridor width is influenced by the transverse dimension of the upper arch
C Up to about 2 mm of gingival display on full smile is generally regarded as acceptable
D Incisor display at rest is unaffected by age and remains constant through adult life
E Midline deviation is more readily perceived by laypeople than a canted occlusal plane

747 A 23-year-old man requires 4 mm of space in the upper labial segment for alignment. His incisors are of normal width with flat broad contact points and healthy interdental papillae. Which single approach to interproximal reduction is most appropriate?

A Remove 1 mm from each of the four incisor contacts using an air rotor without cooling
B Remove up to about 0.5 mm from each of the eight available contacts under water cooling
C Remove up to 2 mm from the mesial surface of each canine only, preserving the incisors
D Remove 0.25 mm from each contact, accepting that only half the space will be created
E Remove enamel from the labial surfaces of the incisors rather than the contact areas

748 A clinician is concerned that interproximal reduction may predispose to caries and periodontal disease. Which single appraisal of the evidence is best supported?

A The evidence is entirely absent and the procedure should be regarded as experimental
B Interproximal reduction doubles the risk of approximal caries and should be avoided in adults
C Interproximal reduction causes irreversible loss of interdental papillae in most cases
D Polished reduced surfaces show no increase in caries or attachment loss on follow-up
E Reduced surfaces require topical fluoride varnish every three months for life

749 A 21-year-old woman has generalised spacing of 6 mm in the upper arch with small conical lateral incisors, a Class I incisor relationship and an average overbite. She dislikes the gaps. Which single plan is most appropriate?

A Redistribute the spaces orthodontically beside the lateral incisors and build them up with composite
B Close all the spaces orthodontically and accept the reduced width of the lateral incisors
C Restore all the spaces with composite additions without any orthodontic movement, accepting the uneven spacing
D Extract the conical lateral incisors and close the resulting spaces with fixed appliances
E Expand the upper arch to increase its perimeter and then close the spaces with elastics

750 A clinician is planning space for a crowded Class I case. Regarding crowding and its management:

TRUE OR FALSE — judge each statement independently; any number may be true

A Crowding of 3 mm can usually be relieved without recourse to extractions
B Extraction of teeth is the only means of creating space in a crowded arch
C Proclination of the lower incisors to create space risks an unstable and recession-prone result
D Expansion of the upper arch increases its perimeter and can contribute space
E Distal movement of the upper molars is a reliable way of creating space in the lower arch

751 A 17-year-old boy has had a severely rotated upper lateral incisor derotated with fixed appliances. It has narrow rounded contact points and a healthy periodontium. Which single measure best protects the result?

A Extended wear of a rigid rectangular archwire for three months before debonding
B Placement of a vacuum-formed retainer worn at night only for the first six months
C Deliberate overcorrection combined with reshaping of the contact points
D Proclination of the tooth by 5 degrees to increase its contact with the neighbours
E Reduction of the incisal edge to match the height of the contralateral incisor

752 A candidate is asked whether a patient with a Class I incisor relationship, a half-unit Class II molar relationship on the right and a Class I molar relationship on the left can be described as a Class I malocclusion. Which single answer is correct?

A Yes, but only if the asymmetry is dental rather than skeletal in its origin
B No, since any molar asymmetry requires the case to be classified as Class II subdivision
C No, since a Class I malocclusion requires Class I molar and canine relationships throughout
D Yes, since the classification of malocclusion follows the incisor relationship in British practice
E The question cannot be answered without a lateral cephalogram to define the skeletal pattern

Block V — Class II malocclusion (Q753–792)

Volume I: Topic 12 (Reducing overjet in a growing child); Chapters 9, 12 and 13
Volume II: **§22** — read it again if you score under 70%
This block: 40 questions (10 of them true/false) · 80 marks · allow 50 minutes

753 A 13-year-old boy has an overjet of 8 mm, upper incisors at 118 degrees to the maxillary plane, lower incisors at 94 degrees to the mandibular plane and an incomplete overbite. The lower lip rests behind the upper incisors at rest. Which single classification is correct?

A Class II division 2, since the overbite is incomplete and the lower incisors are retroclined
B Class II division 1, since the lower incisor edges lie behind the cingulum plateau
C Class I, since the upper incisor inclination is the only abnormal cephalometric value
D Class III, since the lower lip position determines the incisor classification used
E Class II division 1 subdivision, since only the labial segment is involved in the discrepancy

754 A 14-year-old girl has upper central incisors at 96 degrees to the maxillary plane, upper lateral incisors proclined and mesially rotated, a complete overbite onto the palatal mucosa, an overjet of 3 mm and a high lower lip line. Which single classification is correct?

A Class II division 2 subdivision, since only the central incisors are retroclined in the arch
B Class I, since an overjet of 3 mm lies within the normal range for a Class I relationship
C Class II division 1, since the lateral incisors are proclined beyond their normal inclination
D Class III, since the upper central incisors are retroclined relative to the maxillary plane
E Class II division 2, with the laterals proclined by a lower lip behind the centrals

755 A 15-year-old boy has a Class II division 2 incisor relationship. Regarding this malocclusion:

TRUE OR FALSE — judge each statement independently; any number may be true

A The lower face height is typically increased and the mandibular plane angle steep
B The overbite is typically incomplete and the lower incisors contact the palatal mucosa
C The interincisal angle is typically reduced below 120 degrees
D A large overjet is a defining feature of division 2
E Division 2 is more common than division 1 in most populations surveyed

756 A 12-year-old girl has an 8 mm overjet, proclined upper incisors, retroclined lower incisors and a lower lip that rests behind the upper incisors at rest. Her ANB angle is 4 degrees. Which single conclusion about aetiology is best supported?

A The malocclusion is largely soft tissue in origin, with the lip trap maintaining the incisor positions
B The malocclusion is largely skeletal, since an ANB of 4 degrees indicates a marked discrepancy
C The malocclusion is the direct result of a digit habit, which the incisor pattern demonstrates
D The malocclusion arises from mandibular retrognathia that will resolve on its own with continued growth
E The malocclusion cannot be attributed to any cause without a functional analysis of swallowing

757 A 13-year-old boy with a Class II division 1 malocclusion has an ANB of 7 degrees, a well positioned maxilla on cephalometric analysis and a retrognathic mandible. His lower face height is average. Which single statement best describes the basis for treatment planning?

A The discrepancy is dental, so extractions and retraction of the upper labial segment are indicated
B The discrepancy is maxillary, so headgear should be used to restrain forward maxillary growth
C The discrepancy is mandibular, so growth modification should aim to encourage mandibular advancement
D The discrepancy is vertical, so posterior intrusion is required before the sagittal correction
E The site of the discrepancy cannot influence the choice of appliance in a growing child

758 A 12-year-old girl has a 9 mm overjet on a moderate Class II skeletal base with mandibular retrusion. Her cervical stage is CS2 and she is keen to start treatment. Which single management is most appropriate?

A Defer treatment for three years and reassess when growth has finished
B Begin a functional appliance now, since CS2 places the pubertal peak about a year ahead
C Begin fixed appliances with upper first premolar extractions and headgear anchorage
D Begin a functional appliance only when the third molars are radiographically visible
E Refer for orthognathic assessment, since a 9 mm overjet cannot be corrected by growth

759 A functional appliance is proposed for a 12-year-old with a Class II division 1 malocclusion. Regarding functional appliance treatment:

TRUE OR FALSE — judge each statement independently; any number may be true

A The correction achieved is predominantly skeletal, with a dental contribution of less than a fifth
B Compliance is the principal determinant of success with a removable functional appliance
C A functional appliance is equally effective in a patient whose growth has finished
D Proclination of the lower incisors during treatment contributes nothing to the overjet reduction
E Functional appliances have no effect on the vertical dimension of the face

760 A 12-year-old boy has a 7 mm overjet with a prognathic maxilla, a well positioned mandible and an average lower face height. His cervical stage is CS3. Which single appliance is most appropriate for growth modification?

A A removable appliance with a palatal screw to expand the upper arch and reduce the overjet
B A twin block appliance worn full time to posture the mandible forward during the peak
C Fixed appliances with Class II intermaxillary elastics worn full time for twelve months
D Headgear worn 10 to 12 hours a day at 250 to 350 g per side to restrain the maxilla
E Rapid maxillary expansion followed by a facemask to protract the maxilla forwards

761 A 9-year-old boy has a 10 mm overjet, grossly incompetent lips and upper incisors that are already chipped from a playground fall. His mother asks whether treatment should start now or at twelve. Which single consideration is most important?

A Reduction of the overjet now reduces the risk of further incisor trauma
B Starting now will produce a better final occlusal result than starting at twelve
C Starting now will shorten the total duration of appliance wear across both phases
D Starting now will avoid the need for fixed appliances in the permanent dentition
E Starting now will allow a greater skeletal change than treatment at the pubertal peak

762 A trial of early Class II treatment randomised children to a functional appliance at eight to ten years or to treatment in early adolescence, and found no difference in final occlusal outcome, self-esteem or treatment duration between the groups. Which single conclusion is best supported?

A Early treatment should be abandoned entirely, since it confers no benefit of any kind
B Early treatment should be offered to every child with a Class II division 1 malocclusion
C Early treatment should be reserved for specific indications rather than offered routinely
D The trial was too small to support any conclusion about the timing of treatment
E Early treatment shortens the adolescent phase and should be preferred for that reason

763 Regarding the randomised trials of early Class II treatment:

TRUE OR FALSE — judge each statement independently; any number may be true

A Early treatment was shown to produce a better final occlusal result than treatment in adolescence
B The reported risk ratio for incisal trauma with early treatment was about 0.58
C Early treatment was shown to reduce the total time spent in appliances across both phases
D Children treated early showed a durable improvement in self-esteem at the end of treatment
E The group treated later finished with a slightly worse occlusal result than the early group

764 A risk ratio of 0.58 is reported for incisal trauma in children whose overjet is reduced early. A clinician asks how this should be presented to parents. Which single approach is most appropriate?

A Present it only as a statement that trauma is common in children with large overjets
B Express it as a 58 per cent reduction in the chance of the child damaging an incisor
C Express it as a guarantee that early treatment will prevent trauma to the upper incisors
D Omit the figure, since risk ratios cannot be communicated meaningfully to parents without causing confusion
E Express it as a relative reduction of about 42 per cent and give the absolute risk in that child

765 A department is writing a policy on early Class II treatment. Which single policy is best supported by the evidence and most practical to apply?

A Treat early where trauma risk or teasing is marked, and otherwise wait for the peak
B Treat every Class II division 1 child at eight years to obtain a skeletal response
C Treat no child before the permanent dentition, irrespective of overjet or lip competence
D Treat early only where the parents request it, irrespective of the clinical findings
E Treat early only where a cervical stage of CS3 has already been reached at eight years

766 A 25-year-old woman has an 11 mm overjet, an ANB of 9 degrees, a markedly retrognathic mandible, incompetent lips with marked strain on closure and a well aligned upper arch. She is unhappy with her profile. Which single management is most appropriate to discuss first?

A A twin block appliance worn full time for nine months followed by fixed appliances
B Camouflage with upper first premolar extractions and maximum anchorage retraction
C A combined orthodontic and orthognathic approach with mandibular advancement
D Upper fixed appliances alone with Class II elastics to the lower arch for twelve months
E Acceptance and reassurance, since an 11 mm overjet is compatible with normal function

767 A 24-year-old with a Class II division 1 malocclusion is choosing between camouflage and surgery. Regarding the decision:

TRUE OR FALSE — judge each statement independently; any number may be true

A The size of the skeletal discrepancy is a principal determinant of whether camouflage is feasible
B A patient whose main concern is facial appearance is more likely to need a surgical approach
C Camouflage is preferred where the lower incisors are already proclined and the profile is flat
D Class II camouflage improves the skeletal relationship as well as the incisor relationship
E A functional appliance may still produce useful skeletal change in a patient of this age

768 A 16-year-old girl has a 7 mm overjet, a mild Class II skeletal pattern, an average lower face height, a well aligned lower arch and 3 mm of upper crowding. Growth is essentially complete and the profile is acceptable. Which single plan is most appropriate?

A Refer for mandibular advancement surgery to correct the skeletal discrepancy directly
B Extract four first premolars and close all the spaces, accepting the risk of upper incisor retroclination
C Extract upper second molars and distalise the upper arch with a fixed appliance
D Extract lower first premolars only and procline the upper labial segment to meet them
E Extract upper first premolars only and retract the upper labial segment with anchorage support

769 A 15-year-old boy requires 7 mm of upper space closure entirely by retraction of the labial segment after upper first premolar extractions. Which single anchorage plan is most appropriate?

A Accept reciprocal space closure, since the molars and incisors will move about equally
B Reinforce with a palatal miniscrew or headgear to prevent mesial molar movement
C Use a transpalatal arch alone, which will prevent any mesial movement of the molars
D Use Class II elastics to the lower arch, which will hold the upper molars in position
E Tie the upper arch into a single unit with a rigid archwire, which prevents molar movement

770 A 13-year-old girl has a half-unit Class II buccal segment relationship, a 5 mm overjet, no crowding and second molars that are erupted. She declines extractions and headgear. Which single approach is most appropriate to discuss?

A Expansion of the upper arch to generate the space required to reduce the overjet
B Upper removable appliance with finger springs to retract the canines and then the incisors
C Class III elastics from the upper molars to the lower canines worn full time
D Intra-oral distalisation of the upper molars using a miniscrew-supported appliance
E Acceptance of the overjet, since a half-unit discrepancy cannot be corrected without extractions

771 Regarding distalisation of the upper buccal segments:

TRUE OR FALSE — judge each statement independently; any number may be true

A Headgear used for distalisation is worn about 10 hours a day at 250 to 350 g per side
B Intra-oral distalisers anchored on the palatal vault avoid reliance on patient compliance
C Distalisation is easier after the second permanent molars have erupted than before
D Anchorage loss during distalisation shows as proclination of the upper incisors
E Every headgear requires only one safety mechanism to be fitted

772 A 15-year-old boy with a high mandibular plane angle, a reduced overbite and a 6 mm overjet is to be treated with fixed appliances. A colleague suggests full-time Class II elastics for six months. Which single concern is most important?

A Class II elastics will intrude the buccal segments and deepen the overbite excessively
B Class II elastics have no effect on the vertical dimension and the concern is unfounded
C Extrusion of the upper molars and lower incisors will rotate the mandible down and back
D Class II elastics act only on the lower arch and will procline the upper incisors
E Class II elastics cannot reduce an overjet of this size and a functional appliance is needed

773 A clinician argues that Class II elastics are a simple substitute for a functional appliance in a growing patient. Which single appraisal is best supported?

A Their effect is almost entirely dentoalveolar, so they camouflage rather than modify growth
B They produce the same skeletal change as a twin block provided they are worn full time
C They are contraindicated in growing patients because of the risk of condylar damage
D They reliably distalise the upper molars without any reciprocal effect on the lower arch
E Their effect cannot be assessed because no comparative studies have been undertaken

774 Regarding intermaxillary Class II elastics:

TRUE OR FALSE — judge each statement independently; any number may be true

A They have no reciprocal effect on the lower arch
B They are the treatment of choice in a patient with a markedly increased lower face height
C Full-time wear is required for a consistent effect, and compliance is a limiting factor
D The vertical component extrudes the upper molars and the lower incisors
E They correct the underlying skeletal discrepancy when used during the growth spurt

775 A 15-year-old girl has a Class II division 2 malocclusion with a 7 mm complete overbite onto the palatal mucosa, retroclined upper central incisors and mild lower crowding. Which single sequence of treatment is most appropriate?

A Fit a functional appliance to correct the incisor relationship before any fixed appliance
B Retract the upper labial segment first and reduce the overbite during the finishing stage
C Extract upper first premolars at the outset and close the spaces before levelling the arches
D Correct the lower crowding first and leave the upper arch untouched until it is aligned
E Reduce the overbite and torque the upper central incisors before attempting any retraction

776 A 16-year-old boy with a Class II division 2 malocclusion has had his upper central incisors proclined from 96 to 112 degrees to the maxillary plane. The roots are now close to the labial plate and the interincisal angle is 132 degrees. Which single measure is most important for the stability of the result?

A Leave the interincisal angle at 132 degrees, as this is the most stable value
B Retract the upper incisors to re-establish the original crown position at debond
C Extrude the lower incisors to re-establish a complete overbite before debond
D Apply palatal root torque so that the roots are returned into cancellous bone
E Bond a fixed retainer to the upper labial segment and accept the root position

777 A candidate is asked what determines whether a reduced overbite will remain reduced after Class II division 2 treatment. Which single answer is best?

A An interincisal angle left above 150 degrees, which distributes the occlusal load evenly
B An interincisal angle brought towards 135 degrees with the lower incisor edge on the upper cingulum
C A vacuum-formed retainer worn full time for two years after the appliances are removed
D Complete elimination of the overbite so the incisors no longer contact, removing any risk of relapse
E Proclination of the lower incisors so that they clear the palatal surfaces of the uppers

778 A 17-year-old girl has completed treatment of a Class II division 2 malocclusion. The overbite has been reduced from 7 mm to 3 mm, the upper incisors have been torqued and the interincisal angle is 133 degrees. Which single retention plan is most appropriate?

A An upper removable retainer worn full time for two years and then discontinued
B A vacuum-formed retainer in the upper arch worn at night for six months only
C A bonded retainer across the lower labial segment with no upper retainer, relying on the lower arch alone
D No retention, since the interincisal angle achieved is within the stable range
E An upper removable retainer with an anterior bite plane worn at night for the long term

779 Regarding the management of Class II division 2 malocclusion:

TRUE OR FALSE — judge each statement independently; any number may be true

A Conversion of the malocclusion into a division 1 form is an early objective of treatment
B Torque control of the upper central incisors is required if the correction is to be stable
C The deep overbite is corrected before space closure rather than after it
D The interincisal angle should be left as large as possible to protect the incisor edges
E A functional appliance has no place in the management of division 2 in a growing child

780 A twin block is being constructed for a 12-year-old boy with a 10 mm overjet. The technician asks how the working bite should be recorded. Which single instruction is most appropriate?

A Advance the mandible the full 10 mm at once with the incisors edge to edge and no clearance
B Advance the mandible about 7 mm with 2 to 3 mm of interincisal clearance, in stages if necessary
C Record the bite in centric occlusion and rely on the blocks alone to posture the mandible
D Advance the mandible 2 mm only, and reactivate the appliance at every visit thereafter
E Open the bite 10 mm vertically without any anteroposterior advancement of the mandible

781 A parent is told that a twin block will make her son's jaw grow. Which single explanation is the most accurate account of how the overjet reduces?

A The change is produced by restraint of the maxilla alone, with no mandibular effect
B The whole of the change is skeletal, since the appliance stimulates the condylar cartilage
C The change is entirely dental, since no skeletal effect has been demonstrated in any study
D Most of the change is dentoalveolar, with a smaller and less predictable skeletal contribution
E The change occurs only after the appliance is removed, during the retention phase

782 A 13-year-old girl is to wear headgear to reinforce anchorage during retraction of the upper labial segment after premolar extractions. Which single prescription is correct?

A 10 to 12 hours a day at 250 to 350 g per side, with two independent safety mechanisms
B 14 hours a day at 450 to 500 g per side, with two independent safety mechanisms
C 8 hours a day at 150 g per side, with a single safety release incorporated in the facebow
D 24 hours a day at 500 g per side, with safety mechanisms fitted only if requested
E 10 hours a day at 500 g per side, with a snap-away facebow as the sole safety device

783 Regarding headgear in the management of Class II malocclusion:

TRUE OR FALSE — judge each statement independently; any number may be true

A Two independent safety mechanisms must be fitted to every headgear
B The direction of pull has no influence on the vertical position of the upper molars
C Headgear used for anchorage requires more force per side than headgear used for distalisation
D Headgear depends on compliance and objective monitoring of wear is difficult
E Injury to the eye from a facebow is a theoretical risk that has never been reported

784 A 15-year-old boy with a 9 mm overjet and a moderate Class II skeletal pattern is at cervical stage CS5, and his height has not changed over the past year. His mother asks for a twin block because a friend's son did well with one. Which single response is most appropriate?

A Fit the twin block but plan to wear it for twice as long to compensate for reduced growth
B Fit the twin block, since a functional appliance is effective at any age if worn full time
C Explain that growth modification requires growth and discuss camouflage or surgery instead
D Defer all treatment for two years and reassess whether further growth has occurred
E Fit a headgear instead, since maxillary restraint remains effective after the growth peak

785 A 14-year-old girl has a full-unit Class II buccal segment relationship on the right and a Class I relationship on the left, with a 5 mm overjet, coincident upper midline and a lower midline displaced 2 mm to the right. Which single conclusion is best supported?

A The asymmetry is an artefact of recording and the case should be re-examined in six months
B The asymmetry lies in the upper arch, and the upper first premolar should be extracted on the right
C The asymmetry is skeletal, and unilateral mandibular surgery is the only means of correction
D The asymmetry lies in the lower arch, and unilateral mechanics or extraction should be considered
E The case should be treated as a symmetrical Class II with four premolar extractions

786 A 14-year-old boy has had a 9 mm overjet reduced to 2 mm with a functional appliance followed by fixed appliances. At debond his lips are just competent and his lower lip rests below the upper incisor edges. Which single factor is most important for stability?

A The upper incisor inclination is irrelevant once the overjet has been reduced to 2 mm
B The lower lip now lies in front of the upper incisors and will help to maintain the reduced overjet
C The lower lip position has no bearing on the result once the fixed appliances are removed
D Continued mandibular growth will be unfavourable and will tend to increase the overjet again
E A bonded upper retainer alone will maintain the correction whatever the lip behaviour

787 Regarding the aetiology of Class II division 1 malocclusion:

TRUE OR FALSE — judge each statement independently; any number may be true

A A retrognathic mandible is a more common skeletal cause than a prognathic maxilla
B A lower lip trap tends to procline the upper incisors and retrocline the lower incisors
C A persistent digit habit can produce an increased overjet with an asymmetrical open bite
D Incompetent lips are associated with a greater risk of trauma to the upper incisors
E The size of the overjet is determined by the skeletal pattern alone

788 A 14-year-old boy has a 7 mm overjet, an increased lower face height, a steep mandibular plane and an incomplete overbite. Fixed appliances with upper premolar extractions are planned. Which single mechanical consideration is most important?

A Acceptance of the incomplete overbite, since vertical mechanics cannot influence it
B Maximum use of Class II elastics, which will close the bite as the overjet reduces
C Deliberate extrusion of the upper incisors to establish a normal overbite during finishing
D Expansion of the upper arch, which will reduce the lower face height as the bite closes
E Vertical control of the upper molars, since extrusion will rotate the mandible further backwards

789 A clinician asks what should prompt referral for orthognathic assessment rather than camouflage in an adult Class II patient. Which single indication is most useful?

A A severe skeletal discrepancy with a facial concern that tooth movement alone cannot address
B An overjet of more than 6 mm, irrespective of the skeletal pattern or the patient's concerns
C A Class II division 2 incisor relationship, since division 2 cannot be camouflaged
D An ANB angle above 4 degrees, which marks the limit of dental compensation
E A patient who has already worn a functional appliance without complete success as a child

790 A 16-year-old girl has had a 9 mm overjet reduced with a functional appliance and fixed appliances. She is at cervical stage CS4 and the buccal segments are a solid Class I. Which single retention plan is most appropriate?

A No retention, since a solid Class I buccal interdigitation is self-retaining after debond
B A functional appliance worn full time for a further year to hold the sagittal correction
C Standard removable retainers, with review of the sagittal relationship as growth completes
D A bonded upper retainer alone, since the overjet is the only feature likely to relapse
E Class II elastics worn at night for a further year alongside removable retainers

791 Regarding retention after correction of a Class II malocclusion:

TRUE OR FALSE — judge each statement independently; any number may be true

A A stable buccal interdigitation contributes to retention of the sagittal correction
B Continued unfavourable growth can re-express a Class II relationship after debond
C A persisting lower lip trap predisposes the overjet to relapse
D Overbite reduction achieved without an occlusal stop is prone to relapse
E Retention after a Class II correction follows the same indefinite principles as any other case

792 A specialist is asked to summarise when a Class II division 1 child should be treated. Which single summary is best supported by the evidence?

A Treat all children at seven to nine years, since the skeletal response is greatest at that age
B Treat at the pubertal peak for most children, and earlier only where trauma risk or teasing is marked
C Treat all children after the pubertal peak, since the occlusal result is then more predictable

D Treat only when the permanent dentition is complete, whatever the overjet or lip competence

E Treat at whatever age the parents request, since the evidence shows no difference in outcome

Block W — Class III malocclusion (Q793–822)

> **Volume I:** Topic 17 (Correcting a Class III incisor relationship); Chapters 8, 9 and 13
> **Volume II:** **§23** — read it again if you score under 70%
> **This block:** 30 questions (8 of them true/false) · 62 marks · allow 38 minutes

793 A 9-year-old girl attends with an anterior crossbite of both upper central incisors. In the retruded contact position the incisors meet edge to edge, and the mandible then slides forwards into a reverse overjet of 2 mm. The molars are Class I in the retruded position and the profile is straight. Which single description of her malocclusion is most accurate?

A A true Class III on a Class III skeletal base, since the incisors meet in reverse overjet
B A skeletal Class III with a compensating displacement, since the molars are Class I only in retrusion
C A dentoalveolar Class III without displacement, the forward slide being a normal closing arc
D A pseudo-Class III, since the crossbite is reached only after a forward slide from edge to edge
E An unclassifiable case until a lateral cephalogram in the displaced position is obtained

794 A 14-year-old boy has a Class III incisor relationship. Cephalometric analysis gives SNA 76 degrees, SNB 84 degrees and ANB minus 8 degrees, with a maxillary-mandibular planes angle of 22 degrees and an overbite of 5 mm. Which single statement about the skeletal components of his malocclusion is best supported?

A Both bases contribute, the maxilla being retrusive and the mandible prognathic
B The maxilla alone is at fault, the mandible sitting comfortably within the normal range
C Neither base is displaced, so the discrepancy must be entirely dentoalveolar in origin
D The mandible alone is at fault, the maxilla lying within its normal anteroposterior range
E The reduced planes angle excludes any skeletal contribution to the discrepancy

795 A 12-year-old boy with a Class III incisor relationship is assessed for the likely success of orthodontic camouflage. Regarding the features that make the prognosis favourable:

TRUE OR FALSE — judge each statement independently; any number may be true

A A mild Class III or Class I skeletal base is more favourable than a markedly Class III base
B An average or reduced lower face height is more favourable than an increased face height
C An average or increased overbite is more favourable than a reduced overbite
D Proclined lower incisors with upright upper incisors are more favourable than the reverse
E An anterior displacement of the mandible on closing is a favourable finding

796 An 8-year-old boy has a reverse overjet of 2 mm, Class III molars, a retrusive maxilla, a well positioned mandible and no displacement on closing. His mother asks whether a facemask would help. Which single plan is most appropriate?

A Defer treatment until growth is complete, since early protraction relapses in any case
B Defer the facemask until the pubertal peak, when the growth available is greatest
C Begin facemask therapy now, worn 12 to 14 hours a day while the sutures remain responsive
D Fit a chin cup now to restrain mandibular growth, the maxilla being of normal size
E Fit an upper removable appliance with a Z-spring, which needs less cooperation

797 A randomised trial of protraction facemask therapy in children with Class III malocclusion followed participants to fifteen years of age. Over 60 per cent of the treated group had avoided a need for orthognathic surgery, against 36 per cent of the untreated controls. Which single conclusion is best supported?

A Facemask reliably removes the need for surgery, since most treated children avoided it
B The difference is too small to change advice, since both groups largely avoided surgery
C Facemask should be reserved for those already committed to surgical correction later
D The control figure of 36 per cent shows the untreated group was unrepresentative
E Facemask reduces, but does not abolish, the later need for surgical correction

798 A protraction facemask combined with rapid maxillary expansion is planned for a 9-year-old boy with maxillary retrusion. Regarding this treatment:

TRUE OR FALSE — judge each statement independently; any number may be true

A Expansion is used partly to disarticulate the maxilla before protraction begins
B Protraction elastics are directed upwards and backwards from the upper canine region
C Part of the correction achieved is dentoalveolar rather than skeletal
D A backward rotation of the mandible during treatment worsens the Class III appearance
E The result should be monitored into the late teens because Class III worsens with growth

799 A 15-year-old boy with a Class III incisor relationship has a reverse overjet of 4 mm, a moderately Class III skeletal base and 5 mm of lower arch crowding. Surgery at the end of growth is a realistic possibility. Which single plan for the lower arch is most appropriate now?

A Avoid lower arch extractions now, since decompensation before surgery will need that space
B Extract lower first premolars now, since relieving the crowding simplifies later surgery
C Extract lower second premolars now, which costs less anchorage than first premolars
D Extract lower first molars now, whose prognosis is poorest and whose space closes readily
E Extract first premolars in both arches and camouflage the reverse overjet at this stage

800 A 19-year-old woman with a Class III skeletal pattern is listed for bimaxillary surgery. Her upper incisors are proclined to 122 degrees to the maxillary plane and her lower incisors are retroclined to 80 degrees to the mandibular plane. Which single effect should she be warned of before presurgical orthodontics begins?

A The reverse overjet will reduce, because alignment and levelling upright the incisors
B The profile will improve progressively, because the arches are being coordinated
C The reverse overjet will increase, because decompensation removes the dental camouflage
D The lower incisors will be retroclined further, because that is how crowding is relieved
E The upper incisors will be proclined further, because that enlarges the surgical movement

801 A non-growing patient with a Class III malocclusion has a Wits appraisal of minus 12 mm, an ANB of minus 6 degrees and a reverse overjet of 6 mm. Which single interpretation is best supported by the published camouflage thresholds?

A Only the Wits value exceeds its threshold, so camouflage remains a reasonable option
B The thresholds were derived in growing patients and cannot be applied to an adult
C Camouflage remains appropriate, since an ANB of minus 6 degrees is only mildly Class III
D No judgement is possible until the Holdaway angle and gonial angle are measured too
E All three values lie beyond the published camouflage thresholds, favouring surgery

802 The parents of an 11-year-old boy with a mild Class III incisor relationship ask when his treatment will be finished. Regarding growth in Class III malocclusion:

TRUE OR FALSE — judge each statement independently; any number may be true

A Mandibular growth in boys is complete by 16 years, so review may safely stop then
B Class III patients tend to continue mandibular growth later than their Class I peers
C A Class III malocclusion tends to become worse rather than better with growth
D Serial cephalograms superimposed on the cranial base cannot detect this late change
E Definitive camouflage or surgery should be deferred until growth has demonstrably ceased

803 A 7-year-old girl has a Class III incisor relationship with a prognathic mandible and a normally positioned maxilla. Her parents ask about a chin cup, which a relative used as a child. Which single statement best represents current understanding?

A A chin cup reliably inhibits mandibular growth and permanently reduces its final size
B A chin cup rotates the mandible backwards and downwards, and relapse on stopping is usual
C A chin cup is contraindicated because it damages the temporomandibular joint in users
D A chin cup works by protracting the maxilla and should be combined with expansion
E A chin cup is effective only once the pubertal growth spurt has been completed

804 A 9-year-old boy has an anterior crossbite of the upper left central and lateral incisors, a 2 mm forward displacement on closing and an overbite of 4 mm once the incisors are crossed. There is no crowding and the upper incisors are upright. Which single appliance is most appropriate?

A A protraction facemask with rapid maxillary expansion to correct the skeletal component
B A lower removable appliance to retrocline the lower incisors out of the crossbite
C A fixed 2 by 4 appliance with open coil, giving bodily control of the upper incisors
D An upper removable appliance with Z-springs and posterior bite planes to free the bite
E A chin cup worn at night, since the displacement indicates excess mandibular growth

805 A child is being examined for a possible pseudo-Class III malocclusion. Regarding that examination:

TRUE OR FALSE — judge each statement independently; any number may be true

A The incisor relationship must be recorded in the retruded contact position as well as in intercuspation
B An edge-to-edge incisor contact in retrusion followed by a forward slide is the defining finding
C The profile in the retruded contact position is concave in a pseudo-Class III
D Molars Class I in retrusion but Class III in intercuspation support the diagnosis
E A pseudo-Class III is excluded once the reverse overjet exceeds 2 mm

806 A 16-year-old girl with a Class III malocclusion has a lower anterior face height of 68 per cent of total anterior face height, an overbite of 0 mm, lower incisors inclined at 78 degrees to the mandibular plane and no displacement on closing. Which single conclusion about camouflage is best supported?

A Camouflage is reasonable provided the lower incisors are retroclined a further five degrees
B Camouflage is likely to succeed, since the reverse overjet itself is small and disguisable
C Camouflage is unlikely to succeed, since every prognostic feature present is unfavourable
D Camouflage should be attempted first, with surgery held in reserve if the result fails
E Camouflage cannot be judged until a full functional occlusal analysis has been carried out

807 A discriminant analysis of Class III patients treated by camouflage or by surgery identifies a lower incisor inclination of 80 degrees or less to the mandibular plane as favouring surgery. Which single explanation for that threshold is best?

A Incisors retroclined to that degree have no further retroclination to offer as camouflage
B An inclination below 80 degrees is associated with a low maxillary-mandibular planes angle
C Retroclined lower incisors indicate rapid mandibular growth, for which surgery is needed
D Inclination is the only cephalometric variable that predicts the surgical decision reliably
E Incisors at that inclination already lie outside the alveolar bone and will resorb

808 Regarding the aetiology and epidemiology of Class III malocclusion:

TRUE OR FALSE — judge each statement independently; any number may be true

A Class III has the weakest hereditary component of the three incisor classes
B The prevalence of Class III is much the same in all ethnic groups
C A Class III incisor relationship may occur on a Class I skeletal base
D Repair of a cleft lip and palate is a recognised cause of acquired maxillary retrusion
E Acromegaly produces a Class III appearance chiefly by retruding the maxilla

809 An 11-year-old boy with maxillary retrusion has worn a protraction facemask for eight months with no change in the reverse overjet. He reports wearing it every night while asleep. Which single action is most appropriate?

A Increase the elastic force to about 1000 g per side and continue for a further year
B Add a chin cup so that mandibular growth is restrained at the same time
C Refer for surgical assessment now, since failure of protraction marks a severe case
D Change to an upper removable appliance with Z-springs and correct the incisors alone
E Reassess the diagnosis and the hours of wear, then consider monitoring growth instead

810 A 10-year-old girl completed facemask therapy six months ago, finishing with an overjet of 2 mm and an overbite of 3 mm. Her parents ask what should happen next. Which single plan is most appropriate?

A Discharge her now, since a positive overbite will retain the correction indefinitely
B Continue facemask wear at 14 hours a day until the permanent dentition is complete
C Begin fixed appliances now to consolidate the result before the permanent teeth erupt
D Retain the correction and review growth annually into the late teens before deciding
E Refer for orthognathic assessment now so that surgery can be planned well in advance

811 A 22-year-old man has a mild Class III skeletal base, a reverse overjet of 1 mm, an ANB of minus 2 degrees, upper incisors upright at 105 degrees to the maxillary plane, a well aligned upper arch and 5 mm of lower arch crowding. He declines surgery. Which single extraction pattern best supports camouflage?

A Upper second premolars with lower first premolars, favouring lower incisor retraction
B Lower first premolars alone, retracting the lower labial segment and leaving the upper
C Upper and lower first premolars, treating the two arches symmetrically
D Upper first premolars alone, retracting the upper labial segment to relieve crowding
E No extractions, relying on interproximal reduction in both arches to create the space

812 A colleague puts the following propositions about managing Class III malocclusion in the growing child. Regarding each:

TRUE OR FALSE — judge each statement independently; any number may be true

A A facemask is most effective when started after the pubertal growth spurt

B A positive overjet achieved at nine years of age may be regarded as a stable result
C Class III malocclusion in the mixed dentition should be left untreated in every case
D Expansion is contraindicated before protraction because it destabilises the maxilla
E A chin cup produces a permanent reduction in the final size of the mandible

813 Among the cephalometric discriminants used to separate Class III cases suitable for camouflage from those needing surgery, a gonial angle above 120 degrees and a Holdaway angle of 3.5 degrees or less both favour surgery. Which single explanation is best?

A Both reflect a maxillary width deficiency that camouflage is unable to address
B Both measure soft tissue thickness and so predict only the aesthetic outcome
C Both increase with age, so the thresholds apply only to patients over twenty-five
D Both measure mandibular length directly, which is what the surgery itself alters
E Both mark a backward rotating long face pattern in which camouflage is unstable

814 A 17-year-old Class III patient has a lower anterior face height of 68 per cent of total anterior face height, a maxillary-mandibular planes angle of 38 degrees and an overbite of 1 mm. Which single statement about planning is best supported?

A The vertical pattern has no bearing, the decision resting on the anteroposterior values
B The vertical pattern favours surgery, face height above 66 per cent being a discriminant
C An overbite of 1 mm is adequate to retain camouflage, any positive overbite sufficing
D High-pull headgear will control the face height enough to make camouflage workable
E The face height lies within normal limits and does not affect the plan either way

815 A 10-year-old boy has a reverse overjet of 3 mm with no displacement on closing, a moderately Class III skeletal base, an increased lower face height and a reduced overbite. His father underwent orthognathic surgery at nineteen. Which single course is most appropriate?

A Begin facemask therapy with expansion, expecting a stable correction to be achieved
B Begin fixed appliances and camouflage the reverse overjet in the mixed dentition
C Begin a chin cup now, which addresses the mandibular component of the problem directly
D Monitor growth and prepare the family for probable surgical correction after growth
E Refer for surgical assessment now so that the mandible can be set back without delay

816 A 20-year-old woman with a Class III malocclusion is about to begin presurgical orthodontics. Regarding decompensation:

TRUE OR FALSE — judge each statement independently; any number may be true

A The reverse overjet is expected to increase during the presurgical phase
B Lower incisors are retroclined as part of decompensation in a Class III case
C Decompensation is unnecessary when a bimaxillary procedure is planned
D Upper incisors are proclined further as part of decompensation in a Class III case
E Decompensation increases the magnitude of the surgical movement required

817 A 24-year-old man has Class I molars in the retruded contact position, an edge-to-edge incisor contact on closing and a forward slide of 3 mm into a reverse overjet of 2 mm. He complains of his teeth clicking together and of wear on the lower incisal edges. Which single management is most appropriate?

A Orthodontic correction of the incisor relationship, which removes the displacement
B Occlusal splint therapy alone, treating the displacement as a joint disturbance
C Composite build-ups on the lower incisors to replace the tissue already lost
D Orthognathic mandibular setback, which corrects the reverse overjet definitively
E Reassurance and monitoring, since displacements of this size are well tolerated

818 A 30-year-old woman with a mild Class III malocclusion has 2 mm of gingival recession on both lower central incisors and a thin labial plate on palpation. She wishes to avoid surgery. Which single consideration should most influence the plan?

A Recession is unrelated to incisor inclination and does not affect the plan here
B Proclining the lower incisors will thicken the labial plate and improve the recession
C Retroclining the lower incisors risks dehiscence and further recession in thin bone
D The recession should be grafted after orthodontic movement rather than before it
E Camouflage is safe here, since adult bone remodels around roots that are moved

819 A trial randomised children with Class III malocclusion to protraction facemask or to no treatment at seven to nine years of age, and reported outcomes at fifteen years. Which single feature of the design most strengthens its conclusion?

A The cephalometric measurement of ANB, which is objective and reproducible
B The long interval to follow-up, which reduces the number of participants lost to attrition
C The use of a single operator, which removes variation between clinicians
D The untreated control arm, which separates the effect of growth from that of treatment
E The narrow age band at recruitment, which improves generalisability

820 Regarding orthodontic camouflage of Class III malocclusion in the adult:

TRUE OR FALSE — judge each statement independently; any number may be true

A Camouflage is preferred to surgery when the Wits appraisal is minus 13 mm
B Camouflage relies on proclining the upper and retroclining the lower incisors
C A reduced overbite improves the stability of a camouflaged correction
D Camouflage improves the facial profile more predictably than it improves the occlusion
E An anterior displacement on closing is an unfavourable feature for camouflage

821 A 13-year-old boy with a Class III incisor relationship is to be monitored for late mandibular growth before a definitive decision is taken. Which single set of records is most useful at each annual review?

A Panoramic radiographs, which follow condylar morphology from year to year
B Standardised lateral cephalograms superimposed on the anterior cranial base
C Hand-wrist radiographs alone, which date the growth spurt with some precision
D Study models alone, since occlusal change reflects skeletal change reliably
E Clinical photographs alone, since profile change is what concerns the patient

822 A 21-year-old woman has completed camouflage of a mild Class III malocclusion, finishing with an overjet of 2 mm, an overbite of 3 mm and lower incisors at 82 degrees to the mandibular plane. Which single retention regimen is most appropriate?

A Six months of night-time retainer wear, after which the overbite will hold it
B No retention, since the corrected overbite of 3 mm is self-retaining in itself
C A functional appliance worn at night to maintain the jaw relationship achieved
D An upper retainer only, since the lower labial segment was not decompensated
E Long-term night-time retainers in both arches, with a bonded lower retainer

Block X — Transverse problems and facial asymmetry (Q823–847)

Volume I: Topics 16 and 19 (Crossbite correction; facial asymmetry); Chapters 9, 12, 13 and 15
Volume II: **§24** — read it again if you score under 70%
This block: 25 questions (6 of them true/false) · 49 marks · allow 31 minutes

823 An 11-year-old girl has a crossbite of the upper left first permanent molar and second premolar. On closing from the retruded contact position the mandible deviates 2 mm to the left. Study models show the maxillary intermolar width to be 4 mm narrower than the mandibular. Which single classification is most accurate?

A A functional crossbite with displacement, on a mild transverse skeletal discrepancy
B A dental crossbite, arising from tipping of the individual posterior teeth
C A skeletal crossbite without displacement, the deviation being a normal closing arc
D A scissors bite, in which the palatal cusps occlude buccal to the lower teeth
E A true unilateral skeletal crossbite requiring expansion of one side only

824 A 9-year-old boy has a unilateral buccal crossbite of the right posterior segment with a 3 mm displacement of the mandible to the right on closing. He is otherwise Class I with no crowding and no symptoms. Which single reason to intervene is strongest?

A Untreated crossbites of this kind invariably become skeletal asymmetries in adult life
B The crossbite will obstruct eruption of the permanent canine on the affected side
C The displacement carries a risk of attrition and gingival recession on that side
D Correcting now removes the need for any retention later in the permanent dentition
E Posterior crossbite is a recognised cause of temporomandibular disorder in adolescence

825 An 8-year-old girl with a narrow maxilla and a bilateral posterior crossbite is to be treated by slow maxillary expansion. Regarding slow expansion:

TRUE OR FALSE — judge each statement independently; any number may be true

A A removable expansion plate is typically activated 0.25 mm twice a week
B A quadhelix corrects the crossbite roughly three months sooner than a removable plate
C High-certainty evidence supports expansion for posterior crossbite in children aged seven to eleven
D Slow expansion produces a greater proportion of dental tipping than rapid expansion does
E A quadhelix does not depend on cooperation in the way a removable plate does

826 A 10-year-old boy has the upper right lateral incisor in crossbite. It is tipped palatally, 3 mm of space is available in the arch, the overbite once corrected would be 3 mm and there is a 1 mm displacement on closing. Which single appliance is most appropriate?

A A fixed 2 by 4 appliance with a nickel-titanium archwire and open coil spring
B An upper removable appliance with a Z-spring and flat posterior bite planes
C A lower removable appliance to tip the lower incisors lingually out of the way
D Composite build-up of the palatal surface, letting occlusal force jump the tooth
E Rapid maxillary expansion with a bonded acrylic-splint jackscrew appliance

827 A single upper incisor has been tipped over the bite out of an anterior crossbite. Which single factor most determines whether the correction will hold?

A The age of the patient at the time at which the crossbite was corrected
B The length of time for which the appliance is worn after the correction is complete
C The amount of space that was available in the arch before correction began
D The overbite present once the tooth has been carried into a positive relationship
E The inclination of the opposing lower incisor once the correction is complete

828 A 12-year-old girl is to have rapid maxillary expansion with a tooth-borne jackscrew appliance for a bilateral posterior crossbite and a 7 mm transverse deficiency. Which single regimen is most appropriate?

A One to four turns a day for two to four weeks, then retention for three to six months
B Two turns a day for twelve weeks, removing the appliance as the crossbite corrects
C One turn a week for six months, then removal of the appliance immediately
D One turn a day until a midline diastema appears, then removal of the appliance
E Four turns a day for six weeks, followed by two weeks of retention in place

829 Regarding rapid maxillary expansion in a growing patient:

TRUE OR FALSE — judge each statement independently; any number may be true

A Forces generated at the midpalatal suture may reach 10 kg
B About 90 per cent of the expansion achieved is skeletal rather than dental
C Relapse after retention is negligible when measured at four to eight months
D A midline diastema appearing between the upper central incisors indicates sutural opening
E The expansion should be retained for two to four weeks only

830 A 26-year-old man has a bilateral posterior crossbite with a 7 mm transverse maxillary deficiency and no anteroposterior or vertical discrepancy. Tooth-borne rapid expansion has produced buccal tipping and gingival recession but no midline diastema. Which single approach is most appropriate?

A A further course of tooth-borne rapid expansion with a heavier jackscrew
B Slow expansion with a quadhelix over nine months, which tips less than a jackscrew
C Surgically assisted rapid palatal expansion, releasing the resistant articulations
D Segmental Le Fort I osteotomy with mandibular setback to match the arch widths
E Acceptance of the discrepancy, camouflaged with unilateral premolar extractions

831 A 21-year-old woman requires 6 mm of maxillary expansion. A colleague argues that surgical assistance is unnecessary because she is only twenty-one. Which single response is best?

A Chronological age is the accepted criterion and expansion is safe below twenty-five
B Expansion of more than 3 mm always requires surgery, whatever the patient's age
C Expansion in adults is contraindicated because the palate has ossified completely
D A hand-wrist radiograph will settle whether the suture is still capable of opening
E Interdigitation of the circummaxillary articulations, not age in years, decides this

832 A 14-year-old girl has the lower left second premolar and first molar occluding entirely palatal to their upper opposites, with no buccal cusp contact at all. The upper left segment is buccally inclined and the lower lingually inclined. Which single management is most appropriate?

A Cross-elastics from the upper palatal to the lower buccal surfaces on rigid wires
B Extraction of the upper left first molar to eliminate the interference
C A quadhelix expanded on the left to upright the upper buccal segment

D Acceptance and monitoring, since a scissors bite is functionally well tolerated
E Rapid maxillary expansion to widen the upper arch still further

833 A 16-year-old boy is seen because his mother has noticed facial asymmetry. Regarding its assessment:

TRUE OR FALSE — judge each statement independently; any number may be true

A Perfect facial symmetry is the norm, so any detectable asymmetry is pathological
B A single set of records is sufficient to distinguish a progressive from a stable asymmetry
C An occlusal cant is best assessed by measuring the dental midlines on study models
D Bone scintigraphy is of no value in assessing a possibly active condylar hyperplasia
E Surgical correction should be undertaken as soon as an asymmetry is noticed in a growing patient

834 A 19-year-old woman complains that her smile looks crooked. The chin point deviates 5 mm to the left and the maxillary occlusal plane appears tilted. Which single clinical manoeuvre best demonstrates the occlusal cant?

A Compare the two mandibular ramus heights on a panoramic radiograph
B Measure the distance from each canine cusp tip to the facial midline on a photograph
C Ask her to bite on a wooden spatula and compare it with the interpupillary line
D Record the deviation of the dental midlines from the facial midline in intercuspation
E Superimpose a posteroanterior cephalogram on the crista galli and measure the tilt

835 A 17-year-old boy has a chin point deviated 7 mm to the right, no occlusal cant, symmetrical posterior face heights and a unilateral crossbite on the right. Which single diagnosis in the Obwegeser and Makek classification best fits?

A Hemimandibular hyperplasia, in which one side enlarges in all three dimensions
B Condylar resorption, in which the affected condyle progressively loses height
C A functional asymmetry, in which the deviation disappears when the mouth opens
D Hemifacial microsomia, in which the affected side is deficient rather than enlarged
E Hemimandibular elongation, in which the mandible lengthens horizontally on one side

836 A 14-year-old girl is referred with a facial asymmetry that her mother has noticed over the past year. The chin deviates 4 mm to the left and she is still growing. Which single course is most appropriate?

A Perform a mandibular osteotomy now, before the deviation becomes more marked
B Take serial records six to twelve months apart to establish whether it is progressing
C Begin fixed appliances with asymmetric elastics to camouflage the deviation now
D Reassure her, since asymmetry noticed during growth resolves once growth is complete
E Refer for condylectomy, since progression in a growing patient can be assumed

837 A 13-year-old boy has a bilateral posterior crossbite on a narrow maxilla. Regarding the choice of expansion:

TRUE OR FALSE — judge each statement independently; any number may be true

A A displacement on closing strengthens the case for correcting the crossbite
B Rapid expansion is preferred to slow where a true skeletal deficiency is present
C A quadhelix is contraindicated in the mixed dentition
D Expansion should be retained for two weeks after the crossbite has corrected
E Buccal tipping and palatal cusp drop are recognised effects of tooth-borne expansion

838 A 9-year-old girl needs 4 mm of upper arch expansion for a bilateral crossbite. Her mother reports that she lost two mouthguards last season and forgets to wear her spectacles. Which single appliance is most appropriate?

A A removable expansion plate turned 0.25 mm twice a week by her mother
B A bonded rapid maxillary expander turned four times a day for three weeks
C A vacuum-formed expansion appliance exchanged for the next in the series weekly
D A quadhelix, which is fixed and does not depend on her remembering to wear it
E No appliance, since the crossbite will correct as the permanent teeth erupt

839 A 13-year-old boy has the upper right second premolar erupted palatally and in crossbite with its lower opposite. The remaining occlusion is Class I with 2 mm of upper arch crowding. Which single management is most appropriate?

A Expand the whole upper arch with a quadhelix to correct the single tooth
B Extract the premolar, since a single tooth in crossbite is not worth aligning
C Align the tooth into the arch with a fixed appliance, using open coil for space
D Accept the position, since a single-tooth crossbite carries no functional risk
E Use cross-elastics from the upper palatal to the lower buccal surface

840 A Cochrane review of orthodontic treatment for posterior crossbite in children reports high-certainty evidence that a quadhelix corrects the crossbite about three months sooner than a removable expansion plate, in children aged seven to eleven. Which single conclusion is best supported?

A A quadhelix should be preferred where speed of correction matters, in this age group
B A removable plate is ineffective and should no longer be offered for posterior crossbite
C The finding extrapolates to adults, in whom the same two appliances are also used
D High-certainty evidence means the effect will be seen in every child who is treated
E Three months is too small a difference to influence any clinical decision at all

841 Regarding the causes of facial asymmetry:

TRUE OR FALSE — judge each statement independently; any number may be true

A Unilateral condylar fracture in childhood most often enlarges the affected side
B Hemifacial microsomia produces a deficient rather than an enlarged affected side
C Condylar hyperplasia characteristically presents in the sixth decade of life
D A unilateral crossbite with a displacement is a true skeletal asymmetry
E Juvenile idiopathic arthritis spares the temporomandibular joints

842 A 12-year-old girl completed rapid maxillary expansion four months ago, the appliance having been removed as soon as the crossbite corrected. The crossbite has now recurred on the left side. Which single explanation is most likely?

A The expansion was too rapid and produced only dental tipping, which always relapses
B The mandible has grown transversely faster than the maxilla in the intervening months
C The original diagnosis was a dental rather than a skeletal crossbite
D A residual digit habit has narrowed the maxilla again since the appliance came out
E The appliance was removed before the three to six months of retention the suture needs

843 A 22-year-old woman has a maxillary occlusal cant of 4 mm, higher on the right, with the chin deviated 6 mm to the left and matching asymmetry of the ramus heights. Growth is complete. Which single surgical plan is most appropriate?

A Mandibular osteotomy alone, since the asymmetry originates in the mandible not the maxilla
B Unilateral condylectomy to arrest further growth on the affected side

C Genioplasty alone, since the chin point is the feature of which she complains
D Le Fort I osteotomy to level the maxilla with a mandibular procedure to centre the chin
E Orthodontic camouflage with asymmetric elastics and no surgery at all

844 A 17-year-old boy has a facial asymmetry which serial records taken over two years show to be stable, with no change in the deviation of the chin point. Which single statement about the timing of correction is best?

A Correction should be deferred indefinitely, since stable asymmetries never justify surgery
B Definitive correction may be planned once growth is complete, the deformity being stable
C Further scintigraphy is mandatory before any plan, whatever the serial records show
D Correction should proceed at once, since a stable deformity cannot recur after surgery
E Stability over two years excludes any skeletal component to the asymmetry

845 Regarding surgically assisted rapid palatal expansion in the adult:

TRUE OR FALSE — judge each statement independently; any number may be true

A It is generally indicated where more than about 5 mm is needed in a non-growing patient
B The only meaningful resistance to expansion in the adult is the midpalatal suture
C Releasing the lateral maxillary buttresses reduces the dental tipping otherwise seen
D A bone-borne expander costs less periodontal support than a tooth-borne one
E It is contraindicated once the third molars have erupted

846 A 24-year-old man has a bilateral posterior crossbite. The upper buccal segments are already tipped buccally and the lower lingually, so that the crossbite looks milder than the underlying skeletal relationship. Which single consequence follows for planning?

A The compensation indicates that the crossbite is dental rather than skeletal in origin
B The dental compensation means less expansion will be required than the crossbite suggests
C The compensation makes conventional tooth-borne expansion the treatment of choice
D The compensation is irrelevant, since planning is based on the crossbite as it presents
E The true transverse deficiency is larger than the crossbite suggests

847 A 45-year-old woman has had a bilateral posterior crossbite since childhood, with no displacement on closing, no attrition, a stable periodontium and no symptoms. She attends for a routine examination. Which single management is most appropriate?

A Refer for surgically assisted expansion so that the transverse relationship is normalised
B Explain the finding and monitor, a crossbite without displacement needing no correction
C Fit a quadhelix, which corrects the crossbite without surgery at any age
D Selectively grind the buccal cusps to eliminate the crossbite contacts
E Fit an occlusal splint to prevent the crossbite from causing joint symptoms

Block Y — Vertical problems: deep bite and open bite (Q848–877)

Volume I: Topic 11 (Overbite and overbite reduction); Chapter 13, pp. 928–945
Volume II: **§25** — read it again if you score under 70%
This block: 30 questions (7 of them true/false) · 58 marks · allow 38 minutes

848 A 15-year-old girl has upper central incisors overlapping the lower incisors by rather more than half the height of the lower crowns, and the lower incisal edges contact the palatal mucosa behind the upper incisors. The mucosa is healthy. Which single description is most accurate?

A An increased and incomplete overbite, since the incisors do not meet tooth to tooth
B An average overbite, coverage of one third to one half of the lower crown being normal
C A traumatic overbite, which is defined by contact with the palatal mucosa alone
D An increased overbite, complete to the mucosa but not by itself traumatic
E An increased overbite complete to tooth, the mucosal contact being incidental

849 A 28-year-old man has a complete overbite with the lower incisors contacting the palatal gingivae behind the upper central incisors, where the tissue is inflamed and stripped. The labial gingivae of the lower incisors are also receding. Which single statement is most accurate?

A It is a traumatic overbite, and reduction is indicated to arrest the tissue damage
B It is a normal complete overbite needing no more than improved oral hygiene
C The palatal stripping is periodontitis and will respond to scaling and polishing
D The recession is unrelated to the overbite and reflects toothbrush trauma alone
E Overbite reduction is contraindicated in an adult because the result is unstable

850 A colleague lists ways in which an increased overbite may be reduced. Regarding each proposition:

TRUE OR FALSE — judge each statement independently; any number may be true

A Extrusion of the buccal segments is the method of choice in a non-growing adult
B True incisor intrusion is best achieved with a reverse curve in a continuous archwire
C Proclining already proclined lower incisors is a stable way of reducing overbite
D Overbite reduction requires no attention to be paid to the interincisal angle
E A flat anterior bite plane intrudes the lower incisors directly

851 A 12-year-old boy has a Class II division 2 malocclusion with a deep traumatic overbite, a reduced lower anterior face height and a maxillary-mandibular planes angle of 20 degrees. Which single mechanism of overbite reduction is most appropriate?

A True intrusion of the lower incisors with a segmental utility arch of about 60 g
B Proclination of the lower incisors, which advances them out of the deep overbite
C Extrusion of the buccal segments behind an anterior bite plane, with condylar growth
D Extraction of lower second molars, allowing the buccal segments to settle upwards
E Surgical maxillary impaction, which closes the anterior vertical dimension directly

852 A 34-year-old woman has a deep overbite, a maxillary-mandibular planes angle of 34 degrees and an increased lower anterior face height. Which single mechanism of overbite reduction is most appropriate?

A Extrusion of the buccal segments behind an anterior bite plane, as in a child
B A removable appliance with a flat anterior bite plane worn at night only
C Proclination of the lower incisors, which are currently upright at 88 degrees

D True intrusion of the incisors with a utility arch delivering about 60 g in all
E Acceptance of the overbite, since adults tolerate deep bites without consequence

853 A utility arch is to be used to intrude the four lower incisors in an adult. Which single force regimen is most appropriate?

A About 50 g per tooth and 200 g in total, which speeds the intrusion usefully
B About 15 g per tooth and 60 g in total, which limits the risk of root resorption
C About 5 g per tooth and 20 g in total, below the threshold for any root damage
D Force magnitude is immaterial provided a stainless steel wire is used throughout
E About 100 g per tooth and 400 g in total, matching the force used for retraction

854 Regarding the stability of overbite reduction:

TRUE OR FALSE — judge each statement independently; any number may be true

A Houston in 1989 identified an occlusal stop as the condition for a stable result
B An obtuse interincisal angle predisposes the reduced overbite to return
C Overbite reduction achieved by proclining the lower incisors is stable in the long term
D Relapse of a reduced overbite tends to occur in the first year after debond
E A division 2 case treated without correcting the interincisal angle is likely to relapse

855 A 16-year-old girl with a Class II division 2 malocclusion has an interincisal angle of 155 degrees and a complete overbite. Which single treatment objective most affects the long-term stability of overbite reduction?

A Retaining with a bonded lower retainer for a minimum of five years
B Levelling the curve of Spee fully, so that the lower arch finishes flat
C Achieving a Class I molar relationship on both sides at the end of treatment
D Correcting every rotation in the upper labial segment before debonding
E Reducing the interincisal angle by torquing the upper incisor roots palatally

856 A 42-year-old man has localised tooth surface loss of the palatal surfaces of the upper incisors, with no interocclusal space in which to restore them and an otherwise sound posterior occlusion. Which single approach best creates the space required?

A A Dahl appliance or composite build-ups, allowing the posterior teeth to erupt
B Full-mouth rehabilitation carried out at an increased vertical dimension
C Elective devitalisation and post crowns on the upper incisor teeth
D Orthodontic intrusion of the upper incisors with a segmental intrusion arch
E Crown lengthening surgery on the upper incisors to expose more tooth tissue

857 A 9-year-old girl has an asymmetrical anterior open bite of 3 mm, proclined upper incisors, a unilateral posterior crossbite with displacement and a callus on her right thumb. Which single management is most appropriate?

A Correct the posterior crossbite with a quadhelix, leaving the habit to resolve
B Fit a posterior bite plane appliance and close the anterior open bite directly
C Address the digit habit first, then reassess both the open bite and the crossbite
D Refer for assessment of tongue thrust before any orthodontic intervention
E Begin fixed appliances with anterior vertical elastics to close the open bite

858 A 13-year-old boy has an anterior open bite of 4 mm. Regarding its aetiology:

TRUE OR FALSE — judge each statement independently; any number may be true

A An increased planes angle with a backward growth rotation is a skeletal cause
B An adaptive tongue thrust is a primary cause and must be treated before the teeth move

C An open bite produced by a digit habit is symmetrical and centred on the midline
D Nasal obstruction has been shown to be the principal cause of anterior open bite
E An endogenous tongue thrust is associated with a lisp and with relapse after closure

859 A 15-year-old with an anterior open bite is observed to thrust the tongue forwards between the incisors on swallowing. Which single interpretation is best?

A The thrust is endogenous, so closure of the open bite is contraindicated
B The thrust is most likely adaptive and will cease when the open bite closes
C The thrust is irrelevant, since tongue posture has no effect on the dentition
D The thrust proves a neuromuscular disorder requiring speech therapy first
E The thrust indicates a macroglossia for which reduction surgery is indicated

860 A 25-year-old woman has an anterior open bite of 3 mm, a maxillary-mandibular planes angle of 40 degrees and an increased lower anterior face height. She asks for fixed appliances with anterior vertical elastics. Which single reason to decline that plan is strongest?

A Vertical elastics cannot generate enough force to move incisors in an adult
B Vertical elastics would resorb the incisor roots beyond the acceptable limit
C Anterior open bite in an adult cannot be closed by any orthodontic means
D Extruded incisors relapse as they return to where the lips and tongue allow
E The elastics would produce an unacceptable Class III molar relationship

861 A 27-year-old man has an anterior open bite of 4 mm on a long face pattern with an excessive lower anterior face height and no anteroposterior discrepancy. He is unwilling to consider surgery. Which single mechanical approach is most appropriate?

A Intrusion of the upper posterior teeth against buccal and palatal miniscrews
B Extrusion of the upper and lower incisors with anterior vertical elastics
C Extraction of upper and lower second molars to shorten the posterior dentition
D A posterior bite plane worn full time so that the incisors may erupt
E High-pull headgear to the upper molars, worn for 14 hours a day

862 Regarding the mechanics of overbite reduction:

TRUE OR FALSE — judge each statement independently; any number may be true

A A flat anterior bite plane works by freeing the posterior teeth to erupt
B A reverse curve of Spee in a continuous archwire tends to procline the lower incisors
C A utility arch bypasses the buccal segments and loads the incisors alone
D Extrusion of the buccal segments increases the lower anterior face height
E Intrusion carries a greater risk of root resorption than tipping does

863 A 14-year-old boy has a deep overbite with lower incisors retroclined to 84 degrees to the mandibular plane and 3 mm of lower arch crowding. Which single statement about proclining the lower incisors is most accurate?

A Proclination relapses whenever the interincisal angle is left obtuse afterwards
B Proclination has no effect on the overbite and merely relieves the crowding
C Each degree of proclination gains 0.1 to 0.2 mm of space and reduces overbite
D Proclination must be limited to 2 degrees in a growing patient of this age
E Proclination should be preceded by lower first premolar extraction in every case

864 Which single condition best predicts that a reduced overbite will remain reduced?

A The lower arch has been levelled to a completely flat curve of Spee
B The upper incisors have been aligned without any residual rotation
C A bonded retainer has been placed from lower canine to lower canine
D The patient has finished growing before the appliances are removed
E The lower incisor edges occlude against the upper cingulum plateau

865 A 38-year-old woman with a Class II division 2 malocclusion and a deep overbite has a reverse curve of Spee placed in a continuous lower stainless steel archwire. Which single unwanted effect is most likely?

A Molar extrusion rotating the mandible down and back
B Lingual tipping of the lower incisors with loss of arch length
C Forward rotation of the mandible with a reduction in face height
D True intrusion of the lower incisors with consequent root resorption
E Rapid closure of the lower extraction spaces without anchorage loss

866 A 23-year-old man has an anterior open bite of 6 mm, a maxillary-mandibular planes angle of 42 degrees, excessive maxillary vertical development and 5 mm of upper incisor show at rest. Which single surgical plan is most appropriate?

A Mandibular advancement by bilateral sagittal split osteotomy alone
B Anterior segmental mandibular osteotomy to raise the lower labial segment
C Le Fort I with posterior maxillary impaction, letting the mandible autorotate
D Le Fort I osteotomy with maxillary advancement and no vertical change
E Genioplasty to reduce the lower face height without moving either jaw

867 Regarding posterior open bite:

TRUE OR FALSE — judge each statement independently; any number may be true

A Posterior open bite is more common than anterior open bite
B Lateral tongue thrust and primary failure of eruption are both recognised causes
C It resolves reliably once any associated anterior open bite has been closed
D Primary failure of eruption does not respond to orthodontic traction
E Ankylosis of a deciduous molar is unrelated to a localised posterior open bite

868 An 11-year-old girl with a deep overbite has worn an upper removable appliance carrying a flat anterior bite plane for nine months, and the overbite has reduced. Which single mechanism best explains the change?

A Resorption of the alveolar bone supporting the upper incisor roots
B True intrusion of the upper incisors by the force applied by the bite plane
C Backward rotation of the mandible increasing the lower anterior face height
D Proclination of the lower incisors against the acrylic of the anterior bite plane
E Posterior eruption with relative intrusion of the incisors as the face grows

869 A 17-year-old boy has upper incisors overlapping the lower incisors vertically by about two thirds of the lower crown height, but the lower incisal edges contact neither the upper incisors nor the palatal mucosa. Which single term is most accurate?

A An increased and complete overbite, since more than half the crown is overlapped
B An increased and incomplete overbite, since nothing is contacted by the lower edges
C An average and complete overbite, since the coverage lies within normal limits
D An anterior open bite, since there is no contact between the incisor teeth
E A traumatic overbite, since a deep overbite of this size damages the tissues

870 A 15-year-old girl with a Class II division 2 malocclusion, a deep traumatic overbite and retroclined upper central incisors is to be treated with fixed appliances. Which single sequence is most appropriate?

A Reduce the overbite first with a lower reverse curve, then align the upper arch
B Extract upper first premolars first, then align with a light round archwire
C Retract the upper canines first to create space for the central incisors
D Procline the upper central incisors first, then reduce the overbite and retract
E Align the lower arch to completion before placing any upper attachments

871 Regarding the Dahl concept in the management of localised anterior tooth wear:

TRUE OR FALSE — judge each statement independently; any number may be true

A Space is gained by a combination of posterior eruption and anterior intrusion
B The concept requires a general anaesthetic and a full-mouth rehabilitation
C Re-establishment of posterior occlusal contact usually takes several months
D It is less predictable where the posterior teeth are heavily restored or ankylosed
E Patients must be warned that the posterior teeth are out of contact at first

872 A 12-year-old boy has an anterior open bite of 3 mm, a long face, incompetent lips, habitual mouth breathing and grossly enlarged tonsils. Which single course of action is most appropriate first?

A Begin fixed appliances with anterior vertical elastics to close the bite now
B Fit a tongue crib to prevent the tongue from maintaining the open bite
C Refer for ear, nose and throat assessment before orthodontic treatment
D Plan a Le Fort I impaction once growth is complete and defer everything
E Reassure, since lymphoid tissue involutes and the open bite will close itself

873 Long-term follow-up of anterior open bites closed by extrusive mechanics reports relapse in a substantial proportion of patients within three years. Which single explanation is best?

A Extruded incisors return towards the position the lips and tongue dictate
B Relapse reflects inadequate retainer wear rather than the mechanics that were used
C Extrusion is inherently a slower movement and so is never fully expressed
D The open bite recurs because the condyles continue to grow into adult life
E The alveolar bone around extruded incisors resorbs and the teeth re-erupt

874 A 48-year-old man with a treated but reduced periodontium, bone loss of about one third of root length and a deep traumatic overbite requires overbite reduction. Which single approach is most appropriate?

A No orthodontic treatment, a reduced periodontium precluding tooth movement
B Extrusion of the buccal segments behind an anterior bite plane worn at night
C Proclination of the lower incisors so as to advance them out of the deep bite
D Extraction of the lower incisors with prosthetic replacement of the segment
E Light true intrusion of the incisors, once the periodontal disease is stabilised

875 Regarding an increased overbite in a 30-year-old patient:

TRUE OR FALSE — judge each statement independently; any number may be true

A An overbite complete to the mucosa is by definition a traumatic overbite
B Stripping of the palatal gingivae behind the upper incisors indicates trauma
C Labial gingival recession on the lower incisors may accompany a traumatic overbite
D An increased overbite is always accompanied by a reduced planes angle
E Reduction is indicated on aesthetic grounds alone whenever the overbite exceeds 4 mm

876 A 19-year-old man has finished treatment for a Class II division 2 malocclusion, with the overbite reduced to 3 mm and the interincisal angle reduced to 133 degrees. Which single retention plan is most appropriate?

A A bonded lower retainer alone, the corrected angle making upper retention needless
B No retention at all, since the interincisal angle has now been corrected
C Vacuum-formed retainers in both arches for six months and no longer
D An upper removable retainer incorporating a flat anterior bite plane at night
E A functional appliance worn at night for a further two years

877 A 7-year-old boy has an anterior open bite of 2 mm with a continuing thumb habit, no crossbite and an average skeletal pattern in all three dimensions. Which single course is most appropriate?

A Fit a fixed palatal crib at this visit so that the habit stops immediately
B Explain the cause to the child and parents, with a review in six months
C Begin a course of myofunctional appliance therapy directed at the tongue
D Take a lateral cephalogram to quantify the vertical skeletal pattern first
E Refer for speech therapy to address the abnormal swallowing pattern

Block Z — TMD, condylar trauma, the airway and myofunctional therapy (Q878–907)

Volume I: Topic 18 (Malocclusion and TMD); Chapter 2, pp. 102–105
Volume II: **§26** — read it again if you score under 70%
This block: 30 questions (8 of them true/false) · 62 marks · allow 38 minutes

878 Epidemiological surveys report that about three quarters of an adult population have at least one sign of temporomandibular disorder, while about one third report at least one symptom. Which single implication for orthodontic practice is best supported?

A Three quarters of orthodontic patients require joint treatment before appliances
B The gap between signs and symptoms shows the surveys were poorly conducted
C The figures show that malocclusion is the principal cause of these signs
D Screening for signs is unnecessary, since so few of them ever become symptoms
E Signs are common and often silent, so a finding must be read against that base rate

879 Regarding the history of the claim that orthodontic treatment causes temporomandibular disorder:

TRUE OR FALSE — judge each statement independently; any number may be true

A Costen described a syndrome relating ear and sinus symptoms to the dentition in the 1930s
B The gnathological movement of the 1970s held that occlusion was irrelevant to joint function
C In the late 1980s functional orthodontists attributed the disorder to premolar extraction
D Brimm v Malloy in 1987 was decided in favour of the orthodontist
E Studies reported by the American Association of Orthodontists in 1992 found no association

880 In 1987 a jury awarded 850,000 US dollars against an orthodontist in a case alleging that treatment had caused a temporomandibular disorder. Which single lesson is best drawn from that verdict?

A The verdict established that orthodontic treatment causes temporomandibular disorder
B A legal verdict reflects the evidence put to a jury rather than a scientific consensus
C The verdict means orthodontists must obtain written consent for joint symptoms
D The verdict shows that extraction of premolars should now be abandoned
E The verdict was overturned on appeal and therefore carries no lesson at all

881 A 14-year-old girl is about to begin fixed appliance treatment. She reports occasional clicking of the left joint, with no pain and no limitation of opening. Which single course is most appropriate?

A Defer treatment until the clicking has resolved with a course of splint therapy
B Refer to an oral surgeon for arthroscopy before any appliances are placed
C Proceed without recording the finding, since a painless click is of no importance
D Record the finding and the range of movement, explain the natural history, and proceed
E Advise that treatment will cure the clicking by correcting her occlusion

882 A patient asks whether orthodontic treatment will protect her daughter from developing jaw joint problems in later life. Which single answer best reflects the evidence?

A Orthodontic treatment neither causes nor prevents temporomandibular disorder
B Orthodontic treatment substantially reduces the risk by correcting the occlusion
C Orthodontic treatment raises the risk, particularly where premolars are extracted
D The question cannot be answered, since no studies have yet addressed it directly
E Orthodontic treatment prevents disorder only where a Class I occlusion results

883 Regarding the relationship between orthodontic treatment and temporomandibular disorder:

TRUE OR FALSE — judge each statement independently; any number may be true

A Prospective studies have demonstrated that premolar extraction causes the disorder
B Weak statistical associations between occlusal features and the disorder are reported
C Symptoms commonly begin in late adolescence, when orthodontic treatment is usually given
D Orthodontic treatment is an accepted treatment for an established disorder
E Absence of evidence of harm has been read by some as evidence that no harm occurs

884 Symptoms of temporomandibular disorder most often begin in late adolescence and early adulthood, which is also when most orthodontic treatment is carried out. Which single consequence for research is most important?

A Cohort studies of orthodontic patients cannot be conducted in this age group
B The coincidence means randomised trials of orthodontic treatment are unethical
C A temporal association will arise between treatment and symptoms without causation
D Symptoms reported during treatment should be attributed to the appliance by default
E Only case reports can capture the relationship between the two phenomena

885 A 16-year-old boy in the eighth month of fixed appliance treatment develops pain in the right joint and limitation of opening to 30 mm. Which single course is most appropriate?

A Remove the appliances immediately and abandon orthodontic treatment altogether
B Fit a stabilisation splint over the appliances and continue with heavy elastics
C Continue mechanics unchanged, since orthodontics does not cause joint problems
D Pause active mechanics, manage conservatively with analgesia and diet, and review
E Refer for arthrocentesis before any further orthodontic tooth movement is done

886 An adult with a long history of temporomandibular pain asks for orthodontic treatment specifically in order to cure it. Which single response is best?

A Agree to treat, since correcting the occlusion resolves most such cases
B Explain that orthodontics is not a treatment for it and manage the pain separately
C Decline all orthodontic care while any joint symptom remains present
D Agree to treat, provided the patient accepts that the outcome may be uncertain
E Offer occlusal equilibration first, then orthodontics to consolidate the result

887 Kandasamy in 2022 set out practical guidance on temporomandibular disorder for the orthodontist. Regarding that guidance:

TRUE OR FALSE — judge each statement independently; any number may be true

A Joint signs and symptoms should be screened for and documented before treatment begins
B Patients should be told that these conditions fluctuate and often settle spontaneously
C Orthodontic treatment should be offered as a definitive treatment for the disorder
D Active mechanics should be paused if symptoms develop during treatment

E Initial management of symptoms should be conservative and reversible

888 A 5-year-old boy fell on his chin and sustained bilateral intracapsular condylar fractures, which were treated conservatively. Which single sequela should his parents be warned about and monitored for?

A Excessive mandibular growth producing a Class III malocclusion in adolescence
B Premature loss of the deciduous molars from disturbed occlusal loading
C Permanent anaesthesia in the distribution of the inferior alveolar nerve
D Accelerated eruption of the permanent dentition on both affected sides
E Ankylosis with progressive limitation of opening and deficient mandibular growth

889 A 13-year-old girl with juvenile idiopathic arthritis has a retrognathic mandible, an anterior open bite, a steep mandibular plane and flattening of both condylar heads on imaging. Which single explanation is best?

A Inflammatory destruction of the condyles has restricted mandibular growth
B A digit habit has produced the open bite and the condylar change is incidental
C Condylar hyperplasia has rotated the mandible backwards on both sides
D The changes reflect long-term corticosteroid effects on the alveolar bone
E The mandible is normal and the maxilla has grown excessively downwards

890 Regarding progressive condylar resorption:

TRUE OR FALSE — judge each statement independently; any number may be true

A It affects men more often than women
B It is most often recognised in the seventh decade of life
C A high mandibular plane angle and a retrognathic mandible are recognised risk factors
D It may present after orthognathic surgery as a progressively increasing open bite
E Juvenile idiopathic arthritis is one recognised cause of condylar destruction

891 Obstructive sleep apnoea affects about 2 per cent of children. Which single implication for orthodontic practice is best supported?

A It is common enough that every orthodontic patient should have polysomnography
B It is too rare to justify asking about snoring or daytime sleepiness at all
C It is uncommon but important, so suggestive symptoms warrant referral onwards
D Orthodontic appliances should be offered as first-line treatment to such children
E A prevalence of 2 per cent means adenotonsillectomy is rarely of any benefit

892 A 9-year-old boy has a long narrow face, an anterior open bite, incompetent lips, a high palate and habitual mouth breathing. His mother attributes the appearance to the mouth breathing and asks for expansion to correct it. Which single response is best?

A Mouth breathing is the proven cause and expansion will reverse the facial pattern
B The facial pattern and the mouth breathing are associated, but causation is unproven
C The pattern is entirely genetic and the airway has no bearing on it whatever
D Expansion is contraindicated in any child who habitually breathes through the mouth
E Adenotonsillectomy will normalise the facial pattern within about twelve months

893 A patient has read that extracting premolars narrows the airway and causes sleep apnoea, and asks whether her planned extractions are safe. Which single statement best reflects the evidence?

A Extraction has been shown to reduce the oropharyngeal airway and to cause apnoea
B Extraction enlarges the airway by allowing the tongue to move forwards in the arch
C The question is untestable, since airway dimensions cannot be measured reliably

D No clinically meaningful change in airway function follows premolar extraction
E Extraction narrows the airway only where four first premolars are removed

894 Regarding rapid maxillary expansion and the airway:

TRUE OR FALSE — judge each statement independently; any number may be true

A Expansion increases the width of the nasal floor measurably
B Expansion is an established first-line treatment for paediatric obstructive sleep apnoea
C Reduced nasal resistance after expansion reliably converts mouth breathers to nasal breathers
D Any effect on the airway is an additional consideration rather than an indication in itself
E Expansion has been shown to abolish apnoea and hypopnoea events in most treated children

895 A 7-year-old girl snores every night, is witnessed to stop breathing briefly during sleep, wakes unrefreshed and is inattentive at school. Her tonsils are grossly enlarged. Which single course is most appropriate?

A Refer for sleep assessment and an ear, nose and throat opinion now
B Fit a mandibular advancement appliance to be worn every night
C Reassure the parents, since snoring in childhood is entirely benign
D Review in twelve months, since lymphoid tissue involutes with age
E Begin rapid maxillary expansion, which will resolve the obstruction

896 A parent asks about myofunctional therapy, advertised as correcting malocclusion and improving the airway without appliances. Which single statement best reflects the evidence?

A It has no effect of any kind and should never be recommended in any circumstance
B Randomised trials show it corrects skeletal discrepancies as effectively as appliances
C The evidence is of low quality and does not support it as an alternative to orthodontics
D It is established as the treatment of choice for anterior open bite in children
E It is effective only when begun before the permanent incisors have erupted

897 An 11-year-old girl still sucks her thumb and says she wants to stop. She has an anterior open bite of 3 mm and a posterior crossbite with a displacement. Which single plan is most appropriate?

A Rely on explanation alone, since habits cease by twelve years of age
B Close the open bite with vertical elastics while the habit continues
C Apply bitter-tasting nail preparations as the sole intervention for a year
D Defer all treatment until the permanent dentition is fully established
E Fit a fixed palatal crib as an aid, with expansion for the crossbite

898 Regarding condylar fracture in childhood:

TRUE OR FALSE — judge each statement independently; any number may be true

A Intracapsular fractures in young children carry the greatest risk of ankylosis
B Deficient growth on the affected side produces antegonial notching
C Early mobilisation is important in reducing the risk of ankylosis
D Open reduction and internal fixation is the standard treatment in young children
E The condyle has a considerable capacity to remodel in the growing child

899 A retrospective study compares 200 adults who had orthodontic treatment as teenagers with 200 who did not, and finds a higher rate of joint clicking in the treated group. Which single criticism is most important?

A Those who sought orthodontic treatment may differ systematically from those who did not
B Joint clicking is too rare an outcome to be studied in samples of this size
C Retrospective studies cannot establish the temporal order of exposure and outcome
D Two hundred in each arm gives insufficient power for a comparison of this kind
E Clicking should have been measured by magnetic resonance imaging rather than clinically

900 A 25-year-old woman with pre-existing intermittent joint pain wishes to have orthodontic treatment for crowding. Which single step is most important before treatment begins?

A Obtain magnetic resonance imaging of both joints as a medico-legal baseline
B Insist the symptoms be completely resolved before any appliance is fitted
C Record the symptoms and range of movement, and document the discussion
D Warn her that orthodontic treatment is likely to worsen the joint symptoms
E Advise that the treatment will probably relieve the joint symptoms as well

901 Which single statement about the natural history of temporomandibular disorder is most useful in counselling a patient before orthodontic treatment?

A Symptoms progress inexorably to degenerative joint disease if left untreated
B Symptoms once established are permanent and require lifelong splint therapy
C Symptoms are strongly predictive of the need for later joint replacement
D Symptoms resolve permanently as soon as the occlusion has been corrected
E Symptoms fluctuate over time and most improve without invasive treatment

902 Regarding assessment of the airway by the orthodontist:

TRUE OR FALSE — judge each statement independently; any number may be true

A A lateral cephalogram gives a reliable measure of airway volume
B Snoring, witnessed apnoeas and daytime sleepiness are the symptoms to ask about
C Adenoidal size on a cephalogram correlates well with the apnoea-hypopnoea index
D The orthodontist should make the diagnosis of obstructive sleep apnoea directly
E Polysomnography remains the reference standard for diagnosis

903 A systematic review of airway change after premolar extraction reports small and inconsistent dimensional changes with no demonstrated functional consequence. Which single conclusion is best supported?

A Extraction should be avoided wherever a non-extraction plan is remotely possible
B Extraction decisions should continue to be made on dental and skeletal grounds
C Airway dimension should be the deciding factor in every extraction decision
D The review proves that extraction enlarges the airway in most of those treated
E The inconsistency means no conclusion of any kind can be drawn from the data

904 A 30-year-old man with myofascial pain and no joint noise wants fixed appliances for crowding. His pain is worst on waking and he reports clenching at night. Which single first step is most appropriate?

A Immediate orthodontic treatment, since correcting the occlusion stops clenching
B Occlusal equilibration to remove the interferences that provoke the clenching
C Referral for arthroscopy to exclude an internal derangement of the joint
D Conservative management with reassurance, jaw exercises and a stabilisation splint
E Muscle relaxant injections into the masseter before treatment is considered

905 A 48-year-old man with mild obstructive sleep apnoea confirmed on polysomnography cannot tolerate continuous positive airway pressure. Which single option is most appropriate to discuss?

A Rapid maxillary expansion, which enlarges the nasal airway in the adult
B Fixed appliance treatment to procline the upper incisors and widen the arch
C A custom mandibular advancement appliance worn nightly, with dental monitoring
D Extraction of the lower second molars to allow the tongue to move forwards
E Myofunctional therapy alone, which resolves apnoea in most adult patients

906 A colleague makes the following assertions in a lecture on occlusion and the temporomandibular joints. Regarding each:

TRUE OR FALSE — judge each statement independently; any number may be true

A A slide of more than 2 mm from retruded contact to intercuspation requires equilibration
B Failure to finish a case in centric relation is a recognised cause of the disorder
C Orthodontists should routinely mount study casts on a fully adjustable articulator
D Weak statistical associations between occlusal variables and symptoms are reported
E A patient developing symptoms during treatment should be told the appliance caused them

907 Which single form of words best summarises the evidence on orthodontics and temporomandibular disorder for a consent discussion with the parent of a 14-year-old?

A Treatment neither causes nor prevents joint problems, which are common at this age
B Treatment will protect the joints by establishing a stable and balanced occlusion
C Joint problems are so rare that they need not be mentioned at consent at all
D Treatment carries a small but definite risk of causing permanent joint damage
E Any joint symptom arising during treatment will be the result of the appliances

Block AA — Adult orthodontics and orthognathic surgery (Q908–937)

Volume I: Chapters 13 and 15 (adult treatment; orthognathic surgery)
Volume II: **§27** — read it again if you score under 70%
This block: 30 questions (8 of them true/false) · 62 marks · allow 38 minutes

908 A 47-year-old woman with treated and stable periodontitis has 40 per cent horizontal bone loss around the upper incisors, which have drifted labially with spacing. Probing depths are 3 mm or less throughout and there is no bleeding. Which single statement best explains how her reduced periodontal support alters the mechanics of retracting those incisors?

A The centre of resistance moves apically, so a substantially heavier force is needed before any movement begins
B The centre of resistance moves incisally, so a given force at the bracket produces more bodily movement
C The centre of resistance is unaltered, because it sits at the junction of the coronal and middle thirds of the root in every tooth
D The centre of rotation moves apically, so a retraction force produces intrusion rather than tipping
E The centre of resistance moves apically, so a given force at the bracket produces a larger moment and more crown tipping

909 A 38-year-old man asks how his treatment will differ from that of his 13-year-old daughter, who is in fixed appliances. Regarding orthodontic treatment in the adult:

TRUE OR FALSE — judge each statement independently; any number may be true

A Growth modification is not available, so a skeletal discrepancy must be camouflaged or corrected surgically
B Reduced alveolar bone turnover means that tooth movement is typically slower to initiate
C Adults report more discomfort in the first week than adolescents and are more likely to contact the practice about it
D Because adults do not grow, the response of the periodontal ligament to force is qualitatively different from a child's
E Adult motivation is usually extrinsic, driven by a parent or by school, so compliance is less predictable

910 A 52-year-old man lost the lower right first molar eight years ago. The second molar has tipped mesially so that its crown obstructs the space, its mesial marginal ridge lies 4 mm below the distal, and there is a 6 mm pocket mesially. An implant is planned. Which single approach to uprighting is most appropriate?

A No uprighting, because distal crown tipping will deepen the existing mesial pocket further
B Extraction of the second molar and prosthetic restoration of the quadrant, since uprighting is unpredictable at this age
C Mesial root movement of the second molar, so that the crown position and the existing space dimension are preserved
D Distal crown tipping with a segmental uprighting spring, accepting some extrusion and adjusting the occlusal surface afterwards
E Placement of the implant first, uprighting the molar against it as absolute anchorage at the same visit

911 A 44-year-old woman fractures the upper right central incisor obliquely, the palatal margin lying 3 mm subgingivally at the level of the alveolar crest. The tooth is root treated, the root is 14 mm long and the periodontium is sound. The restorative colleague needs 2 mm of sound tooth above the crest for a ferrule. Which single option best preserves both the tooth and the gingival architecture?

A Orthodontic extrusion of about 3 mm without fibreotomy, allowing the crest and gingival margin to follow the tooth coronally
B Orthodontic extrusion of about 3 mm with light continuous force, with supracrestal fibreotomy repeated during the movement
C Extraction and immediate implant placement, since the fracture already extends to the level of the crest
D Restoration over the subgingival margin with monitoring, since biological width re-forms around a well-finished margin
E Surgical crown lengthening of the incisor alone, removing palatal bone to expose the fracture margin

912 An adult has lost the lower left first molar and the upper left first molar has over-erupted by 3 mm, its cusps approaching the lower ridge and preventing any restoration. The upper molar is vital, unrestored and periodontally sound. Which single approach is most appropriate?

A A lower partial denture with a reduced pontic height, leaving the over-erupted upper molar untouched
B Extraction of the over-erupted molar and prosthetic restoration of both arches, since intruded molars relapse completely
C Intrusion using an intrusion arch anchored on the upper labial segment, accepting some proclination of the incisors
D Reduction of the crown by 3 mm with elective root canal treatment, since molar intrusion is unpredictable in adults
E Intrusion using two palatal miniscrews together with a buccal miniscrew, so that the force is balanced across the tooth

913 A 24-year-old woman with a Class III skeletal pattern, a reverse overjet of 4 mm and lower incisors at 78 degrees to the mandibular plane is listed for bimaxillary surgery. Regarding her presurgical orthodontic phase:

TRUE OR FALSE — judge each statement independently; any number may be true

A The lower incisors should be proclined towards 90 degrees, which increases the reverse overjet before surgery
B Decompensation allows the surgeon to move the jaws by the full amount the skeletal discrepancy requires
C The arches should be coordinated in width so that the segments fit together at the planned occlusion
D Further camouflage of the incisors during this phase reduces the surgical movement and so improves stability
E The patient should be warned that her appearance and her bite will worsen before the operation

914 A 21-year-old man has a reverse overjet of 2 mm, an ANB of minus 1 degree, lower incisors at 84 degrees to the mandibular plane and a straight to mildly concave profile that does not trouble him. He wants his crossbite and his crowding corrected. Which single plan is most appropriate?

A Mandibular setback alone, because a negative ANB identifies a mandibular excess that requires surgical correction
B Fixed appliances with lower premolar extractions, because the discrepancy lies within the envelope of orthodontic movement
C No active treatment, because correcting the crossbite would decompensate the incisors and destabilise the result
D Maxillary protraction with a bone-anchored facemask, because skeletal change remains achievable at this age
E Bimaxillary orthognathic surgery, because a reverse overjet in an adult is beyond correction by tooth movement alone

915 A 23-year-old woman shows 8 mm of upper incisor and 4 mm of gingiva on smiling, has a 9 mm overjet, an increased lower face height and lip incompetence. The cephalogram shows vertical maxillary excess with a backwardly rotated mandible. Which single surgical plan best addresses the deformity?

A Reduction genioplasty with vertical shortening, since the increased lower face height is the presenting problem
B Bilateral sagittal split advancement of the mandible alone, correcting the overjet where the skeletal deficiency lies
C Le Fort I advancement of the maxilla, bringing the upper lip forwards and improving the incisor relationship
D Le Fort I osteotomy with superior repositioning of the maxilla, allowing the mandible to autorotate forwards
E Anterior maxillary segmental setback, reducing the overjet and the gingival display in a single procedure

916 A 19-year-old man is being consented for a bilateral sagittal split osteotomy to advance the mandible 8 mm. He asks what is most likely to go wrong. Which single statement best reflects the risk he should be told about?

A Altered sensation of the lower lip and chin, near universal in the early weeks and persisting in a minority at one year
B Unfavourable splitting of the buccal plate, which is common but has no bearing on fixation or recovery
C Loss of vitality of the lower incisors, which follows the split in approximately half of all patients
D Permanent weakness of the lower lip from facial nerve injury, which follows about one in five sagittal splits according to older case series
E Non-union of the segments, which is the commonest complication and usually requires a second operation

917 Regarding the hierarchy of stability of orthognathic procedures:

TRUE OR FALSE — judge each statement independently; any number may be true

A Superior repositioning of the maxilla is among the most stable of the standard movements
B Mandibular advancement is more predictable than mandibular setback
C Surgical widening of the maxilla is the least stable of the common movements
D Downward repositioning of the maxilla relapses readily, because the soft tissues resist lengthening of the face

E Rigid internal fixation has improved the stability of mandibular setback compared with wire fixation

918 A 26-year-old woman has a Class I incisor relationship, a well-positioned maxilla and a retrusive chin point with a deep labiomental fold and mentalis strain on lip closure. Her occlusion is sound and she does not want it altered. Which single procedure is most appropriate?

A Bilateral sagittal split advancement, which corrects the skeletal profile at the level of the true deficiency
B Alloplastic chin implant, which avoids an osteotomy and produces an immediate change in contour
C Advancement genioplasty, which moves the chin point and its soft tissue without disturbing the occlusion
D Orthodontic proclination of the lower incisors, which projects the lower lip and disguises the chin deficiency
E Le Fort I advancement, which projects the midface and improves support for the lower lip

919 A 28-year-old man has a 9 mm transverse maxillary deficiency with bilateral posterior crossbite and no mandibular displacement. Imaging confirms that the midpalatal suture is fused. Which single approach is most appropriate?

A Acceptance of the crossbite with buccal crown torque on the upper molars as dental camouflage
B Slow expansion with a quadhelix at 0.25 mm twice weekly, which is better tolerated by adult patients
C Conventional rapid maxillary expansion at one turn a day, accepting a longer period of activation
D Surgically assisted rapid palatal expansion, releasing the resisting buttresses before the appliance is activated
E Le Fort I osteotomy in three segments, which is the only means of widening a maxilla with a fused suture in adult patients

920 A 25-year-old woman with a Class III deformity is offered a surgery-first approach. Which single statement best describes when this is appropriate?

A Where crowding is severe, because the surgeon can create the space required by segmenting the maxilla
B Where the incisors are minimally compensated and the arches already coordinate, so the postsurgical phase can finish the occlusion
C Where the patient declines orthodontics, because postsurgical appliances can be dispensed with altogether
D Where the discrepancy is large, because the regional acceleratory phenomenon then guarantees a stable result
E Where the patient wants a shorter overall treatment, since the presurgical orthodontics is simply moved to after the operation

921 A 19-year-old woman requests treatment for what she calls a crooked smile. She has a Class I incisor relationship with 1 mm of lower labial segment crowding, has had two courses of aligner treatment elsewhere, checks her teeth in a hand mirror repeatedly during the consultation and says she avoids being photographed. Regarding this presentation:

TRUE OR FALSE — judge each statement independently; any number may be true

A Her preoccupation is proportionate to the clinical finding and requires no further consideration
B Body dysmorphic disorder is rare in orthodontic populations, affecting well under 1 per

cent of patients

C Repeated treatment elsewhere with persisting dissatisfaction is reassuring against a psychological cause

D Undertaking the aligner treatment she requests is the appropriate way to resolve her dissatisfaction

E Referral for psychological assessment before any further irreversible treatment is appropriate

922 A 22-year-old woman with a high mandibular plane angle, an anterior open bite and a retrognathic mandible undergoes a 9 mm sagittal split advancement. Eighteen months later the open bite has returned and the chin point has moved backwards. Which single explanation is most likely?

A A persistent tongue thrust, which has re-established the anterior open bite by dentoalveolar means

B Failure of rigid fixation, which allows the distal segment to slide backwards during the first weeks of healing

C Orthodontic relapse of the incisors, which has reopened the bite at the dentoalveolar level alone

D Continued downward growth of the maxilla, which has rotated the mandible backwards since the operation

E Progressive condylar resorption, to which her facial pattern, her sex and the size of the advancement all predispose

923 A 45-year-old man has generalised periodontitis with bleeding on probing at 60 per cent of sites and 6 to 7 mm pockets in the molar regions. He requests fixed appliances to align his drifted upper incisors. Which single course of action is most appropriate?

A Complete periodontal therapy and demonstrate stability at a re-evaluation before any appliance is placed

B Align the incisors with a removable appliance, since removable appliances carry no periodontal risk

C Decline orthodontic treatment permanently, since periodontitis is a contraindication to tooth movement

D Place appliances now and treat the periodontium in parallel, since alignment itself improves plaque control

E Place appliances now but use only light forces, since light forces do not aggravate periodontal breakdown

924 A 39-year-old woman with a thin gingival biotype has 2 mm of recession on the lower left central incisor with 1 mm of attached gingiva. There is 5 mm of lower labial segment crowding to be relieved. Which single statement best guides the plan?

A Proclination is safe here because the recession is already established and does not progress once present

B A free gingival graft beforehand removes the need to consider the direction in which the incisors are subsequently moved

C Relieving the crowding by proclination risks extending the recession, so extraction or interproximal reduction is preferred

D Expansion of the lower intercanine width is preferable, because it avoids labial movement of the incisors

E Recession is caused by plaque alone, so the direction of incisor movement has no bearing on the outcome

925 A 50-year-old woman with treated and stable periodontitis and generalised 50 per cent bone loss is to have fixed appliances to close pathological spacing between her upper incisors. Regarding her management:

TRUE OR FALSE — judge each statement independently; any number may be true

A Forces should be reduced, because the reduced root surface area concentrates stress within the remaining ligament
B Supportive periodontal therapy at intervals of about three months should continue throughout appliance treatment
C A reduced but healthy periodontium cannot tolerate tooth movement and is an absolute contraindication
D A bonded retainer should be avoided in this patient because it prevents effective interdental cleaning
E Root resorption is less likely in this patient, because the periodontium has already been reduced

926 A 33-year-old man lost the upper right lateral incisor to trauma ten years ago. The residual space measures 4.5 mm, the central incisor has tipped distally and the canine mesially, so that the two roots converge into the site. An implant is planned. Which single orthodontic objective matters most before the fixture is placed?

A Overcorrecting the space by 1 mm to allow for relapse after the appliance is removed
B Uprighting the adjacent roots so that they diverge and leave adequate interradicular bone at the site
C Widening the space to 7 mm so that a standard-diameter fixture can be accommodated
D Levelling the gingival margins by extruding the canine before the implant is placed
E Closing the space entirely and recontouring the canine as a lateral incisor, avoiding an implant

927 An adult requires 7 mm of upper incisor retraction after loss of the upper first premolars. No mesial movement of the molars is acceptable because the upper second molars are already mesially inclined, and headgear has been declined. Which single anchorage strategy is most appropriate?

A A palatal arch carried on the second molars, which recruits the extra root surface behind the extraction site
B Class II elastics to the lower arch, which transfer the anchorage demand away from the upper molars
C Sequential retraction of the canines and then the incisors, which halves the force acting at any one moment
D Bilateral buccal miniscrews placed between the second premolar and first molar, retracting directly from the screws
E A transpalatal arch with a Nance button, which unites the posterior segment into a single anchorage unit

928 A 27-year-old woman has a Class I molar relationship, severe bimaxillary dentoalveolar protrusion, lips competent only on effort and no discrepancy between the jaw bases. The joint clinic has agreed on a surgical approach. Which single procedure is most appropriate?

A Anterior maxillary and mandibular segmental osteotomies with premolar extraction, setting both labial segments back
B Le Fort I setback of the whole maxilla combined with a mandibular sagittal split setback
C Advancement genioplasty, increasing chin projection so that the incisor prominence is balanced
D Bilateral sagittal split advancement, bringing the lower lip forwards to balance the profile

E Posterior maxillary segmental osteotomy, intruding the buccal segments so that the mandible autorotates forwards

929 Regarding adjunctive orthodontic treatment for the adult restorative patient:

TRUE OR FALSE — judge each statement independently; any number may be true

A Uprighting a mesially tipped molar can reduce a mesial pseudo-pocket by altering the crest position relative to the tooth
B Forced eruption with fibreotomy moves the tooth while leaving the gingival margin and alveolar crest in place
C Slow extrusion without fibreotomy can be used deliberately to bring bone and soft tissue coronally before implant placement
D Adjunctive treatment aims at a defined restorative objective and need not correct the occlusion as a whole
E Intrusion of an over-erupted molar relapses completely once the opposing tooth has been restored

930 A 41-year-old woman is referred by her general dental practitioner for correction of a 4 mm overjet that the practitioner has noticed. At consultation she says that her teeth have never troubled her and that she has come because she was told to. Which single response is most appropriate?

A Take records and present a plan now, since motivation generally develops once a plan has been seen
B Recommend aligners, since their appearance makes adult patients more willing to accept treatment
C Explain the findings, make clear that the treatment is elective, and offer review rather than starting an appliance
D Begin treatment, since a 4 mm overjet meets the threshold for correction and a referral from her practitioner has been made
E Discharge her to the practitioner with a note that she is unsuitable for orthodontic treatment

931 In a bimaxillary case in which the maxilla is repositioned first, which single function does the intermediate wafer serve?

A It holds the final planned occlusion once both jaws have been moved and is worn with elastics for about six weeks afterwards
B It records the retruded contact position used to mount the study models before planning begins
C It protects the palatal mucosa and the osteotomy site during the early phase of healing
D It allows the vertical position of the mandible to be checked before the sagittal split is made
E It locates the repositioned maxilla against the unoperated mandible, which serves as the reference for the first movement

932 A 19-year-old man and his parents attend a joint clinic and accept a plan of presurgical orthodontics followed by bimaxillary surgery. Which single element of consent is most important to establish before appliances are placed?

A That he understands his bite and his appearance will worsen for 12 to 18 months before surgery corrects them
B That he understands the plates may be removed under a second anaesthetic at a later date
C That he consents to the general anaesthetic and its specific risks, which is done on the day of admission
D That he accepts the fee structure for the orthodontic and the surgical components sepa-

rately

E That he agrees to wear postsurgical elastics for at least six weeks following the operation

933 Regarding complications of orthognathic surgery:

TRUE OR FALSE — judge each statement independently; any number may be true

A Neurosensory disturbance of the inferior alveolar nerve is more likely with larger advancements and in older patients

B An unfavourable buccal plate fracture during a sagittal split requires the operation to be abandoned and repeated later

C Bleeding requiring transfusion is the commonest complication of Le Fort I, occurring in about one third of cases

D Avascular necrosis of a maxillary segment is more likely where the maxilla is segmented and widened at one operation

E Progressive condylar resorption is substantially more common in men than in women

934 A study reports that 20 per cent of patients undergoing mandibular advancement show more than 2 mm of horizontal relapse at two years, and that relapse correlates with the magnitude of the advancement and with a high mandibular plane angle. Which single conclusion is best supported?

A Rigid internal fixation has no bearing on the stability of advancement and may reasonably be omitted

B Relapse after advancement reflects orthodontic settling and is answered by a longer period of retention with a fixed appliance

C Large advancements in high-angle patients should prompt consideration of a two-jaw plan that shares the movement

D The finding applies only to patients operated upon before skeletal maturity was reached

E Mandibular advancement should be abandoned in favour of maxillary surgery in every patient

935 A 16-year-old girl with a marked Class III skeletal pattern asks when her jaw surgery can be carried out. Her growth is being monitored. Which single approach to timing is most appropriate?

A Defer to 25 years, since the mandible continues to grow slowly throughout the third decade in most patients

B Operate now, since the presurgical orthodontics will occupy the remaining growth period in any case

C Operate at 18 years, the age at which mandibular growth is complete in all patients

D Operate now and plan a second procedure if growth continues, which is routine practice in Class III

E Defer until serial cephalograms a year apart show no change and skeletal maturity indicators confirm cessation

936 Regarding the biological differences that affect orthodontic tooth movement in the adult:

TRUE OR FALSE — judge each statement independently; any number may be true

A Higher forces are required in the adult because the periodontal ligament is wider than in the adolescent

B Hyalinisation does not occur in adults, because the alveolar bone is more densely mineralised

C The absence of growth means that the rate of tooth movement in the adult is about half that of an adolescent

D Adults show a lower prevalence of root resorption than adolescents treated with the same mechanics
E Reduced alveolar bone height in the adult shortens the lag phase and speeds the onset of movement

937 A 30-year-old man has an overjet of 11 mm on a severe Class II skeletal base, with a normal lower face height, a retrognathic mandible and a convex profile that he dislikes. Which single statement best explains why camouflage is unsuitable for him?

A Camouflage would require headgear, which is not an acceptable appliance for an adult patient
B Retracting the incisors far enough would flatten an already convex profile and leave the mandibular deficiency untouched
C Camouflage would be unstable because the lower incisors would have to be proclined beyond 120 degrees
D Camouflage is contraindicated in adults because the periodontal ligament will not tolerate retraction of that magnitude
E An overjet of 11 mm exceeds the maximum retraction obtainable from four premolar extractions in any patient

Block AB — Cleft lip and palate and craniofacial syndromes (Q938–968)

> **Volume I:** Chapter 16 (Cleft Lip and Palate, and Syndromes)
> **Volume II:** **§28** — read it again if you score under 70%
> **This block:** 31 questions (8 of them true/false) · 63 marks · allow 39 minutes

938 A newborn boy has a complete cleft of the lip and alveolus on the left with an intact hard and soft palate. Which single embryological event best accounts for this?

A Failure of the palatal shelves to elevate above the tongue and fuse between the sixth and tenth weeks
B Failure of the two medial nasal processes to merge with one another in the midline of the face
C Failure of the mandibular processes to merge, with secondary distortion of the upper lip above them
D Failure of the maxillary process to fuse with the medial nasal process on the left by about the eighth week
E Breakdown of an epithelial seam that had already formed between the lip and the alveolar process

939 A newborn girl has a wide U-shaped cleft of the hard and soft palate with an intact lip and alveolus. Which single statement best explains how her cleft differs from a cleft of the lip and primary palate?

A It results from mechanical obstruction by the tongue alone and therefore carries no genetic component
B It arises from failure of the maxillary and medial nasal processes to meet, in front of the incisive foramen, by about the eighth week
C It arises from failure of the palatal shelves to elevate or fuse between the sixth and tenth weeks, behind the incisive foramen
D It arises later, at about the sixteenth week, when ossification of the hard palate normally begins
E It shares the same embryological origin, differing only in the extent of one continuous fusion failure

940 Regarding the epidemiology of orofacial clefting:

TRUE OR FALSE — judge each statement independently; any number may be true

A Cleft lip with or without cleft palate occurs in about one in 700 live births
B Cleft lip with or without cleft palate is about twice as common in boys as in girls
C About 70 per cent of cases of cleft lip with or without cleft palate are non-syndromic
D Isolated cleft palate occurs in about one in 2000 live births and is about four times commoner in girls
E About half of all cases of isolated cleft palate occur as part of a recognised syndrome

941 A cleft team auditing its outcomes finds that the Veau classification sorts its patients into only four groups, whereas the LAHSHAL system records seven anatomical sites. Which single advantage of LAHSHAL matters most for this audit?

A It records which side is affected and whether each site is complete or incomplete, allowing like to be compared with like
B It distinguishes syndromic from non-syndromic cases, which the Veau classification does not attempt
C It includes submucous clefts, which are excluded from every other classification in current use
D It records the timing of surgical repair alongside the anatomy, linking the operation to the outcome
E It weights the severity of each cleft on a numerical scale, so that groups of patients can be compared statistically

942 A neonate has bilateral complete clefts of the lip, alveolus and palate with a markedly protruding premaxilla. Which single statement best describes the implications for the first year of care?

A The premaxilla realigns spontaneously once the palate is repaired, so early intervention is unnecessary
B The premaxilla should be excised at the time of lip repair, so that closure is achieved without tension
C Lip repair should be deferred until three years of age to allow the premaxilla to reposition itself
D Feeding support is no different from that in a unilateral cleft, since the palatal defect is comparable
E Presurgical orthopaedics or lip strapping may be used to retract and align the premaxilla before lip repair

943 The parents of a newborn with a complete unilateral cleft of the lip and palate are struggling to feed her. Which single piece of advice is most appropriate in the first days of life?

A Delay oral feeding until a presurgical orthopaedic plate has been fitted to seal the palatal defect
B Thicken the feeds and nurse the infant flat, because this reduces regurgitation through the nose
C Pass a nasogastric tube until the palate is repaired, because oral feeding carries a high risk of aspiration
D Feed upright using a squeezable bottle with a wide-based teat, because the infant cannot generate suction
E Continue direct breastfeeding unmodified, because a cleft does not alter the mechanics of feeding

944 Regarding primary lip repair in a child born with a unilateral cleft lip and palate:

TRUE OR FALSE — judge each statement independently; any number may be true

A Repair is usually undertaken between three and six months of age
B The rule of tens requires a weight of 10 kg, a haemoglobin of 10 grams per decilitre and an age of 10 months
C Lip repair precedes palate repair because the palate must be left open until speech has developed
D Early lip repair has no influence on the position of the underlying alveolar segments
E The scar of the lip repair has no bearing on subsequent maxillary growth

945 A child born with a cleft of the secondary palate is reviewed at nine months of age. Which single statement best describes the aims and timing of palate repair?

A Closure at three years of age, allowing maxillary growth to proceed before the scarring of repair is imposed
B Closure between six and twelve months, establishing velopharyngeal function before speech develops, with the middle ear monitored
C Closure between six and twelve months, undertaken principally to make normal feeding possible for the first time
D Closure in two stages with the hard palate deliberately left open until adolescence, which is now the standard practice in most cleft centres
E Closure at six years of age, timed with eruption of the first permanent molars and the start of school

946 A boy with a repaired left unilateral cleft lip and palate is nine years old. The upper left permanent canine is unerupted and a radiograph shows about half of its root formed. Which single statement best explains the timing of his alveolar bone graft?

A Graft now, principally so that the lateral incisor space can be restored without further delay
B Graft in adulthood, immediately before an implant is placed in the lateral incisor position
C Graft after the canine has erupted, so that the consolidated graft is not disturbed by an erupting tooth
D Graft at 14 years of age, once vertical growth of the alveolar process has been completed
E Graft now, so that the canine erupts through the grafted bone and its eruption maintains the graft volume

947 A 10-year-old girl with a repaired right unilateral cleft has an oronasal fistula at the alveolus, a mobile lesser segment and a supernumerary tooth within the cleft. Which single statement best describes what her alveolar bone graft will achieve?

A Restoration of the nasal airway, since the fistula is the principal cause of her nasal obstruction
B Union of the segments, closure of the fistula and bone through which the canine can erupt and teeth can later be moved
C Stabilisation of the segments only, since a fistula requires a separate soft tissue flap to close it
D Immediate placement of an implant in the lateral incisor site once the graft has consolidated
E Correction of her maxillary retrusion, so that orthognathic surgery in late adolescence can be avoided altogether

948 Regarding secondary alveolar bone grafting:

TRUE OR FALSE — judge each statement independently; any number may be true

A Cancellous bone taken from the iliac crest is the most commonly used graft material
B The graft is best placed after the canine has erupted into the cleft, when the site is most stable
C Orthodontic expansion of a collapsed maxillary arch is usually carried out before the graft is placed
D A successful graft allows orthodontic movement of teeth into the grafted bone
E Grafting is contraindicated where the lateral incisor is absent, since no tooth will erupt through it

949 A 12-year-old with a repaired left unilateral cleft lip and palate attends for assessment. Which single finding at the cleft site would be most expected?

A Normal morphology of all the teeth, with crowding as the only abnormality at the site
B An absent upper left canine with a normally formed lateral incisor standing in the line of the arch
C A supernumerary upper left central incisor displacing the lateral incisor towards the palate
D An absent or malformed upper left lateral incisor with a rotated, distally inclined canine beside it
E Absence of the upper left first premolar, with its deciduous predecessor retained in the arch

950 Adults with unrepaired clefts in populations without access to surgery are reported to have near-normal maxillary projection, whereas adults with repaired clefts frequently show maxillary retrusion. Which single conclusion is best supported?

A Much of the midface deficiency seen in cleft patients is iatrogenic, arising from scarring after lip and palate repair
B Unrepaired clefts represent a milder phenotype, which accounts entirely for the difference observed
C Maxillary retrusion in cleft patients is wholly genetic and unrelated to the technique or the timing of repair, even when repair is delayed into later childhood
D The cleft itself is the principal cause of maxillary retrusion, independent of any surgery performed
E Surgical repair should be withheld until facial growth is complete in every child born with a cleft

951 The parents of a newborn with a wide complete unilateral cleft ask about nasoalveolar moulding. Which single statement best represents the current evidence?

A It produces a permanent correction of the nasal deformity and removes the need for secondary rhinoplasty
B It has been shown in randomised trials to improve maxillary growth measured at adolescence
C It can narrow the cleft and improve nasal symmetry before repair, but a lasting advantage over surgery alone is not established
D It should be offered to every infant with a cleft, because it reliably reduces the total number of operations required over the growth period
E It is contraindicated because plates in the neonate interfere with feeding and with speech development

952 Regarding the multidisciplinary care of a child born with a cleft lip and palate:

TRUE OR FALSE — judge each statement independently; any number may be true

A Care is best delivered by the general dental practitioner coordinating referrals as problems arise
B The orthodontist first becomes involved at the time of alveolar bone grafting, so no earlier records are needed
C Speech and language therapy begins only once all primary surgery has been completed
D Audiological review is required only for those children who have had grommets inserted
E Centralisation of cleft care into fewer, higher-volume centres has not been shown to affect outcomes

953 A newborn has a U-shaped cleft of the soft palate, a markedly retrognathic mandible and episodes of airway obstruction when supine. Which single statement best guides immediate management?

A Reassure that no airway risk exists, since the open palate itself provides an alternative airway
B Perform a tracheostomy in every such infant, since the obstruction is progressive and unrelenting
C Repair the cleft palate within the first month, since closing the palate relieves the airway obstruction
D Nurse prone with graded airway support, since the mandible commonly catches up and the obstruction improves
E Advance the mandible by distraction in the first week, since catch-up growth does not occur in this condition

954 A 13-year-old girl has downward-sloping palpebral fissures, colobomas of the lower eyelids, hypoplastic zygomas, bilateral microtia with conductive hearing loss, a steep mandibular plane and an anterior open bite. Which single statement best describes her condition and its orthodontic implications?

A A condition dominated by supernumerary teeth and failed eruption, requiring staged exposure and traction
B Bilateral first and second arch hypoplasia, producing a steep retrognathic mandible and an open bite that usually needs surgery
C A mild skeletal pattern that is largely self-correcting, so alignment with fixed appliances alone is usually sufficient without surgical input
D A condition dominated by hypodontia with conical teeth, in which prosthetic replacement takes precedence
E A unilateral and progressive condition, so serial asymmetry records and growth monitoring are the priority

955 A 14-year-old boy has retained deciduous teeth, only four erupted permanent teeth, multiple supernumeraries on a panoramic radiograph, a hypoplastic maxilla with a relatively prognathic mandible, and can approximate his shoulders in front of his chest. Which single treatment approach is most appropriate?

A Rapid maxillary expansion with a protraction facemask, since the maxillary hypoplasia is the primary problem
B Extraction of all retained deciduous and supernumerary teeth followed by immediate prosthetic replacement with complete dentures
C Observation until 18 years of age, since the permanent teeth in this condition erupt once growth has ceased
D Alignment of the four erupted permanent teeth alone, since traction of unerupted teeth in this condition fails
E Staged removal of supernumeraries and retained deciduous teeth with surgical exposure and orthodontic traction of the permanent teeth

956 A 9-year-old boy has sparse fine hair, reduced sweating with episodes of unexplained fever, dry skin, and multiple absent teeth with conical crowns on those present. Regarding his condition and its management:

TRUE OR FALSE — judge each statement independently; any number may be true

A The commonest form is X-linked, so boys are more severely affected than girls
B Reduced alveolar development follows the absent teeth and limits the bone available for implants

C Removable prostheses and overdentures are usually the mainstay through childhood, remade as growth proceeds
D Implants in the anterior mandible may be considered before growth is complete where hypodontia is severe
E Orthodontic space closure is generally preferable to prosthetic replacement in this condition

957 A 7-year-old girl has bilateral coronal synostosis with exorbitism, midface retrusion, an anterior open bite and a Class III incisor relationship. Her hands and feet are normal. Which single statement best describes her condition and its management?

A Crouzon syndrome, in which midface advancement by Le Fort III or monobloc distraction is the definitive correction
B Hemifacial microsomia, in which the asymmetry accounts for the open bite and a hybrid appliance is indicated
C A non-syndromic Class III, in which maxillary protraction with a facemask will correct the midface deficiency
D Apert syndrome, in which symmetrical syndactyly would be expected and the midface is managed identically
E Pierre Robin sequence, in which the open bite follows the small mandible and resolves with catch-up growth

958 A 15-year-old boy with Down syndrome attends with his carer requesting orthodontic treatment. He has generalised spacing, a Class III incisor relationship, an anterior open bite, a relatively large tongue and generalised periodontitis with 5 mm pockets. Which single consideration should carry most weight?

A Atlantoaxial instability makes anaesthesia and head positioning the dominant concern in any orthodontic plan
B The relative macroglossia causes the open bite, so tongue reduction should precede any appliance therapy
C Periodontal disease is early and aggressive in this group and appliances will accelerate it unless it is controlled and maintainable
D Congenital cardiac disease makes antibiotic prophylaxis before every orthodontic visit the dominant concern
E The Class III pattern is progressive in this condition, so early orthognathic planning should take precedence over other issues

959 An 8-year-old boy has a deficient right ramus with a small, misshapen but present condyle, an occlusal cant with the maxilla higher on the right, a preauricular skin tag and a small right ear. Which single description and management approach is most appropriate?

A Treacher Collins syndrome, managed by bimaxillary orthognathic surgery once facial growth is complete
B Hemifacial microsomia, Pruzansky grade II, managed with a hybrid functional appliance to guide differential vertical growth
C Hemifacial microsomia, Pruzansky grade III, requiring costochondral graft reconstruction of an absent ramus
D Condylar hyperplasia of the left side, managed by high condylectomy once active growth has been confirmed
E Juvenile idiopathic arthritis of the right joint, managed medically with a distraction splint on the affected side until growth is complete

960 Regarding hemifacial microsomia and Goldenhar syndrome:

TRUE OR FALSE — judge each statement independently; any number may be true

A Hemifacial microsomia is the second commonest facial birth anomaly after cleft lip and palate
B Goldenhar syndrome is distinguished by epibulbar dermoids and vertebral anomalies in addition to the facial features
C The condition is bilateral in the great majority of affected children
D The Pruzansky classification grades the severity of the maxillary deficiency
E The facial asymmetry is static and does not increase as the child grows

961 A 6-year-old with a repaired cleft palate has hypernasal speech with audible nasal emission on plosive sounds. Which single interpretation is most likely?

A Conductive hearing loss from middle ear effusion, which has distorted the acquisition of speech sounds and could account for the nasal emission
B An oronasal fistula at the alveolus, which is the usual cause of hypernasality after a palate repair
C Normal development for her age, since nasal escape resolves spontaneously by about eight years
D Velopharyngeal insufficiency, a structural failure of the sphincter that may require a secondary speech operation
E An articulation disorder learned before repair, which speech and language therapy alone will correct

962 A 4-year-old with a repaired cleft palate has recurrent middle ear effusions and a 30 decibel conductive hearing loss. Which single explanation best accounts for this?

A Abnormal insertion and function of the tensor veli palatini leaves the eustachian tube unable to ventilate the middle ear
B Scarring of the eustachian tube orifice at palate repair, which is an avoidable complication of the operation itself
C Nasal regurgitation of milk during infancy, which has caused permanent damage to the middle ear
D Sensorineural loss, which is usual in clefting and is unrelated to eustachian tube function
E An undiagnosed syndrome, which warrants genetic referral before any treatment of the ears

963 A 9-year-old with a repaired unilateral cleft attends before alveolar bone grafting. Regarding his orthodontic management at this stage:

TRUE OR FALSE — judge each statement independently; any number may be true

A Expansion of the collapsed upper arch is usually undertaken before the graft rather than after it
B Fixed appliances should be placed in the lower arch at this visit to begin levelling
C Fissure sealants and intensive prevention are unnecessary, since caries rates match those of the general population
D Any tooth lying in the line of the cleft should be retained until the graft has fully consolidated
E Definitive alignment of the upper labial segment should be completed before grafting

964 A 19-year-old man with a repaired bilateral cleft has maxillary retrusion with a reverse overjet of 8 mm, a heavily scarred palate and borderline velopharyngeal function. Which single plan is most appropriate?

A Orthodontic camouflage by proclination of the upper incisors, which avoids operating in a scarred field
B Bimaxillary surgery deferred to 25 years, since cleft maxillary growth continues into the third decade
C Maxillary advancement by distraction osteogenesis, with formal speech assessment before and after the movement
D Conventional Le Fort I advancement of 8 mm, which is as stable as distraction in a scarred cleft maxilla
E Mandibular setback alone, which avoids moving the maxilla and so protects velopharyngeal function

965 A cleft team audits the age at alveolar bone grafting against the stage of canine root development. Which single measure is the most appropriate indicator that the timing is correct?

A The proportion of grafts placed between the eighth and the tenth birthdays
B The proportion of patients whose canine subsequently erupted without surgical exposure, reflecting adequate graft timing
C The proportion of grafts placed before the first permanent molars had erupted
D The proportion of patients who required a second graft within two years of the first
E The proportion of grafts placed when the canine root is between one half and two thirds formed

966 Regarding the sequence of care for a child born with a complete unilateral cleft lip and palate:

TRUE OR FALSE — judge each statement independently; any number may be true

A Lip repair is usually undertaken between three and six months of age
B Palate repair is usually undertaken between six and twelve months of age
C Alveolar bone grafting is usually undertaken between eight and ten years of age
D Definitive fixed appliance treatment usually begins before the alveolar bone graft is placed
E Orthognathic surgery, where it is required, is usually undertaken at about 14 years of age

967 A 5-year-old is referred before a planned adenoidectomy for recurrent otitis media. Examination shows a bifid uvula, a bluish midline translucency of the soft palate and a notch at the posterior border of the hard palate. Which single action is most appropriate?

A Advise against adenoidectomy and refer to the cleft team, since these three signs indicate a submucous cleft
B Arrange genetic testing before any surgery, since a submucous cleft is always part of a wider syndrome
C Proceed with adenoidectomy, since a bifid uvula is a normal variant of no functional consequence
D Proceed with adenoidectomy but add a pharyngoplasty at the same operation as a precaution
E Repair the soft palate immediately, since a submucous cleft invariably causes speech problems

968 A 9-year-old girl is referred with a single upper right second premolar bearing a sharply demarcated patch of yellow-brown hypoplastic enamel on its buccal surface. Her primary predecessor was extracted at 5 years after an abscess. She is of short stature with a webbed neck. Which interpretation is correct?

A The tooth defect is molar incisor hypomineralisation presenting in a premolar, and requires no further investigation
B Both the tooth and the stature are features of Turner syndrome, which should now be investigated
C The premolar defect follows the earlier infection and the general findings are unrelated to it
D The tooth defect is an early carious lesion which should be restored and the general findings ignored
E Both findings indicate amelogenesis imperfecta with an associated growth disturbance to be investigated

Block AC — Risks of orthodontic treatment and the medical history

(Q969–998)

Volume I: Topics 9 and 10 (Benefits and complications; Medical conditions); Chapters 2 and 7
Volume II: §29 — read it again if you score under 70%
This block: 30 questions (8 of them true/false) · 62 marks · allow 38 minutes

969 A 26-year-old woman asks what orthodontic treatment will actually do for her. Which single benefit is supported by the strongest evidence?

A Prevention of temporomandibular disorder in later adult life
B Improvement in the articulation of speech, particularly of sibilant sounds
C Enlargement of the pharyngeal airway, with a reduction in snoring
D Reduction in the incidence of dental caries over the decade following treatment
E Improvement in dental appearance, and the psychosocial gains that follow from it

970 A patient asks what orthodontic treatment will do for her health as well as for her appearance. Regarding the evidence for the benefits of orthodontic treatment:

TRUE OR FALSE — judge each statement independently; any number may be true

A Reduction of incisor trauma in children with a large overjet is supported by strong evidence, with a risk ratio of 0.58
B Improvement in dental aesthetics is supported by strong evidence
C Improvement in plaque scores is supported by evidence of moderate strength
D Prevention of temporomandibular disorder is supported by strong evidence from randomised trials
E Reduction in the incidence of caries is supported only by weak evidence

971 A systematic review reports a risk ratio of 0.58 for incisor trauma in children receiving early overjet reduction compared with those who do not. A 9-year-old boy has a 10 mm overjet with incompetent lips. Which single interpretation best applies this figure to him?

A A relative reduction of about 42 per cent, which represents the same absolute gain in every child regardless of overjet
B A relative reduction of about 42 per cent, which in a boy with this overjet and lip posture represents a large absolute gain
C An absolute reduction of trauma risk to 0.58 per cent, which is negligible whatever the presentation
D A finding that applies to populations only and that cannot inform this individual boy's decision at all
E A finding that justifies offering early treatment to every child with a Class II division 1 malocclusion

972 A 34-year-old woman with bilateral joint clicking and intermittent masseter tenderness asks whether fixed appliances will cure her joint problems. Which single response is best supported by the evidence?

A Occlusal equilibration should be carried out first, because a stable occlusion is required before any appliance is placed
B A splint should be worn for two years and the symptoms reassessed, because orthodontics cannot begin before then
C Orthodontic treatment reliably relieves temporomandibular symptoms by removing the

occlusal interferences that cause them

D Orthodontic treatment neither causes nor prevents temporomandibular disorder, and should not be undertaken for that purpose

E Orthodontic treatment is contraindicated here, because appliances worsen temporomandibular symptoms in most patients

973 The parents of a 12-year-old boy have read that extracting premolars narrows the airway and causes sleep apnoea. He requires four first premolar extractions for 10 mm of crowding. Which single response is best supported by the evidence?

A Premolar extraction reliably narrows the oropharynx, so a non-extraction expansion approach should be used instead in this case

B Airway dimensions do fall after extraction, but the change is too small to be detected on a lateral cephalogram

C Premolar extraction and incisor retraction have not been shown to reduce airway dimensions or to cause sleep-disordered breathing

D Rapid maxillary expansion should be added to every extraction plan in order to protect the airway

E Sleep studies should be arranged before and after treatment in every extraction case as a routine precaution

974 Regarding claims commonly made for the benefits of orthodontic treatment:

TRUE OR FALSE — judge each statement independently; any number may be true

A Correction of a malocclusion prevents temporomandibular disorder in later life

B Orthodontic alignment eliminates periodontal disease by removing plaque-retentive irregularity

C Premolar extraction reduces pharyngeal airway dimensions sufficiently to cause obstructive sleep apnoea

D Correcting an anterior open bite reliably resolves an associated lisp

E Orthodontic treatment has been shown to reduce the lifetime incidence of dental caries

975 A 13-year-old boy is four months into fixed appliance treatment. Plaque covers the gingival third of the upper labial segment and there is an early opacity at the gingival margin of the upper left lateral incisor. Which single measure is most appropriate now?

A Reinforce oral hygiene and diet and apply professional fluoride varnish, repeating it every six to eight weeks

B Change to a self-ligating bracket system, which is associated with less plaque accumulation

C Debond the appliance immediately, since visible demineralisation means the risk has already been realised

D Prescribe a chlorhexidine mouthrinse for twelve months as the principal preventive measure

E Microabrade the lesion now, so that the opacity is removed before it becomes established

976 A 15-year-old girl is debonded and has chalky white lesions on the labial surfaces of the upper central incisors and the buccal surfaces of the lower canines. Which single approach is most appropriate immediately after debond?

A Immediate composite veneers on the affected teeth, since the enamel has been irreversibly lost

B High-concentration fluoride varnish applied daily for two weeks, which arrests and reverses cavitation

C Microabrasion at the debond visit itself, which removes the lesions before they take up stain

D Immediate resin infiltration of every lesion, since untreated lesions do not improve after the appliance is removed
E Fluoride and hygiene alone for at least three months, since these lesions remineralise and fade considerably with time

977 Regarding demineralisation during fixed appliance treatment:

TRUE OR FALSE — judge each statement independently; any number may be true

A White spot lesions are seen in up to half of patients treated with fixed appliances
B The commonest sites are the first molars, the upper lateral incisors and the lower canines
C Lesions require at least two years of appliance wear before they become clinically visible
D A daily fluoride mouthrinse has been shown to be more effective than professionally applied varnish in adolescents
E Once a white spot lesion is visible the enamel is cavitated and can no longer remineralise

978 A 14-year-old girl with a Class II division 2 malocclusion has short, blunt upper incisor roots on her pre-treatment radiograph and a history of trauma to the central incisors at eight years. Fixed appliances with 6 mm of incisor retraction are planned. Which single feature most increases her risk of severe root resorption?

A The 6 mm of planned retraction, since apical displacement of this magnitude is the dominant risk factor
B Her age of 14 years, since resorption is consistently more severe in adolescents than in adults
C Her Class II division 2 malocclusion, which is independently associated with root resorption
D The blunt pre-existing root morphology, which is among the strongest predictors of severe resorption
E The history of trauma at eight years, which makes severe resorption a certainty in these teeth

979 Twelve months into fixed appliance treatment a routine radiograph shows that each upper central incisor has lost about 3 mm of root length. Which single course of action is most appropriate?

A Reduce the archwire dimension and continue active retraction, since lighter forces cannot resorb further root
B Remove the active archwire and leave the teeth passive for three to six months before reassessing radiographically
C Splint the incisors to the adjacent teeth and continue, since splinting distributes the applied force
D Continue treatment unchanged, since radiographic shortening is found in the majority of treated patients
E Debond immediately and accept the present occlusion, since resorption continues while brackets remain bonded

980 A patient about to consent to fixed appliances asks how likely it is that her roots will shorten. Which single set of figures best answers her?

A Histological evidence in up to 90 per cent, with severe loss of root length in about 30 per cent of patients
B Radiographic shortening in about 10 per cent, with severe loss in fewer than one patient in a thousand
C Severe loss in about 20 per cent, which is the reason radiographs are repeated at six-monthly intervals

D Radiographic shortening in essentially every patient, with severe loss in about 15 per cent of them
E Radiographic shortening in up to about 73 per cent, with loss of a third or more of the root in 1 to 5 per cent

981 Regarding orthodontically induced inflammatory root resorption:

TRUE OR FALSE — judge each statement independently; any number may be true

A Radiographic evidence is found in up to about 73 per cent of treated patients
B Histological evidence is found in up to about 90 per cent of treated teeth
C Loss of a third or more of the root length occurs in about 1 to 5 per cent of patients
D A pause of three to six months without active force allows cemental repair of resorbed root surfaces
E Previously traumatised, blunt or pipette-shaped roots carry an increased risk

982 A 12-year-old telephones the practice two days after her fixed appliance was placed, saying her teeth are too sore to eat. Which single response is most appropriate?

A Advise chewing exercises alone, since analgesics reduce the rate of tooth movement and should be avoided
B Reassure that discomfort peaks at about 48 hours and settles within a week, and advise simple analgesia and a soft diet
C Arrange an emergency appointment to remove the archwire, since pain at this stage indicates that the force is excessive
D Advise that the pain indicates root resorption and arrange a periapical radiograph without delay
E Prescribe ibuprofen for three weeks, since discomfort continues throughout the alignment phase

983 A 15-year-old boy had the upper right central incisor avulsed and replanted at nine years of age. It is now discoloured, tender to percussion, and a periapical radiograph shows apical radiolucency. Fixed appliances were fitted three months ago. Which single course of action is most appropriate?

A Extract the incisor now and close the space, since a replanted tooth cannot be moved orthodontically
B Increase the force on that tooth to complete its movement before the pulpal change progresses further
C Continue treatment but exclude the tooth from the archwire permanently, avoiding endodontic intervention
D Suspend active movement of that tooth and arrange endodontic assessment before treatment continues
E Continue treatment and review the tooth at debond, since discolouration is common after replantation

984 A 16-year-old girl has 3 mm of recession on the labial surface of the lower left central incisor, which was proclined by 6 degrees during treatment. She is now in retention with a bonded retainer and has no symptoms. Which single approach is most appropriate?

A Monitor with serial records and refer for grafting only if the recession progresses or she becomes symptomatic
B Remove the bonded retainer, since bonded retainers are the principal cause of recession at this site in adolescent patients
C Refer for a connective tissue graft at once, since recession of this depth invariably progresses

D Retrocline the incisor to its pre-treatment position, which allows the recession to resolve
E Reassure that recession in adolescents resolves spontaneously once facial growth is complete

985 Regarding gingival changes during and after fixed appliance treatment:

TRUE OR FALSE — judge each statement independently; any number may be true

A Gingival hyperplasia during treatment usually resolves within a few months of debond once plaque control is restored
B Recession during treatment is most commonly seen on the palatal surfaces of the upper molars
C Labial movement of a lower incisor beyond its alveolar housing reduces recession risk by thickening the biotype
D Gingivectomy should be performed routinely at debond in any patient showing hyperplasia
E Black triangles appearing after alignment of crowded incisors are prevented by proclination

986 A 15-year-old girl reports that costume jewellery brings her out in an itchy rash. She is about to begin fixed appliance treatment. Which single course of action is most appropriate?

A Use standard components but prescribe an antihistamine to cover the first month of treatment
B Decline treatment, since a history of nickel sensitivity is an absolute contraindication to appliances
C Take a detailed history and, if it is consistent with nickel sensitivity, plan nickel-free brackets, wires and bands
D Proceed with standard stainless steel components, since intraoral reactions to nickel do not occur
E Refer for patch testing before any decision, since a history alone can never guide the choice of components in an adolescent patient

987 A 14-year-old boy with spina bifida requires intermaxillary elastics as part of his treatment. Which single precaution is most important?

A Use latex elastics but restrict wear to night-time, so that cumulative exposure is reduced
B Test a single latex elastic in the mouth for 24 hours and observe him for any reaction
C Prescribe a prophylactic antihistamine on the days on which elastics are to be worn
D Use non-latex elastics and latex-free gloves and equipment throughout, since his risk of latex allergy is high
E Avoid elastics altogether and use a fixed Class II corrector, since no acceptable alternative elastic exists

988 A 12-year-old wearing headgear at night attends with a scratch on his cheek and reports that the facebow has come off twice while he was asleep. Which single response is most appropriate?

A Continue as before, since a single scratch is a minor event and occasional detachment is to be expected
B Check that both a safety release and a separate snap-away or locking device are fitted, and rehearse fitting and removal
C Reduce wear to ten hours a day, since repeated detachment indicates that the applied force is too high
D Restrict him to daytime wear from now on, since night-time wear of headgear is unsafe in all patients

E Replace the facebow with a longer outer bow, since repeated detachment results from an outer bow that is too short

989 Regarding the safe prescription of headgear:

TRUE OR FALSE — judge each statement independently; any number may be true

A Two independent safety mechanisms must be fitted whenever headgear is worn
B For reinforcement of anchorage, 10 to 12 hours a day at 250 to 350 g per side is usual
C For distalisation of the buccal segments, 14 hours a day at 450 to 500 g per side is usual
D Ocular injury from headgear occurs only where the appliance has been worn during contact sport
E A patient reporting that the facebow has come loose overnight should be advised to continue unchanged

990 During consent for fixed appliances a 16-year-old asks how long she will have to wear retainers. Which single statement is most appropriate?

A Retention is required for two years, the period over which the periodontal fibres fully remodel
B Retention is unnecessary where the teeth have been moved into a stable position within the alveolus
C Retention is required only in extraction cases, since space closure is the movement that relapses
D Retainers are worn full-time for six months and then discarded, after which the result is stable
E Retention is indefinite if the result is to be held, because relapse remains a lifelong risk

991 A 58-year-old woman requests fixed appliances. She has taken oral alendronate for osteoporosis for six years. Which single statement best informs the plan?

A Tooth movement is likely to be slow and incomplete, and any extraction carries a risk of medication-related osteonecrosis of the jaw
B Oral bisphosphonates do not affect tooth movement, since only intravenous preparations reach the alveolar bone in sufficient quantity
C Treatment may proceed normally provided heavier forces are used to overcome the reduced bone turnover
D Bisphosphonates increase bone turnover, so tooth movement will be faster than would otherwise be expected
E The drug should be stopped for three months before treatment, which restores normal bone turnover

992 A 17-year-old boy with type 1 diabetes, whose physician describes his glycaemic control as poor, requests fixed appliances. He has generalised marginal gingivitis. Which single course of action is most appropriate?

A Decline treatment permanently, since poorly controlled diabetes is an absolute contraindication to fixed appliance therapy
B Proceed using removable appliances only, since these have no effect on the periodontal tissues
C Defer placing appliances until glycaemic control and periodontal health have improved, in liaison with his diabetes team
D Proceed now, since diabetes has no bearing on the periodontal response to orthodontic appliances
E Proceed now but schedule all appointments in the late afternoon, which reduces the risk of hypoglycaemia

993 A 14-year-old girl attends for a new patient assessment. Regarding aspects of her medical history and what they change:

TRUE OR FALSE — judge each statement independently; any number may be true

A Poorly controlled epilepsy is a reason to prefer fixed appliances, because a loose plate may be inhaled during a seizure
B A patient taking warfarin requires an INR check before any extraction is arranged
C Juvenile idiopathic arthritis involving the temporomandibular joints may cause mandibular growth restriction and an anterior open bite
D Previous head and neck radiotherapy including the jaws is a strong relative contraindication to extractions and to tooth movement
E A previous episode of infective endocarditis means antibiotic prophylaxis is required before every band fitting

994 A 13-year-old girl with epilepsy that is not fully controlled, and who takes phenytoin, requires an appliance to correct an anterior crossbite. Which single modification is most appropriate?

A Use the removable appliance as planned but advise that it is left out at night, when seizures are more likely to occur
B Use a fixed appliance rather than a removable one, and reinforce plaque control because of drug-induced gingival overgrowth
C Defer all orthodontic treatment until she has been free of seizures for at least five years
D Use the removable appliance with reduced acrylic bulk, since a thinner baseplate is less hazardous
E Use the removable appliance with additional clasps, since improved retention removes the inhalation risk

995 A 16-year-old boy with mild haemophilia A requires extraction of four first premolars as part of his orthodontic plan. Which single arrangement is most appropriate?

A Arrange the extractions under general anaesthesia in practice, since general anaesthesia reduces bleeding
B Avoid extractions altogether and treat on a non-extraction basis, whatever the degree of crowding
C Carry out the extractions in practice with local haemostatic measures alone, since premolar sockets are small
D Arrange the extractions with his haemophilia centre, which will advise on factor or desmopressin cover and on tranexamic acid
E Carry out the extractions in practice and prescribe aspirin for postoperative pain, which avoids the gastric irritation caused by other analgesics

996 A 15-year-old boy with a repaired ventricular septal defect and a previous episode of infective endocarditis attends for fixed appliance treatment. Regarding his management:

TRUE OR FALSE — judge each statement independently; any number may be true

A NICE does not recommend routine antibiotic prophylaxis before orthodontic procedures
B Excellent oral hygiene and dental health are the most important protection against endocarditis in this patient
C Mortality from an episode of infective endocarditis is around 21 per cent
D Bacteraemia arises from everyday activities such as toothbrushing and chewing as well as from dental procedures
E Any episode of dental infection in this patient should be treated promptly rather than observed

997 A 14-year-old girl with juvenile idiopathic arthritis has limited mouth opening, a retrognathic mandible with a steep mandibular plane, an anterior open bite and bilateral condylar flattening on imaging. Which single statement best guides orthodontic planning?

A Disease activity must be controlled with the rheumatology team first, since continuing joint destruction will undo any correction
B The open bite is a habit effect and will close once the joints have become comfortable again
C Fixed appliances should be avoided entirely, since arthritis is a contraindication to tooth movement
D Orthognathic surgery should be arranged now, since joint destruction has already brought growth to an end
E A functional appliance should be fitted without delay, in order to stimulate condylar growth and close the anterior open bite

998 A 19-year-old man completed radiotherapy for a nasopharyngeal carcinoma two years ago, with the maxilla and mandible within the treatment field. He requests fixed appliances. Which single statement best guides the response?

A Treatment should proceed using heavier forces, since irradiated bone remodels more slowly than normal bone
B Treatment may proceed normally, since the effects of radiotherapy on bone resolve within about two years
C Tooth movement is unpredictable and extractions risk osteoradionecrosis, so treatment is avoided or kept very limited
D Extractions are safe provided that they are carried out at least one year after radiotherapy is completed, since bone vascularity is said to normalise by then
E Hyperbaric oxygen therapy before treatment removes the risk entirely and permits a standard plan

Block AD — Orthodontic practice, infection control and governance

(Q999–1023)

Volume I: Part V, units V.2–V.4 (radiation protection; infection control; governance and audit)
Volume II: **§30** — read it again if you score under 70%

This block: 25 questions (6 of them true/false) · 49 marks · allow 31 minutes

999 A 13-year-old attends with delayed eruption of the upper right permanent canine. Before any radiograph is taken, which single step is required?

A Confirm that the number of films taken this year keeps the patient within her annual dose limit
B Obtain a written referral from the general dental practitioner specifying the projection required
C Calculate the effective dose in microsieverts for the projection that has been chosen
D Obtain written consent for the exposure from both the patient and an accompanying parent
E Record in the notes the specific clinical question that the radiograph is expected to answer

1000 An upper permanent canine is unerupted at 13 years. A panoramic radiograph shows the crown overlapping the root of the lateral incisor, and an upper standard occlusal is then taken. On the occlusal view the canine crown appears to have moved in the same direction as the shift of the tube head. Which single conclusion follows?

A The canine lies palatally, since an object moving with the tube shift lies further from the film
B The canine lies in the line of the arch, since vertical parallax cannot localise in this plane
C No conclusion is possible, since parallax requires two films taken with a horizontal tube shift
D The canine lies buccally, since panoramic views magnify palatal objects and cancel the apparent shift
E The canine lies buccally, since an object moving with the tube shift lies closer to the film

1001 Regarding radiation protection in orthodontic practice:

TRUE OR FALSE — judge each statement independently; any number may be true

A Justification requires that the benefit of an exposure outweighs its detriment for that individual patient
B Optimisation requires that the dose be kept as low as reasonably practicable while still answering the clinical question
C Dose limitation applies to staff and to members of the public but not to patients undergoing diagnostic exposure
D A specific clinical justification should be recorded in the notes before the exposure is made
E The whole of a panoramic radiograph should be examined and reported, not only the region of clinical interest

1002 A 12-year-old has a palatally impacted upper canine. Two-dimensional films have localised it palatally and show no resorption of the lateral incisor root. Which single statement about cone beam computed tomography is most appropriate?

A It should be taken because resorption of the lateral incisor cannot be excluded on any two-dimensional film
B It should be taken in every impacted canine case, since it is the reference standard for localisation
C It is not justified here, because the films have already answered the question that would change management
D It should be taken because its effective dose is comparable to that of a single periapical radiograph
E It should be taken to plan the surgical exposure, which cannot be planned from two-dimensional views

1003 A panoramic radiograph taken to assess third molars in an 18-year-old also shows a well-defined radiolucency at the angle of the right mandible, well away from the region of interest. Which single course of action is required?

A Repeat the radiograph using a different projection before making any written record of the finding
B Refer the patient for the third molars and leave the incidental finding to the receiving unit to address
C Note the finding in the records but take no further action, since such radiolucencies are usually anatomical
D Record a report covering the third molar regions only, since that was the justification for the exposure
E Record a report covering the whole image and arrange appropriate investigation of the radiolucency

1004 A parent asks whether the number of radiographs her son has had during two years of orthodontic treatment has exceeded a safe limit. Which single response is correct?

A The annual patient dose limit is one millisievert, and his exposures have remained comfortably below it
B There is no dose limit for patients, and each exposure must instead be separately justified and optimised
C The annual patient dose limit is 20 millisieverts, the same figure that applies to classified staff
D Patients may undergo up to four panoramic radiographs a year before a limit is reached
E Limits apply only to cone beam imaging, for which a maximum of one scan a year is permitted

1005 Regarding the choice of radiographic view in orthodontics:

TRUE OR FALSE — judge each statement independently; any number may be true

A A lateral cephalogram is the appropriate view for assessing the anteroposterior skeletal relationship before orthognathic planning
B A panoramic radiograph alone reliably localises an unerupted canine in the buccopalatal plane
C A dental panoramic tomogram is the view of choice for the detection of early approximal caries
D Two periapical films taken at the same horizontal and vertical angle allow parallax localisation
E Cone beam computed tomography should be taken routinely before every surgical expo-

sure of a canine

1006 A nurse returns a tray of used instruments on which dried blood and set composite remain at the working ends, and places them directly into the autoclave. Which single statement best describes the problem?

A The problem is cosmetic only, since steam penetrates organic material at 134 degrees Celsius
B The instruments should have been autoclaved first and then cleaned while they were still warm
C The autoclave cycle will remove the debris, provided that a longer holding time is selected
D Sterilisation cannot be achieved through debris, so cleaning and inspection must precede it
E The instruments should have been disinfected chemically rather than processed in an autoclave

1007 An orthodontic practice is classifying its instruments in order to set reprocessing standards. Which single classification is correct?

A A scalpel blade used to expose a canine, which penetrates the mucosa, is semi-critical
B A bracket-holding tweezer used intraorally is non-critical, since it contacts only enamel
C A band seater, which contacts intact mucosa without penetrating it, is semi-critical
D An alginate mixing spatula is critical, because the impression subsequently enters the mouth
E The dental chair headrest is semi-critical, because the patient's hair rests against it

1008 A practice audits the cleanliness of its reprocessed instruments and finds residual debris most often on one group. Which single group is most likely to be implicated?

A Hinged pliers and cutters, whose box joints retain debris that brushing does not reach
B Band pushers, whose serrated working ends cannot be inspected without magnification
C Mirrors and probes, whose long smooth shanks are awkward to hold against a brush
D Impression trays, whose perforations are the commonest site of retained alginate
E Bracket tweezers, whose fine tips are damaged by ultrasonic cleaning and so are cleaned by hand

1009 Regarding decontamination in an orthodontic practice:

TRUE OR FALSE — judge each statement independently; any number may be true

A The cycle runs cleaning, inspection, disinfection, sterilisation and protected storage, in that order
B A vacuum, type B, autoclave is required for wrapped instruments and for hollow or lumened items
C A non-vacuum, type N, autoclave is adequate for unwrapped solid instruments used immediately afterwards
D Instruments may be stored unwrapped on an open bracket table between sessions once they have been sterilised
E An ultrasonic bath sterilises instruments and may replace the autoclave for solid items

1010 A practice wishes to sterilise pouched instrument sets so that they can be stored ready for use. Which single requirement follows?

A A dry heat oven is required, since steam cannot safely be used with paper or plastic pouches
B Chemical disinfection followed by pouching is acceptable, since the pouch then maintains sterility
C Any autoclave is sufficient, since pouches are freely permeable to steam at 134 degrees Celsius
D A non-vacuum, type N, autoclave is sufficient, provided that the holding time is extended accordingly for pouched loads
E A vacuum, type B, autoclave is required, since air must be actively removed for steam to reach a wrapped load

1011 During band selection several molar bands are tried in the mouth and rejected before one is found to fit. Which single action is required for the rejected bands?

A Rinse them under running water and return them to the kit, since saliva alone is not an infection risk
B Wipe them with an alcohol wipe and return them to the band kit, since they were in the mouth only briefly
C Discard them, since bands that have been tried in cannot be sterilised without distorting them
D Return them for full reprocessing through cleaning, inspection and sterilisation before they are used again
E Immerse them in disinfectant solution for ten minutes and then return them to the band kit

1012 An alginate impression is taken and is to be sent to an external laboratory for construction of a removable appliance. Which single statement describes the required practice?

A Disinfect the returned appliance only, since the cast is poured before contamination can occur
B Disinfect the impression before despatch, and disinfect the returned appliance before it is fitted
C Neither step is necessary provided that the work travels in a sealed and clearly labelled container
D Disinfect the impression before despatch, the returned appliance requiring no treatment before fitting
E Send the impression rinsed only, since the laboratory disinfects all work that it receives

1013 Regarding infection control in orthodontic practice:

TRUE OR FALSE — judge each statement independently; any number may be true

A Instruments may be sterilised through a light film of dried blood provided that the holding time is extended
B Bands tried in the mouth and rejected may be returned to the kit after wiping with an alcohol swab
C An ultrasonic bath is a form of sterilisation and may replace autoclaving for solid instruments
D Laboratory work needs to be disinfected only when it is returned, not when it is sent out
E After a sharps injury the wound should be scrubbed vigorously with a brush and antiseptic

1014 A nurse sustains a needlestick injury from a local anaesthetic needle used on a patient whose bloodborne virus status is unknown. Which single sequence of immediate actions is correct?

A Scrub the wound with chlorhexidine, apply a dressing, and report the incident at the end of the working week
B Squeeze the wound dry, seal it immediately with an occlusive dressing, and arrange testing in one month
C Wash the wound, take blood from the nurse alone, and inform occupational health only if symptoms develop
D Apply alcohol to the wound, complete an incident form, and take no further action unless the patient is subsequently found to be positive
E Encourage bleeding, wash under running water without scrubbing, cover the wound, report it and contact occupational health the same day

1015 A department measures its bond failure rate, finds it to be 12 per cent, introduces a new bonding protocol and presents its findings at a departmental meeting. Which single step is required for this to constitute an audit?

A Randomise subsequent patients to the old and the new protocol in order to establish causation directly rather than mere association
B Re-measure the bond failure rate after the change and compare it with both the original figure and the standard
C Obtain ethical approval before the findings are published outside the department
D Calculate whether the reduction in failures reaches statistical significance
E Compare the department's failure rate with that of a neighbouring department

1016 A department wishes to know whether a new self-ligating bracket aligns teeth faster than its current bracket, and plans to allocate patients to one or the other. Which single description is correct?

A This is service evaluation, since it describes what the service currently achieves for its patients
B This is audit, since no treatment new to the department is being introduced by the comparison
C This is service evaluation, and requires ethical approval because patients are being allocated
D This is research, and requires ethical approval and informed consent to participate
E This is audit, since it measures the department's own performance against its current practice

1017 Regarding clinical audit:

TRUE OR FALSE — judge each statement independently; any number may be true

A An audit requires a standard against which practice is to be measured
B An audit is complete only when practice has been re-measured after a change has been made
C Service evaluation describes what a service currently achieves and sets no standard
D Audit does not usually require approval from a research ethics committee
E Audit is the appropriate method for deciding whether a new appliance is more effective than an existing one

1018 A case is scored with the PAR index before and after treatment. The pre-treatment score is 34 and the post-treatment score is 6. Which single interpretation is correct?

A A reduction of 28 points and of 82 per cent, which places the case in the greatly improved category
B A reduction of 28 points, which places the case in the improved but not the greatly improved category
C A reduction of 82 per cent, which is insufficient for greatly improved without a 30-point reduction
D A post-treatment score of 6 indicates a failure to improve, since the threshold for success is zero
E The scores cannot be interpreted without applying the weighting for the patient's age and sex

1019 A department compares aligner cases with fixed appliance cases. PAR reduction is equivalent between the two groups, but the aligner cases score significantly worse on the ABO objective grading system. Which single interpretation is best supported?

A Aligners are as effective as fixed appliances, and the difference in ABO grading is of no clinical relevance to patients
B The ABO result is explained by the greater proportion of aligner cases that required refinement
C The two indices measure different things, and the finishing detail captured by ABO grading is where aligners underperform
D The PAR result is the more valid of the two, since PAR was validated against the opinion of a panel
E The finding indicates an error in scoring, since two occlusal indices applied to the same cases should agree

1020 A commissioner proposes using IOTN to judge whether a provider's completed treatment has been successful. Which single objection is most valid?

A IOTN is too time-consuming to apply to an entire provider caseload in routine practice
B IOTN has poor reproducibility between examiners and so cannot support any comparison
C IOTN measures aesthetics alone and takes no account of the dental health of the patient's malocclusion
D IOTN measures the need for treatment before it begins and says nothing about the quality of the result
E IOTN applies only to patients under 18 years and so cannot describe an adult caseload

1021 Regarding indices used in orthodontics:

TRUE OR FALSE — judge each statement independently; any number may be true

A The Index of Complexity, Outcome and Need combines need, complexity and outcome within a single instrument
B A PAR reduction of at least 22 points or of at least 30 per cent is classified as greatly improved
C The aesthetic component of IOTN is scored by the patient rather than by the clinician
D The ABO objective grading system is less sensitive than PAR to errors in marginal ridge levels
E PAR scores the buccolingual inclination of the posterior teeth

1022 A newly appointed orthodontist wishes to run a first audit in a busy practice. Which single topic is most suitable?

A Whether a new bonding adhesive reduces bond failure compared with the adhesive currently used
B The proportion of new patient records containing a recorded justification for every radiograph taken
C Whether patients treated with self-ligating brackets finish sooner than those treated conventionally
D Patient satisfaction with the decor of the practice, measured by an open-ended questionnaire
E The mean PAR reduction of every case treated in the practice over the past fifteen years

1023 A general dental practitioner with a special interest in orthodontics is asked by a parent to treat a 9-year-old with a repaired cleft lip and palate who has a collapsed upper arch and an unerupted canine. Which single course of action is most appropriate?

A Begin treatment and seek advice from the cleft team by letter should any difficulties arise
B Decline any involvement in the child's dental care, including routine prevention and monitoring
C Refer to a local specialist orthodontist in practice, who can manage the arch and the unerupted canine
D Begin upper arch expansion, since expansion is straightforward and the cleft team can continue afterwards
E Refer to the regional cleft team, since this care is delivered by a commissioned multidisciplinary service

Part 2 — Answers, explanations and references

Each answer gives the key, the reasoning, and a reference to the other two volumes so you can go straight to the material. **Read the explanation for every question in a block, not only the ones you answered incorrectly.** A complete answer-key grid follows at the end of this Part.

References are given as **Vol I** unit · **Vol II** section — the Volume I topic, chapter or Part V unit whose main treatment covers the question, and the Volume II section that revises it against the curriculum.

Block A — Concepts of growth and development (Q1–26 · 50 marks · Vol II §1)

1 A Concavities in all three vertebral bodies with C3 and C4 horizontally rectangular is cervical stage CS4, at which the pubertal peak is one to two years past and the residual growth available for modification is small. The tempting answer is that the peak is within the coming year, but that is CS3, in which C4 has not yet notched and the bodies are still trapezoid. [Vol I Chapter 4 · Vol II §1.4]

2 C On Scammon's curves lymphoid tissue overshoots to roughly 200 per cent of adult size in late childhood and then regresses through adolescence, so obstruction from adenoids and tonsils commonly improves without surgery. The neural curve, not the lymphoid curve, is the one that is 90 per cent complete by six years. [Vol I Chapter 4 · Vol II §1.2]

3 TFTFT

A True — the vault follows the neural curve and is around 90 per cent complete by six years, while the face is not
B False — the genital curve is almost flat through childhood and rises abruptly at puberty
C True — the face is neither purely neural nor purely somatic and its curve lies between the two
D False — girls reach their adult value about two years earlier than boys
E True — lymphoid tissue regresses through adolescence, so mild symptoms often resolve without intervention

[Vol I Chapter 4 · Vol II §1.2]

> MNEMONIC **Neural early, lymphoid overshoots, genital late, general steady** — the four Scammon curves, each on its own timetable.

4 E Pooling children of differing maturity at one moment averages together peaks that occur at different ages, so a cross-sectional curve smooths the spurt away; only repeated measurement of the same child reveals a true velocity peak. Study 1 is in fact the better design for population means at each age, which is the reversal the question invites. [Vol I Chapter 4 · Vol II §1.3]

5 B The mandible has no sutures and the condyle is now regarded as a site rather than a centre of growth, so the deficiency follows restricted function and a scarred capsular matrix rather than the loss of an autonomous centre. The primary growth centre answer is the historical Sicher position and is the distractor most candidates select. [Vol I Chapter 4 · Vol II §1.5]

6 FFTFF

A False — transplanted suture does not grow autonomously, whereas transplanted cartilage does, which is what undermined the theory
B False — Scott placed the primary role in the nasal septal cartilage and regarded the sutures as compensatory
C True — early septal resection is followed by midfacial deficiency, the observation most often used to support Scott
D False — Moss regarded all skeletal units as secondary to their functional matrices
E False — the servosystem model makes condylar growth responsive to peripheral occlusal feedback

[Vol I Chapter 4 · Vol II §1.5]

> MNEMONIC **Sicher-suture, Scott-septum, Moss-matrix, Petrovic-periphery** — the four classical growth theories in one line.

7 D Cervical staging is read from a radiograph taken for an independently justified purpose, so it carries no additional dose, and it agrees closely enough with hand-wrist staging for a timing decision. Taking a hand-wrist film as well cannot be justified when the same information is already available at no extra exposure. [Vol I Chapter 4 · Vol II §1.4]

8 C Dental, skeletal and somatic maturity correlate only loosely, so dental delay alongside normal centiles is a dissociation to be reviewed rather than a disorder to be investigated. Inferring skeletal delay from dental delay is the common error, because the correlation between the two is far too weak to support it. [Vol I Chapter 4 · Vol II §1.4]

9 A The annual increments are 7, 9 and 6 cm, so velocity rises to a maximum in the fourteenth year and then falls; a peak is defined by the turning point of the increments and not by the largest attained height. Choosing the final interval mistakes greatest total height for greatest annual gain. [Vol I Chapter 4 · Vol II §1.1]

10 TTTTT

A True — differentiation and functional maturation continue beyond the cessation of growth in size
B True — a falling velocity still adds size, which is why attained height says nothing about the spurt
C True — this dissociation is precisely why maturity indicators exist
D True — low-grade late growth is well documented and is relevant to long-term stability
E True — chronological age is only a proxy, and a poor one, for the biological state that matters

[Vol I Chapter 4 · Vol II §1.1]

11 E The interval excludes zero so the effect is real, but a lower bound of 0.4 mm is below anything a patient could perceive, and a 2 mm mean is a small fraction of what a large overjet requires. Reading the point estimate as though it applied to every patient is the usual misinterpretation of this literature. [Vol I Chapter 4 · Vol II §1.6]

12 D Girls reach peak height velocity at about 12 years and boys at about 14, so at the same chronological age a girl is on average roughly two years ahead in skeletal maturity. Starting both together because they are the same age is exactly the error that reliance on chronological age produces. [Vol I Chapter 4 · Vol II §1.3]

13 TTFTF

A True — implants are the only markers that remain fixed within remodelling bone
B True — this is the basis of the classical animal work on appositional and resorptive fields
C False — surface remodelling of the lower border masks a large part of the true rotation
D True — structures such as the anterior cranial base are stable enough for clinical purposes
E False — a single measurement of each child cannot yield an individual velocity curve

[Vol I Chapter 4 · Vol II §1.3]

14 B The servosystem model treats condylar growth as responsive to peripheral signals, so a sustained change in mandibular posture shifts the comparator and produces condylar growth with fossa remodelling. Describing the condyle as an autonomous growth centre is the discarded Sicher position and does not explain why the effect depends on wear. [Vol I Chapter 4 · Vol II §1.5]

15 E Growth prediction performs acceptably for group averages and poorly for individuals, and the Class III mandible is the least predictable of all because its late growth is both large

and variable. The belief that the Class III pattern is fixed early is the error that produces surgery planned before growth has finished. [Vol I Chapter 4 · Vol II §1.6]

16 B Superimposition of serial cephalograms a year apart demonstrates directly that the facial skeleton has stopped changing, which is what governs the stability of a surgical result. CS4 is far too early, marking a point only one to two years after the peak with years of growth still to come. [Vol I Chapter 4 · Vol II §1.4]

17 FTTFT

A False — the sesamoid appears shortly before the peak and signals that it is approaching
B True — epiphyseal capping is one of the classical peri-peak hand-wrist signs
C True — menarche is a late event, typically a year or so after the peak
D False — dental maturity correlates poorly with skeletal maturity and cannot substitute for it
E True — every other indicator is judged by how well it predicts the velocity peak

[Vol I Chapter 4 · Vol II §1.4]

MNEMONIC **Sesamoid before, capping at, menarche after** the peak — three maturity landmarks placed in time around PHV.

18 D The mid-growth or juvenile spurt is a small and inconstant acceleration within the pre-pubertal deceleration and has none of the timing implications of the pubertal peak. Mistaking it for the pubertal spurt is the trap, because it would start growth modification three or four years too early. [Vol I Chapter 4 · Vol II §1.1]

19 A The vault follows the neural curve and is largely complete early, whereas the face follows a slower curve that continues through adolescence, so a discrepancy that is trivial at five years can be a surgical problem at fifteen. The vault does not shrink; its growth simply stops sooner. [Vol I Chapter 4 · Vol II §1.2]

20 FFTFT

A False — at CS1 all three lower borders are flat and C3 and C4 are trapezoid
B False — at CS2 the peak is expected about one year later, not earlier
C True — CS3 places the peak within the coming year, which is the window for growth modification
D False — it is read from a lateral cephalogram taken for planning, adding no dose
E True — the stages are defined by concavity and by body shape, not by dimensions

[Vol I Chapter 4 · Vol II §1.4]

MNEMONIC **CS1 two years out, CS2 one year out, CS3 imminent, CS4 one-to-two years past** — the CVM peak timeline.

21 C Sutural interdigitation increases with maturity, so the skeletal proportion of the correction falls and the dental proportion rises as the patient ages; in a child of this age about half the expansion is skeletal. Deferring to 14 years reverses the direction in which the sutural response actually changes. [Vol I Chapter 4 · Vol II §1.5]

22 D Serial height velocity defines the peak directly and is the standard against which the other indicators are validated, but it requires records accumulated over years, whereas cervical staging is available from a cephalogram already justified for planning. Dental age is the weakest of the four because its correlation with skeletal maturity is poor. [Vol I Chapter 4 · Vol II §1.4]

MNEMONIC Best to worst for gauging maturity: **height velocity, cervical stage, dental age, chronological age**.

23 B With growth essentially finished the skeletal discrepancy can no longer be modified, so the realistic options are dental camouflage with anchorage planned for a large retraction, or surgery. A twin block is tempting because the presentation looks like a textbook functional appliance case, but growth modification requires growth. [Vol I Chapter 4 · Vol II §1.4]

24 E Size and maturity are different quantities: a tall child may simply be a large child rather than an early maturing one, and CS1 places the peak roughly two years ahead. Using attained height centile as a maturity indicator is the error, because only velocity, not size, carries timing information. [Vol I Chapter 4 · Vol II §1.3]

25 A Menarche follows the peak, so by two years afterwards the velocity curve has fallen steeply and only slow residual growth remains, which is too little to build a skeletal plan on but is not nothing. Treating her as a finished adult ignores the low-grade growth that continues into the late teens and beyond. [Vol I Chapter 4 · Vol II §1.3]

26 C Moss's functional matrix theory is the framework the examiners expect: bone is laid down and resorbed where the enveloping soft tissues and functional spaces require it, which is why the face keeps adapting while function changes. The condyle as a primary growth centre is the classic trap — condylar cartilage is secondary and adaptive, and both grafting and distraction evidence contradict the idea that it pushes the mandible forwards. [Vol I Chapter 4 · Vol II §1.4]

Scored under 70% of the 50 marks on this block? Re-read Volume II §1, then sit it again in a fortnight.

Block B — Craniofacial growth, prenatal and postnatal (Q27–56 · 58 marks · Vol II §2)

27 B Malar bone, mandible, middle ear ossicles and external ear are all first and second arch derivatives that depend on neural crest ectomesenchyme, so a single migration defect explains a combination that otherwise looks like several unrelated anomalies. Failure of fusion of the medial nasal and maxillary processes produces a cleft lip, which is not part of this picture. [Vol I Chapter 3 · Vol II §2.1]

28 TTFFT

A True — the first arch divides into maxillary and mandibular processes early in the fourth week
B True — each arch takes its own nerve with it, and the trigeminal is the nerve of the first arch
C False — they are second arch derivatives supplied by the facial nerve
D False — the mandible forms by intramembranous ossification lateral to Meckel's cartilage, which largely regresses
E True — the second arch contributes the stapes superstructure, which is why ear and jaw anomalies travel together

[Vol I Chapter 3 · Vol II §2.1]

MNEMONIC **Arch 1 = Meckel's/trigeminal/mastication; Arch 2 = Reichert's/facial nerve/expression plus stapes.**

29 D The lip and the primary palate as far as the incisive foramen are formed where the maxillary process meets the medial nasal process, so a cleft in this position and of this extent is a failure at that junction. Failure of the palatal shelves to fuse produces a cleft behind the incisive foramen, which is a separate event occurring later. [Vol I Chapter 3 · Vol II §2.2]

30 A Shelf fusion begins at the incisive foramen and runs posteriorly, so an interruption late in the process leaves the anterior palate intact and the posterior palate and uvula clefted. The primary palate is formed by a different mechanism at an earlier stage and is intact here, which is precisely why the lip and alveolus are normal. [Vol I Chapter 3 · Vol II §2.2]

31 C Cleft lip with or without palate arises at the lip and primary palate around the sixth week, affects roughly 1 in 700 with a male preponderance of about 2 to 1, whereas isolated cleft palate arises later at shelf fusion, affects about 1 in 2000 and shows a female preponderance of about 4 to 1. Treating them as one condition of varying severity misstates both the embryology and the recurrence risk. [Vol I Chapter 3 · Vol II §2.2]

MNEMONIC **CLP 1:700, male 2:1; CP 1:2000, female 4:1** — opposite rates and opposite sexes for the two cleft types.

32 FTTTT

A False — the facial processes are only beginning to appear at four weeks and the face is not complete until about ten
B True — the placodes appear in the frontonasal process in the fifth week and sink to form the nasal pits
C True — fusion starts at the incisive foramen and runs backwards, ending with the uvula
D True — failure of tongue descent, as in Pierre Robin sequence, obstructs shelf elevation
E True — the three structures meet in the midline, which is why a wide cleft leaves the septum attached on one side only

[Vol I Chapter 3 · Vol II §2.2]

MNEMONIC Weeks in order: **4 face begins, 5 placodes, 6 lip fuses, 6-12 shelves fuse front to back**.

33 E The vault sutures are secondary sites that respond to the expanding brain, so early fusion of one suture blocks growth at right angles to it while compensatory growth continues at the patent sutures, giving a normal circumference with an abnormal shape. Calling the sutures primary growth centres is the historical error that this pattern of compensation disproves. [Vol I Chapter 4 · Vol II §2.3]

34 A The spheno-occipital synchondrosis is an endochondral growth site that stays active until about 13 to 15 years, and its growth carries the glenoid fossa and therefore the mandibular position relative to the maxilla. The anterior cranial base is essentially complete by about seven years, which is exactly why it is used for superimposition. [Vol I Chapter 4 · Vol II §2.4]

35 C Displacement of the whole bone at its sutures and remodelling of its surfaces are independent processes acting at once, so the maxilla advances bodily even while its anterior surface resorbs and its tuberosity region appositionally lengthens the arch. Assuming forward movement must come from anterior apposition is the error the implant data were designed to correct. [Vol I Chapter 4 · Vol II §2.5]

36 E The ramus relocates posteriorly by resorbing its anterior border and depositing on its posterior border, which lengthens the corpus and creates space for the erupting molars. Reversing the appositional and resorptive fields is the standard error, and it would shorten rather than lengthen the dental arch. [Vol I Chapter 4 · Vol II §2.6]

37 FFFFF

A False — it is a syndesmosis between two intramembranous bones, which is why it can be expanded
B False — the condyle is secondary cartilage with a fibrous covering and no organised columnar zone
C False — cartilage is unique in being able to grow interstitially as well as appositionally
D False — the vault forms and grows by intramembranous ossification at its sutures and surfaces
E False — the sutures respond to displacement of the nasomaxillary complex rather than driving it

[Vol I Chapter 4 · Vol II §2.3]

38 B The difference between the two figures is the remodelling that removes bone beneath the chin and deposits it at the gonial angle, so a conventional tracing understates true rotation by a large margin. Trusting the mandibular plane as the accurate record is exactly the assumption that the implant studies overturned. [Vol I Chapter 4 · Vol II §2.6]

39 D Intramatrix rotation is the rotation of the corpus within its soft-tissue matrix, produced by surface remodelling, and it is defined as total rotation minus matrix rotation, which is precisely

the discrepancy described. Matrix rotation is the change in the lower border relative to the cranial base, and here it is the component that is nearly absent. [Vol I Chapter 4 · Vol II §2.6]

40 C Forward rotation brings the symphysis up and forwards and progressively deepens the bite, so overbite reduction in these patients is fighting the direction of growth and relapse is the expected behaviour. Backward rotation gives the opposite picture of increased lower face height and a tendency to open bite. [Vol I Chapter 4 · Vol II §2.6]

41 TTTFT

A True — the symphysis swings upwards and forwards, closing the bite anteriorly
B True — a deep antegonial notch accompanies the backward rotating, vertically growing pattern
C True — that is the definition, and it is measurable only with implants
D False — matrix rotation is measured from the lower border tangent, not from implants
E True — this is why the mandibular plane angle changes so little in a strong forward rotator

[Vol I Chapter 4 · Vol II §2.6]

MNEMONIC **Bjork's 7 signs**: condyle, canal, lower border, symphysis, interincisal angle, intermolar angle, face height.

42 A Flexure of the cranial base brings the glenoid fossa and therefore the whole mandible forwards relative to the maxilla, which can produce a Class III relationship without either jaw being abnormal in size. Excess condylar growth is the obvious answer but is excluded by the finding that mandibular length is normal. [Vol I Chapter 4 · Vol II §2.4]

43 E An osseointegrated implant behaves like an ankylosed tooth and does not follow the continuing vertical development of the alveolus, so the adjacent natural teeth erupt past it and the crown appears to submerge. Assuming maxillary growth is complete at 15 is the error, because vertical alveolar development continues into adult life. [Vol I Chapter 4 · Vol II §2.5]

44 D Deposition on the concave inner aspect with resorption on the convex outer aspect enlarges the V while relocating it towards the wide end, which is how the ramus both grows and moves posteriorly. Reversing the fields is the common error and would make the structure shrink and migrate towards its apex. [Vol I Chapter 4 · Vol II §2.6]

45 TFFFF

A True — growth anterior to sella is largely finished early, making the line reliable for superimposition
B False — it remains active until about 13 to 15 years and may not close fully until 20
C False — the sutures are secondary and respond to separation of the bones by the growing brain
D False — the vault trebles in volume after birth, most of it in the first two years
E False — vault growth follows the neural curve because it is driven by the brain it encloses

[Vol I Chapter 4 · Vol II §2.3]

MNEMONIC **Cranial base clock**: anterior cranial base done by about 7y; spheno-occipital synchondrosis active to about 13-15y.

46 B The vertical pattern described tends to persist or intensify through adolescence, so mechanics that extrude posterior teeth will open the bite further and vertical control becomes the governing principle of the plan. Expecting spontaneous closure by forward rotation reverses the direction of growth in this facial type. [Vol I Chapter 4 · Vol II §2.6]

47 E Movement of point A on cranial base superimposition records displacement of the whole maxilla, while movement of the incisor on maxillary superimposition records eruption and drift within that bone, so the two are additive rather than alternative. Concluding that nothing has grown confuses two different superimpositions measuring two different things. [Vol I Chapter 4 · Vol II §2.5]

48 **B** Nasal and chin growth continue later and by a greater amount than lip growth, so a profile judged acceptable in early adolescence commonly flattens further through the late teens and beyond. Assuming the profile is fixed at 13 is the error, and it leads to plans that leave adults looking flatter than intended. [Vol I Chapter 4 · Vol II §2.7]

49 **TTTFF**

A True — these are the classical associated features of the forward rotating face
B True — the increase in face height is the defining consequence of the backward pattern
C True — low-grade forward rotation continues well beyond the pubertal peak
D False — forward rotation is exactly the pattern in which a reduced overbite relapses
E False — the signs are helpful as a group but their predictive value in the individual is modest

[Vol I Chapter 4 · Vol II §2.6]

50 **D** Vertical separation of the jaw bases is continuously matched by eruption of the teeth carrying their alveolar processes with them, which is why the occlusal relationship is preserved while both bones move apart. Assuming the teeth are static during growth ignores the dentoalveolar compensation that keeps an occlusion intact. [Vol I Chapter 4 · Vol II §2.7]

51 **A** The condyle is secondary cartilage covered by fibrous tissue with an underlying proliferative zone whose activity is modulated by local demand, which is what allows a sustained postural change to alter its growth. Primary cartilage such as the synchondroses grows largely according to intrinsic programming and is far less responsive to the mechanical environment. [Vol I Chapter 4 · Vol II §2.6]

52 **C** Facial development runs from about the fourth to the tenth week, with lip fusion around the sixth week and palatal closure following, so the whole critical period lies within the first trimester and often before a pregnancy is confirmed. Believing the risk period is still ahead at four weeks misplaces it by two months. [Vol I Chapter 3 · Vol II §2.2]

53 **FTFFT**

A False — enamel organ is derived from oral ectoderm, not from neural crest
B True — the ectomesenchyme of papilla and follicle is neural crest in origin
C False — each arch has an ectodermal covering, an endodermal lining and a neural crest derived mesenchymal core
D False — retinoids are potent craniofacial teratogens and are contraindicated in pregnancy
E True — deficient crest population of the arches explains the constellation of jaw, ear and malar defects

[Vol I Chapter 3 · Vol II §2.1]

54 **D** Mandibular growth continues after maxillary growth has largely finished and its late adolescent increment is both larger and more variable, so an established Class III discrepancy typically increases rather than self-corrects. Expecting the maxilla to catch up reverses the actual sequence of jaw growth and is why Class III cases are reviewed to the end of growth. [Vol I Chapter 4 · Vol II §2.7]

55 **B** The neurocranial capsule expands and the vault bones are carried apart passively, with new bone laid down at the sutures to fill the gap, which is the clearest clinical demonstration of a capsular matrix at work. Attributing the enlargement to intrinsic sutural potential cannot explain why the face, with sutures of its own, is unaffected. [Vol I Chapter 4 · Vol II §2.3]

56 **E** The tuberosity lengthens the alveolar process posteriorly and is the site that creates room for the second and third molars, which is why its growth governs whether they erupt or impact. The anterior surface is resorptive during this period, so it removes rather than adds arch length. [Vol I Chapter 4 · Vol II §2.5]

Scored under 70% of the 58 marks on this block? Re-read Volume II §2, then sit it again in a fortnight.

Block C — Development of the dentition (Q57–96 · 80 marks · Vol II §3)

57 **A** The permanent incisors need about 7 mm more room in the maxilla and 5 mm in the mandible than the primary incisors occupy, and most of this is met by pre-existing spacing, so a complete absence of spacing at four years predicts incisor crowding. Reassurance on the grounds that the arches will widen is misplaced, because the increase in intercanine width is only 2 to 3 mm and cannot cover the deficit. [Vol I Chapter 5 · Vol II §3.1]

58 **C** The primate spaces lie mesial to the upper primary canine and distal to the lower primary canine, and their position matters because the lower space is where the lower first permanent molar drifts during the early mesial shift. Reversing the two arches is the standard error and would place the space where it cannot be used. [Vol I Chapter 5 · Vol II §3.1]

59 **TTFTF**

A True — spacing is the principal source of room for the wider permanent incisors
B True — the second primary molars complete the dentition at around two and a half years
C False — primary incisor relationships are a poor predictor, and attrition confounds them further
D True — wear is physiological in the primary dentition and reduces the overbite
E False — the primary molars are described by the terminal plane relationship, not by Angle's classes

[Vol I Chapter 5 · Vol II §3.1]

60 **B** A flush terminal plane leaves the molars cusp to cusp, and the early mesial shift uses the lower primate space to move the lower molar forwards by about 2 mm, which is enough to establish a Class I relationship. Remaining end-to-end is only the outcome where no spacing is available and the late mesial shift into the leeway space has not yet occurred. [Vol I Chapter 5 · Vol II §3.2]

> **MNEMONIC** Two mesial shifts, in order: **early into the primate space at molar eruption, late into the leeway space at 11-12y**.

61 **D** Leeway space is the excess width of the primary canine and molars over their successors, about 2.5 mm per quadrant in the mandible and 1.5 mm in the maxilla, and the lower first molar migrates forwards into part of it unless it is held. Quoting 1.5 mm for the lower arch transposes the maxillary figure and would underestimate the space by 2 mm across the arch. [Vol I Chapter 5 · Vol II §3.3]

62 **A** The combined width of the four permanent upper incisors exceeds that of their predecessors by about 7 mm, which is normal, and the deficit is met by primary spacing, an increase in intercanine width and a more proclined incisor position. Calling it macrodontia and reducing the teeth removes enamel to solve a problem that is a normal stage of development. [Vol I Chapter 5 · Vol II §3.1]

63 **FFTTF**

A False — the widths of the unerupted teeth must be predicted, which adds error absent in the permanent dentition
B False — the methods differ systematically and prediction tables are population specific
C True — the required figure is simply the summed widths of the teeth that must fit
D True — measuring along the existing arch form is meaningless if the plan is to change it
E False — a positive result means there is a surplus of space, not a deficit

[Vol I Chapter 5 · Vol II §3.3]

64 **C** The canine crowns pressing on the lateral incisor roots tip the incisor crowns distally and open a midline space, and the space closes spontaneously once the canines reach the occlusal level, so nothing but explanation and review is needed. Closing the diastema now with an appliance both intervenes in a self-correcting condition and commits the patient to indefinite retention. [Vol I Chapter 5 · Vol II §3.4]

65 **E** When the second molar is ahead of the second premolar in development it erupts first and pushes the first molar mesially, so the leeway space that would have relieved the anterior crowding is absorbed posteriorly. Assuming the premolars will win the race ignores the radiographic evidence of the relative stages of root formation, which is the whole point of the observation. [Vol I Chapter 5 · Vol II §3.3]

66 **A** Lingual eruption of the lower incisors behind retained predecessors is common and the teeth usually move labially into the arch once the primary teeth are removed and the tongue can act on them. Appliance therapy is disproportionate for a situation that corrects itself once the obstruction is removed. [Vol I Chapter 5 · Vol II §3.4]

67 **TTTTF**

A True — both appear at around six years, and the molar is often mistaken for a primary tooth
B True — the sex difference is small but consistent across the dentition
C True — the canine is a late tooth, which is part of why it is so often displaced
D True — this relationship holds well enough to be used in dental age assessment
E False — six months is within normal variation, whereas a difference beyond about six to twelve months merits a film

[Vol I Chapter 5 · Vol II §3.2]

MNEMONIC **6-6 rule**: lower first molar and central incisor erupt together at six; a tooth erupts at 2/3-3/4 root formation.

68 **C** Progressive infraocclusion allows the adjacent teeth to tip over the primary molar and the opposing tooth to overerupt, and a tooth already 2 mm below the plane is behaving as an ankylosed tooth in a growing patient. Good condition of the crown is the finding most likely to prompt retention, but it says nothing about the vertical behaviour that determines the prognosis. [Vol I Chapter 5 · Vol II §3.6]

69 **E** Removing the obstruction while there is still root formation and available space gives a high rate of spontaneous eruption, so surgical exposure with traction should be held in reserve. Exposing and bonding at the same visit is the tempting option but commits the patient to appliance therapy that most such teeth do not need. [Vol I Chapter 5 · Vol II §3.5]

70 **B** Extraction of a first permanent molar of poor prognosis gives the best spontaneous result when the second molar is at the stage of bifurcation calcification, typically around eight and a half to nine and a half years, so the timing rather than the decision is the examinable point. Waiting until the second molars have erupted forfeits the spontaneous mesial movement that makes the extraction work. [Vol I Chapter 5 · Vol II §3.6]

71 **FFFTF**

A False — the great majority are in the maxilla and most of those are in the premaxilla
B False — the tuberculate form is the classic obstructor, whereas conical forms often erupt or cause displacement
C False — the supplemental form most often duplicates the upper lateral incisor
D True — multiple supernumeraries with failure of eruption should prompt that diagnosis
E False — there is a male preponderance of about two to one

[Vol I Chapter 5 · Vol II §3.5]

MNEMONIC **Conical erupts, tuberculate obstructs, supplemental duplicates** — the three supernumerary forms and their behaviour.

72 D Mild infraocclusion of a primary molar with a normal successor usually resolves by normal exfoliation, so review is what is required and intervention is reserved for progressive submergence or tipping of the neighbours. Extracting now creates a space maintenance problem where none yet exists. [Vol I Chapter 5 · Vol II §3.6]

73 C Ectopic eruption of the upper first molar is reversible in roughly two-thirds of cases, with spontaneous disimpaction by about eight years, so observation is the first step and active measures are for the irreversible group. Placing a separator immediately treats a condition that will most often resolve, and it is the answer that looks most decisive. [Vol I Chapter 5 · Vol II §3.6]

74 A Interchange of position between the canine and the first premolar is the commonest maxillary transposition, affecting around 0.3 per cent of the population, and the decision is whether to accept the transposed position or to correct it. Canine and lateral incisor interchange is the characteristic mandibular transposition, not the maxillary one. [Vol I Chapter 5 · Vol II §3.6]

75 TFTTF

A True — figures of between about 4.5 and 6.5 per cent are consistently reported
B False — absence in the primary dentition is rare, at well under 1 per cent
C True — these two account for the majority of absent teeth once third molars are set aside
D True — generalised microdontia and reduced crown form accompany the trait
E False — there is a slight female preponderance

[Vol I Chapter 5 · Vol II §3.5]

MNEMONIC **Hypodontia**: about 5% prevalence, rare in primary teeth, commonest at lower second premolar and upper lateral incisor.

76 E Loss of a second primary molar before the first permanent molar has settled allows rapid mesial drift, and it is the one primary extraction that most reliably costs space, so a space maintainer is the standard response. Balancing is reserved for canines, where the risk is a centreline shift rather than mesial drift. [Vol I Chapter 5 · Vol II §3.7]

77 D Unilateral loss of a primary canine in a crowded arch lets the labial segment collapse to that side, and the resulting centreline shift is difficult to correct later, which is why a balancing extraction is considered. Mesial molar drift is the consequence of losing a primary molar, not a canine. [Vol I Chapter 5 · Vol II §3.7]

78 FTFFF

A False — it depends on the tooth, the age, the crowding and the treatment plan
B True — holding the first molar forwards allows the 2.5 mm per quadrant to be used anteriorly
C False — a band and loop is a unilateral appliance, and a lingual arch is used bilaterally
D False — fixed appliances increase plaque retention and demand more, not less, hygiene support
E False — where a successor is absent the decision about the space is more important, not less

[Vol I Chapter 5 · Vol II §3.7]

MNEMONIC Choose by pattern: **band and loop for one tooth, lingual arch to preserve leeway space bilaterally**.

79 B Counting the teeth in the quadrant is the discriminator: if the double tooth is counted as one and the quadrant is short by a tooth, two germs have joined, which is fusion, whereas gemination leaves a normal count because one germ has divided. Here the rest of the arch is normal but the affected quadrant lacks a separate lateral incisor, which identifies fusion. [Vol I Chapter 5 · Vol II §3.6]

80 **E** Microdontia of one lateral incisor with absence of the other is the classic expression of a variable trait, which is why the anomaly clusters in families and why the contralateral tooth is always examined. Extracting the peg lateral to achieve symmetry destroys tissue that can usually be built up and is the more interventionist reading of a common finding. [Vol I Chapter 5 · Vol II §3.6]

81 **TTFTT**

A True — these are the standard figures and they differ between the arches
B True — the maxillary figure is smaller than the mandibular figure of about 2.5 mm
C False — the increase is of the order of 2 to 3 mm, which is why it cannot solve real crowding
D True — palpation in the buccal sulcus is the routine screening manoeuvre at this age
E True — impalpability at this age is the trigger for a film, not a reason to wait

[Vol I Chapter 5 · Vol II §3.3]

MNEMONIC **7-5, 2.5-1.5**: incisor liability 7mm maxilla / 5mm mandible; leeway 2.5mm mandible / 1.5mm maxilla per quadrant.

82 **B** Serial extraction is designed for exactly this presentation, severe crowding with a Class I skeletal pattern and an average overbite in a full complement, and it exploits eruptive movement to reduce the later appliance burden. Preserving leeway space cannot address a deficiency of 8 mm per arch, which is more than three times the space it can release. [Vol I Chapter 5 · Vol II §3.7]

MNEMONIC Serial extraction order: **C, D, then 4** — primary canines, first primary molars, then first premolars.

83 **D** Crowding of 2 mm at this age is commonly relieved by the increase in intercanine width that accompanies canine eruption, so it does not by itself justify treatment. Extracting the primary canines buys alignment at the cost of lingual collapse of the labial segment and a loss of arch length that is difficult to recover. [Vol I Chapter 5 · Vol II §3.4]

84 **A** Absence of at least one third molar approaches a quarter of the population, so including them raises the figure far above the 4.5 to 6.5 per cent quoted for hypodontia excluding third molars. Selection bias would also inflate the number, but it would not produce a figure so close to the known third molar absence rate. [Vol I Chapter 5 · Vol II §3.5]

85 **FTTFF**

A False — it characteristically affects the posterior teeth, with all teeth distal to the affected tooth involved
B True — this is the defining clinical feature and the reason the diagnosis matters
C True — the association with that gene supports the familial pattern seen
D False — the ankylosed tooth stays where it is while the alveolus grows past it, so it appears to submerge
E False — a dilacerated crown or root often prevents eruption whatever is done, and many such teeth are removed

[Vol I Chapter 5 · Vol II §3.6]

86 **C** Interceptive extraction of the primary canine at about 10 to 11 years improves the position of a palatally displaced canine in a substantial proportion of uncrowded cases, and the result is judged at review a year later. Surgical exposure now bypasses a simple intervention that may render surgery unnecessary altogether. [Vol I Chapter 5 · Vol II §3.5]

87 **D** An object that moves in the same direction as the tube shift lies on the far side of the reference object, so a canine moving with the tube is palatal, which is the position of up to 85 per cent of displaced canines. Concluding that it is buccal reverses the rule and would send the surgeon to the wrong side of the arch. [Vol I Chapter 5 · Vol II §3.5]

MNEMONIC **SLOB** — Same Lingual, Opposite Buccal: a canine moving with the tube shift is palatal, the commoner side.

88 B A sound, minimally infraoccluded primary molar in an uncrowded arch has a useful lifespan and keeping it defers an irreversible decision until the patient is old enough to take part in it. Extracting now and closing the space commits an uncrowded arch to appliance therapy for a problem that is currently causing none. [Vol I Chapter 5 · Vol II §3.6]

89 TFFTT

A True — the increase of 2 to 3 mm accompanies incisor and canine eruption
B False — arch length shortens as the molars drift mesially into the leeway space
C False — intercanine width slowly reduces after the mid-teens in both arches
D True — this is why it is described as a maturational change rather than a treatment failure
E True — the arch narrows and shortens, and the incisors are squeezed together

[Vol I Chapter 5 · Vol II §3.8]

90 E Late lower incisor crowding occurs whether or not third molars are present and is attributed to continuing late mandibular growth with reduction in arch width and length, so removing third molars to prevent it is not supported. Blaming the third molars is the intuitive answer and the one the evidence does not support. [Vol I Chapter 5 · Vol II §3.8]

91 A Holding the lower first molars prevents them consuming the 2.5 mm per quadrant, which gives roughly 5 mm of usable arch space in total and therefore addresses mild to moderate crowding only. Treating it as equivalent to premolar extraction overstates a gain that is a fraction of the 14 mm or so that extractions create. [Vol I Chapter 5 · Vol II §3.7]

92 TTTTT

A True — it erupts distal to the second primary molar and replaces nothing
B True — its eruption at around six years is silent and commonly goes unnoticed
C True — the terminal plane gives way to the molar relationship once it erupts
D True — planned loss at the right stage can allow useful spontaneous space closure
E True — its long exposure and its calcification timing both make it the most affected tooth

[Vol I Chapter 5 · Vol II §3.2]

93 C Spontaneous mesial movement of the upper second molar is more favourable and less likely to leave a tipped, rotated contact than in the lower arch, which is why upper first molars of poor prognosis are extracted more readily. Compensating and balancing extractions are considered on the merits of the individual case rather than being mandatory. [Vol I Chapter 5 · Vol II §3.7]

94 B Eruption normally occurs when about two-thirds of the root is formed, so a lateral incisor with half its root formed at eight years is on schedule and needs only explanation and review. Urgent referral applies to asymmetry beyond about six to twelve months or to an obstruction, neither of which is present here. [Vol I Chapter 5 · Vol II §3.2]

95 D Intrusion of a primary incisor drives its apex towards the developing permanent crown, and an angulated crown will not follow a normal eruptive path, so this is the sequel that changes management rather than merely the restorative plan. Enamel defects are the commoner consequence overall but they do not obstruct eruption, which is what makes dilaceration the answer that matters here. [Vol I Chapter 5 · Vol II §3.6]

96 A Removing the primary canines relieves incisor crowding by letting the labial segment spread into the space, but in the absence of an appliance the buccal segments migrate forwards and the arch length is lost, so the manoeuvre must be planned as part of a treatment rather than as a cure. Believing it will produce a lasting gain in arch length is the error, because nothing has been added to the space in the arch. [Vol I Chapter 5 · Vol II §3.7]

Scored under 70% of the 80 marks on this block? Re-read Volume II §3, then sit it again in a fortnight.

Block D — The biology of tooth movement, eruption and root resorption (Q97–141 · 89 marks · Vol II §4)

97 C The initial phase is rapid displacement of the tooth within the ligament space over a couple of days, after which hyalinised tissue must be removed by undermining resorption before movement resumes, which takes three to four weeks. Increasing the force is the intuitive response and it prolongs the lag phase by producing more hyalinisation, not less. [Vol I Topic 1 · Vol II §4.2]

98 TTFFF

A True — the ligament is only about 0.2 mm wide, which limits how far this phase can carry the tooth
B True — this is the interval during which hyalinised tissue is removed
C False — heavier force produces more hyalinisation and a longer, not shorter, lag
D False — the post-lag phase is the sustained phase in which most of the clinical movement occurs
E False — some hyalinisation occurs even with light forces because the ligament is not loaded uniformly

[Vol I Topic 1 · Vol II §4.2]

MNEMONIC Three phases, in order: **initial (about 2 days), lag (20-30 days), post-lag (sustained)**.

99 E Occluding the capillaries in the compressed ligament kills the local cells and leaves a hyaline zone from which no osteoclasts can be recruited, so resorption has to be undertaken from the marrow behind the lamina dura, which is undermining resorption. Frontal resorption is the appearance seen with light forces, in which the ligament remains vital and osteoclasts line the socket wall directly. [Vol I Topic 1 · Vol II §4.2]

100 A Bodily movement distributes the force over the whole root surface, so around 50 to 100 cN gives an optimal stress in the ligament without producing widespread hyalinisation. Three hundred cN is the figure that comes to mind from studies of the fastest canine retraction, but fastest is not optimal once anchorage loss and root resorption are counted. [Vol I Chapter 6 · Vol II §14.1]

MNEMONIC **Force ladder**: optimal bodily movement 50-100 cN; canine retraction fastest, not optimal, at 272 cN.

101 C Beyond a threshold the ligament is crushed over a wider area, more hyaline tissue must be removed by undermining resorption, and the tooth moves more slowly despite the greater force. Adopting 272 cN as the clinical figure is the trap: the study identified where rate peaks, not where the balance of rate, anchorage and root resorption is best. [Vol I Chapter 6 · Vol II §14.1]

102 FTTTT

A False — osteoprotegerin binds RANKL, not RANK, which is precisely how it blocks the pathway
B True — the peak at around three days corresponds with the onset of active resorption
C True — by sequestering RANKL it prevents ligation of RANK on the precursor
D True — the cells of the ligament and the osteoblast lineage are the principal sources
E True — it is the ratio rather than either concentration alone that governs the response

[Vol I Topic 1 · Vol II §4.3]

MNEMONIC **RANKL-RANK-OPG**: RANKL from PDL and osteoblasts binds RANK to resorb bone; OPG is the decoy blocking it, peaking day 3.

103 E Orthodontic tooth movement proceeds at roughly 1 mm per month, so 8 mm of space closure occupies most of a year even when everything goes well, which is the figure to quote at consent. Two to four months implies a rate of 2 to 4 mm per month, which is not achievable by any biological route and is the promise that produces complaints. [Vol I Topic 1 · Vol II §4.2]

104 **B** A ligament space of around 0.2 mm means that the initial phase can carry the tooth only a fraction of a millimetre, after which further movement depends entirely on remodelling of the socket wall. Imagining a 2 mm space would suggest that much larger movements could occur without any biological response, which is the misconception the figure corrects. [Vol I Topic 1 · Vol II §4.1]

105 **D** Piezoelectric potentials decay rapidly and reverse when the force is removed, so they cannot account for movement sustained over weeks, whereas the vascular and cellular events in the loaded ligament can. Concluding that the pressure-tension account has been disproved overstates the case: the two explanations describe different parts of the same sequence. [Vol I Topic 1 · Vol II §4.3]

106 `FFTFT`

A False — osteocytes are the principal strain sensors in bone and signal to the surface cells
B False — prostaglandin E2 rises and its inhibition slows tooth movement
C True — fluid shear on cell membranes is a leading candidate mechanism
D False — both sides show cytokine activity, with different profiles
E True — deposition on the tension side is what maintains attachment as the tooth moves

[Vol I Topic 1 · Vol II §4.3]

107 **C** Uncontrolled tipping is produced by a single force with no counterbalancing couple, which is exactly what a palatal finger spring delivers, and the tooth rotates about a point roughly one-third of the root length from the apex. A ratio of about 10 to 1 is the requirement for bodily movement, and higher ratios again for root movement. [Vol I Chapter 6 · Vol II §14.2]

MNEMONIC **M:F ladder**: zero force tips, about 10:1 moves bodily, over 12:1 moves the root — rising ratio, rising control.

108 **A** The centre of resistance of a single-rooted tooth lies within the root at roughly one-third to two-fifths of its length from the alveolar crest, and its position determines the moment produced by any force applied at the bracket. Placing it midway between bracket and apex sounds reasonable but misplaces it coronally and would predict the wrong moment for a given force. [Vol I Chapter 6 · Vol II §14.2]

109 **E** Loss of crestal bone lengthens the distance between the bracket and the centre of resistance, so the same force generates a larger tipping moment and a correspondingly larger counter-moment is needed. Assuming the centre of resistance is fixed by root length alone ignores the level of the supporting bone, which is what has changed here. [Vol I Chapter 6 · Vol II §14.2]

110 `TTTFF`

A True — every point moves the same distance in the same direction, which is what a centre of rotation at infinity means
B True — ratios above about 12 to 1 tip the crown back as the root moves
C True — two equal and opposite forces with no resultant produce pure rotation
D False — concentration of stress at the apex is precisely why intrusive forces must be light
E False — extrusion loads the ligament largely in tension and is the gentlest of the movements

[Vol I Chapter 6 · Vol II §14.2]

111 **D** Intrusive force is concentrated at the apex, so the total force on a four-incisor segment is kept to a few tens of centinewtons and directed through the segment's centre of resistance if uncontrolled proclination is to be avoided. Accepting root resorption in exchange for heavier force reverses the reasoning, because intrusion is already the movement most strongly associated with resorption. [Vol I Chapter 6 · Vol II §14.1]

112 **B** Some radiographic shortening is the rule rather than the exception, reported in up to about 73 per cent, with histological evidence in up to 90 per cent, while the clinically important severe form affects only a small minority. Quoting 5 per cent overall understates a near-universal finding and makes the rare severe case look like a treatment failure rather than a recognised risk. [Vol I Topic 1 · Vol II §4.5]

MNEMONIC **73-90-5**: OIIRR is radiographic in up to 73%, histological in up to 90%, severe in only 1-5%.

113 **E** Root morphology and any previous resorption are the strongest predictors, which is why a pre-treatment radiograph of the incisors is examined for them and the finding changes both consent and monitoring. Bracket type is the distractor that appeals because it is under the operator's control, but it has not been shown to alter resorption risk. [Vol I Topic 1 · Vol II §4.5]

MNEMONIC **Root, duration, force, torque**: blunt/resorbed roots, long treatment, heavy force, apical movement raise OIIRR risk.

114 **FFFTT**

A False — lost root length is not regained, although the resorbing surface repairs with cementum
B False — continuing unaltered risks further loss, and the standard response is to pause
C False — duration of treatment is one of the consistently reported treatment-related risk factors
D True — a pause of three to six months allows cemental repair of the resorbed surfaces
E True — this is a recognised complication and both the discussion and the record are required

[Vol I Topic 1 · Vol II §4.5]

115 **B** A passive interval allows the resorbed surfaces to repair with cementum, after which the treatment objectives are revised to whatever can be achieved with the least further movement. Finishing faster with heavier forces is exactly the wrong direction, since force magnitude and treatment duration are both risk factors for further resorption. [Vol I Topic 1 · Vol II §4.5]

116 **D** Patients who are going to resorb significantly show it early, so a film at around six to nine months identifies the susceptible group while there is still time to modify the plan. Waiting until debond identifies the problem when nothing can be done about it, and symptoms are absent in almost all cases. [Vol I Topic 1 · Vol II §4.5]

117 **A** A solid percussion note with absent mobility and a focal loss of the ligament shadow is the classical picture of a localised bony union, and such a tooth resists orthodontic force while its neighbours continue to move. Primary failure of eruption is the distractor, but it characteristically involves the affected tooth and all teeth distal to it rather than a single premolar with normal neighbours. [Vol I Chapter 6 · Vol II §4.6]

118 **TFTFF**

A True — this is the defining behaviour and the reason the diagnosis must be made before traction
B False — the ankylosed tooth stays put while the alveolus grows past it, so it submerges
C True — the distal segment involvement is the pattern that distinguishes it from simple ankylosis
D False — an ankylosed tooth does not move, and the force is expressed on its neighbours instead
E False — the altered percussion note is one of the most useful clinical signs

[Vol I Chapter 6 · Vol II §4.6]

MNEMONIC Tell them apart: **PFE hits the tooth and everything distal to it; ankylosis is focal, with a dull percussion note**.

119 **C** Involvement of a tooth and all teeth distal to it, with a family history, points to primary failure of eruption, in which applied traction arrests movement and ankyloses the tooth, so the plan must be segmental extraction and prosthodontic or restorative management rather than traction. Believing that more force will succeed is the error that turns a difficult problem into an unrestorable one. [Vol I Chapter 6 · Vol II §4.6]

120 **D** The follicle signals the osteoclastic clearance of the eruption pathway and the deposition of bone at the base of the crypt, so removing it abolishes eruption even though the tooth germ is intact. Attributing eruption to root formation is the classical error, disproved by the eruption of rootless teeth and by the failure of teeth whose roots are complete. [Vol I Topic 1 · Vol II §4.4]

121 **B** If root growth pushed the tooth out then eruption should stop when the root is complete and should be impossible without a root, and neither holds, which is why eruption is attributed to the follicle and the surrounding tissues. That teeth erupt with incomplete roots is true but weaker, since a partly formed root could still be doing the pushing. [Vol I Topic 1 · Vol II §4.4]

122 **TTTTT**

A True — the follicle is the essential signalling structure for the eruptive process
B True — resorption above and deposition below are the two sides of the same process
C True — continued eruption through life maintains contact against attrition
D True — overeruption of an unopposed tooth is the everyday demonstration of this
E True — the absence of a cleared pathway is a common reason for permanent impaction

[Vol I Topic 1 · Vol II §4.4]

123 **E** Bisphosphonates are taken up by bone and suppress osteoclastic resorption, which is the rate-limiting step in tooth movement, so movement is slow or incomplete and the risk of osteonecrosis governs any surgical step. Expecting faster movement reverses the drug's action on the cells that must remove bone ahead of the tooth. [Vol I Topic 1 · Vol II §4.3]

124 **A** Reduced cellularity and vascularity in the adult periodontium delay the cellular response, so the lag is longer, but once remodelling is established the rate of movement is broadly comparable. Assuming that heavier forces are needed to overcome denser bone is the common error, and it increases hyalinisation and resorption without increasing speed. [Vol I Topic 1 · Vol II §4.2]

125 **FFFFF**

A False — trials have not shown a clinically meaningful reduction in treatment time
B False — the regional acceleratory phenomenon is transient, lasting a matter of months
C False — it is a surgical procedure with pain, swelling and a risk to the interdental bone
D False — systematic reviews find no significant difference in overall treatment duration
E False — the side effects and the local delivery problem have kept it experimental

[Vol I Topic 1 · Vol II §4.3]

126 **C** Prostaglandin E2 is part of the cascade that recruits osteoclasts, so drugs that inhibit its synthesis may reduce the rate of movement, whereas paracetamol acts by a different route and is the usual recommendation. Aspirin is the trap, since it is the archetypal inhibitor of prostaglandin synthesis rather than an exception to it. [Vol I Topic 1 · Vol II §4.3]

127 **B** The pain curve rises over the first day, peaks at around 48 hours and settles within a week, so the patient is at the worst point of a predictable course and needs an explanation and analgesia rather than an intervention. Removing the appliance at the peak of a self-limiting response abandons treatment at the moment the patient is least able to judge it. [Vol I Topic 1 · Vol II §4.2]

128 **TTFTT**

A True — this is the figure from the anchorage literature and it governs how the space is planned
B True — these are the conventional anchorage figures, distinct from those used for distalisation
C False — tipping reduces the root surface area resisting movement and loses anchorage
D True — the reactive unit is usually the more numerous and responds readily to heavy force
E True — a safety release and a locking facebow, or equivalent, are both required

[Vol I Chapter 6 · Vol II §14.3]

MNEMONIC **2/3 lost; 10-12h at 250-350g; 14h at 450-500g**: unsupported space loss, then headgear anchorage vs distalisation.

129 D Roughly two-thirds of the space is lost forward when the reactive unit is left unsupported, which is why anchorage must be planned from the outset rather than rescued later. Estimating one-third reverses the proportions and would leave a case planned for maximum retraction several millimetres short. [Vol I Chapter 6 · Vol II §14.3]

130 A Distal movement of the molars requires both a heavier force and a longer daily duration than anchorage reinforcement, which is the distinction the two regimens exist to make. The 250 to 350 g regimen is the anchorage prescription and will hold the molars where they are rather than move them back. [Vol I Chapter 6 · Vol II §14.3]

131 FTFTF

A False — two independent safety mechanisms are mandatory and the requirement is not negotiable
B True — a line of pull through the centre of resistance gives bodily movement, above or below it produces tipping
C False — headgear must be removed for sport and for rough play
D True — the high pull vector has a vertical component that counteracts molar extrusion
E False — the serious hazard that drives the safety requirement is ocular injury, not intraoral injury

[Vol I Chapter 6 · Vol II §14.3]

132 C An insertion torque of about 5 to 10 Ncm gives primary stability without generating the compressive stress that causes bone necrosis and later failure, so both too little and too much torque predict loss of the screw. Assuming that higher torque means better stability is the intuitive error, and it is a recognised cause of failure. [Vol I Chapter 6 · Vol II §14.4]

133 E Reported failure is lowest in the median palate, at around 1 per cent, against an overall figure of about 12 per cent and a mandibular buccal shelf figure that reaches 65 per cent, and the palate also suits upper molar distalisation mechanically. The maxillary buccal interradicular site is the familiar default, but its failure rate is several times that of the palate. [Vol I Chapter 6 · Vol II §14.4]

MNEMONIC **1 to 65**: miniscrew failure is lowest at the median palate (about 1%), highest at the mandibular buccal shelf (about 65%).

134 A Keeping the stress below the level that occludes the ligament capillaries allows osteoclasts to work directly on the socket wall, so movement is steady rather than punctuated by lag phases. Claiming that intermittent heavy force avoids resorption altogether is untrue, since force magnitude is among the treatment factors associated with root resorption. [Vol I Topic 1 · Vol II §4.2]

135 TFFTF

A True — deposition on the tension side restores support as the tooth moves away

B False — it is the supracrestal and transseptal fibres that remodel slowly, which is why rotations relapse
C False — the lamina dura is the socket wall and is continually remodelled during movement
D True — the relative resistance of cementum is why bone resorbs preferentially and roots usually survive
E False — attachment is maintained where the periodontium is healthy and the force is appropriate

[Vol I Topic 1 · Vol II §4.1]

136 C The supracrestal and transseptal gingival fibres remodel over years rather than months and are the specific source of rotational relapse, so sectioning them addresses the cause rather than restraining its effect. A bonded retainer is effective and is the usual choice, but it holds the tooth against the fibres rather than dealing with them. [Vol I Topic 1 · Vol II §4.1]

137 E Orthodontically induced inflammatory root resorption is a response to force and is reported with aligners as with fixed appliances, with comparative studies showing at most a small difference that does not change the consent discussion. Saying aligners cannot cause resorption because they do not move roots is simply wrong, since any tooth movement loads the ligament at the apex. [Vol I Topic 1 · Vol II §4.5]

138 B The usable figures are ranges derived from a small number of studies with differing methods, and they vary with the movement and with the root area over which the force is distributed, so a claim to deliver a single optimal value overstates what is known. Equating optimal with fastest ignores anchorage loss, patient discomfort and root resorption, which is precisely why the fastest canine retraction force is not the recommended one. [Vol I Chapter 6 · Vol II §14.1]

139 D Tooth movement in a healthy but reduced periodontium does not cause further attachment loss, whereas moving teeth through inflamed tissues accelerates it, so the governing condition is the state of the gingivae rather than the amount of bone. Assuming that greater force is needed because less bone remains is doubly wrong, since the reduced attachment area means the same force produces a higher stress. [Vol I Topic 1 · Vol II §4.1]

140 C The maxillary incisors are consistently the teeth most affected by orthodontically induced inflammatory root resorption, because they undergo the largest and most prolonged movements, including retraction and torque against the labial plate. The canine also moves a long way but does so bodily along the ridge and is far less often severely affected. [Vol I Topic 1 · Vol II §4.5]

141 A Bone in a long-standing edentulous ridge is denser and narrower, so movement is slow, root control is difficult, and the root can be carried out of the alveolar envelope with dehiscence or fenestration. Expecting the ridge to widen spontaneously ahead of the tooth is optimistic: the alveolus follows the tooth only where there is bone to remodel. [Vol I Topic 1 · Vol II §4.1]

Scored under 70% of the 89 marks on this block? Re-read Volume II §4, then sit it again in a fortnight.

Block E — Occlusion, malocclusion and aetiology (Q142–171 · 58 marks · Vol II §5)

142 E Andrews specified a flat to slightly curved occlusal plane, with a curve of Spee no deeper than about 1.5 mm, so a residual 3 mm curve fails that key while the other five are met. Assuming the curve of Spee is self-correcting is the error: a residual curve tends to deepen with time and carries the overbite back with it. [Vol I Chapter 2 · Vol II §5.1]

MNEMONIC **Andrews' six keys**: molar relationship, crown angulation, crown inclination, no rotations, tight contacts, flat occlusal plane.

143 **D** Angle's classification records the molar relationship and the British Standards classification records the incisor relationship, and the two are independent, so a Class I molar relationship coexists here with a Class II division 2 incisor relationship. Assuming that a Class II incisor relationship must mean a Class II molar relationship conflates two systems that describe different things. [Vol I Chapter 2 · Vol II §5.2]

144 **B** The incisor classification is a description of the relationship of the lower incisor edge to the upper cingulum plateau, and where the sides differ the accepted practice is to classify the more severe relationship and to record the asymmetry explicitly. Averaging conceals the very finding that will determine the diagnosis of a displacement or a skeletal asymmetry. [Vol I Chapter 2 · Vol II §5.2]

145 **TTTFT**

A True — this assumption is the principal weakness of the system, since the molar itself may be displaced
B True — this is the definition and it is the reason the systems are not interchangeable
C True — retroclined upper centrals with proclined laterals can still leave an increased overjet
D False — the reference teeth are absent, which is one reason the incisor classification is preferred
E True — and the overjet is correspondingly reduced or reversed

[Vol I Chapter 2 · Vol II §5.2]

MNEMONIC Two systems: **Angle classifies molars, British Standards classifies incisors** — independent, not interchangeable.

146 **E** Mesial migration of a lower molar after early loss of its primary predecessor shortens that quadrant, producing a Class II molar relationship with a Class I canine relationship and a centreline shift to the affected side. A skeletal asymmetry would be expected to disturb the canine relationship as well, which is precisely what has been preserved here. [Vol I Chapter 2 · Vol II §5.4]

147 **B** An overbite is complete when the lower incisors contact something on closure, and here that contact is with the palatal mucosa rather than with the upper incisors, which is what makes it complete to soft tissue and potentially traumatic. Calling it incomplete is the common error, since incomplete means that the lower incisors contact neither the upper incisors nor the palate. [Vol I Chapter 2 · Vol II §5.3]

MNEMONIC Describe overbite in three pairs: **increased/reduced, complete/incomplete, to tooth/to soft tissue**.

148 **D** When the whole upper tooth lies buccal to the lower so that the palatal cusp misses the lower buccal cusp, the relationship is a lingual crossbite or scissors bite, which is the opposite of the ordinary buccal crossbite. A buccal crossbite describes the far commoner situation in which the upper tooth lies palatal to its partner. [Vol I Chapter 2 · Vol II §5.3]

149 **FFTFF**

A False — the appearance is unilateral but the underlying narrowing is usually symmetrical
B False — the displacement is the single most important finding, since it is what makes correction worthwhile
C True — a symmetrically narrow maxilla forces a lateral slide for the arches to fit together
D False — displacement is the principal indication for treating a posterior crossbite
E False — displacements are associated with signs, even though causation is not established

[Vol I Chapter 2 · Vol II §5.3]

150 **A** Once the lip falls behind the upper incisors it applies a labial force to them and a lingual force to the lowers at rest, which increases the overjet and helps to maintain it, so the soft tissue is both a consequence and a cause. Regarding the lip as a passive consequence ignores the point

that resting soft tissue pressures act for many hours a day and are therefore capable of moving teeth. [Vol I Chapter 2 · Vol II §5.5]

151 C The digit holds the incisors apart asymmetrically while the cheeks act on a maxilla whose tongue support is lowered, producing the characteristic combination of asymmetrical open bite, an increased overjet and a posterior crossbite. Expecting no transverse effect misses the mechanism by which the buccinator narrows the upper arch during sucking. [Vol I Chapter 2 · Vol II §5.5]

152 D A forward tongue posture on swallowing is usually a consequence of an anterior open bite rather than its cause, and the vertical facial pattern described here is the more likely origin, so treatment is directed at the skeletal problem. The endogenous tongue thrust does exist and does predict relapse, but it is rare and should not be diagnosed simply because the tongue comes forward through an existing gap. [Vol I Chapter 2 · Vol II §5.5]

153 TFTTT

A True — duration matters more than magnitude, and resting pressures act for most of the day
B False — swallowing lasts about a second at a time and totals only minutes a day
C True — a threshold of a few hours a day is sufficient, which is the basis of part-time appliance wear
D True — the equilibrium includes intrinsic forces as well as those from the surrounding soft tissues
E True — this is the clinical expression of the theory and the reason arch form is respected

[Vol I Chapter 2 · Vol II §5.5]

MNEMONIC Ranked by duration, not force: **resting lip/cheek/tongue pressure, appliance hours/day, brief swallowing**.

154 B ANB records the difference between the two jaws but not which of them is displaced, so the individual SNA and SNB values are needed, and here an SNA of 88 degrees with a normal SNB identifies the maxilla as the prognathic jaw. Concluding that the mandible is retrognathic is the reflex reading of a Class II skeletal pattern and would send the treatment plan to the wrong jaw. [Vol I Chapter 2 · Vol II §5.4]

155 E Obstruction of a single tooth by a supernumerary is the archetypal local cause, affecting one site while the rest of the occlusion develops normally, and it is remediable by dealing with the obstruction. Calling it a generalised dental cause would imply an arch-wide problem of tooth number or size, which is not what is present. [Vol I Chapter 2 · Vol II §5.6]

MNEMONIC **Skeletal, Soft tissue, Local dental, Habit, Generalised dental** — run the five before naming a single cause.

156 A Class II division 1 affects roughly one in six of a British population while Class III affects around 3 per cent, and getting the order of magnitude right matters because it determines the size of the service required for each. Reversing the two figures would predict a Class III workload six times greater than reality. [Vol I Chapter 2 · Vol II §5.7]

MNEMONIC **One in six, one in thirty**: Class II division 1 about 15-20%, Class III about 3%, in a British population.

157 FTTFF

A False — Class III prevalence is considerably higher in Chinese and other east Asian populations
B True — crowding is the commonest trait and the commonest reason for referral
C True — this is why figures quoted from different sources so often disagree
D False — the figure is of the order of a few per cent, not a quarter
E False — reported figures are of the order of 8 to 16 per cent

[Vol I Chapter 2 · Vol II §5.7]

158 **C** Skeletal relationships, particularly mandibular prognathism, show clear familial aggregation, whereas crowding and many dental traits reflect local and environmental factors, so a family history raises the risk without determining the outcome. Treating the inheritance as absolute forecloses a decision that cannot be made until growth is complete. [Vol I Chapter 2 · Vol II §5.6]

159 **B** An ideal occlusion is an anatomical abstraction against which a case is measured, whereas treatment aims at an occlusion that functions well and looks well within the biological limits of the individual. Defining normal as merely the commonest arrangement is the trap, since the commonest arrangement in most populations is some degree of malocclusion. [Vol I Chapter 2 · Vol II §5.1]

160 **D** Where the lower lip rests high on the upper central incisors it retroclines them, while the lateral incisors, lying outside the main force of the lip, procline to compensate, which produces the characteristic division 2 incisor arrangement. Reversing cause and effect leaves no explanation for why the laterals behave differently from the centrals. [Vol I Chapter 2 · Vol II §5.5]

161 **TTFTT**

A True — the deep overbite is the feature that most often makes treatment necessary
B True — the reduced vertical dimension and forward rotation are typical accompaniments
C False — an increased overjet is permitted by the definition where the centrals are retroclined
D True — the low mandibular plane angle and forward rotation go with the deep bite
E True — the growth direction and the lip position both act to re-deepen the bite

[Vol I Chapter 2 · Vol II §5.3]

MNEMONIC **Class II div 2**: deep bite to soft tissue, Class I or mild II base, reduced face height, forward rotation, prone to relapse.

162 **A** An asymmetrical open bite with a normal vertical facial pattern in a child who has recently stopped a digit habit is the presentation that closes spontaneously as the incisors erupt, so observation is the correct step. Starting appliance therapy at seven years treats a resolving condition and forfeits the spontaneous improvement that is the best evidence of the diagnosis. [Vol I Chapter 2 · Vol II §5.6]

163 **C** A mandible that can achieve an edge-to-edge relationship in the retruded position has a much better prognosis than one that is reversed throughout the arc of closure, and correcting the incisor inclinations can eliminate the displacement entirely. Treating the intercuspal appearance as the skeletal truth is the error, and it overstates the severity of the case by the amount of the slide. [Vol I Chapter 2 · Vol II §5.4]

164 **E** Andrews derived the keys from non-orthodontic ideal occlusions, and a flat to slightly curved plane accompanies a well-controlled overbite, which is why a residual curve at debond predicts overbite relapse. The claim that the keys came from treated cases reverses the origin of the study, which examined 120 untreated ideal occlusions. [Vol I Chapter 2 · Vol II §5.1]

165 **FFFFF**

A False — the maxilla may be prognathic, or both jaws may contribute, which is why SNA and SNB are read separately
B False — loss of a primary incisor after the canines have erupted causes little space change
C False — the association is observed but causation is not established
D False — most malocclusions are multifactorial, which is the central difficulty in aetiology
E False — tooth size and jaw size are both strongly inherited, and crowding follows from their mismatch

[Vol I Chapter 2 · Vol II §5.6]

166 **A** A path of closure that carries the mandible away from its retruded position can convert a Class I relationship into an apparent Class III or a symmetrical arch into an apparent asymmetry, so both positions must be examined before a diagnosis is made. Recording the intercuspal position alone is convenient and is exactly how displacements are missed. [Vol I Chapter 2 · Vol II §5.4]

167 **C** Crowding is the expression of a discrepancy between tooth material and the space available to accommodate it, and both tooth size and jaw size have a substantial inherited component, which is what the family history reflects. Blaming the third molars is a persistent belief that the evidence on late lower arch crowding does not support. [Vol I Chapter 2 · Vol II §5.6]

168 **E** When the lower anterior teeth are narrow relative to the upper, the arches cannot be finished with both a Class I canine relationship and an ideal overjet, and the residual space appears as excess overjet. Retracting the upper incisors further would achieve the overjet only by disturbing the canine relationship, which is why the discrepancy is identified and managed rather than mechanically overridden. [Vol I Chapter 2 · Vol II §5.1]

169 **TFTFT**

A True — this is the convention, and the most prominent incisor is used where they differ
B False — contact with the palate makes the overbite increased and complete to soft tissue
C True — the lower canine occluding in that embrasure defines a Class I canine relationship
D False — a crossbite is any transverse discrepancy in the buccolingual relationship, anterior or posterior
E True — differential drift of the molars means the two often disagree, and both are recorded

[Vol I Chapter 2 · Vol II §5.3]

170 **B** The late and variable mandibular growth increment falls after the maxilla has largely completed its growth, so an existing Class III discrepancy typically increases through adolescence, which is why these patients are reviewed to the end of growth. Expecting the maxilla to catch up reverses the actual order in which the two jaws finish growing. [Vol I Chapter 2 · Vol II §5.4]

171 **D** Angle's system depends on first permanent molars whose position is altered by early loss, drift and extraction, whereas the incisor relationship is both the presenting problem and a more dependable reference. Better correlation with the skeletal pattern is the tempting justification, but neither classification is a reliable guide to the underlying skeletal relationship. [Vol I Chapter 2 · Vol II §5.2]

Scored under 70% of the 58 marks on this block? Re-read Volume II §5, then sit it again in a fortnight.

Block F — Treatment need, indices and epidemiology (Q172–196 · 49 marks · Vol II §6)

172 **C** The 4a band runs above 6 mm up to and including 9 mm, and grade 5a begins only above 9 mm, so a measurement of exactly 9 mm sits at the top of grade 4a. Recording 5a is the trap, because the boundary is exclusive and a case can be lifted a whole grade by careless rounding of a single measurement. [Vol I Chapter 2 · Vol II §6.1]

173 **A** Grade 5i covers impeded eruption of any tooth other than a third molar, whatever the obstruction, and takes precedence over the lesser traits present. The supernumerary category applies where a supernumerary is present without obstructing eruption, which is not the situation described here. [Vol I Chapter 2 · Vol II §6.1]

174 **TTTFF**

A True — the scale is an attractiveness ranking, so morphological similarity to the photograph is irrelevant

B True — scores of 8 to 10 denote definite need, 5 to 7 borderline and 1 to 4 little or none
C True — the aesthetic score is used to resolve the borderline grade 3 cases, not to override grade 4
D False — patients consistently rate themselves more favourably and agreement with clinicians is poor
E False — the photographs show the anterior teeth in occlusion from the front

[Vol I Chapter 2 · Vol II §6.2]

MNEMONIC Aesthetic component bands: **1-4** little/none, **5-7** borderline, **8-10** definite need.

175 E The reduction of 6 points is 25 per cent of the starting score, and any case improving by less than 30 per cent falls into the worse or no different category whatever the absolute change. Calling it improved reads the direction of change without applying the threshold that gives the category its meaning. [Vol I Chapter 2 · Vol II §6.3]

176 D A crossbite with more than 2 mm of discrepancy between retruded contact and intercuspal position is grade 4c, and the worst trait present governs the grade recorded. The 1 to 2 mm crossbite category is the trap, because it describes the same anatomy and differs only in the size of the displacement. [Vol I Chapter 2 · Vol II §6.1]

177 B A gross mismatch between a trivial objective finding and profound distress, with serial consultations and avoidance behaviour, is the presentation in which appearance-related psychological morbidity must be considered before any appliance is discussed. Proceeding on perceived need alone is the trap, because treating a preoccupation that is not proportionate to the defect rarely resolves it and often escalates the demand. [Vol I Chapter 2 · Vol II §6.4]

178 FFFFF

A False — only the single worst trait is recorded, which is why two very different mouths can share a grade
B False — a reverse overjet above 1 mm and up to 3.5 mm without functional difficulty is grade 3b
C False — the aesthetic component is matched from the anterior teeth in occlusion viewed from the front
D False — the index records occlusal traits alone and says nothing about disease risk
E False — displacements above 2 mm and up to 4 mm are grade 3d, and 4d begins above 4 mm

[Vol I Chapter 2 · Vol II §6.1]

MNEMONIC **MOCDO** — Missing, Overjet, Crossbite, Displacement, Overbite. Only the single worst trait is recorded, never summed.

179 E An increased and complete overbite with gingival or palatal trauma is grade 4f, and the tissue damage is what separates it from the lesser overbite categories. The grade for a complete overbite without trauma is the tempting choice, because the overbite itself is identical and only the state of the mucosa distinguishes them. [Vol I Chapter 2 · Vol II §6.1]

180 B The aesthetic component is multiplied by 7 in ICON, more than any other element, so a mouth that looks unremarkable from the front scores low even when posterior crowding is severe. Attributing it to the buccal segment relationship misplaces the weighting, which is the smallest of the five at 3. [Vol I Chapter 2 · Vol II §6.3]

181 D Rationing a scarce resource by an explicit and reproducible measure of need applies the same yardstick to everyone and can be published and audited, which is what distributive justice requires. Ordering by date of referral looks impartial but allocates by who happened to be seen first, which systematically favours the articulate and well-informed. [Vol I Chapter 2 · Vol II §6.5]

182 **A** The worst trait rule applies and a displacement above 4 mm is grade 4d, which outranks both the open bite at grade 3e and the overjet at grade 2a. The open bite category is tempting because it is the more conspicuous feature clinically, but at 3 mm it does not reach the 4 mm threshold that would take it into grade 4. [Vol I Chapter 2 · Vol II §6.1]

183 **TTFTT**

A True — reported figures in East Asian populations run several times the European figure of around 3 per cent
B True — the lower second premolar and upper lateral incisor account for most of it
C False — anterior open bite and bimaxillary proclination are both reported more frequently
D True — up to 85 per cent of impacted canines are palatally positioned
E True — around 3 per cent of the population is the figure usually quoted

[Vol I Chapter 2 · Vol II §6.4]

MNEMONIC Key figures: hypodontia **4-6%**, canine impaction **2-3%**, of which up to **85%** lie palatal.

184 **C** Grade 5h requires hypodontia with restorative implications amounting to more than one missing tooth in any one quadrant, which is exactly what two absent teeth in the upper left quadrant represent. The less extensive hypodontia grade is the trap, since it covers a single missing tooth per quadrant and would understate the need here. [Vol I Chapter 2 · Vol II §6.1]

185 **D** Recording only the single worst trait is what makes the index quick and reproducible, and it is also precisely why two mouths of very different complexity share a grade. Rescoring by summation is the intuitive fix but destroys the calibration and the reproducibility on which the index depends. [Vol I Chapter 2 · Vol II §6.5]

186 **B** The prevalence of Class III malocclusion in Chinese and Japanese populations is several times that reported in white European populations, and this drives the very different case mix in East Asian units. The increased overbite is the tempting choice because it is common everywhere, but its prevalence does not differ between these populations in the way the Class III figure does. [Vol I Chapter 2 · Vol II §6.4]

187 **TFTTT**

A True — it is the professional's judgement, which is exactly why it can diverge from the patient's
B False — agreement is modest, and patients weight anterior appearance far more heavily
C True — unrecognised severe traits are the commonest reason for late presentation
D True — uptake tracks parental education and access as much as it tracks occlusal severity
E True — an explicit, published and reproducible criterion is the classical requirement of justice

[Vol I Chapter 2 · Vol II §6.5]

188 **E** The index was developed and calibrated for use at around 11 to 12 years, and displacements in the early mixed dentition resolve or worsen as the canines and premolars arrive, so a grade recorded now is provisional. Acting on it immediately is the trap, because it commits a child to treatment on a measurement that the developing dentition is likely to change. [Vol I Chapter 2 · Vol II §6.2]

189 **A** An overjet above 3.5 mm and up to 6 mm is grade 3a when the lips are incompetent and grade 2a when they are competent, so lip competence alone decides the grade here. Recording 3a is the common error, because the overjet measurement is identical in both categories and only the soft tissue finding separates them. [Vol I Chapter 2 · Vol II §6.1]

190 **C** The overjet component carries a weighting of 6 in the United Kingdom weightings and crossing from the 7 to 9 mm band to the normal band removes three raw points, giving 18

weighted points. The centreline is the next most heavily weighted at 4, but its maximum raw score of 2 can release only 8 points however completely it is corrected. [Vol I Chapter 2 · Vol II §6.3]

MNEMONIC **OCOB** — Overjet x6, Centreline x4, Overbite x2, Buccal/anterior segments x1: heaviest to lightest PAR weights.

191 B The reduction of 34 points exceeds the 22 point threshold and the 81 per cent change is far above the 30 per cent floor, so the case satisfies the criterion for the highest category on either reading. Classifying it merely as improved understates an outcome that meets the absolute point threshold comfortably. [Vol I Chapter 2 · Vol II §6.3]

192 FTFFT

A False — it is scored on study models or digital equivalents, which is what makes it reproducible
B True — the weighting of 6 is the largest, reflecting the dental health importance of the overjet
C False — the index records alignment and occlusal relationships and takes no account of root position
D False — it is an audit tool for groups of cases, and case difficulty confounds any single result
E True — the 30 per cent floor defines the lowest category whatever the absolute change

[Vol I Chapter 2 · Vol II §6.3]

193 D IOTN grades an occlusal trait and is silent about periodontal support, prognosis and the risk of treatment, all of which dominate the decision in a patient with a periodontally compromised dentition. Treating the grade as the decision is the error, because it was validated as a measure of population need in children rather than as an indication for an individual adult. [Vol I Chapter 2 · Vol II §6.5]

194 A Normative need is what a calibrated examiner records, whereas demand is what families actually pursue, and the gap between them is filled by awareness, cost, travel and the value placed on appearance. Reading the discrepancy as examiner error assumes the two quantities should match, when they measure different things. [Vol I Chapter 2 · Vol II §6.5]

195 FTTFT

A False — occlusal indices record anatomy, and quality of life requires a separate patient-reported instrument
B True — an index that two examiners score differently cannot ration a service fairly
C True — an easy and a very difficult case can share a grade, which is why ICON was devised
D False — it was built from a ranked professional judgement of need, not from disease causation data
E True — the aesthetic score is exactly what resolves the borderline grade 3 cases

[Vol I Chapter 2 · Vol II §6.5]

196 C Prevalence is a snapshot of how many people have the condition at one time, which a single examination of a representative sample measures directly and efficiently. Using consecutive clinic attenders is the trap, because those who reach a department are selected by referral and by demand and will overstate the severity of need in the population. [Vol I Chapter 2 · Vol II §6.4]

Scored under 70% of the 49 marks on this block? Re-read Volume II §6, then sit it again in a fortnight.

Block G — Evidence-based orthodontics (Q197–226 · 62 marks · Vol II §7)

197 E Allocating by chance balances both the recognised and the unrecognised prognostic factors, which is the only way to separate the effect of timing from the characteristics of the

families who choose it. The prospective cohort is the tempting alternative because it collects data forwards in time, but parents who elect early treatment differ systematically from those who do not and that difference is exactly what the comparison is trying to measure. [Vol I Chapter 1 · Vol II §7.1]

198 `TTTTT`

A True — this is the property that no amount of careful matching in an observational study can reproduce
B True — concealment operates before entry, whereas blinding operates after it
C True — once the earlier allocations in a block are known the remainder can be deduced
D True — stratified randomisation forces balance on a factor known to affect outcome
E True — concealment is about the moment of allocation and is achievable whatever happens afterwards

[Vol I Chapter 1 · Vol II §7.2]

199 **A** A sound sequence is worthless if the recruiter can foresee the next assignment, because patients can then be steered towards or away from entry and the groups cease to be comparable from the outset. Blinding is the answer that first suggests itself, but blinding governs what happens after allocation and would not repair groups that were unbalanced at the moment of entry. [Vol I Chapter 1 · Vol II §7.2]

200 **C** Neither patient nor operator can be kept unaware that an appliance is being worn, but a measurer working from anonymised cephalograms or models need not know the allocation, which removes the most important remaining source of detection bias. A placebo appliance is superficially attractive but cannot be made indistinguishable from a working twin block and would not deceive a wearer for long. [Vol I Chapter 1 · Vol II §7.2]

201 **E** Keeping every participant in the arm to which they were allocated preserves the balance that randomisation created, and non-compliance is itself part of what the policy of offering early treatment produces in practice. Analysing only compliers is the trap, because compliance is a patient characteristic and selecting on it reintroduces exactly the confounding randomisation removed. [Vol I Chapter 1 · Vol II §7.3]

202 `FTFTF`

A False — it includes them, analysed in the group to which they were allocated
B True — diluting the treated group with non-compliers moves the estimate towards the null
C False — selecting on adherence is selection after the fact and destroys that balance
D True — missing outcomes cannot be analysed as randomised, so large attrition undermines it
E False — it applies to any randomised comparison, including trials of appliances

[Vol I Chapter 1 · Vol II §7.3]

203 **B** An interval running from a small disadvantage to more than three months of benefit is compatible with a clinically useful difference as well as with none, so the study is uninformative rather than negative. Declaring equivalence is the standard error, because failing to demonstrate a difference in a small sample is not the same as demonstrating that no difference exists. [Vol I Chapter 1 · Vol II §7.4]

204 **D** A very large sample can establish beyond reasonable doubt that a difference exists without that difference being of any practical consequence, and 0.4 percentage points of bond failure will not alter a chairside decision. Reading the small p value as evidence of importance confuses the certainty of an effect with its size. [Vol I Chapter 1 · Vol II §7.4]

205 **C** An interval spanning 1 includes values from a 22 per cent reduction to a 9 per cent increase in relapse, so the study neither establishes benefit nor excludes it. Quoting the point

estimate as the finding is the commonest misreading, because it presents the single most likely value as though the range around it did not exist. [Vol I Chapter 1 · Vol II §7.4]

206 TFTFT

A True — the two are algebraically linked for a simple comparison of means
B False — it is the probability of data this extreme if the null hypothesis were true
C True — this is precisely why intervals are preferred to bare p values in reporting
D False — precision is driven by sample size and variability, not by the size of the effect
E True — with enough participants a trivial difference becomes statistically detectable

[Vol I Chapter 1 · Vol II §7.4]

207 A An I squared of 82 per cent indicates that most of the variation between trials exceeds what chance would produce, so the first task is to look for differences in age, appliance design or outcome definition before a single summary figure is offered. Moving to a fixed effect model is the trap, because it narrows the interval by assuming the very homogeneity that the statistic has just shown to be absent. [Vol I Chapter 1 · Vol II §7.5]

208 E Absence of small studies with null or negative findings is the classical signature of publication bias, and pooling only what was published then exaggerates the benefit. Treating the asymmetry as unimportant is the error, because the whole purpose of the plot is to test whether the retrieved literature is a fair sample of the literature conducted. [Vol I Chapter 1 · Vol II §7.5]

209 D The risk ratio here is 62 divided by 40, which is 1.55, well below the odds ratio of 2.45, because the two measures diverge as the event rate rises above about 10 per cent. Treating them as interchangeable is the habitual shortcut, and it makes a modest effect sound like a large one. [Vol I Chapter 1 · Vol II §7.4]

210 TTFTF

A True — the pre-specified method is what makes the review reproducible by someone else
B True — clinical or methodological diversity can make a single pooled figure meaningless
C False — the narrative review sits near the bottom because its selection of evidence is unstated
D True — both are recognised sources of a systematically distorted retrieved literature
E False — pooling biased trials produces a precise estimate of a biased result

[Vol I Chapter 1 · Vol II §7.5]

211 B The absolute risk reduction is 10 percentage points, and the number needed to treat is the reciprocal of that figure, giving 10. Answering 3 uses the ratio of the two rates rather than their difference, which is the commonest arithmetic slip in this calculation. [Vol I Chapter 1 · Vol II §7.6]

212 E The occlusal endpoint and the total treatment burden were no better with two phases, but trauma to a protruding incisor was reduced by more than 40 per cent in relative terms, which is the one indication these trials support. Citing self-esteem is tempting because the early difference was real, but it had disappeared by the end of treatment in both arms. [Vol I Chapter 1 · Vol II §7.6]

MNEMONIC O'Brien trial, three findings: **no occlusal difference, self-esteem gain not maintained, trauma cut (RR 0.58)**.

213 B The absolute risk reduction is 24 percentage points and its reciprocal is approximately 4, so about four children must be treated for one to avoid surgery. Answering 24 mistakes the percentage point difference itself for the number needed to treat, which inverts the calculation. [Vol I Chapter 1 · Vol II §7.6]

214 FFFFF

A False — reviews find no clinically meaningful difference in overall treatment time

B False — alignment rates are similar, and the differences reported are small and inconsistent
C False — the adult evidence is weak and expansion is not an established treatment for adult apnoea
D False — most are graded low or very low, reflecting small samples and high risk of bias
E False — absence of evidence of effect is not evidence of absence of effect

[Vol I Chapter 1 · Vol II §7.6]

215 **D** Unequal loss between arms means the survivors are no longer the groups that were randomised, and the only honest response is to show how far the conclusion depends on what the missing patients would have done. Reporting the available cases alone is the trap, because it presents a comparison of self-selected survivors as though the randomisation still held. [Vol I Chapter 1 · Vol II §7.3]

216 **A** Pain is a transient subjective experience and reports of it given years later are reshaped by how the treatment turned out, so the measurement is systematically distorted rather than merely imprecise. Confounding by operator is a real worry in such a study but it would bias the comparison of groups, not the accuracy of the remembered rating itself. [Vol I Chapter 1 · Vol II §7.2]

217 **C** Patients are chosen for aligners because their malocclusion is judged suitable, so the two groups differ in starting severity and the comparison measures case selection rather than appliance performance. Random error is always present but it would widen the uncertainty rather than push the result consistently in one direction. [Vol I Chapter 1 · Vol II §7.2]

218 **TTTTF**

A True — this is what allocation concealment is designed to prevent
B True — extra visits or extra oral hygiene instruction in one arm is the classical example
C True — an assessor working from anonymised records cannot be influenced by allocation
D True — which is why outcomes must be registered before the data are seen
E False — a larger sample gives a more precise estimate of a biased result

[Vol I Chapter 1 · Vol II §7.2]

MNEMONIC Bias by stage: **selection, performance, detection, reporting** — at entry, during care, at measurement, at publication.

219 **D** A structured question names the patients, the two retainers being compared and the outcome to be measured, and those terms become the search strategy that makes the retrieval manageable and reproducible. Restricting to familiar journals is tempting because it cuts the number quickly, but it selects the literature by convenience and reintroduces the bias the search was meant to avoid. [Vol I Chapter 1 · Vol II §7.1]

MNEMONIC **PICO** — Population, Intervention, Comparator, Outcome. Structure the question before you search.

220 **B** A split-mouth design works only where the two sides behave independently, and any carry-across effect contaminates the control side with the effect of the test intervention. The inability to blind the patient is a genuine limitation but it is shared with almost every orthodontic trial and does not undermine the design itself. [Vol I Chapter 1 · Vol II §7.2]

221 **E** Indices measure the anatomy of the result and patients judge appearance, comfort and function, so a measurable occlusal gain need not translate into any benefit the patient can perceive. Dismissing the quality of life measure is the tempting move, because it protects the occlusal finding at the cost of ignoring the outcome that matters to the person treated. [Vol I Chapter 1 · Vol II §7.4]

222 **FFTTF**

A False — a pilot tests feasibility, recruitment and procedures, not effect size

B False — alignment is irreversible, so a patient cannot be returned to baseline for the second period
C True — an equivalence or non-inferiority design with a pre-set margin is required for that claim
D True — the registered protocol is the record against which the publication can be checked
E False — choosing the outcome after seeing the data is the definition of selective reporting

[Vol I Chapter 1 · Vol II §7.2]

223 **A** A claim that something is faster is inescapably comparative, and a series without a control cannot say what those cases would have taken with conventional brackets. Small sample size is a real weakness but it would matter only once there was a comparison to make, which is what the design lacks. [Vol I Chapter 1 · Vol II §7.1]

224 **C** Excluding poor attenders improves the chance of demonstrating an effect but produces a group that is more compliant than the patients seen in practice, so the observed benefit is an upper estimate. Treating the exclusions as a flaw in internal validity is the confusion here, since they strengthen the comparison within the trial while narrowing the population it describes. [Vol I Chapter 1 · Vol II §7.1]

225 **TFFFT**

A True — it summarises all the trials and reduces the influence of any single result
B False — size does not remedy confounding, which is the weakness of the observational design
C False — it sits at the base of the hierarchy rather than outside it
D False — rare harms and prognostic questions are often best answered observationally
E True — this downgrading is the basis of the GRADE approach to rating certainty

[Vol I Chapter 1 · Vol II §7.1]

MNEMONIC Hierarchy, strongest first: **systematic review of RCTs, RCT, cohort, case-control, case series, expert opinion**.

226 **B** Sample size follows from a pre-specified minimal clinically important difference, which anchors the study to a change worth finding rather than to whatever happens to reach statistical significance. Recruiting as many as possible is the trap, because a large enough sample will detect differences too small for anyone to notice. [Vol I Chapter 1 · Vol II §7.4]

Scored under 70% of the 62 marks on this block? Re-read Volume II §7, then sit it again in a fortnight.

Block H — Ethics, consent, communication and the psychosocial patient (Q227–257 · 63 marks · Vol II §8)

227 **D** A risk is material if a reasonable person in this patient's position would attach significance to it, and interference with the tongue in a professional wind player is exactly such a risk even though it is immaterial to most patients. Offering a second opinion is a courtesy but it transfers the duty to disclose rather than discharging it. [Vol I Part V.1 · Vol II §8.1]

228 **A** Consent requires the material risks of the proposed procedure together with any reasonable alternative, and for a mandibular osteotomy the prospect of persistent altered lip sensation is the risk patients most often say they were not told about. Restricting disclosure to what the clinician thinks would concern a typical patient is the standard that Montgomery displaced. [Vol I Part V.1 · Vol II §8.1]

MNEMONIC **PARQ** — Procedure, Alternatives, Risks, Questions: what a consent discussion must cover and record.

229 **TTTTT**

A True — the standard is patient-centred rather than set by what a body of clinicians would do

B True — materiality turns on significance to this patient, not on frequency alone
C True — the option of doing nothing is a reasonable alternative in most orthodontic cases
D True — the earlier professional standard no longer governs the disclosure of risk
E True — the form records a process, and a form signed without discussion protects no one

[Vol I Part V.1 · Vol II §8.1]

230 C A child under 16 who demonstrates sufficient understanding of the proposed treatment can consent in his own right, and the finding of competence should be tested and documented rather than assumed. Requiring the mother's signature is the instinctive response but it would make a competent young person's consent conditional on an agreement the law does not require. [Vol I Part V.1 · Vol II §8.2]

231 E Valid consent requires information given in advance, an opportunity to reflect and ask questions, and confirmation at the point of treatment, which a single compressed appointment cannot provide. Countersigning by a colleague addresses the paperwork and leaves the substantive defect, which is the absence of any interval for the patient to think. [Vol I Part V.1 · Vol II §8.1]

MNEMONIC Valid consent has three limbs: **capacity, information, voluntariness**. A same-visit signature denies the last two.

232 A Parental consent may be legally sufficient but a removable functional appliance depends entirely on the child's willingness to wear it, and proceeding over her objection is both ethically poor and practically futile. Relying on adaptation is the tempting compromise, but it treats a refusal as a preference and forfeits the chance to find out what is actually worrying her. [Vol I Part V.1 · Vol II §8.2]

233 TTFTF

A True — a person may be able to decide one matter and not another, and may vary from day to day
B True — these four elements constitute the functional test
C False — an unwise decision is a reason to check understanding, not a finding of incapacity
D True — the presumption sits with the patient and the burden falls on whoever doubts it
E False — capacity must be assessed for the decision in hand and not inferred from a label

[Vol I Part V.1 · Vol II §8.2]

MNEMONIC The functional test for capacity, all four required: **understand, retain, weigh, communicate**.

234 C Preoccupation with a defect that is slight or unobservable, repeated consultations, mirror checking and social withdrawal are the recognised features of body dysmorphic disorder, in which appliance treatment characteristically fails to relieve the distress. Treating the discrepancy is the trap, because it validates the preoccupation and commonly leads to a demand for further correction. [Vol I Part V.1 · Vol II §8.3]

235 E A prevalence several times that of the general population means that a clinician assessing appearance-related requests will meet the condition regularly, so alertness to the pattern belongs in the routine assessment. Assuming that correction will satisfy these patients is the dangerous error, because dissatisfaction typically persists or transfers to another feature. [Vol I Part V.1 · Vol II §8.3]

236 B White spot lesions appear within six months of bonding in a substantial minority of patients and are largely irreversible, so the response must combine intensive prevention with an explicit standard and a defined review date. Applying varnish and continuing regardless is the trap, because it treats the lesions already present while accepting the conditions that will produce more. [Vol I Part V.1 · Vol II §8.4]

237 TTTTF

A True — demineralisation is early and common, which is why prevention precedes bonding
B True — radiographic resorption is near universal but severe loss is confined to a small minority
C True — this must be stated before treatment, not discovered by the patient at debond
D True — warning the patient of the time course reduces both anxiety and unnecessary contact
E False — the evidence does not support a causal relationship in either direction

[Vol I Part V.1 · Vol II §8.1]

MNEMONIC Numbers: white spots **50%** by **6mo**; resorption **1-5%**; pain peak **48h**, gone by **7d**.

238 D Autonomy entitles a patient to choose among options that can actually be offered, so the task is to correct the expectation before any appliance is discussed, since a plan agreed on a false premise cannot be validly consented to. Starting with aligners intending to convert later is the trap, because it defers the difficult conversation until the patient has already invested time and money. [Vol I Part V.1 · Vol II §8.5]

239 C A 16-year-old is entitled to confidentiality in the same way as an adult, and disclosure without consent would require a risk of serious harm, which non-wear of a retainer does not represent. Disclosing because the parent pays is the trap, because payment confers no right of access to another person's clinical information. [Vol I Part V.1 · Vol II §8.6]

240 A A patient may decline any treatment, but the decision to provide one rests with the clinician, who cannot be obliged to deliver an intervention likely to harm. Relying on a disclaimer is the trap, because no signature makes an inappropriate treatment appropriate or removes the clinician's own responsibility. [Vol I Part V.1 · Vol II §8.7]

241 FTFFF

A False — autonomy governs refusal and choice among offered options, not entitlement to any
B True — every intervention carries both expected benefit and foreseeable harm to be balanced
C False — justice concerns fair distribution of a limited resource and equal treatment of like cases
D False — the principles are of equal standing and must be balanced case by case
E False — the clinician remains responsible for the standard of what is provided

[Vol I Part V.1 · Vol II §8.7]

MNEMONIC The four principles, equal in standing: **autonomy, beneficence, non-maleficence, justice**. None overrides the others.

242 E The duty of candour requires prompt and honest explanation when something goes wrong, together with an apology and an offer to put matters right, and an apology is not an admission of legal liability. Explaining the change as late physiological crowding is the trap, because it is superficially plausible and conceals a failure of monitoring. [Vol I Part V.1 · Vol II §8.8]

MNEMONIC Duty of candour, all four: **tell, apologise, explain, offer remedy** — promptly; an apology is not an admission of liability.

243 D The child is the patient and the one whose cooperation determines the outcome, so he must be spoken to directly and given the chance to describe his own concern, with the parent included rather than displaced. Asking the parent to leave is the tempting shortcut, but in a first consultation it can increase the child's anxiety and it excludes the person who must consent. [Vol I Part V.1 · Vol II §8.5]

244 B Consent is specific to what is proposed, so a refusal of part of a plan requires the alternative to be set out with its compromises rather than the original plan to be pursued. Extracting

a single premolar as a compromise is the trap, because it creates an asymmetry that serves neither plan and was not what she agreed to. [Vol I Part V.1 · Vol II §8.1]

245 TFTTT

A True — the right of access extends to the whole record, including images
B False — written authority from the patient is required before release to a third party
C True — the public interest exception is narrow but it exists
D True — consent for use beyond direct care is separate from consent for the treatment
E True — retention periods are set by regulation and by the patient's age at treatment

[Vol I Part V.1 · Vol II §8.6]

246 **E** Consent to photography for the clinical record does not extend to publication or teaching, so a separate and specific consent naming the intended use is required, and the patient must be able to withdraw it. Cropping the eyes is the familiar shortcut, but it does not reliably prevent recognition and does not remove the need for consent. [Vol I Part V.1 · Vol II §8.6]

247 **B** The contemporaneous note of what was discussed is what demonstrates that consent was obtained, since a form records only that a signature was given and not what preceded it. Treating the missing form as the central failure is the trap, because the form without the discussion would add very little to the defence. [Vol I Part V.1 · Vol II §8.1]

248 TTTFT

A True — several times the figure of around 2 per cent reported in the general population
B True — the disproportion between the finding and the distress is the diagnostic clue
C True — serial opinions and dissatisfaction with each are typical of the presentation
D False — dissatisfaction usually persists or transfers to another feature after correction
E True — the condition responds to psychological and pharmacological treatment, not to appliances

[Vol I Part V.1 · Vol II §8.3]

MNEMONIC BDD features: **trivial or unobservable defect, repeated consultations, mirror-checking, social withdrawal**.

249 **D** Signs of temporomandibular dysfunction are present in around three-quarters of the population and symptoms in about a third, and there is no evidence that orthodontic treatment causes them, so the response is examination, explanation and conservative management. Debonding immediately is the trap, because it concedes a causal link that the evidence does not support and abandons the treatment without resolving the symptoms. [Vol I Part V.1 · Vol II §8.4]

250 **A** The patient is entitled to an accurate account of what is present and what needs to be done, while a judgement about another clinician's care cannot be made from the end result alone and risks being both unfair and unfounded. Telling him it was negligent is the trap, because it converts an observation into an opinion on a matter the examining clinician is not in a position to determine. [Vol I Part V.1 · Vol II §8.8]

251 FFTFF

A False — a child with sufficient understanding of the specific proposal may consent
B False — the competent young person's own consent is sufficient in law
C True — overriding a competent refusal for elective treatment is very difficult to justify
D False — assent records willingness but the legal authority rests with the parent or competent child
E False — parental responsibility is acquired by agreement or court order, not by marriage

[Vol I Part V.1 · Vol II §8.2]

252 **C** An expectation anchored to another person's face rather than to his own predicted result is a recognised warning sign, and it must be resolved before irreversible surgery because an anatomically successful result will otherwise be experienced as a failure. Beginning decompensation while deferring the discussion is the trap, because it commits him to a surgical pathway before the central question has been addressed. [Vol I Part V.1 · Vol II §8.3]

253 **D** Words such as rare and sometimes are understood across an enormous range by different patients, whereas figures such as severe resorption in 1 to 5 per cent and white spot lesions in up to half convey the same meaning to everyone. Adding a leaflet is useful but it repeats the same vague terms in print rather than correcting them. [Vol I Part V.1 · Vol II §8.1]

254 **B** Valid consent requires that the information be understood, and an untrained family member cannot be relied on to convey material risks accurately or impartially, particularly where the information is unwelcome. Written translated information is a useful adjunct but it cannot substitute for a discussion in which questions can be asked and answered. [Vol I Part V.1 · Vol II §8.5]

255 **TFFFF**

A True — most complaints are about communication and are defused by being taken seriously
B False — an expression of regret is not an admission of liability and is expected of clinicians
C False — advice may be sought, but an unexplained delay makes a complaint worse
D False — this is a breach of professional obligations and usually escalates the matter
E False — an index score neither explains nor addresses what the patient is actually complaining about

[Vol I Part V.1 · Vol II §8.8]

256 **E** Starting an appliance that cannot be completed or reliably transferred exposes the patient to all the risks of treatment with no certainty of benefit, whereas records and a written plan carry the assessment forward without that exposure. Bonding now with a letter is the tempting response because it feels responsive to the family, but it leaves a boy in another country with active appliances and no identified clinician. [Vol I Part V.1 · Vol II §8.4]

257 **A** A child under 16 who has sufficient understanding and intelligence to appreciate fully what is proposed may consent in her own right, and the clinician's duty is to assess that understanding and record the assessment. Insisting on the mother's written consent is the common error — parental consent is not a legal prerequisite once the child is competent, although involving the family remains good practice and should still be encouraged. [Vol I Part V.1 · Vol II §8.2]

Scored under 70% of the 63 marks on this block? Re-read Volume II §8, then sit it again in a fortnight.

Block I — Clinical examination and records (Q258–287 · 58 marks · Vol II §9)

258 **C** A unilateral crossbite with a centreline shift is a displacement until proved otherwise, and the finding changes both the diagnosis and the treatment aim, so it must be sought before the static occlusion is written down. Recording the intercuspal relationship first is the trap, because a displaced position then becomes the documented malocclusion and the true transverse discrepancy is never identified. [Vol I Chapter 7 · Vol II §9.1]

259 **TTFTT**

A True — a slumped or tilted head alters the apparent prominence of the chin considerably
B True — the proportion is more informative than the absolute measurement
C False — asymmetry is far better appreciated from above and behind the seated patient
D True — which is why the assessment is repeated in the retruded position

E True — assessing them separately is what prevents a vertical problem being read as an antero-posterior one

[Vol I Chapter 7 · Vol II §9.2]

MNEMONIC Assess in three planes, always the same order: **antero-posterior, vertical, transverse**.

260 B Planes that meet beyond the back of the skull are nearly parallel, which is the reduced Frankfort mandibular planes angle that accompanies a low face height and a deep overbite, and the clinical finding and the occlusion agree. Reading the meeting point behind the occiput as divergence inverts the construction, since divergent planes meet in front of or at the occiput. [Vol I Chapter 7 · Vol II §9.2]

261 D A lower lip that lies in front of the upper incisors after reduction contains them and helps to hold the corrected position, which is why the anticipated lip level is assessed before treatment. Mentalis activity on closure is the sign of a lip that still has to be strained into place, and it indicates the opposite of a stable soft tissue environment. [Vol I Chapter 7 · Vol II §9.3]

262 A The overjet is a single measurement taken from the most prominent upper incisor to the labial surface of the lower incisors parallel to the occlusal plane, and a tooth in crossbite is a separate finding recorded in its own right. Recording a negative overjet because one incisor is instanding is the trap, since it describes the whole labial segment as Class III when only one tooth is displaced. [Vol I Chapter 7 · Vol II §9.4]

263 C The overbite is increased because the lower incisors are entirely obscured, and complete because the incisal edges are in contact, with the contact being on the palatal mucosa rather than on the upper incisors. Recording it as complete to tooth is the common error, and it omits precisely the finding that makes the overbite traumatic and raises the IOTN grade. [Vol I Chapter 7 · Vol II §9.4]

MNEMONIC Describe an overbite in parts: **degree** (increased/average/reduced), **completeness**, and **to tooth or soft tissue**.

264 FTTFT

A False — a consonant arc is one in which the incisal curve is parallel to the lower lip
B True — above this the smile is commonly described as gummy and may be a treatment aim
C True — a narrow arch widens the dark spaces at the corners of the smile
D False — incisor show at rest falls with age as the upper lip lengthens and loses tone
E True — and it falls to around 1 to 2 mm by middle age

[Vol I Chapter 7 · Vol II §9.3]

MNEMONIC Key figures: incisor show **3-4mm** at rest, falling to **1-2mm** with age; over **2-3mm** gingival display is excessive.

265 E Upper incisor show of 2 mm is already at the lower limit for her age, so intrusion of the upper incisors would buy overbite reduction at the cost of an ageing smile, and the correction should be taken from the lower incisors and the posterior segments. Intruding the upper incisors is the mechanically obvious answer and is exactly what the soft tissue finding forbids. [Vol I Chapter 7 · Vol II §9.3]

266 A Coincident centrelines in the retruded position with a lateral slide into a unilateral crossbite is the classical presentation of a symmetrically narrow maxilla, in which the child displaces to one side to gain intercuspation. Treating it as a true unilateral discrepancy is the trap, because unilateral correction would leave the underlying bilateral narrowing uncorrected. [Vol I Chapter 7 · Vol II §9.1]

267 C The discrepancy is found by controlling the arc of closure so that the first tooth contact can be seen and the direction and extent of the slide into intercuspation measured. Articulating models in the intercuspal position is the trap, because it reproduces the displaced position rather than revealing it. [Vol I Chapter 7 · Vol II §9.1]

268 **E** The upper centreline is the one the observer sees against the face and is the discrepancy that must be corrected, with the lower following to preserve the Class I buccal relationship. Accepting both because they match each other is the trap, since a coincident pair still reads as an asymmetric smile when both are off the face. [Vol I Chapter 7 · Vol II §9.4]

269 **TFTTF**

A True — the British Standards definition rests on the cingulum plateau, not on the overjet
B False — the molar class is taken from the first permanent molars
C True — a buccal crossbite exists where the lower buccal cusps occlude buccal to the upper
D True — both forms of recording are acceptable provided the method is stated
E False — Angle classified the teeth, and the skeletal pattern is a separate assessment

[Vol I Chapter 7 · Vol II §9.4]

MNEMONIC **Angle's classes I, II, III** — upper 6 mesiobuccal cusp in, mesial to, or distal to the lower 6 buccal groove.

270 **B** An asymmetry unchanged by guiding the mandible into retruded contact is a true skeletal asymmetry, and a long history without progression points away from an active process, so a frontal film is the appropriate record. Isotope scanning is reserved for the case in which the history or serial records suggest that the asymmetry is still increasing. [Vol I Chapter 7 · Vol II §9.2]

271 **D** Digital models agree closely enough with plaster for the linear measurements used in diagnosis and audit, and they remove the storage, breakage and retrieval problems that dominate a plaster archive. The belief that plaster has some special medicolegal standing is the trap, since what matters is that the record is accurate, dated and retrievable. [Vol I Chapter 7 · Vol II §9.6]

272 **C** Photographs are a valuable record but they are not a precondition of treatment, and a patient who refuses a particular record is entitled to do so provided the refusal and its implications are documented. Refusing to treat is the trap, because it makes an adjunct to the record into a condition of care. [Vol I Chapter 7 · Vol II §9.6]

273 **TTTTT**

A True — they are health records and the same rules of access and disclosure apply
B True — without standardisation an apparent change may be an artefact of positioning
C True — they establish the state of the occlusion at the moment care was handed over
D True — consent for the record is distinct from consent for any wider use
E True — otherwise the buccal segment relationship cannot be read from the image

[Vol I Chapter 7 · Vol II §9.6]

274 **A** An already obtuse nasolabial angle with poor lip support signals a face that will look flatter still if the upper incisors are retracted, so space should be managed with minimal anterior retraction. Reading proclination as an automatic indication for extraction and retraction is the trap, because the soft tissue response, not the incisor angle alone, determines whether the face is improved. [Vol I Chapter 7 · Vol II §9.3]

275 **E** A tongue that comes forward to achieve an anterior seal where the teeth do not meet is behaving adaptively, and the pattern resolves once the open bite closes after the habit has ceased. Treating the swallow as the primary cause is the classical error, because it directs treatment at a consequence and leaves the recently ceased habit unacknowledged. [Vol I Chapter 7 · Vol II §9.3]

276 **D** Asymmetry of the open bite together with proclined uppers, retroclined lowers and a posterior crossbite from the altered cheek and tongue balance is the recognised constellation of an active digit habit. A primary tongue thrust is the tempting alternative, but it characteristically produces a symmetrical open bite and does not explain the crossbite. [Vol I Chapter 7 · Vol II §9.5]

MNEMONIC Digit-sucking constellation: **asymmetric open bite, proclined uppers, retroclined lowers, unilateral posterior crossbite**.

277 B Moving an incisor labially through a thin labial plate in a thin biotype is associated with further recession, so the finding must change the discussion and may change the plan, with extraction-based relief of crowding as the alternative. Photographing and proceeding records the starting point but does nothing about the risk, which is what the finding actually demands. [Vol I Chapter 7 · Vol II §9.5]

278 FFFFF

A False — mobility in the permanent dentition warrants investigation of trauma, resorption or periodontal loss
B False — blanching is also produced by a normal frenum and does not establish causation
C False — the canine is not reliably palpable until around ten to eleven years
D False — plaque accumulates heavily around clasps and baseplates and is a common cause of harm
E False — ankylosis gives a sharper, higher pitched note than a tooth with an intact ligament

[Vol I Chapter 7 · Vol II §9.5]

279 E The canine should be palpable buccally from about ten to eleven years, and absence of the bulge at this age with distally tipped laterals is the classical indication for radiographic localisation while interceptive extraction can still help. Waiting a further two years is the trap, because the window in which removal of the primary canine improves the position closes at around 11 years. [Vol I Chapter 7 · Vol II §9.5]

280 B Ankylosis fuses bone to cementum, so the ligament space is lost on the film and the tooth transmits a sharp, high pitched note on percussion while the adjacent teeth continue to erupt past it. Expecting a dull note and a widened space describes a tooth with an intact but inflamed ligament, which is the opposite of the finding sought. [Vol I Chapter 7 · Vol II §9.5]

MNEMONIC Ankylosis triad: **sharper percussion note, lost periodontal ligament space, infra-occlusion**.

281 D A painless click with a normal range of movement is found in a large proportion of the population and neither contraindicates treatment nor predicts deterioration, but it must be recorded before appliances are placed. Deferring for imaging is the trap, because it investigates a finding that is present in around three-quarters of people and would change nothing in the plan. [Vol I Chapter 7 · Vol II §9.5]

282 TFFTF

A True — the slide from first contact into intercuspation is what defines the displacement
B False — displacements above 2 mm are recorded in the dental health component as grade 4c
C False — it is associated with attrition and recession and must always be recorded
D True — which is why the asymmetry assessment is repeated in the retruded position
E False — an uncorrected displacement persists into the permanent dentition and into adult life

[Vol I Chapter 7 · Vol II §9.1]

283 A Natural head position is recorded with the patient looking at a distant point, reproduces the way the face is actually seen and is free of the variation introduced by landmarks such as porion and nasion. Using the Frankfort plane is the traditional choice, but the variability of porion means that a face oriented on it may not correspond to how the patient is seen. [Vol I Chapter 7 · Vol II §9.2]

284 C A lower lip resting high on the upper incisor crowns is the soft tissue force that produced the retroclination, and it continues to act after the incisors have been proclined, which is why

these cases relapse without prolonged retention. Expecting the lip to retain the correction reverses the direction in which it actually works. [Vol I Chapter 7 · Vol II §9.3]

285 **D** Each record must be justified individually by what it will change, which is both a radiation protection obligation for films and a general principle for the rest of the record set. Taking a full set including a cephalogram for everyone is the trap, because it applies an investigation designed for skeletal problems to a child who has none. [Vol I Chapter 7 · Vol II §9.6]

286 **FTFFF**

A False — records taken before an assessment cannot be justified by the findings of that assessment
B True — this is the test that justifies any record and is a legal requirement for radiographs
C False — an accurate, dated and retrievable digital record satisfies the requirement
D False — it is justified by a skeletal or incisor position question, not by the appliance
E False — they frequently settle disputes about the pre-treatment condition of the teeth

[Vol I Chapter 7 · Vol II §9.6]

287 **B** A dated image and model of the lower labial segment before treatment show directly whether the recession was already present and how far it has changed, which is the factual question in dispute. The consent form matters for what was discussed but says nothing about the clinical state of the tissues before treatment began. [Vol I Chapter 7 · Vol II §9.6]

Scored under 70% of the 58 marks on this block? Re-read Volume II §9, then sit it again in a fortnight.

Block J — Imaging and cephalometrics (Q288–327 · 80 marks · Vol II §10)

288 **E** Every exposure must be justified by a net benefit to this individual, and a protocol cannot justify a film for a child in whom no question has been raised by the history or the examination. Reducing the exposure addresses optimisation, which only applies once an exposure has been justified, and an unjustified film at a lower dose remains unjustified. [Vol I Topic 2 · Vol II §10.1]

> **MNEMONIC** The three IRMER principles: **justification, optimisation, limitation**. No protocol substitutes for justifying this exposure.

289 **A** Localisation requires two images taken with the tube in different positions so that the apparent movement of the canine relative to a fixed reference can be read, and a panoramic with an anterior occlusal provides a vertical tube shift. A single panoramic gives no second viewpoint, so the buccopalatal position simply cannot be determined from it. [Vol I Topic 2 · Vol II §10.2]

290 **TTFTT**

A True — the benefit must accrue to the patient exposed, not to the record or the practice
B True — justification cannot be made without the clinical question the film is to answer
C False — routine interval radiography without a clinical question is the classical unjustified exposure
D True — optimisation follows justification and governs technique, collimation and receptor
E True — failure to report an incidental finding outside the area of interest is a recognised source of harm

[Vol I Topic 2 · Vol II §10.1]

> **MNEMONIC** The three IRMER duty holders: **referrer, practitioner, operator** — each carries a distinct, defined responsibility.

291 **C** An object that appears to move in the same direction as the tube shift lies on the far side of the reference object from the tube, which for an intra-oral film means the palatal side. Reading movement with the tube as a buccal position inverts the rule and is the commonest error in applying parallax. [Vol I Topic 2 · Vol II §10.2]

MNEMONIC **SLOB** — Same Lingual, Opposite Buccal. Moves with the tube shift: it lies further from it, i.e. palatal.

292 **B** The same rule applies whether the tube shift is horizontal or vertical, and an object moving with the tube lies further from it, which here is the palatal side. Concluding a buccal position from movement in the same direction is the standard inversion of the rule, and it is the error that sends a surgeon to the wrong side of the arch. [Vol I Topic 2 · Vol II §10.2]

293 **D** Resorption of a lateral incisor root by an adjacent canine is frequently invisible on conventional films because the two structures are superimposed, and a small volume scan shows the contact and the extent of resorption directly. A further angled periapical is the conventional next step for localisation, but it cannot resolve the superimposition that is hiding the resorbed surface. [Vol I Topic 2 · Vol II §10.3]

294 **FTFTF**

A False — the incisor region is poorly resolved on a panoramic and periapicals are required
B True — the resolution and geometry are far better in the anterior maxilla
C False — a lateral film superimposes the two sides and cannot separate buccal from palatal
D True — it allows the transverse relationship of the jaws and the dental midlines to be compared
E False — proximal caries requires bitewings, and a panoramic misses early lesions

[Vol I Topic 2 · Vol II §10.2]

295 **A** Root length must be assessed on an image with minimal distortion and good resolution, which a paralleling periapical provides and which permits a valid comparison with a comparable earlier film. Repeating the panoramic is the tempting choice because a baseline panoramic usually exists, but the variable magnification and poor anterior resolution make small changes in root length unmeasurable. [Vol I Topic 2 · Vol II §10.2]

296 **C** Cone beam doses range widely and a large volume examination can exceed a panoramic several times over, so the scan is justified only where conventional films leave a question that must be answered. Equating a scan with a periapical is the dangerous error, because it treats a volumetric examination as though it carried the dose of a single intra-oral film. [Vol I Topic 2 · Vol II §10.3]

297 **E** An asymmetry of more than six to twelve months in the eruption of contralateral teeth requires investigation, and a panoramic with an anterior occlusal shows whether a supernumerary is present and where the unerupted incisor lies. Waiting further is the trap, because a supernumerary left in place allows the incisor to become progressively more displaced and dilacerated. [Vol I Topic 2 · Vol II §10.2]

298 **TTFTT**

A True — the geometry allows direct measurement, which projection radiography does not
B True — superimposition on conventional films hides precisely this finding
C False — restricting the field to the region of interest is a principal means of reducing dose
D True — the obligation covers the entire volume, including structures outside the dental region
E True — conventional bitewings remain the appropriate caries examination

[Vol I Topic 2 · Vol II §10.3]

299 **A** SNA is 5 degrees above the standard value of 81, so 0.5 degrees is subtracted from the raw ANB of 7 for each degree of excess, giving 4.5. Quoting the raw value of 7 ignores the forward position of nasion, which is precisely what the correction exists to allow for. [Vol I Topic 2 · Vol II §10.4]

MNEMONIC **Eastman correction**: for every degree SNA is above 81, subtract 0.5 degrees from the raw ANB (add back if below).

300 **C** The Eastman correction is valid only where the maxillary plane lies at about 8 degrees to sella nasion, and at 15 degrees the assumption fails, so an appraisal measured on the occlusal plane rather than on the cranial base is required. Applying the correction anyway is the trap, because it produces an authoritative looking figure from an assumption that the film has already contradicted. [Vol I Topic 2 · Vol II §10.4]

301 **E** The male mean is about minus 1 mm, so a value of plus 4 mm places point A well forward of point B on the occlusal plane and indicates a Class II skeletal discrepancy independent of the position of nasion. Calling it normal for a male is the trap, since the normal value is slightly negative and the sign of the reading carries the whole meaning. [Vol I Topic 2 · Vol II §10.4]

302 **FFTFF**

A False — sella is the centre of the pituitary fossa, located by eye within its outline
B False — that describes point B, whereas point A lies on the maxillary alveolar outline
C True — the bisected angle construction is what makes gonion reproducible at all
D False — nasion is the most anterior point of the frontonasal suture
E False — porion is notoriously difficult to identify and is a major source of error in Frankfort based analyses

[Vol I Topic 2 · Vol II §10.5]

MNEMONIC Landmark traps: **A is maxillary, B is mandibular; Nasion is the most anterior point; Porion is the least reliable**.

303 **B** An angle of 34 degrees is well above the mean of about 27 degrees and marks a vertically growing face in which extrusion of posterior teeth is poorly tolerated, so vertical control governs the plan. Treating a shallow bite as an easy case is the trap, because the difficulty in this facial type lies in preventing further opening rather than in reducing the overbite. [Vol I Topic 2 · Vol II §10.6]

304 **D** The upper incisor value is well below the mean of about 109 degrees while the lower incisors are within the normal range, and torquing the upper incisors labially both reduces the interincisal angle and releases arch length. Attributing the deep overbite to the lower incisors is the trap, since at 92 degrees they are essentially normal and the abnormality lies entirely in the upper labial segment. [Vol I Topic 2 · Vol II §10.6]

MNEMONIC Key norms (degrees): **SNA 81, SNB 78, ANB 3, FMPA 27, U1-Mx 109, L1-Mn 93, interincisal 135**.

305 **C** At 103 degrees the incisors are around 10 degrees beyond the mean of 93 and already at the labial limit of the alveolus, so pushing them further invites both dehiscence and relapse, which makes extraction the safer route to space. Treating the value as normal for age is the error, since the mean and its range do not shift upward with adolescence in the way the answer implies. [Vol I Topic 2 · Vol II §10.6]

306 **TFTTF**

A True — which is exactly why the Eastman correction exists
B False — it is measured on the functional occlusal plane and is sensitive to how that plane is drawn
C True — a discrepancy between them usually indicates an unusual mandibular morphology
D True — the mean is about 135 degrees with a range of around 10 degrees either side
E False — it compares the lower incisor and pogonion, each measured to the nasion to point B line

[Vol I Topic 2 · Vol II §10.6]

307 **A** The Holdaway ratio is most favourable when the two distances are approximately equal, and a lower incisor 5 mm ahead of pogonion describes a labial segment sitting well in front of

the bony chin. Reading two positive values as satisfactory ignores the ratio itself, which is the whole point of the measurement. [Vol I Topic 2 · Vol II §10.6]

308 **E** An abnormally positioned nasion distorts every angle measured from sella nasion, including ANB, so a reading taken on the occlusal plane avoids the faulty reference even though it introduces a dependence on how that plane is drawn. Substituting the Frankfort plane is the tempting escape, but it exchanges an atypical nasion for porion, which is the least reliably identified landmark on the film. [Vol I Topic 2 · Vol II §10.4]

309 **D** Paired outlines arise because the two sides of the face are at different distances from the receptor and are magnified differently, which is the signature of a head rotated out of the true lateral position. Genuine asymmetry is the tempting reading, but it cannot account for duplication of the orbital rims as well as the mandible. [Vol I Topic 2 · Vol II §10.7]

310 **FTTFF**

A False — identification error for many landmarks is systematic, biased along a characteristic direction
B True — this is why the two sides of the face are magnified differently on a lateral film
C True — sella is enclosed by a clear outline whereas porion and gonion must be constructed
D False — remeasuring the same tracing detects measurement error only, since the landmarks are already fixed
E False — random error widens the scatter without displacing the mean

[Vol I Topic 2 · Vol II §10.7]

> **MNEMONIC** Houston's three errors: **projection, landmark identification, measurement** — remeasuring a tracing catches only the last.

311 **B** Retracing after an interval captures both landmark identification and measurement error, which together make up the error of the method, and it does so without exposing anyone to a second film. Remeasuring the same tracings several times in one sitting is the tempting shortcut, but it holds the landmarks fixed and therefore misses the larger of the two components. [Vol I Topic 2 · Vol II §10.7]

312 **E** Superimposition on the cranial base records the sum of maxillary displacement and tooth movement within the maxilla, so the two are separated only by a second, regional superimposition on stable structures within the bone. Assuming the whole change is tooth movement is the standard error, because it credits the appliance with the downward and forward growth of the whole maxilla. [Vol I Topic 2 · Vol II §10.8]

313 **B** Individual prediction fails because the variance of late mandibular growth is large, and the Class III mandible carries both the largest late increment and the widest variation, so a plan fixed at 12 years is likely to be wrong. The belief that the Class III pattern is settled early is the error that produces surgery planned before growth is finished. [Vol I Topic 2 · Vol II §10.8]

314 **FTFFT**

A False — the lower border remodels extensively, which is what conceals true mandibular rotation
B True — these structures remodel least and are the basis of the structural method
C False — it shows the total change, of which tooth movement within the maxilla is only a part
D False — natural stable structures serve clinical purposes well, at some cost in accuracy
E True — an observed change in a treated group contains growth as well as treatment effect

[Vol I Topic 2 · Vol II §10.8]

315 **D** The whole of every image must be examined and any finding acted upon, since a lesion discovered incidentally is no less important than the one the film was taken to show. Restricting

the report to the region of interest is the trap, and failing to report an incidental finding is a well recognised source of avoidable harm. [Vol I Topic 2 · Vol II §10.1]

316 **A** An existing film of adequate quality that answers the clinical question makes a further exposure unjustifiable, and obtaining previous images is an explicit part of justification. Discarding it because it was taken elsewhere is the trap, since ownership of the film has no bearing on whether a further dose can be justified. [Vol I Topic 2 · Vol II §10.1]

317 **C** The dose to the fetus from a dental radiograph is negligible, but an examination that will not change management now is not justified, so deferring it is the straightforward application of the principle. Relying on a lead apron is the trap, because the negligible fetal dose comes from scattered radiation within the mother and the apron adds essentially nothing. [Vol I Topic 2 · Vol II §10.1]

318 **TTTTT**

A True — which is of the same order as a few days of natural background radiation
B True — the field is smaller and less radiosensitive tissue is included
C True — restricting the beam to the receptor removes a large fraction of the irradiated tissue
D True — the tissues are more radiosensitive and there are more years for a malignancy to appear
E True — this is one of the principal dose advantages of digital imaging

[Vol I Topic 2 · Vol II §10.1]

MNEMONIC **ALARP** — As Low As Reasonably Practicable. Collimation, digital receptors and correct exposure all serve it.

319 **D** SNA at 78 lies below the mean of 81 and SNB at 82 lies above the mean of 78, so both jaws deviate from the norm in the direction that produces the negative ANB, and the plan must address both. Attributing the pattern to the mandible alone is the habitual assumption in Class III cases and would misdirect a surgical plan towards a single jaw. [Vol I Topic 2 · Vol II §10.6]

320 **B** SNA is 9 degrees above the standard value, so 4.5 degrees is subtracted from the raw ANB, giving minus 1.5 and reconciling the numbers with the concave profile. Accepting the raw ANB is the trap, because a forward nasion flatters the apparent maxillary position and conceals a Class III relationship. [Vol I Topic 2 · Vol II §10.4]

321 **E** A lateral cephalogram captures one instant in one plane and cannot show how the airway behaves through the respiratory cycle or in sleep, so symptoms, examination and where necessary sleep assessment determine management. Ordering a volumetric scan is the tempting escalation, but a larger dose still yields a static image of a dynamic function. [Vol I Topic 2 · Vol II §10.2]

322 **FFTFF**

A False — the anterior region lies at the narrowest part of the focal trough and angulation is distorted
B False — horizontal magnification varies with the position of the object within the trough
C True — the ghost is produced by a structure crossed twice by the beam and is magnified and raised
D False — the spine commonly obscures the anterior midline on a panoramic film
E False — a single projection gives no parallax and cannot separate buccal from palatal

[Vol I Topic 2 · Vol II §10.2]

MNEMONIC Panoramic limitations: **anterior distortion, non-uniform magnification, ghost shadows, midline spine shadow, no parallax**.

323 **A** With every tooth present, erupted and palpable and no skeletal question to answer, there is no clinical question that a film would resolve, and justification requires such a question. Taking a panoramic to document the roots is the common practice error, because a record made for medicolegal comfort is a benefit to the clinician rather than to the patient exposed. [Vol I Topic 2 · Vol II §10.1]

324 **C** Enlarging an image scales every distance but leaves the angles between lines unchanged, so angular values are directly comparable while lengths must be scaled to a common factor before comparison. Assuming both are unaffected is the trap, and it introduces a systematic difference of several per cent into every linear value. [Vol I Topic 2 · Vol II §10.7]

325 **B** Standards derived from white European samples do not transfer, and incisor values well above the Eastman means are common and normal in East Asian populations, so treating to the European figure would flatten a satisfactory face. Retracting both labial segments to the Eastman values is the trap, because it applies a numerical target derived from a different population to a patient whose soft tissues are already balanced. [Vol I Topic 2 · Vol II §10.6]

326 **FFTFT**

A False — it supplements a clinical assessment and cannot replace it
B False — the same ANB arises from a protrusive maxilla, a retrusive mandible or a displaced nasion
C True — the mandible rotates open, increasing the recorded angle, which is why the teeth are together
D False — norms are population specific and misapplied standards produce inappropriate targets
E True — proportions such as the lower face height percentage are among the most useful values

[Vol I Topic 2 · Vol II §10.6]

327 **D** The film is justified because the numbers it yields bear directly on whether growth modification is appropriate and on how the incisor change will be judged, which is a benefit to this patient. Justifying it as a medicolegal record is the trap, because the benefit then accrues to the clinician rather than to the person receiving the dose. [Vol I Topic 2 · Vol II §10.1]

Scored under 70% of the 80 marks on this block? Re-read Volume II §10, then sit it again in a fortnight.

Block K — Study models, space analysis and the digital set-up (Q328–352 · 49 marks · Vol II §11)

328 **A** Crowding and levelling are separate demands on the same arch circumference and are therefore added, giving 5 mm plus 2 mm. Nine millimetres arises from counting the depth of the curve once for the right side and once for the left, which measures a single curve twice. [Vol I Chapter 7 · Vol II §11.2]

329 **C** An anterior ratio of 83.3 per cent against a mean of 77.2 per cent means the lower tooth material exceeds what the upper arch can accommodate by about 2.8 mm, which appears as overjet or lower crowding once both arches are aligned. Reading the excess as maxillary reverses the direction of the ratio, which is calculated as lower over upper. [Vol I Chapter 7 · Vol II §11.3]

330 **TTTTT**

A True — the second molars are excluded because their widths are unreliable and they are often unerupted
B True — with a standard deviation of about 1.65 per cent, which sets the threshold for significance

C True — the convention is arithmetic, and small deviations are within normal biological variation
D True — individual tooth widths must be inspected to localise the excess before it can be treated
E True — the arithmetic surplus has to appear somewhere in the finished occlusion

[Vol I Chapter 7 · Vol II §11.3]

MNEMONIC **Bolton**: overall ratio from 12 teeth (molar-to-molar); anterior mean 77.2%, SD 1.65%; flag beyond 2 SD.

331 **E** Half the summed incisor width is 11.0 mm and the mandibular constant is 10.5 mm, giving 21.5 mm per lower quadrant. The figure of 22.0 mm comes from adding the 11.0 mm constant, which is the one that applies to the maxillary quadrants. [Vol I Chapter 7 · Vol II §11.4]

MNEMONIC **10.5 lower, 11.0 upper** — Tanaka-Johnston constant added to half the incisor sum, by arch.

332 **A** Reading at a higher percentile builds in a deliberate bias towards larger teeth, so any error in the analysis falls on the side of finding more crowding rather than less. Accuracy is not improved by the choice — only the direction of the error is controlled, and the individual patient's teeth remain as variable as before. [Vol I Chapter 7 · Vol II §11.4]

333 **C** A continuous wire levels a deep curve mainly by extruding the premolars and tipping the incisal segment labially, which spends arch circumference and drives the incisors forwards out of the lip. Intrusion is what the clinician intends but not what the mechanics deliver, and it is the least likely of the movements in an arch of this shape. [Vol I Chapter 7 · Vol II §11.2]

334 **FFFFF**

A False — the tables are population specific and misestimate tooth width when applied elsewhere
B False — an erupted tooth visible on the same film is used to calculate and correct the magnification
C False — measuring along an arch form that is to be changed measures a quantity the plan will not produce
D False — the lower molar drifts forwards into the leeway space unless it is actively held
E False — the whole purpose of the analysis is to predict before the successors erupt

[Vol I Chapter 7 · Vol II §11.4]

335 **E** Lower intercanine width narrows rather than widens after about 12 years, and expansion of it is followed by relapse towards the original dimension, which is the classical finding of the long-term follow-up studies. Retaining for 12 months is the tempting compromise, but it postpones rather than prevents the relapse — only indefinite retention holds an expanded intercanine width. [Vol I Chapter 7 · Vol II §11.2]

336 **B** Arch circumference is measured through the contact points of the crowns, so bringing the incisal segment forwards lengthens the perimeter and releases roughly 2 mm of space for each millimetre of advancement. Describing torque as a purely apical movement is the error — the crown swings labially unless it is deliberately restrained, and it is the crown that defines the arch. [Vol I Chapter 7 · Vol II §11.2]

337 **D** A set-up is a geometric rearrangement in unlimited virtual space: roots, cortical plates, gingival biotype and lip support are not modelled, so it can display a result the patient's biology cannot deliver. The absence of occlusal contacts is the tempting objection, but a set-up does show contacts — what it cannot show is whether the teeth can survive where it has put them. [Vol I Chapter 7 · Vol II §11.5]

338 **TFTTF**

A True — stitching error compounds along the arch, which is why cross-arch dimensions deviate most

B False — digital calipers measure interproximal widths as readily as physical ones, and often more reproducibly
C True — no material is placed in the mouth, which matters in gaggers and in the very young
D True — the file can be sent repeatedly, which is one of the principal practical gains
E False — a static scan cannot record lip posture at rest, which is a clinical observation

[Vol I Chapter 7 · Vol II §11.6]

339 **C** Upper central fossae and lower buccal cusps are the surfaces that occlude in a normal relationship, so comparing them is the standard pairing and the 2 mm shortfall is the transverse discrepancy itself. Dismissing a 2 mm difference as necessarily dental confuses the magnitude of a discrepancy with the level at which it arises. [Vol I Chapter 7 · Vol II §11.1]

340 **A** Two millimetres of advancement releases about 4 mm of arch circumference because the gain is made on both sides of the midline, leaving 2 mm of the original 6 mm outstanding. Reporting 4 mm outstanding is the standard arithmetic error, and it comes from treating a bilateral gain as though it were measured on one side only. [Vol I Chapter 7 · Vol II §11.2]

341 **E** Five anterior contacts yield about 0.5 mm each within the available enamel, which covers a 3 mm excess without any restorative commitment and without moving teeth out of soft tissue balance. Extracting a lower incisor removes 5 to 6 mm of tooth material and converts a small excess into a deficiency with a residual overjet that then has to be managed. [Vol I Chapter 7 · Vol II §11.3]

MNEMONIC **0.5 mm per contact** — from about 0.25 mm of enamel per surface, across five contacts for 2.5 mm total.

342 **TFTTT**

A True — the conventional allowance is one millimetre of circumference for each millimetre of depth
B False — retraction costs about 2 mm of circumference; it is proclination that releases it
C True — the arch is a curve, so a transverse gain converts only partly into perimeter
D True — the separate demands act on one circumference and are therefore additive
E True — the lower labial segment position is settled first and the upper arch is built to fit it

[Vol I Chapter 7 · Vol II §11.2]

MNEMONIC **1-2-2-add**: level 1 mm per mm of curve, retract or procline costs/gains 2 mm per mm, and headings simply add.

343 **D** A series of straight chords is always shorter than the arc it replaces, so the segmental method returns less space available and therefore more apparent crowding than a wire laid along the curve. Attributing the difference to observer error overlooks a systematic difference between the two methods that will recur every time they are compared. [Vol I Chapter 7 · Vol II §11.1]

344 **B** Marginal ridge discrepancies and buccolingual inclination errors are both posterior vertical and transverse faults, and together they account for 11 of the deductions, so posterior detailing offers by far the greatest gain. Refining the labial segment is where attention naturally goes but it addresses only 2 of the points recorded. [Vol I Chapter 7 · Vol II §11.1]

345 **E** The index is the sum of the five linear displacements, so it is 8.0 mm, which places the segment well beyond the 6.5 mm above which irregularity is graded as severe. The figure of 1.6 mm is the mean of the five measurements, and reporting the mean instead of the sum is the commonest error with this index. [Vol I Chapter 7 · Vol II §11.1]

MNEMONIC **Sum, not mean** — Little's index adds the five gaps; above 6.5 mm is graded severe.

346 **TFTFT**

A True — the root forms are generic shapes attached to scanned crowns unless a cone beam scan is merged

B False — delivered reduction differs substantially from that planned, and the error accumulates across contacts
C True — the aligner grips the crown, so transverse gain is largely tipping rather than bodily movement
D False — the alveolar boundary is not present in the scan data at all
E True — which is the counterpart of the finding that more than 17 per cent convert to fixed appliances

[Vol I Chapter 7 · Vol II §11.5]

347 **B** A surplus of 0.5 mm in each buccal segment offsets part of the labial crowding, so the arch-wide figure is 3 mm of crowding less 1 mm of surplus. Reporting 3 mm treats the labial segment as though it were a closed compartment, when in fact space migrates along the arch as the successors erupt. [Vol I Chapter 7 · Vol II §11.4]

348 **D** Even the upper bound of 0.14 mm is an order of magnitude smaller than the millimetre or so that would alter an extraction decision, so the significance reflects the precision of the measurement rather than any clinical consequence. Equating statistical significance with importance is the standard misreading of a study of this design. [Vol I Chapter 7 · Vol II §11.6]

349 **A** Referencing the molars to the midpalatal raphe localises the asymmetry within the upper arch itself, which makes the correction a unilateral space-regaining problem rather than a whole-arch one. Blaming the lower centre line is the reflex answer and is excluded here by the finding that the lower arch is symmetrical and aligned. [Vol I Chapter 7 · Vol II §11.1]

350 **FTFFF**

A False — it is applied to the finished records; case difficulty is measured by a separate index
B True — both are posterior criteria and together they account for many of the points lost
C False — the system is deliberately an objective measure of the finished occlusion only
D False — the trimming of the bases is irrelevant to the measurements the gauge records
E False — root angulation is read from the panoramic radiograph, not the lateral view

[Vol I Chapter 7 · Vol II §11.1]

351 **C** The ratio will report a relative mandibular excess, but comparing the two lateral incisors localises the deficiency to the peg tooth, so tooth material is added there rather than removed from a normal lower arch. Lower interproximal reduction is the trap: it corrects the arithmetic while leaving the asymmetry of the upper labial segment untouched. [Vol I Chapter 7 · Vol II §11.3]

352 **D** All three demands draw on the same arch circumference and are added, giving 4 mm plus 3 mm plus 2 mm. Seven millimetres arises from charging the retraction at 1 mm rather than 2 mm, which again treats a change made on both sides of the midline as though it were one-sided. [Vol I Chapter 7 · Vol II §11.2]

Scored under 70% of the 49 marks on this block? Re-read Volume II §11, then sit it again in a fortnight.

Block L — Diagnosis, the problem list and treatment planning (Q353–392 · 80 marks · Vol II §12)

353 **B** Every later decision — extractions, anchorage, the molar relationship and the achievable overjet — is quantified against the lower labial segment once its final position is fixed, so it is settled before anything else. Choosing the extraction pattern first is the common reversal, because it commits the space budget before the target the budget is meant to reach has been defined. [Vol I Topic 3 · Vol II §12.1]

354 **E** The lower labial segment sits where the tongue and lip pressures balance, and driving it forward through a thin plate with minimal attached tissue courts both relapse and recession. Saying proclination cannot relieve crowding is wrong — it demonstrably can, at about 2 mm

of circumference per millimetre; the objection is to where it leaves the teeth, not to whether it works. [Vol I Chapter 8 · Vol II §12.3]

355 TTTTT

A True — the three-plane scheme is what prevents a transverse or vertical problem being missed behind an obvious overjet
B True — the list is a record of findings, and what is treated is decided afterwards
C True — listing and solving in one step is how a favoured appliance comes to define the diagnosis
D True — ranking is what converts a list into a plan with a sequence and a set of compromises
E True — a result that ignores the presenting complaint is a failure whatever the indices say

[Vol I Topic 3 · Vol II §12.2]

MNEMONIC **Record, Separate, Rank, Respect** — list every finding, keep it apart from the plan, prioritise, keep the complaint.

356 A Retraction is bounded by the labial and palatal cortical plates and by the lip and nasolabial support the incisors provide, and these limits are reached well before a 10 mm discrepancy of this severity is disguised. Treating anchorage as the only constraint is the seductive error of the skeletal anchorage era — absolute anchorage lets the clinician reach the biological limit faster, but it does not move it. [Vol I Chapter 8 · Vol II §12.4]

357 C The envelope nests three progressively larger ranges, with tooth movement alone innermost, growth modification adding a modest increment in a growing patient, and surgery the outermost, which is exactly how a plan is escalated. The asymmetry of the envelope is the point most often forgotten: far more retraction than protraction is available in the lower arch, for instance, so the zones are lopsided rather than circular. [Vol I Chapter 8 · Vol II §12.4]

MNEMONIC **Envelope**: tooth movement, then growth modification, then surgery — concentric but lopsided zones, not circles.

358 FFFFF

A False — CS4 is one to two years past the peak, and the residual growth available to modify is small
B False — refusal changes what may be offered, not what a given discrepancy will accept
C False — the vertical pattern, the soft tissues and the skeletal severity all constrain the choice independently
D False — camouflage disguises a skeletal discrepancy by moving teeth relative to their bases and leaves the bases unchanged
E False — failed camouflage costs years, decompensates the arches in the wrong direction and complicates any later surgery

[Vol I Chapter 8 · Vol II §12.4]

359 B CS2 places the pubertal peak within about a year, which is the window in which a functional appliance has its best skeletal effect, and the psychosocial complaint and trauma risk both argue against waiting. Deferring is the safe-sounding answer and is right for many Class II children, but it discards the growth on which the skeletal component of the correction depends. [Vol I Topic 3 · Vol II §12.5]

360 D Tooth movement through an inflamed periodontium accelerates attachment loss, whereas movement through a reduced but healthy and non-inflamed periodontium is safe, so stability is established first and the orthodontic question is then reopened. Refusing treatment outright is the over-cautious error: reduced bone height is not itself a contraindication once the inflammation is controlled, and the mechanics are simply lightened and the centre of resistance recalculated. [Vol I Chapter 8 · Vol II §12.5]

361 **A** A removable appliance or a loose component is an inhalation risk during a seizure, so a bonded fixed appliance with no free-standing parts is the safer choice, with emergency instructions given to the family. Deferring for five years discards the growth period for no benefit; epilepsy that is controlled modifies the appliance choice rather than the decision to treat. [Vol I Chapter 8 · Vol II §12.5]

362 **TTFTF**

A True — demineralisation is the commonest iatrogenic harm of fixed appliances and is prevented before bonding, not afterwards
B True — both figures come from the standard risk data and are quoted at consent
C False — enthusiasm at the consultation predicts very little; observed behaviour over a period of review predicts more
D True — a tooth with a compromised pulp or a resorbing root changes both the prognosis and the force levels used
E False — the parent's wish and the child's are routinely different, and it is the child who wears the appliance

[Vol I Chapter 8 · Vol II §12.5]

363 **C** Demonstrated non-compliance is data, and a plan whose success rests on a behaviour the patient has already failed to produce should be replaced by one using palatal miniscrews or a transpalatal arch. Moving to second premolar extraction is the tempting technical fix, but it increases rather than reduces the anchorage burden by placing the extraction site further back and leaving more space to be closed from the front. [Vol I Chapter 8 · Vol II §12.5]

364 **E** What he has described is a skeletal and soft tissue concern that the small dental discrepancy cannot address, and saying so before treatment is what converts a mismatched expectation into an informed choice. Retracting four premolar spaces would flatten the lips and make the nose and chin relatively more prominent, achieving the opposite of what he asked for. [Vol I Topic 3 · Vol II §12.3]

365 **A** True intrusion of the upper labial segment reduces both the overbite and the amount of gingiva shown, so one mechanical objective serves the occlusal and the facial aim together and neither has to be traded away. Levelling by molar extrusion is easier to deliver but increases lower facial height and worsens the gingival display, which is the very thing she came about. [Vol I Chapter 8 · Vol II §12.3]

366 **FTTTT**

A False — the first heading is the target, and the appliance is chosen afterwards as the means to reach it
B True — a Class II molar finish with a Class I incisor relationship is a recognised and stable objective
C True — the commonest unstated compromise in orthodontics is the profile sacrificed to close a space
D True — a plan that assumes growth without a contingency is an unexamined gamble
E True — aims that cannot be audited against the finished records are not aims but intentions

[Vol I Topic 3 · Vol II §12.6]

MNEMONIC **Position, Priority, Growth plan, Auditable** — what a written treatment aim must state.

367 **C** A displacement carries a demonstrable risk to the lower incisors and is correctable quickly with a simple appliance, whereas the crowding needs the permanent dentition and a different set of decisions, so the two are separated. Waiting to treat everything together leaves an established displacement in place for years, and an anterior crossbite with a displacement does not self-correct. [Vol I Chapter 8 · Vol II §12.7]

368 **E** Consent must cover the material risks and the reasonable alternatives that this particular patient would want to know about, and an irreversible change to her profile from extraction and retraction is exactly such a risk. The generic risks are necessary but not sufficient; they are recited for every patient and do not address what is specific and irreversible about this plan. [Vol I Topic 3 · Vol II §12.7]

369 **TTFTF**

A True — no treatment is always a reasonable alternative and is presented as one
B True — extraction is irreversible, so the discussion precedes it rather than follows it
C False — a material change of plan requires the discussion to be repeated and recorded afresh
D True — a recorded choice between reasonable options is the substance of shared decision making
E False — the signature records a moment within a process that continues as circumstances change

[Vol I Topic 3 · Vol II §12.7]

MNEMONIC **PARQ** — Procedure, Alternatives (including no treatment), Risks, Questions; repeat it if the plan changes.

370 **B** An increased vertical dimension with a steep mandibular plane both restricts the extrusion available for closure and makes any closure prone to reopening, so the vertical problem governs whether the case is orthodontic, surgical, or better left. Attributing the open bite to the tongue inverts the usual relationship: an anterior tongue position is more often an adaptation to the open bite than its cause, and treating it does not close the bite. [Vol I Chapter 8 · Vol II §12.2]

371 **D** Symmetrical arches that meet edge to edge in the retruded position mean the maxilla is narrow on both sides, and the mandible shifts to one side to find intercuspation, producing the appearance of a unilateral crossbite. Planning asymmetric expansion of one side is the trap: it treats the presentation rather than the finding, and leaves the underlying bilateral narrowing uncorrected. [Vol I Chapter 8 · Vol II §12.2]

372 **C** A plan is a set of prioritised decisions and a plan that promises everything conceals the compromises it is actually making, so the untreated problems are named and agreed rather than quietly dropped. Deleting the intractable problems from the list is the more damaging version of the same error, because it removes from the record exactly what the patient most needs to have been told about. [Vol I Topic 3 · Vol II §12.1]

373 **A** Sutural response to protraction falls away as the circummaxillary sutures become more interdigitated, so treatment started before about ten years produces the largest skeletal component. Reaching edge to edge shows there is a displacement, which is a favourable sign in itself, but it says nothing about whether the sutures will still respond — and it is the sutural response that protraction depends on. [Vol I Chapter 8 · Vol II §12.4]

374 **TFTFT**

A True — the whole space and anchorage analysis is referenced to that single decision
B False — it governs the order of planning, not the order of bonding, which is a separate clinical judgement
C True — a Class I overjet of 2 to 3 mm places the upper incisors once the lower ones are placed
D False — it applies with more force in Class II, where the temptation to retract the upper arch to an arbitrary point is greatest
E True — arch form and intercanine width are among the least stable dimensions to alter

[Vol I Chapter 8 · Vol II §12.6]

375 **E** A moderate skeletal discrepancy with a reduced vertical dimension and uprighted lower incisors is well within the camouflage zone of the envelope, and upper premolar extraction supplies exactly the space the overjet requires. Extracting in the lower arch would retract an already upright lower labial segment and increase the overjet, which is the opposite of what the case needs. [Vol I Chapter 8 · Vol II §12.4]

376 **D** Three findings each independently exclude camouflage — a reverse overjet of 5 mm, lower incisors already compensated to 80 degrees, and an increased vertical dimension with an open bite tendency — so the discrepancy lies in the surgical zone of the envelope. Further lower retroclination is the classic failed camouflage: there is no compensation left to spend, and it opens the bite further. [Vol I Chapter 8 · Vol II §12.4]

377 **B** When two approaches produce the same occlusion, the outcome that separates them is the one on which the decision must rest, and here that is the profile, which is precisely what the patient is consenting about. Reading equivalence of PAR as interchangeability ignores the second finding in the same study — the two treatments differ in a way the occlusal index was never designed to capture. [Vol I Chapter 8 · Vol II §12.4]

378 **TTFTT**

A True — separating diagnosis from solution is what stops the favoured appliance dictating the diagnosis
B True — the space analysis is measured against a target, so the target is set first
C False — the extraction pattern determines how much space is created and where, which then sets the anchorage demand
D True — an unplanned retention regime is decided in haste at the worst possible moment
E True — movements are the requirement and the appliance is the means, and choosing the means first constrains the requirement

[Vol I Topic 3 · Vol II §12.1]

MNEMONIC **Problems, Position, Pattern, Anchorage, Appliance, Retention** — the fixed order for building a plan.

379 **E** An adult skeletal pattern is fixed, so the whole of the correction must come from tooth movement or from surgery, and the envelope available to her is the innermost zone only. Recession is a caution that dictates lighter forces and careful movement direction rather than a contraindication, and it is managed rather than treated as a bar to treatment. [Vol I Chapter 8 · Vol II §12.5]

380 **B** A tooth of poor long-term prognosis is the unit to remove where the space it provides is in the right place, and the residual asymmetry is a mechanical problem to solve rather than a reason to sacrifice a sound tooth. Balancing with the contralateral sound first molar removes a healthy tooth to preserve symmetry the mechanics can supply, and is the reflex that therapeutic extraction is intended to displace. [Vol I Chapter 8 · Vol II §12.5]

381 **FTFFF**

A False — strained competence relies on a habitual effort that is abandoned once the appliance is removed
B True — duration matters more than magnitude, which is why light continuous resting pressures prevail
C False — the soft tissue response is variable and considerably less than one to one, nearer one to three for the upper lip
D False — little attached tissue over a thin plate is the classic setting for recession during proclination
E False — the retainer masks the tendency rather than removing it, and the relapse declares itself when it fails

[Vol I Chapter 8 · Vol II §12.3]

MNEMONIC **True competence, resting pressure, 1:3 lip:incisor ratio, thin biotype** — what limits or predicts stability.

382 D A lower lip trapped behind the upper incisors maintains the proclination, and bringing the overjet to normal repositions the lip in front of the incisal edges, where it becomes a retaining rather than a displacing force. Saying the trap is irrelevant to stability misses the mechanism by which soft tissue form determines whether a corrected overjet holds. [Vol I Chapter 8 · Vol II §12.3]

383 A Full lips carried on proclined incisors depend on that dental support, and removing 8 mm of it across both arches can leave a flat and prematurely aged lower face that the patient did not ask for. Saying bimaxillary proclination contraindicates extraction overstates the point: extraction is often exactly right in these cases, but the amount of retraction has to be titrated against the face rather than against the space available. [Vol I Chapter 8 · Vol II §12.3]

384 C Anteroposterior and vertical aims are both stated but nothing is said about arch width, centre lines or crossbite, so the third plane of space has simply been omitted from the objective. Calling it internally inconsistent is wrong on the facts — a Class I incisor relationship on a full unit Class II molar relationship is a recognised and stable finish where space has been taken from the upper arch alone. [Vol I Topic 3 · Vol II §12.6]

385 D CS1 puts the pubertal peak about two years away and the child himself expresses no wish for treatment, so review costs nothing and allows both the growth timing and his own motivation to declare themselves. Starting now to capture the whole growth period is the trap: functional appliance wear begun two years before the peak means a long period of cooperation spent before the growth that would reward it arrives. [Vol I Chapter 8 · Vol II §12.5]

386 TFFTF

A True — this is the general criterion for intervening early rather than waiting for the permanent dentition
B False — the trials found no difference in the final occlusion between the two approaches
C False — a risk ratio of about 0.58 was reported, which is a 42 per cent relative reduction in trauma
D True — more months in appliances is the principal cost of phasing and is disclosed at consent
E False — most cases still require a definitive phase, which is why the first phase must justify itself on its own terms

[Vol I Chapter 8 · Vol II §12.7]

387 B White spot lesions appearing within six months in a patient whose hygiene is deteriorating will progress to cavitation if the appliance stays and the behaviour does not change, so the risk is removed while the cause is addressed. Adding a fluoride rinse is a reasonable adjunct but it treats the consequence and leaves the appliance and the plaque in place, which is where the damage is being done. [Vol I Chapter 8 · Vol II §12.5]

388 E CS3 places the pubertal peak within a year, and a skeletal correction attempted with growth is available on terms that will never recur, whereas crowding, rotations and centre lines can be dealt with at any age. Reducing the overbite first is technically correct as a sequence of mechanics, but it is a step within the plan rather than the problem that sets the plan's timing. [Vol I Topic 3 · Vol II §12.1]

MNEMONIC **CVM ladder**: CS1 peak in 2 y, CS2 in 1 y, CS3 within 1 y, CS4 1-2 y past — growth windows for planning.

389 A Prediction performs tolerably for group means and poorly for individuals, and late mandibular growth in Class III is both large in magnitude and highly variable in amount, so a definitive commitment at nine years rests on a figure that cannot bear it. Claiming that a 4

mm reverse overjet is always camouflageable replaces one premature certainty with another, and in a growing Class III child neither is available. [Vol I Chapter 8 · Vol II §12.4]

390 TTFFF

A True — the arch responsible has to be identified before the correction can be directed at it
B True — the same plane is recorded separately at the two levels, which is the whole point of the scheme
C False — an overjet may be dental in origin on a Class I base, which is why the two levels are recorded separately
D False — distinguishing a true crossbite from a displacement changes the diagnosis and the treatment entirely
E False — the incisor relationship is the dental level; the skeletal level requires ANB or an equivalent measure

[Vol I Topic 3 · Vol II §12.2]

MNEMONIC **3 planes x 2 levels**: anteroposterior, vertical, transverse, each recorded at skeletal and dental level.

391 C Trials of functional appliances report mean increases in mandibular length of the order of 2 mm, with the remainder of a 10 mm reduction coming from upper incisor retroclination and lower incisor proclination. Claiming that the correction is entirely dental overstates the negative finding: the skeletal effect is real but small, and misrepresenting it in either direction misinforms the consent. [Vol I Topic 3 · Vol II §12.4]

392 B A 30-month course with extractions, surgical exposure and a real burden of decalcification and resorption risk requires an engaged patient, and treating an unwilling adolescent to a high standard is rarely achievable. Resorption risk from displaced canines is a genuine consideration and would be a reason to act, but it is a reason to reopen the discussion and monitor rather than to override a competent patient's declared wishes. [Vol I Topic 3 · Vol II §12.7]

Scored under 70% of the 80 marks on this block? Re-read Volume II §12, then sit it again in a fortnight.

Block M — Extraction decisions and space management (Q393–432 · 80 marks · Vol II §13)

393 D Three millimetres per arch is within the reach of about 0.5 mm of enamel reduction at five or six contacts combined with a millimetre of advancement, whereas four premolars would release far more space than the case requires. Extracting four first premolars is the reflex answer for crowding and here it creates a residual space problem that then has to be closed, with all the anchorage and profile cost that involves. [Vol I Topic 4 · Vol II §13.1]

394 A Space is needed only where the crowding and the overjet are, and extracting in a well-aligned lower arch with correctly positioned incisors creates a space problem the case did not have. Extracting all four first premolars is the habitual symmetrical pattern, and here it would retract a lower labial segment that is already where the plan wants it. [Vol I Topic 4 · Vol II §13.2]

395 TTTTT

A True — these four exhaust the ways in which the space available can be made to match the space required
B True — the arch is a curve, so only part of a transverse gain converts into perimeter
C True — the distalised segment must then be held, which is why distalisation is rarely anchorage-free
D True — the gain is made on both sides of the midline, which is the source of the factor of two
E True — enamel removed does not return, and the limits are conventionally set at a fraction of the interproximal thickness

[Vol I Topic 4 · Vol II §13.1]

MNEMONIC **Extract, Expand, Distalise, Reduce+procline** — the four recognised ways to create arch space.

396 **C** Crowding in the canine region is relieved most efficiently by the extraction immediately distal to it, and first premolar spaces are largely consumed by the canines themselves rather than requiring anterior retraction. Second premolar extraction preserves an anterior contact but places the space a unit further back, which raises the anchorage demand and is the wrong choice where the crowding is severe and anterior. [Vol I Topic 4 · Vol II §13.2]

397 **E** A first premolar space is intended to be closed mainly from the front, whereas a second premolar space is deliberately chosen when the molars are to come forward, so it suits minimum and not maximum anchorage cases. The belief that a space nearer the molar is easier on anchorage reverses the logic: what matters is which unit is meant to move into the space, not how far it is from the molar. [Vol I Topic 4 · Vol II §13.4]

398 **FFFFF**

A False — that describes a compensating extraction; balancing is contralateral within the same arch
B False — that describes a balancing extraction; compensating is the opposing arch on the same side
C False — it is undertaken to prevent a centre line shift, not to supply space elsewhere
D False — primary incisor loss rarely shifts the centre line and balancing is not indicated
E False — both are decided case by case on the centre line and the occlusal consequences, not by rule

[Vol I Topic 4 · Vol II §13.3]

MNEMONIC **Balancing = same arch, other side; Compensating = same side, other arch** — never confuse the two.

399 **A** Calcification at the bifurcation of the second permanent molar roots is the radiographic sign of the optimum window, and extraction at this point gives the best chance of the second molar erupting into a good position of its own accord. Waiting until the second molars have erupted at 12 years forfeits the whole benefit, because a tooth that has already erupted will not drift forward into the space without appliances. [Vol I Chapter 8 · Vol II §13.5]

MNEMONIC **Bifurcation calcifying** — the radiographic window for extracting a doomed first molar for spontaneous closure.

400 **C** Where a tooth of poor prognosis exists, it is the unit removed even though its position makes space closure slower and the anchorage demands greater, since a sound premolar should not be sacrificed to save mechanical effort. Balancing a compromised molar against a sound premolar removes a healthy tooth to buy symmetry, which is precisely the trade that therapeutic extraction is meant to avoid. [Vol I Chapter 8 · Vol II §13.6]

401 **E** Canine substitution leaves the gingival margin of the substituted tooth higher and its emergence profile heavier, so a low smile line removes the aesthetic objection that most often defeats the approach. Light canine colour helps, but shade is the easiest of the substitution problems to manage by reshaping and bleaching, whereas gingival display cannot be corrected once the canine is in the lateral position. [Vol I Chapter 8 · Vol II §13.6]

402 **TTFTT**

A True — this is the house figure for unsupported anchorage and the reason reinforcement is planned in advance
B True — Haque and colleagues reported it, and it is quoted at the planning stage rather than discovered later
C False — both arches lose anchorage, and a lower molar brought forward is as real a loss as an upper one

D True — two-stage closure reduces the anchorage demand at each stage compared with en masse retraction
E True — space consumed by alignment is not anchorage loss, and the distinction has to be made before the case is called a failure

[Vol I Topic 4 · Vol II §13.7]

MNEMONIC **Haque's two-thirds** — about two-thirds of an unsupported extraction space is lost to anchorage drift.

403 B Primary canine extraction lets the incisors align into the space, primary first molar extraction then accelerates the first premolars, and the first premolars are removed as they erupt so that the permanent canines descend into their place. Beginning with the first premolars inverts the whole sequence, because at eight years those teeth are unerupted and cannot be removed without surgery. [Vol I Topic 4 · Vol II §13.5]

MNEMONIC **Canines, then Ds, then fours** — serial extraction order: primary canines, primary first molars, first premolars.

404 D Anterior interproximal enamel allows roughly 0.25 mm per surface, so about 0.5 mm per contact across five contacts supplies around 2.5 mm without approaching dentine. Concentrating the whole reduction on the central incisors would remove far more than the enamel there permits and would leave those two teeth visibly narrower than their neighbours. [Vol I Topic 4 · Vol II §13.7]

MNEMONIC **0.25 mm per surface, 0.5 mm per contact** — the interproximal enamel budget for IPR.

405 FTTFF

A False — resistance to sliding is substantial and comprises classical friction, binding and notching
B True — the force system is determined by the loop geometry and activation rather than by sliding resistance
C True — gable bends and loop design are used to raise the ratio towards the value required for bodily movement
D False — friction opposes the moving unit, and it is often used deliberately to protect the posterior anchorage
E False — loops require individual bending and precise activation at each visit, which is slower rather than quicker

[Vol I Topic 4 · Vol II §13.7]

406 C A sound, non-infraoccluded retained primary second molar in an uncrowded and well-interdigitated arch is the best space maintainer available and many survive into the fourth decade, so nothing is gained by removing it now. Extracting and closing the spaces is the tempting definitive answer, but it commits an uncrowded arch to substantial space closure and to the anchorage cost that comes with it. [Vol I Chapter 8 · Vol II §13.6]

407 A Camouflage of a Class III requires the upper labial segment to come forward and the lower to be retracted, so the upper space is placed further back where it will be lost to mesial molar movement while the lower space is placed forward where retraction can use it. Taking first premolars in both arches supplies the space symmetrically and then requires the mechanics to fight the anchorage in the upper arch to avoid retracting the very segment that needs advancing. [Vol I Topic 4 · Vol II §13.2]

MNEMONIC **Class III: 5s above, 4s below** — upper space lost to drift, lower space used for retraction.

408 E Bringing retroclined upper central incisors forward to a normal inclination spends arch circumference, so the apparent 2 mm of upper crowding understates the true upper space requirement and the extraction decision must be made after that allowance. Extracting four first premolars is the trap: in a reduced vertical pattern it invites further retraction of an already

retroclined labial segment and deepens the overbite rather than relieving it. [Vol I Chapter 8 · Vol II §13.2]

409 **TFTTT**

A True — these are the distalisation figures and they are higher and longer than the anchorage regime
B False — an erupted second molar lies in the path of distalisation and makes it substantially harder
C True — the reciprocal force acts forwards on whatever the appliance is seated against, most often the incisors
D True — headgear wear is the classic compliance problem, and intra-oral appliances need it too for retention
E True — the reactive force passes into bone rather than into the dentition, which is the point of skeletal anchorage

[Vol I Topic 4 · Vol II §13.1]

410 **D** Widening the maxilla both corrects the transverse discrepancy and increases arch perimeter, so the residual crowding is reassessed after expansion and the extraction decision is often avoided or reduced. Saying expansion contributes no arch space is wrong on the arithmetic — it contributes less perimeter than the transverse gain, because the arch is a curve, but it certainly contributes some. [Vol I Topic 4 · Vol II §13.1]

411 **B** Removing a lower incisor takes 5 to 6 mm of lower tooth material from an arch that was in balance with the upper, so the arithmetic surplus appears as increased overjet or overbite unless upper interproximal reduction compensates. A large centre line shift is the feared outcome but is readily controlled by appliance mechanics, whereas the tooth size discrepancy is arithmetic and cannot be mechanically avoided. [Vol I Chapter 8 · Vol II §13.2]

412 **E** Where the third molars are present and favourably positioned, removing the impacted second molars lets the third molars come forward into a serviceable position without surgery or appliances. Uprighting the second molar is the more elaborate alternative and is reserved for cases where the third molar is absent or of poor form, since it commits the patient to surgery and fixed mechanics for a result the third molar can supply. [Vol I Chapter 8 · Vol II §13.5]

413 **FFFTF**

A False — the upper second molar drifts forward far more readily; the lower is the one that tips and rolls
B False — an absent third molar leaves a space at the back of the arch and changes the whole calculation
C False — balancing a first molar is not routine and is decided on the centre line and the state of the contralateral tooth
D True — this radiographic sign defines the window in which the second molar erupts forward of its own accord
E False — delay forfeits the spontaneous closure and commits the patient to appliance treatment that might have been avoided

[Vol I Chapter 8 · Vol II §13.5]

414 **B** A high smile line exposes the gingival margin and emergence profile of a substituted canine, which sits higher and heavier than a lateral incisor, and a unilateral substitution also leaves the two sides visibly unmatched. Extracting the sound contralateral lateral incisor to buy symmetry destroys a healthy tooth and doubles the extent of the problem instead of solving it. [Vol I Chapter 8 · Vol II §13.6]

415 **D** Anchorage demand is set by the proportion of the extraction space to be consumed by

the anterior segment, and here more than three-quarters of it is, which is the definition of a maximum anchorage case. Calling it moderate because the space serves two purposes misapplies the classification: moderate anchorage means roughly half the space going forwards, and 6 mm of 7 mm is not half. [Vol I Topic 4 · Vol II §13.4]

416 **A** About two-thirds of an unsupported extraction space is consumed by mesial movement of the posterior segments, so roughly 2 mm of the 7 mm would reach the anterior teeth and the overjet would not close. Saying the loss is unpredictable and cannot be quantified is the counsel of despair: the figure is well established, which is exactly why reinforcement is planned in advance rather than improvised. [Vol I Topic 4 · Vol II §13.7]

417 **C** Space is created where the crowding is, and the centre line is then protected by differential anchorage on the two sides rather than by removing a tooth from a quadrant that does not need one. Extracting bilaterally to preserve symmetry removes a sound premolar to create 6 mm of space on the right where only 1 mm is required, and that surplus then has to be closed with the anchorage cost that entails. [Vol I Topic 4 · Vol II §13.3]

418 **TTTTF**

A True — which is why the working figure is about 0.5 mm per contact rather than per surface
B True — polishing and fluoride application after reduction are part of the procedure, not optional extras
C True — a flattened contact resists the displacing forces that produce late lower incisor crowding
D True — delivered reduction departs from the plan and the error accumulates across successive contacts
E False — enamel is acellular and does not regenerate; the change is permanent

[Vol I Topic 4 · Vol II §13.7]

419 **D** Severe lower crowding requires space delivered forward in the lower arch, and a matching upper first premolar extraction supplies what the overjet and the upper crowding need while keeping the buccal segments in step. Taking upper second premolars would place the upper space too far back for a 5 mm overjet and would invite the upper molars forward into a Class II relationship the plan is trying to correct. [Vol I Topic 4 · Vol II §13.2]

420 **B** Removing second molars offers space at the back of the arch, but the distalisation is slow, the anchorage demanding, and the result rests on a third molar whose eventual position cannot be guaranteed. Saying it provides no space overstates the case — it does provide space, but at the end of the arch furthest from the crowding, which is the real objection. [Vol I Chapter 8 · Vol II §13.5]

421 **E** A mean of 0.9 mm with wide intervals and marked heterogeneity says that the group average is small and that individuals vary widely, so the response of a particular face is judged from its own soft tissue characteristics. Reading the direction of the mean as a reliable rule for each patient is the standard misuse of a heterogeneous pooled estimate. [Vol I Topic 4 · Vol II §13.1]

422 **FTTFT**

A False — infraocclusion of 3 mm indicates ankylosis and a tooth that is failing rather than one to be kept
B True — crowding elsewhere in the arch means the space is wanted, which settles the open-or-close question
C True — this is a minimum anchorage situation and the plan exploits the mesial drift rather than fighting it
D False — an ankylosed tooth cannot erupt and the infraocclusion worsens as the alveolus around it grows

E True — transplantation is a recognised option where a donor with roughly two-thirds root formation is available

[Vol I Chapter 8 · Vol II §13.6]

423 **A** Eight millimetres of crowding against 14 mm of extraction space leaves about 6 mm to be shared between anterior retraction and mesial molar movement, which is the definition of a moderate anchorage case. Planning maximum anchorage is the cautious error: it adds appliances and cost to resist a movement the space budget can comfortably accommodate. [Vol I Topic 4 · Vol II §13.4]

424 **C** Lower incisors at 88 degrees are retroclined and have room to be advanced, and 2 mm of proclination releases about 4 mm of circumference in a patient whose lips and attached tissue will tolerate it. Extracting a lower incisor is the direct-sounding answer, but it removes 5 to 6 mm of tooth material for a 4 mm problem and creates a tooth size discrepancy that then appears as overjet or overbite. [Vol I Topic 4 · Vol II §13.1]

425 **B** A mobile, non-vital, apically infected incisor with a mid-root fracture will not survive orthodontic force and its space is in the part of the arch where the crowding lies, so it is the logical unit to remove. Root treating it and proceeding as planned is the conservative-sounding option, but it applies force to a tooth with a compromised root and asks a sound premolar to pay for it. [Vol I Chapter 8 · Vol II §13.6]

426 **TFFFT**

A True — this is the principle of therapeutic extraction and it takes precedence over mechanical convenience

B False — proximity is only one consideration, and the anchorage plan may deliberately place the space further back

C False — it is chosen when the posterior segments are meant to come forward, which is the opposite case

D False — at 15 years the second molar is erupted or nearly so and will not drift forward unaided

E True — a grossly ectopic canine or premolar may be removed in preference to a well-positioned sound tooth

[Vol I Topic 4 · Vol II §13.2]

427 **D** Closing bilateral lateral incisor spaces brings the whole upper buccal segment forward by about a lateral incisor width, which converts a half unit Class II into a Class I finish without touching the lower arch. Symmetry of the absence makes either plan easier to execute but does not favour one over the other, since space opening on both sides is equally symmetrical. [Vol I Chapter 8 · Vol II §13.6]

428 **A** Anchorage is established and the arches aligned before the extraction spaces are opened, so that no space is lost to uncontrolled drift during the months when nothing is holding the posterior segments. Waiting until anchorage loss becomes apparent is the commonest practical failure, because by the time it is visible on the models the space it consumed cannot be recovered. [Vol I Topic 4 · Vol II §13.7]

429 **C** The classification is a statement about how the space is to be shared, with maximum anchorage meaning that the great majority of it goes to the anterior teeth and minimum meaning that the posterior teeth are meant to come forward. Using the absolute amount of space is the intuitive error: 7 mm of space is a maximum anchorage problem if the overjet needs all of it and a minimum anchorage problem if crowding consumes it. [Vol I Topic 4 · Vol II §13.4]

MNEMONIC **Anchorage = proportion, not amount** — the share of extraction space claimed by the anterior teeth.

430 **FTTTF**

A False — the technique commits the patient to four premolar extractions and is reserved for severe crowding
B True — serial extraction addresses crowding and does nothing for an anteroposterior or vertical problem
C True — a missing or ectopic successor invalidates the whole sequence, so the radiograph comes first
D True — the spontaneous result is rarely a finished occlusion, and this is disclosed before starting
E False — supervision runs from about eight years to the end of fixed appliances, which is a long commitment

[Vol I Topic 4 · Vol II §13.5]

431 E A horizontally placed canine actively resorbing an incisor root is both very unlikely to align and is causing continuing damage, so removing it stops the resorption and leaves the restorative options open. Exposing and aligning it is the instinct in a young patient, but a horizontal canine across two incisor roots carries a poor prognosis for alignment and prolongs the very contact that is destroying the lateral incisor. [Vol I Chapter 8 · Vol II §13.6]

432 A Closing a second premolar space brings the molars forward along the occlusal plane, which tends to close rather than open an anterior open bite in a long-faced patient, and it supplies the space the crowding requires. First premolar extraction is the default for 7 mm of crowding, but the anterior retraction it invites lengthens the arch anteriorly and works against the vertical control this face needs. [Vol I Topic 4 · Vol II §13.2]

Scored under 70% of the 80 marks on this block? Re-read Volume II §13, then sit it again in a fortnight.

Block N — Biomechanics, force control and anchorage (Q433–485 · 109 marks · Vol II §14)

433 C The reactive force is not a side effect but a necessary consequence of applying force at all, and the only question is whether the reactive unit is held; here it was not. Ascribing the movement to natural mesial drift misses the mechanism entirely — drift over four months is a fraction of a millimetre, whereas 2 mm under a 150 cN reactive load is straightforward anchorage loss. [Vol I Topic 5 · Vol II §14.1]

434 E Anchorage demand is the proportion of the extraction space claimed by the anterior teeth, and 6 mm out of 7 mm leaves almost nothing for the posterior segment, which is the maximum anchorage definition. Calling it moderate because a millimetre remains misreads the classification: moderate means roughly half the space going each way, not a token millimetre. [Vol I Topic 5 · Vol II §14.1]

435 TTTTT

A True — a single force applied away from the centre of resistance always produces translation plus rotation
B True — this is the definition of a moment and the reason bracket position matters so much
C True — for uncontrolled tipping the centre of rotation sits roughly at or just apical to the centre of resistance
D True — raising the ratio progressively shifts the centre of rotation apically towards and beyond the apex
E True — about 10 to 1 in millimetre-newton terms is the conventional figure for incisor translation

[Vol I Topic 5 · Vol II §14.3]

MNEMONIC **M:F ladder**: 0 = uncontrolled tip, 7:1 = controlled tip (apex), 10:1 = translation, >12:1 = root movement.

436 **B** Anchorage reinforcement needs only enough force and duration to oppose the reactive load, and the conventional regime is 250 to 350 g per side for 10 to 12 hours daily. The heavier and longer regime of 450 to 500 g for 14 hours is the distalisation prescription, which is asked for when the molars are to be moved back rather than merely held. [Vol I Topic 5 · Vol II §14.2]

MNEMONIC **Anchorage 250-350g/10-12h; Distalise 450-500g/14h** — more force and time to move, not just hold.

437 **D** Two independent safety mechanisms are required so that the failure of one does not permit the facebow to disengage, and written and verbal instruction accompanies rather than replaces them. A single snap-away release is the commonest shortfall in practice, and it leaves no protection at all once that one mechanism fails. [Vol I Topic 5 · Vol II §14.2]

MNEMONIC **Two independent safety devices, always** — e.g. locking facebow plus snap-release neckstrap; one is not enough.

438 **C** A high pull line of action lies above the molar centre of resistance, so the resultant carries an intrusive component that controls or reduces vertical development of the posterior maxilla. Blaming cervical pull is the reverse error: a cervical line passes below the centre of resistance and extrudes the molars, which increases rather than reduces lower facial height. [Vol I Topic 5 · Vol II §14.2]

439 **FTTTT**

A False — it reduces mesial movement but does not abolish it, and it is not a substitute for skeletal anchorage
B True — controlling molar rotation is a substantial part of the anchorage value it provides
C True — the acrylic button transfers part of the reactive load to the palatal mucosa and bone
D True — asymmetric activation is a recognised and useful property of the appliance
E True — holding the lower molars preserves the leeway space of about 2.5 mm per quadrant

[Vol I Topic 5 · Vol II §14.2]

MNEMONIC **Leeway space**: about 1.5 mm/quadrant maxilla, 2.5 mm/quadrant mandible — held by an arch or lost to drift.

440 **A** Bodily movement requires the net moment about the centre of resistance to be zero, so the counter-moment must equal the force multiplied by the distance, which is 0.8 N times 9 mm. The figure of 240 newton-millimetres corresponds to a ratio of about 3 to 1 at this force, which produces controlled tipping rather than translation. [Vol I Topic 5 · Vol II §14.3]

441 **E** Pure root movement places the centre of rotation at the incisal edge, which requires a moment larger than that needed for translation, so the ratio exceeds the 10 to 1 of bodily movement. Choosing 10 to 1 is the natural error because it is the best-known figure, but it puts the centre of rotation at infinity and moves crown and root together. [Vol I Topic 5 · Vol II §14.3]

442 **TFTTF**

A True — this is the conventional location for a tooth with normal periodontal support
B False — bone loss moves the centre of resistance apically, which is why the moments generated by the same force increase
C True — with no perpendicular distance there is no moment and the tooth translates
D True — splinting creates a single unit whose centre of resistance is determined by the combined root support
E False — the centre of rotation depends on the force system applied and moves as that system changes

[Vol I Topic 5 · Vol II §14.3]

MNEMONIC **CR facts**: 1/3-2/5 root length from the crest; bone loss moves it apically; splinted teeth share one CR.

443 **D** The palate carries thick cortical bone, keratinised mucosa and no roots, and the median failure rate reported for it is about 1.3 per cent, which is the lowest of any site. The buccal shelf

is at the opposite extreme, with reported failure rates as high as 65 per cent, and choosing it for a maxillary retraction case compounds an unnecessary risk. [Vol I Topic 5 · Vol II §14.5]

MNEMONIC **Failure rates**: palate 1.3% (lowest), mandibular buccal shelf up to 65% (highest), pooled 12.3%.

444 **B** Primary stability depends on cortical engagement, and 5 to 10 Ncm gives a firm screw without compressing the bone beyond its tolerance. Driving to the highest torque the instrument allows is the intuitive answer and is the recognised cause of microfracture, cortical ischaemia and late failure. [Vol I Topic 5 · Vol II §14.5]

MNEMONIC **5-10 Ncm** — enough for primary stability without microfracture or cortical ischaemia from over-torquing.

445 **E** Proximity to or contact with a root is the most consistently reported risk factor for failure, and the mobile screw is removed so that both it and the root can be left to heal. Predicting endodontic consequences overstates the harm: periodontal ligament and cementum repair reliably once the screw is withdrawn, provided the contact is recognised early. [Vol I Topic 5 · Vol II §14.5]

446 **TFTTT**

A True — immediate and early loading are both established, and this is the essential difference from a prosthetic implant
B False — that regime belongs to osseointegrated implants and would waste months of treatment time
C True — about 12.3 per cent is the pooled figure across sites
D True — denser cortical bone and thinner attached mucosa in some mandibular sites both contribute
E True — the forces used are within the conventional orthodontic range and do not need to be reduced

[Vol I Topic 5 · Vol II §14.5]

447 **B** Newton's third law is not suspended by skeletal anchorage; the equal and opposite force is simply delivered into bone, which does not move in the way a tooth does. Saying that no reactive force exists is the loose way this is often expressed, and it conceals the fact that the screw itself is loaded and may migrate slightly or fail. [Vol I Topic 5 · Vol II §14.1]

448 **D** A force applied at the tube acts well away from the centre of resistance and generates a moment, so the crown tips distally while the apices swing forward and the roots are left worse angulated than before. Expecting translation from a single force is the persistent error: without a counter-moment from an uprighting spring or a rigid archwire, tipping is the only available outcome. [Vol I Topic 6 · Vol II §14.3]

449 **FTTFT**

A False — equal forces do not produce equal movement, since the response depends on the stress in the ligament
B True — the same force over a smaller root area gives higher stress and faster movement
C True — this is the principle of differential moments and the basis of segmented mechanics
D False — splinting several teeth together gives essentially the resistance of their combined root area, not more
E True — a higher moment-to-force ratio on the posterior unit resists tipping and so preserves its position

[Vol I Topic 5 · Vol II §14.4]

450 **A** The classical work on canine retraction identified about 272 cN as the force producing the fastest movement, which is appreciably above the 50 to 100 cN quoted as optimal for bodily movement generally. Choosing 100 cN is defensible for tissue safety but it is not the figure associated with the fastest rate, which is what the question asks. [Vol I Topic 5 · Vol II §14.7]

MNEMONIC **272 cN = fastest, 50-100 cN = optimal** — fastest is not the same as safest for the periodontium.

451 C Loss of attachment carries the centre of resistance apically, so the same bracket force generates a larger moment and produces more tipping, and both the force and the required counter-moment must be recalculated. Keeping the same ratio is the error that produces the characteristic flared and tipped result in a periodontally compromised adult. [Vol I Topic 5 · Vol II §14.7]

452 D Pure rotation requires a couple, which produces a moment with no net force and therefore turns the tooth about its own centre of resistance without displacing it. A single palatal force would rotate the molar but would also push it buccally, which is why a couple rather than a force is specified for a pure rotational correction. [Vol I Topic 6 · Vol II §14.3]

453 FFFFF

A False — first order is the in-and-out dimension, delivered by offsets and by bracket base thickness
B False — second order is mesiodistal angulation or tip; rotation belongs to the first order
C False — third order is torque, which controls labiopalatal inclination
D False — the prescription is built into the bracket bases and slots, and bends are added only to individualise it
E False — play between a rectangular wire and the slot means much of the prescribed torque is never expressed

[Vol I Topic 6 · Vol II §14.6]

MNEMONIC **1st = in-out, 2nd = tip/angulation, 3rd = torque** — the three orders, in order.

454 B A wire narrower than the slot must rotate through several degrees before its corners engage, and that angle of play is subtracted from the prescription before any torque is delivered. Assuming full expression because the wire is rectangular is the standard misconception, and it is why clinicians add torque by hand when a particular inclination matters. [Vol I Topic 6 · Vol II §14.6]

455 E Intrusive force is concentrated at the root apex, where the periodontal ligament area bearing it is small, so the force per incisor is kept to about 10 to 20 cN to limit the risk of root resorption. Applying 50 to 60 cN is the level appropriate for tipping a whole crown and is several times more than an apex should be asked to bear. [Vol I Topic 5 · Vol II §14.7]

456 A Elastomeric materials undergo rapid stress relaxation, losing a large fraction of their force within 24 hours and continuing to decay thereafter, which is why nickel-titanium coil springs are preferred where a constant force matters. Believing the initial figure persists is what leads to long intervals between appointments and stalled movement. [Vol I Topic 5 · Vol II §14.7]

457 TFFFF

A True — which is why cervical pull tends to extrude the molars as well as move them distally
B False — high pull passes above the centre of resistance and produces an intrusive component
C False — straight pull is the intermediate direction and is chosen where vertical change is to be minimised
D False — that is the distalisation regime; anchorage requires 10 to 12 hours at 250 to 350 g per side
E False — two independent safety mechanisms are mandatory, and instructions accompany rather than replace them

[Vol I Topic 5 · Vol II §14.2]

458 **C** As the crown tips, the rectangular wire contacts opposite corners of the slot and generates a couple that opposes the tipping, and it is this couple that raises the moment-to-force ratio towards translation. Friction is a resistance to sliding along the wire and produces no moment about the centre of resistance, so it cannot supply what bodily movement requires. [Vol I Topic 6 · Vol II §14.3]

459 **B** A Class II elastic delivers equal and opposite forces to both arches, so the anterior component proclines the lower labial segment while the vertical component extrudes the lower molars and rotates the occlusal plane. Attributing the change to growth over six months is the comfortable explanation and it ignores a force system that predicts exactly what was observed. [Vol I Topic 5 · Vol II §14.4]

460 **D** Moving the molars distally requires both a higher force and a longer daily duration than merely holding them, and 450 to 500 g per side for 14 hours is the conventional distalisation prescription. The lighter and shorter regime is the anchorage prescription: it will hold the molars where they are but will not carry them backwards. [Vol I Topic 5 · Vol II §14.2]

461 **FFTFF**

A False — the Nance button bears on the anterior palate, and there is no lower equivalent because of the tongue
B False — it reduces mesial movement substantially but absolute anchorage requires a skeletal unit
C True — this is the house figure for the mandibular leeway space and the reason the arch is fitted early
D False — the reciprocal force proclines the lower incisors and extrudes the lower molars
E False — the combined root surface area is what resists, so more teeth means more resistance

[Vol I Topic 5 · Vol II §14.2]

462 **A** The moment applied to upright the canine root is matched by an equal and opposite moment delivered into the archwire and thence to the anchor teeth, which is why late root uprighting is a recognised source of anchorage loss. Treating root movement as anchorage-free is the error: the crown may stay still, but the moment has to be resisted somewhere. [Vol I Topic 5 · Vol II §14.4]

463 **C** Raising the point of force application towards the level of the centre of resistance of the six anterior teeth reduces the moment generated and therefore the tendency to tip the crowns lingually. Expecting more tipping from a longer hook reverses the geometry: it is a short hook, low on the archwire, that leaves the longest perpendicular distance and produces the greatest tipping moment. [Vol I Topic 5 · Vol II §14.3]

464 **E** The infrazygomatic crest lies above the first molar roots and adjacent to the maxillary sinus, so the angle and depth of insertion are planned to engage the crest without entering the sinus or contacting a root. The greater palatine foramen is a real hazard but for palatal screws, not for a buccal site in the zygomatic buttress. [Vol I Topic 5 · Vol II §14.5]

465 **TTTTT**

A True — a small perforation by a screw of this size heals spontaneously and rarely produces sinus symptoms
B True — which is one of the reasons the palate, with its keratinised mucosa, has the lowest failure rate
C True — thin screws are more prone to torsional fracture, which is why insertion torque is limited
D True — the bundle runs in the lateral palatal groove, so paramedian placement keeps clear of it

E True — failure of one screw is not failure of the strategy, provided a sound adjacent site is available

[Vol I Topic 5 · Vol II §14.5]

466 **A** Retraction at bracket level generates a lingual crown tipping moment, and adding palatal root torque supplies the counter-moment that keeps the incisor inclination as the segment moves back. Shortening the hook increases the perpendicular distance from the centre of resistance and so makes the tipping worse, which is the opposite of what is wanted. [Vol I Topic 6 · Vol II §14.6]

467 **C** The anterior intrusive force is matched by an equal extrusive force on the molars, and the activation bend generates a moment that tips the molar crowns distally and the roots mesially. Expecting an intrusive reaction on the molars inverts Newton's third law, which requires the reaction to oppose the action delivered anteriorly. [Vol I Topic 5 · Vol II §14.4]

468 **E** Absolute anchorage on one side with active protraction on the other separates the two movements entirely, so the reactive force from the left-sided protraction passes into bone rather than into the right molar. A transpalatal arch connects the two molars and therefore transmits the reactive force from one to the other, which is precisely the coupling the plan needs to avoid. [Vol I Topic 5 · Vol II §14.5]

469 **FFFFF**

A False — the optimal range is 50 to 100 cN; 300 to 400 cN is well into the territory of hyalinisation

B False — beyond the optimum, heavier force crushes the ligament and movement slows rather than accelerates

C False — the lag phase runs from about 20 to 30 days, following an initial phase of roughly two days

D False — the ligament is about 0.2 mm wide, an order of magnitude less than stated

E False — RANKL expression peaks at about three days

[Vol I Topic 5 · Vol II §14.7]

MNEMONIC **50-100 cN, lag 20-30 d, PDL 0.2 mm, RANKL peaks day 3** — the house figures for tooth movement biology.

470 **B** Removing 2 mm of posterior vertical dimension allows the mandible to autorotate forwards and upwards about the condyle, which closes the anterior open bite and brings the chin forward. Predicting backward rotation reverses the direction: backward rotation follows posterior extrusion, which is what intrusion is being used to avoid. [Vol I Topic 5 · Vol II §14.5]

471 **D** A ratio of about 7 to 1 places the centre of rotation at the apex, so the crown moves while the apex is held, which is controlled tipping. Expecting translation is the common error: that requires a ratio nearer 10 to 1, and the difference of three units is the difference between an upright and a retroclined finish. [Vol I Topic 5 · Vol II §14.3]

472 **C** Tissue impingement under an acrylic button progresses to ulceration and can produce necrosis of the palatal mucosa, so the appliance is removed or remade and the tissue allowed to recover. Interpreting the embedding as evidence that the anchorage is working is the dangerous reading: the reactive load is being transmitted through inflamed tissue rather than through bone. [Vol I Topic 5 · Vol II §14.2]

473 **A** Levelling a deep curve of Spee with a continuous archwire extrudes the posterior teeth, and in a Class II patient that rotates the mandible down and back, increasing the overjet the plan is trying to reduce. Encouraging extrusion is defensible in a short face with reduced vertical proportions, but it is the wrong choice where the anteroposterior correction depends on forward rotation. [Vol I Topic 5 · Vol II §14.1]

474 `FFFFT`

A False — splinting increases resistance but never abolishes movement; only a skeletal unit approaches absolute
B False — equal forces produce unequal movement, because the ligament stress depends on root area
C False — the reaction exists for as long as the force does, which is why anchorage is monitored throughout
D False — vertical and transverse anchorage are planned in exactly the same way as antero-posterior
E True — the higher stress in the smaller unit's ligament makes it the one that moves, which is the basis of differential anchorage

[Vol I Topic 5 · Vol II §14.1]

475 **E** Reduced attachment lowers the root surface area over which the force is distributed and moves the centre of resistance apically, so the force is reduced and the counter-moment increased to keep the stress and the movement under control. Using the same figures as for an intact periodontium is the error that produces flaring and accelerated attachment loss in exactly the patients who can least afford it. [Vol I Topic 5 · Vol II §14.7]

476 **D** Penetrating ocular injury from a displaced facebow can progress to intraocular infection and loss of sight, so urgent specialist assessment takes precedence over any orthodontic consideration. Continuing with a modified facebow treats the incident as a fitting problem, when what has occurred is a recognised and sight-threatening complication. [Vol I Topic 5 · Vol II §14.2]

477 **B** Turning a tooth about its long axis without displacing it requires a couple, and the elastic supracrestal collagen fibres remodel very slowly and drive the rotation back, which is why a bonded retainer or a pericision is used. Attributing relapse to bone remodelling misplaces the mechanism: alveolar bone remodels within months, whereas the supracrestal fibres remain under tension for years. [Vol I Topic 6 · Vol II §14.6]

478 `TFFTF`

A True — the palate is the most reliable site and this is the figure quoted for it
B False — reported failure at the buccal shelf reaches 65 per cent, which is the extreme of the range
C False — site selection, angulation and torque control all improve with experience, and failure falls accordingly
D True — about 12.3 per cent is the pooled figure and it is the one quoted at consent
E False — most failures occur within the first weeks to months, when primary stability is being lost

[Vol I Topic 5 · Vol II §14.5]

479 **E** The force of 0.5 N acting at 12 mm generates 600 newton-millimetres about the centre of resistance, and the 300 newton-millimetre counter-moment gives a net ratio of about 6 to 1 at the bracket, which is the controlled tipping range. Translation would require the counter-moment to equal the full 600 newton-millimetres, which is double what has been delivered. [Vol I Topic 5 · Vol II §14.3]

480 **B** An intra-oral distalising appliance must brace against something, and the palate and premolars take the reactive load, so anterior proclination and premolar mesial drift are its characteristic side effects. Saying it produces bodily distal movement is the opposite of the usual finding: the molar tips distally and its roots lag behind, which is why uprighting is needed afterwards. [Vol I Topic 5 · Vol II §14.2]

481 **D** Protraction against a skeletal unit placed behind the molar delivers the reactive force

into bone, so the lower incisors are not called upon to act as anchorage at all. A push coil between the canine and the molar is mechanically simple, but its reaction drives the whole anterior segment forwards, which is exactly what the plan forbids. [Vol I Topic 5 · Vol II §14.5]

482 **FTFFT**

A False — prescriptions differ substantially between systems, particularly in incisor torque values
B True — bracket placement error is the commonest reason a prescription is not expressed as intended
C False — base thickness alters the in-and-out position and therefore the effective torque delivered
D False — rotation belongs to the first order; the third order is torque
E True — a full-size wire will seat the crown to the level dictated by the slot, and the error is carried into the finish

[Vol I Topic 6 · Vol II §14.6]

483 **A** A bracket placed gingivally and rotated changes the vertical position, the angulation and the effective torque simultaneously, because all three are referenced to where the slot sits on the crown. Blaming the archwire sequence is the reflex explanation at debond, but a stiff wire expresses a misplaced prescription faithfully rather than preventing it. [Vol I Topic 6 · Vol II §14.6]

484 **C** Proffit's ladder runs from intraoral reinforcement — adding teeth, differential force, transpalatal and lingual arches — through extraoral traction to skeletal anchorage, so headgear is the last purely dental rung before a miniscrew or a miniplate. Adding the second molars is genuine reinforcement but sits well down the ladder, because root surface area alone will not hold a maximum anchorage case against a 7 mm retraction. [Vol I Topic 5 · Vol II §14.4]

MNEMONIC **Proffit's ladder**: add teeth, differential force, transpalatal/lingual arch, headgear, then skeletal anchorage.

485 **TTFTF**

A True — only the fixation screws engage cortical bone, so the interradicular constraint that governs miniscrew siting does not apply
B True — placement and removal are both surgical, which is the main disadvantage against miniscrews
C False — miniplate survival is high and compares favourably with most miniscrew sites, the palate excepted
D True — bone anchored protraction avoids the incisor proclination and mesial molar movement of a tooth borne facemask
E False — miniplates rely on mechanical retention rather than osseointegration, and a short healing period precedes loading

[Vol I Topic 5 · Vol II §14.6]

Scored under 70% of the 109 marks on this block? Re-read Volume II §14, then sit it again in a fortnight.

Block O — Orthodontic materials, wires, bonding and instruments (Q486–530 · 89 marks · Vol II §15)

486 **D** Alignment needs low stiffness and a long range so that a light force is delivered over a large deflection, while finishing and space closure need high stiffness, good formability and dimensional control, and these requirements cannot coexist in one alloy, which is why a sequence is used. Corrosion is a real but minor limitation and does not explain the need for a sequence at all. [Vol I Topic 7 · Vol II §15.1]

487 **B** A grossly displaced tooth requires a wire that can be deflected far into the bracket and still return a light continuous force, and the unloading plateau of a superelastic nickel titanium wire does exactly that. Stainless steel of any dimension will take a permanent set when tied into a 3 mm displacement and then delivers nothing. [Vol I Topic 7 · Vol II §15.2]

488 **TTTTT**

A True — stiffness is proportional to Young's modulus for a given geometry, which is why nickel titanium is springier than steel of the same size
B True — beyond it the wire is permanently bent and the force it was intended to deliver is lost
C True — resilience is the working capacity of the wire and is what the clinician draws on between visits
D True — formability is what allows loops and finishing bends to be placed without fracture
E True — range and stiffness together determine how far a wire can be tied in and what it then delivers

[Vol I Topic 7 · Vol II §15.1]

MNEMONIC In curve order: **stiffness (modulus), elastic limit, resilience, range, formability**.

489 **E** Bending stiffness varies with the fourth power of the diameter, so a ratio of twenty to sixteen raised to the fourth power is about two and a half, while range varies inversely with diameter and therefore falls. The answer that keeps range unchanged ignores that a stiffer wire reaches its elastic limit after a smaller deflection. [Vol I Topic 7 · Vol II §15.2]

490 **A** Bending stiffness falls with the cube of the length between attachments, so adding wire in the form of a loop or a helix is the classical way of softening a wire without changing its material or size. Bonding brackets more incisally moves them apart only marginally and alters the force system in other, less predictable ways. [Vol I Topic 7 · Vol II §15.2]

491 **C** Beta titanium has a modulus about two-fifths that of stainless steel, so it delivers moderate forces, and unlike nickel titanium it is formable and weldable, so torquing and stepping bends can be placed in it. Superelastic nickel titanium of the same dimension cannot be bent at all, which is precisely what this stage requires. [Vol I Topic 7 · Vol II §15.3]

MNEMONIC Relative stiffness, low to high: **NiTi < beta-Ti < CoCr < stainless steel** (beta-Ti is about two-fifths of steel).

492 **TFTFT**

A True — the chromium oxide film is what confers corrosion resistance and it re-forms if scratched
B False — steel is the stiffer of the two, beta titanium having roughly two-fifths of its modulus
C True — work hardening raises the yield point at the cost of ductility, and annealing reverses it
D False — stainless steel is readily soldered with silver solder and is also weldable
E True — the small diameter of each strand gives the bundle a low bending stiffness and a long range

[Vol I Topic 7 · Vol II §15.3]

MNEMONIC **18-8 stainless** — 18% chromium forms the passivating oxide, 8% nickel; the classic composition name.

493 **B** A multistranded wire behaves as a bundle of small-diameter strands, so its bending stiffness is a fraction of that of a solid wire of the same overall size while its range is long, which suits initial alignment. Solid steel of 0.016 inch will be permanently deformed by a 20 degree rotation and thereafter delivers no useful force. [Vol I Topic 7 · Vol II §15.3]

494 **D** Heat-activated wires have a transition temperature range near or a little above mouth temperature, so warming converts martensite to austenite and the wire stiffens and works

harder, while cooling reverses this. Permanent deformation cannot explain a change that reverses itself, and annealing would soften rather than stiffen the wire. [Vol I Topic 7 · Vol II §15.2]

495 **A** Cobalt chromium is supplied in tempers that can be bent while soft and then heat treated at around 480 degrees Celsius for several minutes, which raises the yield strength and stiffness without disturbing the bends. Hard-tempered stainless steel is already at its working stiffness and cannot be softened for forming and then hardened again in the same way. [Vol I Topic 7 · Vol II §15.3]

496 **FFFFF**

A False — it arises from a stress-induced martensitic transformation, not from work hardening
B False — unloading occurs at a lower force than loading, and that hysteresis is what makes the wire kind to the periodontium
C False — across the plateau the force is nearly constant despite changing deflection, which is the whole point
D False — nickel titanium cannot be reliably soldered or welded, which is why auxiliaries are tied rather than joined
E False — the advantage is a light constant force over a large deflection, not any change in the length of wire needed

[Vol I Topic 7 · Vol II §15.2]

497 **C** A glossy surface after drying means the etch pattern has been lost, almost always through salivary contamination during washing or drying, and the only reliable remedy is to re-isolate and re-etch briefly. Proceeding regardless is the commonest cause of an early bond failure on that one tooth. [Vol I Chapter 10 · Vol II §15.4]

> **MNEMONIC** Bonding sequence: **etch, wash, dry, prime, bond** — a glossy surface means step 2 or 3 was compromised, so redo from etch.

498 **E** A repeated failure confined to one region and one arch in a patient with a deep bite points to the brackets being knocked off in function rather than to a technique error, and the remedy is to disclude the bite with a bite plane or posterior build-ups. Salivary contamination would not spare the upper arch and would not affect only premolars. [Vol I Chapter 10 · Vol II §15.4]

499 **A** Glass ionomer bonds chemically to enamel, tolerates a slightly moist surface and releases fluoride at the band margin, which matters in a patient at risk of demineralisation. Composite resin gives the highest bond strength of these but is the least forgiving of moisture, which is the constraint in this case. [Vol I Chapter 10 · Vol II §15.4]

500 **TTFTF**

A True — surface enamel is lost during etching and a further layer is lost at debond
B True — micromechanical interlocking with resin tags is the mechanism, not chemical adhesion
C False — a chemically cured adhesive sets on its own schedule, whereas light curing is what lets the operator decide when setting begins
D True — hydrophilic primers were introduced precisely for the partially erupted or poorly isolated tooth
E False — a light-cured adhesive reaches most of its final strength within minutes, not weeks

[Vol I Chapter 10 · Vol II §15.4]

501 **C** Etching, the fracture of the resin-enamel interface and the clean-up together remove a layer measured in tens of micrometres, which is clinically acceptable but not zero, and the surface is left with fine scratches. The claim that the resin separates cleanly at the interface

is wrong because a proportion of the adhesive fails cohesively and takes enamel with it. [Vol I Chapter 10 · Vol II §15.5]

502 **E** White spot lesions appear in up to half of patients and can be seen within six months of bonding, and the response is to control the cause with hygiene, diet advice and high-concentration fluoride while reassessing whether the appliance load is sustainable. Continuing unchanged is the trap, since these lesions progress quickly and are permanent once cavitated. [Vol I Chapter 10 · Vol II §15.5]

503 **B** A band is retained mechanically by its circumferential fit and cemented with glass ionomer, so the presence of amalgam on the surface that would otherwise carry the bond is irrelevant, and it withstands heavy load and auxiliaries. Bands are not free of demineralisation risk; lesions occur at the band margins, particularly gingivally. [Vol I Chapter 10 · Vol II §15.4]

504 **FTFFF**

A False — once the tooth tips, binding and notching dominate and classical friction becomes a small part of the total
B True — below that angle only classical friction acts, and above it binding rises steeply
C False — notching is permanent deformation at the bracket corner and it arrests movement until the wire is changed
D False — classical friction depends on the normal force, not on the apparent area of contact
E False — a stretched elastomeric module presses harder on the wire than a deliberately loose steel ligature

[Vol I Topic 7 · Vol II §15.6]

MNEMONIC Resistance to sliding, as tipping increases: **classical friction, then binding, then notching**.

505 **D** Elastomeric modules lose a large proportion of their force in the first day and continue to decay thereafter, so by twelve weeks they are holding the wire only loosely and rotational and torque control are lost. The idea that they stiffen with age reverses what actually happens, which is stress relaxation and water absorption. [Vol I Topic 7 · Vol II §15.6]

506 **C** A rotation persists when the wire is not fully seated in the slot, and a steel ligature tied in a figure of eight applies a much higher and more durable seating force than a decayed module, with a wedge adding the couple needed. A stiffer steel wire would not engage any better and risks a heavy force on a tooth that is not yet aligned. [Vol I Topic 7 · Vol II §15.6]

507 **TTFTT**

A True — this is the accepted figure, and it is higher in atopic individuals and in healthcare workers
B True — ear piercing and jewellery account for much of the difference between the sexes
C False — intraoral reactions to nickel are uncommon despite high skin prevalence, and oral ulceration is not the usual picture
D True — type I reactions are immunoglobulin E mediated and can be immediate and severe
E True — synthetic elastics perform acceptably and remove the exposure at once

[Vol I Chapter 10 · Vol II §15.6]

MNEMONIC Allergy numbers: **latex 4.3% general; nickel 10-30% women, 1-3% men**.

508 **A** A distal end cutter retains the severed fragment in its beaks, which is a straightforward airway and soft-tissue safety measure at every archwire placement. Weingart pliers are for carrying and placing the archwire and have no cutting edge at all. [Vol I Chapter 10 · Vol II §15.6]

509 **E** Sliding mechanics need a wire that resists deformation and controls torque during retraction, and a rectangular steel wire close to slot size does both while providing a low-friction track. A rectangular nickel titanium wire is too flexible to resist the tipping moments generated during retraction and will allow the anchorage to collapse. [Vol I Chapter 10 · Vol II §15.2]

510 **D** Torque is delivered by a couple between the flat surfaces of a rectangular wire and the walls of the slot, so a round wire can rotate freely within the slot and expresses no third order information whatever its diameter. Tip is delivered by the bracket in the plane of the slot and is expressed on round wires as well. [Vol I Topic 7 · Vol II §15.2]

511 **FFTTF**

A False — range varies inversely with diameter, so larger wires can be deflected less before permanent deformation
B False — stiffness varies with the inverse cube of length, so halving the span makes it about eight times as stiff
C True — bending stiffness varies with the fourth power of the diameter
D True — stiffness varies with the cube of the dimension in the plane of bending and only linearly with the other
E False — that combination leaves several degrees of torsional play before torque begins to be expressed

[Vol I Topic 7 · Vol II §15.2]

MNEMONIC Beam rules: **stiffness prop. to diameter to the fourth, inversely to length cubed; range inversely to diameter.**

512 **B** Springback is proportional to yield strength divided by the modulus of elasticity, so a low-modulus wire of moderate strength recovers from a larger deflection than a stiff strong one. Reading springback from yield strength alone ignores the denominator, which is exactly why nickel titanium outperforms steel here. [Vol I Topic 7 · Vol II §15.1]

513 **E** A wire retains a permanent set once the stress applied has taken it beyond its elastic limit, after which the stored energy it was intended to release has been spent in plastic deformation. Resilience describes the energy the wire can store within the elastic range and is not itself something that can be exceeded. [Vol I Topic 7 · Vol II §15.1]

514 **B** The surface oxide and the phase behaviour of nickel titanium make conventional soldering and welding unreliable, and heating destroys the superelastic behaviour, so hooks are crimped or the auxiliary is tied. Heat treatment at 480 degrees is the cobalt chromium procedure and would ruin a nickel titanium wire. [Vol I Topic 7 · Vol II §15.3]

515 **FTTTT**

A False — these springs are activated to deliver a stated force at a modest extension, and gross overstretching is neither necessary nor safe
B True — open coil is compressed between attachments and recovers by pushing, which is how space is regained
C True — the superelastic plateau of the spring holds force where an elastomeric chain decays
D True — most of the decay occurs in the first day, and this is the argument for coil springs in space closure
E True — steel does not undergo the stress relaxation that affects elastomerics

[Vol I Topic 7 · Vol II §15.6]

516 **D** Resin-modified glass ionomer retains the fluoride release and moisture tolerance of glass ionomer while its resin component raises the bond strength to a level that will hold a bracket through treatment. Conventional glass ionomer releases more fluoride but its bond strength is too low for routine bracket bonding and the failure rate is unacceptable. [Vol I Chapter 10 · Vol II §15.4]

517 **A** The acidic monomer conditions and primes in one step and produces a less aggressive etch pattern with lower laboratory bond strengths, yet clinical failure rates are similar and less enamel is removed. Claiming identical bond strengths overstates the case; the trade is a measurable reduction in strength for a clinically acceptable result. [Vol I Chapter 10 · Vol II §15.4]

518 **C** A non-significant difference in a trial of this size is a statement about precision rather than about equivalence, and a two-point difference in failure rate would matter across a practice. Declaring the techniques equivalent is the classical misreading of a negative result from an underpowered trial. [Vol I Chapter 10 · Vol II §15.4]

519 **TFTTT**

A True — a peeling or shearing force at the base is the intended mode and is kinder to enamel
B False — fine scratches and a loss of surface enamel persist, and the prism-free layer is not recoverable
C True — failure within the adhesive or at the bracket leaves the enamel intact and the remnant is then removed under control
D True — the adhesive remnant index records the proportion of adhesive left on the tooth
E True — the higher bond strength and lower fracture toughness of ceramic increase the risk, particularly with monocrystalline brackets

[Vol I Chapter 10 · Vol II §15.5]

520 **D** Latex elastics lose a substantial proportion of their force within the first 24 hours, which is why they are changed at least daily and often more frequently, and a weekly regimen delivers a decaying and finally negligible force. Elastics do not stiffen with time in the mouth; they relax. [Vol I Topic 7 · Vol II §15.6]

521 **B** Separators placed a few days to a week beforehand wedge the contacts open by a fraction of a millimetre, which is all that is needed to seat a band without trauma, and they are removed at the fitting visit. Interproximal reduction removes enamel permanently to solve a problem that resolves by itself once the band is cemented. [Vol I Chapter 10 · Vol II §15.6]

522 **E** Beta titanium is formable enough to take torquing and offset bends and stiff enough to hold them while delivering forces well below those of a comparable steel wire, which suits detailing in an adult. A round wire of any alloy cannot express torque and so can deliver only the first order component. [Vol I Topic 7 · Vol II §15.3]

523 **FTTFT**

A False — a Mitchell's trimmer is a hand instrument for removing cement and debris, not for placing wires
B True — the square beak grips and the round beak forms the curve, which is why they suit arrowhead clasps
C True — heavy cutters are made for thick hard wire, and using light cutters on it destroys them
D False — a full-dimension steel archwire needs heavy cutters or a distal end cutter, not light wire cutters
E True — peeling the base is the intended debonding action and protects enamel

[Vol I Chapter 10 · Vol II §15.6]

MNEMONIC Tool-to-job: **Mitchell=cement, Adams=clasps, Mauns=heavy wire, light cutters=thin wire, pliers=peel.**

524 **A** A 0.016 by 0.022 inch steel wire nearly fills the 0.018 inch slot, so torsional play is small and torque is expressed while the wire remains rigid enough for sliding. A round steel wire of the same diameter slides well but cannot express any torque, which is the requirement that decides this case. [Vol I Topic 7 · Vol II §15.2]

525 **C** A type I reaction is immunoglobulin E mediated and can be severe, so exposure must be eliminated rather than reduced, and non-latex elastics with a latex-free environment achieve this without changing the mechanics. Reducing exposure by changing elastics more often still leaves the patient exposed and misunderstands the nature of the reaction. [Vol I Chapter 10 · Vol II §15.6]

526 **B** The undersized wire must rotate within the slot until its corners engage the slot walls before a couple can develop, and that engagement angle accounts for a loss of the order of ten degrees in this combination. Stainless steel has the highest modulus of the common alloys, so a low modulus cannot be the explanation. [Vol I Topic 7 · Vol II §15.2]

527 **FFTTF**

A False — the first wire is chosen for range and low stiffness, and holding arch form is a task for the later rigid wires
B False — the two demands are opposed, and a flexible wire allows the buccal segments to collapse under retraction
C True — a round wire rotates freely in the slot and expresses no third order information
D True — rigidity preserves anchorage and arch form while the teeth slide along the wire
E False — formability, not stiffness, is what allows a bend to be placed without fracture and then held

[Vol I Topic 7 · Vol II §15.2]

MNEMONIC Progression: **round NiTi (align) -> rigid rectangular steel (work) -> formable wire (finish)**.

528 **D** The properties that make nickel titanium ideal for alignment make it useless for the working and finishing stages, where bends must be placed, arch form held and anchorage preserved against retraction forces. Nickel release is measurable but the quantities are far below dietary intake and do not constitute the argument against using the wire throughout. [Vol I Topic 7 · Vol II §15.3]

529 **A** Modules lose much of their force within a day and continue to relax, so towards the end of a long interval the wire may not be fully seated and expression of the prescription is incomplete. Demineralisation relates to plaque retention around any ligation and is not the trade-off that decides this question. [Vol I Topic 7 · Vol II §15.6]

530 **C** A diode laser gives a bloodless soft tissue excision at the chairside with immediate re-bonding, which is exactly what a localised hyperplastic cuff over healthy bone needs. Referring for a flap and osseous recontouring is the over-treatment trap — there is no pocketing and no bone loss here, so the bone does not need touching. [Vol I Topic 7 · Vol II §15.5]

Scored under 70% of the 89 marks on this block? Re-read Volume II §15, then sit it again in a fortnight.

Block P — Removable and functional appliances (Q531–571 · 81 marks · Vol II §16)

531 **E** A removable appliance prescription is written under four headings, active components, retention, anchorage and baseplate, and an appliance with active components but no specified clasping will be displaced by the reaction force and will not work. Baseplate thickness is a laboratory convention and its omission changes nothing about whether the appliance functions. [Vol I Chapter 9 · Vol II §16.1]

MNEMONIC Write every RA prescription under: **active components, retention, anchorage, baseplate**.

532 **A** An Adams clasp on a permanent molar is made in 0.7 mm hard stainless steel wire, and a clasp made in 0.6 mm is too flexible to hold and deforms in function; 0.6 mm is reserved for premolars and primary molars. Reducing the spring activation would slow the intended movement without correcting a clasp that is simply the wrong gauge. [Vol I Chapter 9 · Vol II §16.1]

MNEMONIC Adams clasp gauge: **0.7 mm on permanent molars, 0.6 mm on premolars/primary molars** — swap them and it fails.

533 **FFFFF**

A False — the arrowheads engage the mesiobuccal and distobuccal undercuts, which is why the bridge stands clear of the buccal surface

B False — a labial bow of that gauge is stiff, has a very short range and is a retentive rather than an active component over any useful distance
C False — the baseplate is a major contributor to anchorage, transmitting the reaction to the palatal vault and the whole dentition
D False — the Southend clasp is made in 0.7 mm wire and engages the undercuts of the upper central incisors
E False — a removable appliance delivers a single point of force through a spring and can therefore tip a tooth only

[Vol I Chapter 9 · Vol II §16.1]

534 C A light force of the order of 25 to 40 cN is optimal for tipping a single tooth and is obtained from about 1 to 2 mm of activation of a 0.5 mm palatal finger spring of conventional length. A 4 mm activation delivers a heavy force that causes pain, displaces the appliance and does not move the tooth any faster. [Vol I Chapter 9 · Vol II §16.2]

535 E A single retroclined incisor to be tipped labially over the bite is the classical indication for a Z-spring, and posterior bite planes are needed to free the occlusion while the tooth passes through the crossbite. A palatal finger spring moves a tooth mesiodistally around the arch and cannot push an incisor labially. [Vol I Chapter 9 · Vol II §16.2]

536 B A T-spring seated against the palatal surface delivers a buccally directed force to a single posterior tooth and is the standard component for this movement. A midline screw divided sagittally moves a whole buccal segment and would be an over-treatment where only one tooth is in crossbite. [Vol I Chapter 9 · Vol II §16.2]

537 TFTTF

A True — halving the length of a cantilever makes it about eight times stiffer, which is why spring length is the principal design variable
B False — force varies with the fourth power of the diameter, so that change increases it roughly fourfold
C True — the coil adds wire length without adding bulk, which is exactly why it is incorporated
D True — the guard both supports the spring and keeps it in its intended plane of action
E False — the target for tipping a single tooth is of the order of 25 to 40 cN, and 150 cN would be grossly excessive

[Vol I Chapter 9 · Vol II §16.2]

MNEMONIC Spring rules: **force falls with length cubed, rises with diameter to the fourth; target 25-40 cN to tip a tooth.**

538 D A quarter turn opens the screw by about 0.25 mm and the conventional regimen is one quarter turn once or twice weekly, giving slow dental expansion; turning twice daily delivers about 3.5 mm of screw opening in a week and produces exactly this picture. One quarter turn twice a week is within the accepted range and would not cause it. [Vol I Chapter 9 · Vol II §16.2]

539 C A flat anterior bite plane discludes the posterior teeth, which then erupt while the incisors are prevented from over-erupting, so the overbite reduces by an increase in lower face height and this depends on continuing vertical growth. True incisor intrusion is not achievable with a removable appliance, which can deliver only tipping forces. [Vol I Chapter 9 · Vol II §16.2]

540 A Posterior bite planes disclude the buccal segments so that the teeth can move transversely past the opposing cusps, which a deep overbite would otherwise prevent. A flat anterior bite plane holds the incisors apart but leaves the buccal teeth in contact and therefore locked, which is the reverse of what is required. [Vol I Chapter 9 · Vol II §16.2]

541 TTTTT

A True — a single point of force application produces a moment about the root and therefore tipping
B True — the mechanism is posterior eruption, which is why bite planes are unreliable in the non-growing patient
C True — the same appliance design serves all three purposes with or without active components
D True — the palatal coverage spreads the reaction over a large area and the clasps resist displacement
E True — the appliance is out of the mouth whenever the patient chooses, which no fixed appliance permits

[Vol I Chapter 9 · Vol II §16.1]

MNEMONIC One baseplate design serves three roles: **active appliance, space maintainer, passive retainer**.

542 E The Southend clasp is made in 0.7 mm wire, runs across both central incisors and engages their cervical undercuts, and is the standard anterior retentive component. A labial bow is primarily an anterior stop and, if activated, becomes an active component that would retrocline the incisors rather than retain the appliance. [Vol I Chapter 9 · Vol II §16.1]

543 D Repeated midline fracture indicates that the acrylic is being flexed beyond its strength, and a buried palatal strengthening wire distributes that load without adding bulk. Making the baseplate thinner reduces the section that is already failing and would make the fractures more frequent. [Vol I Chapter 9 · Vol II §16.1]

544 B Relieving 7 mm of crowding and bringing buccally excluded canines into a finished arch requires bodily movement, rotational control and controlled space closure, none of which a removable appliance can deliver. The Class I molar relationship is irrelevant to the choice and is offered only as a distraction. [Vol I Chapter 9 · Vol II §16.1]

545 FTTFF

A False — moving more teeth at once increases the reaction carried by a smaller anchor unit, so demand rises
B True — a large area of palatal coverage spreads the reaction and is one of the appliance's advantages
C True — the classical sign is that the appliance ceases to fit as the anchor teeth drift forwards
D False — heavier forces increase the reaction and accelerate anchorage loss rather than overcoming it
E False — intermaxillary elastics transfer the reaction to the lower arch and produce proclination and extrusion there

[Vol I Chapter 9 · Vol II §16.1]

546 E Removable appliance treatment cannot succeed without wear, and the honest course is to show the patient the objective signs, explore the reasons and make a shared decision about continuing, deferring or changing approach. Increasing the activation punishes the few hours the appliance is worn and makes wear less likely still. [Vol I Chapter 9 · Vol II §16.1]

547 B The twin block is designed to be worn while eating because the bite blocks are what hold the mandible forward, and full-time wear is the regimen on which the published results are based. Removing it for meals is the commonest instruction error and substantially reduces the hours of effective posturing. [Vol I Topic 8 · Vol II §16.3]

548 D Concavities at C2 and C3 with C3 and C4 still trapezoid is CS2, at which the pubertal peak is expected within about a year, so starting now places the active phase across the peak where the skeletal response is greatest. Reading this as CS4 would require concavities in all three bodies with C3 and C4 horizontally rectangular. [Vol I Topic 8 · Vol II §16.3]

MNEMONIC CVM peak timing: **CS1 2y, CS2 1y, CS3 <1y away (now), CS4 1-2y past** — CS2-3 is when to start.

549 `TTFTT`

A True — the reported split is of the order of two-thirds dentoalveolar to one-third skeletal, and the skeletal part is small
B True — the appliance transmits the reaction to the lower labial segment, and this is one of its principal side effects
C False — the upper incisors are retroclined, which contributes to the overjet reduction
D True — the posterior teeth have been held apart by the blocks and have not been in occlusion
E True — progressive trimming allows differential eruption to settle the buccal segments

[Vol I Topic 8 · Vol II §16.4]

MNEMONIC Twin block change: **about two-thirds dentoalveolar, one-third skeletal** — plus lower proclination, upper retroclination.

550 A Posturing the mandible forward removes the occlusal interlock that is holding the arches apart and produces a combination of upper incisor retroclination, lower incisor proclination, differential eruption and a small skeletal change. The claim of a lasting true increase in condylar growth is the historical justification and is not supported by the long-term trial evidence. [Vol I Topic 8 · Vol II §16.4]

551 C The Frankel is largely tissue borne, its buccal shields and lip pads standing away from the alveolus so that the resting pressures of cheek and lip are removed and the arches widen and the labial segments are re-postured. The Herbst is the opposite in concept, being rigidly tooth borne and fixed, with no soft-tissue shielding at all. [Vol I Topic 8 · Vol II §16.5]

552 D The Herbst is cemented or bonded in place, so its effect does not depend on the patient choosing to wear it, and it is the standard answer to repeated non-compliance with a removable functional appliance. A wear diary addresses the measurement of compliance rather than its cause and has already failed twice in this patient. [Vol I Topic 8 · Vol II §16.5]

553 `FTFFF`

A False — the two-phase and one-phase groups reach the same final occlusal outcome, which is the central finding of these trials
B True — reduced trauma is the one clear advantage of treating early, and this is the reported figure
C False — two phases means more appointments over more years for the same end result
D False — the dentoalveolar component predominates, with the skeletal contribution around one-third at most
E False — the early psychosocial advantage is not sustained once the control group is treated

[Vol I Topic 8 · Vol II §16.6]

554 B The posterior open bite is the expected consequence of holding the buccal teeth apart, and it is closed by trimming the blocks so that the lower molars erupt, after which fixed appliances detail the occlusion. Removing the appliance and simply waiting risks relapse of the overjet while the open bite settles unpredictably. [Vol I Topic 8 · Vol II §16.4]

555 E Lower incisor proclination is the routine reaction to the appliance and reduces the overjet, but teeth moved beyond the zone of soft-tissue balance tend to relapse, so the finding shapes the retention plan rather than condemning the appliance. Calling it desirable and seeking more of it trades a short-term overjet gain for a long-term instability. [Vol I Topic 8 · Vol II §16.4]

556 A The corrected position depends on the mandible continuing to be postured and on the buccal occlusion settling, so part-time wear of the functional appliance or an appliance with

an inclined plane holds the correction until fixed appliances take over. Assuming the overjet is stable on its own is the commonest reason these cases relapse in the interval. [Vol I Topic 8 · Vol II §16.6]

557 **TFTTT**

A True — the shields stand away from the alveolus and alter the resting soft-tissue environment
B False — the bionator is a removable one-piece tooth and tissue borne appliance
C True — the telescopic mechanism makes it fixed and independent of patient cooperation
D True — the FR-3 carries lip pads in the upper sulcus and is the Class III member of the series
E True — reduced acrylic bulk with a moderate opening is the design compromise it represents

[Vol I Topic 8 · Vol II §16.5]

MNEMONIC ID by design: **Frankel = tissue-borne shields, Bionator = one-piece, Herbst = fixed telescopic, FR-3 = Class III**.

558 **C** In a patient who is already vertically excessive with a reduced overbite, further posterior eruption would rotate the mandible backwards and worsen both, so the posterior teeth are capped and controlled and high-pull headgear adds a vertical restraining vector. Trimming to encourage posterior eruption is correct for the deep bite case and is precisely wrong here. [Vol I Topic 8 · Vol II §16.5]

559 **B** Two independent safety devices are mandatory for all headgear, because the hazard is the facebow disengaging and recoiling towards the eyes, and that hazard is unchanged by what the bow is attached to. The idea that a removable appliance is inherently safer confuses the appliance with the facebow, which is the dangerous component. [Vol I Topic 8 · Vol II §16.5]

560 **D** Two millimetres against untreated controls is at the margin of what can be measured and is not what corrects an 8 mm overjet, so the appliance earns its place through dentoalveolar correction and, where the overjet and lip competence put the incisors at risk, through reduced trauma. Concluding that the appliance is ineffective ignores the overjet reduction that these trials also demonstrate. [Vol I Topic 8 · Vol II §16.6]

561 **FFTFF**

A False — it settles with trimming and subsequent eruption or with fixed appliances, and surgery has no place
B False — teeth moved out of the soft-tissue balance zone tend to return, and retention is planned around it
C True — transient joint and muscle discomfort in the first weeks is common and usually settles
D False — the upper incisors are retroclined, and that retroclination is part of the correction
E False — any headgear-like maxillary restraint is small and inconsistent, and is not comparable to headgear

[Vol I Topic 8 · Vol II §16.4]

562 **A** Adaptation to a bulky appliance depends on continuous wear, and warning the patient in advance that salivation and lisping are normal and short-lived is what carries her through the period when most non-compliance begins. Part-time wear at the start prolongs the adaptation and is the pattern from which patients rarely progress to full-time wear. [Vol I Topic 8 · Vol II §16.3]

563 **C** The FR-3 carries lip pads and buccal shields in the upper sulcus, relieving soft-tissue pressure on the maxilla while restraining the lower labial segment, and it is the removable

functional appliance designed for Class III. A conventional twin block postures the mandible forwards, which is the last thing a Class III patient needs. [Vol I Topic 8 · Vol II §16.5]

564 **E** The trials show identical final occlusal outcomes for one-phase adolescent and two-phase early treatment, so the default is the shorter single course in early adolescence, with early treatment reserved for the trauma indication where the risk ratio of about 0.58 applies. Treating everyone early doubles the appointments and the duration for no occlusal gain. [Vol I Topic 8 · Vol II §16.6]

565 **TTTFT**

A True — the anchorage regimen is lighter and shorter than the distalising regimen
B True — distalisation needs both a heavier force and longer daily wear
C True — high pull resists the downward and backward rotation that a cervical pull encourages
D False — a force through a facebow on a removable appliance tips the molars distally, and bodily movement requires a fixed attachment
E True — this is an absolute requirement because of the risk of ocular injury

[Vol I Topic 8 · Vol II §16.5]

MNEMONIC Headgear numbers: **10-12 h at 250-350 g for anchorage; 14 h at 450-500 g for distalisation**.

566 **A** A bite plane reduces overbite by allowing the posterior teeth to erupt while the incisors are held apart, and this depends on continuing vertical alveolar development, which is why the technique is a growing patient's treatment. Joint damage is not the reason; the mechanism simply is not available once growth has ceased. [Vol I Chapter 9 · Vol II §16.2]

567 **C** A distorted spring delivers an uncontrolled force and traumatises the palate, so it must be corrected or remade and the guard wire checked, since inadequate guarding is usually why the spring distorted. An antifungal rinse treats a condition that is not present and leaves the mechanical cause untouched. [Vol I Chapter 9 · Vol II §16.2]

568 **E** Denture stomatitis under an orthodontic baseplate is a plaque-associated candidal problem, and it responds to cleaning the appliance and the palate, with antifungal treatment reserved for persistent cases. True acrylic allergy is rare, and a remake in another material does nothing if hygiene is the cause. [Vol I Chapter 9 · Vol II §16.1]

569 **FFFTT**

A False — the patient turns the screw at home, a quarter turn of about 0.25 mm once or twice a week
B False — the expansion obtained is almost entirely dental tipping, the skeletal contribution being negligible
C False — sectioning the acrylic behind one tooth allows a screw to drive that segment alone
D True — the tooth is carried by the acrylic, so the path is defined by the baseplate rather than by a wire that may distort
E True — this is the accepted slow expansion regimen and contrasts with 1 to 4 turns a day in rapid maxillary expansion

[Vol I Chapter 9 · Vol II §16.2]

MNEMONIC Screw rates: **slow expansion = 1/4 turn x1-2/week (0.25 mm); RME = 1-4 turns/day** — confusing them overactivates.

570 **B** Tipping one retroclined incisor labially over an adequate overbite, in an uncrowded arch, is the movement a removable appliance performs well and retains, and it is the classic indication. Space closure after the loss of a molar needs bodily movement of the buccal segment, which no removable appliance can deliver. [Vol I Chapter 9 · Vol II §16.1]

571 **D** The blocks no longer meeting means the mandible is no longer being held forward, and the appliance is reactivated by advancing the bite so that posturing resumes for the remainder of the correction. Abandoning the appliance at a half-corrected stage loses the growth window that the appliance was chosen to exploit. [Vol I Topic 8 · Vol II §16.3]

Scored under 70% of the 81 marks on this block? Re-read Volume II §16, then sit it again in a fortnight.

Block Q — Fixed appliances — labial, lingual and aligners (Q572–612 · 81 marks · Vol II §17)

572 **C** Inclination of the long axis in the labiolingual plane is third order information, delivered by a couple between a rectangular wire and the slot walls, and it is what determines whether the root apex is correctly positioned behind the crown. Second order describes mesiodistal angulation, which the stem states is already correct. [Vol I Topic 6 · Vol II §17.1]

573 **TTTTT**

A True — in-and-out bends compensate for differing crown thicknesses when every bracket base is the same
B True — the slot is cut at an angle to the bracket base, so a straight wire delivers the tip
C True — a round wire rotates freely within the slot and can generate no torquing couple
D True — this is the defining characteristic of standard edgewise and the reason it demands wire-bending skill
E True — prescriptions are averages, and individual variation still requires finishing bends

[Vol I Topic 6 · Vol II §17.1]

MNEMONIC Orders by number: **1st = in-out, 2nd = tip (angulation), 3rd = torque (inclination).**

574 **A** Andrews transferred the first, second and third order information from the wire to the bracket, so that a flat wire of adequate dimension expresses the prescription, which is why the appliance is called straight wire. Slot size, bonding and wire alloy are historical accompaniments and none of them is the defining difference. [Vol I Topic 6 · Vol II §17.1]

575 **E** Andrews examined a collection of non-orthodontic casts with occlusions judged to be excellent and abstracted six keys to normal occlusion from them, and the crown angulation and inclination values in his prescription come from those casts. Averaging treated cases would build the limitations of the treatment of the day into the prescription, which is precisely what he avoided. [Vol I Topic 6 · Vol II §17.2]

MNEMONIC Andrews' base data: **120 naturally ideal casts, six keys to normal occlusion.**

576 **D** Tip built into a canine bracket must be expressed by root movement into the extraction space, which costs anchorage, and MBT reduced canine angulation for exactly this reason. The claim that MBT has more palatal root torque on the lower incisors reverses the actual change, which was towards more labial root torque to resist proclination. [Vol I Topic 6 · Vol II §17.2]

577 **TTTFF**

A True — Roth's values anticipate the small relapse and settling that follow debond
B True — the increase in upper incisor palatal root torque is one of the defining MBT changes
C True — values derived from a sample are averages and individual variation is the rule
D False — Andrews referenced angulation and inclination to the facial axis of the clinical crown
E False — bracket position determines whether the prescription is delivered where it was intended

[Vol I Topic 6 · Vol II §17.2]

578 **B** The larger slot accepts a 0.019 by 0.025 inch steel wire, which is rigid enough to keep arch form and resist the tipping moments of retraction while the teeth slide, and that is the sliding

mechanics argument. The smaller slot is the one that expresses torque more completely at a given wire size, so that consideration favours 0.018 inch rather than 0.022 inch. [Vol I Topic 6 · Vol II §17.3]

579 **E** An undersized rectangular wire must rotate within the slot until its corners engage before any couple develops, and for this wire and slot combination the engagement angle is of the order of ten degrees, so a substantial part of the prescription is lost. Stainless steel has the highest modulus of the routine alloys and transmits torque better than any of them. [Vol I Topic 6 · Vol II §17.3]

580 **B** The reviews find no difference in total treatment duration, pain, arch dimensions or final occlusal quality, but do find that ligation and de-ligation are quicker, which translates into shorter chairside time and sometimes fewer visits. The expansion claim is the most heavily marketed and the least supported; measured arch widths do not differ. [Vol I Chapter 10 · Vol II §17.4]

581 **TFFTT**

A True — the active clip seats the wire and contributes third order control, unlike a passive slide
B False — classical friction falls, but binding and notching remain and dominate once teeth tip
C False — the trials consistently find no clinically or statistically significant difference in duration
D True — the quality of the finished result does not differ between the two systems
E True — this is the one claim that survives systematic review, and it is an advantage to the clinician rather than the patient

[Vol I Chapter 10 · Vol II §17.4]

582 **D** Ceramic brackets bond strongly and are brittle, so they may fracture rather than peel and can take enamel with them, and the appropriate debonding instrument together with informed warning is the accepted management. Sectioning every bracket with a high-speed handpiece introduces a far greater risk of iatrogenic enamel damage than the debond itself. [Vol I Chapter 10 · Vol II §17.4]

583 **A** Ceramic is harder than enamel, so an opposing incisor striking an upper ceramic bracket wears the tooth and may fracture the bracket, and this is the standard contraindication in a deep bite unless the bite is opened first. Friction with ceramic is higher than with steel, not lower, so that objection is the wrong way round. [Vol I Chapter 10 · Vol II §17.4]

584 **C** A difference of under a month with an interval spanning zero in both directions is neither statistically nor clinically meaningful, and it sits alongside a body of trials reaching the same conclusion. Calling the systems identical overstates a single small trial, which can exclude only large differences. [Vol I Chapter 10 · Vol II §17.4]

585 **FFFFF**

A False — the lingual surface is irregular and variable, which is why indirect bonding with laboratory or digital set-ups is the norm
B False — speech disturbance is common, often persists for weeks and is a recognised reason for discontinuation
C False — the force system acts at a different distance from the centre of resistance, so tipping and torque behave differently
D False — a shorter span makes the wire stiffer, which is one of the technical difficulties of lingual mechanics
E False — a deep overbite with heavy contacts is a relative contraindication because the lower incisors strike the upper attachments

[Vol I Chapter 14 · Vol II §17.5]

586 **D** A lingual appliance is fixed, so its effect does not depend on daily wear hours, and it is genuinely invisible from the front, which is what the patient has specified. Clear aligners would meet the aesthetic requirement but depend on precisely the wear compliance she has ruled out. [Vol I Chapter 14 · Vol II §17.5]

587 **B** Tongue irritation and altered speech are the characteristic early problems of lingual appliances and the great majority of patients adapt within weeks with supportive measures. Converting to labial brackets abandons the aesthetic requirement that led to the lingual appliance for a problem that usually resolves. [Vol I Chapter 14 · Vol II §17.5]

588 **E** Each aligner is thermoformed to a set-up a fraction of a millimetre ahead of the current tooth position, so seating it strains the polymer and the recovery of that strain is the force delivered, which decays over the wear period. The shell is not passive; the attachments improve the grip on the tooth but do not themselves generate the force. [Vol I Chapter 14 · Vol II §17.6]

589 **TFTTT**

A True — this is the case type in which aligner outcomes approach those of fixed appliances
B False — extrusion is one of the least predictable aligner movements and requires attachments and even then often fails
C True — there is no undercut for the shell to grip, so the tooth simply slips within it
D True — the shell has no way to generate a true couple comparable with a rectangular wire in a slot
E True — the occlusal coverage acts as a bite block and posterior intrusion is a recognised side effect

[Vol I Chapter 14 · Vol II §17.6]

MNEMONIC Aligners: good for **tipping + IPR**; poor for **extrusion, rotation of round teeth, root torque**.

590 **A** Attachments create the undercut geometry that allows the shell to grip and deliver a couple or an extrusive force, and slower staging keeps each step within what the polymer can deliver. Changing aligners every three days outruns the biological response and produces tracking failure rather than faster movement. [Vol I Chapter 14 · Vol II §17.6]

591 **C** A visible space between the aligner and the tooth means the planned movement has not been achieved and the series is no longer matching the dentition, and the standard response is to check fit and wear, back-track if necessary and take new records for a refinement series. Calling it relapse misplaces the problem in time; nothing has yet been achieved that could relapse. [Vol I Chapter 14 · Vol II §17.6]

592 **B** Peer assessment rating captures gross improvement and aligners perform respectably on it, whereas objective grading examines the details of marginal ridges, buccolingual inclination and occlusal contacts, and it is there that aligner cases fall short. Claiming equality on both indices ignores the consistent deficit in detailed finishing. [Vol I Chapter 14 · Vol II §17.6]

593 **FTFTF**

A False — the reported figure is that about 6 per cent need no refinement, which means the great majority do
B True — this conversion rate is one of the more sobering figures in the aligner literature
C False — large bodily movements across extraction sites are among the least predictable aligner movements
D True — creating space by enamel reduction rather than by arch expansion is a standard part of the plan
E False — refinement reflects the gap between the predicted and the achieved position, which is a biological phenomenon

[Vol I Chapter 14 · Vol II §17.6]

MNEMONIC Aligner numbers: **about 6% need no refinement; over 17% convert to fixed appliances**.

594 **D** The reported proportion completing without any refinement series is of the order of 6 per cent, so the honest consent discussion is that additional aligners are the expectation rather than the exception. The idea that refinement reflects poor compliance alone is wrong; the gap between predicted and achieved movement occurs in well-motivated patients. [Vol I Chapter 14 · Vol II §17.6]

595 **A** Eighteen per cent sits alongside the published conversion rate of more than 17 per cent, so the audit is reassuring rather than alarming and the figure belongs in the consent discussion. Reading it as a local technique failure misattributes a limitation of the appliance to the operator. [Vol I Chapter 14 · Vol II §17.6]

596 **C** A bracket placed too far towards the occlusal surface holds the tooth in a relatively extruded position once the wire is flat, and the marginal ridge discrepancy is the classical sign at the finishing stage. A wire of full dimension that is passive has already expressed what the brackets dictate, so wire rigidity cannot be the cause. [Vol I Chapter 10 · Vol II §17.7]

597 **TFTTF**

A True — the wire seats the bracket centrally, and an off-centre bracket therefore rotates the tooth
B False — indirect bonding transfers the positioning decision to the laboratory or the software, where it can still be wrong
C True — the slot defines the vertical position, which is why height gauges are used
D True — the slot angulation is referenced to the bracket base, so a tilted bracket tilts the tooth
E False — a heavier wire expresses the error more completely rather than correcting it

[Vol I Chapter 10 · Vol II §17.7]

598 **E** A bracket bonded off-centre holds the tooth rotated once the wire is fully seated, and no wire or auxiliary will correct a position that the bracket itself is dictating; the bracket must be moved. A rotation wedge fights the bracket rather than correcting it and the rotation returns as soon as the wedge is removed. [Vol I Chapter 10 · Vol II §17.7]

599 **A** Cutting the torque into the base rather than the slot changes the thickness of the base at different points, so the crown sits further labially or palatally as well as being inclined, which is a first order consequence of a third order design decision. Friction is a function of slot and wire surfaces and ligation and is not altered by where the torque is machined. [Vol I Topic 6 · Vol II §17.3]

600 **TTFFT**

A True — planning on the target occlusion is the principal argument for the digital workflow
B True — the positioning decision is moved out of the mouth, where access and vision are poor
C False — the adhesive still bonds to etched enamel and contamination still causes failure
D False — the set-up is a prediction, and biological response and compliance determine what actually happens
E True — the saving in chairside time is real, at the cost of laboratory or software time

[Vol I Chapter 10 · Vol II §17.7]

601 **C** The tray reproduces the planned bracket positions only if it seats fully and in one position, and a tray that rocks transfers every bracket a fraction out of place across the whole arch at once. Adhesive colour is a convenience for clean-up and has no bearing on positional accuracy. [Vol I Chapter 10 · Vol II §17.7]

602 **E** The six keys were an observational description of naturally excellent occlusions published in 1972, and the straight wire appliance later in the decade was the engineering answer to delivering them. Reversing the order implies the keys were written to justify a product, which is the opposite of what happened. [Vol I Topic 6 · Vol II §17.2]

603 **B** Applying the retraction force on the palatal side places it further from the centre of resistance in the labiolingual plane, so the same force generates a larger tipping moment and additional palatal root torque must be built in to counter it. The lingual interbracket span is shorter, not longer, so wire flexibility is not the explanation. [Vol I Chapter 14 · Vol II §17.5]

604 **TTTFT**

A True — the irregularity of lingual enamel makes accurate direct bonding impractical
B True — the lingual surfaces lie on a shorter arc, which makes the wire stiffer between attachments
C True — the force acts further from the centre of resistance and the prescription compensates for it
D False — access for cleaning is worse, and the patient cannot see the attachments to clean them
E True — access and vision are poorer, and both patient and clinician find these appointments harder

[Vol I Chapter 14 · Vol II §17.5]

MNEMONIC Lingual quirks: **indirect bonding, shorter/stiffer span, extra palatal root torque needed**.

605 **D** Mild to moderate crowding relieved by tipping and enamel reduction in a reliable adult is precisely the case in which aligner outcomes approach those of fixed appliances, and it meets the aesthetic requirement fully. A lingual appliance would also be invisible but commits a patient with a straightforward problem to the most technically demanding and uncomfortable option available. [Vol I Chapter 14 · Vol II §17.6]

606 **C** Worn aligners lose clarity, develop wear facets and fit snugly, and loose unmarked aligners with no movement point to wear rather than to manufacture, so the conversation must precede new records. Taking a refinement scan first simply reproduces the same failure with a new series. [Vol I Chapter 14 · Vol II §17.6]

607 **A** Second order information is referenced to the bracket base, so brackets placed at the wrong angulation on the crowns produce well-aligned crowns with divergent or convergent roots, which is exactly what the radiograph shows. Blaming projection is the easy answer, but marked convergence between two named adjacent teeth on an otherwise satisfactory film is a real finding. [Vol I Chapter 10 · Vol II §17.7]

608 **FTTFF**

A False — a convertible tube has a removable cap that converts the tube itself into a bracket, and it has nothing to do with the second molar erupting
B True — a tube engages the wire at one point only and rotational control is correspondingly weaker
C True — these auxiliary attachments are the reason molar bands are still used in many cases
D False — a wider bracket shortens the free span and makes the wire stiffer between teeth
E False — a band is retained circumferentially and is the stronger option on a heavily restored molar

[Vol I Chapter 10 · Vol II §17.3]

MNEMONIC Know the job: **convertible tube = cap on/off, auxiliary tube = headgear/TPA, wider bracket = stiffer span**.

609 **E** Resistance to sliding in the clinical situation is dominated by the wire binding at the bracket corners once the tooth tips, and that component is unaffected by how the wire is held

in the slot, which is why the measured advantage does not appear as faster treatment. Saying space closure does not involve sliding is simply untrue of the mechanics under discussion. [Vol I Chapter 10 · Vol II §17.4]

610 **D** Objective grading of finished aligner cases consistently shows deficits in the detailing criteria, because the shell cannot deliver the fine vertical and third order corrections that a rectangular wire in a slot delivers. Alignment of crowded incisors is the movement aligners perform best and is not what is lost. [Vol I Chapter 14 · Vol II §17.6]

611 **B** Torque begins to be expressed only once the wire has rotated far enough for its corners to meet the slot walls, and the proportionately better fit in the smaller slot leaves fewer degrees of play to be taken up. Modulus is a property of the alloy and does not change with wire dimension, so that explanation fails at once. [Vol I Topic 6 · Vol II §17.3]

612 **E** Begg's appliance descended from the ribbon arch: a single point of contact in the bracket let the crowns tip freely on light round wires, which is why alignment and space closure were quick, and the roots were then uprighted and torqued in a separate third stage using auxiliary springs. The preadjusted description belongs to Andrews' straight wire appliance, which built tip and torque into the slot precisely so that the tipping and uprighting stages were no longer needed. [Vol I Topic 6 · Vol II §17.2]

MNEMONIC Begg's three stages, in order: **tip, upright, torque**.

Scored under 70% of the 81 marks on this block? Re-read Volume II §17, then sit it again in a fortnight.

Block R — Treatment mechanics: Stage I, II and III (Q613–647 · 71 marks · Vol II §18)

613 **B** Sliding a tooth or retracting a segment requires a rigid wire that the teeth can move along, and a wire that is still expressing alignment is neither flat nor rigid, so retraction on it produces tipping, binding and loss of anchorage. Patient adaptation is a real benefit of the sequence but is not the mechanical reason for it. [Vol I Chapter 10 · Vol II §18.1]

614 **FFFFF**

A False — flattening a curve requires about 1 mm of arch circumference for each millimetre of depth
B False — a round wire expresses no third order information whatever its size
C False — unwanted mesial movement of the buccal segments occurs during alignment, particularly where lacebacks are omitted
D False — the mandibular arch is the template, and the upper arch is built to it
E False — a severely displaced tooth requires a wire of low stiffness and long range, and a heavy steel wire would be permanently deformed

[Vol I Chapter 10 · Vol II §18.1]

MNEMONIC Stage I facts: **1 mm space per 1 mm curve of Spee; round wire = no torque; mandible = the template**.

615 **D** A light flexible wire aligns the crowded labial segment while lacebacks control the crowns of the mesially angulated canines and stop them tipping forwards into the arch as the wire levels. Elastomeric chain on a rigid wire at this stage retracts before the arch is aligned and squanders anchorage on tipping. [Vol I Chapter 10 · Vol II §18.1]

616 **A** A laceback is a ligature wire run from the molar to the canine, holding the canine crown back so that an upright or distally tipped canine is achieved as the wire levels rather than a crown that has tipped mesially. Tying the wire back against the molar tube is the bendback, which is a different auxiliary with a different purpose. [Vol I Chapter 10 · Vol II §18.2]

617 **C** A wire that is not bent back behind the molar tube slides forward as the labial segment unravels, and the increase in wire length anteriorly proclines the incisors, which is exactly the

picture described with the wire ends protruding. Lacebacks control the canine crowns and would not by themselves prevent the wire from sliding forward. [Vol I Chapter 10 · Vol II §18.2]

618 **TTFTT**

A True — the bend traps the wire and stops it feeding forward as the arch aligns
B True — a soft ligature is used so that it can be tightened at each visit
C False — the trials show a small effect on incisor position, and lacebacks act principally on canine crown angulation
D True — this is the classical indication, and it conserves the space that the extraction created
E True — the wire shortens relative to the arch as crowding resolves, so the bendback needs renewing

[Vol I Chapter 10 · Vol II §18.2]

619 **D** Crowding and levelling both draw on the same arch circumference and are added, giving 3 mm plus 2 mm. Counting the curve once for each side gives 7 mm and measures a single curve of Spee twice, which is the commonest arithmetic error in this calculation. [Vol I Chapter 13 · Vol II §18.1]

MNEMONIC Space sum: **crowding + curve of Spee depth, counted once** — doubling the curve is the classic error.

620 **B** The mandibular arch is the template because intercanine width and lower arch form are the least stable dimensions to alter, so the lower arch is respected and the upper is built to occlude with it. Choosing the upper arch first inverts this and commits the lower arch to a width the soft tissues will not maintain. [Vol I Chapter 13 · Vol II §18.1]

621 **E** In a non-growing patient with an average lower face height, posterior extrusion is neither available nor desirable, so the overbite must be reduced by genuine intrusion of the incisors using a segmental or auxiliary mechanism that puts a light force through the incisor segment. A bite plane depends on posterior eruption, which requires the alveolar growth she no longer has. [Vol I Chapter 13 · Vol II §18.3]

622 **TFFTF**

A True — the reverse curve acts through the brackets as a couple that flares the labial segment, which is why bendbacks matter
B False — separating the jaws by posterior eruption rotates the mandible down and back and increases the lower anterior face height
C False — posterior extrusion depends on alveolar growth and relapses in the adult, in whom true intrusion is preferred
D True — the intrusive force passes labial to the centre of resistance, and this is why the labial segment flares unless it is restrained
E False — levelling consumes arch circumference rather than releasing it

[Vol I Chapter 13 · Vol II §18.3]

623 **A** Space closure requires a rigid full-dimension wire to preserve arch form and torque and a force system that decays little between visits, and where anchorage is critical the posterior unit must be reinforced as well. Elastomeric chain on a flexible aligning wire delivers a decaying force along a wire that cannot resist the tipping moments generated, which is the classical way of losing anchorage. [Vol I Chapter 10 · Vol II §18.4]

624 **C** A closing loop moves the teeth by activating the loop rather than by dragging brackets along a wire, which removes friction and binding but places the whole burden on the design of the loop and the differential moments at its two ends. Loops do not deliver a constant force; the force falls as the loop deactivates, which is why the activation interval matters. [Vol I Chapter 10 · Vol II §18.4]

625 **B** Bodily movement of a segment requires a force in the region of 50 to 100 cN at the tooth, and a coil spring rated at about 150 g per side delivers this once the moments needed to prevent tipping are taken into account, giving movement of the order of 1 mm a month. Twenty-five centinewtons is the figure for tipping a single tooth with a removable appliance and is far too light for bodily retraction of a segment. [Vol I Chapter 10 · Vol II §18.4]

MNEMONIC Sliding force numbers: **50-100 cN bodily, 150 g coil spring, about 1 mm/month** — not the 25-40 cN tipping force.

626 **TTTTT**

A True — this is the figure reported for unsupported anchorage and it is why reinforcement is planned from the outset
B True — spreading the reaction across more roots reduces the movement of each
C True — two-stage retraction moves fewer teeth at a time, which is the classical anchorage-conserving approach
D True — one uses the palate as a connector, one as a bearing surface and one as skeletal anchorage
E True — the extraction choice and the anchorage plan are decided together, since each constrains the other

[Vol I Chapter 10 · Vol II §18.4]

MNEMONIC Anchorage: **about two-thirds lost if unsupported; TPA (connector), Nance (bearing), miniscrew (skeletal)**.

627 **D** Class II elastics pull the lower arch forwards and the upper arch back while extruding the teeth at each end of the elastic, which proclines the lower incisors, extrudes the lower molars and upper incisors and rotates the occlusal plane, and the buccal open bite follows from that rotation. Excessive maxillary restraint is a headgear effect and does not explain the lower incisor proclination. [Vol I Chapter 13 · Vol II §18.5]

628 **A** Class III elastics run from the lower anterior teeth to the upper posterior teeth, so they procline the upper labial segment, retrocline the lower labial segment and extrude the upper molars, which opens the bite anteriorly in a high-angle patient. Retroclination of the upper labial segment with proclination of the lower is the Class II elastic effect, which is the reverse of what these elastics produce. [Vol I Chapter 13 · Vol II §18.5]

MNEMONIC Elastics tilt: **Class II procline lower / retrocline upper; Class III procline upper / retrocline lower**.

629 **C** Unsupported anchorage allows the posterior teeth to come forward for about two-thirds of the space, which in this case would leave insufficient room to relieve the crowding and would change the molar relationship. Expecting the space to close entirely by retraction assumes an anchorage value that the posterior segment does not possess. [Vol I Chapter 10 · Vol II §18.4]

630 **TFTTT**

A True — each end of the elastic carries an extrusive component, which is why the occlusal plane rotates
B False — the extrusion and the backward rotation of the mandible make this the worst pattern for Class II elastics
C True — light vertical elastics on flexible finishing wires are the standard settling mechanic
D True — most of the force is lost within the first day, which is why daily replacement is instructed
E True — asymmetric elastic wear is a recognised, if anchorage-expensive, way of shifting a midline

[Vol I Chapter 13 · Vol II §18.5]

631 **E** Differential moments at the two ends of a closing loop determine which segment translates and which tips, and a larger moment posteriorly holds the molars upright while the anterior teeth move, which is how a loop conserves anchorage. Making the loop small reduces its range and forces frequent heavy reactivation, which is the opposite of what anchorage conservation requires. [Vol I Chapter 10 · Vol II §18.4]

632 **A** Tipping of the teeth on either side of a closing extraction space produces an exaggerated curve in one arch and a reverse curve in the other, with lateral open bites, and the correction is to re-level on a rigid wire before resuming closure. Increasing the closing force would deepen the tipping and make the picture worse rather than better. [Vol I Chapter 13 · Vol II §18.6]

633 **C** The open bite is being produced by extrusion of the posterior teeth and backward rotation of the mandible, which is the predictable consequence of Class II elastics in a high-angle patient, so the mechanic must be changed rather than continued. Adding anterior vertical elastics masks the sign while the mechanism that is causing it continues to operate. [Vol I Chapter 13 · Vol II §18.5]

634 **TFFFF**

A True — a notched or deformed wire arrests sliding completely and is a common and easily missed cause
B False — the chain loses most of its force within a day and by eight weeks is delivering very little
C False — a fibrous ridge of tissue in a long-standing extraction site is a recognised mechanical obstruction
D False — an old site may be markedly sclerosed, which slows movement through it
E False — an immobile tooth warrants a radiograph to exclude ankylosis or root contact with an adjacent tooth

[Vol I Chapter 13 · Vol II §18.6]

MNEMONIC Stalled closure, check: **notched wire, decayed chain, fibrous tissue, sclerosed bone, ankylosis**.

635 **E** The space and the buccal segment relationship both lie on the right, so retracting the upper right anterior teeth into the residual space moves the centreline across and corrects the molar relationship at the same time. Protracting the right buccal segment would close the space but would drive the centreline further to the right and worsen the Class II relationship on that side. [Vol I Chapter 13 · Vol II §18.6]

636 **B** A 0.019 by 0.025 inch wire leaves about ten degrees of play in a 0.022 inch slot, so the prescribed torque is incompletely expressed and the remedy is either a torquing bend in the wire or a wire that fills the slot more completely. Changing to a round wire removes the only means by which torque can be delivered at all. [Vol I Chapter 13 · Vol II §18.7]

637 **D** Converging roots over a closed extraction site indicate that the crowns have been closed together while the roots have tipped towards one another, and this is corrected by uprighting bends or bracket repositioning before debond because it is a recognised cause of relapse and space reopening. Dismissing it as projection is the easy option, but convergence over a named extraction site is the classical real finding on this film. [Vol I Chapter 13 · Vol II §18.7]

638 **TTFTF**

A True — rigid full-dimension wires prevent the vertical settling that these elastics are intended to produce
B True — this is the standard check before debond and it is the film on which convergence is detected
C False — torsional play must be taken up first, and additional torque can be bent into the wire thereafter

D True — beta titanium is formable and holds its bends while delivering moderate forces
E False — contacts that have been closed are the desired outcome, and it is open or deficient contacts that require attention

[Vol I Chapter 13 · Vol II §18.7]

639 C The objective grading system scores eight named criteria on the finished models and radiographs and is designed to examine the detail of the finish rather than the gross improvement. The aesthetic and dental health components belong to the index of orthodontic treatment need, which measures need before treatment rather than quality afterwards. [Vol I Chapter 13 · Vol II §18.7]

MNEMONIC **Align, ridges, BL tip, occlusal rel'n, occlusal contacts, overjet, IP contacts, root angle** — ABO's 8 criteria.

640 A Greatly improved requires a reduction of at least 22 points or at least 30 per cent, and both figures are comfortably exceeded. The peer assessment rating measures gross occlusal improvement and says nothing about the quality of the detailing, which is what the objective grading system examines. [Vol I Chapter 13 · Vol II §18.7]

641 TTTFF

A True — these are the two thresholds and either satisfies the definition
B True — marginal ridges, buccolingual inclination and root angulation are graded in detail
C True — gross improvement and fine detailing are different things, which is why both indices exist
D False — root angulation is an objective grading criterion and requires a radiograph, which the peer assessment rating does not use
E False — need is measured by the index of orthodontic treatment need, and these two indices measure outcome

[Vol I Chapter 13 · Vol II §18.7]

642 E A smile arc that does not follow the curvature of the lower lip and wide dark spaces laterally are two of the aesthetic features assessed at the finish, and both are recorded because both can be influenced by bracket positioning and arch width. Incisor display at rest is a separate measurement taken with the lips relaxed and is not what the stem describes. [Vol I Chapter 13 · Vol II §18.8]

643 D A vertical discrepancy between adjacent marginal ridges is a second order problem, corrected either by a step bend in a wire that will hold it or, more reliably, by moving the bracket that put the tooth in the wrong vertical position. A first order offset moves the tooth labially or palatally and leaves the vertical discrepancy untouched. [Vol I Chapter 13 · Vol II §18.7]

644 B Treatment duration and demineralisation are the two risks that rise together, and once active white spot lesions appear the balance shifts decisively towards finishing, since the residual imperfections are small and some will settle. Continuing for another six months buys a marginal improvement in detail at the cost of enamel that will not recover. [Vol I Chapter 13 · Vol II §18.8]

645 FTFTF

A False — overjet reduction on an unlevelled arch produces tipping and binding, which is why levelling comes first
B True — the first stage is defined by the need for range and low stiffness rather than rigidity
C False — levelling precedes closure because closure needs a rigid, flat wire to slide along
D True — finishing bends require formability, and settling requires an occlusion that is nearly correct
E False — combining the stages produces uncontrolled tipping, anchorage loss and a poorer finish

[Vol I Chapter 10 · Vol II §18.1]

646 **E** Elastics applied to an aligning wire that has no rigidity produce proclination of the lower labial segment, extrusion and tipping rather than the controlled correction intended, and the space needed for the crowding is consumed in the process. Saying elastics cannot be worn with nickel titanium wires is a rule of thumb rather than an explanation, and it is the wire's stiffness, not its alloy, that matters. [Vol I Chapter 10 · Vol II §18.1]

647 **B** Rigid full-dimension wires hold the teeth in the position the brackets dictate and prevent the small vertical movements needed for intercuspation, so the wires are softened or sectioned and light vertical elastics are used to settle the occlusion before debond. Debonding and leaving it to a retainer abandons the one stage at which the intercuspation can still be controlled. [Vol I Chapter 13 · Vol II §18.7]

Scored under 70% of the 71 marks on this block? Re-read Volume II §18, then sit it again in a fortnight.

Block S — Retention, relapse and stability (Q648–682 · 71 marks · Vol II §19)

648 **D** The supracrestal and transseptal fibre networks are the slowest periodontal component to remodel and are the principal driver of rotational relapse, so dividing them at debond addresses the cause. Prolonging the appliance is the tempting answer, but holding a tooth for longer does not shorten the reorganisation time of those fibres. [Vol I Topic 20 · Vol II §19.2]

649 **TTTFT**

- A True — the principal fibres of the ligament reorganise over a few months and are not the limiting tissue
- B True — the supracrestal elastic and transseptal fibres remodel very slowly and drive late rotational change
- C True — the rate of relapse is highest immediately after debond and falls thereafter
- D False — fiberotomy reduces rotational relapse but does not address soft tissue pressures or growth
- E True — bone remodels faster than the gingival fibre network, which is why the fibres are the problem

[Vol I Topic 20 · Vol II §19.2]

650 **A** Houston in 1989 showed that reduced overbite is retained by the occlusal stop of the lower incisor edge against the upper cingulum plateau, and that without such a stop the bite redeepens whatever retainer is worn. Intruding the upper incisors is a legitimate way of reducing the bite but says nothing about whether the finished relationship has a stop. [Vol I Topic 20 · Vol II §19.3]

651 **C** Full occlusal coverage of every crown is what resists reopening of closed extraction sites, which is the commonest early change after space closure. The Hawley is tempting because it allows the buccal occlusion to settle, but it grips only a few teeth and leaves the closed sites free to reopen. [Vol I Topic 20 · Vol II §19.4]

652 **D** Wire syndrome is the expression of a residual activation in a fully attached retainer wire, which torques or displaces a single tooth while every pad remains bonded, and the isolated root-labial crown-lingual pattern is characteristic. A partial debond is the other cause of single-tooth movement, but it produces passive drift and the loose pad can usually be detected. [Vol I Topic 20 · Vol II §19.4]

653 **FFTTF**

- A False — bonded retention is indefinite and the commitment to review and repair must be consented at the outset

B False — a canine-only wire leaves the incisors unsupported and they can rotate between the pads
C True — silent failure is the characteristic problem, which is why scheduled review is part of the regimen
D True — the wire is a plaque trap and localised calculus with gingivitis is the usual hygiene consequence
E False — wires debond, fracture and develop unintended activations, so no such guarantee exists

[Vol I Topic 20 · Vol II §19.4]

654 **B** Poor long-term wear is a behavioural finding, so the useful response is to set the expectation of an indefinite commitment before treatment starts and to design the regimen around it. Bonding everyone is the tempting answer, but bonded wires fail silently and bring their own hygiene and torque problems. [Vol I Topic 20 · Vol II §19.5]

655 **E** Late lower incisor crowding occurs in dentitions that have never been treated and continues into the third and fourth decades, so it is better understood as an ageing change than as relapse. Blaming the third molars is the common explanation offered to patients, but their removal does not reliably prevent the change. [Vol I Chapter 13 · Vol II §19.6]

656 **TTTTT**

A True — untreated samples show the same progressive lower labial crowding with age
B True — prophylactic removal of third molars has not been shown to prevent the change
C True — the increase in irregularity continues long after growth has ceased
D True — nothing arrests the change except holding the teeth where they are
E True — it is a physiological age change and is described as such in Chapter 5, page 284

[Vol I Chapter 13 · Vol II §19.6]

MNEMONIC **Ageing, not relapse** — happens untreated too, third molars do not prevent it, worsens into the 30s-40s, and only retention holds it.

657 **A** The opened midpalatal suture must fill with bone before the appliance is removed, which takes three to six months, and premature removal is followed by the greater part of the roughly ten per cent relapse reported at four to eight months. Relying on the corrected interdigitation alone is tempting because the occlusion looks settled, but it does not protect an unossified suture. [Vol I Topic 20 · Vol II §19.3]

MNEMONIC **1-4, 2-4, 3-6**: turn the screw 1 to 4 times a day for 2 to 4 weeks, then keep the appliance passive 3 to 6 months so the suture ossifies.

658 **C** Closed median diastemas are among the least stable results and the risk is higher still with a fibrous frenal attachment, so a fixed retainer maintained indefinitely is the realistic plan and should be consented before treatment begins. A night-time vacuum-formed retainer for a year is the usual reflex, but stopping it at twelve months is when the space reopens. [Vol I Chapter 13 · Vol II §19.4]

659 **B** The index sums the five anterior contact point displacements measured parallel to the occlusal plane, which produces a single number that can be compared between records and between studies. The claim that it also captures rotation and inclination is the usual misunderstanding, since only linear displacement of the contact points is recorded. [Vol I Topic 20 · Vol II §19.6]

660 **FTFTT**

A False — the displacements are measured parallel to the occlusal plane, not perpendicular to it
B True — zero means every anatomical contact point is in contact with its neighbour

C False — only linear displacement of contact points is recorded, which is a known limitation
D True — the measurement is made directly from models or digital scans with calipers
E True — it is the standard outcome measure in post-retention samples and allows comparison

[Vol I Topic 20 · Vol II §19.6]

661 **D** The mandible grows later and for longer than the maxilla, and a boy at CS3 has his peak still to come, so the corrected incisor relationship is at risk until growth has finished and monitoring must continue into the late teens. A positive overjet at the end of treatment is reassuring but does not confer stability while the skeletal pattern is still changing. [Vol I Topic 20 · Vol II §19.3]

662 **A** A commitment to lifelong retention, to its maintenance costs and to the risk of silent failure is material to the decision to be treated at all, so it belongs in the consent taken before the appliances are placed. Raising it at debond is the usual practice, but by then the patient has no opportunity to decline the treatment that created the need. [Vol I Topic 20 · Vol II §19.5]

663 **C** A Hawley leaves the posterior occlusal surfaces uncovered, so the buccal segments can settle into intercuspation while the bow and baseplate hold the anterior teeth. Full occlusal coverage is the reflex choice after debond, but it is precisely what prevents the settling that is wanted here. [Vol I Topic 20 · Vol II §19.4]

664 **FFFFF**

A False — it is the Hawley that leaves the posterior occlusal surfaces free and permits settling
B False — full occlusal coverage resists reopening of closed sites better than a labial bow does
C False — thin thermoplastic material wears, cracks and is lost more readily than acrylic and wire
D False — a rotated tooth needs good retention, but nothing about a Hawley contraindicates it
E False — trials comparing regimens have not shown full-time wear to be superior at two years

[Vol I Topic 20 · Vol II §19.4]

> **MNEMONIC** **Settle vs seal**: the Hawley bow frees the buccal segments to settle; the vacuum-formed retainer seals every crown but wears out sooner.

665 **E** Calculus and gingivitis around a bonded wire respond to professional debridement with instruction in superfloss or interdental brush use, and there is no attachment loss here to justify losing a retainer that is doing its job. Removing it outright is the tempting response to a plaque trap, but it exchanges a reversible gingival problem for irreversible crowding. [Vol I Topic 20 · Vol II §19.4]

666 **A** A tooth held outside the equilibrium between tongue and lip pressures relapses when the appliance is removed, and 9 degrees of lower incisor proclination against a lip that rests on the upper incisors is a classic unstable position. The absence of extractions is reassuring to patients but has no bearing on whether the finished incisor position is within the soft tissue envelope. [Vol I Topic 20 · Vol II §19.2]

667 **C** The corrected relationship depends on the mandible being held forward while the occlusion is detailed and while the remaining growth expresses itself, so night-time wear of the functional appliance or a comparable device bridges the two phases. Class II elastics will hold the overjet but extrude the molars and open the bite if used continuously, which is a poor substitute for retention. [Vol I Topic 20 · Vol II §19.3]

668 **TFTTT**

A True — full-time wear is usually confined to the first weeks or months and then reduced

B False — randomised comparisons of wear regimens have not demonstrated such a difference
C True — indefinite commitment is the honest message and belongs in the pre-treatment discussion
D True — self-reported and objectively monitored wear both fall markedly over that period
E True — bonded wires fail without symptoms, so scheduled review is part of the regimen

[Vol I Topic 20 · Vol II §19.5]

MNEMONIC **FIRM**: Full-time wear only at first, Indefinite commitment consented up front, Reported wear falls by 4-5 years, Monitor bonded wires.

669 **E** Expansion is retained by the corrected teeth dropping into a positive interdigitation, and a cusp-to-cusp finish is an unstable endpoint that tends to return towards crossbite. The absence of a displacement is reassuring for the joints but is not what holds an expanded arch, so it does not protect this result. [Vol I Topic 20 · Vol II §19.3]

670 **B** A rise of this size in two thirds of cases is the consistent post-retention finding and it is the reason that indefinite retention has replaced fixed courses of retention. Calling 3.1 mm acceptable ignores that this is the threshold at which most patients and clinicians judge the labial segment to be visibly crowded again. [Vol I Topic 20 · Vol II §19.6]

MNEMONIC **0.5 to 3.1 mm, one in three**: the ten-year post-retention picture — Irregularity Index rises from 0.5 to 3.1 mm and only a third stay aligned.

671 **D** Open bites reopen where the resting soft tissue pattern that produced them persists, because the teeth are being held against an unchanged equilibrium, and a tongue-to-lip resting relationship is the specific finding here. The supracrestal fibres drive rotational relapse rather than vertical relapse, so they are not the mechanism in an open bite. [Vol I Topic 20 · Vol II §19.3]

672 **TFFTT**

A True — late mandibular growth is a recognised contributor to post-treatment lower crowding
B False — the equilibrium of tongue, lip and cheek pressures determines where a tooth will stay
C False — bone remodels relatively quickly and it is the gingival fibre network that recoils
D True — the pre-treatment lower incisor position is the most reliable guide to a stable position
E True — an obtuse interincisal angle without an occlusal stop allows the overbite to return

[Vol I Topic 20 · Vol II §19.2]

MNEMONIC **GRIN**: Growth (late mandibular), Rotation via fibre recoil, Incisors proclined beyond original position, Non-ideal interincisal angle.

673 **C** A retainer that no longer seats has been outgrown by a small relapse, and the safe response is a new retainer made to the position the teeth now occupy so that further change is arrested. Forcing an ill-fitting retainer is the common instruction, but it delivers uncontrolled force to the tooth and risks unwanted movement or fracture. [Vol I Topic 20 · Vol II §19.5]

674 **A** Similar and substantial failure rates in both arms mean the practical message is that any retention regimen needs monitoring rather than that one device is superior. Reading a five percentage point difference as a reason to abandon bonded retainers over-interprets a small difference in a composite outcome. [Vol I Topic 20 · Vol II §19.5]

675 **TTFTF**

A True — the corrected teeth must drop into intercuspation for the transverse change to hold
B True — the closed contact between the central incisors is among the least stable results
C False — rotations are among the least stable movements because of the supracrestal fibres

D True — the original intercanine width is the most reliable guide to a stable arch form
E False — a lower lip that traps behind the upper incisors continues to procline them after debond

[Vol I Topic 20 · Vol II §19.3]

MNEMONIC **RIDE**: Rotations, Intercanine width increase, Diastema closure, Extrusion or proclination against the lips — the classically unstable moves.

676 E Broad flat contact points resist rotational relapse mechanically, so reshaping small rounded contacts is a recognised adjunct where crown form is the reason the rotation will not hold. Overcorrection has a place but 30 degrees of additional rotation is excessive and simply moves the problem to the other side. [Vol I Chapter 13 · Vol II §19.2]

677 D Reduced but healthy attachment gives no resistance to the forces that caused the migration, so permanent splinting with a bonded retainer, combined with continuing periodontal maintenance, is the only realistic way to hold the result. Treating the periodontal disease removes the cause of the bone loss but not the occlusal and soft tissue forces that displaced the teeth. [Vol I Chapter 13 · Vol II §19.4]

678 B Randomised comparisons of full-time and night-only vacuum-formed wear have not demonstrated a clinically important difference in alignment, so a night-only regimen that the patient will actually follow is defensible. The claim that no such comparison exists is the tempting escape, but several trials have addressed exactly this question. [Vol I Topic 20 · Vol II §19.5]

679 FFFFT

A False — reported wear falls steadily and is low by four to five years after debond
B False — silent failure makes scheduled review essential for the life of the retainer
C False — the commitment is material to the decision to be treated and belongs in the initial consent
D False — cost is a recognised barrier to replacing lost or broken retainers and to continued wear
E True — understanding the reason for the regimen is consistently associated with better adherence

[Vol I Topic 20 · Vol II §19.5]

680 E Late lower incisor crowding occurs whether or not third molars are present, so their removal cannot be recommended as a means of preventing it and the decision must rest on pathology or its risk. The mesial force argument is intuitive and widely repeated, but the observational evidence does not support it. [Vol I Chapter 13 · Vol II §19.6]

681 B Bonding a passive wire back onto a tooth that has already moved retains the relapse, so the sequence is to correct the rotation first and only then place a new passive retainer made to the corrected position. Rebonding the existing wire is quick and tempting, but it cements the very change the retainer was placed to prevent. [Vol I Topic 20 · Vol II §19.4]

682 D Alignment is maintained only while the teeth are held, and the physiological changes that crowd a lower labial segment continue for decades, so the honest answer is indefinite wear with a cautious reduction in frequency. Stopping and watching for crowding is superficially reasonable, but by the time a patient notices a change it has already occurred and needs retreatment. [Vol I Topic 20 · Vol II §19.5]

Scored under 70% of the 71 marks on this block? Re-read Volume II §19, then sit it again in a fortnight.

Block T — Interceptive orthodontics and the developing dentition (Q683–727 · 89 marks · Vol II §20)

683 **A** The permanent canines should be palpable in the buccal sulcus from about nine to ten years, and failure to palpate them together with mobile proclined lateral incisors is the classic sign of a palatally displaced canine that warrants imaging. Reviewing in two years is the commonest error, because the interceptive window closes and root resorption of the incisors may proceed unseen. [Vol I Topic 14 · Vol II §20.4]

684 **C** An object that appears to move in the same direction as the tube shift lies further from the film, which for an intraoral film means palatally, and this is the basis of the same lingual opposite buccal rule. The reasoning that movement with the tube indicates a buccal position reverses the rule and is the commonest error made with parallax. [Vol I Topic 14 · Vol II §20.4]

MNEMONIC **SLOB**: Same Lingual, Opposite Buccal — the crown moves with the tube shift only when the canine lies palatal to the incisor roots.

685 **FTTFT**

A False — the accepted figure is about 3 per cent, not 10 per cent
B True — the large majority of displacements are palatal rather than buccal
C True — routine palpation from nine to ten years is the standard screening manoeuvre
D False — palatal displacement is rather more common in girls
E True — a missing or diminutive lateral incisor is a recognised association with palatal displacement

[Vol I Topic 14 · Vol II §20.4]

686 **D** Removing the deciduous canine at ten to eleven years in an uncrowded arch allows a substantial proportion of displaced canines to improve their position, and a review radiograph at twelve months determines whether the interceptive measure has worked. Proceeding straight to exposure and traction is the tempting response, but it commits the child to surgery that may prove unnecessary. [Vol I Topic 14 · Vol II §20.4]

MNEMONIC **10-11-12**: extract the deciduous canine at 10 to 11 years and take the review radiograph at 12 months to judge the canine's progress.

687 **B** The three prognostic features are vertical height, horizontal overlap with the incisor roots and angulation to the midsagittal plane, and an angle beyond about 30 degrees is the strongest single predictor that the canine will not right itself. A high crown position is also unfavourable, but a crown as low as the cervical third of the lateral root is a favourable rather than an adverse finding. [Vol I Topic 14 · Vol II §20.4]

688 **E** Interceptive extraction works by allowing the canine to drift into available space, so crowding and a severely angulated crown both remove the conditions under which it succeeds, and space maintenance is often needed alongside it. The claim that no controlled evidence exists is the tempting escape, but controlled studies of this intervention have been published. [Vol I Topic 14 · Vol II §20.4]

689 **TTTTT**

A True — exposure and bonded gold chain traction is the standard approach for an alignable canine
B True — traction into a crowded arch produces impaction against the neighbouring teeth
C True — accepting the first premolar in the canine position is a legitimate plan in a grossly displaced case
D True — transplantation into a prepared socket is an established alternative with a reasonable survival
E True — a severely resorbed lateral incisor may be the tooth sacrificed rather than the canine

[Vol I Topic 14 · Vol II §20.4]

MNEMONIC **Traction, extraction, transplant**: align it, accept its loss, or move it whole — incisor root resorption may decide which tooth goes.

690 **A** Resorption progresses while the canine remains in contact with the incisor root, so the priority is to move the canine away from it, and exposure with traction achieves this while preserving both teeth. Extracting the lateral incisor is sometimes necessary, but with two thirds of the root remaining the tooth is usually worth saving and removal is premature. [Vol I Topic 14 · Vol II §20.4]

691 **C** A canine lying high and near the midline will not erupt unaided within a reasonable time, so an attachment bonded at operation allows controlled traction along a planned path from the moment healing is complete. Open palatal exposure is reasonable for a canine lying low and close to the arch, but at this height spontaneous eruption is unlikely and the tooth will need traction anyway. [Vol I Topic 14 · Vol II §20.5]

692 **B** Transplantation places the tooth in one operation and avoids prolonged traction, and it is the recognised option where the displacement is severe and the patient will not accept a long course of appliances. Prolonged traction is the alternative, but a horizontal canine above the incisor apices is exactly the position in which the time and failure risk make the patient's reluctance decisive. [Vol I Topic 14 · Vol II §20.5]

693 **FFTFF**

A False — the mesiodens is the commonest supernumerary of the permanent dentition
B False — the great majority arise in the anterior maxilla
C True — conical forms often erupt while tuberculate forms typically obstruct the incisor
D False — supernumeraries are about twice as common in boys
E False — many incisors erupt spontaneously once the obstruction and adequate space are provided

[Vol I Topic 13 · Vol II §20.2]

MNEMONIC **Conical erupts, tuberculate blocks**: it is usually a mesiodens, about twice as common in boys, and the tuberculate form obstructs eruption.

694 **D** A vertical incisor with incomplete root formation and adequate space will usually erupt once the obstruction is removed, so removal with space maintenance and a defined review period is the least invasive effective plan. Exposing and bonding at the same operation is tempting for efficiency, but it commits a favourably placed tooth to traction it probably does not need. [Vol I Topic 13 · Vol II §20.2]

695 **A** About two thirds of ectopic first molars disimpact spontaneously, and a separator or brass wire is the simplest measure that encourages this while causing minimal disturbance to the arch. Extracting the deciduous molar is the tempting shortcut, but it allows the first molar to drift mesially and converts a self-limiting problem into a space loss that must then be regained. [Vol I Topic 13 · Vol II §20.3]

696 **C** Once the impaction is irreversible and the deciduous tooth is failing, its removal is unavoidable and the clinical task becomes managing the mesial drift that follows, whether by a space maintainer or by planning extractions in the permanent dentition. Persisting with separation beyond six months in a tooth this firmly locked simply delays a decision that has already been made. [Vol I Topic 13 · Vol II §20.3]

697 **TTTTF**

A True — the condition is uncommon overall and affects the upper arch far more often
B True — the majority release themselves without treatment, which justifies a period of observation
C True — the impacted marginal ridge resorbs the distal root of the deciduous molar
D True — separation is the simplest intervention and is appropriate before the impaction is fixed

E False — the first molar drifts mesially and the space must be held or planned for

[Vol I Topic 13 · Vol II §20.3]

MNEMONIC **SAME**: Separator first, About two-thirds self-correct, Mesial drift needs space management if the E is lost, Erosion of its root is the tell.

698 **E** Early loss of a second deciduous molar allows the first permanent molar to drift mesially and lose the leeway space needed by the premolars, and a band and loop on the erupted first molar holds that space simply. Leaving the space is the tempting course in an uncrowded arch, but the loss occurs regardless of crowding and is difficult to regain later. [Vol I Topic 13 · Vol II §20.1]

699 **A** A balancing extraction on the opposite side of the same arch is the recognised means of preventing a centreline shift when a canine or first deciduous molar is lost early from a crowded arch. A compensating extraction in the opposing arch addresses the occlusal relationship rather than the centreline, which is the confusion this question tests. [Vol I Topic 13 · Vol II §20.1]

700 **C** The deciduous molars are wider than their premolar successors by roughly 1.5 mm per quadrant in the upper arch and 2.5 mm in the lower, which is the space a lingual arch is placed to preserve. Reversing the two arches is the standard error, and the liability figures of about 7 mm upper and 5 mm lower belong to the incisors rather than to the leeway space. [Vol I Topic 13 · Vol II §20.1]

701 **TTTFT**

A True — the band and loop is the standard unilateral maintainer against a single erupted abutment
B True — holding the molars back preserves the leeway space for the permanent canines and premolars
C True — deciduous incisor space is rarely lost and replacement is usually for appearance only
D False — maintainers debond, break and are outgrown, so review at each recall is required
E True — unilateral canine loss into a crowded arch allows the labial segment to shift to that side

[Vol I Topic 13 · Vol II §20.1]

702 **E** Severe crowding in an otherwise normal Class I dentition, with spontaneous early loss of the deciduous canines already signalling the space deficit, is the classic indication for serial extraction and it allows the incisors to align as the premolars are guided out. Waiting for the permanent dentition is defensible but forfeits the spontaneous improvement that serial extraction buys in a case selected this carefully. [Vol I Topic 15 · Vol II §20.1]

703 **B** The canines are removed at about nine years so that the incisors align, the first deciduous molars at about ten years to hasten eruption of the first premolars, and the first premolars themselves as they appear so that the permanent canines have a path. Extracting the first premolars first is impossible because they have not yet erupted, which is the sequencing point being tested. [Vol I Topic 15 · Vol II §20.1]

MNEMONIC **CD4**: (deciduous) Canines, then (first) Deciduous molars, then premolar-4s — each stage inviting the next tooth forward in turn.

704 **D** Serial extraction is a Class I procedure with average or reduced overbite, and applying it to a deep bite Class II division 2 case with mild crowding deepens the bite and wastes space that was not needed. The claim about treatment time is the opposite of the truth, since serial extraction lengthens the period under supervision rather than shortening it. [Vol I Topic 15 · Vol II §20.1]

705 **FFFFF**

A False — the procedure is confined to Class I cases with an average or reduced overbite
B False — it is reserved for severe crowding, since mild crowding does not justify four extractions
C False — the child is kept under supervision for several years, which lengthens the period
D False — most cases still need appliances to detail the occlusion and close residual spaces
E False — a reduced lower face height with a deep bite is a contraindication rather than an indication

[Vol I Topic 15 · Vol II §20.1]

MNEMONIC **Class I, severe crowding, average overbite**: serial extraction's three gatekeepers — a deep bite or mild crowding rules it out.

706 C The management of an impacted second molar is an escalating ladder that begins with the simplest measures, and a separator can release a mild mesioangular impaction without appliances or surgery. Extracting the second molar is the last step of that ladder and is unjustified at 25 degrees of angulation when simpler measures have not been tried. [Vol I Topic 13 · Vol II §20.3]

707 A When simpler measures have failed and a well formed vertical third molar is available to take its place, removal of the second molar is the recognised endpoint of the ladder and protects the distal surface of the first molar. Removing the third molar first is the intuitive move, but it sacrifices the tooth that provides the solution and does not itself upright the second molar. [Vol I Topic 13 · Vol II §20.3]

MNEMONIC **Separate, upright, extract**: the escalating ladder for an impacted lower second molar, ending in extraction once a sound vertical eight can replace it.

708 E Impaction of the lower second molar occurs in a little over 2 per cent of people, which is uncommon but frequent enough that every practice will see cases and should recognise the escalating management ladder. Calling it rare at fewer than one in a thousand understates the prevalence by an order of magnitude and leads to late diagnosis. [Vol I Topic 13 · Vol II §20.3]

709 TTFFT

A True — the accepted figure is just over 2 per cent of individuals
B True — lower second molar impaction is considerably more common than upper
C False — management escalates from observation and separation through uprighting to extraction
D False — although plaque stagnates at the contact, the recognised risk is to the second molar and the first molar distal surface is usually accessible
E True — where uprighting fails and the third molar is well formed, removal of the seven is accepted

[Vol I Topic 13 · Vol II §20.3]

710 D A high percussion note with progressive infra-occlusion and no response to force is replacement resorption with ankylosis, and because the ankylosed tooth cannot follow the vertical growth of the alveolus the discrepancy increases through adolescence. Attributing the failure to inadequate force is the common error, since no orthodontic force will move a tooth fused to bone. [Vol I Topic 13 · Vol II §20.6]

711 B A missing upper central incisor is difficult to substitute convincingly because of its size and its dominance in the smile, so space is usually opened and held with a temporary pontic until growth is complete and an implant can be placed. Closing the space and substituting is attractive because it avoids a prosthesis, but a lateral incisor makes a poor central and the midline is hard to control. [Vol I Topic 13 · Vol II §20.6]

712 E An osseointegrated fixture does not follow the vertical development of the alveolar process, so an implant placed before growth is complete ends up below the level of the adjacent

teeth and looks progressively shorter. The idea that the fixture would drift with the arch is the opposite of the problem, since it is precisely the failure to move that causes the discrepancy. [Vol I Topic 13 · Vol II §20.6]

713 `FTFTT`

A False — the space narrows and the alveolus resorbs, so a plan must be made and the space held
B True — the size and prominence of a central incisor make substitution unconvincing
C False — the fixture must wait until vertical facial growth is complete, not until root formation is
D True — an interim pontic on a retainer maintains space and appearance through the growing years
E True — decoronation preserves alveolar bone for a later implant where the tooth is ankylosed

[Vol I Topic 13 · Vol II §20.6]

714 **B** A traumatised tooth needs a rest period of about three months with radiographic and sensibility review before force is applied, because pulpal and periodontal healing is still in progress and early loading increases the risk of resorption. Normal sensibility at six weeks is reassuring but does not establish that healing is complete, which is why the interval rather than the test governs the decision. [Vol I Topic 13 · Vol II §20.6]

715 **D** A sound, well-aligned second deciduous molar with no infra-occlusion may last for decades and there is no advantage in removing it while it is serving, so monitoring with the option of later reduction or replacement is the conservative plan. Removing them to close space is the tempting active option, but 2 mm of crowding does not justify sacrificing two functioning teeth. [Vol I Topic 13 · Vol II §20.2]

716 **A** Crowding, a Class I pattern, pale narrow canines and a low smile line all favour bringing the canines forward, which uses the crowding constructively and avoids a lifetime of prosthetic maintenance. Opening the spaces for implants is the alternative, but it requires space to be created in an already crowded arch and commits the patient to restorations in the most visible part of the mouth. [Vol I Topic 13 · Vol II §20.2]

> MNEMONIC **Crowded, Class I, pale narrow canines, low smile**: the four features favouring canine substitution over opening space for absent laterals.

717 `TTTTF`

A True — together with third molars these are the teeth most frequently missing
B True — ankylosis and infra-occlusion of retained deciduous molars is a common late finding
C True — the canine is narrowed and its cusp reduced, and the premolar is torqued and reshaped
D True — crowding provides the space that closure consumes and a low lip line hides the colour mismatch
E False — definitive implant restoration must wait until vertical growth is complete

[Vol I Topic 13 · Vol II §20.2]

718 **C** Mild infra-occlusion of a deciduous molar with a permanent successor present is usually self-limiting and the tooth exfoliates in the normal way, so observation at recall is all that is required. Extraction with a space maintainer is the interventionist response and creates a need for an appliance where no space has yet been lost. [Vol I Topic 13 · Vol II §20.2]

719 **D** Severe infra-occlusion with tipping of the neighbouring teeth and over-eruption of the opposing molar has already produced occlusal damage, so the tooth is removed and the space

is dealt with by closure or by opening and prosthetic replacement. Monitoring is appropriate for mild infra-occlusion with intact contacts, but once the adjacent teeth have tipped the situation is no longer stable. [Vol I Topic 13 · Vol II §20.2]

720 **B** Loss of the contact points allows the adjacent teeth to tip and the opposing tooth to over-erupt, so submergence to that level marks the point at which observation has become harmful. Treating 1 mm of infra-occlusion as a threshold would lead to the removal of many teeth that would have exfoliated uneventfully. [Vol I Topic 13 · Vol II §20.2]

721 **FFTTF**

A False — the mechanism is ankylosis of the affected tooth while the alveolus around it develops
B False — lower deciduous molars are affected considerably more often than upper ones
C True — mild cases with a successor present generally exfoliate at the expected time
D True — once the neighbours tip the space is compromised and the tooth should be removed
E False — infra-occlusion is commonly associated with absence of the permanent successor

[Vol I Topic 13 · Vol II §20.2]

722 **E** A single incisor tipped over the bite in a Class I case with space and an adequate overbite after correction is the classic removable appliance case, and the posterior bite planes clear the occlusion while the spring tips the tooth. Deferring treatment is the trap, because the displacement risks recession and attrition on the opposing lower incisor and the crossbite will not self-correct. [Vol I Chapter 12 · Vol II §20.7]

723 **A** A unilateral crossbite with a displacement is nearly always a bilaterally narrow maxilla forced to one side, so symmetrical expansion of the whole upper arch removes the interference and the displacement resolves. Expanding only the crossbite side treats the appearance of the problem and leaves the underlying transverse discrepancy uncorrected. [Vol I Chapter 12 · Vol II §20.7]

724 **C** A motivated child of this age is the right candidate for a habit deterrent, and since the posterior crossbite will not correct itself when the habit stops, expansion is combined with it in the same appliance. Expecting everything to resolve spontaneously is only partly right, because the incisor changes often recover but the transverse discrepancy does not. [Vol I Topic 13 · Vol II §20.7]

725 **TFFFT**

A True — the classic picture is an asymmetrical open bite with proclined upper and retroclined lower incisors
B False — the narrowed maxilla persists and requires expansion in its own right
C False — cooperation is essential and an appliance imposed on an unwilling child is a poor use of resources
D False — it is the total daily duration rather than the force of each episode that determines the effect
E True — changes produced before the permanent incisors erupt are largely self-correcting

[Vol I Topic 13 · Vol II §20.7]

> **MNEMONIC** **Duration, not force; crossbite, not open bite**: total daily hours predict the damage, and a crossbite needs its own correction even after the habit stops.

726 **B** A displacement carries a demonstrable risk of attrition and gingival recession on the opposing lower incisor and the correction at this age is simple, which is the combination that defines a genuine interceptive indication. A large overjet is a real concern but with competent lips the trauma risk is lower and treatment can wait for the growth spurt. [Vol I Topic 15 · Vol II §20.1]

727 **D** Distally inclined spaced upper central incisors during eruption of the lateral incisors is the ugly duckling stage, and the crowns upright themselves as the canines erupt and press on the lateral incisor roots. Closing the spaces with an appliance is the tempting response to parental concern, but it treats a normal developmental stage and risks retaining the result unnecessarily. [Vol I Topic 13 · Vol II §20.1]

Scored under 70% of the 89 marks on this block? Re-read Volume II §20, then sit it again in a fortnight.

Block U — Class I malocclusion and bimaxillary protrusion (Q728–752 · 49 marks · Vol II §21)

728 **A** The incisor classification depends solely on the relationship of the lower incisor edge to the cingulum plateau of the upper incisor, so a case with both segments proclined and severely crowded is still Class I. Calling it Class II division 1 confuses absolute incisor inclination, which is a separate matter, with the sagittal incisor relationship. [Vol I Chapter 13 · Vol II §21.1]

729 **C** A 3 mm deficit in a healthy adult labial segment with broad contact points can be resolved by removing a fraction of a millimetre of enamel from each contact, which avoids extractions and leaves the incisors in their pre-treatment position. Two premolar extractions would provide far more space than 3 mm of crowding requires and would commit the case to anchorage-demanding space closure. [Vol I Chapter 13 · Vol II §21.2]

730 **TTFTT**

- A True — unilateral or bilateral molar discrepancies occur with a normal incisor relationship
- B True — the conventional bands are mild, moderate and severe at roughly these thresholds
- C False — dentoalveolar compensation can produce Class I incisors on a discrepant skeletal base
- D True — arch length and tooth size discrepancies are independent between the arches
- E True — the need for space depends on the crowding, not on the anteroposterior class

[Vol I Chapter 13 · Vol II §21.1]

731 **E** Severe crowding concentrated in the canine region calls for space adjacent to that region and for most of the space to be used anteriorly, which is what first premolar extraction in all four quadrants provides while preserving the Class I buccal relationship. Second premolar extraction places the space further back and favours mesial movement of the molars, which is the opposite of what 10 mm of anterior crowding needs. [Vol I Chapter 13 · Vol II §21.2]

732 **A** A diastema of this size at this stage of eruption is a normal transient finding, and the eruption of the canines uprights the lateral and central incisor roots and closes the space without intervention. Frenectomy is the intervention that is most often performed too early, and where the attachment is not fibrous and does not blanch the papilla there is nothing to justify it. [Vol I Chapter 13 · Vol II §21.3]

733 **C** Closing the space first allows the scar tissue from the frenectomy to form across a closed contact rather than an open one, which helps to hold the result, and a bonded retainer is needed because diastema closure is among the least stable movements. Operating first is the traditional sequence but leaves the scar between the incisors, where it can obstruct the final closure. [Vol I Chapter 13 · Vol II §21.3]

734 **FFFFF**

- A False — a diastema at that stage is a normal developmental finding that commonly closes spontaneously
- B False — surgery removes the attachment but does not move the teeth together
- C False — relapse is high and bonded retention maintained indefinitely is the realistic plan
- D False — blanching of the papilla on traction is the standard clinical test of a fibrous attachment

E False — closed diastemas are notoriously prone to reopening and need permanent retention

[Vol I Chapter 13 · Vol II §21.3]

735 **E** Bimaxillary proclination is a normal variant in several populations and the commonly quoted incisor norms come from white European samples, so the values must be interpreted against appropriate standards and against the patient's own soft tissues. Treating the inclinations as pathological is the error that leads to unnecessary retraction and a flattened profile. [Vol I Chapter 13 · Vol II §21.4]

736 **B** The soft tissue of the upper lip follows incisor retraction at roughly seven tenths of the dental movement, which converts 5 mm of retraction into about 3.5 mm of lip change, although lip thickness and tonicity modify the ratio in individuals. Assuming a one to one relationship is the common error and leads to over-promising the profile change to the patient. [Vol I Chapter 13 · Vol II §21.4]

> MNEMONIC **7 in 10**: the upper lip retracts about 0.7 mm for every 1 mm of incisor retraction — a population average, not a per-patient guarantee.

737 **D** Soft tissue response ratios are derived from group means with wide individual variation, so they guide discussion but should be presented to the patient as an approximation rather than a guarantee. Treating the figure as precise to half a millimetre is the error that generates complaints when the profile change does not match the prediction. [Vol I Chapter 13 · Vol II §21.4]

738 **TFFTF**

A True — moving roots against a thin labial plate with a thin phenotype is a recognised recession risk
B False — it is a normal variant in many populations and treatment is driven by the patient's concerns
C False — population-appropriate standards must be used or the diagnosis will be wrong
D True — the soft tissue envelope determines where the incisors will remain after debond
E False — lip competence and lip strain at rest are central to deciding how far to retract

[Vol I Chapter 13 · Vol II §21.4]

739 **C** Labial movement of an incisor through a thin plate with little attached gingiva worsens recession, so the tissue is augmented first where indicated and the space is then found by reduction or extraction rather than by proclination. Declining to treat at all is over-cautious, since orthodontics can be delivered safely once the phenotype has been addressed and the mechanics planned to avoid labial movement. [Vol I Chapter 13 · Vol II §21.5]

740 **A** Correcting a complete transposition requires moving roots past one another and carries a substantial risk of resorption and bone loss, so where the alignment and occlusion are otherwise good the teeth are accepted in place and recontoured. Attempting to move them through each other is the intuitive answer, but it is reserved for incomplete transpositions where the roots are not yet crossed. [Vol I Chapter 13 · Vol II §21.6]

> MNEMONIC **Accept complete, correct incomplete**: transpositions with roots already crossed are recontoured in place; the rest can still be moved through.

741 **E** Derotation is produced by a force couple and the crown sweeps a wider mesiodistal path as it turns, so adequate space either side is a prerequisite and a rotated tooth in a tight arch will not turn. Starting on a rigid steel wire is the opposite of good practice, since rotations need the low forces and long range of a flexible wire. [Vol I Chapter 13 · Vol II §21.6]

742 **FTFFF**

A False — a severely rotated tooth occupies less mesiodistal space, so correction usually demands space
B True — moving roots past one another risks resorption and loss of the bone between the teeth
C False — canine and first premolar transposition in the maxilla is the form most often reported
D False — broad flat contacts resist the rotational recoil of the supracrestal fibre network
E False — deliberate overcorrection remains a recognised measure for an unstable rotation

[Vol I Chapter 13 · Vol II §21.6]

743 **D** A reduced but healthy periodontium tolerates orthodontic movement provided inflammation is controlled and forces are light, because the centre of resistance has moved apically and the same force produces greater stress at the crest. Assuming treatment is contraindicated is the traditional position, but it denies the patient a well-established means of correcting pathological migration. [Vol I Chapter 13 · Vol II §21.5]

744 **B** Dark spaces between the buccal segments and the commissures on smiling are the buccal corridors, and where they are judged excessive the usual orthodontic response is to widen the upper arch transversely. Excessive gingival display is the other common smile complaint but describes tissue shown above the incisors rather than space lateral to the buccal teeth. [Vol I Chapter 13 · Vol II §21.7]

745 **E** With a normal clinical crown height and no skeletal vertical excess the excess display arises from the vertical position of the teeth themselves, so intrusion of the labial segment is the appropriate and least invasive correction. Crown lengthening is the right answer when the crowns are short because of altered passive eruption, which the normal 10 mm crown height here excludes. [Vol I Chapter 13 · Vol II §21.7]

746 **TTTFT**

A True — a consonant smile arc parallels the lower lip and is a recognised aesthetic objective
B True — a narrow maxilla widens the dark spaces lateral to the buccal segments on smiling
C True — a small band of gingiva on full smile is within the range most observers accept
D False — incisor display at rest decreases with age as the upper lip lengthens and loses tone
E True — studies of lay perception consistently find midline shifts and cants both noticed, with cants tolerated less readily than commonly assumed

[Vol I Chapter 13 · Vol II §21.7]

> **MNEMONIC** **Arc, corridor, 2 mm**: the smile arc follows the lower lip, buccal corridor width tracks arch width, and up to about 2 mm gingival show is normal.

747 **B** Roughly half a millimetre can be taken from each contact in the labial segment without breaching the enamel, so spreading the reduction across the available contacts yields the space required while keeping each site within safe limits. Taking 0.25 mm from each contact is cautious but produces only about 2 mm, leaving the case short of the space it needs. [Vol I Chapter 13 · Vol II §21.2]

748 **D** Follow-up studies of reduced contacts that have been finished and polished do not show higher rates of caries or of attachment loss, which is why the technique is accepted as a routine means of creating small amounts of space. The claim that no evidence exists is the tempting escape, but long-term follow-up of reduced surfaces has been published. [Vol I Chapter 13 · Vol II §21.2]

749 **A** Where spacing arises from a tooth size discrepancy the orthodontic task is to gather the space into the sites where the teeth are too small, so that a restorative build-up produces proportionate crown widths. Closing everything ignores the cause of the spacing and leaves

the lateral incisors narrow and the contacts poorly shaped. [Vol I Chapter 13 · Vol II §21.3]

750 **TFTTF**

A True — interproximal reduction, minor arch development or slight proclination usually suffice at this level
B False — reduction, expansion, distalisation and proclination all create space by other means
C True — labial movement takes the incisors out of the soft tissue balance and thins the labial plate
D True — widening the arch increases the circumference available for aligning the teeth
E False — distalisation of the upper molars creates space in the upper arch only

[Vol I Chapter 13 · Vol II §21.2]

MNEMONIC **PIED**: Proclination, IPR, Expansion, Distalisation — four non-extraction ways to find space; distalising the uppers helps that arch only.

751 **C** Rotations are among the least stable movements and both overcorrection and the creation of broad flat contacts act directly against the recoil of the supracrestal fibres. A nighttime retainer is necessary but is not sufficient on its own for a severely rotated tooth with small rounded contacts, which is why the tooth itself is also modified. [Vol I Chapter 13 · Vol II §21.6]

752 **D** The British Standards classification is defined by the relationship of the lower incisor edge to the upper cingulum plateau, so the case is Class I with a molar asymmetry that is recorded separately. The subdivision terminology belongs to Angle's molar classification and describes the buccal segments rather than the incisor relationship being asked about here. [Vol I Chapter 13 · Vol II §21.1]

Scored under 70% of the 49 marks on this block? Re-read Volume II §21, then sit it again in a fortnight.

Block V — Class II malocclusion (Q753–792 · 80 marks · Vol II §22)

753 **B** Class II division 1 requires the lower incisor edge to lie posterior to the cingulum plateau with the upper central incisors of average inclination or proclined, both of which are present here. Division 2 is excluded because the upper central incisors are proclined rather than retroclined, which is the defining feature of that division. [Vol I Topic 12 · Vol II §22.1]

754 **E** Retroclined upper central incisors with proclined and rotated laterals and a complete deep overbite is the characteristic division 2 pattern, produced by a high lower lip line acting on the centrals while the laterals escape it. A 3 mm overjet is normal in size, but in division 2 the overjet is often small and the classification depends on incisor inclination rather than on overjet. [Vol I Topic 12 · Vol II §22.1]

755 **FFFFF**

A False — a reduced lower face height with a flat mandibular plane is the characteristic pattern
B False — the overbite is complete and traumatic contact is onto the palatal mucosa or labial gingivae
C False — the interincisal angle is increased, commonly beyond 140 degrees
D False — the overjet in division 2 is typically small or normal in size
E False — division 1 is considerably more prevalent than division 2 in survey data

[Vol I Topic 12 · Vol II §22.1]

MNEMONIC **RIFT**: Retroclined upper centrals, Increased interincisal angle (over 140), Flat mandibular plane, Traumatic deep overbite — normal or small overjet.

756 **A** An ANB of 4 degrees is only mildly Class II, so a discrepancy of 8 mm is being maintained by the lower lip trapping behind the upper incisors and proclining them while retro-

clining the lowers. Assuming a skeletal cause is the standard error, since the cephalometric value here does not account for the size of the overjet. [Vol I Topic 12 · Vol II §22.2]

757 **C** Where the maxilla is well positioned and the mandible is retrusive the appropriate growth modification is a functional appliance that postures the mandible forward, rather than restraint of a maxilla that is not at fault. Headgear is the tempting alternative but it addresses maxillary protrusion, which cephalometric analysis has specifically excluded here. [Vol I Topic 12 · Vol II §22.3]

758 **B** CS2 indicates that the pubertal growth peak is expected within about a year, so beginning a functional appliance now places the active phase across the peak, which is when the mandibular response is greatest. Deferring until growth has finished forfeits exactly the growth the appliance depends upon, and converts a growth modification case into a camouflage or surgical one. [Vol I Topic 12 · Vol II §22.3]

759 **FTFFF**

A False — the greater part of the change is dentoalveolar, with a smaller skeletal contribution
B True — these appliances work only while worn, and wear is the main variable in outcome
C False — growth modification requires growth, and after the peak the response is much reduced
D False — lower incisor proclination is one of the ways the overjet reduces and is often unwelcome
E False — posterior teeth are commonly allowed to erupt, which affects the vertical dimension

[Vol I Topic 12 · Vol II §22.3]

760 **D** Where the maxilla is protrusive and the mandible well positioned, restraint of forward maxillary growth is the logical growth modification, and headgear worn for the stated hours and forces achieves it during the pubertal peak. A twin block is the reflex answer in any Class II child, but it works by posturing a retrusive mandible forward, which is not the problem in this patient. [Vol I Topic 12 · Vol II §22.3]

761 **A** Trauma risk is the one outcome that early treatment reliably improves, with a reported risk ratio of about 0.58 for incisal trauma, and a child with a 10 mm overjet, incompetent lips and an existing injury sits squarely in that group. The expectation of a better final occlusion is the belief the early treatment trials specifically refuted. [Vol I Topic 12 · Vol II §22.2]

762 **C** Two well conducted multicentre trials found no occlusal, psychological or duration advantage for the early phase, so the remaining justifications are specific ones such as trauma risk or severe teasing. Abandoning early treatment entirely goes further than the evidence allows, because the reduction in incisor trauma is a genuine finding. [Vol I Topic 12 · Vol II §22.2]

763 **FTFFF**

A False — final occlusal outcomes were equivalent between early and adolescent treatment
B True — this is the reported figure, a relative reduction of about 42 per cent in trauma
C False — two phases of treatment lengthened rather than shortened the total time in appliances
D False — early differences in self-esteem were not maintained once both groups were treated
E False — the later group in fact finished marginally better than the two-phase group

[Vol I Topic 12 · Vol II §22.2]

764 **E** A risk ratio of 0.58 means the risk is 58 per cent of that in the untreated group, so the reduction is about 42 per cent, and the parent also needs the baseline risk to judge what that reduction is worth. Reading 0.58 as a 58 per cent reduction is the standard misinterpretation and overstates the benefit by a considerable margin. [Vol I Topic 12 · Vol II §22.2]

MNEMONIC **0.58 down, 42 up**: a risk ratio of 0.58 is about a 42 per cent relative reduction, not 58 per cent — always quote it with the baseline risk.

765 A A selective policy captures the trauma and psychosocial benefits for the children in whom they are real, while avoiding a second phase of treatment that the trials show adds time without improving the occlusion. A blanket policy of treating no child early ignores a genuine and reproducible reduction in incisor trauma in the high-risk group. [Vol I Topic 12 · Vol II §22.2]

766 C With growth complete, a severe skeletal discrepancy and a profile complaint, surgical advancement of the mandible is the only approach that addresses the facial concern as well as the occlusion. Camouflage is a legitimate option in moderate discrepancies, but at this magnitude it would flatten the profile further and demand more retraction than the anchorage will provide. [Vol I Chapter 9 · Vol II §22.5]

767 TTFFF

A True — the larger the discrepancy the less of it can be concealed by moving teeth alone
B True — only surgery alters the underlying skeletal relationship that produces the profile
C False — proclined lower incisors and a flat profile are precisely the findings that argue against camouflage
D False — camouflage disguises the skeletal pattern by altering the dentition and leaves the bases unchanged
E False — growth modification requires growth and has no role once the patient is skeletally mature

[Vol I Chapter 9 · Vol II §22.5]

MNEMONIC **Size, face, profile**: a large discrepancy, a facial complaint, or proclined lowers with a flat profile all argue against camouflage.

768 E A well aligned lower arch in a mild Class II case should be left alone, and upper premolar extraction with anchorage reinforcement uses the space for overjet reduction while accepting a Class II buccal segment relationship. Extracting in both arches would create space in a lower arch that needs none and would make the overjet harder rather than easier to correct. [Vol I Chapter 13 · Vol II §22.4]

769 B If anchorage is not reinforced roughly two thirds of the extraction space is consumed by mesial molar movement, so an absolute anchorage need of this size requires a miniscrew or headgear. A transpalatal arch resists rotation and some transverse change but does not prevent mesial drift, which is the demand being made here. [Vol I Chapter 13 · Vol II §22.4]

MNEMONIC **Two-thirds unguarded**: unreinforced extraction space loses about two-thirds to mesial molar drift, so reinforce with a miniscrew or headgear.

770 D An intra-oral distalising appliance anchored on miniscrews produces molar distal movement without depending on compliance or on headgear, which fits a patient who has declined both. Class III elastics would move the arches in the wrong direction, worsening rather than correcting the sagittal relationship. [Vol I Chapter 13 · Vol II §22.4]

771 FTFTF

A False — those figures are the anchorage prescription, whereas distalisation needs about 14 hours at 450 to 500 g per side
B True — a palatally anchored appliance works whether or not the patient cooperates
C False — distalisation becomes considerably more difficult once the second molars are in the arch
D True — the reciprocal effect of a distalising force is felt as forward movement of the anterior teeth
E False — two independent safety devices are mandatory on every headgear

[Vol I Chapter 13 · Vol II §22.4]

772 **C** The vertical component of a Class II elastic extrudes the upper molars and lower incisors, which opens the mandibular plane angle and reduces the overbite further, and in a high-angle patient with an already reduced overbite this is the wrong direction. Assuming they have no vertical effect is the error that produces open bites in exactly this facial type. [Vol I Chapter 13 · Vol II §22.4]

773 **A** Class II elastics move teeth rather than bones, producing overjet reduction by retracting the upper arch and proclining the lower incisors, so they camouflage a discrepancy rather than alter the skeletal relationship. The claim that they match a twin block skeletally overstates their action, since even functional appliances achieve only a modest skeletal component. [Vol I Chapter 13 · Vol II §22.4]

774 **FFTTF**

A False — the reciprocal force proclines the lower incisors, which is often the limiting side effect
B False — the extrusive component opens the bite and worsens a high-angle case
C True — they act only while worn and are among the most compliance-dependent mechanics
D True — this vertical effect is why they are used cautiously in high-angle patients
E False — their action is dentoalveolar and does not alter the skeletal bases

[Vol I Chapter 13 · Vol II §22.4]

775 **E** Division 2 treatment begins by converting the malocclusion to a division 1 form, since the retroclined upper incisors must be proclined and torqued and the deep overbite reduced before any space closure or retraction is possible. Retracting first is the error that drives the upper incisor roots against the labial plate and deepens the bite further. [Vol I Chapter 12 · Vol II §22.6]

776 **D** Proclination by tipping leaves the roots against the labial plate and the crowns unsupported, so palatal root torque is needed to bring the roots back into bone and place the crowns in a position the lip will maintain. Simply bonding a retainer accepts a root position that predisposes to relapse and to dehiscence, and retention cannot compensate for an unstable inclination. [Vol I Chapter 12 · Vol II §22.6]

777 **B** Overbite reduction is held by the combination of a reasonable interincisal angle and a positive occlusal stop of the lower incisor edge on the upper cingulum plateau, which is Houston's point. A retainer worn diligently will not hold a deep bite that has been finished with an obtuse interincisal angle and no stop. [Vol I Chapter 12 · Vol II §22.6]

778 **E** Deep overbites tend to re-establish themselves over years rather than months, so an upper retainer incorporating an anterior bite plane worn long term at night both holds the incisor inclination and resists the return of the overbite. A six month course of night-time wear is the usual prescription but is far too short for a corrected division 2 bite. [Vol I Chapter 12 · Vol II §22.6]

779 **TTTFF**

A True — the retroclined upper incisors are proclined and torqued before anything else is attempted
B True — tipping alone leaves the roots against the labial plate and the result relapses
C True — a deep bite prevents retraction and must be reduced before the spaces are closed
D False — an excessively obtuse interincisal angle predisposes the overbite to re-deepen
E False — once converted to a division 1 form, a functional appliance may be used in a growing patient

[Vol I Chapter 12 · Vol II §22.6]

780 **B** A working bite of about 7 mm of advancement with a few millimetres of vertical clearance is tolerated by the muscles and joints and leaves scope for reactivation, which is why large overjets are advanced in stages. Advancing the full amount at once is the tempting shortcut but produces discomfort, poor wear and a higher failure rate. [Vol I Topic 12 · Vol II §22.3]

> MNEMONIC **7 and 2-3**: stage a large overjet by advancing the twin block bite about 7 mm with 2 to 3 mm of vertical clearance, not the whole distance at once.

781 **D** Trials consistently show that the greater part of the overjet reduction is dentoalveolar, from retroclination of the upper incisors and proclination of the lower, with a smaller skeletal component that varies between patients. Claiming the change is entirely dental goes too far, since a modest and real skeletal contribution is measurable. [Vol I Topic 12 · Vol II §22.3]

782 **A** Anchorage reinforcement requires less force and fewer hours than distalisation, and the accepted prescription is 10 to 12 hours daily at 250 to 350 g per side with two independent safety devices in every case. The 14 hour regimen at 450 to 500 g per side is the distalisation prescription and delivers more force than anchorage support requires. [Vol I Chapter 13 · Vol II §22.4]

> MNEMONIC Two headgear prescriptions: **10-12 h at 250-350 g for anchorage, 14 h at 450-500 g for distalisation** — two safety devices always.

783 **TFFTF**

- A True — the requirement for two safety devices is absolute and applies to every prescription
- B False — a high-pull direction is chosen precisely to control the vertical position of the molars
- C False — anchorage uses 250 to 350 g per side and distalisation 450 to 500 g per side
- D True — wear is self-reported and timers are seldom used, so compliance is hard to verify
- E False — serious ocular injuries have been reported, which is why safety devices are mandatory

[Vol I Chapter 13 · Vol II §22.4]

784 **C** CS5 with no measurable height change over a year indicates that the pubertal peak is long past, so a functional appliance has almost nothing to work with and the realistic options are dental camouflage or orthognathic surgery. Prolonging the wear does not substitute for the growth that is no longer available, which is the reasoning behind the tempting alternative. [Vol I Topic 12 · Vol II §22.3]

785 **D** A coincident upper midline with the lower displaced towards the Class II side localises the discrepancy to the lower arch, so the correction is directed there by unilateral mechanics, a unilateral extraction or asymmetric elastics. Extracting an upper premolar on the Class II side treats the arch that is not at fault and would displace the upper midline. [Vol I Chapter 13 · Vol II §22.4]

786 **B** An overjet reduced far enough for the lower lip to come to lie in front of the upper incisors converts the lip from a destabilising trap into a force that holds the correction. Assuming that lip position no longer matters after debond is the error that explains overjet relapse in cases finished with a persisting lip trap. [Vol I Topic 12 · Vol II §22.7]

787 **TTTTF**

- A True — mandibular retrusion is the commonest skeletal contribution to a Class II pattern
- B True — the trapped lip acts behind the uppers and in front of the lowers
- C True — the digit produces proclination of the uppers with a localised anterior open bite
- D True — unprotected prominent incisors sustain trauma more often than protected ones

E False — soft tissue behaviour and habits modify the dental expression of any skeletal pattern

[Vol I Topic 12 · Vol II §22.2]

MNEMONIC **Mandible, lip trap, digit, incompetent lips**: the four levers behind a Class II division 1 overjet and its raised trauma risk.

788 E In a high-angle patient any extrusion of the upper molars rotates the mandible down and back, increasing the face height and worsening both the overjet and the overbite, so vertical control is the governing principle. Relying on Class II elastics is the tempting answer but their extrusive component is precisely what this facial pattern cannot tolerate. [Vol I Chapter 13 · Vol II §22.4]

789 A The decision turns on whether the patient's concern is facial and whether the discrepancy is beyond what dental compensation can disguise without damaging the profile or the periodontium. Using a fixed overjet threshold ignores that a large overjet on a mild skeletal pattern is often the easiest of camouflage cases. [Vol I Chapter 9 · Vol II §22.5]

790 C A solid Class I buccal interdigitation gives good sagittal retention, so standard retainers suffice, but the residual growth at CS4 means the relationship should still be reviewed as it completes. Continuing a functional appliance full time for a further year is unnecessary once the occlusion is interdigitated and the growth peak has passed. [Vol I Topic 12 · Vol II §22.7]

791 TTTTT

A True — interdigitated buccal segments resist the return of the sagittal discrepancy
B True — late growth in the original pattern is a recognised cause of post-treatment change
C True — a lip resting behind the upper incisors continues to procline them after debond
D True — without a positive stop on the cingulum plateau the overbite re-deepens
E True — alignment is maintained only while the teeth are held, whatever the original malocclusion

[Vol I Topic 12 · Vol II §22.7]

792 B The trials show equivalent occlusal outcomes whenever treatment is started, which leaves the growth peak as the efficient moment for the single phase, with early intervention reserved for the specific indications that early treatment does improve. Treating everyone early is the position the trials were designed to test and did not support. [Vol I Topic 12 · Vol II §22.2]

Scored under 70% of the 80 marks on this block? Re-read Volume II §22, then sit it again in a fortnight.

Block W — Class III malocclusion (Q793–822 · 62 marks · Vol II §23)

793 D Edge-to-edge contact in the retruded contact position with a forward slide into reverse overjet, Class I molars in retrusion and a straight profile together define a pseudo-Class III, in which the skeletal bases are essentially Class I. Calling it a true Class III is the trap, because it reads the displaced position as the diagnostic one when it is the retruded contact position that reveals the underlying relationship. [Vol I Topic 17 · Vol II §23.2]

794 A An SNA of 76 degrees is below the normal range and an SNB of 84 degrees above it, so maxillary retrusion and mandibular prognathism are contributing in roughly equal measure, which is the commonest pattern in Class III. Attributing the whole discrepancy to the mandible is the traditional assumption, and it is why maxillary deficiency is so often overlooked at assessment. [Vol I Topic 17 · Vol II §23.1]

795 TTTTT

A True — the smaller the underlying skeletal discrepancy, the more of it the teeth can be asked to disguise

B True — a reduced face height accompanies a forward growth rotation and helps an overbite to be maintained
C True — overbite is what retains the corrected incisor relationship once the crossbite has been eliminated
D True — already proclined lower and upright upper incisors leave room for the movement camouflage requires
E True — a displacement means part of the reverse overjet is postural and disappears once the incisors clear

[Vol I Chapter 8 · Vol II §23.3]

MNEMONIC **SHOPS** — Skeletal mild, Height average/reduced, Overbite average/increased, Proclined lower/upright uppers, Slide present.

796 C Protraction acts on the circummaxillary sutures, which interdigitate progressively with age, so the skeletal response is greatest before about ten years and the regimen asks for 12 to 14 hours of wear a day. Waiting for the pubertal peak is the trap, because it is the right instinct for a mandibular functional appliance and the wrong one for maxillary protraction. [Vol I Topic 17 · Vol II §23.4]

MNEMONIC **12–14 h/day, before 10 y** — protraction wear and the age window for a sutural response.

797 E A rise from 36 to over 60 per cent in those avoiding surgery is a real and useful effect, yet nearly 40 per cent of treated children still came to surgery and more than a third of untreated children never did, so consent must carry both figures. Reading the trial as reliable elimination of surgery is the overstatement parents are most often given. [Vol I Topic 17 · Vol II §23.4]

MNEMONIC **60 vs 36** — Mandall 2016: 60% avoided surgery with facemask, only 36% of untreated controls, to age 15.

798 TFTFT

A True — expansion loosens the circummaxillary sutures and is widely used to improve the protraction response
B False — the elastics run downwards and forwards from the canine region, at about 30 degrees to the occlusal plane
C True — roughly half the overjet change comes from forward movement and proclination of the upper incisors
D False — backward rotation reduces the prominence of the chin and helps the Class III, at a cost in face height
E True — late mandibular growth is the commonest reason an early correction is subsequently lost

[Vol I Topic 17 · Vol II §23.4]

799 A If surgery is likely, the lower incisors must later be proclined back to their true position, and it is that proclination which relieves the crowding, so extracting in the lower arch now consumes the space decompensation would have used. Extracting lower first premolars to deal with the crowding is exactly the intuitive move that makes the surgical case harder. [Vol I Chapter 8 · Vol II §23.8]

800 C Decompensation proclines the retroclined lower incisors and uprights the proclined upper incisors, which strips away the dental disguise so that both the reverse overjet and the profile look worse before the operation. Expecting the appearance to improve during the presurgical phase is the misunderstanding that most often unsettles a patient's commitment part way through. [Vol I Chapter 8 · Vol II §23.8]

801 E The published discriminants place surgery beyond a Wits of minus 11 mm, an ANB of minus 4 degrees and an overjet of minus 5 mm, and this patient exceeds all three, so camouflage would demand tooth movement the periodontium cannot deliver. Waiting for the Holdaway

and gonial values is the cautious-sounding answer, but those refine a borderline case rather than overturn three concordant ones. [Vol I Chapter 8 · Vol II §23.7]

MNEMONIC **WAO: 11-4-5** — Wits beyond -11 mm, ANB beyond -4 deg, overjet beyond -5 mm: camouflage thresholds exceeded, favour surgery.

802 **FTTFT**

A False — mandibular growth in boys continues well into the late teens and sometimes beyond twenty
B True — the late growth increment in Class III is both larger and more prolonged than in Class I
C True — the mandible outgrows the maxilla, so the reverse overjet and the profile deteriorate
D False — cranial base superimposition of serial films is precisely how the increment is quantified
E True — correcting before growth has stopped invites a relapse that only a second treatment can address

[Vol I Chapter 8 · Vol II §23.6]

803 **B** The measurable effects of a chin cup are a backward and downward rotation of the mandible with some remodelling at the gonial region, rather than any true reduction in mandibular size, and the change tends to be lost once wear stops. Believing that it permanently restrains growth is the historical claim that longer follow-up failed to support. [Vol I Topic 17 · Vol II §23.5]

804 **D** Two upper incisors tipped over the bite, an adequate overbite to hold them afterwards and a displacement to eliminate is the classic removable appliance case, with posterior bite planes clearing the occlusion while the springs work. Facemask with expansion is the trap, because it treats a skeletal problem that a displacement and an upright upper labial segment argue he does not have. [Vol I Topic 17 · Vol II §23.2]

805 **TTFTF**

A True — the displaced position flatters the mandible, so both positions must be recorded and compared
B True — the slide is the whole of the apparent discrepancy in a pseudo-Class III
C False — the profile is straight in retrusion and appears concave only in the displaced position
D True — a molar relationship that changes between the two positions is the corroborating sign
E False — the size of the reverse overjet does not distinguish the two, only the presence of a slide does

[Vol I Topic 17 · Vol II §23.2]

MNEMONIC **SIMS** — Slide forward, edge-to-edge Incisors in RCP, Molars Class I in RCP, Straight profile in RCP: pseudo-Class III.

806 **C** An increased lower anterior face height, an overbite that is already absent, lower incisors compensated to 78 degrees and no displacement to recover are four unfavourable entries in the prognostic table, and together they leave the teeth nothing to give. Attempting camouflage first and reserving surgery sounds prudent, but it costs years and worsens the periodontal position of the lower incisors. [Vol I Chapter 8 · Vol II §23.3]

807 **A** Camouflage of a Class III depends on retroclining the lower labial segment, so incisors already compensated to 80 degrees or less have exhausted that reserve and any further movement carries the roots out of the alveolus. Reading the threshold as a marker of bone already lost overstates it, since it describes the movement still available rather than damage already done. [Vol I Chapter 8 · Vol II §23.7]

808 FFTTF

A False — the familial pattern in Class III is the strongest of the three, the Hapsburg jaw being the caricature
B False — prevalence varies widely, being several times higher in some East Asian populations
C True — dentoalveolar compensation can carry the incisors into reverse overjet on an essentially Class I base
D True — scarring after early palatal repair restricts maxillary growth in all three dimensions
E False — the mechanism is mandibular overgrowth at the condyle, not retrusion of the maxilla

[Vol I Topic 17 · Vol II §23.1]

MNEMONIC **GRACE** — Genetic (strongest of the 3), Racial variation, Alveolar compensation, Cleft-repair scarring, Endocrine (acromegaly).

809 **E** Wear confined to sleeping hours falls short of the 12 to 14 hours a day the regimen requires, and a boy of eleven is past the age at which the sutures protract readily, so both should be re-examined before more of an ineffective treatment is prescribed. Escalating the force is the instinctive response to a stalled appliance, but it overcomes neither sutural interdigitation nor inadequate wear. [Vol I Topic 17 · Vol II §23.4]

810 **D** The positive overjet and overbite gained will be lost if late mandibular growth goes unnoticed, so the correction is retained and the patient followed annually until growth has finished and a definitive decision can be made. Discharging her on the strength of a positive overbite ignores that Class III deteriorates with growth, which is the commonest way an early success unravels. [Vol I Topic 17 · Vol II §23.9]

811 **B** Camouflage of a mild Class III works by retroclining the lower labial segment and proclining the upper, so the space is needed in the lower arch, which is also where the crowding lies, while the upper arch is left intact to advance. Extracting in both arches is the symmetrical-looking choice, but upper extraction space would be used to retract the very incisors that need to come forward. [Vol I Chapter 8 · Vol II §23.7]

812 FFFFF

A False — the circummaxillary sutures interdigitate with age and protraction works best before about ten years
B False — late mandibular growth commonly reverses an early correction, so review must run into the late teens
C False — over 60 per cent of treated children avoided surgery against 36 per cent of controls in Mandall 2016
D False — expansion disarticulates the maxilla and is routinely used to improve the protraction response
E False — the effect is backward rotation with gonial remodelling, and it is largely lost once wear stops

[Vol I Topic 17 · Vol II §23.4]

813 **E** A gonial angle above 120 degrees marks the backward growth rotation and increased face height that accompany a poor overbite, while a Holdaway angle of 3.5 degrees or less records a soft tissue profile already concave, so each identifies a pattern tooth movement cannot disguise. Reading the Holdaway angle as purely aesthetic misses that it is used here as a marker of the underlying skeletal pattern. [Vol I Chapter 8 · Vol II §23.7]

814 **B** A lower anterior face height at or above 66 per cent sits on the surgical side of the published discriminants, and here it comes with a high planes angle and a minimal overbite, which is the combination that leaves nothing to retain a camouflaged correction. Treating 1 mm of

overbite as sufficient is the error, since it is the overbite after correction that holds the incisors and there is none to spare. [Vol I Chapter 8 · Vol II §23.3]

815 **D** Every prognostic feature here sits on the unfavourable side, with a moderate skeletal discrepancy, an increased face height, a reduced overbite and no displacement to recover, and the family history adds the likelihood of late growth, so honest preparation is worth more than a treatment that growth will overtake. Facemask therapy is not wrong in principle at this age, but the features that predict its success are all absent. [Vol I Chapter 8 · Vol II §23.3]

816 **TFFFT**

A True — removing the dental camouflage exposes the full skeletal discrepancy and the profile worsens with it
B False — Class III lower incisors are already retroclined, and decompensation proclines them
C False — the arches must still be made to fit at the planned position, whichever jaws are moved
D False — upper incisors compensated by proclination are uprighted rather than proclined further
E True — that larger movement is the point, since it allows the jaws to be placed where they belong

[Vol I Chapter 8 · Vol II §23.8]

817 **A** The underlying relationship is Class I and the reverse overjet exists only after a forward slide, so altering the incisor inclinations removes the interference and the cause of the attrition together. A mandibular setback would treat a skeletal discrepancy that the retruded contact position shows he does not have. [Vol I Topic 17 · Vol II §23.2]

818 **C** Camouflage of a Class III depends on retroclining the lower labial segment, and driving roots against a labial plate that is already thin and recessed is the movement most likely to produce dehiscence and further loss of attachment. The notion that adult bone simply remodels around the roots overlooks that cortical plate does not follow a tooth pushed beyond it. [Vol I Chapter 9 · Vol II §23.7]

819 **D** Class III changes with growth whether or not it is treated, so only a randomised untreated arm can show that the difference at fifteen belongs to the facemask rather than to the passage of time. A narrow age band tightens the internal comparison but narrows generalisability rather than improving it, which is the reversal contained in that option. [Vol I Topic 17 · Vol II §23.4]

820 **FTFFF**

A False — a Wits beyond minus 11 mm is on the surgical side of the published discriminants
B True — that pair of movements is the whole mechanism by which the reverse overjet is disguised
C False — overbite is what retains the correction, so a reduced overbite makes relapse more likely
D False — camouflage can produce a good occlusion while leaving the concave profile unaltered
E False — a displacement is favourable, since part of the reverse overjet disappears with it

[Vol I Chapter 8 · Vol II §23.7]

821 **B** Serial cephalograms superimposed on stable cranial base structures quantify how far the mandible has come forward between reviews, and that increment is what decides whether surgery is needed and when it can safely be done. Hand-wrist films date maturation but say nothing about how the jaw relationship is actually changing, which is the question being asked

at review. [Vol I Chapter 8 · Vol II §23.6]

822 **E** The lower incisors have been retroclined out of their neutral zone position and the skeletal pattern that produced the malocclusion is unaltered, so retention must be long term and the lower labial segment held by a bonded retainer. Relying on 3 mm of overbite is the trap, because an overbite helps but does not offset the recoil of decompensated incisors on an uncorrected Class III base. [Vol I Topic 17 · Vol II §23.9]

Scored under 70% of the 62 marks on this block? Re-read Volume II §23, then sit it again in a fortnight.

Block X — Transverse problems and facial asymmetry (Q823–847 · 49 marks · Vol II §24)

823 **A** A maxilla narrower than the mandible meets it symmetrically at first contact, and the mandible then slides to one side to gain intercuspation, which produces an apparently unilateral crossbite with a displacement on a bilaterally narrow base. Calling it a true unilateral skeletal crossbite is the trap, because genuinely one-sided skeletal narrowing is rare and the model measurements show the deficiency is bilateral. [Vol I Topic 16 · Vol II §24.1]

824 **C** A mandibular displacement loads the teeth on the crossbite side outside their intended path and is associated with localised attrition and buccal gingival recession, which is the demonstrable harm that justifies correcting an otherwise symptom-free crossbite. Invoking temporomandibular disorder is the tempting justification, but the association there is weak and cannot carry a treatment decision. [Vol I Topic 16 · Vol II §24.2]

825 TTTTT

- A True — a quarter of a millimetre twice weekly is the standard activation for a slow expansion plate
- B True — this is the difference reported in the Cochrane review of treatment for posterior crossbite
- C True — the review rates the evidence for correcting crossbite in this age group as high certainty
- D True — the lighter continuous forces move teeth more and the suture less than a jackscrew does
- E True — a fixed appliance works whether or not the child remembers it, which is often the deciding point

[Vol I Topic 16 · Vol II §24.3]

MNEMONIC **0.25 mm x2/wk** — removable-plate activation; a quadhelix corrects about 3 months sooner (Cochrane, high certainty, age 7–11).

826 **B** A single tooth tipped palatally, with space to receive it and an overbite that will hold it afterwards, is the textbook indication for a Z-spring, the posterior bite planes clearing the occlusion while the spring works. A fixed 2 by 4 would also succeed but applies bodily control and laboratory cost to a problem that simple tipping resolves. [Vol I Topic 16 · Vol II §24.1]

827 **D** An overbite is the natural retainer of a corrected anterior crossbite, since the lower incisor edge occludes behind the upper and physically prevents its return; without one, relapse follows whatever else is done. Prolonged appliance wear is the answer candidates reach for, but retention cannot substitute for an occlusal relationship that holds the tooth by itself. [Vol I Topic 16 · Vol II §24.1]

828 **A** Rapid expansion delivers its 7 mm over two to four weeks at one to four quarter-turns a day, and the appliance is then left passive for three to six months while the suture fills with bone. Removing the appliance as soon as the crossbite corrects is the commonest error, because the correction is present long before the new bone that holds it. [Vol I Topic 16 · Vol II §24.4]

MNEMONIC **1–4 turns/day, 2–4 weeks, retain 3–6 months** — the rapid maxillary expansion regimen in full.

829 TFFTF

A True — heavy orthopaedic forces of this order are generated and dissipate as the suture separates
B False — roughly half the expansion is skeletal, the remainder being buccal tipping of the teeth
C False — about 10 per cent relapse is reported at four to eight months even after proper retention
D True — the diastema is the classic clinical sign that the suture rather than the teeth has moved
E False — two to four weeks is the active phase, and retention runs for three to six months

[Vol I Topic 16 · Vol II §24.4]

830 **C** Beyond about 5 mm in a patient whose circummaxillary articulations have interdigitated, conventional expansion is spent on tipping teeth and stripping bone rather than on separating the suture, so the resistant buttresses must be released surgically first. Persisting with a heavier jackscrew is the trap, since more force against an unyielding suture only worsens the tipping and the recession. [Vol I Topic 19 · Vol II §24.5]

831 **E** Resistance to expansion comes from the degree to which the midpalatal, zygomaticomaxillary and pterygoid articulations have interdigitated, and that varies widely between individuals of the same age, so the decision rests on the anatomy rather than the birth date. Reaching for a hand-wrist film is the plausible-sounding alternative, but skeletal maturity of the hand says nothing about the state of the palatal articulations. [Vol I Topic 19 · Vol II §24.5]

832 **A** A scissors bite needs the upper segment brought palatally and the lower buccally at the same time, which is exactly what a cross-elastic run from upper palatal to lower buccal attachments achieves once rigid archwires control the tipping. Expanding the upper arch is the reflex response to a crossbite and here would drive the segments further apart. [Vol I Topic 16 · Vol II §24.6]

833 FFFFF

A False — mild asymmetry is present in almost everyone and matters only when it is noticeable or progressive
B False — serial records six to twelve months apart are what demonstrate whether anything is changing
C False — a cant is judged clinically against the interpupillary line, the spatula test making it visible
D False — differential uptake on scintigraphy is a standard way of showing that one condyle is still active
E False — operating on a deformity that is still developing invites recurrence and a second procedure

[Vol I Topic 19 · Vol II §24.7]

MNEMONIC **STIR** — Scintigraphy (active growth), Timing (after growth stops), Interpupillary/spatula (cant), Records serial 6–12mo apart.

834 **C** A spatula held between the posterior teeth extends the occlusal plane out to the face, where any tilt relative to the interpupillary line becomes obvious to the patient as well as the clinician. Midline deviation is a different measurement altogether, recording lateral displacement rather than the vertical tilt that the question asks about. [Vol I Topic 19 · Vol II §24.7]

835 **E** Elongation lengthens one side of the mandible horizontally and carries the chin across the midline while leaving the vertical dimension and the occlusal plane untouched, which is precisely the pattern described. Hyperplasia is the tempting alternative, but it enlarges the

affected side vertically as well and so produces a cant and an increased face height on that side. [Vol I Topic 19 · Vol II §24.8]

MNEMONIC **Hyperplasia (3-D enlarge), Elongation (horizontal only), Microsomia (deficient), Functional (opens out)** — asymmetry types.

836 B Only standardised records repeated at an interval can separate the mild stable asymmetry that most people carry from a genuinely progressive one needing investigation and eventual correction. Operating now is the error, because a deformity corrected while it is still developing recurs and commits the patient to a second operation. [Vol I Topic 19 · Vol II §24.9]

837 TTFFT

A True — eliminating a displacement removes the attrition and recession risk that goes with it
B True — rapid activation directs more of the effect to the suture and less to the teeth
C False — the mixed dentition is its usual field of use, with bands on the first molars
D False — the retention period is three to six months, while new bone fills the opened suture
E True — these are the dental side effects that consume part of every tooth-borne expansion

[Vol I Topic 16 · Vol II §24.3]

838 D Four millimetres in a child of nine is comfortably within the range of slow expansion, and a fixed quadhelix removes compliance from the equation in a child whose history makes compliance the main risk to the result. A removable plate would work equally well in a reliable child, which is exactly what this history says she is not. [Vol I Topic 16 · Vol II §24.3]

839 C A single palatally displaced tooth in an otherwise well related arch needs space and a buccal force applied to that tooth alone, which a fixed appliance with open coil provides. A cross-elastic from the upper palatal surface to the lower buccal surface would pull the tooth further palatally, which is the direction that made the crossbite in the first place. [Vol I Topic 16 · Vol II §24.1]

840 A The review compares two effective appliances and finds one faster, so the finding informs the choice between them within the age range studied rather than condemning either. Concluding that the plate is ineffective is the overreading, since a difference in speed between two working appliances is not evidence that one does not work. [Vol I Topic 16 · Vol II §24.3]

841 FTFFF

A False — growth is restricted on the injured side, producing deficiency with antegonial notching
B True — the first and second branchial arch derivatives on the affected side are hypoplastic
C False — it presents in the second and third decades, when active condylar growth persists
D False — the asymmetry is postural, disappearing when the mandible is guided into retrusion
E False — the joints are involved in a substantial minority and condylar destruction can follow

[Vol I Topic 19 · Vol II §24.8]

842 E The opened suture is filled initially with blood clot and immature tissue, and a passive retention phase of three to six months is what allows bone to form across it, so removing the appliance at the moment of correction guarantees a loss well beyond the usual 10 per cent. Blaming excessive speed inverts the point, since rapid activation is what maximises the skeletal component. [Vol I Topic 16 · Vol II §24.4]

843 D A cant in the maxillary occlusal plane cannot be levelled by moving the mandible, so the maxilla is repositioned first and the mandible then autorotates and is set to centre the chin, which is why asymmetry of this kind is a bimaxillary problem. Operating on the mandible

alone is the tempting simplification and leaves the tilted maxillary plane exactly as it was. [Vol I Topic 19 · Vol II §24.9]

844 **B** Records unchanged over two years establish that the deformity is no longer developing, so the only remaining condition is the completion of growth, after which the correction can be planned with confidence that it will hold. Proceeding at once is the trap, since stability of the deformity is not the same as completion of facial growth in a boy of seventeen. [Vol I Topic 19 · Vol II §24.9]

845 **TFTTF**

A True — beyond that amount conventional expansion in an adult tips teeth rather than opening the suture
B False — the zygomaticomaxillary and pterygoid articulations contribute at least as much resistance
C True — once the buttresses are cut the force separates bone instead of being spent on the teeth
D True — anchoring in the palatal bone spares the buccal plate of the posterior teeth
E False — third molar status has no bearing on whether the articulations will yield

[Vol I Topic 19 · Vol II §24.5]

846 **E** Dentoalveolar compensation disguises part of a transverse discrepancy, so measuring the skeletal bases rather than the cusp relationship reveals a larger deficiency and a correspondingly larger correction, once the compensation is removed. Reading the compensation as evidence of a dental crossbite reverses the logic, since it is the skeletal deficiency that provoked the teeth to compensate. [Vol I Chapter 12 · Vol II §24.1]

847 **B** The harm attributed to posterior crossbite comes from the displacement, and where there is none, and the dentition has functioned without damage for four decades, correction offers nothing that justifies its cost or its risk. Referring for surgical expansion treats a radiographic and model finding rather than a problem the patient has. [Vol I Topic 16 · Vol II §24.2]

Scored under 70% of the 49 marks on this block? Re-read Volume II §24, then sit it again in a fortnight.

Block Y — Vertical problems: deep bite and open bite (Q848–877 · 58 marks · Vol II §25)

848 **D** Completeness records whether the lower incisal edges meet anything at all, tooth or mucosa, and here they meet mucosa, so the overbite is increased and complete; trauma is a separate judgement resting on evidence of tissue damage, which is absent. Calling it traumatic on the strength of mucosal contact alone is the commonest error, since many healthy mouths make that contact without harm. [Vol I Topic 11 · Vol II §25.1]

849 **A** Stripping of the palatal tissue behind the upper incisors with matching labial recession on the lower incisors is the definition of a traumatic overbite, and the damage continues for as long as the incisal edges load the soft tissue. Treating it as a hygiene problem is the trap, because plaque control does not remove the mechanical cause and the tissue continues to recede. [Vol I Topic 11 · Vol II §25.2]

850 **FFFFF**

A False — without condylar growth to compensate the mandible rotates down and back, and the result relapses
B False — a reverse curve in a continuous wire tips the incisors labially and extrudes the molars
C False — proclination gains only 0.1 to 0.2 mm per degree and relapses once the lip recovers
D False — an obtuse interincisal angle predicts relapse and correcting it is part of the plan

E False — it frees the posterior teeth to erupt, so the intrusion achieved is relative rather than true

[Vol I Topic 11 · Vol II §25.3]

851 **C** In a growing child with a reduced face height, holding the incisors apart while the buccal segments erupt increases the face height in a direction that suits him, and continuing condylar growth keeps the mandible from rotating backwards. True intrusion is not wrong but is unnecessary here, since it asks segmental mechanics to do what growth will do free of charge. [Vol I Topic 11 · Vol II §25.3]

852 **D** With growth finished and the face height already increased, extruding the buccal segments rotates the mandible down and back, lengthens the face further and relapses, so the overbite must be reduced by moving the incisors themselves with light continuous force. Extrusion behind a bite plane is the method learned in children and is precisely what this vertical pattern will not tolerate. [Vol I Topic 11 · Vol II §25.4]

853 **B** Intrusion concentrates stress at the root apex, so the force is kept to about 15 g a tooth and 60 g across the labial segment, which is an order of magnitude below the forces used for retraction. Matching retraction forces is the intuitive error and is the fastest route to the apical resorption that intrusion already risks. [Vol I Topic 11 · Vol II §25.4]

MNEMONIC **15 g/tooth, 60 g total** — the intrusion force ceiling for a lower incisor utility arch; far below retraction forces.

854 **TTFTT**

A True — unless the lower incisal edges come to rest against something, nothing holds the correction
B True — an obtuse angle leaves the incisors sliding past one another with no stop
C False — proclination moves the incisors out of the soft tissue balance and the lip returns them
D True — most of the vertical relapse that occurs is seen within twelve months of debond
E True — the retroclined upper incisors re-trap the lower ones and the deep bite re-forms

[Vol I Topic 11 · Vol II §25.5]

MNEMONIC **Houston 1989: occlusal stop = stability** — obtuse angle relapses, proclination alone fails, most relapse is within year 1.

855 **E** At 155 degrees the incisors lie so nearly parallel that the lower edges have nothing to occlude against, and only palatal root torque brings the upper crowns forward far enough to create the cingulum stop that holds the overbite. A flat curve of Spee is necessary but not sufficient, since a levelled lower arch still relapses if the incisors meet at an obtuse angle. [Vol I Topic 11 · Vol II §25.5]

856 **A** Restoring the worn surfaces at an increased height discludes the posterior teeth, which then erupt while the incisors intrude a little, and the posterior occlusion re-establishes itself over several months with the space created localised where it was needed. Full-mouth rehabilitation achieves the same space at vastly greater biological and financial cost for a problem confined to four teeth. [Vol I Topic 11 · Vol II §25.6]

857 **C** The habit is producing both the asymmetrical open bite and the proclination, and the open bite will close as the incisors continue to erupt once sucking stops, so stopping the habit comes first and determines what is left to treat. Expanding straight away is the trap, since the crossbite does need correcting in its own right but will simply be re-created while the thumb is still in the mouth. [Vol I Topic 11 · Vol II §25.7]

858 **TFFFT**

A True — the divergent jaw bases leave the incisors short of one another however they erupt

B False — the adaptive thrust is a consequence of the open bite and stops when the bite closes
C False — a digit habit produces a characteristically asymmetrical bite, biased to the side sucked
D False — the association is inconsistent and obstruction is at most one contributor among several
E True — the rare endogenous thrust persists after closure and is the recognised predictor of relapse

[Vol I Topic 11 · Vol II §25.7]

MNEMONIC **Skeletal, Adaptive thrust, Digit (asymmetric), Endogenous thrust (lisp, relapses)** — causes of anterior open bite.

859 B The great majority of anterior tongue thrusts are adaptive, the tongue sealing a gap the teeth cannot close, and the behaviour disappears once the incisors meet; only a small minority are endogenous. Assuming an endogenous thrust and declining to treat is the cautious error, since it withholds treatment from patients whose thrust would have resolved with the open bite. [Vol I Topic 11 · Vol II §25.7]

860 D Vertical elastics close the bite by extruding incisors into a space the soft tissues and the skeletal pattern do not allow them to occupy, so the correction is held only by the retainer and is lost when it stops. Saying that adult open bites cannot be closed overstates the case, since posterior intrusion and surgery both close them durably. [Vol I Topic 11 · Vol II §25.8]

861 A Intruding the maxillary posterior teeth allows the mandible to autorotate forwards and upwards, which closes the anterior open bite and reduces the face height at the same time, addressing the cause rather than the symptom. Anterior vertical elastics are the low-cost alternative and produce exactly the extrusive correction that this vertical pattern will not retain. [Vol I Topic 11 · Vol II §25.8]

862 TTTTT

A True — the plane holds the incisors apart and the unopposed molars erupt into the space
B True — the wire straightens as it is tied, expressing itself labially at the incisors
C True — the bypass is the whole point, since force applied to the incisors alone gives true intrusion
D True — the mandible rotates downwards and backwards as the buccal segments lengthen
E True — stress concentrates at the apex during intrusion, which is why forces are kept near 15 g a tooth

[Vol I Topic 11 · Vol II §25.3]

MNEMONIC **Bite-plane = eruption, Reverse curve = procline, Utility arch = true intrusion** — three ways to reduce overbite.

863 C Advancing the lower labial segment buys arch length at roughly 0.1 to 0.2 mm for every degree and lowers the incisal edges relative to the upper incisors, which is why it appears in the list of ways to reduce an overbite. The claim that it does nothing to the overbite is wrong, though the related warning about stability is fair, since incisors proclined beyond their soft tissue balance return. [Vol I Topic 11 · Vol II §25.3]

864 E An occlusal stop, with the lower incisal edges meeting the palatal surfaces of the upper incisors, is what physically prevents the overbite from deepening again, and it was identified as the governing condition by Houston in 1989. A bonded retainer is valuable for alignment but resists rotation and irregularity rather than the vertical relapse in question. [Vol I Topic 11 · Vol II §25.5]

865 A A reverse curve in a continuous wire acts as much on the molars as on the incisors, and without growth to absorb the molar extrusion the mandible rotates downwards and backwards, which lengthens the face and pushes the chin back. Expecting true intrusion is the

misconception, since only a bypass or segmental arch confines the force to the incisors. [Vol I Topic 11 · Vol II §25.3]

866 **C** Impacting the posterior maxilla shortens the excessive vertical development and lets the mandible autorotate upwards and forwards, which closes the open bite, reduces the face height and improves the chin position in one movement. Advancing the mandible alone leaves the vertical excess untouched and so leaves the open bite where it was. [Vol I Topic 11 · Vol II §25.8]

867 **FTFTF**

A False — posterior open bite is distinctly uncommon and anterior open bite far outnumbers it
B True — both a habitual lateral tongue posture and a genuine eruption failure produce it
C False — the two have different causes and closing one does nothing for the other
D True — traction on an affected tooth ankyloses it further and the segment must be managed prosthetically
E False — an ankylosed and submerging deciduous molar is a classic local cause

[Vol I Topic 11 · Vol II §25.9]

MNEMONIC **PFE + lateral thrust cause it; rarer than anterior; closing one does not close the other** — posterior open bite facts.

868 **E** The plane holds the buccal segments out of contact so that they erupt while the incisors do not, and in a growing child the face height accommodates that eruption, so the incisors are intruded relative to the occlusal plane rather than absolutely. True intrusion is the tempting reading, but a removable plane cannot deliver a controlled apical force to the incisors. [Vol I Topic 11 · Vol II §25.3]

869 **B** Depth and completeness are separate descriptions: two thirds coverage makes the overbite increased, while the absence of any contact by the lower incisal edges makes it incomplete. Reading deep coverage as automatically complete is the usual confusion, and it matters because an incomplete overbite has no occlusal stop and is correspondingly unstable. [Vol I Topic 11 · Vol II §25.1]

870 **D** The retroclined upper central incisors are what trap the lower labial segment, so converting the case to a division 1 pattern first releases the lower incisors and makes overbite reduction possible at all. Beginning with a reverse curve in the lower arch is the trap, since the lower incisors have nowhere to go while the upper crowns still lie behind them. [Vol I Chapter 13 · Vol II §25.3]

871 **TFTTT**

A True — the axial movement is shared, with most of it coming from eruption of the posterior teeth
B False — it is carried out chairside with composite or a removable appliance and no surgery
C True — most cases re-establish contact within about six months, and a few take longer
D True — restored or ankylosed teeth erupt unpredictably and the occlusion may not close down
E True — the temporary posterior separation is disconcerting and consent depends on explaining it

[Vol I Topic 11 · Vol II §25.6]

MNEMONIC **DAHL** — Discludes posteriors, Axial share (eruption+intrusion), Half-year to reclose, Less predictable if restored/ankylosed.

872 **C** Grossly enlarged tonsils with habitual mouth breathing are a treatable contributor to the posture that maintains the open bite, and assessment of the airway should precede a decision about mechanics that would otherwise be fighting the posture. Relying on lymphoid involu-

tion alone is the passive error, since the tissue may involute long after the vertical pattern has established itself. [Vol I Topic 11 · Vol II §25.7]

873 **A** Extrusion places the incisors where the soft tissue envelope and the vertical skeletal pattern do not permit them to stay, so the teeth drift back once the appliance stops holding them. Blaming retainer wear moves the fault to the patient and obscures the point that the mechanics chose a position the case could not maintain. [Vol I Topic 11 · Vol II §25.8]

874 **E** Intrusion of periodontally involved incisors is feasible and can improve the attachment level provided inflammation is controlled first and the force is reduced, because the centre of resistance has moved apically. Declining to treat is the safe-seeming choice and leaves the traumatic loading, which is itself the reason the lower incisors are being lost. [Vol I Chapter 13 · Vol II §25.4]

875 **FTTFF**

A False — many patients contact the palatal mucosa without any damage, and trauma must be seen
B True — inflamed and stripped tissue behind the upper incisors is the cardinal sign
C True — the upper incisal edges load the lower labial gingivae in the same way
D False — deep overbites occur with average and even increased planes angles as well
E False — the indication rests on tissue damage and on stability, not on a numerical threshold

[Vol I Topic 11 · Vol II §25.2]

876 **D** A reduced overbite in a division 2 case tends to deepen again, and a retainer with an anterior bite plane holds the incisors apart at night so that the buccal segments cannot settle the bite back down. Abandoning retention on the strength of a corrected interincisal angle is the trap, since the corrected angle improves the prognosis without guaranteeing it. [Vol I Topic 11 · Vol II §25.5]

877 **B** A small open bite maintained by a digit habit in a child with an average skeletal pattern closes spontaneously once sucking stops, and most children of this age stop when the reason is explained to them and their parents. Fitting a crib immediately is the trap, since an appliance imposed on a child who has not yet chosen to stop wastes the one intervention held in reserve. [Vol I Topic 11 · Vol II §25.7]

Scored under 70% of the 58 marks on this block? Re-read Volume II §25, then sit it again in a fortnight.

Block Z — TMD, condylar trauma, the airway and myofunctional therapy (Q878–907 · 62 marks · Vol II §26)

878 **E** A sign found in three quarters of the population and a symptom in a third means that most joint noise and tenderness detected at examination is a background finding rather than a disease, and it must be documented and interpreted rather than treated. Concluding that screening is therefore pointless is the opposite error, since the record made before treatment is what distinguishes pre-existing findings from new ones. [Vol I Topic 18 · Vol II §26.1]

> **MNEMONIC** **75/33** — about three quarters of adults have a TMD sign, about a third a symptom: read findings against that base rate.

879 **TFTFT**

A True — Costen's syndrome is the origin of the idea that the bite drives joint and ear symptoms
B False — the gnathologists of that era placed occlusion at the centre of joint function, not outside it
C True — retraction and extraction were said to drive the condyle back, a claim never substantiated

D False — the jury found against the orthodontist and awarded 850,000 US dollars in damages
E True — the studies commissioned after that verdict found no association between treatment and disorder

[Vol I Topic 18 · Vol II §26.2]

MNEMONIC **Costen to AAO**: 1930s Costen syndrome, 1970s gnathology blames occlusion, 1987 Brimm v Malloy lost, 1992 AAO: no link.

880 **B** A civil court decides on the balance of probabilities from the testimony placed before it, which is a different exercise from establishing causation, and the research commissioned afterwards found no association at all. Treating the award as scientific proof is the error the case itself caused, and it drove a generation of defensive practice on no evidential basis. [Vol I Topic 18 · Vol II §26.2]

881 **D** A painless click is among the commonest findings in the population and does not contraindicate treatment, but recording it and the range of movement before appliances are placed is what protects both patient and clinician if symptoms later appear. Proceeding without recording it is the practical trap, since an undocumented pre-existing sign becomes, in retrospect, a complication of treatment. [Vol I Topic 18 · Vol II §26.4]

882 **A** The accumulated evidence shows weak statistical associations between occlusal variables and joint symptoms but no causal effect in either direction, so treatment cannot honestly be offered or refused on joint grounds. Promising prevention is the more attractive answer to give a parent and is exactly the claim the 1992 studies were commissioned to test and did not support. [Vol I Topic 18 · Vol II §26.3]

883 `FTTFT`

A False — no prospective study has shown this, and the claim rests on case series and assertion
B True — the associations are real but small, and far too weak to support a causal reading
C True — that coincidence of timing is the chief confounder in the whole literature
D False — it is not a treatment for the disorder and should not be offered as one
E True — the two are not the same, and the distinction matters when advising a patient

[Vol I Topic 18 · Vol II §26.3]

884 **C** If the disorder begins spontaneously at exactly the ages at which appliances are worn, a proportion of patients will develop symptoms during treatment however inert the treatment is, and only a comparison with an untreated group of the same age can separate the two. Attributing such symptoms to the appliance by default is the reasoning that produced the litigation of the 1980s. [Vol I Topic 18 · Vol II §26.3]

885 **D** Symptoms arising during treatment are managed as they would be in any other patient, with the active mechanics suspended so that the appliance is not loading a painful joint, and most settle with simple reversible measures. Abandoning treatment is the over-reaction, since it leaves the patient with an unfinished occlusion and does nothing the pause would not have achieved. [Vol I Topic 18 · Vol II §26.4]

886 **B** Orthodontic treatment has never been shown to resolve temporomandibular disorder, so offering it for that purpose misrepresents what it can do and leaves the patient with an unmet expectation and an irreversible change. Agreeing to treat with a caveat about uncertainty is the softer version of the same error, since the treatment is still being offered for an indication it does not have. [Vol I Topic 18 · Vol II §26.4]

887 `TTFTT`

A True — a baseline record is what distinguishes a pre-existing finding from a new complication
B True — counselling on the natural history is the single most useful thing said at the first visit
C False — the guidance is explicit that orthodontics is not a treatment for the disorder
D True — suspending mechanics stops the appliance loading a painful joint while the episode settles
E True — reassurance, analgesia, soft diet and exercises precede anything irreversible

[Vol I Topic 18 · Vol II §26.4]

MNEMONIC **Screen, Counsel (fluctuates, often settles), Not a cure, Reversible/conservative first** — Kandasamy 2022.

888 **E** Intracapsular fractures in a young child carry the highest risk of bony ankylosis, and the resulting restriction of function produces a short retrognathic mandible with antegonial notching over the following years. Expecting excessive growth inverts the effect, since the injured condyle grows less rather than more. [Vol I Topic 18 · Vol II §26.5]

889 **A** Synovial inflammation destroys the condylar surface in a substantial minority of children with this disease, and loss of the condyle as a growth site produces exactly this picture of a short mandible rotated backwards with an anterior open bite. Blaming a digit habit is the tempting simple answer and does not account for the bilateral condylar flattening. [Vol I Topic 18 · Vol II §26.6]

890 **FFTTT**

A False — young women are affected far more often, in ratios commonly quoted above eight to one
B False — it is characteristically a condition of the second and third decades
C True — these are the classic predisposing skeletal features quoted in every case series
D True — the open bite develops months after surgery as the condyles lose height
E True — inflammatory arthropathy is among the established causes of condylar loss

[Vol I Topic 18 · Vol II §26.6]

MNEMONIC **F>M 8:1, 2nd–3rd decade, high MPA + retrognathic mandible, may follow orthognathic surgery, JIA a cause** — resorption risk.

891 **C** A prevalence of about one child in fifty is too low to justify investigating everybody and far too high to ignore, so the orthodontist's role is to ask the screening questions and refer the children whose answers are suggestive. Offering appliances as first-line treatment is the overreach, since the diagnosis and its primary management belong elsewhere. [Vol I Chapter 2 · Vol II §26.7]

892 **B** The long face with open bite and incompetent lips travels with mouth breathing often enough to have been named after the adenoids, but the experimental and longitudinal evidence for one producing the other is weak, and expansion cannot be promised as a cure for the face. Denying any connection at all is the opposite overstatement and is equally unsupported. [Vol I Chapter 2 · Vol II §26.7]

893 **D** The studies that have measured airway dimensions after premolar extraction report small and inconsistent changes with no demonstrated effect on breathing or on sleep, which is why extraction decisions remain dental and skeletal decisions. Claiming that extraction enlarges the airway is the mirror-image overstatement and rests on no better evidence than the claim it contradicts. [Vol I Chapter 2 · Vol II §26.8]

894 **TFFTF**

A True — separating the halves of the maxilla widens the nasal floor, and this is readily measured

B False — adenotonsillectomy and positive airway pressure remain the first-line treatments
C False — habitual mouth breathing frequently persists after the mechanical resistance has fallen
D True — expansion is prescribed for a transverse discrepancy, with any airway benefit a secondary gain
E False — reported reductions in the apnoea-hypopnoea index are modest and inconsistent between studies

[Vol I Chapter 2 · Vol II §26.7]

895 **A** Witnessed apnoeas with unrefreshing sleep and daytime behavioural change in a child with obstructing tonsils is the classic presentation of obstructive sleep apnoea, and it needs diagnosis and surgical assessment rather than an orthodontic appliance. Waiting for lymphoid involution is the passive error, since that process takes years and the daytime consequences are accruing now. [Vol I Chapter 2 · Vol II §26.7]

896 **C** The trials are small, heterogeneous and largely unblinded, and while there is a defensible adjunctive role in habit control and in adult sleep-disordered breathing, nothing supports it as a substitute for orthodontic correction. Dismissing it entirely overstates the case in the other direction, since the weakness of the evidence is not the same as evidence of no effect. [Vol I Topic 18 · Vol II §26.9]

897 **E** A habit persisting after the permanent incisors have erupted in a child who wants to stop is the situation in which a reminder appliance earns its place, and the crossbite has to be expanded on its own merits whatever the habit does. Relying on explanation alone is right for a younger child but has already been tried by the time a habit reaches eleven. [Vol I Topic 18 · Vol II §26.9]

898 **TTTFT**

A True — haemarthrosis within the capsule organises and the joint may fuse
B True — the notch marks the altered muscle pull on a mandible that has grown short on that side
C True — immobilisation is the main modifiable risk factor, so function is restored early
D False — conservative management with early function is standard, surgery being reserved for few
E True — substantial regeneration of condylar form is well documented after conservative care

[Vol I Topic 18 · Vol II §26.5]

MNEMONIC **Intracapsular = ankylosis risk, notching, early mobilise to prevent it, condyle can remodel** — childhood condylar fracture.

899 **A** Patients who present for orthodontic treatment differ from those who do not in malocclusion, in dental attendance and in their readiness to report symptoms, so the two groups were never comparable and the difference in clicking may belong entirely to that selection. Power is not the difficulty here, since clicking is common enough for 200 in each arm to detect a real difference. [Vol I Topic 18 · Vol II §26.3]

900 **C** A written baseline of what the joints did before treatment, together with a record of what she was told about the natural history, is what allows any later change to be interpreted rather than disputed. Imaging both joints adds cost and incidental findings without altering the plan, which is why it is not part of routine orthodontic assessment. [Vol I Topic 18 · Vol II §26.4]

901 **E** Episodes wax and wane and the majority settle with reassurance and simple reversible measures, which is the single most useful thing a patient can be told before appliances are placed. Warning of inexorable degeneration is the opposite distortion and frightens patients

into irreversible treatment they do not need. [Vol I Topic 18 · Vol II §26.1]

902 **FTFFT**

A False — a two-dimensional film of a collapsible three-dimensional tube cannot give a volume
B True — these three questions are the practical screen and take under a minute to ask
C False — anatomical size on a film predicts the physiological severity poorly
D False — the orthodontist screens and refers, and the diagnosis is made in a sleep service
E True — overnight study remains the standard against which every screening tool is judged

[Vol I Chapter 2 · Vol II §26.7]

MNEMONIC **Snore, Apnoea, Sleepy** — the 3-question airway screen; refer on, don't diagnose; PSG is the standard.

903 **B** Changes that are small, inconsistent in direction and without measurable effect on breathing give no reason to alter a decision that rests on crowding, incisor position and the facial profile. Concluding that nothing can be drawn from the data is the counsel of despair, since a consistent absence of functional effect is itself an informative finding. [Vol I Chapter 2 · Vol II §26.8]

904 **D** Muscle pain worst on waking in a self-reported clencher is managed first with the reversible measures that have the best evidence behind them, and the orthodontic problem can be addressed once the symptoms have settled. Equilibration is the tempting alternative because it feels like treating the cause, but it is irreversible and rests on an occlusal model of the disorder that the evidence does not support. [Vol I Topic 18 · Vol II §26.4]

905 **C** A custom-made appliance that holds the mandible forward during sleep is the accepted second-line treatment for mild to moderate disease when positive airway pressure fails, and the occlusal changes it causes over years are the reason dental review is part of the package. Expansion in an adult would need surgical assistance and has no established role in this disease. [Vol I Chapter 2 · Vol II §26.7]

906 **FFFTF**

A False — slides of that size are common in symptom-free people and are not in themselves treated
B False — no study has shown this, and the assertion is a survival of the gnathological era
C False — routine mounting on a fully adjustable articulator has never been shown to change outcomes
D True — the associations exist but are far too weak to support a causal claim
E False — the coincidence of timing does not establish causation and misleads the patient

[Vol I Topic 18 · Vol II §26.3]

907 **A** The honest summary is that the evidence shows no causal effect in either direction, coupled with a warning that joint signs and symptoms are common in adolescence and may appear during treatment for reasons unconnected with it. Saying nothing at all is the practical trap, because a symptom that arrives unannounced during treatment will be attributed to the appliance by the family. [Vol I Topic 18 · Vol II §26.3]

Scored under 70% of the 62 marks on this block? Re-read Volume II §26, then sit it again in a fortnight.

Block AA — Adult orthodontics and orthognathic surgery (Q908–937 · 62 marks · Vol II §27)

908 **E** Attachment loss shifts the centre of resistance apically, which lengthens the perpendicular distance from the bracket and raises the moment-to-force ratio delivered by any given force, so lighter forces with added root torque are required. The answer that fixes the centre of resistance at the coronal-middle third junction is the trap — that figure describes an intact periodontium and does not survive bone loss. [Vol I Chapter 13 · Vol II §27.2]

909 TTTFF

A True — the growth that a functional appliance or facemask exploits has ended, leaving camouflage or surgery as the options
B True — cellularity and vascularity of the alveolus fall with age, which lengthens the lag phase rather than preventing movement
C True — adults report greater initial discomfort and seek reassurance about it more often
D False — the sequence of hyalinisation, resorption and remodelling is the same, only slower to begin
E False — adult motivation is characteristically intrinsic, which is why adult compliance with hygiene and appliances is generally good

[Vol I Chapter 13 · Vol II §27.1]

910 **D** Distal crown tipping uprights the molar, decompensates the mesial osseous defect and widens the space to a restorable dimension, the price being extrusion that is managed by occlusal adjustment at the end. Moving the root mesially instead is the tempting alternative, but it preserves an already inadequate space and demands anchorage a single quadrant cannot supply. [Vol I Chapter 13 · Vol II §27.3]

911 **B** Extruding the tooth while dividing the supracrestal fibres every one to two weeks moves the root coronally and leaves the bone and gingival margin behind, which delivers the ferrule without altering the gingival zenith or the neighbouring crown lengths. Extrusion without fibreotomy is the trap — dragging bone and soft tissue coronally is what you want when developing an implant site, not when the aim is to expose a margin. [Vol I Chapter 13 · Vol II §27.3]

912 **E** A molar pulled from the buccal side alone rotates about its long axis so that the palatal cusp extrudes, and anchorage on both sides of the tooth is what makes 3 mm of true intrusion achievable and stable in the reported series. Anchoring an intrusion arch on the incisors is the trap — the reciprocal force proclines and extrudes the labial segment, which is exactly the anchorage loss that skeletal anchorage exists to avoid. [Vol I Chapter 13 · Vol II §27.3]

913 TTTFT

A True — removing the retroclination that masked the discrepancy unmasks it, and the reverse overjet characteristically increases
B True — decompensated incisors allow the jaws to be set at the planned skeletal position rather than a compromise
C True — transverse coordination on the planning models is what allows a passive and reproducible fit at operation
D False — more camouflage shrinks the surgical movement but leaves the skeleton uncorrected and the incisors unstable
E True — the temporary worsening is a consent point and a common reason for patients to disengage mid-treatment

[Vol I Chapter 15 · Vol II §27.5]

MNEMONIC **Decompensate, coordinate, warn** — presurgical orthodontics unmasks the discrepancy, matches the arches, and the bite worsens before surgery fixes it.

914 **B** A 2 mm reverse overjet with the lower incisors already at 84 degrees leaves room inside the envelope of discrepancy for a dental correction, and the absence of any facial concern removes the principal indication for surgery. Offering surgery because the ANB is negative is the trap — the cephalometric value describes the skeleton, but the decision rests on the size of the movement needed and on what the patient wants changed. [Vol I Chapter 15 · Vol II §27.4]

915 **D** Superior repositioning corrects the gingival display and shortens the lower face, and the mandible then autorotates forwards about the condyle, which closes much of the overjet

without a second osteotomy. Advancing the mandible alone is the tempting answer — it reduces the overjet but leaves the vertical maxillary excess and the gingival display untouched. [Vol I Chapter 15 · Vol II §27.6]

916 **A** The inferior alveolar nerve lies directly in the path of the split, so early neurosensory disturbance is the rule and a minority — more with larger advancements and older age — retain some permanent alteration. Facial nerve weakness is the trap because it is the nerve patients ask about, but the facial nerve is not at risk during a sagittal split. [Vol I Chapter 15 · Vol II §27.7]

917 `TTTTT`

A True — impaction is consistently placed at the top of the hierarchy, with relapse rates lower than any other movement
B True — advancement is stable at moderate magnitudes, whereas setback depends on fixation and relapses more often
C True — transverse expansion of the maxilla is the movement most likely to relapse, whatever the technique
D True — the pterygomasseteric sling and the palatal soft tissues pull the maxilla back up after downgrafting
E True — immobilisation with plates and screws removed much of the relapse seen with wire fixation and intermaxillary wiring

[Vol I Chapter 15 · Vol II §27.8]

> MNEMONIC **Up stable, setback shaky, wide worst** — impaction is most stable; setback is less predictable than advancement; expansion relapses most.

918 **C** A horizontal osteotomy of the chin carries the pogonion forwards with its attached musculature and soft tissue and leaves the dentition and the occlusion untouched, which is precisely what the patient has asked for. Sagittal split advancement is the tempting answer because the mandible looks deficient, but moving the whole mandible in a sound Class I occlusion creates a malocclusion that then needs orthodontics. [Vol I Chapter 15 · Vol II §27.6]

919 **D** In a skeletally mature patient the resistance lies in the fused suture and the zygomatic buttresses, and surgical release before activation allows a 9 mm correction that is largely skeletal rather than dentoalveolar. Conventional rapid maxillary expansion is the trap — in the adult it produces buccal tipping, dehiscence and pain instead of sutural separation. [Vol I Chapter 15 · Vol II §27.6]

920 **B** Surgery-first works only when the presurgical occlusion is close enough to fit at operation, which in practice means little compensation, coordinated arches and mild crowding, and the early facial change and accelerated postsurgical movement are then genuine gains. Choosing it simply to shorten treatment is the trap — without a planned and reproducible interim occlusion the result is unpredictable and relapse is common. [Vol I Chapter 15 · Vol II §27.5]

921 `FFFFT`

A False — 1 mm of crowding cannot account for this degree of distress, and the mismatch is the cardinal feature
B False — body dysmorphic disorder is reported in 5 to 13 per cent of orthodontic and orthognathic populations
C False — repeated treatment with persisting dissatisfaction is one of the strongest warning signs
D False — treatment does not relieve the preoccupation and commonly transfers it to another feature
E True — assessment before irreversible treatment is the correct step, and the orthodontist should not treat

[Vol I Chapter 15 · Vol II §27.9]

922 **E** Progressive condylar resorption characteristically declares itself months to years after surgery in a young woman with a high-angle retrognathic pattern after a large advancement, and produces exactly this combination of recurrent open bite and receding chin. Fixation failure is the trap — it relapses in the first weeks after operation, not at eighteen months. [Vol I Chapter 15 · Vol II §27.7]

923 **A** Moving teeth through an inflamed periodontium converts gingival inflammation into attachment loss, so disease control with a documented stable re-evaluation comes first and the appliance follows it. Treating in parallel is the trap — it sounds efficient and is how the patient will press for it, but it adds plaque-retentive appliances to an active lesion. [Vol I Chapter 13 · Vol II §27.2]

924 **C** Labial movement thins or dehisces an already thin labial plate, and with only 1 mm of attached gingiva that is where recession progresses, so space must be created within the arch rather than in front of it. Grafting first is the tempting answer — it may well be needed, but it does not license proclination that the bony housing cannot support. [Vol I Chapter 13 · Vol II §27.2]

925 **TTFFF**

A True — the same force across less attachment raises stress per unit area, so forces are lowered accordingly
B True — a three-monthly maintenance interval through treatment is what keeps the reduced attachment stable
C False — a treated and inflammation-free periodontium tolerates movement well, whatever its height
D False — long-term bonded retention is usually essential here, and hygiene is managed around it rather than by removing it
E False — reduced attachment does not protect the root, and resorption risk is if anything higher in adults

[Vol I Chapter 13 · Vol II §27.2]

MNEMONIC **Lighter forces, 3-monthly SPT, lifelong retention** — the three-part plan for moving teeth through a reduced but healthy periodontium.

926 **B** Converging roots leave too little bone between them to receive a fixture and are the commonest orthodontic reason an implant site will not accept a standard placement, so root parallelism is the objective the restorative colleague depends on. Widening the space to 7 mm is the trap — a lateral incisor site takes a narrow fixture, and over-widening produces an unaesthetic emergence profile. [Vol I Chapter 13 · Vol II §27.3]

927 **D** Only skeletal anchorage is genuinely absolute; intra-arch devices reduce but do not abolish loss, and unsupported posterior segments migrate into about two thirds of the extraction space (Haque 2017). The transpalatal arch with a Nance button is the tempting answer — it is a real improvement on nothing, but the case specifies that no molar movement at all is acceptable. [Vol I Chapter 13 · Vol II §27.3]

928 **A** Where the jaw bases are correct and the deformity lies in the labial segments, anterior segmental osteotomies using premolar spaces move both blocks back directly and preserve the existing molar relationship. Setback of both whole jaws is the trap — it moves bases that are not at fault and flattens the entire midface. [Vol I Chapter 15 · Vol II §27.6]

929 **TTTTF**

A True — the defect that reads as a pocket is largely a consequence of the tipped crown and improves as the tooth is uprighted

B True — dividing the supracrestal fibres repeatedly prevents the attachment following the tooth coronally
C True — this is site development, in which the attachment is deliberately dragged coronally with the root
D True — the objective is restorative and limited, which is what distinguishes adjunctive from comprehensive treatment
E False — intrusion achieved with skeletal anchorage is largely stable once an opposing contact is restored

[Vol I Chapter 13 · Vol II §27.3]

930 **C** Adult treatment is elective, lasts about two years and depends on intrinsic motivation, and a patient attending on somebody else's concern has no complaint for the treatment to answer. Starting because the overjet meets an index threshold is the trap — indices describe need across a population and do not supply an individual's reason for wanting treatment. [Vol I Chapter 13 · Vol II §27.1]

931 **E** With the maxilla moved first, the only stable reference remaining in the operating field is the unoperated mandible, so the intermediate wafer transfers the planned maxillary position to it; the final wafer then sets the mandible to the new maxilla. Confusing it with the final wafer is the commonest error — that one carries the planned occlusion and is used after the second osteotomy. [Vol I Chapter 15 · Vol II §27.5]

932 **A** The presurgical phase deliberately worsens both the occlusion and the profile, and a patient who has not grasped this is the one who abandons treatment decompensated and worse off than when he started. Anaesthetic consent matters greatly but is taken by the surgical and anaesthetic teams at admission, not before the orthodontics begins. [Vol I Chapter 15 · Vol II §27.9]

933 **TFFTF**

A True — both the magnitude of the movement and increasing age raise the chance of persistent alteration
B False — an unfavourable split is recognised and managed at the time with additional fixation or a bone graft
C False — haemorrhage is a recognised risk, particularly from the descending palatine vessels, but is far less common than this
D True — segmenting and widening at once compromises the palatal blood supply on which the segments depend
E False — it is characteristically a condition of young women with high-angle retrognathic patterns

[Vol I Chapter 15 · Vol II §27.7]

934 **C** Relapse that scales with the size of the movement and with a high-angle pattern is the observation underlying the stability hierarchy, and it argues for dividing a large correction between the jaws rather than making one very large movement. Abandoning mandibular advancement altogether over-generalises the finding — it remains among the more stable procedures at moderate magnitudes. [Vol I Chapter 15 · Vol II §27.8]

935 **E** Class III mandibular growth continues later and far less predictably than Class II, and operating during residual growth is the classic route to a second operation, so cessation must be demonstrated rather than inferred from age. Fixing the date at 18 is the trap — it is an average, and the variation around it is precisely what matters in this diagnosis. [Vol I Chapter 15 · Vol II §27.4]

936 **FFFFF**

A False — the ligament narrows with age, and the adult requires lighter rather than heavier forces
B False — hyalinisation still occurs and clears more slowly, which is why the lag phase lengthens
C False — once the lag phase is past the rate approaches the usual 1 mm per month; it is the onset that is delayed
D False — resorption is if anything more frequent in adults, which is why interim radiographs matter
E False — reduced bone height moves the centre of resistance apically and has no effect on the lag phase

[Vol I Chapter 13 · Vol II §27.1]

MNEMONIC **Narrower ligament, longer lag, same rate, more resorption** — how adult PDL biology truly differs from an adolescent's, not simply slower movement.

937 B The envelope of discrepancy limits both how far teeth can be moved and how far they should be, and in a convex profile with a deficient mandible the dental correction makes the face worse even where the movement itself is achievable. The claim that 11 mm exceeds what extractions can deliver is the trap — the space exists, but spending it produces a dished upper lip and no chin projection. [Vol I Chapter 15 · Vol II §27.4]

Scored under 70% of the 62 marks on this block? Re-read Volume II §27, then sit it again in a fortnight.

Block AB — Cleft lip and palate and craniofacial syndromes (Q938–968 · 63 marks · Vol II §28)

938 D The lip and primary palate are formed when the maxillary process meets the medial nasal process, and failure of that fusion by about the eighth week produces a cleft of the lip and alveolus in front of the incisive foramen. Failure of shelf elevation is the trap — that is the mechanism of isolated cleft palate, a separate event with a different epidemiology. [Vol I Chapter 16 · Vol II §28.1]

939 C Formation of the secondary palate is a separate and later event from formation of the lip and primary palate, which is why the two conditions differ in sex ratio, frequency and syndromic association. Treating them as extents of a single defect is the trap — clefts in front of and behind the incisive foramen do not share an embryological mechanism. [Vol I Chapter 16 · Vol II §28.1]

940 TTTTT

A True — this is the figure quoted for populations of European origin, with higher rates in Asian populations
B True — the male predominance is consistent and is the reverse of that seen in isolated cleft palate
C True — the remaining 30 per cent are syndromic, which is why every affected infant is examined for other anomalies
D True — both the lower incidence and the female predominance distinguish it from cleft lip
E True — the far higher syndromic proportion is the reason genetic assessment matters more in this group

[Vol I Chapter 16 · Vol II §28.2]

MNEMONIC **700/2/70 vs 2000/4/50** — CL/P is 1 in 700, male 2:1, 70% non-syndromic; isolated CP is 1 in 2000, female 4:1, 50% non-syndromic.

941 A LAHSHAL is a striped code covering lip, alveolus, hard palate, soft palate and the mirror sites, with capitals for complete and lower case for incomplete clefting, so laterality and severity both survive into the dataset. Numerical severity weighting is the trap — LAHSHAL is descriptive rather than a score and yields no number to analyse. [Vol I Chapter 16 · Vol II §28.2]

MNEMONIC **Veau I-IV** — soft palate only, hard and soft palate, unilateral complete lip and palate, bilateral complete: four groups by anatomy alone.

942 **E** A protruding premaxilla beneath a bilateral lip repair places the repair under tension and distorts it, so the segment is brought back with strapping, a passive plate or nasoalveolar moulding before surgery. Excising the premaxilla is the historical answer and the trap — it removes the tooth-bearing segment and produces severe midface deficiency. [Vol I Chapter 16 · Vol II §28.3]

943 **D** A cleft palate prevents the negative intraoral pressure on which sucking depends, so milk must be delivered by compression with the infant upright to limit nasal escape, supported by the specialist cleft nurse from diagnosis. Nasogastric feeding is the trap — most cleft infants feed orally with modified equipment, and tube feeding is reserved for airway problems such as Pierre Robin sequence. [Vol I Chapter 16 · Vol II §28.3]

944 **TFFFF**

A True — this is the conventional window, with the rule of tens used to confirm fitness for anaesthesia
B False — the figures are 10 pounds in weight, 10 weeks of age and a haemoglobin of 10 grams per decilitre
C False — the palate is closed between six and twelve months, before speech develops rather than after
D False — the repaired lip moulds the segments together and is a deliberate part of early management
E False — scarring around the lip and across the palate is a recognised cause of later maxillary retrusion

[Vol I Chapter 16 · Vol II §28.3]

MNEMONIC **Rule of tens** — 10 weeks old, 10 lb, 10 g/dL haemoglobin: fitness for lip repair, not 10 kg and not 10 months.

945 **B** Repair before the first words gives the palate a chance to function for speech, and because the tensor veli palatini is abnormal in a cleft, otitis media with effusion is sought at the same time and grommets considered. Deferring repair to three years to protect growth is the tempting answer and was once practised, but the cost to speech outweighs the growth that is gained. [Vol I Chapter 16 · Vol II §28.3]

946 **E** Grafting when the canine root is between one half and two thirds formed, usually between eight and ten years, allows the tooth to erupt into the graft, and it is that eruption which preserves the bone. Waiting until an adult implant is planned is the trap — a graft with no tooth erupting through it resorbs, and the segments remain mobile in the meantime. [Vol I Chapter 16 · Vol II §28.4]

947 **B** A successful secondary graft converts two mobile segments into one continuous alveolar arch, closes the oronasal fistula and supplies bone into which the canine can erupt and teeth can subsequently be moved. Expecting it to correct the midface is the trap — the graft is alveolar and does nothing for maxillary retrusion, which may still require a Le Fort I. [Vol I Chapter 16 · Vol II §28.4]

948 **TFTTF**

A True — cancellous bone is preferred for its osteogenic cells and rapid revascularisation
B False — the graft is placed before eruption, and eruption through it is what maintains its volume
C True — expanding a collapsed arch afterwards would risk splitting the consolidated graft
D True — movement of the canine or an incisor into grafted bone is one of the aims of the procedure

E False — the canine usually erupts through the graft, and the absence of a lateral incisor does not alter the indication

[Vol I Chapter 16 · Vol II §28.4]

949 **D** The lateral incisor lies in the line of the cleft and is absent, duplicated or hypoplastic in the great majority of cases, and the canine erupts through distorted bone, so rotation and abnormal inclination are the rule. Expecting the canine to be the missing tooth is the trap — it is usually present but mispositioned, which is exactly why grafting is timed to its eruption. [Vol I Chapter 16 · Vol II §28.5]

950 **A** The comparison isolates the effect of the operations: scar tissue across the palate and around the lip restricts sutural and appositional growth of the maxilla, which is why technique and timing are judged partly on growth. The milder-phenotype explanation is the serious alternative, but it fails because retrusion in repaired cases tracks the extent of surgery rather than the extent of the cleft. [Vol I Chapter 16 · Vol II §28.6]

951 **C** Presurgical orthopaedics can bring the alveolar segments and the alar base into a better position for repair and is valued by many parents, but randomised evidence has not demonstrated a lasting gain in growth, speech or operations avoided. The claim of a permanent nasal correction is the trap — the nose is improved at repair, but the deformity commonly recurs and secondary rhinoplasty remains common. [Vol I Chapter 16 · Vol II §28.3]

952 **FFFFF**

A False — care is delivered by a commissioned multidisciplinary team from diagnosis onwards

B False — the orthodontist contributes records and planning from the mixed dentition and often earlier

C False — assessment and therapy begin in infancy, long before the last operation

D False — hearing is monitored in every child with a cleft palate because eustachian tube function is abnormal

E False — national audits of outcome were the reason cleft services were centralised in the first place

[Vol I Chapter 16 · Vol II §28.7]

MNEMONIC **Surgeon, orthodontist, SLT, audiology, psychology, genetics, specialist nurse** — the commissioned multidisciplinary cleft team, from diagnosis onward.

953 **D** In Pierre Robin sequence the small mandible displaces the tongue base backwards, so prone positioning and graded airway support carry most infants through while mandibular catch-up occurs, and airway and feeding take precedence over the palate. Early palate repair is the trap — closing the palate narrows the airway further, which is why repair is deferred until the airway is secure. [Vol I Chapter 16 · Vol II §28.8]

MNEMONIC **Pierre Robin triad** — micrognathia, glossoptosis, airway obstruction, often with a cleft palate: the small jaw pushes the tongue back to block it.

954 **B** Treacher Collins syndrome is a bilateral autosomal dominant hypoplasia of the first and second branchial arches, and the malar and mandibular deficiency with a high angle produces a convex profile and anterior open bite beyond orthodontic correction. Confusing it with hemifacial microsomia is the trap — that condition is characteristically unilateral and is graded by the Pruzansky classification of the ramus. [Vol I Chapter 16 · Vol II §28.9]

MNEMONIC **Treacher Collins triad** — malar hypoplasia, mandibular hypoplasia, ear anomalies: bilateral first and second arch dysplasia, unlike unilateral HFM.

955 **E** In cleidocranial dysplasia the permanent teeth fail to erupt because of supernumeraries and a defective eruption mechanism, so a planned sequence of removal, exposure and traction

is what brings them into the arch. Waiting for spontaneous eruption is the trap — the delay here reflects a failure of mechanism rather than of timing, and waiting achieves nothing. [Vol I Chapter 16 · Vol II §28.9]

MNEMONIC **Cleidocranial triad** — absent or hypoplastic clavicles (shoulders touch), open sutures and fontanelles, multiple unerupted supernumeraries.

956 TTTTF

A True — hypohidrotic ectodermal dysplasia is most often X-linked recessive, with carrier girls mildly affected
B True — the alveolus develops with the teeth, so severe hypodontia leaves a knife-edge ridge
C True — prostheses are provided early for function and appearance and replaced as the child grows
D True — the anterior mandible is the one site where early implants are accepted in severe cases
E False — with so few teeth there is nothing to close space with, and replacement rather than closure is the aim

[Vol I Chapter 16 · Vol II §28.9]

957 **A** Craniosynostosis with exorbitism and midface retrusion but normal extremities is Crouzon syndrome, and because the deficiency extends to and above the orbital level a Le Fort III or monobloc advancement, often by distraction, is required. Apert syndrome is the trap — the craniofacial picture overlaps closely, but it is distinguished by symmetrical syndactyly of the hands and feet. [Vol I Chapter 16 · Vol II §28.9]

958 **C** Periodontitis in Down syndrome begins early, progresses rapidly and is the main threat to the dentition, so the practical question is whether plaque control and maintenance can be sustained before any appliance is added. Antibiotic prophylaxis is the trap — cardiac lesions are indeed common in this group, but NICE does not recommend routine prophylaxis for dental procedures. [Vol I Chapter 16 · Vol II §28.9]

959 **B** A small and malformed but present condyle and ramus is Pruzansky grade II, and in a growing child a hybrid functional or unilateral bite-block appliance is used to encourage vertical development on the affected side while the maxilla erupts down with it. Grade III is the trap — that grade denotes an absent ramus and condyle, which is what makes a costochondral graft necessary. [Vol I Chapter 16 · Vol II §28.9]

960 TTFFF

A True — it is quoted as second only to clefting in frequency among congenital facial anomalies
B True — these two features are what convert hemifacial microsomia into the oculo-auriculovertebral spectrum
C False — it is characteristically unilateral, which is the feature that separates it from Treacher Collins syndrome
D False — Pruzansky grades the mandibular ramus and condyle, from mildly small to entirely absent
E False — the asymmetry increases with growth, which is why growing children are monitored and treated actively

[Vol I Chapter 16 · Vol II §28.9]

961 **D** Hypernasality with nasal emission on pressure consonants means the velopharyngeal sphincter is not sealing, and structural insufficiency is assessed by nasendoscopy or videofluoroscopy with pharyngoplasty or palate re-repair considered. Attributing it to learned articulation is the trap — compensatory articulations such as glottal stops do occur and respond to therapy, but they do not produce nasal escape. [Vol I Chapter 16 · Vol II §28.6]

962 **A** In a cleft the tensor veli palatini has no functional sling to act against, so the eustachian tube fails to open, effusion accumulates and conductive loss follows, which is why hearing is monitored from birth and grommets are often placed at palate repair. Blaming the repair itself is the trap — the dysfunction is intrinsic to the cleft and is present before any surgery is undertaken. [Vol I Chapter 16 · Vol II §28.6]

963 **TFFFF**

A True — expanding afterwards risks splitting the graft, so the collapsed segments are corrected first
B False — lower arch alignment belongs to the definitive phase in adolescence, not to the pre-graft phase
C False — caries risk is higher in cleft children, and prevention is a formal part of the team protocol
D False — a malformed or supernumerary tooth compromising the graft site is removed at or shortly before grafting
E False — definitive alignment waits until the segments are united and the canine has erupted

[Vol I Chapter 16 · Vol II §28.5]

964 **C** Scar tissue resists a large single-stage advancement and relapse is high, so gradual distraction with soft tissue accommodation is preferred, and because advancing the maxilla stretches the velopharynx it is planned jointly with the speech and language therapist. Mandibular setback is the tempting compromise — it protects speech but worsens an already flat midface and answers the wrong deficiency. [Vol I Chapter 16 · Vol II §28.6]

965 **E** Root development rather than chronological age defines the window, because the graft must be in place for the canine to erupt through it and children reach that stage at widely differing ages. Using the eighth to tenth birthdays is the tempting proxy — it describes the usual age range but will mis-time the graft in early and late developers. [Vol I Chapter 16 · Vol II §28.4]

966 **TTTFF**

A True — the lip is closed first, with the rule of tens used to confirm fitness for anaesthesia
B True — closure before the first words gives the palate a chance to function for speech
C True — the window is set by canine root development rather than by the birthday itself
D False — definitive alignment follows the graft and the eruption of the canine, in adolescence
E False — it waits until growth has ceased, which in a cleft patient is late teens at the earliest

[Vol I Chapter 16 · Vol II §28.3]

MNEMONIC **Lip 3-6 months, palate 6-12 months, graft 8-10 years, surgery late teens** — the cleft care timeline in order, each step waiting on the last.

967 **A** Bifid uvula, a zona pellucida and a notched hard palate together constitute a submucous cleft, and removing the adenoid pad against which such a palate has been sealing commonly precipitates velopharyngeal insufficiency. Treating the bifid uvula as an incidental variant is the trap — in isolation it may be, but with the other two signs it marks a palate dependent on the adenoid for closure. [Vol I Chapter 16 · Vol II §28.8]

968 **C** The trap is the shared eponym, and the two findings are unrelated. A localised hypoplastic defect on a single successor whose predecessor was infected is a Turner tooth, the periapical infection having disturbed the developing enamel; short stature with a webbed neck in a girl points instead to Turner syndrome, a chromosomal disorder with no causal link to the enamel defect. Each needs its own referral and its own management. [Vol I Chapter 16 · Vol II §28.4]

Scored under 70% of the 63 marks on this block? Re-read Volume II §28, then sit it again in a fortnight.

Block AC — Risks of orthodontic treatment and the medical history (Q969–998 · 62 marks · Vol II §29)

969 **E** Improved dental aesthetics is the one benefit resting on strong evidence, and it is the reason most patients seek treatment, whereas the remaining claims range from weak to entirely unsupported. Caries reduction is the tempting answer because aligned teeth look easier to clean, but the evidence for it is weak and plaque scores improve only moderately. [Vol I Chapter 2 · Vol II §29.1]

970 **TTTFT**

A True — a risk ratio of 0.58 is a 42 per cent relative reduction and is the best-supported health benefit
B True — this is the benefit with the strongest evidence base and the one patients most often seek
C True — plaque scores improve, though the size of the effect is modest
D False — there is no evidence that orthodontic treatment either causes or prevents temporomandibular disorder
E True — a caries benefit is plausible but the evidence supporting it remains weak

[Vol I Chapter 2 · Vol II §29.1]

971 **B** A risk ratio of 0.58 is a 42 per cent relative reduction, and because his baseline risk is high — a 10 mm overjet with the lips apart — the absolute number of injuries prevented is correspondingly large. Applying the same absolute gain to every child is the trap — a relative reduction delivers very little to a child whose overjet and lip coverage already make trauma unlikely. [Vol I Chapter 2 · Vol II §29.1]

972 **D** At least one sign is found in about three quarters of the population and one symptom in about a third, and no causal link with malocclusion or its correction has been established, so treatment is planned on other grounds and the symptoms managed conservatively. Promising relief is the trap — it is what the patient hopes to hear and it commits the clinician to an outcome the evidence will not support. [Vol I Chapter 2 · Vol II §29.1]

973 **C** Studies of airway dimension after premolar extraction and incisor retraction show no clinically meaningful reduction and no association with sleep-disordered breathing, so the crowding should be treated on its own merits. Substituting expansion for extraction is the trap — it concedes a mechanism that has never been demonstrated and imposes a plan the crowding does not support. [Vol I Chapter 2 · Vol II §29.1]

974 **FFFFF**

A False — no causal relationship between malocclusion, its treatment and temporomandibular disorder has been shown
B False — plaque scores improve modestly, but periodontal disease is driven by plaque control and host response
C False — airway dimensions are not meaningfully reduced by extraction and retraction
D False — the evidence for any speech benefit is weak, and articulation commonly adapts to the malocclusion
E False — the evidence for a caries benefit is weak and no lifetime reduction has been demonstrated

[Vol I Chapter 2 · Vol II §29.1]

975 **A** White spot lesions appear in up to half of patients and can be seen within six months of bonding, and the effective response combines reinforced hygiene, dietary advice and professionally applied fluoride at six to eight weekly intervals. Debonding at once is the tempting

safe option — it is reserved for lesions progressing despite this regimen, because abandoning treatment brings problems of its own. [Vol I Topic 9 · Vol II §29.2]

MNEMONIC **Up to half by six months, varnish every 6-8 weeks** — white spot prevalence during fixed appliances and the interval for reinforcing fluoride.

976 **E** Lesions improve substantially over the first months after appliance removal as saliva and fluoride remineralise the surface layer, so any irreversible intervention should wait until that improvement has plateaued. Resin infiltration is the tempting immediate answer — it is reasonable for lesions that persist, but performing it at debond treats enamel that would have improved on its own. [Vol I Topic 9 · Vol II §29.2]

977 **TTFFF**

A True — this is the figure quoted for conventional fixed appliances without an intensive preventive regimen
B True — these sites combine bracket or band margins with poor access for the toothbrush
C False — lesions can appear within six months of bonding, which is why early review matters
D False — rinses depend on daily compliance, which is exactly what fails in the patients most at risk
E False — the surface layer is intact in an early lesion, and it is this that allows remineralisation

[Vol I Topic 9 · Vol II §29.2]

978 **D** Blunt, pipette-shaped or previously resorbed roots identified before treatment are the best available predictor of severe resorption and justify a shorter plan, lighter forces and an interim radiograph. The amount of planned retraction is the serious alternative — large apical displacement and long treatment do raise the risk, but they modify it far less than root form visible before a bracket is placed. [Vol I Topic 9 · Vol II §29.3]

979 **B** Three millimetres on a central incisor exceeds the threshold at which a pause is advised, and three to six months without active force allows the resorbed surfaces to repair with cementum before deciding how much further treatment to attempt (Levander 1994). Continuing unchanged is the trap — minor radiographic resorption is indeed common, but this is not minor. [Vol I Topic 9 · Vol II §29.3]

980 **E** Minor radiographic shortening is common at up to about 73 per cent and histological evidence is found in up to 90 per cent, but the figure that governs consent is the 1 to 5 per cent who lose a third or more of a root. Quoting the histological figure alongside a severe rate of 30 per cent is the trap — it conflates a microscopic finding with clinically significant loss. [Vol I Topic 9 · Vol II §29.3]

981 **TTTTT**

A True — minor apical changes are the rule rather than the exception on post-treatment films
B True — microscopic resorption lacunae are almost universal, which is why the histological figure is so high
C True — this is the clinically significant group and the figure that belongs in the consent discussion
D True — the resorbed surface repairs with cementum once the force is removed (Levander 1994)
E True — pre-existing root form is the strongest predictor available before treatment begins

[Vol I Topic 9 · Vol II §29.3]

MNEMONIC **90 histological, 73 radiographic, 1-5 severe** — OIIRR figures, and only the last one belongs in a consent discussion.

982 **B** Appliance pain follows a predictable course, peaking at around 48 hours and largely resolved by seven days, so a call at this point is expected and is answered with explanation and simple analgesia. Removing the archwire is the trap — it treats a normal inflammatory response as a complication and simply restarts the same sequence at the next visit. [Vol I Topic 9 · Vol II §29.4]

> MNEMONIC **Peaks at 48 hours, gone by 7 days** — the expected course of fixed appliance pain; reassurance and simple analgesia is all it needs.

983 **D** Percussion tenderness with an apical radiolucency in a previously traumatised incisor indicates pulpal necrosis, and moving a tooth with an inflamed apex increases the risk of resorption, so the endodontic problem is addressed first. Continuing to debond is the trap — discolouration alone might reasonably be watched, but tenderness with a radiolucency is a diagnosis rather than a cosmetic finding. [Vol I Topic 9 · Vol II §29.4]

984 **A** Most post-treatment recession is stable and asymptomatic, so the decision to graft turns on documented progression, sensitivity or an unacceptable appearance rather than on the depth measured at one visit. Removing the bonded retainer is the trap — a plaque-retentive retainer needs attention, but removing it here invites relapse of the very proclination that produced the recession. [Vol I Topic 9 · Vol II §29.5]

985 **TFFFF**

A True — appliance-related hyperplasia is inflammatory and regresses once the plaque trap is removed
B False — recession follows labial movement and is seen on labial surfaces, typically of lower incisors
C False — labial movement thins the plate and increases recession risk in a thin biotype
D False — most hyperplasia resolves without surgery, so gingivectomy at debond is premature
E False — black triangles follow the divergent root form and lost papilla height, and proclination does not prevent them

[Vol I Topic 9 · Vol II §29.5]

986 **C** Cutaneous nickel sensitivity affects 10 to 30 per cent of females and 1 to 3 per cent of males and is far commoner than intraoral reaction, but a clear history justifies titanium or coated components, which are readily available. Insisting on patch testing first is the tempting answer — it is useful in doubtful cases, but the practical response to a convincing history is simply to avoid the metal. [Vol I Topic 10 · Vol II §29.6]

> MNEMONIC **10-30% female, 1-3% male** — cutaneous nickel sensitivity rates; a convincing history, not a patch test, is what changes the appliance choice.

987 **D** Latex allergy affects about 4.3 per cent of the general population and 9.7 per cent of healthcare workers, and repeatedly exposed groups such as children with spina bifida carry a substantially higher risk, so the whole episode of care is managed latex-free. Test exposure in the mouth is the trap — deliberately challenging a possibly sensitised patient risks the reaction the precaution exists to prevent. [Vol I Topic 10 · Vol II §29.6]

> MNEMONIC **4.3% general population, 9.7% healthcare workers** — latex sensitisation rates; repeatedly exposed groups run higher still.

988 **B** Two independent safety features are mandatory — a mechanism that releases if the bow is pulled and a device that prevents recoil or accidental detachment — and a bow that has come off twice is a warning that both must be checked and the patient re-instructed. Dismissing it as a minor event is the trap — the injuries that matter are ocular and they follow exactly this pattern of repeated night-time detachment. [Vol I Topic 10 · Vol II §29.7]

989 **TTTFF**

A True — a release mechanism and an anti-recoil or locking device are both required, not one or the other
B True — this regimen holds the molars without moving them distally
C True — distalisation requires both longer daily wear and a heavier force than anchorage reinforcement
D False — the injuries of greatest concern have occurred at night, when the bow disengages during sleep
E False — repeated detachment is the warning sign that precedes injury and requires the safety devices to be checked

[Vol I Topic 10 · Vol II §29.7]

990 E Late lower incisor crowding occurs whether or not treatment was ever carried out, so the honest position is that holding a result requires retention for as long as the result is wanted, and this is said before treatment begins rather than at debond. The two-year figure is the trap — supracrestal fibres do remodel slowly, but they are only one of several mechanisms and none of them stops at two years. [Vol I Topic 9 · Vol II §29.8]

991 A Bisphosphonates bind avidly to bone and suppress osteoclastic resorption for years, so movement is slow, extraction spaces fail to close and surgical sites may not heal, which makes extraction-based plans particularly hazardous. A drug holiday is the tempting fix — the skeletal reservoir means that stopping the drug does not restore turnover within any useful period. [Vol I Chapter 7 · Vol II §29.9]

992 C Poorly controlled diabetes impairs the periodontal response and healing, so appliances placed on an unstable patient risk attachment loss for which the treatment will be blamed; well-controlled diabetic patients are treated normally. Timing appointments around hypoglycaemia is sound practice but is the trap here — it addresses the visit rather than the periodontal risk that governs the decision. [Vol I Chapter 7 · Vol II §29.9]

993 TTTTF

A True — an acrylic plate displaced during a convulsion is an inhalation and trauma hazard
B True — the INR must be known and within range close to the day of the extraction
C True — condylar destruction restricts mandibular growth and rotates the mandible backwards
D True — irradiated bone is hypovascular and hypocellular, so extraction risks osteoradionecrosis
E False — NICE does not recommend routine antibiotic prophylaxis for dental procedures, prevention resting on oral health

[Vol I Chapter 7 · Vol II §29.9]

MNEMONIC **MJTHREADS** — Myocardial infarction, Jaundice, TB, Hypertension/rheumatic fever, Epilepsy, Asthma, Diabetes, Stroke: the medical history checklist.

994 B A loose acrylic plate is an inhalation and trauma hazard during a seizure, so fixed appliances are preferred, and phenytoin-associated gingival overgrowth makes reinforced hygiene part of the same decision. Adding clasps is the trap — better retention reduces but does not remove the hazard, and the plate can still be displaced during a convulsion. [Vol I Chapter 7 · Vol II §29.9]

995 D Extractions in a patient with a factor deficiency are planned jointly with the haemophilia centre, which determines the cover required and the local haemostatic measures; the surgery itself is then straightforward. Relying on local measures alone is the trap — premolar sockets are indeed small, but mild haemophilia still produces delayed and prolonged bleeding that local measures will not control. [Vol I Chapter 7 · Vol II §29.9]

996 TTTTT

A True — routine prophylaxis for dental procedures is not recommended, and this applies to orthodontic care
B True — cumulative everyday bacteraemia far outweighs that of any single dental procedure
C True — this is the figure that explains why prevention is taken so seriously in this group
D True — which is precisely the reasoning behind the shift away from procedure-based prophylaxis
E True — untreated infection is a far greater risk than any appliance placed in a healthy mouth

[Vol I Chapter 7 · Vol II §29.9]

997 **A** Condylar resorption in active juvenile idiopathic arthritis progresses independently of any appliance, so orthodontic or surgical correction undertaken during active disease is built on a moving foundation and treatment is staged around medical control. Fitting a functional appliance is the trap — a damaged condyle has no reliable growth response to stimulate, and the open bite is skeletal rather than habitual. [Vol I Chapter 7 · Vol II §29.9]

998 **C** Irradiated bone is hypocellular, hypovascular and hypoxic, so tooth movement is slow and unpredictable and any surgical insult carries a lifelong risk of osteoradionecrosis; the dose and the fields must be known before anything is offered. Waiting a year before extracting is the trap — the risk does not diminish with time and is not abolished by a delay. [Vol I Chapter 7 · Vol II §29.9]

Scored under 70% of the 62 marks on this block? Re-read Volume II §29, then sit it again in a fortnight.

Block AD — Orthodontic practice, infection control and governance (Q999–1023 · 49 marks · Vol II §30)

999 **E** Justification is the first of the three principles and requires a recorded, patient-specific clinical reason showing that the benefit of the exposure outweighs its detriment, so a routine or protocol-driven film is not justified. A patient dose limit is the trap — limits apply to staff and to members of the public, never to a patient undergoing a diagnostic exposure for their own benefit. [Vol I Part V.2 · Vol II §30.1]

1000 **A** Under the same-lingual-opposite-buccal rule an object appearing to move with the tube lies on the lingual or palatal side, which fits the majority of impacted canines, up to 85 per cent of which are palatal. Reading it as buccal is the trap — that is the correct conclusion when the object moves in the opposite direction to the tube shift. [Vol I Part V.2 · Vol II §30.1]

> MNEMONIC **SLOB** — Same Lingual, Opposite Buccal: the canine moved with the tube shift, so it lies palatal to the lateral incisor root.

1001 TTTTT

A True — justification is patient-specific, which is why routine screening films cannot be justified
B True — this is the practical meaning of ALARP in the selection of view, exposure factors and equipment
C True — a patient's exposure is undertaken for that patient's own benefit and is controlled differently
D True — the entry is the evidence that the exposure was justified and is required at the time
E True — the duty attaches to the whole image, and incidental findings must be acted upon

[Vol I Part V.2 · Vol II §30.1]

> MNEMONIC **Justification, optimisation, limitation** — the three IRMER principles: benefit outweighs harm, dose kept ALARP, limits bind staff/public, not patients.

1002 **C** Cone beam imaging is justified only where it will alter management, and here both the localisation and the incisor roots have already been assessed at a small fraction of the dose. The argument that resorption can never be excluded on plain films is the tempting one — cone beam imaging is indeed more sensitive, and that is exactly why it is reserved for cases in which resorption is suspected. [Vol I Part V.2 · Vol II §30.1]

1003 **E** Every exposure carries a duty to examine and report the whole image, and a finding discovered in that process must then be acted upon; failure to do so is a recurring source of complaint. Reporting only the region of interest is the trap — justification defines why the film was taken, not how much of it the clinician is responsible for. [Vol I Part V.2 · Vol II §30.1]

1004 **B** Dose limits are set for occupational exposure and for the public but not for patients, because a patient's exposure is made for that patient's own benefit and is controlled by justification and optimisation instead. Quoting one millisievert is the trap — it is a real figure, but it is the limit for members of the public rather than for a patient being investigated. [Vol I Part V.2 · Vol II §30.1]

1005 **TFFFF**

A True — it is the only standard view that relates the jaw bases to each other and to the cranial base
B False — a single view gives no parallax and cannot separate buccal from palatal position
C False — bitewings are the appropriate view, panoramic films being insensitive to early approximal lesions
D False — parallax requires a deliberate change of tube position between the two exposures
E False — it is justified only where the two-dimensional films leave a question that would change management

[Vol I Part V.2 · Vol II §30.1]

1006 **D** The decontamination cycle runs cleaning, inspection, disinfection, sterilisation and protected storage in that order, and organic debris shields micro-organisms from the steam so that nothing beneath it is sterilised. Extending the holding time is the trap — time and temperature govern the steam that reaches a surface, not the surfaces the steam never contacts. [Vol I Part V.3 · Vol II §30.2]

> **MNEMONIC** **Clean, inspect, disinfect, sterilise, store** — the decontamination cycle in strict order; organic debris blocks every step after it.

1007 **C** Semi-critical items contact intact mucosa or non-intact skin without penetrating it and require sterilisation or at minimum high-level disinfection, which is where most intraoral orthodontic instruments sit. Calling a scalpel semi-critical is the trap — anything penetrating tissue or entering the vascular system is critical and must be sterile at the point of use. [Vol I Part V.3 · Vol II §30.2]

> **MNEMONIC** **Critical, semi-critical, non-critical** — Spaulding's classes: penetrates tissue, touches mucosa only, touches intact skin only, each its own standard.

1008 **A** Box joints and hinges are the hardest feature of an orthodontic instrument set to clean, which is why pliers are opened for processing, passed through an ultrasonic bath or washer-disinfector and inspected under illuminated magnification. Impression trays are the tempting answer — retained alginate is conspicuous, but being conspicuous is exactly why it is removed. [Vol I Part V.3 · Vol II §30.2]

1009 **TTTFF**

A True — each step depends on the one before it, and none of them can be taken out of sequence
B True — an active vacuum phase is needed to draw air out of pouches and lumens so that steam can enter

C True — passive air displacement is sufficient for a solid unwrapped load used straight away
D False — sterility is lost on storage unless the instruments are wrapped or kept in a protected environment
E False — ultrasonic cleaning removes debris and is a cleaning step, not a sterilisation process

[Vol I Part V.3 · Vol II §30.2]

1010 E A type N machine relies on passive displacement of air and cannot evacuate the air trapped inside a pouch or a lumen, so wrapped and hollow loads require an active vacuum cycle. Extending the holding time on a type N machine is the trap — it is residual air, not insufficient time, that prevents steam reaching the instrument. [Vol I Part V.3 · Vol II §30.2]

1011 D A band that has entered the mouth is contaminated however briefly it was there, and it re-enters the decontamination cycle exactly as any other reusable instrument does. Chemical immersion is the tempting shortcut — it is disinfection rather than sterilisation, and the band will next be seated on another patient's tooth. [Vol I Part V.3 · Vol II §30.2]

1012 B Laboratory items carry contamination in both directions, so work is disinfected as it leaves the surgery and again on return, with the method used recorded and communicated to the laboratory. Assuming the laboratory will deal with it is the trap — the duty is shared, and an undisinfected impression exposes laboratory staff before anyone there has the chance to act. [Vol I Part V.3 · Vol II §30.2]

1013 FFFFF

A False — nothing can be sterilised through debris, whatever the time or temperature selected
B False — anything that has entered the mouth re-enters the full decontamination cycle
C False — ultrasonic processing is a cleaning step that precedes sterilisation rather than replacing it
D False — disinfection is required in both directions, because contamination travels both ways
E False — the wound is washed under running water without scrubbing, which would abrade and inoculate it

[Vol I Part V.3 · Vol II §30.2]

1014 E The immediate measures are to encourage bleeding, wash without scrubbing, cover, report and be risk-assessed by occupational health the same day, because post-exposure prophylaxis where indicated is time-critical. Deferring the report to the end of the week is the trap — the paperwork can wait but the risk assessment cannot, and prophylaxis loses its effect with delay. [Vol I Part V.3 · Vol II §30.3]

1015 B An audit is a loop, and it is closed by re-measuring against the same standard after the change, which is the step that shows whether practice has actually improved. Randomisation is the trap — it is the right answer to a research question, but audit measures practice against an existing standard rather than creating new knowledge. [Vol I Part V.4 · Vol II §30.4]

> MNEMONIC **Standard, measure, change, re-measure** — the audit cycle; it only closes when practice is re-measured against the same standard.

1016 D Allocating patients in order to answer a question about which treatment is better creates new knowledge rather than measuring existing practice, and that makes it research with the governance requirements that follow. Calling it audit is the trap — audit compares what is done against an agreed standard and never allocates patients in order to find out. [Vol I Part V.4 · Vol II §30.4]

1017 TTTTF

A True — without a standard there is nothing to measure against and the exercise is description, not audit
B True — the re-measure closes the cycle and is the step most often omitted
C True — this is the feature that distinguishes service evaluation from audit
D True — it measures existing practice rather than creating new knowledge
E False — comparing two appliances to find out which is better is research and needs a research design

[Vol I Part V.4 · Vol II §30.4]

1018 **A** Greatly improved requires a reduction of at least 22 points or of at least 30 per cent, and this case clears both thresholds comfortably. Insisting on a 30-point reduction is the trap — the percentage and the absolute criteria are alternatives, and the absolute threshold is 22 points rather than 30. [Vol I Part V.4 · Vol II §30.5]

1019 **C** PAR is a relatively coarse index weighted towards alignment and overjet, whereas ABO grading scores marginal ridges, buccolingual inclination, occlusal contacts and root angulation, which is precisely the detail aligners struggle to deliver. Concluding that the two must agree is the trap — indices are built for different purposes, and a disagreement between them is informative rather than an error. [Vol I Part V.4 · Vol II §30.5]

1020 **D** The dental health and aesthetic components of IOTN grade the severity of an untreated malocclusion in order to establish priority, and a treated occlusion is measured instead with an outcome index such as PAR, ABO grading or ICON. Reproducibility is the tempting objection — IOTN is reasonably reproducible with training, and even a perfectly reproducible index of need would still not measure outcome. [Vol I Part V.4 · Vol II §30.5]

1021 **TTFFF**

A True — this is what the index was designed to do and why it is quoted as a single-instrument alternative
B True — either criterion is sufficient, which is why both figures must be known
C False — the ten-point photographic scale is applied by the clinician, though patient perception may also be recorded
D False — marginal ridge levels are scored by ABO grading and are among the details PAR does not capture
E False — buccolingual inclination is an ABO grading component and is absent from the PAR weighting

[Vol I Part V.4 · Vol II §30.5]

1022 **B** A good audit topic has an existing standard, data that are already recorded, a manageable sample and a change the team can actually make, and radiographic justification satisfies all four. Comparing two appliances or two adhesives is the trap — those are research questions, because no agreed standard exists against which practice could be measured. [Vol I Part V.4 · Vol II §30.4]

1023 **E** Cleft care is commissioned through centralised multidisciplinary teams whose timing of expansion, grafting and definitive orthodontics is planned as a whole, so the referral is to that service rather than to any individual clinician. Starting the expansion first is the trap — it looks helpful, but it pre-empts a graft that must follow expansion on the team's own timetable. [Vol I Part V.4 · Vol II §30.6]

Scored under 70% of the 49 marks on this block? Re-read Volume II §30, then sit it again in a fortnight.

Answer key grid

Eight answers per row. A single letter is a single best answer; a five-character string is a true/false question, read A to E, one mark per statement.

	1	2	3	4	5	6	7	8
1	A	C	TFTFT	E	B	FFTFF	D	C
9	A	TTTTT	E	D	TTFTF	B	E	B
17	FTTFT	D	A	FFTFT	C	D	B	E
25	A	C	B	TTFFT	D	A	C	FTTTT
33	E	A	C	E	FFFFF	B	D	C
41	TTTFT	A	E	D	TFFFF	B	E	B
49	TTTFF	D	A	C	FTFFT	D	B	E
57	A	C	TTFTF	B	D	A	FFTTF	C
65	E	A	TTTTF	C	E	B	FFFTF	D
73	C	A	TFTTF	E	D	FTFFF	B	E
81	TTFTT	B	D	A	FTTFF	C	D	B
89	TFFTT	E	A	TTTTT	C	B	D	A
97	C	TTFFF	E	A	C	FTTTT	E	B
105	D	FFTFT	C	A	E	TTTFF	D	B
113	E	FFFTT	B	D	A	TFTFF	C	D
121	B	TTTTT	E	A	FFFFF	C	B	TTFTT
129	D	A	FTFTF	C	E	A	TFFTF	C
137	E	B	D	C	A	E	D	B
145	TTTFT	E	B	D	FFTFF	A	C	D
153	TFTTT	B	E	A	FTTFF	C	B	D
161	TTFTT	A	C	E	FFFFF	A	C	E
169	TFTFT	B	D	C	A	TTTFF	E	D
177	B	FFFFF	E	B	D	A	TTFTT	C
185	D	B	TFTTT	E	A	C	B	FTFFT
193	D	A	FTTFT	C	E	TTTTT	A	C
201	E	FTFTF	B	D	C	TFTFT	A	E
209	D	TTFTF	B	E	B	FFFFF	D	A
217	C	TTTTF	D	B	E	FFTTF	A	C
225	TFFFT	B	D	A	TTTTT	C	E	A
233	TTFTF	C	E	B	TTTTF	D	C	A
241	FTFFF	E	D	B	TFTTT	E	B	TTTFT
249	D	A	FFTFF	C	D	B	TFFFF	E
257	A	C	TTFTT	B	D	A	C	FTTFT
265	E	A	C	E	TFTTF	B	D	C
273	TTTTT	A	E	D	B	FFFFF	E	B
281	D	TFFTF	A	C	D	FTFFF	B	E
289	A	TTFTT	C	B	D	FTFTF	A	C
297	E	TTFTT	A	C	E	FFTFF	B	D
305	C	TFTTF	A	E	D	FTTFF	B	E
313	B	FTFFT	D	A	C	TTTTT	D	B
321	E	FFTFF	A	C	B	FFTFT	D	A
329	C	TTTTT	E	A	C	FFFFF	E	B
337	D	TFTTF	C	A	E	TFTTT	D	B
345	E	TFTFT	B	D	A	FTFFF	C	D

	1	2	3	4	5	6	7	8
353	B	E	TTTTT	A	C	FFFFF	B	D
361	A	TTFTF	C	E	A	FTTTT	C	E
369	TTFTF	B	D	C	A	TFTFT	E	D
377	B	TTFTT	E	B	FTFFF	D	A	C
385	D	TFFTF	B	E	A	TTFFF	C	B
393	D	A	TTTTT	C	E	FFFFF	A	C
401	E	TTFTT	B	D	FTTFF	C	A	E
409	TFTTT	D	B	E	FFFTF	B	D	A
417	C	TTTTF	D	B	E	FTTFT	A	C
425	B	TFFFT	D	A	C	FTTTF	E	A
433	C	E	TTTTT	B	D	C	FTTTT	A
441	E	TFTTF	D	B	E	TFTTT	B	D
449	FTTFT	A	C	D	FFFFF	B	E	A
457	TFFFF	C	B	D	FFTFF	A	C	E
465	TTTTT	A	C	E	FFFFF	B	D	C
473	A	FFFFT	E	D	B	TFFTF	E	B
481	D	FTFFT	A	C	TTFTF	D	B	TTTTT
489	E	A	C	TFTFT	B	D	A	FFFFF
497	C	E	A	TTFTF	C	E	B	FTFFF
505	D	C	TTFTT	A	E	D	FFTTF	B
513	E	B	FTTTT	D	A	C	TFTTT	D
521	B	E	FTTFT	A	C	B	FFTTF	D
529	A	C	E	A	FFFFF	C	E	B
537	TFTTF	D	C	A	TTTTT	E	D	B
545	FTTFF	E	B	D	TTFTT	A	C	D
553	FTFFF	B	E	A	TFTTT	C	B	D
561	FFTFF	A	C	E	TTTFT	A	C	E
569	FFFTT	B	D	C	TTTTT	A	E	D
577	TTTFF	B	E	B	TFFTT	D	A	C
585	FFFFF	D	B	E	TFTTT	A	C	B
593	FTFTF	D	A	C	TFTTF	E	A	TTFFT
601	C	E	B	TTTFT	D	C	A	FTTFF
609	E	D	B	E	B	FFFFF	D	A
617	C	TTFTT	D	B	E	TFFTF	A	C
625	B	TTTTT	D	A	C	TFTTT	E	A
633	C	TFFFF	E	B	D	TTFTF	C	A
641	TTTFF	E	D	B	FTFTF	E	B	D
649	TTTFT	A	C	D	FFTTF	B	E	TTTTT
657	A	C	B	FTFTT	D	A	C	FFFFF
665	E	A	C	TFTTT	E	B	D	TFFTT
673	C	A	TTFTF	E	D	B	FFFFT	E
681	B	D	A	C	FTTFT	D	B	E
689	TTTTT	A	C	B	FFTFF	D	A	C
697	TTTTF	E	A	C	TTTFT	E	B	D
705	FFFFF	C	A	E	TTFFT	D	B	E
713	FTFTT	B	D	A	TTTTF	C	D	B
721	FFTTF	E	A	C	TFFFT	B	D	A
729	C	TTFTT	E	A	C	FFFFF	E	B
737	D	TFFTF	C	A	E	FTFFF	D	B
745	E	TTTFT	B	D	A	TFTTF	C	D
753	B	E	FFFFF	A	C	B	FTFFF	D
761	A	C	FTFFF	E	A	C	TTFFF	E

	1	2	3	4	5	6	7	8
769	B	D	FTFTF	C	A	FFTTF	E	D
777	B	E	TTTFF	B	D	A	TFFTF	C
785	D	B	TTTTF	E	A	C	TTTTT	B
793	D	A	TTTTT	C	E	TFTFT	A	C
801	E	FTTFT	B	D	TTFTF	C	A	FFTTF
809	E	D	B	FFFFF	E	B	D	TFFFT
817	A	C	D	FTFFF	B	E	A	C
825	TTTTT	B	D	A	TFFTF	C	E	A
833	FFFFF	C	E	B	TTFFT	D	C	A
841	FTFFF	E	D	B	TFTTF	E	B	D
849	A	FFFFF	C	D	B	TTFTT	E	A
857	C	TFFFT	B	D	A	TTTTT	C	E
865	A	C	FTFTF	E	B	D	TFTTT	C
873	A	E	FTTFF	D	B	E	TFTFT	B
881	D	A	FTTFT	C	D	B	TTFTT	E
889	A	FFTTT	C	B	D	TFFTF	A	C
897	E	TTTFT	A	C	E	FTFFT	B	D
905	C	FFFTF	A	E	TTTFF	D	B	E
913	TTTFT	B	D	A	TTTTT	C	D	B
921	FFFFT	E	A	C	TTFFF	B	D	A
929	TTTTF	C	E	A	TFFTF	C	E	FFFFF
937	B	D	C	TTTTT	A	E	D	TFFFF
945	B	E	B	TFTTF	D	A	C	FFFFF
953	D	B	E	TTTTF	A	C	B	TTFFF
961	D	A	TFFFF	C	E	TTTFF	A	C
969	E	TTTFT	B	D	C	FFFFF	A	E
977	TTFFF	D	B	E	TTTTT	B	D	A
985	TFFFF	C	D	B	TTTFF	E	A	C
993	TTTTF	B	D	TTTTT	A	C	E	A
1001	TTTTT	C	E	B	TFFFF	D	C	A
1009	TTTFF	E	D	B	FFFFF	E	B	D
1017	TTTTF	A	C	D	TTFFF	B	E	

Mark sheet

One row per block. Record the date and the score out of the block total in marks — a true/false question carries five, one per statement. Anything under 70 per cent sends you back to the Volume II section in the last column before you sit it again.

Block	Subject	Qs	Marks	1st: date / score	2nd	3rd	Vol II
A	Concepts of growth and development	26	50				§1
B	Craniofacial growth, prenatal and postnatal	30	58				§2
C	Development of the dentition	40	80				§3
D	The biology of tooth movement, eruption and root resorption	45	89				§4
E	Occlusion, malocclusion and aetiology	30	58				§5
F	Treatment need, indices and epidemiology	25	49				§6
G	Evidence-based orthodontics	30	62				§7
H	Ethics, consent, communication and the psychosocial patient	31	63				§8
I	Clinical examination and records	30	58				§9
J	Imaging and cephalometrics	40	80				§10
K	Study models, space analysis and the digital set-up	25	49				§11
L	Diagnosis, the problem list and treatment planning	40	80				§12
M	Extraction decisions and space management	40	80				§13
N	Biomechanics, force control and anchorage	53	109				§14
O	Orthodontic materials, wires, bonding and instruments	45	89				§15
P	Removable and functional appliances	41	81				§16
Q	Fixed appliances — labial, lingual and aligners	41	81				§17
R	Treatment mechanics: Stage I, II and III	35	71				§18
S	Retention, relapse and stability	35	71				§19
T	Interceptive orthodontics and the developing dentition	45	89				§20
U	Class I malocclusion and bimaxillary protrusion	25	49				§21
V	Class II malocclusion	40	80				§22
W	Class III malocclusion	30	62				§23
X	Transverse problems and facial asymmetry	25	49				§24
Y	Vertical problems: deep bite and open bite	30	58				§25
Z	TMD, condylar trauma, the airway and myofunctional therapy	30	62				§26

Block	Subject	Qs	Marks	1st: date / score	2nd	3rd	Vol II
AA	Adult orthodontics and orthognathic surgery	30	62	_____	_____	_____	§27
AB	Cleft lip and palate and craniofacial syndromes	31	63	_____	_____	_____	§28
AC	Risks of orthodontic treatment and the medical history	30	62	_____	_____	_____	§29
AD	Orthodontic practice, infection control and governance	25	49	_____	_____	_____	§30

Part 3 — 207 viva questions with model answers

How to use this bank. Read the question, close the book, and answer it *out loud* to the end before reading the model. Answering in your head is not practice; the oral examination tests whether you can produce a structured spoken answer under pressure, and that is a separate skill from knowing the material.

The model answers are written in the first person, at the length and register of a good spoken answer — roughly twenty to forty seconds. They are not the only correct answers, and an examiner would accept a good deal of variation. What they are meant to demonstrate is *shape*: a direct opening sentence, the reasoning, the number where a number exists, and a close that says what you would actually do. Where a model answer names a study, name it or say nothing; a mis-attributed reference is worse than none.

The italic note beneath each answer is the examiner's point — what the question is really testing, or the trap it is set to catch.

The sections

Questions	Handbook	Section, and where it sits in *Notes In Orthodontics*
V1–V10	§1–§2	Growth, development and the craniofacial complex Chapters 3 and 4; Appendix A, Topic 1 background
V11–V20	§3	Development of the dentition Chapter 5; crash notebook *The arithmetic of the developing arch*
V21–V30	§4	The biology of tooth movement and root resorption Topic 1; Chapter 6; Appendix A, Topic 1
V31–V38	§5	Occlusion, aetiology and epidemiology Chapter 2
V39–V48	§6–§7	Treatment need, indices and the evidence base Chapters 1 and 2; Appendix D; crash notebook *Ten things the evidence overturned*
V49–V58	§8	Ethics, consent and communication Part V, unit V.1; Topic 9 in part; Chapter 7
V59–V68	§9–§10	Clinical examination and records Topic 2; Chapter 7; Appendix A, Topic 2
V69–V80	§11–§12	Space analysis, diagnosis and treatment planning Topic 3; Chapter 8; Appendix A, Topic 3
V81–V90	§13	Extraction decisions and space management Topic 4; Chapter 8; Appendix A, Topic 4
V91–V102	§14	Biomechanics and anchorage Topics 5 and 6; Chapters 6 and 11; Appendix A, Topic 5
V103–V111	§15	Materials, wires and bonding Topic 7; Chapter 10; Appendix A, Topic 7
V112–V121	§16–§17	Removable, functional and fixed appliances Topics 6 and 8; Chapters 9, 10 and 14; Appendix A, Topics 6 and 8
V122–V129	§18	Treatment mechanics and finishing

		Chapters 10 and 13; Supplements E and F
V130–V139	§19	Retention and stability
		Topic 20; Chapter 13; Appendix A, Topic 20
V140–V151	§20	Interceptive orthodontics and impacted teeth
		Topics 13, 14 and 15; Chapter 12; Appendix A, Topics 13–15
V152–V161	§22–§23	Class II and Class III management
		Topics 12 and 17; Chapters 9, 12 and 13; Appendix A, Topics 12 and 17
V162–V173	§24–§25	Transverse and vertical problems
		Topics 11, 16 and 19; Chapters 9, 12, 13 and 15
V174–V183	§26, §29	TMD, the airway and the medically complex patient
		Topics 10 and 18; Chapter 2, pp. 102–105; Chapter 7
V184–V193	§27, §28, §30	Adults, orthognathic surgery, clefts and governance
		Chapters 15 and 16; Part V, units V.2–V.4 for governance
V194–V207	Handbook Part 5	The questions that test judgement
		Crash notebook *The last hour*; the twelve traps

Growth, development and the craniofacial complex (V1–V10)

Notes In Orthodontics: Chapters 3 and 4; Appendix A, Topic 1 background
Supplements: Revision Handbook **§1–§2**
Practise this: 10 questions · answer each one aloud before reading the model

V1 Why does it matter whether this patient is still growing?

"Because it determines which of the three treatment routes is available to me. If the patient is growing I can attempt growth modification, overbite reduction by buccal segment extrusion is stable because condylar growth compensates, sutural expansion and facemask protraction are possible, and I can expect spontaneous improvement after well-timed extraction. If growth has finished, every one of those becomes either a dentoalveolar compromise or a surgical problem. So growth status is the first thing I establish and the last thing I return to.

The single most powerful discriminator in orthodontic planning. Say it in the first sentence of any planning answer.

V2 How would you assess whether a 12-year-old girl is at her growth spurt?

"I would use several indices rather than one. Clinically, serial standing height gives me height velocity, which is the direct measure, and I would ask about menarche, because menarche follows peak height velocity by about a year. If I already had a lateral cephalogram taken for another reason, I would stage the cervical vertebrae: CS3, at a mean age of 12, means the peak occurs during this year. What I would not do is take a radiograph purely to stage growth, and I would add that none of these indices is universally accepted.

Tests whether you will justify a radiograph. The examiner is listening for the caveat as much as for the staging.

V3 Why is chronological age of so little use in orthodontics?

"Because it correlates poorly with everything that matters. Two 11-year-olds can be two years apart in dental age and three years apart in skeletal age. Every clinically useful index of readiness is a maturational one: dental age from tooth formation, skeletal age from cervical vertebral maturation or a hand-wrist film, sexual maturation, and morphological age from height velocity.

A short answer. The mark is for naming the five ages, not for elaborating.

MNEMONIC The five ages: **chronological, dental, skeletal, sexual, morphological**. Chronological is the one to distrust.

V4 Is the mandible's growth modifiable?

"Within limits, yes. The condyle is a secondary, adaptive cartilage rather than a primary growth centre, which is precisely why the functional environment can influence it. But the randomised evidence is that the effect is a few millimetres at best and largely transient. So my honest position is Petrovic's: growth is modifiable within genetically determined limits, and those limits are individual.

A deliberately open question. The examiner wants the qualification, not enthusiasm.

V5 What is the difference between a site and a centre of growth?

"A site is simply a location where growth occurs. A centre is a location where growth occurs independently, under its own genetic control. The distinction matters because the whole nasal-septum-versus-sutures argument turns on it: Scott held that cartilage was a true centre and the sutures merely responded, whereas Moss held that neither was a centre and that both responded to the functional matrix.

Precision of language. It is a two-sentence answer if you know it and a rambling one if you do not.

V6 Describe Bjork's three mandibular rotations.

"Total, or true, rotation is rotation of the mandibular body relative to the cranial base, and it can only be measured on implants. Matrix, or apparent, rotation is what a cephalogram shows without implants, which is rotation of the soft tissue matrix relative to the cranial base. Intramatrix rotation is the difference between them, and it represents angular remodelling of the lower border. Total equals matrix plus intramatrix.

Frequently asked, rarely answered cleanly. Learn the equation.

MNEMONIC Bjork's equation: **Total = Matrix + Intramatrix**. Only total rotation needs implants to measure.

V7 What are the clinical consequences of a backward-rotating growth pattern?

"It is the pattern that will defeat me. Anterior face height grows more than posterior, so I get an increased lower face height and a tendency to anterior open bite. Overbite reduction by extrusion is unstable, anchorage is harder, and any open bite recurs. Practically it means no cervical headgear, no Class II elastics if I can avoid them, and no extrusive mechanics at all. I would want that pattern identified before I chose the mechanics, not after.

A grading question. Recognising the high-angle face and changing the mechanics is what separates a pass from a good pass.

V8 Where do clefts of the lip and of the palate arise developmentally?

"They are different failures at different times. Cleft lip, with or without the primary palate, is a failure of the maxillary process to fuse with the medial nasal process, and it happens by week 8. Isolated cleft palate is a failure of the palatine shelves to elevate above the tongue or to fuse, between weeks 6 and 10. The clearest evidence that they are distinct is that the sex ratios are opposite: cleft lip is male 2 to 1 and isolated cleft palate female 4 to 1.

The opposite sex ratios are the point. Give them unprompted.

V9 What continues to change in the face after the age of eighteen?

"Rather a lot, and slowly. Behrents showed continued rotational growth throughout adult life, in the individual's original pattern, and described the cumulative effect as surprisingly large. Practically it means continued uprighting of the lower labial segment, which is why late lower incisor crowding happens in untreated people too, continued vertical growth, and soft tissue change: the nose and upper lip lengthen and incisor show at rest reduces. It is also why an implant placed in a young adult can end infra-occluded.

Connects growth to retention, to late crowding and to implant timing in one answer. Examiners like that.

V10 Which theory of growth control would you defend, and why?

> *"The modern synthesis rather than any single classical theory. Moss's functional matrix explains the environmental influence and is the intellectual basis of functional appliances, but on its own it overstates the case. Petrovic's servosystem model is the most defensible: growth as a cybernetic system with local and systemic feedback, in which the condylar cartilage is a secondary and adaptive tissue. That formulation lets me say that growth is modifiable and that the modification is limited, which is what the evidence actually shows.*

Do not simply list the theories. Take a position and justify it.

Development of the dentition (V11–V20)

Notes In Orthodontics: Chapter 5; crash notebook *The arithmetic of the developing arch*
Supplements: Revision Handbook **§3**
Practise this: 10 questions · answer each one aloud before reading the model

V11 What are incisor liability and leeway space, and which is larger where?

"Incisor liability is the fact that the permanent incisors are wider than their predecessors, so there is a deficit to be met: about 7 mm in the maxilla and 5 mm in the mandible. Leeway space is the fact that the primary canine and molars are wider than the permanent canine and premolars, so there is an excess to be used: about 1.5 mm per quadrant in the maxilla and 2.5 mm in the mandible. Note the asymmetry: the larger liability is maxillary and the larger leeway is mandibular.

Two numbers do most of the work in this chapter. Get the asymmetry the right way round.

MNEMONIC **Liability's big up top, leeway's big down below** — 7/5 mm liability, 1.5/2.5 mm leeway per quadrant.

V12 How is the incisor liability met?

"Three ways. First, the primate and developmental spacing already present in the primary dentition, which is why generalised spacing there is normal and necessary. Second, an increase in intercanine width, about 3 mm in the mandible and 4.5 mm in the maxilla. Third, the permanent incisors erupt more labially than their predecessors. If the primary dentition had no spacing at all, I would expect the permanent dentition to be crowded.

The question behind the question is whether you will reassure a parent about spaced primary teeth.

V13 A parent is worried that the lower permanent incisors are crowded in an 8-year-old. What do you say?

"That this is normal and I expect it to improve. Space for the mandibular incisors is negative for about two years after they erupt, so transient early crowding is the rule rather than the exception, and the leeway space becomes available as the primary molars are shed. I would reassure, review at a defined date rather than vaguely, and do nothing at this stage. What I would not do is expand the lower arch, because the mandibular arch is the template and lower expansion relapses.

Tests restraint. The wrong answer is an appliance.

V14 Why does the second premolar end up crowded in the mandible and the canine in the maxilla?

"Because the last tooth to erupt in each arch is the one that ends up short of space. In the mandible the sequence is canine, then first premolar, then second premolar, so the second premolar is left. In the maxilla it is first premolar, second premolar, then canine, so the canine is left. That is one reason the maxillary canine is the tooth most often displaced.

A simple mechanism that explains two clinical observations. Give the mechanism, not just the fact.

V15 What does a distal step terminal plane in the primary dentition tell you?

> *"That the molar relationship will be Class II, and that it will not self-correct. A mesial step becomes Class I. A flush terminal plane, which is the commonest at around 60 to 75 per cent, is a coin toss: it may correct to Class I using the leeway space or remain end to end. So the distal step is the one that gives me advance warning.*

Examiners like this because it rewards understanding rather than recall of a percentage.

V16 You find a peg-shaped upper lateral incisor in a 10-year-old. What does that make you do?

> *"Look for the others. Dental anomalies co-occur far more often than chance, which reflects a shared genetic field effect. So I would palpate both canines in the buccal sulcus, check for hypodontia elsewhere, particularly the lower second premolars, look for infra-occluded primary molars, and consider a radiograph if the clinical picture warranted it. Anomalies travel in packs.*

A one-line answer that demonstrates a habit of mind. It scores well.

V17 When would you extract a first permanent molar of poor prognosis?

> *"Ideally when the bifurcation of the second permanent molar is just calcifying, which is about 8 to 10 years of dental age, so that the second molar erupts into the space. Before I committed, I would confirm the presence and position of all the permanent teeth including the third molars, and I would consider whether a compensating extraction of the opposing molar was needed to prevent over-eruption. Spontaneous closure is more reliable in the maxilla, so timing matters most in the lower arch.*

Three elements: the timing, the radiographic check, and the compensating extraction. Miss any one and the answer is incomplete.

MNEMONIC **Time it, check it, compensate it** — bifurcation calcifying at 8–10y, confirm all teeth, consider the opposing molar.

V18 What is the difference between a balancing and a compensating extraction?

> *"A balancing extraction is removal of the contralateral tooth in the same arch, to prevent a centre line shift, and it is most relevant for canines. A compensating extraction is removal of the opposing tooth in the same quadrant, to preserve the buccal occlusion and prevent over-eruption, and it is considered mainly for first permanent molars. Neither is automatic; both are decisions I would justify from the centre line and the occlusion in the individual case.*

Definitions plus the refusal to apply them mechanically. That last sentence is the mark.

V19 How does a tooth erupt?

> *"The dental follicle is the essential structure. It coordinates the osteoclastic resorption that creates the eruption pathway through bone, and the tooth follows. The evidence is elegant: remove the follicle and the tooth does not erupt, but remove the tooth and leave the follicle and a pathway still forms. Eruption begins at about two thirds to three quarters of root formation and continues to occlusal contact, then slowly thereafter to compensate for wear.*

The follicle experiment is the detail that separates a good answer from an average one.

V20 A lower second molar has failed to erupt and does not respond to traction. What is your differential?

> *"Two possibilities that must be separated. Mechanical obstruction, usually from the third molar, does respond to intervention, and I would work up the escalating ladder from observation, through disimpaction and third molar removal, to surgical uprighting. Primary failure of eruption, which is a loss-of-function mutation in PTH1R, does not respond, and orthodontic traction on such a tooth causes ankylosis. So I would not apply traction until I was reasonably confident which I was dealing with.*

The trap is answering with mechanics. The examiner is testing whether you know when not to pull.

The biology of tooth movement and root resorption (V21–V30)

Notes In Orthodontics: Topic 1; Chapter 6; Appendix A, Topic 1
Supplements: Revision Handbook **§4**
Practise this: 10 questions · answer each one aloud before reading the model

V21 Why does an ankylosed tooth not move, and what does that tell you?

"Because orthodontic tooth movement is mediated by the periodontal ligament, and an ankylosed tooth has none. The same is true of an osseointegrated implant. That single observation is the cleanest available proof that the site of the response is the ligament and not bone alone, and it is why I would open any answer on this topic with biology rather than mechanics: I apply force, but the movement is produced by remodelling of the ligament and the alveolar bone.

The classic opening question. Lead with the PDL and the rest of the viva is easier.

V22 What is hyalinisation, and why does it delay tooth movement?

"It is a cell-free, glass-like zone of necrotic tissue that forms where the ligament is compressed and the blood supply is cut off. Osteoclasts cannot resorb bone beneath it, so direct frontal resorption stops and the tooth cannot move until the necrotic tissue is removed by undermining resorption from the adjacent marrow spaces. That process takes 20 to 30 days, which is the lag phase. Reitan showed it occurs even at 30 grams, and more with tipping than with bodily movement, so it is not simply a complication of heavy force.

The trap is calling hyalinisation a complication of excessive force. It happens at 30 g.

V23 If lighter forces are biologically preferable, why not use the lightest possible force?

"Because below a threshold nothing happens: the response is a dose-response curve with a plateau, as Quinn and Yoshikawa showed, not a straight line. And what matters is not the absolute force but the differential stress generated in the ligament, which depends on the root surface area loaded and on the type of movement. That is why intrusion needs 10 to 20 grams per tooth, since the force is dissipated through the small apical area, while bodily movement needs 50 to 100 centinewtons. The honest position is that no universal optimum exists.

Tests whether you will over-simplify. The examinable position is that there is no single number.

MNEMONIC Force by job: **intrusion 10–20 g, bodily movement 50–100 cN** — small root area needs less force.

V24 Describe the RANKL and osteoprotegerin axis.

"RANKL is expressed by osteoblasts, osteocytes and periodontal ligament cells, and it binds RANK on pre-osteoclasts. It is essential for osteoclastic resorption: delete it in transgenic mice and tooth movement stops. Osteoprotegerin is a soluble decoy receptor that binds RANKL and prevents it reaching RANK, and gene transfer of osteoprotegerin reduces tooth movement. Clinically, RANKL rises on the compression side and osteoprotegerin on the tension side. RANKL peaks at three days and returns to baseline between days six and 58, which implies other pathways sustain the post-lag phase.

Reversing RANKL and OPG is one of the twelve costly slips. Say compression and tension explicitly.

V25 How would you explain the risk of root resorption to a patient at consent?

> *"I would say that some root shortening happens in most people treated with fixed appliances, that it is usually a fraction of a millimetre and of no consequence, but that in about one to five per cent of teeth it exceeds a third of the root, and that it is commonest in the upper front teeth. I would then say what I do about it: light forces, keeping treatment as short as I can, and a radiograph part-way through so that I can pause treatment for three to six months if I see anything developing. And I would record that I had said it.*

Quantified, honest, and closes with the mitigation and the record. That is the shape of a good consent answer.

V26 Which risk factors for root resorption are actually supported by evidence?

> *"Only heavy force has substantive evidence, from Weltman's work. Everything else on the usual list is suggested rather than established: continuous rather than intermittent force, intrusion, the total amount of movement, Class II elastics, abnormal root morphology, treatment duration, roots driven into cortical bone, allergy and asthma, and teeth without occlusal contact. I would mention previous trauma and previously identified resorption as the two I would look for before starting.*

Separating what is proved from what is asserted. That distinction is itself the mark.

MNEMONIC Three buckets: **force, tooth, patient** factors — continuous force/elastics; morphology/cortical contact; allergy/asthma.

V27 What happens if you apply a force greater than the anchorage unit's threshold?

> *"The anchorage unit moves. That is Newton's third law and there is no way around it: the reaction has to go somewhere, and if the force exceeds what the anchor unit will resist, the space closes from the wrong end. In practice I would see the molars coming forward, the overbite deepening and the centre line shifting. It is also why heavier forces are not simply faster: they hyalinise the ligament on the active side and they move the anchorage on the reactive side.*

Links force levels to anchorage. A complete answer joins the two rather than treating them separately.

V28 Do acceleration techniques work?

> *"Not in the sense patients mean. Corticotomy, micro-osteoperforation and piezocision can increase the local rate through the regional acceleratory phenomenon, but the effect appears short-lived and there is sparse evidence that any of them shortens overall treatment time. El-Angbawi found the same for photobiomodulation: it may accelerate discrete phases but does not influence overall treatment time. What actually shortens treatment is good bracket positioning, an efficient wire sequence, light continuous force, avoiding breakages, and the patient attending.*

A commercially attractive claim with weak evidence. The examiner wants the last sentence.

V29 How does periodontal bone loss change the mechanics?

> *"It moves the centre of resistance apically. The bracket is where it always was, so the moment arm is now longer, and the same force produces a bigger moment and more tipping. So I would use lighter forces and aim for a higher moment-to-force ratio to get controlled movement, and I would want the periodontal disease treated and stable before I placed anything. The reduced periodontium changes the mechanics, not merely the risk.*

That final sentence is the one examiners are listening for.

V30 What is the difference between frontal and undermining resorption, and which do you want?

> *"Frontal resorption is direct resorption of the bone surface adjacent to the compressed ligament by osteoclasts recruited from the ligament itself. It is what happens with appropriate force, and it is what I want. Undermining resorption proceeds from the adjacent marrow spaces because a hyalinised zone prevents osteoclasts acting at the surface. It is slower, it produces the lag phase, and it is what heavy force forces me to accept. Proffit frames the whole mechanism as the choice between the two.*

A clean conceptual answer that shows you understand why light force is preferred.

Occlusion, aetiology and epidemiology (V31–V38)

Notes In Orthodontics: Chapter 2
Supplements: Revision Handbook §5
Practise this: 8 questions · answer each one aloud before reading the model

V31 What are the limitations of Angle's classification?

"It assumes the first permanent molar is in a stable and correct anteroposterior position, which is often untrue, and it cannot be used at all if that molar is missing or displaced. It describes only the anteroposterior plane, so it ignores the vertical and the transverse. And it says nothing about the skeletal pattern. That is why the British Standards incisor classification is used alongside it, and why I would describe a malocclusion in all three planes rather than by class alone.

Tests whether you will describe a malocclusion properly rather than label it.

V32 What proportion of malocclusion has a specific identifiable cause?

"Five per cent or fewer. That short list is the syndromes and single-gene disorders, embryological defects such as clefting, trauma including condylar fracture and incisor avulsion, pathology such as cysts and tumours, functional deviations, and local dental causes such as early tooth loss. Everything else is multifactorial, the product of a genetically determined skeletal pattern and tooth size interacting with the soft tissue environment.

A number and a list. Both are examinable.

V33 Explain equilibrium theory and why it matters clinically.

"Teeth occupy the position where the resting soft tissue forces balance. The critical point is that it is light, sustained forces that move teeth, not heavy brief ones, so resting lip and tongue posture matters and chewing does not. Clinically it explains why a lower lip trap perpetuates a Class II division 1, why a tooth moved outside the neutral zone returns towards it, and therefore why lower incisor advance of more than about two millimetres very probably needs permanent retention.

Links aetiology to retention. The best answers in orthodontics usually connect two chapters.

V34 How would you describe an overbite properly?

"In three respects. Whether it is increased, average or reduced, remembering that normal is 2 to 4 millimetres or about a third to a half of the lower incisor crown. Whether it is complete, to tooth or to tissue, or incomplete. And whether it is traumatic, which means looking specifically at the palatal mucosa behind the upper incisors and at the labial gingiva of the lower incisors. All three go in the notes.

A pure examination-technique answer. Say all three or lose the mark.

MNEMONIC Three things to record: **degree, completeness, trauma** — 2–4 mm normal; complete to tooth or tissue; check for mucosal trauma.

V35 What is a pseudo-Class III?

> *"A case that looks Class III in intercuspation but manipulates back to an edge-to-edge incisor relationship in centric relation, because there is an anterior displacement on closing. The true relationship is Class I or mild Class III with a displacement, and the prognosis is entirely different from a true skeletal Class III. It is the reason I seat every patient into centric relation before I diagnose anything, and the reason the path of closure is one of the most important findings I can record.*

If you do not mention centric relation unprompted in a Class III question, expect to be asked this.

V36 Does the airway cause long face syndrome?

> *"The association is real but weak, and causation is not established. Experimental nasal obstruction in primates produced facial change, but the human evidence is observational and confounded. So what I would say is that chronic nasal obstruction is associated with mouth breathing, a low tongue posture and lips apart, and with a narrower maxilla and increased face height; but I would not assert that it causes it, and I would not offer rapid maxillary expansion as a treatment for breathing, because that is not supported.*

Stating an association without asserting a cause is a mark of a careful candidate.

V37 How common is Class III, and does that vary?

> *"In United States adolescent samples it is about 5 to 6 per cent, and the worldwide figure is about 6.8 per cent, but the regional variation is striking: it is considerably more common in East Asian populations, and in those populations maxillary retrusion is the more frequent component rather than mandibular prognathism. That matters practically, because it changes the balance between protraction and setback.*

Epidemiology with a clinical consequence attached. That is what earns the mark.

V38 What are Andrews' six keys, and what were they derived from?

> *"Correct molar relationship, correct crown angulation, correct crown inclination, no rotations, tight contacts with no spaces, and a flat or minimal curve of Spee. He derived them from 120 non-orthodontic normal occlusions, which is the point: they describe what an ideal occlusion actually looks like in nature, and that is what made the pre-adjusted prescription possible seven years later. A seventh key covering crown size and shape is often added.*

The sample size and the link to the prescription are the details that mark a prepared candidate.

MNEMONIC Six keys, in Andrews' order: **molar relation, angulation, inclination, no rotations, tight contacts, flat Spee**.

Treatment need, indices and the evidence base (V39–V48)

Notes In Orthodontics: Chapters 1 and 2; Appendix D; crash notebook *Ten things the evidence overturned*
Supplements: Revision Handbook **§6–§7**
Practise this: 10 questions · answer each one aloud before reading the model

V39 How does IOTN work, and what is the commonest error in applying it?

"It has two independent components. The Dental Health Component grades 1 to 5 and is hierarchical: you work down the MOCDO list, which is missing teeth, overjet, crossbite, displacement of contact points and overbite, and record only the single worst trait. The commonest error is to treat it as cumulative and add traits together. The Aesthetic Component is a separate ten-photograph scale, and the photograph is chosen for the degree of aesthetic impairment rather than for morphological similarity.

Hierarchical, not cumulative. That is the whole question.

MNEMONIC **MOCDO** — Missing, Overjet, Crossbite, Displacement, Overbite. Take the first that applies, never add them up.

V40 What is the difference between IOTN and PAR?

"IOTN measures need for treatment, before treatment, and it is hierarchical. PAR measures the outcome of treatment, is scored from casts before and after, is accumulative and has no maximum. A reduction of more than 30 per cent counts as improved and 22 points or more as greatly improved. PAR is a coarse instrument best used to audit an operator or a service over many cases, rather than to judge a single case.

Confusing the two is a common and avoidable error.

V41 Which index detected the difference between aligners and fixed appliances?

"The ABO Objective Grading System, not PAR. PAR shows no difference between them, whereas ABO-OGS shows aligners significantly worse, particularly for buccolingual inclination, occlusal contacts and relationship, overjet and root angulation. PAR is the coarser measure and the two do not correlate, which is itself worth knowing.

A favourite question because it rewards understanding of what an index actually measures.

V42 Name three orthodontic beliefs that the evidence has overturned.

"Early Class II treatment does not improve the final result: O'Brien's Manchester trial in 2009 found no long-term difference, with the late group finishing slightly better. Self-ligation is not faster: Scott and Fleming found no difference in rate, treatment time, visits or occlusal outcome. And orthodontics neither causes nor cures temporomandibular disorder: the American Association of Orthodontists studies of 1992. I would add that third molars are not a primary cause of late lower incisor crowding, and that interceptive canine extraction has very weak evidence.

Naming the study is the fastest way to demonstrate scholarship. Cite accurately or not at all.

MNEMONIC **EST** overturned: Early Class II treatment (O'Brien 2009), Self-ligation speed (Scott/Fleming), TMD causation (AAO 1992).

V43 What does an I squared of 80 per cent tell you about a meta-analysis?

> *"That there is high heterogeneity: 80 per cent of the variability between studies is attributable to real differences between them rather than to chance. The conventional thresholds are 25 per cent low, 50 per cent moderate and 75 per cent high. Practically it makes me cautious about the pooled estimate, and it makes me want to know what differed between the trials — population, intervention, outcome measure or follow-up.*

Do not just recite the thresholds. Say what you would do with the information.

MNEMONIC **25/50/75** = low/moderate/high heterogeneity — climbs in steps of 25 per cent.

V44 A colleague tells you a new bracket system reduces treatment time by four months. How do you respond?

> *"By asking what the evidence is. Penning's randomised trial of a customised appliance found comparable outcomes with no reduction in treatment time, more breakages, more complaints and more planning time, and found that treatment time was dictated by case difficulty and by the treating orthodontist rather than by the appliance. I would want to see a randomised comparison with a similar case mix before I accepted the claim, and I would ask who funded it.*

Tests scepticism without cynicism. The named trial does the work.

V45 What is intention-to-treat analysis and why does it matter?

> *"It means analysing participants in the group to which they were randomised, whatever treatment they actually received. It matters because it preserves the benefit of randomisation: if you analyse only those who completed the treatment, you have reintroduced selection bias through the back door, and in orthodontics the people who drop out are systematically different from those who do not. It gives the more conservative and the more realistic estimate of effect.*

The orthodontic angle — dropouts are not random — is what lifts this above a textbook definition.

V46 Why is blinding difficult in orthodontic trials, and what can be done?

> *"The operator obviously cannot be blinded to whether they have placed a functional appliance, and neither can the patient. What can and should be blinded is the outcome assessor, and that is the blinding that matters most, because it protects against detection bias in the measurement of the outcome. That is why orthodontic Cochrane reviews so often downgrade for risk of bias and for imprecision from small unblinded trials.*

Knowing which blinding is achievable, and why it matters, is more useful than lamenting that trials are unblinded.

V47 Where does a case series sit in the hierarchy of evidence?

> *"Below cross-sectional and cohort studies and above expert opinion. The critical limitation is that without a control group it cannot establish causation, only describe. That matters in orthodontics because a great deal of what is asserted about new appliances rests on case series, and a case series of consecutively treated successful cases tells you almost nothing about the failures.*

A short answer with one sharp observation is better than a long recitation of the hierarchy.

V48 How would you explain to a patient that the evidence for something is weak?

> *"Plainly, and without pretending otherwise. I would say that this is something we do because it seems reasonable and because experience supports it, but that it has not been tested properly, and then I would say what the alternatives are and which of them is easiest to reverse if it does not work. Patients handle uncertainty better than they handle being told something confidently that later turns out not to be true.*

The examiner is testing honesty and communication together. The reversibility point is the clinical one.

Ethics, consent and communication (V49–V58)

Notes In Orthodontics: Part V, unit V.1; Topic 9 in part; Chapter 7
Supplements: Revision Handbook §8
Practise this: 10 questions · answer each one aloud before reading the model

V49 What would you discuss when consenting a 13-year-old and her mother for fixed appliances?

"I would cover what I propose and why; all reasonable alternatives, including doing nothing and what happens if we do nothing; and the material risks. Those are demineralisation, which affects up to half of fixed appliance patients and can be visible within six months; root resorption, which exceeds a third of the root in one to five per cent of teeth; recession; relapse, and that retention is lifelong if the result is to be held; and pain, which peaks at 48 hours. I would cover the time commitment and the cost including replacement retainers, and the possibility that I do not achieve the aim. I would give it in writing, check understanding by asking her to tell me back what we had agreed, and record it. And I would satisfy myself that the child assents, not only that her mother consents.

A set-piece. Rehearse it until it is automatic and the rest of the viva is easier.

MNEMONIC **PARQ** — Procedure, Alternatives, Risks, Questions. Cover all four, then check she can say it back, and record it.

V50 What is the Montgomery standard?

"That the risks to be disclosed are those to which a reasonable patient in this patient's position would attach significance, rather than those a responsible body of clinicians would disclose. It shifted the test from a professional standard to a patient-centred one. Practically it means I have to think about what matters to this particular patient: a professional singer will attach significance to a transient speech disturbance that most patients would not.

The example is what shows you have understood it rather than memorised it.

V51 A parent consents but the child clearly does not want treatment. What do you do?

"I would not start. Legally the parent's consent may be sufficient, but clinically the child's assent is decisive, because a technically ideal plan the child will not wear is not a plan. I would explore why: often it is fear, or a friend's account, or simply that the parent wants this more than the child does. I would offer to see the child again, defer the decision, and record the discussion. Elective treatment on an unwilling child is bad orthodontics before it is anything else.

Tests the distinction between legal sufficiency and clinical wisdom.

V52 How do you assess capacity?

"Capacity is decision-specific and time-specific, so I assess it for this decision at this time rather than as a general attribute. The functional test is whether the person can understand the information, retain it long enough to decide, weigh it in the balance, and communicate the decision. In a child under 16 I would additionally consider Gillick competence, which asks whether they have sufficient

understanding and intelligence to understand fully what is proposed.

Two frameworks, correctly distinguished. Do not conflate them.

MNEMONIC The capacity test, in order: **understand, retain, weigh, communicate**. Miss one step and capacity fails.

V53 An adult self-refers for aligners, dissatisfied with two previous courses of treatment, and shows you extensive photographs of a discrepancy you cannot see. What do you do?

"I would not commit to treatment at that visit. Serial dissatisfaction with a defect that is invisible or trivial to me is a red flag for body dysmorphic disorder, which affects 5 to 13 per cent of orthodontic patients and up to 17 per cent of those seeking orthognathic surgery. I would explore what she expects treatment to achieve, ask what success would look like to her, and if I remained concerned I would say honestly that I did not think I could deliver what she was hoping for, and consider whether onward referral for psychological assessment was appropriate.

The mark is for not treating. Recognising the red flags and declining is the safe answer.

V54 Is orthodontic treatment worth having? What are the benefits, honestly stated?

"The benefits are narrower than they are often claimed to be. Dental aesthetics and reduction in incisor trauma have strong evidence. Improvement in the emotional and social domains of oral health related quality of life has moderate evidence. Caries reduction, periodontal health, masticatory efficiency and speech all have weak evidence or none, and long-term psychological benefit has little objective evidence. I would say that plainly rather than overclaim, because overclaiming is both dishonest and, when the patient discovers it, corrosive.

Overclaiming the benefits is one of the twelve traps. State the confidence level for each.

V55 A 14-year-old wants treatment but her plaque control is poor. What do you do?

"I would not start. Demineralisation affects up to half of fixed appliance patients, appears within six months, and in elective treatment I chose to begin, it is my responsibility. I would put a documented oral hygiene programme in place with a defined review, explain in writing that appliances will not be placed until plaque control is adequate, and re-assess. That is not a punishment; it is the single most effective preventive measure available to me.

The wrong answer is to start and reinforce hygiene at each visit. Say why.

V56 How would you handle a patient who wants a treatment you think is wrong for them?

"I would find out first why they want it, because the reason is usually reasonable even when the request is not. Then I would explain plainly what I think it would and would not achieve in their case, offer what I would recommend and why, and give them the alternatives including doing nothing. If they still wanted it and I could deliver it safely and defensibly, I would; if I could not, I would say so, offer a second opinion, and record the whole discussion.

Neither capitulation nor refusal. The examiner wants a process.

V57 When does consent need to be revisited during treatment?

"Whenever the plan changes materially. A decision to extract that was not part of the original plan, a decision to proceed to surgery, a change of appliance, or a decision to accept a compromise rather

than the original aim all require the discussion to be repeated and recorded. Consent is a process rather than an event, and a form signed at the assessment visit does not cover a decision made eighteen months later.

A frequently missed point, and a common source of complaint.

V58 What would make you decline to treat a patient?

"Inadequate plaque control that has not improved despite a documented programme; expectations I cannot meet, including the body dysmorphic picture; a case beyond my competence that I should refer rather than attempt; an active eating disorder or a medical situation where treatment should be deferred; and a patient who will not agree to the retention plan, because without retention the result is temporary and the treatment is not worth its risks.

Five reasons, one of them about your own limits. That last is the one examiners are waiting for.

Clinical examination and records (V59–V68)

Notes In Orthodontics: Topic 2; Chapter 7; Appendix A, Topic 2
Supplements: Revision Handbook **§9–§10**
Practise this: 10 questions · answer each one aloud before reading the model

V59 Take me through your orthodontic examination.

"In a fixed order. The presenting complaint in the patient's own words and what they expect treatment to achieve, then the dental, medical, social and family history and any habits. Then extra-oral, with the patient upright in natural head position, teeth in occlusion and lips at rest: the three planes of space, lips and interlabial gap, incisor show at rest and on smiling, the smile arc, the nasolabial angle and the joints. Then intra-oral: oral health first, then each arch with the crowding quantified in millimetres, then the occlusion in three planes, then the path of closure. Then a provisional problem list, and only then do I decide which records I need.

The last clause is the one that scores: the examination decides the records, not the other way round.

MNEMONIC Fixed order: **complaint and history, extra-oral, intra-oral, problem list, then records** — records come last, not first.

V60 Why does the path of closure matter so much?

"Because it can change the diagnosis, not merely the detail. An anterior displacement from retruded contact position to intercuspal position may mean this is a pseudo-Class III rather than a true skeletal Class III, which is an entirely different prognosis. A lateral displacement means an apparently unilateral crossbite is really a bilateral constriction, which changes both the appliance and the urgency. And a displacement is associated with attrition, with recession on the loaded tooth and with asymmetric growth. Posselt showed that a discrepancy of nought to two millimetres is normal and that only about ten per cent of people have the two positions coincident.

One finding, three consequences. That structure is worth borrowing.

V61 From what age do you palpate for the maxillary canines, and why?

"From nine to ten years, at every examination. The canine has the longest eruptive path in the dentition, over twenty millimetres, and it fails to erupt in about two to three per cent of children with up to 85 per cent of those palatal. If I cannot palpate it buccally, I assume it is palatal and investigate. The real threat is not the impaction but resorption of the adjacent incisor roots, which is silent and which I will only find if I look.

A screening habit with a stated rationale. Give both.

V62 What radiographs would you take for a 12-year-old with crowding and an unerupted upper canine?

"A panoramic radiograph as the survey film, and periapicals with a parallax shift to localise the canine, using the rule that the object moving with the tube is palatal. I would consider a CBCT only if conventional imaging had not established the relationship of the canine to the lateral incisor root, or whether that root was resorbing, and only because the answer would change whether I

retained the lateral incisor. I would not take a lateral cephalogram unless I intended significant incisor movement, growth modification or surgery.

Justification for each film, and refusal of the one that is not needed.

V63 Justify a lateral cephalogram to me.

"I would take one if I intended to move the incisors significantly, to modify growth, or if surgery was a possibility, because in each of those the skeletal pattern and incisor inclination change the plan. I would not take one routinely, and specifically not if only minor tooth movement is planned. I would not take one to predict growth, because cephalometric growth prediction has been shown to be at best unreliable. And I would record the justification and report the film.

Three indications, two refusals, and the record. Radiation justification is easy marks and routinely omitted.

V64 What are the four sources of error in cephalometric analysis?

"Reproducibility, from landmark identification error, where each landmark has its own non-circular envelope of error and where the error is cumulative because landmarks are reused across an analysis. Validity, because points A and B are convenient rather than anatomical and both remodel as the incisors move. Projection, because a three-dimensional object is rendered in two dimensions and every machine has its own enlargement factor. And superimposition, because it assumes stable reference structures and no method has been proved completely accurate. Houston's summary is that cephalometrics is definitely not an exact science.

Four headings and a quotation. Weight the limitations equally with the uses.

MNEMONIC Four sources of error: **reproducibility, validity, projection, superimposition**. Weight them equally.

V65 What is the Eastman correction and when would you apply it?

"It corrects ANB for the fact that ANB is distorted by the anteroposterior position of nasion and by the inclination of the sella-nasion plane. If the sella-nasion to maxillary plane angle differs from eight degrees, or if SNA differs from 81 degrees, ANB is adjusted by half a degree for each degree of difference. I would apply it whenever SNA is markedly outside the normal range, because otherwise I may call a Class I pattern Class II purely because of where nasion happens to sit.

A correction with a reason attached, and a clinical consequence. Do not just recite the arithmetic.

V66 When would a CBCT be justified in orthodontics?

"Where there is a specific question that conventional imaging has not answered and where the answer will change management. In practice that is impacted or ectopic teeth where the relationship to adjacent roots is unclear, suspected resorption of an adjacent incisor root, clefts and alveolar bone graft planning, complex asymmetry, orthognathic planning, and occasionally assessment of a temporary anchorage device site. The dose is far higher than a panoramic, incidental findings are reported in a quarter to nearly all scans, and it cannot image the disc or muscle, which needs MRI.

Indications, dose, incidental findings and the limitation. All four should appear.

V67 Why can you not apply two-dimensional cephalometric norms to a three-dimensional analysis?

"Because the landmark definitions are not the same. A landmark defined on a projected two-dimensional image is not the same point as the equivalent three-dimensional anatomical landmark, and there is no agreed normative database for three-dimensional measurements. Three-dimensional imaging removes projection and superimposition error in principle, but the dose is higher and the norms are not transferable, so it does not simply replace the lateral cephalogram.

A question that separates candidates who have thought about imaging from those who have memorised norms.

V68 How would you record incisor display?

"At rest and on smiling, photographed both times. For rest position I ask the patient to say Emma or Mississippi and then relax, and I measure the incisor show, which is about two to four millimetres in a young adult and reduces with age. For the posed smile I ask them to bite together and say cheese, and I record how much of the incisor crown shows, which is 75 to 100 per cent in an average smile, together with the gingival display, the smile arc and the buccal corridors. I would do this before planning anything vertical.

The instructions are the detail. Anyone can say measure the incisor show; few say how.

Space analysis, diagnosis and treatment planning (V69–V80)

Notes In Orthodontics: Topic 3; Chapter 8; Appendix A, Topic 3
Supplements: Revision Handbook **§11–§12**
Practise this: 12 questions · answer each one aloud before reading the model

V69 Take me through your treatment planning sequence.

"Nine stages, each constraining the next. Diagnosis, from the examination, the records and the patient's own concerns. A ranked problem list. Aims, both facial and occlusal, reconciled against the skeletal pattern. Patient factors: age and growth, dental and periodontal health, medical history, compliance and expectations. The space analysis. The anchorage requirement, which follows directly from it. The appliance. The phasing. And retention, planned and agreed before treatment begins. The point is that it is a sequence rather than a list: the problem list sets the aims, the aims set the space requirement, the space requirement sets the extraction pattern, and the extraction pattern sets the anchorage demand.

The last sentence converts a list into an argument. That is the difference between a pass and a good pass.

MNEMONIC Nine stages, in order: **diagnosis, problem list, aims, patient factors, space, anchorage, appliance, phasing, retention**.

V70 Where does treatment planning begin?

"With the position of the dentition, and particularly the maxillary incisors, within the face. That is a deliberate shift from the classical approach of planning around mandibular incisor position and isolated static occlusal goals. The soft tissues define the limits of orthodontic treatment in each individual, so I start from the face and work inwards, rather than achieving a Class I occlusion and hoping the face tolerates it.

Facial aims are routinely forgotten. Volunteer them.

V71 Your space analysis shows 6 mm of crowding in the lower arch. Take me through how that changes your plan.

"Six millimetres is moderate crowding, so it is a judgement rather than an automatic extraction. First I would ask what else needs space: levelling the curve of Spee consumes roughly a millimetre of arch length per millimetre of depth, and any change in incisor position costs two millimetres of space per millimetre of movement. Then I would ask where the space can come from: expansion yields only about half its width and the lower arch is the template, so I would not expand it; interproximal reduction offers a modest amount. If the total requirement approached eight millimetres I would extract, and I would then compare the lower requirement with the upper, because the difference between the arches determines the molar relationship I finish to and therefore the extraction pattern.

The examiner is testing whether you can convert a number into a decision.

MNEMONIC Space arithmetic: **Spee = 1 mm per mm of depth, incisor move = 2 mm per mm, expansion buys about half**.

V72 How do you decide between growth modification, camouflage and surgery?

"Growth first. If the patient is growing and the discrepancy is skeletal, growth modification is available, though I would be honest that the skeletal gain is a few millimetres at best and largely transient, and that it is far more effective in Class II than Class III. If the patient is not growing and the discrepancy is mild to moderate, camouflage, limited by the alveolar envelope, by the soft tissue response of about 0.7 millimetres of lip retraction per millimetre of incisor retraction, and by how much dental compensation is already present. If the discrepancy is severe or the facial aims cannot be met by moving teeth, surgery. And that decision has to be made before treatment starts, because pre-surgical orthodontics decompensates in the opposite direction from camouflage.

A set-piece. Prepare it, because it comes up in every oral in one form or another.

V73 What are Proffit's thresholds for Class II surgery?

"An overjet greater than ten millimetres, pogonion to the nasion perpendicular of eighteen millimetres or more, a mandibular body length under seventy millimetres, or an anterior face height over 125 millimetres. I would add that these are indicators rather than rules: a patient at one threshold whose chief concern is the occlusion rather than the face may still be a reasonable camouflage case, and one below all four with a severe aesthetic concern may not be.

Quote the numbers, then refuse to apply them mechanically.

MNEMONIC Proffit's Class II surgery numbers: **OJ >10 mm, pog–Npog >=18 mm, body <70 mm, AFH >125 mm**.

V74 What are the thresholds that suggest surgery in Class III?

"A Wits appraisal of minus eleven millimetres or worse, an ANB of minus four degrees or worse, a reverse overjet of minus five millimetres or worse, a Holdaway angle of 3.5 degrees or less, a gonial angle of 120 degrees or less, a lower anterior face height of 66 per cent or more, and a lower incisor inclination of eighty degrees or less. Discriminant analysis on those variables predicts the need for surgery with over ninety per cent accuracy. In practice the one that decides it for me is usually the lower incisor inclination, because it tells me how much decompensation has already happened and therefore how little room is left.

Seven numbers, then the one that matters clinically. That last step is the judgement.

V75 What is the envelope of discrepancy?

"It is a description of how far teeth can be moved in each direction by orthodontics alone, by orthodontics with growth modification, and by surgery. Three things about it are worth saying. It is asymmetric: maxillary incisors can be retracted considerably further than they can be advanced. Growth modification extends it only modestly, and only in a grower. And the soft tissue limits are frequently tighter than the hard tissue limits, so the fact that a tooth can be moved somewhere does not mean the lip will tolerate it.

Three observations rather than a diagram. The third is the clinically useful one.

V76 Which parts of your plan would change if this patient were 32 rather than 12?

"Growth modification disappears, so a skeletal discrepancy becomes camouflage or surgery. Overbite reduction reverses: in a child I can level by relative intrusion and let the buccal segments extrude, whereas in an adult the same manoeuvre rotates the mandible down and back and relapses, so I would need true intrusion. Movement is slower, hyalinisation is more prolonged and treatment

takes longer. I would probe every tooth and record pre-existing recession. And I would plan the restorative endpoint with a restorative colleague first, rather than moving teeth and consulting afterwards.

Four or five specific differences beat a general statement that adults are different.

V77 How do you assess motivation and compliance at the first visit?

"There is no reliable predictor, so I use proxies. Whether the patient can articulate their own concern in their own words, or whether the concern is entirely the parent's. Their attendance history. Their plaque control at the assessment visit, which is a behaviour rather than a promise. And whether they can describe what they would be willing to do — wearing something at night, avoiding certain foods. I would say plainly that compliance is a planning factor rather than an afterthought, because a technically ideal plan the patient will not wear is not a plan.

Honesty about the absence of predictors, then the proxies you actually use.

V78 A colleague's diagnostic case lists twelve findings and no argument. What is missing?

"A ranking, and a decision. A problem list has to be ranked in order of importance to the patient, with the chief concern high, because that is what determines which problems I treat if I cannot treat all of them. Then I would want to see aims stated so that a colleague could measure me against them, rather than a rubber-stamp objective such as achieving a good functional occlusion, and I would want at least three options with reasons for rejecting two of them.

The commonest failure in the diagnostic case component. Rank, and choose.

V79 How does the space analysis determine the extraction pattern?

"Through the difference between the arches. If the requirement is equal in both, the case finishes to Class I molars and I extract corresponding premolars. If the maxillary requirement is greater, the case finishes to Class II molars and I extract further back in the mandible: upper first premolars with lower second premolars. If the mandibular requirement is greater, as in a Class III camouflage, I extract further forward in the mandible. The principle is simply to put the space where the movement is needed.

Reversing the Class II and III patterns is one of the twelve costly slips. Give the principle so you can reconstruct it under pressure.

V80 What would make you refer rather than treat?

"A severe skeletal discrepancy, or any case where surgery is a realistic possibility. An impacted tooth needing surgical exposure and complex traction. Cleft and craniofacial cases. Significant asymmetry, particularly if it is progressive. Severe hypodontia. A reduced periodontium needing complex mechanics. A suspected primary failure of eruption. And in general, any case where the aims I have set cannot be delivered by the mechanics I am competent to use. I would rather say that at the assessment visit than discover it eighteen months in.

Volunteering the boundary before you are asked is the strongest single move in an oral examination.

Extraction decisions and space management (V81–V90)

Notes In Orthodontics: Topic 4; Chapter 8; Appendix A, Topic 4
Supplements: Revision Handbook **§13**
Practise this: 10 questions · answer each one aloud before reading the model

V81 What factors govern your choice of extraction?

"Four. The condition of the teeth, which comes first: a compromised tooth should always be removed in preference to a healthy one, even if it sits in a less convenient place. The site and degree of crowding, because space is most useful where the movement is needed. The type of malocclusion, which determines the differential requirement between the arches. And the age, because a well-timed extraction in the mixed dentition permits spontaneous improvement whereas in the permanent dentition appliances become obligatory.

Prognosis first. Say it first.

MNEMONIC **PSTA**, in priority order: Prognosis (condition of the tooth), Site/degree of crowding, Type of malocclusion, Age.

V82 Why do you extract further back in the mandible in a Class II case?

"Because I need more space in the maxilla than in the mandible. Extracting upper first premolars and lower second premolars gives a large anterior space in the upper arch to retract the labial segment, while the smaller and more posteriorly sited lower space allows the lower molars to come forward, so the case finishes to a Class I canine relationship on a Class II molar relationship. The general principle is to put the space where the movement is needed, and in Class III it is exactly reversed.

Give the principle as well as the pattern, so you can reconstruct it if you blank.

V83 When would you extract a second premolar rather than a first?

"When the crowding is mild to moderate and sited more posteriorly, when I want the anchorage balance to favour the anterior segment, or when the second premolar itself is absent, hypoplastic or displaced. In a Class II case, second premolar extraction in the lower arch also eases the mechanics. Anchorage loss, incidentally, is the same for first and second premolar extraction, so that is not the discriminator people sometimes think it is.

The Haque finding is a useful corrective to a common assumption.

V84 When would you extract a lower incisor?

"Rarely. Severe lower labial segment crowding, a lower incisor of genuinely poor prognosis, or occasionally mild Class III camouflage in a non-growing patient. The costs are real: a tooth-size discrepancy that shows up on a Bolton analysis, residual spacing that may need restorative closure, and a deepened overbite. I would want a diagnostic set-up or at least a careful Bolton calculation before committing.

Knowing the costs is what makes this a safe answer rather than a reckless one.

V85 Do extractions damage the face?

"The comparative evidence says not, in the way the claim is usually made. Rushing, Paquette and Iared found minimal soft tissue differences between comparable cases treated with and without extraction, with both groups perceived to benefit. Bowman and Johnston found that lip retraction must be considerable before a profile is judged unattractive. Meyer and Christou found that extraction does not widen the buccal corridors or produce a less attractive smile. What decides it for me is the space requirement in millimetres and the nasolabial angle, not a general position for or against extraction.

Three named findings and then a clinical criterion. That is the shape.

V86 What about the claim that extractions narrow the airway?

"There is no evidence for it. Larsen and Orabi looked specifically at this and found no evidence that extraction-based treatment reduces the oropharyngeal airway or increases sleep apnoea, and the claim has been described in the literature as scaremongering and marketing. If a parent raises it I would say that plainly, and offer to discuss the actual evidence rather than dismiss the concern.

Answer the claim and address the person raising it. Both are marked.

V87 How much space does expansion actually give you?

"About half the amount of expansion. That is the O'Higgins and Lee figure, and it is why expansion is a poor substitute for extraction when the crowding is significant. It is also why I would not expand the lower arch at all: the mandibular arch is the template, lower intercanine width is among the least modifiable dimensions, and lower expansion relapses, so it buys me a small amount of space at the cost of permanent retention.

The number, then the reason it does not solve the problem.

V88 When would you extract a second permanent molar?

"Where I need space at the back of the arch, to ease Class II mechanics, or to allow a crowded third molar to erupt. Before committing I would want the third molar present, of good size, correctly angulated at thirty degrees or less to the second molar long axis, and at an appropriate stage of development. It is a decision that depends entirely on a tooth I cannot yet see erupt, so I would be cautious and I would explain the uncertainty.

The conditions are the answer. Anyone can name the extraction.

V89 How much interproximal reduction is available across an arch?

"A modest amount. Roughly half a millimetre per anterior contact and up to a millimetre per contact in the buccal segments, which across a full arch comes to somewhere in the order of four to eight millimetres, and rather less than that if I want to stay comfortable. Enamel is finite, the technique matters, and I would want good fluoride protection afterwards. It is a useful adjunct rather than a substitute for extraction in significant crowding.

Quantify it and then say what it cannot do.

V90 A first permanent molar is heavily restored and hypomineralised, and there is 5 mm of lower crowding. Which tooth comes out?

> *"The first permanent molar. Prognosis beats convenience of position, and a compromised molar should be removed in preference to a healthy premolar. I would time it against the development of the second permanent molar, ideally when its bifurcation is just calcifying at about eight to ten years of dental age, confirm all the permanent teeth including third molars are present on a radiograph, and consider a compensating extraction of the opposing molar to prevent over-eruption. Spontaneous closure is less reliable in the mandible, so I would plan for appliance treatment.*

Prognosis, timing, radiographic check, compensating extraction. Four elements.

MNEMONIC **Prognosis, Timing, Check, Compensate** — the four things this extraction decision turns on.

Biomechanics and anchorage (V91–V102)

Notes In Orthodontics: Topics 5 and 6; Chapters 6 and 11; Appendix A, Topic 5
Supplements: Revision Handbook **§14**
Practise this: 12 questions · answer each one aloud before reading the model

V91 What is the moment-to-force ratio and why does it matter?

> *"It is the single number that decides what kind of movement I get. The centre of resistance sits inside bone and is inaccessible, so I have to apply force at the crown, which means a moment is always produced. A ratio of zero gives uncontrolled tipping, with crown and root moving in opposite directions. About seven to one gives controlled tipping, with the centre of rotation near the apex. About ten to one gives translation, with the centre of rotation at infinity. Twelve to one and above gives root movement.*

The four ratios and the four movements. Learn them as a set.

MNEMONIC **0:1, 7:1, 10:1, >=12:1** — tip, controlled tip, translation, root movement (torque). Ratio climbs, control increases.

V92 Why does a removable appliance only tip teeth?

> *"Because it achieves only two-point contact on a tooth, so it cannot generate a couple. Without a couple the moment-to-force ratio is effectively zero, and the tooth tips about a centre of rotation near its centre of resistance. That is why bodily movement, rotation and torque are beyond a removable appliance, and it is one of its two fatal limitations. The other is compliance.*

A mechanical explanation of a clinical limitation. Give the mechanism.

V93 What is a couple?

> *"Two equal, opposite, parallel and non-collinear forces. The important properties are that it produces no net translational force, and that it produces pure rotation about the centre of resistance regardless of where on the tooth it is applied. That is why a rectangular wire engaged in a rectangular slot can torque a root without translating the crown, and it is the whole basis of third order control.*

The two properties are the answer; the clinical application shows you understand them.

V94 What is the difference between a determinate and an indeterminate force system?

> *"A one-couple system, such as a cantilever or a utility arch engaged in one bracket and tied as a point contact at the other end, is statically determinate: the force system can be calculated and measured. A two-couple system, which is what a continuous archwire engaged in several brackets is, is statically indeterminate: the force system can only be estimated. Most of clinical orthodontics is indeterminate, which is precisely why clinical experience matters and why segmental mechanics are used when the force system needs to be known.*

A distinction with a reason for caring about it.

V95 How do you plan anchorage?

> *"I start from the space analysis, because the anchorage requirement follows from it rather than the other way round. I classify the demand as minimum, moderate or maximum. Then I recognise that anchorage is spent or saved during levelling and aligning, before any active mechanics begin, so I use lacebacks, I bend the wire back distal to the molar, and in a maximum anchorage case I keep the anchor teeth out of the initial aligning wire. Then I reinforce: second molars in the appliance, a transpalatal or lingual arch, headgear, interarch elastics, or skeletal anchorage. And the number I would quote is that unsupported, about two-thirds of the space created is lost.*

Sequence, classification, timing, reinforcement, number. Five elements.

V96 How much anchorage does a transpalatal arch actually give you?

> *"Very little in the sagittal plane. Diar-Bakirly found it provides very little additional anteroposterior anchorage, and Rebellato found much the same for a lower lingual arch, which additionally can procline the lower incisors. They are useful for molar derotation, for transverse control and, in the case of the lingual arch, for preserving leeway space; but if I need real anchorage I would not rely on them.*

A useful corrective. Examiners like candidates who know which received practices have weak evidence.

V97 What is Proffit's anchorage ladder?

> *"A way of quantifying the trade-off as a ratio of anterior retraction to posterior protraction. Reciprocal space closure without any reinforcement is about one to one, and that is determined by periodontal ligament area rather than by tooth number. A stabilising lingual or palatal arch shifts it to about two to one. Headgear combined with interarch elastics reaches three or four to one. And skeletal anchorage is effectively absolute. Putting numbers on it makes the conversation with the patient about compliance a good deal easier.*

The ratios convert a vague concept into a plan.

MNEMONIC **1:1, 2:1, 3–4:1, absolute** — reciprocal, stabilising arch, headgear+elastics, skeletal anchorage. The ladder climbs.

V98 How does a temporary anchorage device work?

> *"It does not change the force system at all. Newton's third law still applies and the reaction still has to go somewhere; the screw simply places the reaction in bone, where nothing can move. So what changes is the biological response, not the mechanics. In direct anchorage the line of force must pass through the screw. In indirect anchorage I ligate the anchor teeth to the screw with a large rectangular stainless steel wire, rigid enough not to flex.*

The opening sentence is the one that demonstrates understanding rather than familiarity.

V99 Would you place a miniscrew yourself?

> *"In some sites, with appropriate training. The maxillary buccal interradicular region has a pooled failure rate of about 7.5 per cent, and the palate is the most reliable site at about 1.3 per cent in the median palate, because placement there is not restricted by the dentition. The mandibular buccal shelf fails in about 65 per cent of cases, and that is not somewhere I would be placing screws. More generally, if a case depends on a screw in a difficult site, I would refer the whole case rather than the*

screw.

The site-specific failure rates, then the judgement about referral.

V100 What insertion torque do you aim for, and why does it matter?

"Five to ten newton centimetres. Below that the screw has poor primary stability and is likely to be mobile from the outset. Above it, I risk microfracture of the surrounding bone and therefore poor secondary stability. I would also avoid screws shorter than eight millimetres or narrower than 1.2 millimetres, because failure rates rise below those dimensions, and the commonest single cause of failure is root contact.

A number with a mechanism at each end of the range.

V101 Your miniscrew is mobile at four weeks. What do you do?

"Remove it. A mobile screw will not become stable and it perpetuates inflammation in the tissue around it. Then I would work out why: root contact is the commonest cause, so I would look again at the radiograph and at where I placed it. I would re-site at least a few millimetres away, or change the anchorage plan altogether. And I would reflect that I should not have designed a case in which the failure of one screw leaves me with nothing to fall back on.

The last sentence is about planning rather than about screws, and it is the one that scores.

V102 What force and duration would you prescribe for headgear, and what safety precautions?

"It depends on the purpose. For anchorage support, 250 to 350 grams per side for 10 to 12 hours a day. For distalisation or an orthopaedic effect, 450 to 500 grams per side for 14 hours a day. Two independent safety mechanisms are mandatory: a self-releasing snap-away neck strap and a locking facebow. I would warn the patient in writing about the two injury mechanisms Samuels described, the catapult injury and nocturnal disengagement, tell them to stop wearing it and attend if the bow ever comes out, and record that I had done so.

Quoting one headgear figure is one of the twelve traps. Give both, and the safety devices unprompted.

MNEMONIC Headgear: **250–350 g/10–12 h anchorage, 450–500 g/14 h distalisation** — plus two safety devices, always.

Materials, wires and bonding (V103–V111)

Notes In Orthodontics: Topic 7; Chapter 10; Appendix A, Topic 7
Supplements: Revision Handbook **§15**
Practise this: 9 questions · answer each one aloud before reading the model

V103 What are the properties of an ideal archwire?

"High strength, low stiffness, high range and high formability, plus biocompatibility, environmental stability, the ability to be welded or soldered, low friction and reasonable cost. No single wire has all of them, which is exactly why a wire sequence exists. The useful identity is Proffit's: strength equals stiffness multiplied by range, so I cannot raise one without paying somewhere else.

The identity is what turns a list into an understanding.

MNEMONIC Proffit's identity: **strength = stiffness x range**. Raise one, pay in another.

V104 Compare stainless steel and nickel titanium.

"Stainless steel is the 18-8 austenitic alloy, about 70 per cent iron with 17 to 20 per cent chromium and 8 to 12 per cent nickel. It is stiff and strong, has excellent formability, low friction and low cost, and it can be welded and soldered, which makes it the working and finishing wire and the wire that carries arch form. Nickel titanium has a modulus about 0.26 that of steel, giving very low stiffness and a very large range, with shape memory and superelasticity, which makes it the alignment wire. But it cannot be bent, welded or soldered, so its role ends when alignment ends.

A comparison organised by what each is used for, not just by properties.

V105 What is superelasticity?

"A real, measurable phenomenon rather than an advertising term. It arises from a stress-induced transformation from the austenitic to the martensitic phase, which produces a plateau in the load-deflection curve so that the wire delivers a roughly constant force over a range of deflection, with hysteresis between loading and unloading. Clinically that means a relatively constant light force as the tooth moves, which is the property I want during alignment.

Mechanism, curve, clinical consequence. Three sentences.

V106 Which archwire would you choose for initial alignment, and does the choice matter?

"A light, flexible, large-range wire: 0.012 to 0.018 inch round nickel titanium or a multistrand steel. As to whether it matters, the honest answer is less than the marketing suggests. The Cochrane evidence found insufficient evidence that any archwire material is superior, with the single exception that coaxial superelastic nickel titanium produced greater movement over twelve weeks than single strand. Papageorgiou found copper nickel titanium slightly slower to reach a working wire and more painful.

A choice, then the evidence that the choice matters less than people think.

V107 Describe your bonding procedure.

> *"Clean with pumice and no fluoride, isolate, etch with 37 per cent phosphoric acid for 15 to 30 seconds, rinse and dry to a frosted appearance, prime, place the adhesive and the bracket, remove excess, and cure. A self-etching primer combines the etch and prime steps, gives a shallower etch pattern and has comparable clinical bond failure rates. For molars carrying headgear or a palatal arch, or for heavily restored teeth, I would band and cement with glass ionomer for the fluoride release.*

A sequence with the numbers in it, and the decision about banding.

MNEMONIC Bonding sequence: **clean, isolate, etch (15–30s), rinse/dry, prime, bond, cure**.

V108 Why do ceramic brackets cause problems at debond?

> *"Because they bond too well. They bond by silane coupling, which is a chemical bond rather than a purely mechanical one, and the ceramic is stiffer and more brittle than the enamel. So at debond the weakest point can be the enamel rather than the adhesive, and enamel fracture is a recognised risk. They also have higher friction and they abrade opposing enamel. I would use the manufacturer's debonding instrument, work slowly, and warn the patient in advance.*

The mechanism explains the risk. Give both.

V109 How do you minimise demineralisation during fixed appliance treatment?

> *"Patient selection first: I do not start until plaque control is demonstrably adequate, because demineralisation affects up to half of fixed appliance patients and can be visible within six months, and in elective treatment that makes it my responsibility. Then daily high-fluoride toothpaste at 2800 or 5000 parts per million, professional high-concentration fluoride every six to eight weeks, glass ionomer for band cementation, steel ligatures rather than elastomeric modules in a poor brusher, diet advice, hygiene reinforcement recorded at every visit, and a willingness to debond early if it is not improving.*

Selection first, then the regimen, then the willingness to stop. All three.

V110 A patient reports a nickel allergy. What do you do?

> *"I would not regard it as a contraindication to orthodontic treatment. Nickel allergy affects about 10 to 30 per cent of females and 1 to 3 per cent of males, and it is a type IV delayed hypersensitivity. Stainless steel is safe, because the nickel is bound within the austenitic lattice, and intra-oral reactions are in fact rare. What I would avoid is nickel titanium, and I would avoid extra-oral nickel-containing components such as headgear, where reactions are commoner. I would record the discussion.*

Saying nickel allergy contraindicates orthodontics is one of the twelve traps.

V111 What is the diode laser used for in orthodontics, and what are its limits?

> *"It operates at 810 to 980 nanometres, is absorbed by melanin and haemoglobin, and cuts soft tissue while sparing enamel and metal. I would use it for exposing a partially erupted tooth, for gingival recontouring, for a frenectomy or an operculectomy, with the advantages of no sutures, no dressings and controlled bleeding. Its limit is that it is a finishing instrument: I would diagnose the cause of any gingival disproportion — altered passive eruption, hyperplasia from poor hygiene, or simply*

> *an unerupted tooth — before reaching for it.*

The last sentence is the clinical judgement the question is really asking about.

Removable, functional and fixed appliances (V112–V121)

Notes In Orthodontics: Topics 6 and 8; Chapters 9, 10 and 14; Appendix A, Topics 6 and 8
Supplements: Revision Handbook **§16–§17**
Practise this: 10 questions · answer each one aloud before reading the model

V112 What does a functional appliance actually do?

"Mostly dentoalveolar change. The upper incisors retrocline, the lower incisors procline, buccal segment eruption is guided, and the lip trap is eliminated so lip competence is gained. There is a skeletal effect but it is a few millimetres at best and essentially transient, and there is usually a headgear effect restraining the maxilla. The more sophisticated explanation of the mechanism is that a Class II relationship is partly locked by intercuspation, and the appliance unlocks it and lets normal mandibular growth express itself alongside the dentoalveolar change.

Saying the appliance grows the mandible is overclaiming. The unlocking formulation reads far better.

V113 When would you start functional appliance therapy?

"At the circumpubertal period, around cervical vertebral maturation stage CS3, which has a mean age of about twelve years. Starting earlier wastes compliance on a phase that will not deliver a lasting skeletal effect; starting later wastes the growth. The exception is a child at high risk of incisor trauma with a very large overjet and incompetent lips, where I might start earlier, accepting that the benefit is trauma reduction rather than a better final result.

Timing, the reason, and the exception.

V114 Twin block or Herbst?

"It depends principally on compliance. Pacha's randomised trial found 37 per cent failure with the twin block against 17 per cent with the Herbst, and a larger mean overjet reduction with the Herbst at 7.1 against 5.8 millimetres, with the twin block taking about a month and a half longer. Against that, the Herbst required more visits and produced more emergencies. So for a compliant patient I would use a twin block; where compliance is the limiting factor I would consider a fixed functional.

One trial with four numbers, then a decision rule.

MNEMONIC Pacha: **37% vs 17% failure, 5.8 vs 7.1 mm reduction** — twin block vs Herbst. Choose by compliance, not the numbers.

V115 A patient in a twin block develops lateral open bites. What do you do?

"Nothing alarming, because I expect them. Overjet reduction outpaces overbite reduction, so posterior open bites are the rule during twin block treatment. I would trim the bite blocks progressively to allow the posterior teeth to erupt, and I would make sure the case is settled before the appliance is abandoned. What I would not do is stop treatment or tell the patient something has gone wrong.

Recognising a normal finding as normal is itself a mark.

V116 Was early Class II treatment worth it?

> *"Not for the reasons it was traditionally offered. The trials — Florida, North Carolina and the Manchester twin block trial — showed that early treatment reduces the overjet dentoalveolarly and gives a transient rise in self-esteem, but does not improve the final skeletal or occlusal outcome, does not shorten overall treatment, and results in more attendances. O'Brien found the late-treated group finished slightly better. The one durable benefit is a reduction in incisor trauma, with a relative risk of 0.58. So I would treat early only for trauma risk or significant teasing.*

The trial names and the number. Then the policy.

V117 Do self-ligating brackets treat faster?

> *"No. Scott, Fleming and others found no difference in rate of movement, treatment time, number of visits or occlusal outcome, the pain benefit was marginal, and the expansion claims are unsupported. The distinction between active clips that store energy against the wire and passive slides that do not is a design difference rather than a demonstrated clinical advantage. What decides treatment time is case difficulty, bracket positioning and the operator, which is also what Penning found for customised appliances.*

A short, well-referenced no. Do not hedge.

V118 What can aligners not do well?

> *"Root torque, extrusion, rotation of round-sectioned teeth such as canines and premolars, bodily space closure at extraction sites, molar uprighting and large anteroposterior correction. The reason is that a viscoelastic plastic shell persuades whereas a wire in a slot overpowers, and the reaction forces are accepted rather than designed because the appliance acts on every tooth at once. Attachments help with extrusion and rotation, but they do not change the underlying limitation.*

A list with a mechanism. The plastic persuades formulation is memorable and accurate.

MNEMONIC Aligners struggle: **torque, extrusion, round-tooth rotation, extraction-site closure, molar uprighting, big AP moves**.

V119 What would you tell a patient asking for aligners?

> *"That they are less visible, less painful in the first few days and easier to keep clean, and that in a mild to moderate Class I case they can produce a good result. Then the two numbers: only about six per cent of cases finish without a refinement, and over seventeen per cent convert to fixed appliances to complete treatment. I would say that fixed appliances outperform aligners on every occlusal measure on the ABO grading system, that they need to be worn 20 to 22 hours a day, and that compliance is the weakest element in the system.*

The two numbers make the consent honest and are what the examiner is listening for.

V120 Why did the pre-adjusted appliance not solve everything?

> *"Because it moved the error rather than removing it. Angle's edgewise appliance required all three orders to be bent into the wire tooth by tooth, which is why results were traditionally left ten to twenty per cent undertreated. Andrews built the prescription into the bracket, but the prescription can only express itself if the bracket is in the right place, so the error moved from wire bending to bracket positioning. That is why bracket position, not prescription, decides the result, and why*

bracket repositioning at the working wire stage is the commonest thing that rescues a case.

A historical answer with a present-day clinical consequence.

V121 When would you consider a lingual appliance?

"For an adult whose sole objection to treatment is visibility and who will accept the trade-offs. Those are speech disturbance, usually transient over one to three weeks, tongue discomfort and ulceration, more difficult access and hygiene, longer chairside time, cost and a steep learning curve for me. Biomechanically the bracket sits palatal to the centre of resistance and closer to it vertically, so the moment systems differ, and the anterior interbracket span is shorter, which raises effective stiffness. The comparative evidence is limited, so I would not claim a better outcome, only a less visible one.

Indication, trade-offs, mechanics, and an honest statement about the evidence.

Treatment mechanics and finishing (V122–V129)

Notes In Orthodontics: Chapters 10 and 13; Supplements E and F
Supplements: Revision Handbook **§18**
Practise this: 8 questions · answer each one aloud before reading the model

V122 When is anchorage actually spent in a case?

"Earlier than most people think — during levelling and aligning, before any active mechanics begin. As the initial wire straightens it pushes the incisors labially and the molars mesially, so by the time I reach a working wire the anchorage may already be gone. That is why I use lacebacks, which distalise the canine without tipping it, why I bend the wire back distal to the molar tube, and why in a maximum anchorage case I keep the anchor teeth out of the initial aligning wire. Without lacebacks the lower incisor advances about 1.4 millimetres; with them it retracts about 1.0.

The timing is the insight. The numbers make it concrete.

V123 Take me through your wire sequence in a 0.022 slot.

"Alignment on a light round nickel titanium, 0.012 to 0.018, with second molars bonded as early as I can because they are needed for bite opening and anchorage. Then levelling on round stainless steel, 0.016 to 0.020. Then a 0.019 by 0.025 stainless steel working wire, left passive for about a month so the slots line up, because level slot line-up is what makes sliding mechanics possible. Then space closure on that wire with tiebacks reactivated every four to six weeks at about 150 grams. Then finishing on a wire that fills or nearly fills the slot, 0.021 by 0.025, varying stiffness by changing material rather than dimension.

A sequence with the reasons attached. The passive month is the detail few candidates give.

MNEMONIC Five stages, in order: **align, level, work, close, finish**.

V124 Why 150 grams for space closure?

"Because more is not faster. The MBT work found that attempts to transfer traditional edgewise force levels of 500 to 600 grams to a pre-adjusted appliance produced the roller coaster effect, and that 200 grams gave no advantage over 150. Biologically it makes sense: heavier force hyalinises the ligament on the active side, lengthening the lag phase, and exceeds the anchorage threshold on the reactive side, so the space closes from the wrong end.

A number with both a clinical and a biological justification.

V125 What is the commonest reason a case does not finish well?

"Bracket positioning. The pre-adjusted prescription can only express itself if the bracket is where it should be, so pre-adjustment moved the error from wire bending to bracket placement. Practically I reassess every bracket at the working wire stage, against marginal ridge levels and the gingival contour, and reposition before I go into finishing wires. It is far more efficient than putting a compensating bend in a wire.

Honest and specific. It is also true, which examiners appreciate.

V126 How do you finish a case?

"Deliberately rather than by luck. I reposition brackets at the working wire stage, confirm root parallelism on a panoramic radiograph particularly at extraction sites, then finish in a wire that fills the slot with first, second and third order detailing bends. I check the occlusion in five planes — horizontal, vertical, transverse, dynamic and cephalometric or aesthetic — against the ABO criteria. Then I settle in light round wires with vertical elastics for about six weeks, keeping every space closed, and only then go to retainers. I never go straight from rectangular wires to retention.

Six steps. The settling period is the one candidates omit.

V127 What aesthetic criteria would you check before debonding?

"The midline against the face, the smile arc following the lower lip, incisor show at rest and on smiling, gingival levels with the central and canine zeniths on a line and the lateral half a millimetre below it, incisal embrasures increasing in size from the midline distally, and connector areas following the fifty, forty, thirty proportion. Also the buccal corridors and the vertical position and torque of the canines, which are the teeth most often left slightly extruded or under-torqued.

The appearance of the teeth, not just the occlusion, is the target. Nanda's criteria are all checkable.

V128 What arch form would you use?

"The patient's own. The mandibular arch is the template, and intercanine width is among the least modifiable dimensions, so I preserve the original form rather than imposing a standard one. Changing arch form results in relapse in up to seventy per cent of cases. I would also be aware that preformed superelastic nickel titanium archwires can expand the lower intercanine width by up to 5.9 millimetres if allowed to express fully, so arch form has to be a deliberate decision rather than something the wire does to me.

The last sentence is the practical warning that turns a principle into an action.

V129 When would you use loop mechanics rather than sliding mechanics?

"Where friction is the limiting factor and I need a force system I can predict — for instance, a segmented approach in a case with reduced periodontal support, or where I want differential space closure. The trade-off is that a closing loop avoids friction but produces a force system that is harder to control and requires accurate wire bending, whereas sliding mechanics are simpler and more forgiving but depend on level slot line-up and on the friction being manageable. For most cases in general practice, sliding mechanics on a 0.019 by 0.025 steel wire is the safer choice.

The trade-off and then an honest statement of what you would actually do.

Retention and stability (V130–V139)

Notes In Orthodontics: Topic 20; Chapter 13; Appendix A, Topic 20
Supplements: Revision Handbook **§19**
Practise this: 10 questions · answer each one aloud before reading the model

V130 Why do teeth relapse?

"Three causes on three different timescales. Elastic recoil of the gingival and supracrestal fibres, which takes six to twelve months to reorganise and which is why rotations relapse most readily. Soft tissue pressures, which act continuously and for life, so any tooth moved outside the zone of equilibrium returns towards it. And continued growth, which is slow, lifelong and in the individual's original pattern. Only the first is time-limited, which is why retention has no natural endpoint.

Three causes, three timescales, and the conclusion that follows.

MNEMONIC Three causes, three timescales: **recoil, pressure, growth** — only recoil (6–12 months) is time-limited.

V131 What do you tell a patient about how long they need to wear a retainer?

"The honest sentence: if you want the teeth to stay where they are, you wear a retainer for as long as you want that, and there is no endpoint. I would say it before treatment starts rather than at debond, because it is part of the decision to have treatment at all. I would also tell them that by four to five years after treatment most patients have stopped wearing their retainers, and that relapse is the commonest reason people return.

Honest, specific and delivered at the right time. All three are marked.

V132 Which retainer would you choose for the lower arch?

"In most cases a bonded retainer from three to three, with a vacuum-formed retainer over the top for night wear. The evidence is that fixed retainers are marginally more effective in the lower arch for the first twelve months, particularly for rotations, and equivalent thereafter, and that vacuum-formed retainers outperform Hawleys in the lower labial segment. The trade-off with a bonded retainer is that it needs lifelong monitoring: bond failure, wire fracture, unwanted torque or so-called wire syndrome, calculus and caries.

A choice with the evidence and the trade-off. The monitoring point is essential.

MNEMONIC Five things to check at every bonded-retainer review: **bond, wire, torque, calculus, caries**.

V133 What is Houston's rule about overbite stability?

"That overbite reduction relapses unless the interincisal angle is corrected and an occlusal stop is established. It is the reason a Class II division 2 case cannot simply have the upper incisors tipped labially: without genuine palatal root torque the interincisal angle remains increased and the overbite comes back. And it is why, whichever of the three mechanisms I use to reduce the overbite, I have to end with the lower incisors contacting something.

One rule with two clinical consequences.

V134 Is late lower incisor crowding relapse?

> *"No — it is ageing. It occurs in untreated people as well as treated ones. Sinclair and Little showed mandibular arch width and length decreasing with age in untreated subjects, with increasing anterior crowding, and Thilander found the same. The major driver is continued mandibular growth rotation uprighting the lower labial segment. No extraction pattern and no archwire sequence prevents it, and third molars are not a primary cause — NICE advises against removing them for this indication. The honest answer is long-term bonded retention, and saying so at consent.*

Confusing relapse with age-related change is one of the twelve traps. Say ageing, and say it early.

V135 How far can you advance lower incisors and still expect stability?

> *"About two millimetres, and beyond that permanent retention is very likely to be needed. The reason is equilibrium: the tooth has been moved out of the zone where the resting soft tissue forces balance, and it will return towards it. The same logic applies to lower intercanine expansion, which relapses reliably, and it is one of the reasons the mandibular arch is treated as the template rather than as something to be developed.*

A number and the principle behind it.

V136 What is circumferential supracrestal fiberotomy, and when would you do it?

> *"A surgical division of the supracrestal gingival fibres around a rotated tooth, performed before appliance removal, to reduce the elastic recoil that drives rotational relapse. The evidence for it is weak but it is not harmful. I would consider it for a severely rotated tooth, alongside a bonded retainer rather than instead of one, and I would perform it before debond rather than after, as with a frenectomy for a diastema.*

The timing and the fact that it supplements rather than replaces retention.

V137 What retention would you plan after closing a midline diastema?

> *"A permanent bonded retainer. A closed diastema relapses reliably, and this is one of the situations where I would say to the patient at the outset that the retention is permanent rather than time-limited. If a frenectomy is indicated, I would perform it after the space is closed and before the appliance is removed, because operating first leaves scar tissue that resists closure.*

Two specifics: permanent retention, and the sequence with the frenectomy.

V138 A patient returns after a week without wearing their retainer and says it feels tight. What do you say?

> *"Not to force it, and to come in. Tightness means the teeth have moved, and forcing an ill-fitting retainer can fracture it or move teeth unpredictably. I would check the fit, assess how much movement has occurred, and if it is minor consider an active retainer or a short course of aligners to recover it. Then I would revisit the conversation about why retention has no endpoint, without making the patient feel blamed for a very common lapse.*

Clinical management plus the communication. Both are marked.

V139 When can retention be stopped?

> *"Strictly speaking it cannot, because the three causes of relapse do not all resolve. Fibre recoil settles over six to twelve months, but soft tissue pressures act for life and growth continues slowly throughout adult life in the original pattern. So the answer I give patients is that retention becomes less demanding rather than unnecessary, and that a night-time retainer indefinitely is the price of a stable result. What I would not do is tell a sixteen-year-old that two years of retention will be enough.*

The refusal to give a comforting but false answer is what scores.

Interceptive orthodontics and impacted teeth (V140–V151)

Notes In Orthodontics: Topics 13, 14 and 15; Chapter 12; Appendix A, Topics 13–15
Supplements: Revision Handbook **§20**
Practise this: 12 questions · answer each one aloud before reading the model

V140 Which interceptive treatments have good evidence?

"Two clearly. Correction of a posterior crossbite with a displacement at ages seven to eleven, where the Cochrane evidence is high certainty that a quadhelix or expansion plate beats observation, and that the quadhelix beats a plate and is about three months faster. And maintenance of the leeway space with a lower lingual arch where the space is needed. Beyond those the evidence weakens quickly: arch expansion purely to relieve crowding is weak, and two-phase Class II treatment buys nothing but cost and time except where trauma risk or teasing justify it.

Grading interventions by evidence is exactly what this question is for.

V141 A seven-year-old has a unilateral crossbite with a 2 mm displacement. What do you do?

"Treat it on diagnosis. The displacement tells me this is really a bilateral maxillary constriction that the child is closing out of, which is why it looks unilateral in intercuspation. It is one of the few interceptive interventions with high-certainty evidence, and untreated it causes attrition, is associated with recession on the loaded tooth and with asymmetric mandibular growth. I would use a quadhelix, because it beats a plate and compliance is not a variable in a seven-year-old, overcorrect, and retain for three to six months.

Diagnosis, evidence, consequence of not treating, appliance, overcorrection, retention. Six elements in one answer.

MNEMONIC **DECOR** — Diagnose (bilateral), Evidence (high-certainty), Correct (quadhelix), Overcorrect, Retain 3–6 months.

V142 When would you extract a primary canine to intercept a displaced permanent canine?

"Between ten and eleven years, or ten to thirteen on the Royal College guidance, provided there is space in the arch, the canine crown lies distal to the midline of the lateral incisor root and its angulation to the midsagittal plane is twenty to thirty degrees. I would review at twelve months with a defined endpoint and abandon it if there is no improvement. I would also be honest about the evidence: the 2021 Cochrane review found only very weak evidence that it increases eruption and no evidence at all that it reduces the need for surgery.

The criteria, the review interval, and the honest statement about the evidence.

V143 What actually worries you about a palatally displaced canine?

"Not the impaction — resorption of the lateral and sometimes the central incisor root. It is silent, it can be rapid, and by the time it is obvious the incisor may be lost. That is why I palpate from nine or ten years, why I localise carefully with parallax, and why the one justification I would readily accept for a CBCT here is establishing the relationship of the canine to the incisor root and whether that root is resorbing, because the answer changes whether I keep the lateral incisor.

Naming the real threat rather than the obvious one.

V144 What determines the prognosis for aligning an impacted canine?

"Three positional factors. Height: prognosis worsens above the apical third of the incisor roots. Horizontal position: worse at or across the midline of the lateral incisor. And angulation: worse above about 55 degrees to the midsagittal plane. Ericson and Kurol's split is useful — overlap of less than half the lateral incisor root gives about 91 per cent normalisation after primary canine extraction, and more than half about 64 per cent. Age matters too: the older the patient, the poorer the prognosis.

Three factors, each with a threshold, plus the Ericson and Kurol numbers.

V145 Open or closed exposure?

"The Cochrane review by Parkin found no difference in success or other outcomes, on low to very low quality evidence. Cassina subsequently reported that open exposure was associated with just over two months less time to alignment and lower odds of ankylosis. For a buccal canine lying below the mucogingival junction I would use an apically repositioned flap, following Kokich, so the tooth erupts through attached tissue. In practice the decision is usually made with the surgeon and depends on the depth and position of the tooth.

Evidence, then the one situation where the technique is determined by the anatomy.

V146 What are the alternatives to exposing and aligning an impacted canine?

"Leaving it and monitoring, if it is asymptomatic, there is no resorption of adjacent roots, and the patient accepts retaining the primary canine — with a defined review, because monitoring is not the same as ignoring. Extracting the canine and either closing the space or restoring it. Autotransplantation, with a success rate of around 88 per cent, splinted semi-rigidly for no more than three weeks and then root treated. And of course accepting the primary canine while it lasts, with the honest statement that it will not last for ever.

Four options with the conditions attached to each.

MNEMONIC **LEAP** — Leave (monitor), Extract, Autotransplant, Primary canine (accept, knowing it won't last).

V147 Take me through your management of an impacted lower second molar.

"A ladder, with a defined endpoint at each step. Observe for six to twelve months; no progress means an established impaction. Disimpact with a separator, brass wire or an open coil placed below the contact point. Remove the third molar if it is contributing to the obstruction. Surgically upright, exposing, removing the third molar, luxating and bonding a bracket at surgery. And extract the second molar as a last resort, which is the least successful outcome because it commits the patient to unpredictable third molar eruption. Throughout I would keep primary failure of eruption in mind, because traction on such a tooth causes ankylosis.

Five steps and the differential diagnosis. The last is what stops you doing harm.

V148 A nine-year-old has lost an upper central incisor by avulsion. What do you do?

"Attempt re-implantation, always, even if the long-term prognosis is poor, because it maintains both the space and the alveolar bone and buys years. Then the decision is between maintaining the space

with a denture or a retainer with a pontic, which risks alveolar resorption, and allowing the space to close and reopening it later, which preserves bone but commits the patient to further orthodontics. Definitively, a resin-retained bridge or eventually an implant — and implants must wait until growth is complete, because continued vertical alveolar growth leaves the crown in infra-occlusion.

Re-implantation first. The alveolar bone argument runs through the whole answer.

V149 When can orthodontic force be applied to a previously traumatised tooth?

"It depends on the injury. Three months after an uncomplicated crown fracture, a subluxation or a minor lateral luxation. Twelve months after a moderate or severe lateral luxation, an intrusion, an extrusion or a re-implantation, and usually twelve months after endodontic treatment following trauma. After a complicated crown fracture, until a hard tissue barrier is visible. A root fracture healing with dentine and cementum needs twelve to 24 months, and one healing with connective tissue should be treated as a short-rooted tooth with light forces.

A table learned as a list. It is frequently examined and easily forgotten.

V150 How do you monitor a traumatised tooth during orthodontic treatment?

"Before: vitality testing and periapical radiographs, recorded as a baseline. During: light forces, avoiding excessive movement, vitality testing every three months and a repeat radiograph six to nine months after starting movement. If resorption appears, I pause for three months, which is shorter than the three to six months quoted for orthodontically induced resorption generally. And I would have consented specifically for the three complications of moving such a tooth: loss of vitality, pulpal calcification and resorption where resorption has already occurred.

Baseline, monitoring interval, action, and the three complications named at consent.

V151 When would you use serial extraction?

"Rarely, and only in severe crowding with a Class I skeletal pattern and no vertical problem. The sequence is primary canines at about eight or nine, first primary molars at nine or ten, then the first premolars as they erupt. The disadvantages are substantial: a long period of unattractive spacing, a deepening overbite, retroclination of the lower incisors, a fixed appliance usually still needed to finish, and an irreversible commitment made at a young age. It is contraindicated in Class II and Class III, in deep or open bite, and in hypodontia.

The conditions and the disadvantages together. Anyone can recite the sequence.

MNEMONIC Serial extraction order: **primary canines (8–9y), first primary molars (9–10y), first premolars (as they erupt)**.

Class II and Class III management (V152–V161)

Notes In Orthodontics: Topics 12 and 17; Chapters 9, 12 and 13; Appendix A, Topics 12 and 17
Supplements: Revision Handbook **§22–§23**
Practise this: 10 questions · answer each one aloud before reading the model

V152 How would you treat a 12-year-old with an 11 mm overjet and a skeletal Class II?

"I would establish first that he is growing, and where he is in the growth curve. Given the size of the overjet and the trauma risk I would not wait long. A twin block worn full time, timed to about CS3, expecting the correction to be largely dentoalveolar, with lateral open bites appearing and managed by trimming the blocks, over about six to nine months. Overcorrect, then transition into a fixed appliance to align, level and detail. Retain with a vacuum-formed retainer and, if the skeletal correction is at risk, night-time functional appliance wear for at least a year. At consent I would cover the trauma risk while the overjet persists, that the skeletal gain is small and largely transient, and the possibility of surgery if growth is unfavourable.

A full case answer. Growth first, appliance, expected findings, transition, retention, consent.

V153 Which direction of headgear pull would you choose, and why?

"From the vertical dimension rather than from the anteroposterior discrepancy. High pull or occipital for a hyperdivergent patient with increased vertical proportions, because it is intrusive on the molar. Cervical or low pull for a hypodivergent patient with reduced vertical proportions and a deep bite, because it is extrusive. Straight or combination pull for average proportions where I want pure distalisation. The single commonest error is to choose the headgear from the Class II and ignore the face.

The rule is one sentence. The error it prevents is the reason for the question.

V154 Why is the high-angle Class II the difficult case?

"Because everything that corrects the anteroposterior discrepancy tends to open the bite. Buccal segment extrusion, cervical headgear and Class II elastics all rotate the mandible down and back, which increases face height, worsens the Class II and relapses. So I have to correct the sagittal relationship without any extrusion: high-pull headgear or skeletal anchorage, possibly molar intrusion, and consideration of extraction to allow the mandible to rotate forwards. Anchorage is also at greater risk in these patients.

Naming the mechanism, then listing what you would do instead.

V155 What is the mechanical problem in Class II division 2?

"The increased interincisal angle combined with a deep, often complete, overbite. Retroclined upper central incisors have to be torqued labially at the root rather than merely tipped, or the interincisal angle stays increased and the overbite relapses — that is Houston's rule. So the sequence is to procline the upper labial segment to convert the case to a division 1 pattern, reduce the overbite, then retract if extraction space has been created. And because there is often little or no overjet, many division 2 cases are treated non-extraction in the upper arch, since extraction space there has

nowhere to go.

Problem, sequence, and the extraction consequence.

V156 How would you retain a corrected deep overbite?

"A Hawley retainer with an anterior bite plane, worn long-term and often into the early twenties. The reason is that overbite reduction is easy and holding it is difficult: the correction relapses unless the interincisal angle is genuinely corrected and an occlusal stop established, and the growth pattern that produced the deep bite continues. A vacuum-formed retainer holds alignment but does not provide the bite plane, so I would use both.

The appliance and the reason it is not a vacuum-formed retainer.

V157 Why is Class III harder than Class II?

"Because the growth works against me. In Class II residual mandibular growth helps to correct the discrepancy; in Class III it worsens it, and mandibular growth in Class III patients persists beyond adolescence and later in males. So an early correction can be undone, monitoring is often the right answer, and surgical decisions have to wait for serial superimposition to confirm growth has ceased.

One sentence of mechanism that explains the whole chapter.

V158 When would you treat a Class III child early?

"When the case is favourable: a Class I or mild Class III skeletal base, average or reduced lower face height, an average or increased overbite, relatively uncompensated incisors, and an anterior displacement on closing. In those, a facemask with or without expansion at 12 to 14 hours a day before age ten is worth attempting, and Mandall found over sixty per cent avoided surgery against 36 per cent untreated. In an unfavourable case — moderate to severe base, increased face height, reduced overbite, heavily compensated incisors, no displacement — I would monitor into the late teens and avoid mandibular extractions in case surgery is needed.

The favourable and unfavourable lists, then the two policies. This is a set-piece.

MNEMONIC **SHOPS** — Skeletal mild, Height average or reduced, Overbite average or increased, Proclined lowers, Slide on closing. All five means favourable.

V159 What do you tell the parents of a Class III seven-year-old who want treatment now?

"I would separate two things. If there is an anterior crossbite with a displacement, that is treated on diagnosis, because it causes attrition and recession on the lower incisor and can drive asymmetric growth, and it corrects in three or four weeks with an incline plane. The skeletal Class III is a different question: early protraction improves the chance of avoiding surgery, sixty per cent against 36, but most of the skeletal effect is lost with growth and Proffit reports a 25 to 33 per cent recurrence of the crossbite at eight years. So I would treat the displacement now, and be honest that the skeletal picture may still declare itself in the teens.

Separating the two problems is the whole answer.

V160 Does a chin cup work?

"Not in the way it is often described. There is no agreement that it inhibits mandibular growth. What it largely does is redirect the mandible downwards and backwards, which improves the incisor relationship in the short term at the cost of increasing face height, and the short-term gain is not maintained after growth. So I would not offer it as a means of controlling mandibular prognathism.

A short, evidence-based no with the mechanism attached.

V161 How would you camouflage a mild Class III?

"Extract further forward in the mandible — lower first premolars with upper second premolars — procline the uppers and retrocline the lowers. Occasionally a single lower incisor extraction. I would be cautious with Class III elastics, and would avoid them altogether in a high-angle case because they extrude the lower molars and rotate the mandible down and back. And I would check first how much compensation is already present: lower incisors at eighty degrees or less tell me there is very little room left, which usually means camouflage is not the answer.

The pattern, the caution, and the check that decides whether camouflage is possible at all.

MNEMONIC Camouflage extracts **forward below, back above** — lower first premolars with upper second premolars. Lower incisors at 80 deg or less: no room left.

Transverse and vertical problems (V162–V173)

Notes In Orthodontics: Topics 11, 16 and 19; Chapters 9, 12, 13 and 15
Supplements: Revision Handbook **§24–§25**
Practise this: 12 questions · answer each one aloud before reading the model

V162 How do you classify a crossbite?

"By site, cause and displacement. Site: anterior or posterior, unilateral or bilateral, buccal or lingual. Cause: dental, where individual teeth are displaced; skeletal, where there is a genuine transverse discrepancy of the bases; or functional, where a bilateral discrepancy produces a lateral displacement into an apparently unilateral crossbite. And with or without a mandibular displacement on closing, which is the single most important of the three because it determines urgency and prognosis.

Three axes, with a statement of which matters most.

MNEMONIC Three axes: **site, cause, displacement** — and displacement is the one that decides urgency.

V163 Which expansion appliance would you choose?

"For a dental or mild skeletal discrepancy in the mixed dentition, a quadhelix: it beats an expansion plate on the Cochrane evidence, is about three months faster, and takes compliance out of the equation. For a true transverse skeletal deficiency, rapid maxillary expansion at one to four turns a day over two to four weeks, generating forces up to ten kilograms, achieving up to four millimetres at the suture which is about half the screw expansion, and retained in situ for three to six months. Beyond about five millimetres in a non-growing patient, surgically assisted expansion.

Three scenarios, each with the appliance and the numbers.

V164 How much of rapid maxillary expansion is skeletal, and how much relapses?

"Up to four millimetres at the suture, about fifty per cent of the total screw expansion, with roughly ten per cent of the skeletal change relapsing at four to eight months. Dental expansion is better retained with slow expansion at about 95 per cent than with rapid at about 85 per cent. I would add Proffit's dissent: at four months rapid and slow expansion have produced about the same skeletal to dental split, so rapid activation is not actually an effective way of minimising tooth movement.

The numbers, then the dissenting position. Naming a disagreement shows you have read more than one source.

MNEMONIC **4–50–10** — up to 4 mm at the suture, about 50 per cent of screw expansion is skeletal, about 10 per cent of that relapses by 4 to 8 months.

V165 Would you use expansion to treat a child's breathing?

"No. Rapid maxillary expansion is not recommended as a treatment for nasal obstruction or sleep-disordered breathing on current evidence. If the parents are worried about the breathing — and snoring, witnessed apnoeas, restless sleep or daytime behavioural problems are real concerns — I would refer to ENT or paediatrics for proper assessment including a sleep study, rather than treat it with an orthodontic appliance. I would expand if there were an orthodontic indication, and say plainly that any airway benefit is not the reason.

Refusing an attractive but unsupported indication, while taking the underlying concern seriously.

V166 In what order do you correct crossbites?

"Posterior before anterior, always. If I correct the anterior crossbite first and then expand, I lose the correction. And the anterior correction is self-retaining only if a positive overbite results, which is why I check that there is room and that the overbite will be positive before proclining. An anterior crossbite corrected with a positive overbite is usually stable and needs no retention.

A rule, a reason, and the condition for stability.

V167 How do you reduce an overbite, and which method is stable?

"Three mechanisms: intrude the incisors, extrude the buccal segments, or procline the labial segments. Which is stable depends entirely on growth. In a grower, buccal segment extrusion gives relative incisor intrusion and is stable, because condylar growth compensates for the increased posterior vertical dimension. In a non-grower the same manoeuvre rotates the mandible down and back, worsens the Class II, increases face height and relapses, so I need true intrusion — a utility or bypass arch, segmental mechanics, or miniscrews, at about sixty grams total or fifteen per tooth. Proclination gives about 0.1 to 0.2 millimetres of overbite change per degree, and is unstable unless the interincisal angle is genuinely corrected.

Three mechanisms and the growth-dependent stability. This is one of the twelve traps.

MNEMONIC **Intrude, extrude, procline** — the three mechanisms. Extrusion is stable only in a grower; in a non-grower it must be true intrusion.

V168 How much intrusion can you actually achieve?

"Modest amounts. Segmental mechanics achieve about 1.5 millimetres in the maxilla and 1.9 in the mandible in non-growing patients, on Ng's figures. Miniscrews give about 0.67 to 0.95 millimetres more than segmental mechanics with less molar extrusion, though on low-quality evidence. Segmental techniques give greater true intrusion than a continuous reverse curve of Spee, and a Ricketts utility arch produces more lower incisor proclination. So I would be realistic with the patient about how much of the overbite reduction will come from intrusion.

Quantified expectations. Being realistic with the patient is the clinical point.

V169 What is the key sign of a skeletal anterior open bite?

"An interlabial gap of more than three millimetres at rest, in a face with an increased lower face height, a high mandibular plane angle and a backward mandibular rotation. That distinguishes it from a habit-related open bite, which is typically asymmetric, often localised to the digit, and associated with a posterior crossbite, proclined uppers and retroclined lowers. The distinction matters because a habit-related open bite usually closes spontaneously if the habit stops before the permanent incisors are established, and a skeletal one will not.

One sign, the differential, and the consequence for management.

V170 How would you correct a skeletal anterior open bite?

"Vertical control throughout, and no extrusive mechanics. High-pull headgear, posterior bite blocks, and above all molar intrusion with temporary anchorage devices, because one millimetre of molar intrusion gives about three millimetres of open bite reduction, which is the most efficient lever available. Extraction may help by allowing the mandible to rotate forwards. If that is not enough,

posterior maxillary impaction with mandibular autorotation once growth is complete. And I would say at consent that about twenty per cent of open bite corrections relapse, surgical and non-surgical alike.

The lever, the alternatives, and the relapse number.

V171 Is the tongue the cause of an anterior open bite?

"Usually not. Tongue posture is generally adaptive to the open bite rather than causative — the tongue fills the space that exists. A genuinely endogenous tongue thrust is rare, and those cases are unstable and need permanent retention. So I would not blame the tongue reflexively, I would not promise that exercises will close the bite, and I would look for the actual aetiology: habit, skeletal pattern, or iatrogenic extrusion in a high-angle patient.

Reversing the usual assumption. The iatrogenic cause is the one to volunteer.

V172 How would you assess a facial asymmetry in a growing child?

"Clinically first: from in front, from above and behind, and with the patient biting on a spatula to reveal a true occlusal cant. Then the path of closure, because a deviation from retruded contact position means a functional shift rather than a true skeletal asymmetry. Mild asymmetry is normal — clinically significant asymmetric jaw development affects fewer than 0.5 per cent, and the mandible is involved in 85 to 90 per cent of cases, most often after a childhood condylar fracture. In a grower with mild asymmetry the answer is usually to watch, but with a defined review interval and records I can superimpose.

Clinical assessment, the prevalence, and the restraint.

MNEMONIC Look **from in front, from above and behind, then on a spatula** — the spatula reveals a true cant; the path of closure separates a shift from a skeletal asymmetry.

V173 What would make you refer an asymmetry?

"Progression. Accelerating deviation of the chin, a new or worsening occlusal cant, or new temporomandibular symptoms appearing in the teens. Those suggest an active condylar hyperplasia, and operating on an active process invites relapse, so growth has to be monitored before committing to surgery. Scintigraphy is sensitive but non-specific with false positives; serial three-dimensional facial superimposition, using the forehead, mid-nose and infraorbital regions, is a reasonable non-invasive alternative.

Progression is the trigger. The imaging caveat is the detail.

TMD, the airway and the medically complex patient (V174–V183)

Notes In Orthodontics: Topics 10 and 18; Chapter 2, pp. 102–105; Chapter 7
Supplements: Revision Handbook **§26, §29**
Practise this: 10 questions · answer each one aloud before reading the model

V174 What is the relationship between orthodontics and temporomandibular disorder?

"Orthodontics is TMJ-neutral. Weak statistical associations exist between certain occlusal traits and TMD signs, but there is no evidence of causation in either direction: orthodontics neither causes nor prevents temporomandibular disorder. The studies I would name are the American Association of Orthodontists funded work of 1992, following the Brimm versus Malloy award of 850,000 dollars in 1987. The principal confounder is that symptoms commonly begin in late adolescence, which is exactly when we treat.

State it, defend it with the named studies and the confounder, and stop there.

V175 A patient develops jaw pain and clicking during fixed appliance treatment. What do you do?

"Stop active mechanics and manage conservatively. I would take a proper history, examine the joints and muscles, record range of opening, deviation, sounds and tenderness, and reassure the patient about the natural history, because most symptoms remit. Soft diet, jaw rest, avoiding wide opening and parafunction, analgesia, and physiotherapy or a stabilisation splint if needed. What I would not do is reach for an occlusal explanation, perform an occlusal adjustment, or promise that completing the treatment will resolve it.

Suspend, assess, manage conservatively, and resist the occlusal explanation.

MNEMONIC Conservative five: **soft diet, jaw rest, no parafunction, analgesia, physiotherapy or splint** — and no occlusal adjustment.

V176 Which imaging would you use for the temporomandibular joint?

"MRI for the disc, the retrodiscal tissues and any effusion — it is the investigation of choice for internal derangement. CT or CBCT for bone, including condylar morphology and degenerative change. A panoramic radiograph gives a screening view of condylar form only. The point to make is that CBCT cannot image the disc or muscle, so a CBCT ordered to investigate clicking will not answer the question.

The last sentence is the one that shows you understand what each modality does.

V177 What do you tell a parent worried about their child's snoring?

"That it is worth taking seriously, and that it is not an orthodontic problem in the first instance. Snoring, witnessed apnoeas, restless sleep, mouth breathing, daytime somnolence or behavioural problems all warrant referral to ENT or paediatrics, with a sleep study if indicated. Paediatric obstructive sleep apnoea affects about two per cent of children, and adenotonsillectomy is the first-line treatment in most. I would not diagnose or treat it orthodontically, and I would not offer expansion as a treatment for it.

Taking the concern seriously while declining the orthodontic role.

V178 What is the relationship between the airway and facial form?

> *"An association, not an established cause. Chronic nasal obstruction is associated with mouth breathing, a low tongue posture and lips apart, and with a narrower maxilla, increased face height and backward mandibular rotation — the so-called adenoid facies. Experimental nasal obstruction in primates produced facial change, but human evidence is observational and confounded. Adenoid tissue peaks at three to five years and involutes through adolescence, so many children grow out of the obstruction. I would state the association and refuse to assert the cause.*

Precision about the strength of a claim is itself examinable.

V179 How does a childhood condylar fracture affect orthodontic management?

> *"Considerably, because the condyle is the mandible's principal growth site and a fracture there is the commonest cause of significant mandibular asymmetry. Acutely, the management is conservative and functional — early mobilisation, soft diet, analgesia, physiotherapy — because prolonged immobilisation risks ankylosis, which is the outcome to avoid at all costs. Orthodontically, I would monitor growth with serial records I can superimpose, consider a functional appliance to encourage condylar growth and maintain symmetry, and expect that definitive correction of an established asymmetry will be orthognathic.*

Acute management, the ankylosis risk, and the orthodontic role.

V180 Which medical conditions change your force levels?

> *"Sickle cell disease, thalassaemia, a history of childhood malignancy and its treatment, diabetes, previous radiotherapy, and any patient with short roots. In all of those I would use lighter forces, keep the aims simple, and keep treatment short. I would also keep in mind that bisphosphonates slow or inhibit tooth movement and delay eruption, and that stopping the drug does not help, so in those patients the aims have to be modest from the outset.*

Group them by what they change — force, extraction, appliance, timing — rather than listing conditions.

MNEMONIC Lighter forces for **blood, cancer, sugar, rays, short roots** — sickle cell and thalassaemia, childhood malignancy, diabetes, radiotherapy, short roots.

V181 Which conditions change your appliance choice?

> *"Epilepsy, where I would avoid removable appliances because of the risk of displacement during a seizure. Phenytoin therapy or transplant immunosuppression, where gingival hyperplasia makes bonded tubes preferable to bands. Asthma, where I would avoid a palate-covering removable. Nickel allergy, where I would avoid nickel titanium but use stainless steel freely. And a patient likely to need MRI, where ceramic brackets or aligners may be preferable.*

Five conditions, each with the appliance consequence.

V182 Does a patient with a heart murmur need antibiotic cover for band placement?

> *"Not under NICE guidance, which advises that antibiotic prophylaxis solely to prevent infective endocarditis should not routinely be provided for dental procedures. I would add that this conflicts with the American Heart Association and European Society of Cardiology positions, which still*

recommend cover for the highest-risk patients, and that mortality from infective endocarditis is around 21 per cent, which is why the position remains contested. In an individual high-risk patient I would liaise with the cardiologist rather than decide alone.

Give the guidance, acknowledge the conflict, and say what you would actually do.

V183 When would you defer orthodontic treatment on medical grounds?

"At least six months after an organ transplant. Until a malignancy is in remission. While an eating disorder is active, with liaison, prevention and management of erosion in the interim. During pregnancy for anything elective, providing urgent care ideally in the second trimester. And where a medical condition is uncontrolled and its control would change the risk — poorly controlled diabetes being the obvious example. In each case I would explain the reason and set a review rather than simply refusing.

Five situations and the principle that deferral is explained and reviewed rather than imposed.

Adults, orthognathic surgery, clefts and governance (V184–V193)

Notes In Orthodontics: Chapters 15 and 16; Part V, units V.2–V.4 for governance
Supplements: Revision Handbook **§27, §28, §30**
Practise this: 10 questions · answer each one aloud before reading the model

V184 What is different about treating an adult?

"No growth, so growth modification is unavailable and every millimetre comes from tooth movement or from surgery. Overbite mechanics reverse: buccal segment extrusion, which is stable in a child, rotates the mandible down and back in an adult. The periodontium is often reduced, which moves the centre of resistance apically and changes the mechanics rather than just the risk. Movement is slower. There are restorations to bond to and missing teeth to plan around. Aesthetic expectations are higher and more specific. And I would probe every tooth and record pre-existing recession before I start.

Seven differences. Growth, mechanics, periodontium and records are the four that matter most.

V185 What is decompensation, and why does it matter?

"It is the deliberate removal of the dental compensation that has developed around a skeletal discrepancy, and it is what pre-surgical orthodontics is for. In a Class III that means retroclining the uppers and proclining the lowers, so the reverse overjet gets worse before surgery. It matters because it is the opposite of camouflage: I cannot start camouflage and change my mind, which is why the camouflage-or-surgery decision has to be made before treatment begins. And it matters at consent, because the patient has to understand that the bite will look worse for eighteen months.

Calling pre-surgical orthodontics aligning the arches is one of the twelve traps.

V186 What is the hierarchy of stability in orthognathic surgery?

"Direction of movement matters more than fixation, which matters more than surgical technique. Most stable are maxillary impaction, mandibular advancement in a low- or average-angle patient, and genioplasty in any direction. Then maxillary advancement, and maxilla up with mandible forward. Mandibular setback is problematic. Least stable are maxillary down-grafting and surgical maxillary expansion. Very stable means about a ninety per cent chance of no clinically significant change.

The ordering principle first, then the list. The definition of very stable is a nice detail.

V187 What would you consent a patient for before orthognathic surgery?

"Nerve injury, particularly the inferior alveolar nerve in a sagittal split, with altered sensation common transiently and occasionally permanent. Infection, bleeding, unfavourable split, relapse, and the possibility of further orthodontics afterwards. Temporomandibular symptoms, and condylar resorption, which is a particular risk in young women with a high-angle pattern and pre-existing joint disease. Plate removal. The eighteen months of pre-surgical orthodontics during which the bite gets worse. The psychological adjustment to a changed face. And the possibility of dissatisfaction even with a technically excellent result, which is why I would screen for body dysmorphic disorder.

A long list, and the last item is the one that distinguishes a thoughtful answer.

V188 Take me through the cleft care timeline.

> *"At birth, diagnosis, feeding advice, hearing assessment and parental support. Lip repair at three to six months, by the rule of tens. Palate repair at six to twelve months, before speech develops, with hearing monitored because glue ear is near-universal. Prevention and fissure sealing through the mixed dentition, with arch expansion. Alveolar bone graft at eight to ten years, before the canine erupts and when about two thirds of its root is formed, using cancellous bone usually from the iliac crest. Definitive orthodontics in adolescence, and orthognathic surgery in the late teens if the maxillary hypoplasia warrants it. Throughout, multidisciplinary team care with national audit.*

A timeline learned as a sequence of ages. Expand before you graft is the detail examiners listen for.

MNEMONIC Lip **3–6 months**, palate **6–12 months**, graft **8–10 years**; fitness by the **rule of tens** — 10 weeks, 10 lb, 10 g/dL.

V189 Why is the alveolar bone graft timed to the canine?

> *"Because the purpose is to provide bone for the canine to erupt into. Grafting before eruption, at about two thirds of root formation, means the canine erupts through the graft and maintains it, whereas grafting after eruption leaves a graft with no functional stimulus. The graft also stabilises the maxillary segments, closes the oronasal fistula, supports the alar base and makes later implant placement possible. And I would expand first, because the graft needs surgical access and the arch needs its form established.*

Purpose first, then the other four benefits, then the sequencing point.

V190 What dental problems would you expect in a unilateral cleft?

> *"A missing lateral incisor on the cleft side, most commonly, together with supernumerary and malformed teeth often on both sides of the cleft. The central incisor adjacent to the cleft is characteristically rotated and tipped towards it and often hypoplastic. Anterior and posterior crossbites from transverse and sagittal deficiency and from surgical scarring. Higher caries risk and enamel defects, with difficult hygiene. Dental anomalies are commoner in bilateral cases. And retention is prolonged, usually permanent.*

The list, plus the two generalisations about bilateral cases and retention.

V191 How would you distinguish Apert from Crouzon syndrome?

> *"By the hands and feet. Both involve FGFR2 mutations and both produce craniosynostosis with midface hypoplasia, exophthalmos and a relative mandibular prognathism. Apert additionally has syndactyly, and its midface hypoplasia is generally more severe. Crouzon has no limb anomalies. I would also be careful to distinguish a Turner tooth, which is a localised enamel hypoplasia caused by infection or trauma to the primary predecessor, from Turner syndrome, which is 45,X — examiners do use both terms in the same paper.*

One discriminating feature, plus the terminological trap.

V192 What would you record in the notes at an orthodontic assessment?

> *"The complaint in the patient's own words and what they expect treatment to achieve. The history. The examination in three planes with the crowding quantified. The IOTN. A provisional problem*

list. Which records I justified and took, with the justification for any radiograph and a report on it. Then the plan: facial and occlusal aims, the options discussed including no treatment, the extraction pattern, the anchorage plan, the appliance, the estimated duration and the agreed retention plan. And the consent discussion: what was said, what was warned about, what was given in writing, who was present, and that the patient agreed.

A structured list. It is also the answer to any complaint that arises three years later.

MNEMONIC **Complaint, expectation, history, examination, IOTN, problem list, records, plan, consent** — in that order, the complaint in the patient's own words.

V193 What would you audit in your own orthodontic practice?

"Case outcomes using PAR or ICON, because those are designed for auditing an operator over many cases. Failure and breakage rates. Decalcification rates at debond, which is the one that tells me most about my patient selection. Treatment duration. Discontinuation rates. Radiograph reject rates and dose. And decontamination cycle records. If I found a decalcification rate I was unhappy with, I would change my selection criteria rather than my fluoride regimen first.

An audit list with one worked inference. That last sentence is the mark.

The questions that test judgement (V194–V207)

Notes In Orthodontics: Crash notebook *The last hour*; the twelve traps
Supplements: Revision Handbook **Handbook Part 5**
Practise this: 14 questions · answer each one aloud before reading the model

V194 Present this case to me.

"A 13-year-old growing girl with a mild skeletal Class II pattern, increased vertical proportions and a symmetrical face. Her lips are incompetent with an interlabial gap of about five millimetres and a lower lip trap. Intra-orally she is caries-free with fair hygiene, all teeth present, with six millimetres of upper crowding and four of lower. She has a Class II division 1 incisor relationship with an overjet of eight millimetres and an incomplete overbite, on a half-unit Class II molar relationship, with no displacement on closing. My working diagnosis is a Class II division 1 on a mild skeletal Class II base with a vertical tendency, and the decision in this case turns on whether the vertical pattern will tolerate the mechanics needed to reduce that overjet.

Sixty seconds, in a fixed order, ending with what the decision turns on. Rehearse it until it is automatic.

V195 What is the first thing you want to know about any orthodontic patient?

"Two things, and I would give them in this order. What the patient themselves is concerned about, because that becomes the outcome against which the treatment will be judged and it ranks highly in the problem list. And whether they are still growing, because that determines which of the three treatment routes is available and it changes the mechanics for overbite reduction, expansion and protraction.

The two opening moves. Both should be automatic.

V196 You do not know the answer to a question. What do you do?

"Say so, and then say how I would find out. Something like: I am not certain of that figure, and I would rather not guess; what I would do is look it up in the Handbook before I saw the patient again, and in the meantime the principle I would work from is this. That scores; a guess defended does not, and a mis-attributed reference is worse than no reference at all.

Examiners are testing safety, not omniscience. This answer is worth practising out loud.

V197 An examiner pushes you towards a plan you would not carry out. How do you respond?

"Politely, and by saying where my limit is and why. Something like: I can see the argument for that, but it is beyond what I would attempt myself, and what I would actually do is refer the case at this point. Being led step by step into a plan I would not deliver is a worse outcome than disagreeing with an examiner, because the examination is testing whether I know my own limits.

Being led is a recognised failure mode. Practise the sentence that stops it.

V198 You realise mid-answer that you have said something wrong. What do you do?

> *"Correct it immediately and move on. Something like: I said 150 grams for intrusion, and that is wrong — intrusion needs ten to twenty grams per tooth, because the force is concentrated at the apex. Correcting myself costs one mark; leaving it costs the examiner's confidence for the rest of the viva.*

A recoverable error, handled well, can leave a better impression than no error at all.

V199 How would you structure an answer to a broad question?

> *"By signposting. Something like: there are three things I would want to say about that — first, second, third — and then finishing in the clinic with what I would actually do. Structure earns breadth, numbers earn depth, and weighing the evidence earns judgement. What loses marks is a stream of associations with no shape.*

The single most useful technique in the oral. It also buys thinking time.

MNEMONIC Signpost, then land: **name the number, first, second, third, then what you would actually do**. The clinical close is what earns the mark.

V200 What would you do if you disagreed with a colleague's treatment plan for a shared patient?

> *"Speak to the colleague first, not to the patient. I would ask what their reasoning was, because there is usually something I do not know, and I would put my concern in terms of the specific finding rather than the general plan. If I still disagreed and thought the patient was at risk of harm, I would say so clearly, document the discussion, and if necessary suggest a second opinion. What I would not do is undermine a colleague in front of a patient.*

Professionalism questions appear in every oral. Colleague first, patient's interests throughout, record it.

V201 A patient is unhappy with their result. How do you handle it?

> *"Listen first, and find out specifically what they are unhappy about, because it is often not what I would have predicted. Then look at it with them against the records and the aims we agreed at the start. If they are right and something is correctable, I would say so and offer to correct it. If it is a limitation we discussed and agreed at consent, I would show them that discussion in the notes and explain it again. Either way I would record the conversation, and I would treat it as audit data about my own practice.*

Listening, records, honesty, and the audit loop. Four elements.

V202 How do you decide whether a case is within your competence?

> *"By asking whether the aims I have set can be achieved by the mechanics I am competent to deliver, and whether I would be comfortable managing the foreseeable complications. If the answer to either is no, the case goes. In practice that means referring severe skeletal discrepancies and likely surgical cases, complex impactions needing exposure and traction, cleft and craniofacial cases, significant or progressive asymmetry, severe hypodontia, reduced periodontium cases needing complex mechanics, and suspected primary failure of eruption. And I would rather decide that at the assessment visit than eighteen months in.*

A test rather than a list, followed by the list. The test is what generalises.

V203 What are the commonest reasons candidates lose marks in an oral like this?

> *"Not saying whether the patient is growing. Quoting a number confidently and wrongly. Presenting tooth movement as mechanics rather than biology. Reversing the Class II and Class III extraction patterns, or RANKL and OPG. Forgetting the facial aims. Calling pre-surgical orthodontics aligning the arches. Overclaiming the benefits of treatment. Omitting radiation justification. And leaving retention out of the plan, or mentioning it as a sentence in the conclusion rather than as a paragraph in the plan.*

The twelve traps, condensed. Read them on the morning of the examination.

MNEMONIC The four that cost most marks: **growth unstated, a confident wrong number, mechanics without biology, retention as an afterthought**.

V204 Tell me about a case that went wrong.

> *"I would choose one honestly and describe it in the same order I would describe any case: what I planned, what happened, when I noticed, what I did, and what I would do differently. For example, a case where I underestimated the anchorage demand, saw the molars coming forward at the working wire stage, and had to add reinforcement mid-treatment rather than having planned it at the start. What I learned was to do the space analysis formally rather than by eye, because the anchorage requirement follows from it.*

Every candidate has one. Preparing it turns a difficult question into an easy one.

MNEMONIC Five beats, in order: **planned, happened, noticed, did, would do differently**.

V205 What is the single most important thing you have learned about orthodontics?

> *"That growth status decides almost everything, and that most of what is presented as skeletal correction is dentoalveolar and largely transient. The corollary is that the profession has often been more confident than the evidence justified — early Class II treatment, self-ligation, third molars and crowding, orthodontics and TMD — and that being able to say which is which is more useful than knowing any individual technique.*

An open question. A prepared, specific answer beats a general one.

V206 Why should a general dentist with an interest in orthodontics treat orthodontic cases at all?

> *"Because a great deal of orthodontic need is straightforward, because access to specialist care is limited, and because interceptive treatment in particular is most effectively delivered by the clinician who sees the child regularly — palpating for canines from nine or ten years, spotting a crossbite with a displacement, timing a first molar extraction. What makes that safe is knowing the boundary: treating what I can treat well, referring what I cannot, and recognising the difference early.*

The question behind the question is whether you understand the level the Diploma is pitched at.

V207 If you had one sentence to give a colleague sitting this examination, what would it be?

> *"Say whether the patient is growing in your first sentence, say where you would refer before you are asked, and finish every answer with what you would actually do.*

The three moves that carry an oral examination. Everything else is detail.

Viva tracker

One row per section. Tick a box each time you have answered every question in that section aloud, *to the end, before reading the model. Three ticks is the target. The last column is the Handbook section to re-read if the words would not come.*

Questions	Section	No.	1st	2nd	3rd	Hbk
V1–V10	Growth, development and the craniofacial complex	10	☐	☐	☐	§1–§2
V11–V20	Development of the dentition	10	☐	☐	☐	§3
V21–V30	The biology of tooth movement and root resorption	10	☐	☐	☐	§4
V31–V38	Occlusion, aetiology and epidemiology	8	☐	☐	☐	§5
V39–V48	Treatment need, indices and the evidence base	10	☐	☐	☐	§6–§7
V49–V58	Ethics, consent and communication	10	☐	☐	☐	§8
V59–V68	Clinical examination and records	10	☐	☐	☐	§9–§10
V69–V80	Space analysis, diagnosis and treatment planning	12	☐	☐	☐	§11–§12
V81–V90	Extraction decisions and space management	10	☐	☐	☐	§13
V91–V102	Biomechanics and anchorage	12	☐	☐	☐	§14
V103–V111	Materials, wires and bonding	9	☐	☐	☐	§15
V112–V121	Removable, functional and fixed appliances	10	☐	☐	☐	§16–§17
V122–V129	Treatment mechanics and finishing	8	☐	☐	☐	§18
V130–V139	Retention and stability	10	☐	☐	☐	§19
V140–V151	Interceptive orthodontics and impacted teeth	12	☐	☐	☐	§20
V152–V161	Class II and Class III management	10	☐	☐	☐	§22–§23
V162–V173	Transverse and vertical problems	12	☐	☐	☐	§24–§25
V174–V183	TMD, the airway and the medically complex patient	10	☐	☐	☐	§26, §29
V184–V193	Adults, orthognathic surgery, clefts and governance	10	☐	☐	☐	§27, §28, §30
V194–V207	The questions that test judgement	14	☐	☐	☐	Handbook Part 5

SPEAK IT, TIME IT, SHORTEN IT

A good spoken answer runs twenty to forty seconds. On your first pass most answers will run to ninety, because you will say everything you know. On the second pass, say the same answer in half the time by leading with the conclusion and keeping only the reasoning that supports it. The third pass is for the sections you still dread — and those are the ones the examiner will find.

Index

By question number*, not page. Where a whole block is that term's subject the block is named in* ***bold*** *with its question range — that is the block to sit if you want to test the term deliberately; the plain numbers are the individual questions elsewhere in the bank that turn on it. Headwords are the same as Volume II's index, so a term looked up in one volume is found under the same word in the other.*

A

ABO Objective Grading System 639, 1019, 1021
Adams clasp 523, 532, 533, 542
Adenoid hypertrophy, adenoid facies **892**, 2, 967
Adjunctive treatment **929**, 272, 387, 896
Aesthetic Component, IOTN **174**, **178**, **193**, 282, 1020, 1021
Ageing versus relapse **655**, **656**
Aligners, clear **Block Q** (572–612), 137, 930
Alveolar bone graft **Block AB** (938–968), 933
Alveolar bone, created by eruption 50, 121, 710, 712, 868, 873, 929, 946, 947, 948, …
Anchorage **Block N** (433–485), 23, 100, 101, 103, 128, 129, 130, 133, 138, 226, …
— ladder, Proffit's 484
— loss, two-thirds 128, 402, 416, 695, 927
— skeletal **Block N** (433–485), 23, 252, 353, 356, 359, 375, 396, 407, 409, 626, …
Andrews' six keys **142**, **164**, 575, 602
Angle's classification **Block E** (142–171), 59, 269, 752
Ankylosis **Block T** (683–727), 43, 68, 72, 74, 85, 95, 99, 117, 118, 119, …
Antibiotic prophylaxis, NICE 958, 993, 996
Apert syndrome 957
Arch form **Block R** (613–647), 28, 63, 64, 65, 69, 76, 153, 334, 343, 351, …
Archwires **Block O** (486–530), 97, 100, 108, 115, 127, 138, 333, 368, 405, 416, …
Assent, child's 251
Asymmetry, facial **Block X** (823–847), 268, 270, 294, 309, 349, 635, 960
Audit **Block AD** (999–1023), 175, 181, 192, 271, 285, 286, 366, 595, 640, 941, …
Autotransplantation 406, 422, 689, 692

B

Balancing extraction **Block M** (393–432), 76, 77, 93, 341, 380, 666, 699, 750, 863, 880, …
Begg appliance 612
Bendback **Block R** (613–647)
Bimaxillary protrusion **738**, 928
Bisphosphonates 123, 991
Björk's rotations **41**, 13, 314
Body dysmorphic disorder **234**, **235**, **248**, 177, 921
Bolton analysis **Block K** (328–352), 57
Bonded retainer **Block S** (648–682), 136, 219, 242, 250, 381, 414, 477, 503, 516, 556, …
Bonding **Block O** (486–530), 69, 116, 220, 231, 236, 237, 256, 257, 281, 287, …
British Standards classification **143**, **145**, **171**, 269, 752
Buccal corridors 264, 642, 744, 746

C

Camouflage **Block L** (353–392), 23, 407, 758, 766, 767, 773, 784, 789, 795, 799, …
Canine, impacted maxillary **685**, 183
Canine substitution 351, 401, 414, 427, 667, 711, 713, 716, 717, 740
Capacity 231, 233, 488, 898
Centre of resistance / rotation **Block N** (433–485), 109, 622, 908
Cephalometrics **Block J** (288–327), 47, 283, 357, 459, 735, 745, 753, 756, 757, 794, …
— errors, the four **Block J** (288–327), 735, 756
— landmarks and norms 283, 302, 310, 311
Cervical vertebral maturation **Block A** (1–26), 117, 309, 322, 358, 359, 385, 388, 391, 437, 438, …
Chin cup **Block W** (793–822)
Circumferential supracrestal fiberotomy 648
Class I malocclusion **730**, **742**, **752**, 25, 93, 113, 145, 157, 158, 159, 165, 166, 183, …
Class II malocclusion **Block V** (753–792), 25, 113, 145, 165, 186, 361, 363, 375, 447, 459, …
— division 2 **Block V** (753–792), 113, 145, 447, 851, 855, 865, 870, 876, 978
— elastics 459
Class III malocclusion **Block W** (793–822), 145, 157, 158, 166, 183, 186, 557, 888
— elastics **Block R** (613–647), 359, 416, 459, 461, 468, 481, 507, 520, 545, 571, …
Cleft lip and palate **Block AB** (938–968), 29, 30, 31, 52, 808, 1023
Cleidocranial dysplasia 71, 955
Compensating extraction **398**, **411**, 76, 77, 93, 359, 376, 698, 699, 730
Compliance 181, 201, 224, 362, 363, 409, 480, 520, 541, 552, …
Condylar cartilage 5, 6, 26, 37, 42, 51, 781
Condylar hyperplasia 270, 833, 835, 841, 889, 959
Condylar trauma 993
Consent **Block H** (227–257), 103, 113, 137, 272, 273, 287, 317, 362, 368, 369, …
Cortical anchorage 356, 443, 485
Couple **Block N** (433–485), 31, 97, 107, 110, 506, 510, 526, 572, 573, 579, …
Crossbite **Block X** (823–847), 21, 148, 149, 151, 157, 169, 172, 176, 178, 180, …
Crowding, classification 415, 429, 728
— late lower incisor **Block S** (648–682), 67, 89, 90, 242, 418, 791, 990
— transient early 561, 727, 732
Curve of Spee **Block Y** (848–877), 142, 164, 328, 333, 342, 352, 473, 614, 619, 622, …

D

Dahl concept 871
Decompensation **Block AA** (908–937), 252, 358, 375,

376, 799, 800, 816, 822
Demineralisation **975**, **977**, 237, 362, 499, 503, 529, 644
Dental age **Block A** (1–26), 54, 67, 188, 373, 383, 656, 669, 712, 784, 825, ...
Diastema, midline 64, 278, 538, 658, 726, 732, 733, 828, 829, 830
Digit habit 152, 162, 276, 725, 756, 787, 842, 857, 858, 877, ...
Digital set-up **337**, **346**, 286, 418, 585, 600, 604
Distalisation **Block V** (753–792), 128, 133, 393, 395, 409, 420, 424, 436, 457, 460, ...
Down syndrome 958, 959
E
Eastman correction **Block J** (288–327)
Ectodermal dysplasia 956
Ectopic first permanent molar 73, 426, 695, 697
Edgewise appliance **Block Q** (572–612), 482
Envelope of discrepancy **Block L** (353–392), 914, 937
Equilibrium theory 153
Eruption **Block D** (97–141), 36, 43, 47, 50, 56, 60, 61, 62, 64, 65, ...
— primary failure of 74, 85, 117, 118, 119, 867, 955
Ethics 1017
Evidence, hierarchy of **210**, **225**
Evidence-based orthodontics 846
Expansion **Block X** (823–847), 21, 23, 82, 153, 214, 266, 335, 346, 371, 380, ...
Extraction **Block M** (393–432), 23, 57, 60, 61, 62, 63, 64, 66, 68, 69, ...
— pattern by class **Block M** (393–432), 82, 238, 353, 370, 380, 716, 722, 731, 768, 788, ...
— serial **Block T** (683–727), 82, 367, 385, 403, 430
F
Facemask **Block W** (793–822), 213, 485, 563, 760, 909, 914, 955, 957
Fibre recoil 655, 671, 672, 742, 751
Fiberotomy see *Circumferential*
Finishing **Block R** (613–647), 15, 23, 25, 26, 45, 54, 115, 168, 170, 217, ...
Fixed appliances **Block Q** (572–612), 23, 64, 66, 73, 78, 83, 112, 114, 116, 126, ...
Fluoride regimen 975, 977
Frankel appliance **Block P** (531–571)
Frankfort plane 259, 260, 283, 308
Frenectomy 64, 727, 732, 733, 734
Functional appliances **Block P** (531–571), 1, 7, 11, 12, 20, 22, 23, 25, 51, 199, ...
Functional matrix theory 5, 6, 26, 55
G–H
Genioplasty 364, 843, 866, 915, 918, 928
Gillick competence 257
Gingival hyperplasia 530, 985
Goldenhar syndrome 960
Governance **1016**, 241
Growth, adult 2, 3, 19, 25, 43, 48, 49, 357, 379, 566, ...
— concepts and control **6**, **11**, **26**, 46, 303, 438, 711, 804, 812, 819, 997
— craniofacial **37**, 6, 13, 26, 957
— modification 357, 379, 909
— prediction, discredited 19, 49, 850, 909
— spurt, adolescent 4, 717
Headgear **Block N** (433–485), 23, 128, 130, 131, 359, 363, 373, 376, 393, 397, ...
— safety, two devices 437, 559, 771, 782, 783, 989
Hemifacial microsomia **Block AB** (938–968), 835, 841
Hemimandibular elongation / hyperplasia 835
Herbst appliance **Block P** (531–571)
Holdaway angle 306, 801, 813
Houston's rule 310, 650, 777, 854, 864
Hyalinisation **Block D** (97–141), 433, 469, 909, 936
Hypodontia 8, 75, 84, 95, 183, 184, 186, 715, 717, 721, ...
I–J
ICON index 195, 1020
Impacted teeth see *Canine*, *Second molar*
Incisor liability **81**, 242, 700
Index, Little's Irregularity 345, 659
Indices **195**, 221, 355, 592, 641, 930, 1019, 1021
Infection control **1013**
Infra-occlusion 280, 710, 715, 717, 718, 719, 720, 721
Intention to treat 202
Interceptive orthodontics **705**, 74, 386
Interincisal angle **Block Y** (848–877), 41, 304, 306, 650, 666, 672, 755, 776, 777, 778, ...
Interproximal reduction **Block M** (393–432), 62, 136, 341, 346, 351, 521, 589, 591, 593, 605, ...
Intrusion, force and amount **Block Y** (848–877), 110, 111, 118, 438, 452, 455, 467, 479, 539, 616, ...
IOTN **174**, **178**, **193**, 263, 282, 1020, 1021
Juvenile idiopathic arthritis 841, 889, 890, 959, 993, 997
K–L
Laceback **Block R** (613–647)
Laser, diode 530
Latex allergy 507, 525, 987
Leeway space **Block C** (57–96), 334, 439, 461, 698, 700, 701, 702
Lingual appliances **Block Q** (572–612), 66, 78, 82, 83, 136, 227, 423, 439, 461, 484, ...
Lingual arch, lower **Block T** (683–727), 66, 78, 82, 83, 91, 423, 439, 461, 481, 652, ...
Lip trap **Block V** (753–792), 172, 176, 274, 325, 382, 675, 854, 916, 928, 937, ...
Lips, competence 172, 176, 179, 182, 189, 325, 354, 381, 382, 391, ...
M
Malocclusion, aetiology **165**, 756, 787, 808
— classification **145**, **159**, 752
— prevalence 808
Mandibular arch as template 342, 374, 614, 620
Materials **490**, **516**, 167, 168, 204, 220, 227, 228, 229, 254, 329, 330, ...
MBT prescription 576, 577
Medical history **993**
Miniplates 484, 485
Miniscrews **Block N** (433–485), 132, 133, 356, 363, 409, 423, 428, 626, 706, 769, ...
— failure by site **Block N** (433–485), 133
MIH 70, 92, 968
Moment-to-force ratio **Block N** (433–485), 107, 110, 405, 908
Montgomery standard 228, 229
Moyers analysis 332
Myofunctional therapy 152, 162, 370, 877, 896, 905
N–O
Nance button **Block N** (433–485), 626, 927
Nasolabial angle 274
Nickel allergy 507

Numbers, the 237, 320, 327, 507, 565, 593, 625
Obwegeser and Makek 835
Occlusion **Block E** (142–171), 6, 14, 23, 50, 64, 68, 72, 85, 88, 115, ...
Open bite, anterior **Block Y** (848–877), 40, 46, 151, 152, 157, 162, 182, 183, 260, 275, ...
Oral examination 278, 296
Orthognathic surgery **914**, **933**, 16, 213, 238, 358, 376, 377, 784, 797, 810, 815, ...
Osteoprotegerin **102**
Overbite **Block Y** (848–877), 40, 41, 46, 49, 50, 59, 82, 111, 142, 143, ...
Overjet, reduction **Block V** (753–792), 48, 190, 192, 211, 212, 261, 329, 341, 344, 386, ...

P

PAR index **Block F** (172–196), 221, 377, 1018, 1019, 1020, 1021
Parallax 33, 87, 169, 260, 262, 264, 291, 292, 322, 637, ...
Periodontal support, reduced 109, 193, 475, 748, 908
Photobiomodulation 125, 139, 208, 360, 469, 475, 545, 920
Pierre Robin sequence 32, 943, 953, 957
Planning, treatment **Block L** (353–392), 1, 7, 18, 20, 23, 46, 68, 96, 119, 170, ...
Pressure–tension theory **105**
Primary failure of eruption 74, 85, 117, 118, 119, 867, 955
Primate space 58, 60
Problem list **Block L** (353–392)
Pruzansky classification 954, 960
Pseudo-Class III 793, 805

Q–R

Radiation protection **1001**, 285
RANKL **102**, 469
Records **Block I** (258–287), 15, 17, 22, 24, 47, 143, 154, 169, 177, 178, ...
Recession 277, 282, 287, 354, 379, 381, 722, 726, 738, 739, ...
Referral, knowing your limits **Block AD** (999–1023), 7, 13, 17, 18, 24, 62, 64, 66, 72, 73, ...
Relapse **Block S** (648–682), 40, 49, 135, 136, 152, 153, 161, 164, 205, 219, ...
Removable appliances **Block P** (531–571), 66, 82, 83, 107, 136, 232, 278, 361, 577, 586, ...
Retention **Block S** (648–682), 8, 64, 68, 78, 82, 237, 245, 250, 257, 284, ...
Risks of treatment **987**, 114, 230, 237, 247, 256, 368, 369, 392, 554, 644, ...
Root resorption **Block D** (97–141), 73, 227, 237, 253, 293, 295, 368, 392, 431, 455, ...
Rotations, mandibular **Block B** (27–56), 13, 161, 314, 473, 660, 672, 726, 798, 803, 812, ...
Rule of tens 944, 966

S

SARPE 830, 845, 919
Second molar, impacted mandibular **Block T** (683–727), 56, 65, 93, 183, 412, 851
Self-ligation **Block Q** (572–612), 102, 113, 125, 214, 458, 490, 504, 505, 506, 515, ...
Serial extraction **Block T** (683–727), 82, 367, 385, 403, 430
Skeletal anchorage **Block N** (433–485), 23, 252, 353, 356, 359, 375, 396, 407, 409, 626, ...
Sleep apnoea **Block Z** (878–907), 214, 973, 974
Smile analysis 264, 746
Soft tissue response, 0.7 mm 736
Space analysis **Block K** (328–352), 63, 264, 304, 374, 378, 746
Stability, hierarchy of (surgical) **917**
Stainless steel **Block O** (486–530), 72, 405, 454, 466, 532, 543, 579, 615, 618, 623, ...
Superimposition **Block J** (288–327), 13, 16, 34, 35, 45, 47, 459, 802
Supernumerary teeth **Block C** (57–96), 297, 693, 947, 949, 954, 955

T

TADs see *Miniscrews*
Terminal plane 59, 60, 92
TMD **Block Z** (878–907), 149, 163, 227, 237, 249, 281, 561, 566, 803, 824, ...
Tooth movement, biology of **Block D** (97–141), 153, 208, 312, 314, 336, 346, 354, 357, 360, 370, ...
— optimal force 101, 138, 469, 534
— three phases 97, 98, 134, 469, 909, 936
Torque, third order **Block Q** (572–612), 453, 482, 510, 527, 614
Transposition 74, 740, 742
Transpalatal arch **Block N** (433–485), 363, 428, 521, 591, 623, 626, 657, 702, 769, 927
Trauma, dental 179, 359, 787, 970, 993
Treacher Collins syndrome 954, 959, 960
Turner tooth versus Turner syndrome 968
Twin block **Block P** (531–571), 14, 23, 48, 200, 667, 760, 766, 773, 780, 781, ...

U–Z

Veau classification 941
Vertical control 46, 303, 432, 438, 558, 610, 783, 788, 814
White spot lesions **975**, **977**, 209, 236, 237, 253, 362, 387, 502, 516, 644
Wire syndrome 652
Wits appraisal **Block J** (288–327), 801, 820